# Global Smoothness for the Three-Dimensional Incompressible Navier–Stokes Equations

with a Refutation of OpenAI's Finite-Time Blowup Claim

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September 2026

**Abstract**

Fix $\nu > 0$ and a smooth divergence-free finite-energy datum $u_0$. For the unforced equations, the datum generates one maximal pressure-normalized history. At every finite temporal depth and spatial height, the datum construction places the corresponding pressure-complete rectangle on the whole interval leading to a proposed finite terminal time. The compatible rectangles exhaust the mixed Sobolev intersection, while the completed critical-height identities recover every spatial row from those same finite coordinates. The adjacent rows have a common $H^\infty_\sigma$ endpoint; the equation recovers its complete velocity and pressure jet; and the local restart joins to the incoming history, excluding a finite maximal time. For a prescribed $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$, the finite-row construction is applied anew to the complete forced graph. The source pairing and pressure-potential bounds control every added affine force coordinate, the forced rectangles yield that endpoint regularity, and the restart retains the original absolute-time force. This gives a unique global smooth solution in both the unforced and forced settings.

The proposed field in OpenAI's *Finite Time Blowup for Navier–Stokes* is then tested against the forced theorem. Its full localization residual has uniformly bounded finite jets with compatible compactly supported terminal values; an explicit Borel construction extends it to a force in $C^\infty_c(\mathbb{R}^3 \times (0,\infty))$. The forced theorem supplies a global smooth solution for that zero datum, viscosity, and prescribed force. Strict-subinterval uniqueness identifies every claimed classical candidate with that solution on $[0,1)$. The resulting finite velocity and mixed rectangle bounds contradict both the asserted velocity blowup and the divergent positive exterior contribution to $R_{0,1}$ calculated from the proposed field. The field cannot solve the stated forced initial-value problem.

This manuscript was prepared with assistance from OpenAI's ChatGPT and Codex.

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## Part I

# How the Proof Works

The three-dimensional Navier–Stokes regularity problem asks whether a smooth fluid flow can lose its smoothness in finite time. We start with smooth, divergence-free initial velocity $u_0$, finite energy, and positive viscosity. The fluid evolves according to

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \qquad \nabla \cdot u = 0.$$

### Critical Height and Rescaling

The problem starts with a vortex.

As it lengthens, it thins and spins faster and faster as it gets smaller.

The concern is that it could keep doing this, with each contraction taking less time than the one before, until the core reaches zero width and the speed becomes unbounded. Infinitely many contractions could then fit into a finite time. This is a Zeno cascade ("Zeno" for infinitely many stages in finite time; "cascade" for the progression to smaller scales).

The equations have no minimum vortex width. To follow this possibility down through the scales, we need to compare a vortex with a smaller, faster version that still satisfies the equation. Reducing lengths by $\lambda > 1$ requires multiplying velocity by $\lambda$ and running the evolution $\lambda^2$ times as fast:

$$u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t), \qquad p_\lambda(x,t) = \lambda^2 p(\lambda x, \lambda^2 t).$$

The chain rule gives, with every right-hand side evaluated at $(\lambda x, \lambda^2 t)$,

$$\partial_t u_\lambda = \lambda^3 \partial_t u, \quad (u_\lambda \cdot \nabla)u_\lambda = \lambda^3 (u \cdot \nabla)u,$$
$$\nabla p_\lambda = \lambda^3 \nabla p, \qquad \nu \Delta u_\lambda = \lambda^3 \nu \Delta u.$$

Every term acquires a factor $\lambda^3$, leaving the equation unchanged at fixed viscosity $\nu$. If a flow repeated this contraction, its stage durations would satisfy

$$\tau_j = \lambda^{-2j} \tau_0, \qquad \sum_{j=0}^{\infty} \tau_j = \frac{\tau_0}{1 - \lambda^{-2}} < \infty.$$

For the energy measurement, squared speed grows by $\lambda^2$, but volume shrinks by $\lambda^3$, giving

$$E[u_\lambda](t) = \lambda^{-1} E[u](\lambda^2 t), \qquad E[u] = \frac{1}{2} \int_{\mathbb{R}^3} |u|^2 \, dx.$$

The smaller, faster vortex has less energy. A bound on total energy can miss precisely the concentration we are concerned about. The measurement has to include how sharply the velocity changes across the shrinking core.

A Sobolev norm gives these finer changes more weight. In the Fourier description of the velocity, a large spatial frequency $|\xi|$ means variation across a short distance. Weighting each frequency by $|\xi|^s$ gives the homogeneous Sobolev norm of order $s$:

$$\|u(t)\|_{\dot{H}^s}^2 = \int_{\mathbb{R}^3} |\xi|^{2s} |\widehat{u}(\xi,t)|^2 \, d\xi, \qquad \Lambda = (-\Delta)^{1/2}.$$

Here $\widehat{u}$ is the Fourier transform of the velocity, and $\Lambda^s$ multiplies it by $|\xi|^s$. Integer orders measure ordinary spatial derivatives; fractional orders give intermediate weights to fine structure.

The order $s$ determines how much weight the fine scales receive. We can find the point where that weight exactly offsets the loss of volume. In the Fourier transform, the velocity factor contributes $\lambda$ and the change of spatial variables contributes $\lambda^{-3}$. Substituting the resulting transform into the norm and then setting $\xi = \lambda\eta$ gives

$$\begin{aligned}
\widehat{u_\lambda}(\xi,t) &= \lambda^{-2}\widehat{u}(\xi/\lambda, \lambda^2 t), \\
\|u_\lambda(t)\|_{\dot{H}^s}^2 &= \lambda^{-4} \int_{\mathbb{R}^3} |\xi|^{2s} |\widehat{u}(\xi/\lambda, \lambda^2 t)|^2 \, d\xi \\
&= \lambda^{-4}\lambda^{2s}\lambda^3 \|u(\lambda^2 t)\|_{\dot{H}^s}^2 = \lambda^{2s-1}\|u(\lambda^2 t)\|_{\dot{H}^s}^2.
\end{aligned}$$

The frequency weight contributes $\lambda^{2s}$, balancing the remaining factor $\lambda^{-1}$ precisely when $2s - 1 = 0$. The dividing height is $s = \frac{1}{2}$. For $s < \frac{1}{2}$, contraction reduces the norm; for $s > \frac{1}{2}$, it increases the norm. At the dividing height $s = \frac{1}{2}$, the norm is unchanged by the equation's rescaling. This lets us use one bound at every vortex width, without shrinking the region of fast-moving fluid making the bound easier to satisfy.

This is why the critical height matters: concentration cannot make this measurement smaller simply by occupying less space. But its value can still change as the fluid evolves. Transport and viscosity both acquired a factor $\lambda^3$, so the rescaling has not decided which controls that change.

The equation gives their contributions to the critical energy directly. Pair it with $\Lambda u$ and integrate the viscous term by parts. The pressure pairing vanishes because $\nabla \cdot \Lambda u = 0$, giving

$$\frac{1}{2}\frac{d}{dt}\|u\|_{\dot{H}^{1/2}}^2 + \nu\|u\|_{\dot{H}^{3/2}}^2 = -\operatorname{Re}\langle \Lambda u, (u\cdot\nabla)u\rangle.$$

The positive term on the left measures viscous dissipation. The transport term on the right can increase the critical energy. To compare their sizes, use Hölder's inequality and the classical Sobolev inequality $\|w\|_{L^3} \le C_3\|w\|_{\dot{H}^{1/2}}$ on each of the three factors:

$$\begin{aligned}
|\langle \Lambda u, (u\cdot\nabla)u\rangle| &\le \|\Lambda u\|_{L^3}\|u\|_{L^3}\|\nabla u\|_{L^3} \\
&\le C\|u\|_{\dot{H}^{1/2}}\|u\|_{\dot{H}^{3/2}}^2, \qquad C = C_3^3.
\end{aligned}$$

Both contributions contain the $\dot{H}^{3/2}$ size. Their relative strength is governed by the critical $\dot{H}^{1/2}$ norm itself. Substitution into the balance makes this explicit:

$$\frac{1}{2}\frac{d}{dt}\|u\|_{\dot{H}^{1/2}}^2 + \big(\nu - C\|u\|_{\dot{H}^{1/2}}\big)\|u\|_{\dot{H}^{3/2}}^2 \le 0.$$

When $C\|u\|_{\dot{H}^{1/2}} < \nu$, the coefficient is positive and the critical energy decreases. If it starts below this threshold, the decrease keeps it there. For arbitrary initial size, the coefficient need not be positive, so this estimate cannot establish viscous control of the nonlinear contribution. The classical difficulty is to bound the critical norm throughout the evolution without assuming it is already small.

### Following One Fluid Motion

If a singularity were to actually form, the velocity would have to blow up, which means the acceleration would have to blow up, followed by its jerk, and so on, and so on.

The equation already contains the velocity and its time derivative. Differentiate the whole equation in time, once and then again:

$$\begin{aligned}
\partial_t u &= \nu\Delta u - (u\cdot\nabla)u - \nabla p,\\
\partial_t^2 u &= \nu\Delta\partial_t u - (\partial_t u\cdot\nabla)u - (u\cdot\nabla)\partial_t u - \nabla\partial_t p,\\
\partial_t^3 u &= \nu\Delta\partial_t^2 u - (\partial_t^2 u\cdot\nabla)u - 2(\partial_t u\cdot\nabla)\partial_t u - (u\cdot\nabla)\partial_t^2 u - \nabla\partial_t^2 p.
\end{aligned}$$

The derivatives distribute across the two velocity factors, producing the binomial coefficients. With $V_n = \partial_t^n u$ and $Q_n = \partial_t^n p$, the general form is

$$V_{n+1} = \nu\Delta V_n - \sum_{a=0}^{n}\binom{n}{a}(V_a\cdot\nabla)V_{n-a} - \nabla Q_n,$$

$$-\Delta Q_n = \partial_i\partial_j\sum_{a=0}^{n}\binom{n}{a}V_{a,i}V_{n-a,j}.$$

Taking the divergence gives the second equation; decay at spatial infinity fixes the pressure. Every $V_n$ is a derivative of the original velocity, and every $Q_n$ is determined by its velocity products. All of these derivatives belong to one fluid.

But the vortex is also narrowing across space. Each time derivative is a spatial field, with its own spatial derivatives. Retaining those gives

$$J_{n,\alpha} = \partial_x^\alpha\partial_t^n u, \qquad \Pi_{n,\alpha} = \partial_x^\alpha\partial_t^n p.$$

Here $\alpha$ records the spatial directions and derivative orders. Together the fields form the mixed jet. The product rule acts in both time and space:

$$J_{n+1,\alpha} = \nu\Delta J_{n,\alpha} - \nabla\Pi_{n,\alpha} - \sum_{a=0}^{n}\sum_{\beta\le\alpha}\binom{n}{a}\binom{\alpha}{\beta}(J_{a,\beta}\cdot\nabla)J_{n-a,\alpha-\beta}.$$

The two binomial factors count where those derivatives land. Pressure, viscosity, and transport remain together in every equation. These time derivatives are taken at fixed positions; along a moving fluid particle, the derivative is $D_t = \partial_t + u\cdot\nabla$. The transport already present in the equations accounts for that motion.

Return to the critical energy. We left it in a balance involving the spatial order 3/2. Write the energy at a general height $s$ as

$$E_s = \frac{1}{2}\|\Lambda^s u\|_2^2.$$

Its time derivative comes from the velocity equation. Substitution and integration by parts give

$$\begin{aligned} E_s' &= \operatorname{Re}\langle \Lambda^s u, \Lambda^s \partial_t u\rangle \\ &= -\nu\|\Lambda^{s+1}u\|_2^2 + \mathcal{P}_s \\ &= -2\nu E_{s+1} + \mathcal{P}_s, \end{aligned}$$

with the transport and pressure contributions collected in

$$\mathcal{P}_s = -\operatorname{Re}\langle \Lambda^s u, \Lambda^s((u\cdot\nabla)u + \nabla p)\rangle.$$

The viscous term contains $E_{s+1}$: integration by parts put one of the Laplacian's two derivatives on each velocity factor. Starting from $H = E_{1/2}$, differentiation of that next energy gives

$$\begin{aligned} H' &= -2\nu E_{3/2} + \mathcal{P}_{1/2}, \\ H'' &= -2\nu E_{3/2}' + \partial_t \mathcal{P}_{1/2} \\ &= 4\nu^2 E_{5/2} - 2\nu\mathcal{P}_{3/2} + \partial_t\mathcal{P}_{1/2}, \\ H^{(n)} &= (-2\nu)^n E_{n+1/2} + \sum_{j=0}^{n-1}(-2\nu)^j \partial_t^{n-1-j}\mathcal{P}_{j+1/2}. \end{aligned}$$

The higher we go in time derivatives, the higher we go in Sobolev space. Every step brings another factor $-2\nu$, with the complete transport and pressure derivatives alongside it. All of those terms are determined by the mixed jet of this fluid. The spatial heights run through

$$H^{1/2}, \qquad H^{3/2}, \qquad H^{5/2}, \qquad H^{7/2}, \qquad \ldots.$$

We began with a vortex getting smaller and spinning faster, with its velocity changing more sharply across the core. The higher Sobolev orders measure precisely those sharp changes. Above height $k + \frac{3}{2}$, Sobolev embedding bounds the ordinary spatial derivatives through order $k$:

$$H^s(\mathbb{R}^3) \hookrightarrow C_b^k(\mathbb{R}^3), \qquad s > k + \tfrac{3}{2}.$$

Any chosen derivative is covered by a sufficiently high finite level.

Continuing through all finite orders reaches the full smoothness condition:

$$L^2 \cap \bigcap_{n\geq 0} \dot{H}^{n+1/2}(\mathbb{R}^3) = \bigcap_{m\geq 0} H^m(\mathbb{R}^3) = H^\infty(\mathbb{R}^3) \hookrightarrow C_b^\infty(\mathbb{R}^3).$$

The $L^2$ term retains the large spatial scales; the increasing Sobolev orders retain the small ones. This is the full spatial regularity of the fluid.

See, the intuition here is what matters: when you follow a vortex down to its smallest scales, highest frequencies, and fastest velocities, you are following its successive time derivatives. For a singularity to occur, you would expect the spatial regularity to break down as you go up in those time derivatives. But what we see here is that viscosity relates that faster motion to more spatial regularity, not less. This is the opposite of what you would expect from a singularity or blowup.

## Proof and Application

Establishing this regularity through a proposed singular time requires bounds on the coupled derivatives throughout the motion.

Fix one $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$ and one $\nu > 0$. The subscript $\sigma$ records incompressibility. Suppose the smooth motion ends at a finite time $T_*$, and write $I = (0, T_*)$. The initial velocity determines the pressure and, through the equation, the subsequent changes in velocity:

$$u(t) = u_0 + \int_0^t \big[\nu\Delta u(s) - (u(s)\cdot\nabla)u(s) - \nabla p(s)\big]\, ds,$$
$$p(t) = R_i R_j\big(u_i(t)u_j(t)\big), \qquad 0 \le t < T_*.$$

Here $R_iR_j$ solves the pressure Poisson equation with its decaying normalization. Substitution gives the first time derivative,

$$V_1(t) = \nu\Delta u(t) - (u(t)\cdot\nabla)u(t) - \nabla Q_0(t),$$

then differentiation of the pressure product gives $Q_1$, and the next velocity equation gives $V_2$. At $t = 0$, this recurrence generates every initial derivative from $u_0$. Throughout the motion, it gives the mixed fields

$$J_{n,\alpha}(t,x) = \partial_x^\alpha V_n(t,x), \qquad \Pi_{n,\alpha}(t,x) = \partial_x^\alpha Q_n(t,x).$$

Arrange the time derivatives down the rows of a matrix and the spatial orders across its columns. Any chosen calculation uses a finite depth $N$ and height $M$, which the paper calls a rectangle. Each entry still depends on all of space and on the entire interval $I$. Its equation retains the neighbouring derivatives in the Laplacian, both factors of every transport product, and the pressure determined by those factors.

The viscous relationship is an exact relation between entries of this matrix. Rearranging the time equation gives

$$\nu\Delta V_n = V_{n+1} + \sum_{a=0}^{n} \binom{n}{a} (V_a\cdot\nabla)V_{n-a} + \nabla Q_n.$$

The next time response is tied to two more spatial derivatives, together with the complete transport and pressure sum. The initial jet fixes where every row begins. The construction retains these equations together on $I$, with pressure fixed by its velocity products, and takes finite records of their compatible Sobolev realization. Increasing a rectangle reads more derivatives of this motion.

A bound that deteriorates as $t$ approaches $T_*$ would still leave the possible singularity unresolved. The whole-interval construction result in Theorem 3.9 establishes the paired memberships on $I = (0, T_*)$:

$$V_n \in L^2(I; H^{M+1}),$$
$$V_{n+1} \in L^2(I; H^{M-1}), \qquad 0 \le n \le N,$$

for every finite $N, M$. The paired heights $M+1$ and $M-1$ differ by the Laplacian's two derivatives. The rectangle $\mathcal{R}_{N,M}[u_0; T_*]$ has finite size

$$\mathfrak{R}_{N,M}(u; T_*) = \sum_{n=0}^{N} \Big( \|V_n\|^2_{L^2(I;H^{M+1})} + \|V_{n+1}\|^2_{L^2(I;H^{M-1})} \Big) < \infty.$$

Each finite depth and height has its own bound. Restricting the integral to an earlier time can only decrease it.

Time integration alone can allow a tall, narrow spike. Here the adjacent row also controls the rate of change. Differentiating the squared $H^M$ norm and splitting its spatial weights gives

$$\frac{d}{dt}\|V_n(t)\|^2_{H^M} = 2\,\mathrm{Re}\langle (1-\Delta)^{(M+1)/2} V_n, (1-\Delta)^{(M-1)/2} V_{n+1} \rangle_{L^2}.$$

The two factors are precisely the pair in the rectangle. Integration, Cauchy–Schwarz, and $2ab \le a^2 + b^2$ give

$$\sup_{0 \le t < T_*} \|V_n(t)\|^2_{H^M} \le \|V_n(0)\|^2_{H^M} + \|V_n\|^2_{L^2(I;H^{M+1})} + \|V_{n+1}\|^2_{L^2(I;H^{M-1})}.$$

The initial value is fixed by $u_0$, and the integrals remain finite through $T_*$. The row is bounded at every time, including the approach to the proposed singularity. Increasing the rectangle gives this control at each further derivative order.

Pressure follows the velocity products in its Poisson equation. A spatial product estimate, followed by Cauchy–Schwarz in time, gives

$$\|Q_n\|_{L^1(I;H^M)} \le C^{\mathrm{p}}_M \sum_{a=0}^{n} \binom{n}{a} \|V_a\|_{L^2(I;H^{M+1})} \cdot \|V_{n-a}\|_{L^2(I;H^{M+1})} < \infty,$$

where $C^{\mathrm{p}}_M$ is the spatial product constant. Spatial differentiation distributes the derivatives across those factors, so a sufficiently large rectangle covers every chosen pressure derivative too.

The rectangles agree where they overlap: an entry $(n, \alpha)$ always means $\partial_x^\alpha \partial_t^n u$, beginning with the derivatives generated by $u_0$. Enlarging the matrix preserves its existing entries. This compatibility lets us collect their bounds without changing the fluid being described.

Choose any time order $r$ and spatial order $m$. A large enough rectangle contains $V_0, \dots, V_r$ in $L^2(I; H^m)$, giving

$$\|u\|^2_{H^r(I;H^m)} = \sum_{n=0}^{r} \|V_n\|^2_{L^2(I;H^m)} < \infty.$$

Since every finite pair $(r, m)$ is covered,

$$u \in \bigcap_{r,m \ge 0} H^r(I; H^m(\mathbb{R}^3)).$$

The full intersection means that each finite number of time and spatial derivatives has a finite norm through $T_*$. Every membership concerns the fluid that began at $u_0$.

The critical-height equations express the spatial side of this construction. Rearranging the identity already derived isolates each higher spatial energy:

$$(-2\nu)^k E_{k+1/2} = H^{(k)} - \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} \mathcal{P}_{j+1/2}, \qquad k \geq 1.$$

Positive $\nu$ makes its coefficient nonzero. Finite selections form the critical-height rectangles, obtained by differentiating the scalar energy. The mixed rectangles come from differentiating the velocity and pressure fields. Their common entries agree because both calculations use the original equation for $u$.

On the constructed whole-interval jet, these identities recover its half-integer spatial energies. Kinetic energy covers low frequencies; a weight of order $k + \frac{1}{2} \geq m$ covers the high frequencies needed for any chosen integer order $m$. Together they give every finite spatial Sobolev order.

The equation also lets us recover the other rows from this complete spatial information. At any chosen height $m$, two additional spatial orders give

$$\|(u \cdot \nabla)u\|_{L^1(I;H^m)} \leq C_m \|u\|^2_{L^2(I;H^{m+2})},$$
$$\|\Delta u\|_{L^1(I;H^m)} \leq \sqrt{T_*} \|u\|_{L^2(I;H^{m+2})},$$

with $C_m$ depending on the spatial order. The pressure formula gives an integrable pressure gradient as well. The right-hand side of the velocity equation is now integrable through $T_*$, so its integral gives a continuous terminal velocity at every height. The recurrences then give $Q_0, V_1$, followed by $Q_1, V_2$, and so on. The time and spatial descriptions recover one another with viscosity, pressure, and transport present throughout.

There is still a distinction between boundedness before $T_*$ and reaching a state from which the fluid can continue. The next time row settles it. Integrating $V_{n+1}$ between two late times gives

$$\begin{aligned} \|V_n(t) - V_n(s)\|_{H^m} &\leq \int_s^t \|V_{n+1}(\tau)\|_{H^m} \, d\tau \\ &\leq (t-s)^{1/2} \|V_{n+1}\|_{L^2(s,t;H^m)}, \qquad s < t < T_*. \end{aligned}$$

The remaining change tends to zero. Each row reaches a limit in every finite $H^m$, and the limits at different heights agree. The velocity reaches a smooth divergence-free state $u_* \in H^\infty_\sigma(\mathbb{R}^3)$. The pressure formula and differentiated equations pass to these limits, giving its complete velocity and pressure jet.

Local existence starts a smooth solution from $u_*$. Its derivatives match the incoming ones, and uniqueness joins the continuation to the incoming motion. The fluid passes through $T_*$. It cannot be a finite maximal time.

Each finite part of the matrix also gives concrete bounds on any finite interval $[0, T]$. For a time derivative of order $r$ and a spatial derivative of order $k$, choose $m > k + \frac{3}{2}$. Sobolev embedding then bounds its largest size anywhere in the fluid:

$$\sup_{0 \leq t \leq T} \|\partial_t^r u(t)\|_{W^{k,\infty}} \leq C_{m,k} \sup_{0 \leq t \leq T} \|\partial_t^r u(t)\|_{H^m} < \infty.$$

Here $C_{m,k}$ is the embedding constant. In particular, the velocity gradient has a finite time integral. That gradient bounds how quickly neighbouring fluid particles approach each other. For particles starting at $a$ and $b$, integration along their trajectories gives

$$|X(a,t) - X(b,t)| \geq |a-b| \exp\left( - \int_0^t \|\nabla u(s)\|_\infty \, ds \right) > 0.$$

The vortex can stretch and narrow, but a positive material separation retains a positive lower bound throughout each finite interval.

So, for example, choose the rectangle with $N = 0$ and $M = 1$. It contains the velocity in $L^2(0,T;H^2)$ and its time derivative in $L^2(0,T;L^2)$. Write their sizes as

$$A_T = \left( \int_0^T \|u(t)\|_{H^2}^2 \, dt \right)^{1/2},$$
$$B_T = \left( \int_0^T \|\partial_t u(t)\|_2^2 \, dt \right)^{1/2}.$$

The adjacent-row calculation now bounds the $H^1$ norm at every time. Pair $u_t$ with $(1-\Delta)u$ and integrate:

$$\begin{aligned} \|u(t)\|_{H^1}^2 &= \|u_0\|_{H^1}^2 + 2 \operatorname{Re} \int_0^t \langle (1-\Delta)u(s), \partial_s u(s) \rangle_{L^2} \, ds \\ &\leq \|u_0\|_{H^1}^2 + 2 A_T B_T. \end{aligned}$$

These two entries give three familiar regularity estimates. First, the $L^3$ velocity norm is unchanged by the vortex rescaling, so it measures concentration across scales. The spatial embedding

$$\|f\|_{L^3} \leq \frac{\|f\|_{H^1}}{2^{5/6} \pi^{1/6}}$$

turns the $H^1$ bound into the Escauriaza–Seregin–Šverák endpoint estimate. The numerical constants here use the paper's unitary Fourier normalization:

$$\sup_{0 \leq t \leq T} \|u(t)\|_{L^3} \leq \frac{\left( \|u_0\|_{H^1}^2 + 2 A_T B_T \right)^{1/2}}{2^{5/6} \pi^{1/6}}.$$

Second, the $H^2$ entry bounds the largest speed anywhere in the fluid:

$$\|u(t)\|_{L^\infty} \leq \frac{\|u(t)\|_{H^2}}{\sqrt{8\pi}}.$$

Its squared time integral is already contained in $A_T^2$, giving the Serrin estimate,

$$\left( \int_0^T \|u(t)\|_{L^\infty}^2 \, dt \right)^{1/2} \leq \frac{1}{\sqrt{8\pi}} \left( \int_0^T \|u(t)\|_{H^2}^2 \, dt \right)^{1/2} = \frac{A_T}{\sqrt{8\pi}}.$$

Third, the spatial $H^2 \hookrightarrow W^{1,3}$ estimate bounds the velocity gradient:

$$\|\nabla u(t)\|_{L^3} \le \frac{\sqrt{3}\,7^{1/6}}{2^{5/3}\pi^{1/6}}\|u(t)\|_{H^2}.$$

Its squared time integral is again bounded by $A_T^2$, giving the Beirão da Veiga estimate [2]:

$$\left(\int_0^T \|\nabla u(t)\|_{L^3}^2\,dt\right)^{1/2} \le \frac{\sqrt{3}\,7^{1/6}}{2^{5/3}\pi^{1/6}}\left(\int_0^T \|u(t)\|_{H^2}^2\,dt\right)^{1/2} = \frac{\sqrt{3}\,7^{1/6}}{2^{5/3}\pi^{1/6}}A_T.$$

The memberships measured by $\mathfrak{R}_{0,1}$ give all three estimates. Choosing higher rectangles extends the calculation to higher derivatives.

An external force changes the acceleration of the fluid. It enters both the velocity equation and the pressure that enforces incompressibility:

$$u_t + (u\cdot\nabla)u + \nabla p = \nu\Delta u + f, \qquad \nabla\cdot u = 0,$$
$$-\Delta p = \partial_i\partial_j(u_i u_j) - \nabla\cdot f.$$

Fix the initial velocity, positive viscosity, and the prescribed force $f$. The forced theorem takes $f$ smooth through each finite time, with its mixed derivatives decaying rapidly in space, uniformly on finite intervals: $f \in C^\infty([0,\infty);\mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$. A smooth force with compact support in space and time belongs to this class. Its size enters the bounds. Writing $F_n = \partial_t^n f$, the differentiated equations become

$$V_{n+1} = \nu\Delta V_n - \sum_{a=0}^{n}\binom{n}{a}(V_a\cdot\nabla)V_{n-a} - \nabla Q_n + F_n,$$
$$-\Delta Q_n = \partial_i\partial_j\sum_{a=0}^{n}\binom{n}{a}V_{a,i}V_{n-a,j} - \nabla\cdot F_n.$$

Spatial differentiation adds $\partial_x^\alpha F_n$ to each mixed row. The initial recurrence uses $F_n(0)$ alongside $u_0$, and the pressure responds to the force through its divergence.

In the energy balance, the additional term is the work done by the force:

$$E_s' = -2\nu E_{s+1} + \mathcal{P}_s + W_s, \qquad W_s = \operatorname{Re}\langle\Lambda^s u, \Lambda^s f\rangle.$$

The viscous coefficient stays $-2\nu$. Repeated differentiation now includes the derivatives of the force work:

$$H^{(k)} = (-2\nu)^k E_{k+1/2} + \sum_{j=0}^{k-1}(-2\nu)^j\partial_t^{k-1-j}(\mathcal{P}_{j+1/2} + W_{j+1/2}).$$

Every derivative of $W_s$ pairs velocity derivatives with known derivatives of $f$. Their contribution to a selected row can be bounded directly. Split the spatial weights between $V_n$ and $F_n$, then apply Cauchy–Schwarz and Young's inequality:

$$2\left|\langle V_n, F_n\rangle_{H^M}\right| \le \nu\|V_n\|_{H^{M+1}}^2 + \nu^{-1}\|F_n\|_{H^{M-1}}^2.$$

The first term is absorbed at the spatial order of the viscous estimate. The second has a finite time integral because the force is prescribed and smooth through $T_*$. Its pressure contribution is controlled by the Poisson formula. Retaining these source terms, the forced construction supplies the paired whole-interval bounds

$$\mathfrak{R}^f_{N,M}(u;T_*) = \sum_{n=0}^{N} \Big( \|V_n\|^2_{L^2(I;H^{M+1})} + \|V_{n+1}\|^2_{L^2(I;H^{M-1})} \Big) < \infty$$

for every finite $N, M$. The bounds now depend on the force as well as $u_0$ and $\nu$. The adjacent-row calculation again gives bounds at every time, and compatibility collects the finite orders into

$$u \in \bigcap_{r,m\geq 0} H^r(I;H^m(\mathbb{R}^3)).$$

Integrating the next time derivative makes the remaining change tend to zero at $T_*$. The fluid reaches a smooth state $u_*$, with pressure and higher derivatives fixed by the forced recurrence and the values $F_n(T_*)$. Local existence continues from that state with the prescribed $f(t)$ at its original time. The derivatives agree across $T_*$, and uniqueness joins the incoming and outgoing motion. The proposed singular time is again an ordinary time of the flow.

The bounds reach each term of the momentum equation individually:

$$\sup_{0\leq t\leq T} (\|u_t\|_{H^m} + \|(u\cdot\nabla)u\|_{H^m} + \|\nabla p\|_{H^m} + \nu\|\Delta u\|_{H^m}) < \infty, \qquad m < \infty.$$

The time derivative is a row of the matrix; the Laplacian uses two more spatial orders; transport is a product of controlled derivatives; pressure is determined by the velocity and force. Sobolev embedding bounds their pointwise sizes too. Thus each term stays finite, even when the terms partly cancel in the equation.

OpenAI's proposed blowup starts from rest, uses a smooth force with compact support, and concentrates increasing velocity into a shrinking region while keeping kinetic energy finite.[1] This returns to the opening problem: total energy can remain finite while velocity concentrates into a smaller region. Write $v$ for the proposed velocity and put its proposed singular time at $t = 1$. With $\tau = 1 - t$, its claimed growth in (10.21) implies

$$\|v(1-\tau)\|_{L^\infty} \geq c\,\tau^{-A} \longrightarrow \infty, \qquad c > 0, \quad A > 0.$$

The relevant bound is on the largest speed. If the proposed force has its stated smooth extension through $t = 1$, denote it by $f_{\mathrm{OA}}$. The forced theorem gives a smooth solution $(U, P)$ from rest for that force and viscosity, through $t = 1$.

Before $t = 1$, both $v$ and $U$ solve the prescribed initial value problem. Their difference $z = v - U$ starts at zero. Subtracting the equations cancels the force; pairing with $z$ and using incompressibility gives

$$\frac{1}{2}\frac{d}{dt}\|z\|_2^2 + \nu\|\nabla z\|_2^2 \leq \|\nabla U\|_\infty \|z\|_2^2, \qquad z(0) = 0.$$

The coefficient has a finite time integral, so Gronwall's inequality keeps the difference zero. On every closed interval ending before 1, $v = U$.

[1] OpenAI, *Finite Time Blowup for Navier–Stokes*, Theorem 1.1 and Section 10, especially (10.20)–(10.21).

The rectangle $N = 0, M = 2$ contains $U$ at spatial height 3 and $U_t$ at height 1. Their adjacent-row trace bounds $H^2$ at every time, which bounds the speed. With zero initial velocity,

$$\mathfrak{R}^f_{0,2}[U] = \|U\|^2_{L^2(0,1;H^3)} + \|U_t\|^2_{L^2(0,1;H^1)} < \infty,$$

$$\sup_{0\le t<1} \|v(t)\|^2_{L^\infty} = \sup_{0\le t<1} \|U(t)\|^2_{L^\infty} \le \frac{\mathfrak{R}^f_{0,2}[U]}{8\pi} < \infty.$$

The proposed divergence contradicts this finite bound. Under the stated smooth-force hypothesis, the flow continues smoothly through $t = 1$; the divergent field cannot solve that initial value problem.

The vortex can stretch, narrow, and gain energy from the force. Its velocity, pressure, and derivatives remain coupled throughout that motion. The finite bounds obtained from their viscous relationship survive down through the scales, and the fluid reaches a smooth state from which it continues.

# Part II
# The Unforced Equations

**Theorem.** *For every $\nu > 0$ and every $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, there exists a pressure-normalized pair*

$$(u, p) \in C^\infty([0,\infty) \times \mathbb{R}^3; \mathbb{R}^3 \times \mathbb{R})$$

*satisfying*

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0, \qquad u(0) = u_0, \qquad \lim_{|x|\to\infty} p(t,x) = 0,$$

*and*

$$\frac{1}{2}\|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u(s)\|_{L^2}^2 \, \mathrm{d}s = \frac{1}{2}\|u_0\|_{L^2}^2 \qquad (t \geq 0).$$

*Every global Leray–Hopf pair $(v, q)$ at viscosity $\nu$ with datum $u_0$, in the sense of Definition 8.9, has $v = u$, and $q = p + c(t)$ distributionally for a scalar distribution $c$ in time. If $q$ is represented by a locally integrable function with $q(t,\cdot) \in C_0(\mathbb{R}^3)$ for almost every $t$, then $q = p$ almost everywhere.*

Every solenoidal Schwartz datum belongs to $H^\infty_\sigma(\mathbb{R}^3)$. Restricted to that subclass, the theorem gives the global smooth pair and bounded kinetic energy required by the whole-space positive formulation in [5]; the theorem's datum class is broader.

## 1 Introduction

Fix $\nu > 0$ and a divergence-free datum $u_0 \in H^\infty(\mathbb{R}^3; \mathbb{R}^3)$. Smooth-data local theory gives a unique classical solution on a maximal interval $[0, T_*)$ [6, 8]. Throughout Part II, $(u, p)$ denotes that solution. The problem is to decide whether $T_*$ can be finite. On each strict subinterval $[0, T]$, smoothness already makes every finite Sobolev norm finite. That fact alone says nothing about what happens as $T \uparrow T_*$: the bound may depend on $T$ and diverge at the endpoint. The proof must instead control the approach to $T_*$ with constants that do not depend on the cutoff.

Leray constructed global finite-energy weak solutions [10], and Caffarelli, Kohn, and Nirenberg bounded the size of the possible singular set for suitable weak solutions [3]. Serrin proved regularity under scale-balanced spacetime integrability hypotheses [13]; Escauriaza, Seregin, and Šverák treated the endpoint $L^\infty_t L^3_x$ case [4]. Fujita–Kato and Kato supply the strong local theory used below at the terminal restart [6, 8], and Koch–Tataru place small-data global well-posedness in the critical space $\mathrm{BMO}^{-1}$ [9]. Tao gives a localization and compactness formulation of the global problem [15]. The regularity criteria show that control of an appropriate critical quantity excludes singularity. The argument below derives the required control from the compatible finite rectangles of the maximal velocity–pressure history. Mixed-index exhaustion and the completed critical-height relationships use those same rectangles and lead to that all-orders Sobolev membership.

Pressure is part of that derivative structure. With the decay normalization, incompressibility determines it from the current velocity:

$$-\Delta p = \partial_i \partial_j (u_i u_j), \qquad p = R_i R_j (u_i u_j).$$

Here $R_i$ are the Riesz transforms, and repeated indices are summed. Write

$$V_n := \partial_t^n u, \qquad Q_n := \partial_t^n p.$$

Differentiating the equation gives

$$Q_n = R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i} V_{n-a,j}, \tag{1}$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^{n} \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n. \tag{2}$$

Thus the time derivatives, spatial derivatives, and pressure derivatives are coordinates of one solution. A time derivative raises the order of the viscous term by two spatial derivatives and distributes the transport and pressure terms among earlier time derivatives. The resulting estimates must track temporal depth and spatial depth independently.

The Sobolev consequences used below have a simple form. For any prescribed spatial derivative order $k$, a Sobolev order $s > k + 3/2$ gives a continuous derivative. Applying this at every finite order gives the spatial column

$$L^2 \cap \bigcap_{j \in \mathbb{N}_0} \dot{H}^{j+1/2} = \bigcap_{m \in \mathbb{N}_0} H^m \hookrightarrow C^\infty.$$

The corresponding one-dimensional embedding, $\bigcap_r H^r(0, T_*) \hookrightarrow C^\infty(0, T_*)$, applies to the temporal-height column of $H(t) = \frac{1}{2} \left\| \Lambda^{1/2} u(t) \right\|_2^2$. The viscous contribution to $H'$ is $-\nu \left\| \Lambda^{3/2} u \right\|_2^2$. Each further time derivative exposes the next spatial energy together with differentiated transport and pressure terms. Subtracting those terms from $H^{(k)}$ and dividing by $(-2\nu)^k$ recovers $\frac{1}{2} \left\| \Lambda^{k+1/2} u \right\|_2^2$, as shown in Proposition 4.3.

For finite $N, M \in \mathbb{N}_0$, let

$$\mathcal{R}_{N,M}[u_0; T_*] := \left( (V_n)_{n=0}^N, (V_{n+1})_{n=0}^N; (Q_n)_{n=0}^N \right)$$

be the corresponding pressure-complete rectangle of the maximal history. The datum-generated rectangle theorem places these selected rows on the whole interval $(0, T_*)$, and enlargement preserves every coordinate already present. Reading the compatible family by its temporal and spatial indices gives

$$u \in \bigcap_{r,m \in \mathbb{N}_0} H^r(0, T_*; H^m) \tag{3}$$

on a proposed finite maximal interval. Reading this family through the recurrences (1)–(2) and the completed heights $R_k = E_{k+1/2}$ first gives $u \in \bigcap_m L^2(0, T_*; H^m)$; the recurrence then recovers every temporal and pressure row. Both readings use one datum-generated compatible family.

The rectangle and its norm are different objects. The rectangle retains $V_0, \ldots, V_N$, their next time derivatives, and their pressure rows. The scalar sum of its selected velocity norms on a strict interval is

$$\mathfrak{R}_{N,M}(u; T) := \sum_{n=0}^{N} \|V_n\|^2_{L^2(0,T;H^{M+1})} + \sum_{n=0}^{N} \|V_{n+1}\|^2_{L^2(0,T;H^{M-1})}. \tag{4}$$

Each term measures one coordinate of that compatible rectangle. Its full-interval membership, written in mixed-row coordinates, gives

$$\sup_{0<T<T_*} \mathfrak{R}_{N,M}(u; T) \le \sum_{n=0}^{N} \left( \|V_n\|^2_{L^2(0,T_*;H^{M+1})} + \|V_{n+1}\|^2_{L^2(0,T_*;H^{M-1})} \right) < \infty. \tag{5}$$

Theorem 4.6 proves the finiteness of this rectangle norm. The constant may depend on $N, M$, but is fixed independently of the cutoff $T$. We do not take a supremum over derivative orders.

Larger rectangles literally restrict to identical rows in every smaller rectangle:

$$\mathcal{R}_{N,M}[u_0;T_*]\big|_{(N',M')} = \mathcal{R}_{N',M'}[u_0;T_*] \qquad (N' \le N,\ M' \le M). \tag{6}$$

Every finite mixed row occurs in a sufficiently large rectangle. Theorem 5.2 proves this compatibility and exhaustion. The mixed-index route gives (3) directly. The completed-height identities are evaluated on these rectangles; exhausting their heights gives the spatial Sobolev column and then, by the equation, that mixed intersection.

For the velocity-time column, time Sobolev embedding applies to $H^m$-valued functions: fixing $m$ in (3) gives $u \in C^\infty(0,T_*;H^m)$. Taking every finite $m$ then gives joint smoothness in time and space. The pressure formula and its differentiated recurrences determine the pressure rows from these velocity rows. Applying the temporal and spatial embeddings to that full pressure-explicit jet gives joint smoothness of $(u,p)$. Each complete representation thus has its own smoothness implication; their compatibility identifies the derivatives used throughout the proof.

The zeroth time row of (3) gives, for each finite $m$, $u \in L^2(0,T_*;H^{m+2})$. Sobolev multiplication gives

$$\|\mathbb{P}\nabla\cdot(u\otimes u)\|_{H^m} \le C_m \|u\|^2_{H^{m+2}}.$$

Since $T_* < \infty$, the nonlinear term and $\Delta u$ belong to $L^1(0,T_*;H^m)$; for the linear term, explicitly,

$$\|\Delta u\|_{L^1(0,T_*;H^m)} \le T_*^{1/2} \|u\|_{L^2(0,T_*;H^{m+2})}.$$

The equation

$$u_t = \nu\Delta u - \mathbb{P}\nabla\cdot(u\otimes u)$$

now gives

$$u \in W^{1,1}(0,T_*;H^m) \hookrightarrow C([0,T_*];H^m) \qquad (m<\infty).$$

The endpoint values agree under the Sobolev inclusions because they are limits of one function $u(t)$. They define one terminal state

$$u_* := u(T_*) \in \bigcap_{m<\infty} H^m.$$

Since $u_* \in H^\infty_\sigma$, local well-posedness restarts the equation at $T_*$. Equations (1)–(2) recover the endpoint pressure and time derivatives. Uniqueness joins the restarted solution to $u$, producing an extension past $T_*$. This contradicts maximality and proves global smoothness.

## 2 Analytic framework

The argument has two analytic joints. Products and Fourier multipliers must keep every differentiated velocity–pressure row in a common Sobolev scale; at the terminal time, an $H^3$ local theory must restart that same pressure-normalized history. We record the estimates needed at those two joints, with the constants later used in the quantitative bounds.

**Definition 2.1.** For $f \in \mathcal{S}(\mathbb{R}^3)$, set

$$\widehat{f}(\xi) := (2\pi)^{-3/2} \int_{\mathbb{R}^3} e^{-ix\cdot\xi} f(x)\,\mathrm{d}x, \qquad \langle\xi\rangle := (1+|\xi|^2)^{1/2}.$$

This integral defines $\widehat{f}$ for $f \in L^1(\mathbb{R}^3)$. The Fourier transform on $\mathcal{S}'(\mathbb{R}^3)$ is its extension by duality. For $s \in \mathbb{R}$, define

$$H^s(\mathbb{R}^3) := \left\{ f \in \mathcal{S}'(\mathbb{R}^3) : \langle\xi\rangle^s \widehat{f} \in L^2(\mathbb{R}^3) \right\},$$

and define the finite homogeneous-norm domain by

$$\dot{H}^s(\mathbb{R}^3) := \left\{ f \in \mathcal{S}'(\mathbb{R}^3) : \begin{array}{l} \widehat{f} \text{ is represented by a measurable function on } \mathbb{R}^3, \\ |\xi|^s \widehat{f} \in L^2(\mathbb{R}^3) \end{array} \right\}.$$

On these domains set

$$\|f\|_{H^s}^2 := \int_{\mathbb{R}^3} \langle\xi\rangle^{2s} \left|\widehat{f}(\xi)\right|^2 \mathrm{d}\xi, \qquad \|f\|_{\dot{H}^s}^2 := \int_{\mathbb{R}^3} |\xi|^{2s} \left|\widehat{f}(\xi)\right|^2 \mathrm{d}\xi.$$

Set $A := (1-\Delta)^{1/2}$, $\Lambda := (-\Delta)^{1/2}$,

$$H^s_\sigma := \{v \in H^s(\mathbb{R}^3;\mathbb{R}^3) : \nabla\cdot v = 0\}, \qquad H^\infty := \bigcap_{m\in\mathbb{N}_0} H^m, \qquad H^\infty_\sigma := \bigcap_{m\in\mathbb{N}_0} H^m_\sigma.$$

$$\mathcal{S}_\sigma(\mathbb{R}^3) := \{v \in \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3) : \nabla\cdot v = 0\}.$$

For a vector or tensor $F = (F_I)_I$, all Sobolev norms are componentwise:

$$\|F\|_{H^s}^2 := \sum_I \|F_I\|_{H^s}^2, \qquad \|F\|_{\dot{H}^s}^2 := \sum_I \|F_I\|_{\dot{H}^s}^2.$$

Repeated Latin indices are summed from 1 to 3.

**Lemma 2.2** (Sobolev nesting and embedding)**.** *If $s_2 \geq s_1$, then*

$$\|f\|_{H^{s_1}} \leq \|f\|_{H^{s_2}}.$$

*If $k \in \mathbb{N}_0$ and $s > k + 3/2$, then*

$$\max_{|\alpha|\leq k} \|\partial^\alpha f\|_{L^\infty} \leq C^{\mathrm{emb}}_{s,k} \|f\|_{H^s},$$

*where*

$$C^{\mathrm{emb}}_{s,k} := (2\pi)^{-3/2} \max_{0\leq j\leq k} \left( \int_{\mathbb{R}^3} |\xi|^{2j} \langle\xi\rangle^{-2s}\,\mathrm{d}\xi \right)^{1/2} < \infty.$$

*In particular,*

$$H^\infty(\mathbb{R}^3) \hookrightarrow C^\infty_b(\mathbb{R}^3).$$

*Proof.* The first assertion follows from $\langle\xi\rangle^{2s_1} \le \langle\xi\rangle^{2s_2}$. For $|\alpha| \le k$, Fourier inversion and Cauchy–Schwarz give

$$|\partial^\alpha f(x)| \le (2\pi)^{-3/2} \left( \int_{\mathbb{R}^3} |\xi|^{2|\alpha|} \langle\xi\rangle^{-2s} \, \mathrm{d}\xi \right)^{1/2} \|f\|_{H^s} .$$

The integral is finite exactly when $s > |\alpha| + 3/2$. Approximation by Schwartz functions and dominated convergence give continuous representatives. For each $k$, choose $s = k + 2$. □

**Lemma 2.3** (Explicit three-dimensional homogeneous Sobolev constants)**.** *For $3 \le p \le 6$, set*

$$s(p) := 3\left(\frac{1}{2} - \frac{1}{p}\right)$$

*and*

$$\mathsf{S}(p) := 2^{-s(p)} \pi^{-s(p)/2} \left( \frac{\Gamma((3 - 2s(p))/2)}{\Gamma((3 + 2s(p))/2)} \right)^{1/2} \left( \frac{\Gamma(3)}{\Gamma(3/2)} \right)^{s(p)/3} . \tag{7}$$

*Then every scalar, vector, or finite tensor field $F$ in $\dot{H}^{s(p)}(\mathbb{R}^3)$ satisfies*

$$\|F\|_{L^p(\mathbb{R}^3)} \le \mathsf{S}(p) \, \|F\|_{\dot{H}^{s(p)}(\mathbb{R}^3)} . \tag{8}$$

*In particular,*

$$\mathsf{S}(3) = 2^{-1/6} \pi^{-1/3}, \qquad \mathsf{S}(6) = \frac{2^{2/3}}{\sqrt{3}\, \pi^{2/3}}, \qquad \mathsf{S}(6)\mathsf{S}(3) = \frac{1}{\pi}\sqrt{\frac{2}{3}}.$$

*Proof.* For scalar fields, (8) is the sharp three-dimensional homogeneous Sobolev inequality obtained from the sharp Hardy–Littlewood–Sobolev inequality [11]. After duality, insertion of the Riesz-potential kernel constant, and conversion to the unitary Fourier transform fixed above, the squared sharp Sobolev constant for the multiplier $\widehat{\Lambda^s f}(\xi) = |\xi|^s \widehat{f}(\xi)$ is

$$2^{-2s} \pi^{-s} \frac{\Gamma((3 - 2s)/2)}{\Gamma((3 + 2s)/2)} \left( \frac{\Gamma(3)}{\Gamma(3/2)} \right)^{2s/3} . \tag{9}$$

Taking the square root in (9) and putting $p = 6/(3 - 2s)$ gives (7)– (8). For a finite component array $F = (F_I)_I$, Minkowski's inequality in $L^{p/2}$ gives

$$\|F\|_p^2 = \left\| \sum_I |F_I|^2 \right\|_{p/2} \le \sum_I \|F_I\|_p^2 \le \mathsf{S}(p)^2 \sum_I \|F_I\|_{\dot{H}^{s(p)}}^2 .$$

The two endpoint formulas follow by evaluating the gamma factors at $s = 1/2$ and $s = 1$. □

**Lemma 2.4** (Sobolev multiplication)**.** *Set*

$$\mathsf{H}_4 := (2\pi)^{-3/4}\left(\int_{\mathbb{R}^3}\langle\xi\rangle^{-4}\,\mathrm{d}\xi\right)^{1/4}, \qquad C_0^{\mathrm{alg}} := \mathsf{H}_4^2.$$

*For $m \geq 1$, define*

$$\mathsf{E}_m := \max_{0\leq j\leq \lfloor m/2\rfloor} C^{\mathrm{emb}}_{j+2,j}, \qquad \mathsf{C}_m^{\mathrm{mix}} := 13^{m/2}\mathsf{E}_m,$$

$$\mathsf{C}_m^{\mathrm{F}} := 2^m(2\pi)^{-3/2}\left(\int_{\mathbb{R}^3}\langle\xi\rangle^{-2m}\,\mathrm{d}\xi\right)^{1/2} \qquad (m\geq 2),$$

*and set*

$$C_1^{\mathrm{alg}} := \mathsf{C}_1^{\mathrm{mix}}, \qquad C_m^{\mathrm{alg}} := \max\{\mathsf{C}_m^{\mathrm{mix}}, \mathsf{C}_m^{\mathrm{F}}\} \quad (m\geq 2).$$

*All these constants are finite and explicit. For every $m \in \mathbb{N}_0$,*

$$\|fg\|_{H^m} \leq C_m^{\mathrm{alg}}\,\|f\|_{H^{m+1}}\,\|g\|_{H^{m+1}}.$$

*For $m \geq 2$,*

$$\|fg\|_{H^m} \leq C_m^{\mathrm{alg}}\,\|f\|_{H^m}\,\|g\|_{H^m}.$$

*Proof.* Hausdorff–Young and Hölder give

$$\|f\|_4 \leq (2\pi)^{-3/4}\left\|\widehat{f}\right\|_{4/3} \leq \mathsf{H}_4\,\|f\|_{H^1}.$$

Thus $\|fg\|_2 \leq C_0^{\mathrm{alg}}\,\|f\|_{H^1}\,\|g\|_{H^1}$.

Let $m \geq 1$. The exact integer-order identity

$$\|h\|_{H^m}^2 = \sum_{|\alpha|\leq m}\frac{m!}{(m-|\alpha|)!\,\alpha!}\,\|\partial^\alpha h\|_2^2$$

follows by expanding $(1+|\xi|^2)^m$. In each Leibniz term for $\partial^\alpha(fg)$, place the factor carrying at most $\lfloor|\alpha|/2\rfloor$ derivatives in $L^\infty$. The definition of $\mathsf{E}_m$, Sobolev nesting, and $\sum_{\beta\leq\alpha}\binom{\alpha}{\beta} = 2^{|\alpha|}$ give

$$\|\partial^\alpha(fg)\|_2 \leq \mathsf{E}_m 2^{|\alpha|}\,\|f\|_{H^{m+1}}\,\|g\|_{H^{m+1}}.$$

The multinomial identity

$$\sum_{|\alpha|\leq m}\frac{m!}{(m-|\alpha|)!\,\alpha!}4^{|\alpha|} = 13^m$$

proves the first estimate with $\mathsf{C}_m^{\mathrm{mix}}$.

For $m \geq 2$, the inequalities

$$\langle\eta+\zeta\rangle^m \leq 2^{m-1}\big(\langle\eta\rangle^m + \langle\zeta\rangle^m\big), \qquad \left\|\widehat{f}\right\|_1 \leq \left(\int_{\mathbb{R}^3}\langle\xi\rangle^{-2m}\,\mathrm{d}\xi\right)^{1/2}\|f\|_{H^m},$$

combined with $\widehat{fg} = (2\pi)^{-3/2}\widehat{f}*\widehat{g}$ and Young's convolution inequality give

$$\|fg\|_{H^m} \leq \mathsf{C}_m^{\mathrm{F}}\,\|f\|_{H^m}\,\|g\|_{H^m}.$$

The chosen maximum proves both asserted estimates with this constant. □

**Lemma 2.5** (Riesz and Leray multipliers)**.** *Let*

$$r_j(\xi) := \begin{cases} i\xi_j/\left|\xi\right|, & \xi \neq 0, \\ 0, & \xi = 0, \end{cases} \qquad P(\xi) := \begin{cases} I - \dfrac{\xi \otimes \xi}{\left|\xi\right|^2}, & \xi \neq 0, \\ I, & \xi = 0, \end{cases}$$

*and*

$$\widehat{R_j f}(\xi) = r_j(\xi)\widehat{f}(\xi), \qquad \widehat{\mathbb{P}f}(\xi) = P(\xi)\widehat{f}(\xi).$$

*For all* $s \in \mathbb{R}$*,*

$$\left\|R_j f\right\|_{H^s} \leq \left\|f\right\|_{H^s}, \qquad \left\|\mathbb{P}f\right\|_{H^s} \leq \left\|f\right\|_{H^s}.$$

*If* $m \in \mathbb{N}_0$ *and* $v \in H^{m+1}_\sigma$*, then*

$$q := \sum_{i,j=1}^{3} R_i R_j (v_i v_j) \in H^m, \qquad -\Delta q = \sum_{i,j=1}^{3} \partial_i \partial_j (v_i v_j),$$

*and*

$$\mathbb{P}\nabla \cdot (v \otimes v) = (v \cdot \nabla)v + \nabla q.$$

*Here*

$$(v \otimes v)_{ij} := v_i v_j, \qquad (\nabla \cdot (v \otimes v))_i := \sum_{j=1}^{3} \partial_j (v_i v_j), \qquad ((v \cdot \nabla)v)_i := \sum_{j=1}^{3} v_j \partial_j v_i.$$

*Proof.* The multiplier symbols have operator norm at most one. Plancherel proves the first two estimates. Lemma 2.4 gives $v_i v_j \in H^m$. The remaining identities follow after multiplication by $\left|\xi\right|^2$ and by the symbol of $\mathbb{P}$. □

**Lemma 2.6** (Mixed Leibniz identities)**.** *For* $N \in \mathbb{N}_0$ *and* $\alpha \in \mathbb{N}_0^3$*,*

$$\partial_t^N \partial_x^\alpha (fg) = \sum_{a=0}^{N} \sum_{\beta \leq \alpha} \binom{N}{a} \binom{\alpha}{\beta} (\partial_t^a \partial_x^\beta f)(\partial_t^{N-a} \partial_x^{\alpha-\beta} g).$$

*Proof.* Apply the one-variable Leibniz identity successively in $t, x_1, x_2, x_3$. □

**Lemma 2.7** (Adjacent Hilbert-scale trace)**.** *Let $0 < T < \infty$, $m \in \mathbb{R}$, and*

$$v \in L^2(0,T;H^{m+1}), \qquad \partial_t v \in L^2(0,T;H^{m-1}).$$

*Then $v$ has a unique representative in $C([0,T];H^m)$, and, for $0 \le r \le t \le T$,*

$$\|v(t)\|_{H^m}^2 - \|v(r)\|_{H^m}^2 = 2\,\mathrm{Re}\int_r^t \left\langle A^{m-1}\partial_t v(s), A^{m+1}v(s)\right\rangle_{L^2}\,\mathrm{d}s, \tag{10}$$

$$\left|\|v(t)\|_{H^m}^2 - \|v(r)\|_{H^m}^2\right| \le 2\,\|\partial_t v\|_{L^2(r,t;H^{m-1})}\,\|v\|_{L^2(r,t;H^{m+1})}\,, \tag{11}$$

$$\sup_{0\le t\le T}\|v(t)\|_{H^m}^2 \le T^{-1}\,\|v\|_{L^2(0,T;H^m)}^2 + 2\,\|\partial_t v\|_{L^2(0,T;H^{m-1})}\,\|v\|_{L^2(0,T;H^{m+1})}\,. \tag{12}$$

*Proof.* Let $S_R$ denote Fourier restriction to $\{|\xi| \le R\}$. For $v_R = S_R v$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\|v_R(t)\|_{H^m}^2 = 2\,\mathrm{Re}\left\langle A^{m-1}\partial_t v_R(t), A^{m+1}v_R(t)\right\rangle_{L^2}.$$

Integration and Cauchy–Schwarz prove (10)–(11) for $v_R$. Applied to $S_R v - S_S v$, this estimate shows that $(v_R)_R$ is Cauchy in $C([0,T];H^m)$; choose one Lebesgue time for the initial Cauchy term. Its limit equals $v$ in $L^2(0,T;H^m)$. Passage to the limit proves the first two formulas. Choose $r \in (0,T)$ with $\|v(r)\|_{H^m}^2 \le T^{-1}\,\|v\|_{L^2H^m}^2$ and apply (11). □

**Lemma 2.8** (Explicit integer transport commutator)**.** *Set*

$$\Gamma_{1,1} := 1,$$

$$\Gamma_{\ell,k} := (2\ell - 4 + \sqrt{3})^{(k-1)(\ell-k)} \qquad (\ell \ge 2,\ 1 \le k \le \ell), \tag{13}$$

$$K_m := \sum_{\ell=1}^{m} 3^\ell \binom{m}{\ell} \sum_{k=1}^{\ell} \binom{\ell}{k} \Gamma_{\ell,k} \qquad (m \ge 3). \tag{14}$$

*Every $v \in H^{m+1}_\sigma(\mathbb{R}^3)$ satisfies*

$$|\mathrm{Re}\,\langle A^m \mathbb{P}\nabla\cdot(v\otimes v), A^m v\rangle_{L^2}| \le K_m\,\|\nabla v\|_{L^\infty}\,\|v\|_{H^m}^2\,. \tag{15}$$

*In particular,*

$$K_m \le (7^m - 1)(2m - 4 + \sqrt{3})^{(m-1)^2/4}. \tag{16}$$

*Proof.* We first prove the interpolation inequality used in the commutator. For the full ordered derivative tensor $\nabla^j F$, fix $M \ge 1$ and set

$$p_{M,0} := \infty, \qquad p_{M,j} := \frac{2M}{j} \quad (1 \le j \le M), \qquad A_M := 2M - 2 + \sqrt{3}.$$

Then

$$\left\|\nabla^j F\right\|_{L^{p_{M,j}}} \le A_M^{j(M-j)/2}\,\|F\|_\infty^{1-j/M}\,\left\|\nabla^M F\right\|_2^{j/M} \qquad (0 \le j \le M). \tag{17}$$

The endpoint cases are identities. For $1 \le j \le M-1$, put

$$G := \nabla^{j-1}F, \qquad W := \nabla G, \qquad p := p_{M,j}.$$

All contractions below are the Euclidean contractions of the corresponding ordered tensors. Integration by parts gives

$$\int_{\mathbb{R}^3} |W|^p = -\int_{\mathbb{R}^3} |W|^{p-2} G : \Delta G \\ - (p-2)\int_{\mathbb{R}^3} |W|^{p-4} \sum_{q=1}^{3} (G : W_{\cdot q})(W : \partial_q W).$$

The pointwise bounds

$$|\Delta G| \le \sqrt{3}\,|\nabla^2 G|, \qquad \left|\sum_{q=1}^{3} (G : W_{\cdot q})(W : \partial_q W)\right| \le |G|\,|W|^2 |\nabla^2 G|$$

give

$$\int_{\mathbb{R}^3} |W|^p \le (p - 2 + \sqrt{3}) \int_{\mathbb{R}^3} |G|\,|W|^{p-2} |\nabla^2 G|.$$

Since

$$\frac{1}{p_{M,j-1}} + \frac{p-2}{p} + \frac{1}{p_{M,j+1}} = 1,$$

Hölder's inequality, with $X_j := \left\|\nabla^j F\right\|_{L^{p_{M,j}}}$, gives

$$X_j^2 \le \left(\frac{2M}{j} - 2 + \sqrt{3}\right) X_{j-1} X_{j+1} \le A_M X_{j-1} X_{j+1}. \tag{18}$$

The numbers $d_j := j(M-j)/2$ obey

$$2d_j - d_{j-1} - d_{j+1} = 1, \qquad d_0 = d_M = 0.$$

The discrete maximum principle applied to (18) gives

$$X_j \le A_M^{d_j} X_0^{1-j/M} X_M^{j/M},$$

which is (17). The integration by parts is justified first with $(|W|^2 + \varepsilon)^{(p-2)/2}$, followed by $\varepsilon \downarrow 0$; smooth approximation handles zero tensor norms.

For $|\alpha| \le m$, set

$$\alpha! := \alpha_1!\alpha_2!\alpha_3!, \qquad c_{m,\alpha} := \frac{m!}{(m - |\alpha|)!\,\alpha!}.$$

The multiplier $A^{2m} = (1-\Delta)^m$ and the multinomial identity give the exact pairing formula

$$\langle A^m f, A^m g\rangle_{L^2} = \sum_{|\alpha|\le m} c_{m,\alpha}\,\langle \partial^\alpha f, \partial^\alpha g\rangle_{L^2}\,. \tag{19}$$

The Leray projection is self-adjoint, commutes with $A^m$, and fixes $A^m v$. Since $\nabla\cdot(v\otimes v) = (v\cdot\nabla)v$, it follows that

$$\langle A^m \mathbb{P}\nabla\cdot(v\otimes v), A^m v\rangle_{L^2} = \sum_{|\alpha|\le m} c_{m,\alpha}\,\langle \partial^\alpha((v\cdot\nabla)v), \partial^\alpha v\rangle_{L^2}\,. \tag{20}$$

For each $\alpha$,

$$\partial^\alpha((v\cdot\nabla)v) = (v\cdot\nabla)\partial^\alpha v + \sum_{0<\beta\le\alpha} \binom{\alpha}{\beta} (\partial^\beta v\cdot\nabla)\partial^{\alpha-\beta} v.$$

The divergence-free transport term has zero real pairing with $\partial^\alpha v$, because

$$\operatorname{Re}\langle (v\cdot\nabla)\partial^\alpha v, \partial^\alpha v\rangle_{L^2} = \frac{1}{2}\int_{\mathbb{R}^3} v\cdot\nabla\,|\partial^\alpha v|^2\,\mathrm{d}x = 0.$$

For a remaining term, put

$$\ell := |\alpha|\,, \qquad k := |\beta|\,.$$

If $\ell \geq 2$, apply (17) to $F = \nabla v$, with

$$M = \ell - 1, \qquad j = k - 1.$$

The two Hölder exponents are $p_{M,j}$ and $p_{M,M-j}$, whose reciprocal sum is $1/2$. The two factors are subarrays of the corresponding ordered derivative tensors, so

$$\begin{aligned}\left\|(\partial^\beta v\cdot\nabla)\partial^{\alpha-\beta}v\right\|_2 &\leq \left\|\partial^\beta v\right\|_{L^{p_{M,j}}}\left\|\nabla\partial^{\alpha-\beta}v\right\|_{L^{p_{M,M-j}}}\\ &\leq \Gamma_{\ell,k}\,\|\nabla v\|_\infty\left\|\nabla^\ell v\right\|_2.\end{aligned} \tag{21}$$

For $\ell = 1$, this formula follows from the direct $L^\infty$-$L^2$ estimate and $\Gamma_{1,1} = 1$. Since

$$\left\|\nabla^\ell v\right\|_2 = \|v\|_{\dot H^\ell} \leq \|v\|_{H^m}\,, \qquad \|\partial^\alpha v\|_2 \leq \|v\|_{H^m}\,,$$

each remaining pairing is bounded by

$$\Gamma_{\ell,k}\,\|\nabla v\|_\infty\,\|v\|_{H^m}^2\,.$$

The multinomial identities

$$\sum_{|\alpha|=\ell} c_{m,\alpha} = 3^\ell\binom{m}{\ell}, \qquad \sum_{\substack{\beta\leq\alpha\\ |\beta|=k}}\binom{\alpha}{\beta} = \binom{\ell}{k}$$

now give (15) with the constant (14). Finally,

$$\sum_{k=1}^{\ell}\binom{\ell}{k} \leq 2^\ell, \qquad (k-1)(\ell-k) \leq \frac{(\ell-1)^2}{4},$$

and summation of $\binom{m}{\ell}3^\ell 2^\ell$ over $1\leq\ell\leq m$ proves (16). Approximation by divergence-free Schwartz fields justifies all integrations by parts for $v\in H^{m+1}_\sigma$. □

The preceding estimates control the differentiated rows as they approach an endpoint. To continue their common limit as the original solution, we now fix a time-translation-invariant local class that admits both late preterminal slices and the endpoint restart, and in which uniqueness holds on every overlap.

**Definition 2.9** (Shifted pressure-normalized $H^3$ mild class)**.** Let $0\leq s<b<\infty$ and $a\in H^3_\sigma(\mathbb{R}^3)$. Define $\mathscr{M}^3_\nu(s,b;a)$ to be the set of pairs $(w,q)$ such that

$$w\in C([s,b];H^3_\sigma(\mathbb{R}^3)), \qquad q\in C([s,b];H^3(\mathbb{R}^3)), \tag{22}$$

$$w(s) = a, \qquad q(t) = R_iR_j\big(w_i(t)w_j(t)\big), \tag{23}$$

The remaining defining condition is the Duhamel identity: for every $t \in [s, b]$,

$$w(t) = e^{\nu(t-s)\Delta} a - \int_s^t e^{\nu(t-r)\Delta} \mathbb{P}\nabla \cdot (w \otimes w)(r)\, \mathrm{d}r \quad \text{in } H^3_\sigma(\mathbb{R}^3). \tag{24}$$

The integral is an $H^3_\sigma$-valued Bochner integral: multiplication in $H^3(\mathbb{R}^3)$ places $\mathbb{P}\nabla \cdot (w \otimes w)$ in $C([s, b]; H^2_\sigma)$, and the heat multiplier from $H^2$ to $H^3$ has an integrable $(t-r)^{-1/2}$ singularity. The maximal lifespan in this class is

$$\tau^{\max}_{\nu,3}(s, a) := \sup \big\{ b - s : \ b > s, \ \mathscr{M}^3_\nu(s, b; a) \neq \varnothing \big\} \in [0, \infty]. \tag{25}$$

**Lemma 2.10** (Restriction and classical admission)**.** *If $(w, q) \in \mathscr{M}^3_\nu(s, b; a)$ and $s \le r < c \le b$, then*

$$(w, q)|_{[r,c]} \in \mathscr{M}^3_\nu(r, c; w(r)). \tag{26}$$

*If $(u, p)$ is a pressure-complete classical history in the sense of Definition 3.2 and $[s, b]$ is contained in its interval of existence, then*

$$(u, p)|_{[s,b]} \in \mathscr{M}^3_\nu(s, b; u(s)). \tag{27}$$

*Proof.* Split (24) at $r$ and use the heat semigroup identity. This gives, for $r \le t \le c$,

$$w(t) = e^{\nu(t-r)\Delta} w(r) - \int_r^t e^{\nu(t-\rho)\Delta} \mathbb{P}\nabla \cdot (w \otimes w)(\rho)\, \mathrm{d}\rho.$$

The pressure formula is unchanged under restriction. For a classical history, apply $\mathbb{P}$ to the momentum equation and use variation of constants; the decaying pressure formula gives (23). □

**Theorem 2.11** (Quantitative local restart)**.** *Set $C_{\mathrm{q}} := \sqrt{3}\, C_3^{\mathrm{alg}}$. For $\nu > 0$, set*

$$L_\nu := 4(1 + \nu + \nu^{-1}) \exp\!\left( \frac{\nu + \nu^{-1}}{2} \right), \qquad c_\nu := (8 C_{\mathrm{q}} L_\nu^2)^{-2}.$$

*For every $s \ge 0$, every $a \in H^3_\sigma(\mathbb{R}^3)$, and every*

$$0 < \delta \le \delta_\nu(a) := \min\{1, c_\nu (1 + \|a\|_{H^3})^{-2}\},$$

*the class $\mathscr{M}^3_\nu(s, s+\delta; a)$ contains exactly one pair $(w, q)$. Its velocity satisfies*

$$w \in C([s, s+\delta]; H^3_\sigma) \cap L^2(s, s+\delta; H^4), \qquad w_t \in L^2(s, s+\delta; H^2).$$

*If $a \in H^\infty_\sigma$, then*

$$\partial_t^n w, \ \partial_t^n q \in C([s, s+\delta]; H^m) \qquad (n, m \in \mathbb{N}_0).$$

*For every finite $b > s$, the class $\mathscr{M}^3_\nu(s, b; a)$ contains at most one pair. Its compatible members on strict subintervals of $[s, s + \tau^{\max}_{\nu,3}(s, a))$ define the maximal shifted pressure-normalized $H^3$ mild history from $a$.*

*Proof.* By time translation it is enough to prove the assertions with $s = 0$. Let $0 < \delta \le 1$, let

$$z_t - \nu\Delta z = F, \qquad z(0) = a, \qquad a \in H^3_\sigma, \quad F \in L^2(0,\delta; H^2_\sigma),$$

and set

$$A_0 := \|a\|_{H^3}, \qquad G := \|F\|_{L^2(0,\delta;H^2)}.$$

The $H^3$ energy identity is

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|z\|^2_{H^3} + \nu\|\nabla z\|^2_{H^3} = \operatorname{Re}\langle A^2F, A^4 z\rangle_{L^2}.$$

Since $\|z\|^2_{H^4} = \|z\|^2_{H^3} + \|\nabla z\|^2_{H^3}$, Young's inequality gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\|z\|^2_{H^3} + \nu\|\nabla z\|^2_{H^3} \le \nu\|z\|^2_{H^3} + \nu^{-1}\|F\|^2_{H^2}.$$

Consequently,

$$\|z\|_{C([0,\delta];H^3)} \le e^{\nu/2}\big(A_0 + \nu^{-1/2}G\big). \tag{28}$$

Pairing $A^2 z_t - \nu\Delta A^2 z = A^2 F$ with $-\Delta A^2 z$ gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\big\|\nabla A^2 z\big\|_2^2 + \nu\big\|\Delta A^2 z\big\|_2^2 = \operatorname{Re}\langle A^2F, -\Delta A^2 z\rangle_{L^2}.$$

Integration and Young's inequality yield

$$\|\Delta z\|_{L^2(0,\delta;H^2)} \le \nu^{-1/2}A_0 + \nu^{-1}G. \tag{29}$$

The pointwise multiplier inequality

$$\langle\xi\rangle^8 \le 2\big(\langle\xi\rangle^6 + |\xi|^4\langle\xi\rangle^4\big)$$

and $\delta \le 1$ imply

$$\|z\|_{L^2(0,\delta;H^4)} \le \sqrt{2}\left(\|z\|_{C([0,\delta];H^3)} + \|\Delta z\|_{L^2(0,\delta;H^2)}\right). \tag{30}$$

The equation and (29) give

$$\|z_t\|_{L^2(0,\delta;H^2)} \le \nu^{1/2}A_0 + 2G. \tag{31}$$

Equations (28)–(31) give

$$\begin{aligned}
&\|z\|_{CH^3} + \|z\|_{L^2H^4} + \|z_t\|_{L^2H^2} \\
&\quad\le e^{\nu/2}(A_0 + \nu^{-1/2}G) + \sqrt{2}\left(e^{\nu/2}(A_0 + \nu^{-1/2}G) + \nu^{-1/2}A_0 + \nu^{-1}G\right) + \nu^{1/2}A_0 + 2G \\
&\quad\le L_\nu(A_0 + G).
\end{aligned}$$

The last inequality follows from

$$\nu^{1/2} \le \frac{1+\nu}{2}, \qquad \nu^{-1/2} \le \frac{1+\nu^{-1}}{2}, \qquad e^{\nu/2} \le e^{(\nu+\nu^{-1})/2}.$$

Fourier truncation and passage to the limit justify the preceding linear estimates for the stated data class.

For $v, z \in H^3_\sigma$, Lemmas 2.4 and 2.5 give

$$\|\mathbb{P}\nabla \cdot (v \otimes z)\|_{H^2} \le C_{\mathrm{q}} \|v\|_{H^3} \|z\|_{H^3}, \tag{32}$$

$$\|\mathbb{P}\nabla \cdot (v \otimes v - z \otimes z)\|_{H^2} \le C_{\mathrm{q}} (\|v\|_{H^3} + \|z\|_{H^3}) \|v - z\|_{H^3}. \tag{33}$$

Indeed,

$$\|\nabla \cdot (v \otimes z)\|^2_{H^2} \le 3 \sum_{i,j=1}^{3} \|v_i z_j\|^2_{H^3} \le 3(C_3^{\mathrm{alg}})^2 \|v\|^2_{H^3} \|z\|^2_{H^3}.$$

Define

$$X_\delta := \left\{ v \in C([0,\delta]; H^3_\sigma) \cap L^2(0,\delta; H^4_\sigma) : v_t \in L^2(0,\delta; H^2_\sigma) \right\},$$

$$\|v\|_{X_\delta} := \|v\|_{CH^3} + \|v\|_{L^2H^4} + \|v_t\|_{L^2H^2}, \qquad X_\delta(a) := \{v \in X_\delta : v(0) = a\}.$$

$X_\delta$ is a Banach space with this norm, and $X_\delta(a)$ is closed. For $A_0 > 0$, set $R := 2L_\nu A_0$ and

$$\mathbb{B}_R := \{v \in X_\delta(a) : \|v\|_{X_\delta} \le R\}.$$

The set $\mathbb{B}_R$ is nonempty: the zero-forcing heat trajectory $v^{\mathrm{lin}}(t) = e^{\nu t\Delta} a$ belongs to $X_\delta(a)$, and the linear estimate with $F = 0$ gives $\|v^{\mathrm{lin}}\|_{X_\delta} \le L_\nu A_0 < R$. For $v \in \mathbb{B}_R$, define

$$\Phi(v)(t) := e^{\nu t\Delta} a - \int_0^t e^{\nu(t-s)\Delta} \mathbb{P}\nabla \cdot (v \otimes v)(s)\, \mathrm{d}s.$$

If $\delta \le \delta_\nu(a)$, then

$$\delta^{1/2} \le \frac{1}{8C_{\mathrm{q}} L_\nu^2 (1 + A_0)}.$$

The linear estimate and (32) imply

$$\begin{aligned} \|\Phi(v)\|_{X_\delta} &\le L_\nu \big(A_0 + C_{\mathrm{q}} \delta^{1/2} R^2\big) \\ &\le L_\nu A_0 + \frac{L_\nu A_0^2}{2(1 + A_0)} < R. \end{aligned}$$

For $v, z \in \mathbb{B}_R$,

$$\begin{aligned} \|\Phi(v) - \Phi(z)\|_{X_\delta} &\le L_\nu C_{\mathrm{q}} \delta^{1/2} (\|v\|_{CH^3} + \|z\|_{CH^3}) \|v - z\|_{CH^3} \\ &\le \frac{A_0}{2(1 + A_0)} \|v - z\|_{X_\delta} \le \frac{1}{2} \|v - z\|_{X_\delta}. \end{aligned}$$

Thus $\Phi$ has a unique fixed point in $\mathbb{B}_R$. If $A_0 = 0$, the zero function is the fixed point. Let $b > 0$ and let

$$(v, q_v), (z, q_z) \in \mathscr{M}^3_\nu(0, b; a).$$

Set

$$M_b := \|v\|_{C([0,b];H^3)} + \|z\|_{C([0,b];H^3)}.$$

On every subinterval $I \subset [0, b]$ of length $h \le 1$ on which the two velocities have a common left trace, their shifted Duhamel identities and the zero-data linear estimate give

$$\|v - z\|_{C(I;H^3)} \le L_\nu C_{\mathrm{q}} h^{1/2} M_b \|v - z\|_{C(I;H^3)}.$$

A finite partition with $L_\nu C_{\mathrm{q}} h^{1/2} M_b < 1$ proves $v = z$ on $[0, b]$. The pressure formula gives $q_v = q_z$. This proves the at-most-one assertion for the full shifted mild class. For the fixed point, (32) places the forcing in $L^2(0, \delta; H^2)$; the linear estimates give

$$w \in L^2(0, \delta; H^4), \qquad w_t \in L^2(0, \delta; H^2).$$

Lemma 2.5 gives the normalized pressure $q = R_i R_j (w_i w_j)$.

Fix $m \geq 3$ and assume $a \in H^m_\sigma$. Let $S_k$ be Fourier restriction to $\{|\xi| \leq k\}$, set $a_k = S_k a$, and replace the fixed-point map above by

$$\Phi_k(v)(t) := e^{\nu t \Delta} a_k - \int_0^t e^{\nu (t-s)\Delta} S_k \mathbb{P} \nabla \cdot (v \otimes v)(s) \, \mathrm{d}s.$$

The projection $S_k$ is self-adjoint, contractive on every Sobolev space, and commutes with the heat flow, $\mathbb{P}$, and spatial derivatives. The estimates that gave $\Phi$ its contraction factor at most $1/2$ apply to $\Phi_k$ on the closed radius-$R$ ball in $X_\delta(a_k)$. Its fixed point $w^{(k)} = S_k w^{(k)}$ consequently exists on $[0, \delta]$ and satisfies

$$\sup_k \left\| w^{(k)} \right\|_{X_\delta} \leq R. \tag{34}$$

Because its Fourier support is bounded, $w^{(k)}$ belongs to every spatial Sobolev space. Self-adjointness of $S_k$ removes the cutoff from the energy pairing, so Lemma 2.8 gives

$$\frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \left\| w^{(k)} \right\|^2_{H^m} + \nu \left\| \nabla w^{(k)} \right\|^2_{H^m} \leq K_m \left\| \nabla w^{(k)} \right\|_{L^\infty} \left\| w^{(k)} \right\|^2_{H^m}.$$

Set

$$C_{\nabla,3} := (2\pi)^{-3/2} \left( \int_{\mathbb{R}^3} |\xi|^2 \langle \xi \rangle^{-6} \, \mathrm{d}\xi \right)^{1/2}.$$

Fourier inversion and Cauchy–Schwarz give $\|\nabla v\|_\infty \leq C_{\nabla,3} \|v\|_{H^3}$. From (34),

$$\int_0^\delta \left\| \nabla w^{(k)}(s) \right\|_\infty \, \mathrm{d}s \leq C_{\nabla,3} \delta R.$$

Put

$$F_k(t) := \left\| w^{(k)}(t) \right\|^2_{H^m} + 2\nu \int_0^t \left\| \nabla w^{(k)}(s) \right\|^2_{H^m} \, \mathrm{d}s.$$

The differential inequality above gives

$$F_k(t) \leq \|a\|^2_{H^m} + 2K_m \int_0^t \left\| \nabla w^{(k)}(s) \right\|_\infty F_k(s) \, \mathrm{d}s.$$

Gronwall's inequality now gives the uniform bound

$$\sup_{0 \leq t \leq \delta} \left( \left\| w^{(k)}(t) \right\|^2_{H^m} + 2\nu \int_0^t \left\| \nabla w^{(k)}(s) \right\|^2_{H^m} \, \mathrm{d}s \right) \leq e^{2K_m C_{\nabla,3} \delta R} \|a\|^2_{H^m}. \tag{35}$$

Since $\|f\|^2_{H^{m+1}} = \|f\|^2_{H^m} + \|\nabla f\|^2_{H^m}$, this also bounds $w^{(k)}$ in $L^2 H^{m+1}$. The cutoff equations and Lemma 2.4 give, with a constant depending only on $m$,

$$\left\| \partial_t w^{(k)} \right\|_{L^2 H^{m-1}} \leq \nu \left\| w^{(k)} \right\|_{L^2 H^{m+1}} + \sqrt{3}\, C^{\mathrm{alg}}_m \delta^{1/2} \left\| w^{(k)} \right\|^2_{L^\infty H^m},$$

so the time derivatives are uniformly bounded in $L^2H^{m-1}$.

It remains to identify the higher-order weak limit with the fixed point already constructed in $X_\delta(a)$. The contraction estimate gives

$$\left\|w^{(k)} - w\right\|_{X_\delta} \le 2\,\|\Phi_k(w) - \Phi(w)\|_{X_\delta}\,.$$

The linear bound proved above controls the right-hand side by a constant times

$$\|(S_k - I)a\|_{H^3} + \|(S_k - I)\mathbb{P}\nabla\cdot(w\otimes w)\|_{L^2H^2}\,,$$

which tends to zero. Hence $w^{(k)} \to w$ in $X_\delta$. Weak compactness in the three uniformly bounded higher-order spaces and uniqueness of the distributional limit place

$$w \in L^\infty(0,\delta;H^m_\sigma)\cap L^2(0,\delta;H^{m+1}_\sigma), \qquad w_t \in L^2(0,\delta;H^{m-1}_\sigma).$$

Lemma 2.7 supplies the representative $w \in C([0,\delta];H^m_\sigma)$; its trace at zero is $a$, since it is the distributional trace of the already constructed $CH^3$ solution. Assume now that $a \in H^\infty_\sigma$. Set $W_0 = w$ and define recursively

$$Q_n = R_iR_j\sum_{a=0}^{n}\binom{n}{a}W_{a,i}W_{n-a,j},$$

$$W_{n+1} = \nu\Delta W_n - \sum_{a=0}^{n}\binom{n}{a}(W_a\cdot\nabla)W_{n-a} - \nabla Q_n.$$

If $W_0,\dots,W_n \in CH^{m+2}$, then Lemmas 2.4 and 2.5 give

$$Q_n \in CH^{m+1}, \qquad W_{n+1}\in CH^m.$$

Since $m$ is arbitrary, induction gives $W_n, Q_n \in CH^m$ for every $n,m\in\mathbb{N}_0$. The equation first gives $W_0\in C^1H^m$ and $\partial_tW_0 = W_1$. If $W_a = \partial_t^a w$ for $0\le a\le n$, distributional differentiation of $q = R_iR_j(w_iw_j)$ and Lemma 2.6 give $\partial_t^n q = Q_n$. Differentiation of the momentum equation then gives $\partial_tW_n = W_{n+1}$. Induction gives

$$W_n\in C^1H^m, \qquad \partial_tW_n = W_{n+1}, \qquad W_n = \partial_t^n w, \qquad Q_n = \partial_t^n q.$$

Time translation proves the result for every $s\ge 0$. Lemma 2.10 and the at-most-one assertion make all nonempty classes below $\tau^{\max}_{\nu,3}(s,a)$ compatible; their union is the maximal shifted pressure-normalized $H^3$ mild history. □

# 3 The pressure-complete setting and mixed jet

**Proposition 3.1** (Pressure normalization)**.** *Let $u,p$ be smooth on $I\times\mathbb{R}^3$, let $\nabla\cdot u = 0$, and suppose*

$$u(t)\in H^1_\sigma(\mathbb{R}^3), \qquad p(t)\in H^0(\mathbb{R}^3) \qquad (t\in I),$$

*and*

$$u_t + (u\cdot\nabla)u + \nabla p = \nu\Delta u. \tag{36}$$

*Then*

$$-\Delta p = \partial_i \partial_j (u_i u_j), \tag{37}$$
$$p = R_i R_j (u_i u_j), \tag{38}$$
$$(u \cdot \nabla) u + \nabla p = \mathbb{P} \nabla \cdot (u \otimes u). \tag{39}$$

*The condition $p(t) \in H^0(\mathbb{R}^3)$ fixes the pressure uniquely.*

*Proof.* Incompressibility gives

$$(u \cdot \nabla) u_i = \partial_j (u_j u_i).$$

Taking the divergence of (36) cancels $\nabla \cdot u_t$ and $\nabla \cdot \Delta u$, which yields (37). Lemma 2.5, applied with $m = 0$, gives $q := R_i R_j (u_i u_j) \in H^0$, this Poisson identity, and

$$(u \cdot \nabla) u + \nabla q = \mathbb{P} \nabla \cdot (u \otimes u).$$

If $p_1, p_2 \in H^0$ satisfy (37), then $h = p_1 - p_2$ satisfies $|\xi|^2 \widehat{h} = 0$. Since $\widehat{h} \in L^2$ and is supported at $\{0\}$, it vanishes. Hence $p = q$, which proves (38) and (39). □

**Definition 3.2** (Maximal pressure-complete classical history)**.** Fix $\nu > 0$ and $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$. A pressure-complete classical history generated by $(u_0, \nu)$ is a triple

$$(T_*, u, p), \qquad 0 < T_* \le \infty,$$

such that, for every $0 < T < T_*$ and $n, m \in \mathbb{N}_0$,

$$\partial_t^n u \in C([0,T]; H^m_\sigma(\mathbb{R}^3)),$$
$$\partial_t^n p \in C([0,T]; H^m(\mathbb{R}^3)),$$

and

$$u_t + (u \cdot \nabla) u + \nabla p = \nu \Delta u, \tag{40}$$
$$\nabla \cdot u = 0, \tag{41}$$
$$u(0) = u_0, \tag{42}$$
$$p(t) = R_i R_j (u_i(t) u_j(t)). \tag{43}$$

The history is maximal if no pressure-complete classical history $(\widetilde{T}, \widetilde{u}, \widetilde{p})$ generated by $(u_0, \nu)$ satisfies

$$\widetilde{T} > T_*, \qquad (\widetilde{u}, \widetilde{p})|_{[0,T_*)} = (u, p).$$

**Definition 3.3** (Temporal and mixed velocity–pressure jets)**.** Let $(T_*, u, p)$ be a maximal pressure-complete classical history. For $n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$, define

$$V_n := \partial_t^n u, \qquad Q_n := \partial_t^n p, \qquad J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \qquad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p.$$

For $\beta \le \alpha$, define

$$\binom{\alpha}{\beta} := \prod_{\ell=1}^{3} \binom{\alpha_\ell}{\beta_\ell}, \qquad |\alpha| := \alpha_1 + \alpha_2 + \alpha_3.$$

**Proposition 3.4** (Pressure-complete binomial recurrences)**.** *For every $n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,*

$$Q_n = R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i} V_{n-a,j}, \tag{44}$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^{n} \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n, \tag{45}$$

$$\Pi_{n,\alpha} = R_i R_j \sum_{a=0}^{n} \sum_{\beta \le \alpha} \binom{n}{a} \binom{\alpha}{\beta} J_{a,\beta,i} J_{n-a,\alpha-\beta,j}, \tag{46}$$

$$J_{n+1,\alpha} + \sum_{a=0}^{n} \sum_{\beta \le \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} + \nabla \Pi_{n,\alpha} = \nu \Delta J_{n,\alpha}, \tag{47}$$

$$\nabla \cdot J_{n,\alpha} = 0. \tag{48}$$

*Proof.* Apply $\partial_t^n$ to (43) and use Lemma 2.6 with $\alpha = 0$ to obtain (44). Apply $\partial_t^n$ to (40) to obtain (45). Apply $\partial_x^\alpha$ to both identities, use Lemma 2.6, and commute the derivatives with $R_i R_j$ to obtain (46) and (47). Differentiation of (41) gives (48). □

**The unforced datum and its derivative rows.** The initial velocity and viscosity specify all inputs to these recurrences. Pressure is determined by the velocity products, and there is no prescribed force jet. A smooth forced equation can still be differentiated; its rows also contain the corresponding force derivatives, and the pressure equation contains the divergence of the force. In the unforced kinetic-energy calculation of Proposition 4.1, transport and pressure have zero global pairing with the velocity, and no external work term remains. At higher Sobolev orders, the full internal transport–pressure contribution is retained in Proposition 4.3. These are the governing rows and energy identities used in the construction below.

**Definition 3.5** (Canonical pressure constant)**.** For vector fields $f, g$, set

$$\mathcal{Q}(f,g) := \sum_{i,j=1}^{3} R_i R_j (f_i g_j).$$

For $m \in \mathbb{N}_0$, define

$$C_m^{\mathrm{p}} := \sup_{f,g \in H^{m+1}(\mathbb{R}^3;\mathbb{R}^3)\setminus\{0\}} \frac{\|\mathcal{Q}(f,g)\|_{H^m} + \|\nabla \mathcal{Q}(f,g)\|_{H^{m-1}}}{\|f\|_{H^{m+1}} \|g\|_{H^{m+1}}}.$$

**Lemma 3.6.** *For every $m \in \mathbb{N}_0$,*

$$C_m^{\mathrm{p}} \le 2 C_m^{\mathrm{alg}}.$$

*Proof.* Regard $(f_i g_j)_{i,j}$ as one tensor. The contraction $F \mapsto R_i R_j F_{ij}$ has Fourier-multiplier norm one from the tensor $H^m$ norm to scalar $H^m$. Lemma 2.4 and $\|\nabla h\|_{H^{m-1}} \le \|h\|_{H^m}$ give

$$\|\mathcal{Q}(f,g)\|_{H^m} + \|\nabla \mathcal{Q}(f,g)\|_{H^{m-1}} \le 2 C_m^{\mathrm{alg}} \|f\|_{H^{m+1}} \|g\|_{H^{m+1}}.$$

□

**Lemma 3.7** (Initial pressure-complete jet)**.** *Set*

$$V_n^0 := V_n(0), \qquad Q_n^0 := Q_n(0), \qquad J_{n,\alpha}^0 := J_{n,\alpha}(0), \qquad \Pi_{n,\alpha}^0 := \Pi_{n,\alpha}(0).$$

*Then*

$$V_0^0 = u_0, \tag{49}$$

$$Q_n^0 = R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i}^0 V_{n-a,j}^0, \tag{50}$$

$$V_{n+1}^0 = \nu \Delta V_n^0 - \sum_{a=0}^{n} \binom{n}{a} (V_a^0 \cdot \nabla) V_{n-a}^0 - \nabla Q_n^0. \tag{51}$$

*For every $n, m \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,*

$$V_n^0, Q_n^0, J_{n,\alpha}^0, \Pi_{n,\alpha}^0 \in H^m(\mathbb{R}^3).$$

*Proof.* The initial velocity fixes the pressure before the first temporal response is calculated:

$$Q_0^0 = R_i R_j (u_{0,i} u_{0,j}), \qquad V_1^0 = \nu \Delta u_0 - (u_0 \cdot \nabla) u_0 - \nabla Q_0^0.$$

Differentiating the pressure product places the first temporal response on either velocity factor. Substitution into the next velocity row gives

$$\begin{aligned} Q_1^0 &= R_i R_j \big(V_{1,i}^0 u_{0,j} + u_{0,i} V_{1,j}^0\big), \\ V_2^0 &= \nu \Delta V_1^0 - (V_1^0 \cdot \nabla) u_0 - (u_0 \cdot \nabla) V_1^0 - \nabla Q_1^0. \end{aligned}$$

At temporal order $n$, the product rule distributes the $n$ derivatives between the two velocity factors in $\binom{n}{a}$ ways. Thus the already determined values $V_0^0, \dots, V_n^0$ determine $Q_n^0$ through (50), and then determine $V_{n+1}^0$ through (51).

Fix any finite spatial order $m$. The datum belongs to every higher Sobolev space needed by these finitely many products and derivatives. Lemmas 2.4 and 2.5 place $Q_n^0$ and $V_{n+1}^0$ in $H^m$. Induction from $V_0^0 = u_0$ proves the assertion at every finite $n, m$. Spatial differentiation gives the initial mixed coordinates $J_{n,\alpha}^0 = \partial_x^\alpha V_n^0$ and $\Pi_{n,\alpha}^0 = \partial_x^\alpha Q_n^0$. The original classical history obeys these differentiated equations, so its initial traces are exactly the values just calculated. □

**Theorem 3.8** (Finite-row pressure estimates)**.** *Let $I = (0,T) \subset (0,T_*)$, let $n, M \in \mathbb{N}_0$, and suppose*

$$V_a \in L^2(I; H^{M+1}(\mathbb{R}^3)) \qquad (0 \le a \le n).$$

*Then*

$$\begin{aligned} &\|Q_n\|_{L^1(I;H^M)} + \|\nabla Q_n\|_{L^1(I;H^{M-1})} \\ &\quad \le C_M^{\mathrm{p}} \sum_{a=0}^{n} \binom{n}{a} \|V_a\|_{L^2(I;H^{M+1})} \, \|V_{n-a}\|_{L^2(I;H^{M+1})}. \end{aligned} \tag{52}$$

*If*

$$\mathcal{V}_{n,M}(I)^2 := \sum_{a=0}^{n} \|V_a\|_{L^2(I;H^{M+1})}^2,$$

*then*

$$\|Q_n\|_{L^1(I;H^M)} + \|\nabla Q_n\|_{L^1(I;H^{M-1})} \le C_M^{\mathrm{p}} 2^n \mathcal{V}_{n,M}(I)^2. \tag{53}$$

*For $\alpha \in \mathbb{N}_0^3$, assume*

$$J_{a,\beta} \in L^2(I; H^{M+1}(\mathbb{R}^3)) \qquad (0 \le a \le n, \ \beta \le \alpha).$$

*Then*

$$\begin{aligned}&\|\Pi_{n,\alpha}\|_{L^1(I;H^M)} + \|\nabla\Pi_{n,\alpha}\|_{L^1(I;H^{M-1})}\\&\quad\le C_M^{\mathrm{p}} \sum_{a=0}^{n}\sum_{\beta\le\alpha} \binom{n}{a}\binom{\alpha}{\beta} \|J_{a,\beta}\|_{L^2(I;H^{M+1})} \|J_{n-a,\alpha-\beta}\|_{L^2(I;H^{M+1})}\,. \end{aligned}\tag{54}$$

*If*

$$\mathcal{J}_{n,\alpha,M}(I)^2 := \sum_{a=0}^{n}\sum_{\beta\le\alpha} \|J_{a,\beta}\|^2_{L^2(I;H^{M+1})}\,,$$

*then*

$$\|\Pi_{n,\alpha}\|_{L^1(I;H^M)} + \|\nabla\Pi_{n,\alpha}\|_{L^1(I;H^{M-1})} \le C_M^{\mathrm{p}} 2^{n+|\alpha|} \mathcal{J}_{n,\alpha,M}(I)^2.\tag{55}$$

*Proof.* Lemmas 2.4 and 2.5 give

$$\|R_iR_j(f_ig_j)\|_{H^M} + \|\nabla R_iR_j(f_ig_j)\|_{H^{M-1}} \le C_M^{\mathrm{p}} \|f\|_{H^{M+1}} \|g\|_{H^{M+1}}\,.\tag{56}$$

Equations (44) and (56) give, for almost every $t \in I$,

$$\|Q_n(t)\|_{H^M} + \|\nabla Q_n(t)\|_{H^{M-1}} \le C_M^{\mathrm{p}} \sum_{a=0}^{n} \binom{n}{a} \|V_a(t)\|_{H^{M+1}} \|V_{n-a}(t)\|_{H^{M+1}}\,.$$

Integration and Cauchy–Schwarz prove (52). For nonnegative $x_a$,

$$\sum_{a=0}^{n} \binom{n}{a} x_a x_{n-a} \le \sum_{a=0}^{n} \binom{n}{a} x_a^2 \le 2^n \sum_{a=0}^{n} x_a^2.$$

This proves (53).

Apply (56) to each summand in (46). Integration and Cauchy–Schwarz give (54). The involution

$$(a,\beta) \longmapsto (n-a, \alpha-\beta)$$

preserves the weights $\binom{n}{a}\binom{\alpha}{\beta}$, and

$$\sum_{a=0}^{n}\sum_{\beta\le\alpha} \binom{n}{a}\binom{\alpha}{\beta} = 2^{n+|\alpha|}.$$

In particular, every individual weight is at most $2^{n+|\alpha|}$. The inequality $2xy \le x^2 + y^2$ proves (55). □

### 3.1 The datum-generated jet and its compatible finite records

Start with $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$ and $\nu > 0$. Theorem 2.11 constructs a pressure-normalized classical history on a positive interval. Uniqueness identifies the local solutions on their overlaps; their compatible union gives the maximal history of Definition 3.2.

Its derivative rows are determined by the equation. The datum specifies the initial zeroth velocity row. The pressure formula in Proposition 3.4 fixes its initial pressure, and the velocity recurrence fixes its first temporal row. Repeating the recurrences determines every finite initial

temporal row. Spatial differentiation gives the initial mixed jet. Along the history, those rows are the actual distributional derivatives of $u, p$, with the full transport and pressure terms retained.

On every strict subinterval $(0, T)$, $T < T_*$, persistence of the classical history identifies these rows and gives finite norms at each fixed pair of indices. For a proposed finite maximal time, put $I = (0, T_*)$. The construction uses the datum, fixed viscosity, complete initial jet, and pressure-complete equations on $I$.

The derivative order leaves the governing variables free. For example, $V_1(t, x)$ is determined by the velocity, its spatial derivatives, and its normalized pressure at $(t, x)$; evaluating that identity at $t = 0$ gives the initial value calculated above. Repeated differentiation retains $(t, x)$ in every row. It describes the responses of the evolving field throughout its interval.

The viscous term makes the temporal and spatial descriptions meet in each complete equation:

$$\nu \Delta V_n = V_{n+1} + \sum_{a=0}^{n} \binom{n}{a} (V_a \cdot \nabla) V_{n-a} + \nabla Q_n. \tag{57}$$

The two additional spatial derivatives occur together with the next temporal response and the full transport–pressure sum. This is the field relation behind the spatial orders exposed by the critical-height calculation in Proposition 4.3.

There are two constructions from these equations. The direct mixed-jet construction retains the compatible temporal and spatial coordinates and their Sobolev memberships. Differentiating the critical height and substituting the mixed rows instead produces relationships among those coordinates; finite selections of the relationships form rectangles. Their common origin fixes every overlapping velocity and pressure entry. Either representation can be used at its stated scope without making the other construction its prerequisite.

**Theorem 3.9** (Datum-generated whole-terminal finite rectangles)**.** *Let $\nu > 0$, $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and let $(T_*, u, p)$ be the maximal pressure-complete classical history generated by $(u_0, \nu)$. Assume, toward contradiction, that $T_* < \infty$, and put $I = (0, T_*)$. For finite $N, M$, the selected pressure-complete record is*

$$\mathcal{R}_{N,M}[u_0; T_*] := \left( (V_n)_{n=0}^N, (V_{n+1})_{n=0}^N; (Q_n)_{n=0}^N \right), \tag{58}$$

*and the datum-generated construction produces, for $0 \le n \le N$,*

$$V_n \in L^2(I; H^{M+1}), \qquad V_{n+1} \in L^2(I; H^{M-1}). \tag{59}$$

*Define its velocity norm on $I$ by*

$$\mathfrak{R}_{N,M}(u; T_*) := \sum_{n=0}^{N} \left( \|V_n\|^2_{L^2(I;H^{M+1})} + \|V_{n+1}\|^2_{L^2(I;H^{M-1})} \right). \tag{60}$$

*The theorem's whole-terminal output is*

$$\mathfrak{R}_{N,M}(u; T_*) < \infty \qquad (N, M < \infty). \tag{61}$$

*Taking $N \ge n$ and $M = m$ gives the corresponding statement for every finite pair $(n, m)$. Every row is the corresponding distributional derivative of the single maximal history. For every finite $n, m, \alpha$,*

$$J_{n,\alpha} \in L^2(I; H^m), \qquad \Pi_{n,\alpha} \in L^1(I; H^m),$$

*and enlargement of a finite rectangle preserves every common velocity and pressure coordinate.*

*Datum construction and finite-coordinate realization.* Put $I = (0, T_*)$, and fix finite $N, M$. The datum first generates the complete initial jet

$$A_0 = u_0, \qquad P_n = R_i R_j \sum_{a=0}^{n} \binom{n}{a} (A_a)_i (A_{n-a})_j,$$

$$A_{n+1} = \nu \Delta A_n - \sum_{a=0}^{n} \binom{n}{a} (A_a \cdot \nabla) A_{n-a} - \nabla P_n.$$

At step $n$, the right-hand side uses only the already determined rows $A_0, \dots, A_n$. Lemma 3.7 then places every required $A_n$ and $P_n$ in every finite spatial Sobolev order. Spatial differentiation supplies

$$J_{n,\alpha}(0) = \partial_x^\alpha A_n, \qquad \Pi_{n,\alpha}(0) = \partial_x^\alpha P_n.$$

The selected equations retain their complete finite dependency cover. For an equation indexed by $(n, \alpha)$, retain the adjacent temporal coordinate $J_{n+1,\alpha}$, the viscous coordinates $J_{n,\alpha+2e_\ell}$, and the pressure coordinates $\Pi_{n,\alpha+e_\ell}$, for $1 \le \ell \le 3$. For every $a \le n$, $\beta \le \alpha$, and velocity component $i$, retain both transport factors

$$J_{a,\beta,i}, \qquad J_{n-a,\alpha-\beta+e_i}.$$

Here $e_i$ is the $i$-th spatial unit multi-index. Set $\Gamma_\alpha := \{\alpha\} \cup \{\alpha + e_\ell : 1 \le \ell \le 3\}$. For every $\gamma \in \Gamma_\alpha$, every $a \le n$, and every $\beta \le \gamma$, also retain the two pressure-product factors $J_{a,\beta,i}$ and $J_{n-a,\gamma-\beta,j}$. Each retained normalized pressure coordinate then carries its full product formula

$$\Pi_{n,\gamma} = R_i R_j \sum_{a=0}^{n} \sum_{\beta \le \gamma} \binom{n}{a} \binom{\gamma}{\beta} J_{a,\beta,i} J_{n-a,\gamma-\beta,j} \qquad (\gamma \in \Gamma_\alpha).$$

Thus the retained equation is

$$\begin{aligned} \mathscr{E}_{n,\alpha}(J, \Pi) := J_{n+1,\alpha} + \sum_{a=0}^{n} \sum_{\beta \le \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} \\ + \nabla \Pi_{n,\alpha} - \nu \Delta J_{n,\alpha} = 0. \end{aligned} \tag{62}$$

There are finitely many retained coordinates for the fixed selected equations. The dependency cover is not an autonomous truncated fluid: every entry is the corresponding derivative of the original maximal history. Applying $\mathbb{P}$ reads the solenoidal evolution, and applying $I - \mathbb{P}$ reads the normalized pressure equation. The initial face, every nonlinear factor, the viscous coordinates, and the pressure coordinates are present together.

The datum-generated whole-terminal jet already contains its compatible Sobolev coordinates on $I$. Selecting this finite record projects that constructed jet onto the rows $V_n, V_{n+1}, Q_n$, while retaining the original time interval. Its adjacent velocity coordinates read

$$V_n \in L^2(I; H^{M+1}), \qquad V_{n+1} \in L^2(I; H^{M-1}).$$

For every $n = 0, \dots, N$, these are components of the constructed whole-terminal jet, as recorded in (59). The finite selection preserves their memberships and the identity $\partial_t V_n = V_{n+1}$. Equation (62) retains their complete transport, pressure, and viscous coupling.

For this fixed finite pair, form the sum of the defining squared norms:

$$C_{N,M}[u_0, \nu; I] := \sum_{n=0}^{N} \left( \|V_n\|^2_{L^2(I;H^{M+1})} + \|V_{n+1}\|^2_{L^2(I;H^{M-1})} \right) < \infty. \tag{63}$$

Each displayed membership asserts finiteness of its defining squared norm, and the sum has only finitely many terms. The whole-terminal norm is

$$\mathfrak{R}_{N,M}(u;T_*) = C_{N,M}[u_0,\nu;I] < \infty. \tag{64}$$

The constant belongs to the selected finite pair; no constant common to the infinite tower is asserted. This calculation writes the constructed jet's whole-terminal memberships as the finite norm in (61). Its finiteness is retained under the coordinate projection on $I$.

Taking $N \geq n$ and $M = m$ gives the two memberships in (59). Taking $M \geq m + |\alpha|$ and differentiating spatially gives $J_{n,\alpha} \in L^2(I;H^m)$. The pressure formula and Theorem 3.8 give

$$\|\Pi_{n,\alpha}\|_{L^1(I;H^m)} \leq C_{n,m,\alpha} \sum_{a=0}^{n} \|V_a\|_{L^2(I;H^{m+|\alpha|+1})} \|V_{n-a}\|_{L^2(I;H^{m+|\alpha|+1})} < \infty.$$

When $N' \geq N$ and $M' \geq M$, both rectangles begin with one initial jet. The temporal recurrence, spatial Leibniz rule, and normalized pressure formula give identical expressions on every common coordinate; uniqueness identifies their realizations. Deleting the outer coordinates recovers the smaller rectangle. Time restriction preserves those same coordinates, while (63) and (64) identify their whole-terminal norms. □

Its two cofinal readings are consequences of that finite-row output, rather than consequences of one another.

**Corollary 3.10** (Mixed-index exhaustion of the compatible rectangles)**.** *Under the hypotheses of Theorem 3.9, compatibility of the finite rows gives*

$$u \in \bigcap_{r,m\in\mathbb{N}_0} H^r(I;H^m_\sigma(\mathbb{R}^3)), \qquad I = (0,T_*). \tag{65}$$

*Its mixed velocity–pressure coordinates are*

$$J_{n,\alpha} = \partial_x^\alpha \partial_t^n u, \quad \Pi_{n,\alpha} = \partial_x^\alpha \partial_t^n p, \qquad n \in \mathbb{N}_0, \quad \alpha \in \mathbb{N}_0^3. \tag{66}$$

*These coordinates obey the pressure-complete recurrences of Proposition 3.4 and agree under every temporal and spatial restriction.*

*Proof.* Fix finite $r, m$. A rectangle with temporal depth at least $r$ and spatial height at least $m$ contains $V_n \in L^2(I;H^m)$ for $0 \leq n \leq r$. The recurrence identifies these rows with the weak derivatives $\partial_t^n u$, so

$$\|u\|^2_{H^r(I;H^m)} = \sum_{n=0}^{r} \|V_n\|^2_{L^2(I;H^m)} < \infty.$$

Taking every finite $(r,m)$ proves (65). The coordinate memberships and restriction compatibility in Theorem 3.9 give (66). □

The return from the spatial column to the mixed jet uses only the original equation and the one fixed history.

**Proposition 3.11** (Spatial-column reconstruction of the mixed intersection)**.** *Let $(T_*,u,p)$ be the pressure-normalized maximal history with $T_* < \infty$, put $I = (0,T_*)$, and assume*

$$u \in \mathscr{S}_0(I) := \bigcap_{m\in\mathbb{N}_0} L^2(I;H^m_\sigma(\mathbb{R}^3)). \tag{67}$$

*Then $V_n, Q_n \in C([0,T_*];H^m)$ for every finite $n,m$, the adjacent rows satisfy* (59)*, and* (65) *follows for this same history.*

*Proof.* Fix $m$. Lemmas 2.4 and 2.5 give

$$\|\mathbb{P}\nabla\cdot(u\otimes u)\|_{L^1(I;H^m)} \le C_m \|u\|^2_{L^2(I;H^{m+2})} < \infty,$$
$$\|\Delta u\|_{L^1(I;H^m)} \le \sqrt{T_*}\,\|u\|_{L^2(I;H^{m+2})} < \infty.$$

Substitution into the unforced equation

$$u_t = \nu\Delta u - \mathbb{P}\nabla\cdot(u\otimes u)$$

gives $u_t \in L^1(I;H^m)$. Also $u \in L^1(I;H^m)$, since $T_* < \infty$. The Banach-valued fundamental theorem gives

$$u \in W^{1,1}(I;H^m) \hookrightarrow C([0,T_*];H^m) \qquad (m \in \mathbb{N}_0).$$

Integrating the equation from the initial datum makes the trace bound explicit:

$$\sup_{0\le t\le T_*} \|u(t)\|_{H^m} \le \|u_0\|_{H^m} + \nu\sqrt{T_*}\,\|u\|_{L^2(I;H^{m+2})} + C_m\|u\|^2_{L^2(I;H^{m+2})} < \infty.$$

These continuous representatives agree under Sobolev embedding because they extend the original velocity. Their endpoint values define one $u_* \in H^\infty_\sigma$.

Now assume the temporal rows $V_0,\dots,V_n$ are continuous on $[0,T_*]$ in every finite Sobolev space. This has just been proved for $V_0 = u$. The pressure and velocity recurrences are

$$Q_n = R_iR_j\sum_{a=0}^{n}\binom{n}{a}(V_a)_i(V_{n-a})_j,$$
$$V_{n+1} = \nu\Delta V_n - \sum_{a=0}^{n}\binom{n}{a}(V_a\cdot\nabla)V_{n-a} - \nabla Q_n.$$

Products, spatial differentiation, and the Riesz operators are continuous between the required finite Sobolev spaces. Choosing sufficiently high spatial orders in the available rows makes both right-hand sides continuous in any prescribed $H^m$. On $I$, they equal the actual derivatives $Q_n = \partial_t^n p$ and $V_{n+1} = \partial_t V_n$ by Proposition 3.4. The identity

$$V_n(t) - V_n(s) = \int_s^t V_{n+1}(\tau)\,d\tau \qquad \text{in } H^m$$

extends from interior times to $0 \le s \le t \le T_*$ by continuity. It identifies the continuous extension with the temporal derivative, including the one-sided derivatives at the endpoints. Induction gives $V_n, Q_n \in C([0,T_*];H^m)$ for every finite $n,m$. Finite $T_*$ now gives the corresponding $L^2(I;H^m)$ memberships. In particular, selecting adjacent temporal coordinates gives

$$V_n \in L^2(I;H^{m+1}), \qquad V_{n+1} \in L^2(I;H^{m-1}) \qquad (n,m \in \mathbb{N}_0).$$

For fixed finite $r,m$, these actual weak derivatives give

$$\|u\|^2_{H^r(I;H^m)} = \sum_{n=0}^{r}\|V_n\|^2_{L^2(I;H^m)} < \infty.$$

Thus the full-interval spatial component propagates through the original equation to every finite mixed Sobolev membership. The argument uses finitely many temporal orders at each step, with no bound required to be uniform over $r,m$. The identities in Proposition 3.4 give compatibility: restricting the indices retains the corresponding derivatives of $u,p$. These are the finite coordinates used in Theorem 4.6 and the derivative-sum calculation of Theorem 5.2. □

The adjacent memberships also give a bound valid up to the proposed terminal time. Apply Lemma 2.7 with $\partial_t V_n = V_{n+1}$. It gives

$$V_n \in C([0,T_*];H^m)$$

and, using the initial value determined in Lemma 3.7,

$$\sup_{0\le t\le T_*} \|V_n(t)\|_{H^m}^2 \le \|V_n(0)\|_{H^m}^2 + 2\,\|V_{n+1}\|_{L^2(I;H^{m-1})}\,\|V_n\|_{L^2(I;H^{m+1})} < \infty.$$

Both time norms on the right are supplied on the entire interval $I$, so the bound remains fixed as $t$ approaches $T_*$. The trace at each spatial order is a continuous representative of the original row. Uniqueness of these representatives under Sobolev embedding makes their terminal values compatible. The detailed endpoint calculation in Theorem 5.5 uses precisely these rows and norms.

The pressure spaces in the statement retain the original normalization. For fixed $n, m$, the compatible finite-row family gives $V_a \in L^2(I;H^{m+1})$ for $0 \le a \le n$. The pressure formula and the finite-row product estimate give

$$\|Q_n\|_{L^1(I;H^m)} \le C_m^{\mathrm{p}} \sum_{a=0}^{n} \binom{n}{a} \|V_a\|_{L^2(I;H^{m+1})}\,\|V_{n-a}\|_{L^2(I;H^{m+1})} < \infty.$$

Choosing the spatial order higher by $|\alpha|$ gives the asserted mixed pressure coordinate. Differentiation and restriction act on the fixed distributions $u, p$, so their overlapping coordinates agree. This pressure completion is already carried by the compatible finite rows of Theorem 3.9; it is retained in both descriptions.

Fixing temporal and spatial orders selects a finite rectangle from the mixed jet. Adding its squared row norms gives Theorem 4.6, and monotone convergence gives its value on $(0,T_*)$ in Proposition 4.7. Letting the rectangle indices increase exhausts every finite mixed velocity and pressure row. The next section derives the completed-height exhaustion on these finite rectangles: all finite $R_k$ give the complete spatial Sobolev column, and the recurrence above reconstructs every temporal and pressure row. The two routes meet at that direct mixed Sobolev membership.

The endpoint argument uses adjacent temporal coordinates of this family. Theorem 5.5 applies the Hilbert-scale trace estimate to obtain their compatible limits at $T_*$. Theorem 5.6 passes the pressure formula and velocity recurrence to those limits. The resulting smooth endpoint and its jet are the data used in Theorem 5.8; local existence and overlap uniqueness continue the original history past the proposed finite maximal time. The small-critical-data calculation in Theorem 8.3 gives a separate quantitative result under its stated smallness condition.

# 4 Critical-height relationships on these rectangles

A finite rectangle selects temporal and spatial orders from the datum-generated mixed jet, retaining their velocity and pressure coordinates and governing relationships. Across all finite $N, M$, every larger rectangle restricts to identical derivatives and pressure coordinates in each smaller rectangle, and the family exhausts every finite mixed index. Independently, the completed critical-height identities are evaluated on those same finite rectangles. Exhausting their heights gives every spatial Sobolev row, and the Navier–Stokes recurrence reconstructs those temporal and pressure rows.

**Proposition 4.1** (Kinetic-energy identity)**.** *For every* $0 \le t < T_*$,

$$\frac{1}{2}\,\|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \,\mathrm{d}s = \frac{1}{2}\,\|u_0\|_2^2 . \tag{68}$$

*Consequently, for* $0 < T < T_*$,

$$\|u\|_{L^\infty(0,T;L^2)} \le \|u_0\|_2 , \tag{69}$$

$$\|u\|^2_{L^2(0,T;H^1)} \le \left(T + \frac{1}{2\nu}\right) \|u_0\|_2^2 . \tag{70}$$

*Proof.* Take the $L^2$ pairing of the momentum equation with $u$. Integration by parts and $\nabla \cdot u = 0$ give

$$\langle (u \cdot \nabla)u, u\rangle_{L^2} = 0, \qquad \langle \nabla p, u\rangle_{L^2} = 0, \qquad \langle \Delta u, u\rangle_{L^2} = -\|\nabla u\|_2^2 .$$

Integration in time proves (68). The two estimates follow by discarding the nonnegative dissipation and adding $\int_0^T \|u(s)\|_2^2 \, \mathrm{d}s$. □

Pairing a differentiated velocity equation with a spatial derivative of that row expresses the change of its Sobolev energy through viscous dissipation and the full transport–pressure contribution. Repeated time differentiation then exposes higher spatial energies, with the differentiated nonlinear terms retained in each identity.

**Definition 4.2** (Completed height rows)**.** For $r \in \mathbb{N}_0$ and $s \ge 0$, set

$$E_{r,s}(t) := \frac{1}{2} \|\Lambda^s V_r(t)\|_2^2 , \tag{71}$$

$$\mathcal{P}_{r,s}(t) := -\operatorname{Re}\left\langle \Lambda^s V_r(t), \Lambda^s \left( \sum_{a=0}^{r} \binom{r}{a} (V_a \cdot \nabla) V_{r-a} + \nabla Q_r \right)(t) \right\rangle_{L^2} . \tag{72}$$

For the base row write $E_s := E_{0,s}$, $\mathcal{P}_s := \mathcal{P}_{0,s}$, and $H := E_{1/2}$.

Differentiating this quadratic height along the classical history gives

$$H^{(r)}(t) = \frac{1}{2} \sum_{a=0}^{r} \binom{r}{a} \left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{r-a}(t) \right\rangle_{L^2} .$$

On a proposed finite maximal interval $I = (0, T_*)$, the compatible finite datum rectangles give the continuous $H^1$ row traces

$$\left\| H^{(r)} \right\|_{L^\infty(I)} \le \frac{1}{2} \sum_{a=0}^{r} \binom{r}{a} \|V_a\|_{C([0,T_*];H^1)} \|V_{r-a}\|_{C([0,T_*];H^1)} < \infty.$$

Every finite derivative of $H$ belongs to $L^2(I)$, since $T_* < \infty$. Thus $H \in \bigcap_{r\in\mathbb{N}_0} H^r(I)$, and temporal Sobolev embedding gives its smoothness. The identities below retain the complete transport–pressure terms and expose, at every finite height, the spatial energy carried by that rectangle.

**Proposition 4.3** (Completed height recurrence in every temporal row)**.** *For every strict classical interval* $(0, T)$*, every* $r \in \mathbb{N}_0$*, and every* $s \ge 0$*,*

$$\partial_t E_{r,s} = \mathcal{P}_{r,s} - 2\nu E_{r,s+1}. \tag{73}$$

*For every* $k \in \mathbb{N}$*,*

$$\partial_t^k E_{r,1/2} = (-2\nu)^k E_{r,k+1/2} + \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} \mathcal{P}_{r,j+1/2}. \tag{74}$$

*In the base row $r = 0$,*

$$H^{(k)} = (-2\nu)^k E_{k+1/2} + \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} \mathcal{P}_{j+1/2}, \tag{75}$$

*and the highest spatial energy is the exact coordinate*

$$E_{k+1/2} = \frac{H^{(k)} - \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} \mathcal{P}_{j+1/2}}{(-2\nu)^k}. \tag{76}$$

*Proof.* Pair the temporal recurrence for $V_r$ with $\Lambda^{2s} V_r$. The pressure-complete nonlinear sum is exactly $\mathcal{P}_{r,s}$, while

$$\nu \operatorname{Re} \langle \Lambda^s V_r, \Lambda^s \Delta V_r \rangle_{L^2} = -\nu \left\| \Lambda^{s+1} V_r \right\|_2^2 = -2\nu E_{r,s+1}.$$

This proves (73). Induction on $k$ gives the stronger identity

$$\partial_t^k E_{r,s} = (-2\nu)^k E_{r,s+k} + \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} \mathcal{P}_{r,s+j}. \tag{77}$$

Indeed, differentiation of the order-$k$ identity followed by $\partial_t E_{r,s+k} = \mathcal{P}_{r,s+k} - 2\nu E_{r,s+k+1}$ produces the order-$k+1$ identity. Taking $s = 1/2$ proves (74). Taking $r = 0$ gives (75); since $\nu > 0$, division by $(-2\nu)^k$ gives (76). □

**Definition 4.4** (Completed critical-height coordinates)**.** Set

$$R_0 := H = E_{1/2}, \qquad R_k := \frac{H^{(k)} - \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} \mathcal{P}_{j+1/2}}{(-2\nu)^k} \quad (k \in \mathbb{N}). \tag{78}$$

Proposition 4.3 gives the exact identity

$$R_k = E_{k+1/2} = \frac{1}{2} \left\| \Lambda^{k+1/2} u \right\|_2^2 \qquad (k \in \mathbb{N}_0). \tag{79}$$

**Definition 4.5** (Finite mixed rectangle)**.** Let $0 < T < T_*$ and $N, M \in \mathbb{N}_0$. Restrict the whole-terminal record in (58) to $(0, T)$:

$$\mathcal{R}_{N,M}[u_0; T] := \left( (V_n|_{(0,T)})_{n=0}^N, (V_{n+1}|_{(0,T)})_{n=0}^N; (Q_n|_{(0,T)})_{n=0}^N \right).$$

Its associated velocity and pressure norms are

$$\mathfrak{R}_{N,M}(u; T) := \sum_{n=0}^N \|V_n\|_{L^2(0,T;H^{M+1})}^2 + \sum_{n=0}^N \|V_{n+1}\|_{L^2(0,T;H^{M-1})}^2, \tag{80}$$

$$\mathfrak{Q}_{N,M}(p; T) := \sum_{n=0}^N \left( \|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})} \right). \tag{81}$$

The pressure-gradient coordinate is retained through (81) and with every row constrained by Proposition 3.4.

**Theorem 4.6** (Compatible finite rectangle norms)**.**

*Let $\nu > 0$, $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and let $(T_*, u, p)$ be the maximal pressure-complete classical history generated by $(u_0, \nu)$. If $T_* < \infty$, then, for every $N, M \in \mathbb{N}_0$,*

$$\mathfrak{R}^*_{N,M}[u_0, \nu, T_*] := \sup_{0<T<T_*} \mathfrak{R}_{N,M}(u; T) < \infty. \tag{82}$$

*This assertion is coordinatewise in $(N, M)$; no bound uniform in all derivative orders is asserted. The quantity $\mathfrak{R}^*_{N,M}$ is the exact supremum for this history; for general data the theorem does not replace it by a formula involving only finitely many norms of $u_0$. Equivalently, for every $n, m \in \mathbb{N}_0$,*

$$V_n \in L^2(0, T_*; H^{m+1}), \qquad V_{n+1} \in L^2(0, T_*; H^{m-1}). \tag{83}$$

*Every coordinate in* (83) *is the corresponding distributional derivative of the single maximal history.*

*Proof.* Apply the compatible finite-rectangle statement of Theorem 3.9 on $I = (0, T_*)$. For fixed $N, M$, (59) gives

$$V_n \in L^2(I; H^{M+1}), \qquad V_{n+1} \in L^2(I; H^{M+1}) \hookrightarrow L^2(I; H^{M-1}), \qquad 0 \le n \le N.$$

Thus the finitely many selected coordinates have the finite total

$$C_{N,M} := \sum_{n=0}^{N} \left( \|V_n\|^2_{L^2(I;H^{M+1})} + \|V_{n+1}\|^2_{L^2(I;H^{M-1})} \right) < \infty.$$

Restriction decreases each norm, so

$$\mathfrak{R}_{N,M}(u; T) \le C_{N,M} \qquad (0 < T < T_*).$$

Taking the supremum proves (82). The number $C_{N,M}$ is fixed before the cutoff $T$ is chosen. Its finiteness is coordinatewise and does not require a bound common to all derivative orders. □

The displayed sum is the velocity norm of $\mathcal{R}_{N,M}$. The next proposition proves its equivalence with the adjacent-row memberships and identifies its whole-interval value by monotone convergence.

**Proposition 4.7** (Norm equivalence and full-interval realization)**.**

*For the fixed maximal history, the family of bounds* (82) *is equivalent to the adjacent-row statement* (83)*. Whenever either formulation holds, for every $N, M \in \mathbb{N}_0$,*

$$\sum_{n=0}^{N} \left( \|V_n\|^2_{L^2(0,T_*;H^{M+1})} + \|V_{n+1}\|^2_{L^2(0,T_*;H^{M-1})} \right) = \lim_{T \uparrow T_*} \mathfrak{R}_{N,M}(u; T) = \mathfrak{R}^*_{N,M}[u_0, \nu, T_*]. \tag{84}$$

*Every row is the corresponding distributional derivative of the single pressure-normalized maximal history.*

*Proof.* Assume first (83). For fixed $N, M$, set

$$C_{N,M} := \sum_{n=0}^{N} \left( \|V_n\|^2_{L^2(0,T_*;H^{M+1})} + \|V_{n+1}\|^2_{L^2(0,T_*;H^{M-1})} \right).$$

The sum is finite because it contains finitely many instances of (83). Restriction from $(0, T_*)$ to $(0, T)$ gives

$$\mathfrak{R}_{N,M}(u; T) \le C_{N,M} \qquad (0 < T < T_*),$$

and hence (82).

Conversely, assume (82). Every term in $\mathfrak{R}_{N,M}(u;T)$ is the integral of a nonnegative function. Monotone exhaustion of $(0,T_*)$ gives (84). Taking $N=n$ and $M=m$ yields (83) for each finite $(n,m)$. Proposition 3.4 identifies every entry with the corresponding distributional derivative of the fixed pressure-normalized history. □

**Corollary 4.8** (Critical-height exhaustion and Sobolev recovery)**.** *For the datum-generated compatible finite rectangles of Theorem 3.9, every completed critical coordinate belongs to $L^1(0,T_*)$, and*

$$\mathscr{H}_k[u_0,\nu,T_*] := \|R_k\|_{L^1(0,T_*)} = \frac{1}{2}\left\|\Lambda^{k+1/2}u\right\|^2_{L^2(0,T_*;L^2)} \le \frac{1}{2}\mathfrak{R}^*_{0,k}[u_0,\nu,T_*] \qquad (k\in\mathbb{N}_0). \tag{85}$$

*For every $m\in\mathbb{N}_0$, the completed heights recover the spatial Sobolev row through*

$$\|u\|^2_{L^2(0,T_*;H^m)} \le 2^m\left(T_*\|u_0\|_2^2 + 2\|R_m\|_{L^1(0,T_*)}\right) \le 2^m\left(T_*\|u_0\|_2^2 + \mathfrak{R}^*_{0,m}[u_0,\nu,T_*]\right). \tag{86}$$

*Thus the compatible critical-height exhaustion gives*

$$(R_k)_{k\in\mathbb{N}_0} \in \prod_{k\in\mathbb{N}_0} L^1(0,T_*), \qquad u\in\bigcap_{m\in\mathbb{N}_0} L^2(0,T_*;H^m(\mathbb{R}^3)). \tag{87}$$

*Proof.* Take the $n=0, m=k$ finite rectangle in (59); equivalently, take $N=0$ and $M=k$ in (84). Since $|\xi|^{2k+1}\le\langle\xi\rangle^{2k+2}$,

$$\begin{aligned} 2\|R_k\|_{L^1(0,T_*)} &= \left\|\Lambda^{k+1/2}u\right\|^2_{L^2(0,T_*;L^2)} \\ &\le \|u\|^2_{L^2(0,T_*;H^{k+1})} \le \mathfrak{R}^*_{0,k}[u_0,\nu,T_*], \end{aligned}$$

which proves (85). For $|\xi|\le 1$, $\langle\xi\rangle^{2m}\le 2^m$; for $|\xi|>1$, $\langle\xi\rangle^{2m}\le 2^m|\xi|^{2m+1}$. Hence

$$\|f\|^2_{H^m} \le 2^m\left(\|f\|_2^2 + \left\|\Lambda^{m+1/2}f\right\|_2^2\right).$$

Integrating this inequality, using (69) and (85), proves (86). Since $k$ and $m$ are arbitrary finite indices, (87) follows. □

The preceding calculation recovers that all-orders membership through the completed heights. For each finite $k$, the quantity $R_k = E_{k+1/2}$ is computed from the velocity row of a sufficiently high rectangle. Compatibility under enlargement identifies all such evaluations with one history. Taking every finite $k$ gives (87). Applying Proposition 3.11 to that complete spatial row produces all temporal and pressure rows and hence (65). Conversely, the mixed-index exhaustion contains the velocity rows from which every finite $R_k$ is computed. Both descriptions use that datum-generated rectangle $\mathcal{R}_{N,M}$ at every finite pair $(N,M)$.

**Proposition 4.9** (Pressure completion)**.** *Under the hypotheses of Theorem 4.6, for every $N,M\in\mathbb{N}_0$,*

$$\mathfrak{Q}^*_{N,M}[u_0,\nu,T_*] := \sup_{0<T<T_*}\mathfrak{Q}_{N,M}(p;T) \le C^{\mathrm{p}}_M(2^{N+1}-1)\mathfrak{R}^*_{N,M}[u_0,\nu,T_*]. \tag{88}$$

*For every $0 \le n \le N$, the identities*

$$Q_n = R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i} V_{n-a,j}, \tag{89}$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^{n} \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n \tag{90}$$

*hold respectively in $L^1(0,T_*;H^M)$ and $L^1(0,T_*;H^{M-1})$, including when $M=0$.*

*Proof.* For every $0 < T < T_*$, Theorem 3.8 gives

$$\|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})} \le C_M^{\mathrm{p}} 2^n \mathfrak{R}^*_{N,M}[u_0,\nu,T_*].$$

Sum over $0 \le n \le N$, then let $T \uparrow T_*$. Monotone convergence in the nonnegative norm integrals proves (88). Since $\nabla \cdot V_a = 0$,

$$(V_a \cdot \nabla) V_{n-a} = \nabla \cdot (V_{n-a} \otimes V_a).$$

Lemma 2.4 and Cauchy–Schwarz in time give

$$\|V_{n-a} \otimes V_a\|_{L^1(0,T_*;H^M)} \le C_M^{\mathrm{alg}} \|V_a\|_{L^2(0,T_*;H^{M+1})} \|V_{n-a}\|_{L^2(0,T_*;H^{M+1})}.$$

Thus every transport term belongs to $L^1(0,T_*;H^{M-1})$, also for $M=0$. Moreover,

$$\|\Delta V_n\|_{L^1 H^{M-1}} \le T_*^{1/2} \|V_n\|_{L^2 H^{M+1}}.$$

Theorem 4.6 places $V_{n+1}$ in $L^2(0,T_*;H^{M-1})$, hence in $L^1(0,T_*;H^{M-1})$. The strict-interval identities are identities of distributions; their full-interval representatives belong to the stated spaces. This proves the two identities on $(0,T_*)$. □

**Definition 4.10** (Finite rectangles and their restrictions)**.** For $N, M \in \mathbb{N}_0$, the full-interval rectangle already defined in (58) contains the rows

$$\begin{aligned} V_n &\in L^2(0,T_*;H^{M+1}), \\ V_{n+1} &\in L^2(0,T_*;H^{M-1}), \qquad 0 \le n \le N. \\ Q_n &\in L^1(0,T_*;H^M), \end{aligned} \tag{91}$$

If $N' \le N$ and $M' \le M$, restricting to $(N',M')$ means discarding the temporal coordinates with index larger than $N'$ and applying

$$H^{M+1} \hookrightarrow H^{M'+1}, \qquad H^{M-1} \hookrightarrow H^{M'-1}, \qquad H^M \hookrightarrow H^{M'}$$

to the retained coordinates.

**Proposition 4.11** (Restriction compatibility)**.** *For $N' \le N$ and $M' \le M$,*

$$\mathcal{R}_{N,M}[u_0;T_*]|_{(N',M')} = \mathcal{R}_{N',M'}[u_0;T_*]. \tag{92}$$

*If $(N'',M'') \le (N',M') \le (N,M)$, then restricting first to $(N',M')$ and then to $(N'',M'')$ gives the rows obtained by restricting directly to $(N'',M'')$. Thus the datum-generated rectangles form a compatible family.*

*Proof.* All retained entries are the fixed distributions $\partial_t^n u$, $\partial_t^{n+1} u$, and $\partial_t^n p$. The Sobolev embeddings preserve those distributions. Deletion of rows and composition of canonical embeddings are associative. □

# 5 Intersection coordinates, endpoint jet, and restart

The mixed-index and critical-height exhaustions have now identified that all-orders membership of the maximal history. The endpoint argument no longer distinguishes their coordinates. For every finite spatial order, the common family contains the adjacent velocity and pressure rows needed to reach $T_*$, recover one endpoint jet, and restart the original history.

**Definition 5.1** (Compatible exhaustive rectangle family)**.** The datum-generated rectangles form the family

$$\{\mathcal{R}_{N,M}[u_0;T_*] : N, M \in \mathbb{N}_0\}, \qquad \mathcal{R}_{N,M}[u_0;T_*]|_{(N',M')} = \mathcal{R}_{N',M'}[u_0;T_*] \quad (N' \le N,\ M' \le M). \tag{93}$$

It is exhaustive in the direct finite-row sense: for every $r, m \in \mathbb{N}_0$, a rectangle with $N \ge r$ and $M \ge m$ contains

$$V_n \in L^2(0,T_*;H^m), \qquad Q_n \in L^1(0,T_*;H^m), \qquad 0 \le n \le r. \tag{94}$$

**Theorem 5.2** (Compatible exhaustive rectangle family)**.**

*Let $\nu > 0$, $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and let $(T_*, u, p)$ be the maximal pressure-complete classical history generated by $(u_0, \nu)$. If $T_* < \infty$, then*

$$\mathcal{R}_{N,M}[u_0;T_*]|_{(N',M')} = \mathcal{R}_{N',M'}[u_0;T_*] \qquad (N' \le N,\ M' \le M), \tag{95}$$

*and every row is the corresponding derivative of the original pressure-complete history. Every pair $(V_n, Q_n)$ belongs to a sufficiently large rectangle $\mathcal{R}_{N,M}[u_0;T_*]$ and retains the memberships*

$$V_n \in \bigcap_{m\in\mathbb{N}_0} L^2(0,T_*;H^m_\sigma(\mathbb{R}^3)), \tag{96}$$

$$Q_n \in \bigcap_{m\in\mathbb{N}_0} L^1(0,T_*;H^m(\mathbb{R}^3)), \tag{97}$$

*The family exhausts every finite pair of temporal and spatial orders. Its velocity hence has the direct membership*

$$u \in \bigcap_{r,m\in\mathbb{N}_0} H^r(0,T_*;H^m(\mathbb{R}^3)), \tag{98}$$

*and, for every $r, m \in \mathbb{N}_0$,*

$$\sum_{n=0}^{r} \|\partial_t^n u\|^2_{L^2(0,T_*;H^m)} \le \mathfrak{R}^*_{r,m}[u_0,\nu,T_*]. \tag{99}$$

*Proof.* The mixed-index exhaustion in Theorem 3.10 and the critical-height exhaustion in Corollary 4.8, followed by Proposition 3.11, give those velocity memberships in (96) and (98). The pressure recurrence then gives (97). For each finite $N, M$, the rectangle is selected directly from the mixed jet. Theorem 4.6 and Proposition 4.9 give its finite row bounds. Proposition 4.11 gives

$$\mathcal{R}_{N,M}[u_0;T_*]|_{(N',M')} = \mathcal{R}_{N',M'}[u_0;T_*] \qquad (N' \le N,\ M' \le M).$$

Hence the rectangles are compatible. Every $V_n, Q_n$ appears in a sufficiently large rectangle, and every spatial derivative is fixed by its distributional velocity or pressure row. The family consequently exhausts all finite mixed velocity and pressure rows in either description.

Fix finite $r, m$, put $I = (0, T_*)$, and choose $N \geq r$, $M \geq m$. The upper rows of that finite record belong to $L^2(I; H^{M+1}) \hookrightarrow L^2(I; H^m)$ for $0 \leq n \leq r$. Proposition 3.4 identifies them as the distributional derivatives $V_n = \partial_t^n u$. Thus the finite record contains every term in the derivative-sum norm of $H^r(I; H^m)$. Taking $N = r$, $M = m$ gives

$$\begin{aligned}\|u\|^2_{H^r(I;H^m)} &= \sum_{n=0}^{r} \|V_n\|^2_{L^2(I;H^m)} \\ &\leq \sum_{n=0}^{r} \|V_n\|^2_{L^2(I;H^{m+1})} \leq \mathfrak{R}^*_{r,m}[u_0, \nu, T_*].\end{aligned}$$

This proves (99) and shows how each mixed Sobolev norm is recovered from a finite record. The mixed-index and completed-height descriptions agree on that record and hence on the full compatible family. □

**Proposition 5.3** (Complete spatial row and common endpoint)**.** *Under the hypotheses of Theorem 5.2, for every $m \in \mathbb{N}_0$,*

$$u \in W^{1,1}(0, T_*; H^m_\sigma(\mathbb{R}^3)) \hookrightarrow C([0, T_*]; H^m_\sigma(\mathbb{R}^3)). \tag{100}$$

*The endpoint values supplied at the different Sobolev orders are the images of one common state*

$$u_* := u(T_*) \in \bigcap_{m \in \mathbb{N}_0} H^m_\sigma(\mathbb{R}^3) = H^\infty_\sigma(\mathbb{R}^3). \tag{101}$$

*Proof.* Fix $m$. The zeroth temporal coordinate of (98) gives

$$u \in L^2(0, T_*; H^{m+2}).$$

The pressure-completed nonlinearity satisfies

$$\|\mathbb{P}\nabla\cdot (u \otimes u)\|_{H^m} \leq C_m \|u\|^2_{H^{m+2}}$$

by Lemmas 2.4 and 2.5. It belongs to $L^1(0, T_*; H^m)$. This spatial row and $T_* < \infty$ give $\Delta u \in L^1(0, T_*; H^m)$, and the projected equation

$$u_t = \nu\Delta u - \mathbb{P}\nabla\cdot (u \otimes u)$$

places $u_t$ in $L^1(0, T_*; H^m)$. Also $u \in L^1(0, T_*; H^m)$, so the Banach-valued fundamental theorem gives (100).

If $k \geq m$, the $H^k$- and $H^m$-continuous representatives are the original preterminal function after the embedding $H^k \hookrightarrow H^m$. Uniqueness of continuous representatives makes their endpoint values agree under that embedding. These compatible values define the single state in (101). Continuity of divergence from $H^{m+1}$ to $H^m$ gives $\nabla \cdot u_* = 0$. □

**Corollary 5.4** (Full-interval pressure-complete mixed jet)**.** *Under the hypotheses of Theorem 5.2, for every $r, m, n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,*

$$u \in H^r(0, T_*; H^m_\sigma(\mathbb{R}^3)), \tag{102}$$

$$J_{n,\alpha} := \partial_x^\alpha V_n \in L^2(0, T_*; H^{m+1}(\mathbb{R}^3)), \tag{103}$$

$$J_{n+1,\alpha} \in L^2(0, T_*; H^{m-1}(\mathbb{R}^3)), \tag{104}$$

$$\Pi_{n,\alpha} := \partial_x^\alpha Q_n \in L^1(0, T_*; H^m(\mathbb{R}^3)). \tag{105}$$

*The following finite bounds hold:*

$$\sum_{j=0}^{r} \|V_j\|^2_{L^2(0,T_*;H^m)} \le \mathfrak{R}^*_{r,m}[u_0,\nu,T_*], \tag{106}$$

$$\|J_{n,\alpha}\|^2_{L^2(0,T_*;H^{m+1})} + \|J_{n+1,\alpha}\|^2_{L^2(0,T_*;H^{m-1})} \le \mathfrak{R}^*_{n,m+|\alpha|}[u_0,\nu,T_*], \tag{107}$$

$$\|\Pi_{n,\alpha}\|_{L^1(0,T_*;H^m)} \le C^{\mathrm{p}}_{m+|\alpha|}(2^{n+1}-1)\mathfrak{R}^*_{n,m+|\alpha|}[u_0,\nu,T_*]. \tag{108}$$

*Every mixed row satisfies the recurrences in Proposition 3.4 on* $(0,T_*)$.

*Proof.* Equations (102) and (106) follow from Theorem 5.2. The Fourier multiplier bound

$$\|\partial_x^\alpha f\|_{H^s} \le \|f\|_{H^{s+|\alpha|}}$$

and the rectangle $(n, m+|\alpha|)$ give (103)–(107). Apply this multiplier bound to the pressure completion in Proposition 4.9 to obtain (105) and (108). Proposition 3.4 supplies the stated identities. □

**Theorem 5.5** (Compatible full-interval Sobolev traces)**.** *Assume the hypotheses of Theorem 5.2. For* $N, m \in \mathbb{N}_0$*, define*

$$\mathfrak{E}_{N,m} := (1+T_*^{-1})^{1/2}\big(\mathfrak{R}^*_{N,m}[u_0,\nu,T_*]\big)^{1/2}. \tag{109}$$

*If* $0 \le n \le N$ *and* $\alpha \in \mathbb{N}_0^3$*, then*

$$V_n \in C([0,T_*];H^m_\sigma(\mathbb{R}^3)), \qquad \sup_{0\le t\le T_*}\|V_n(t)\|_{H^m} \le \mathfrak{E}_{N,m}, \tag{110}$$

$$J_{n,\alpha} \in C([0,T_*];H^m(\mathbb{R}^3)), \qquad \sup_{0\le t\le T_*}\|J_{n,\alpha}(t)\|_{H^m} \le \mathfrak{E}_{N,m+|\alpha|}. \tag{111}$$

*There are unique values*

$$V_n^* := \lim_{t\uparrow T_*} V_n(t) \in H^\infty(\mathbb{R}^3), \tag{112}$$

$$J^*_{n,\alpha} := \lim_{t\uparrow T_*} J_{n,\alpha}(t) = \partial_x^\alpha V_n^* \in H^\infty(\mathbb{R}^3), \tag{113}$$

*such that, for* $k \ge m$*, the image of the* $H^k$ *trace under* $H^k \hookrightarrow H^m$ *is the* $H^m$ *trace. Quantitatively,*

$$\|V_n^*\|_{H^m} \le \mathfrak{E}_{n,m}, \tag{114}$$

$$\|V_n(t)-V_n^*\|_{H^m} \le (T_*-t)\mathfrak{E}_{n+1,m}, \qquad 0\le t\le T_*, \tag{115}$$

$$\big\|J^*_{n,\alpha}\big\|_{H^m} \le \mathfrak{E}_{n,m+|\alpha|}, \tag{116}$$

$$\big\|J_{n,\alpha}(t)-J^*_{n,\alpha}\big\|_{H^m} \le (T_*-t)\mathfrak{E}_{n+1,m+|\alpha|}. \tag{117}$$

*In particular, the zeroth trace agrees with the common endpoint in Proposition 5.3, and*

$$u_* = V_0^* \in H^\infty_\sigma(\mathbb{R}^3), \qquad u \in \bigcap_{a,m\in\mathbb{N}_0} C^a([0,T_*];H^m_\sigma(\mathbb{R}^3)), \qquad \partial_t^a u(T_*) = V_a^*. \tag{118}$$

*Proof.* For $0 \le n \le N$, Theorem 5.2 provides a finite rectangle containing the adjacent rows

$$V_n \in L^2(0,T_*;H^{m+1}), \qquad \partial_t V_n = V_{n+1} \in L^2(0,T_*;H^{m-1}).$$

That rectangle gives both finite norms; the critical-height description recovers those coordinates. With $A = (1-\Delta)^{1/2}$, the adjacent-row identity is

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\|V_n(t)\|_{H^m}^2 = 2\,\mathrm{Re}\,\big\langle A^{m-1}V_{n+1}(t), A^{m+1}V_n(t)\big\rangle_{L^2}\,.$$

Integrating from the initial time and applying Cauchy–Schwarz gives

$$\sup_{0\le t\le T_*}\|V_n(t)\|_{H^m}^2 \le \|V_n(0)\|_{H^m}^2 + 2\,\|V_{n+1}\|_{L^2(I;H^{m-1})}\,\|V_n\|_{L^2(I;H^{m+1})}\,, \qquad I=(0,T_*). \tag{119}$$

The initial norm is finite by Lemma 3.7. Lemma 2.7 supplies the continuous representative on $[0,T_*]$, justifies the identity by Fourier cutoff, and gives the first membership in (110) and

$$\begin{aligned}\sup_{0\le t\le T_*}\|V_n(t)\|_{H^m}^2 &\le T_*^{-1}\,\|V_n\|_{L^2H^m}^2 + 2\,\|V_n\|_{L^2H^{m+1}}\,\|V_{n+1}\|_{L^2H^{m-1}}\\ &\le (1+T_*^{-1})\mathfrak{R}^*_{N,m}[u_0,\nu,T_*].\end{aligned}$$

This proves the bound in (110). Apply this lemma to $J_{n,\alpha}$, using (107), to prove (111).

For $k\ge m$, the $H^k$-continuous and $H^m$-continuous representatives equal the original preterminal distribution. After applying $H^k\hookrightarrow H^m$, uniqueness of continuous representatives makes them equal on $[0,T_*]$. Their endpoint values are compatible and define (112)–(113). Spatial differentiation is continuous between the corresponding finite Sobolev spaces, so $J^*_{n,\alpha}=\partial_x^\alpha V_n^*$.

Corollary 5.4 gives $V_n, V_{n+1}\in L^1(0,T_*;H^m)$ and $\partial_t V_n = V_{n+1}$ in distributions. The Banach-valued fundamental identity gives

$$V_n(t)-V_n(s) = \int_s^t V_{n+1}(\tau)\,\mathrm{d}\tau \quad \text{in } H^m, \qquad 0\le s\le t\le T_*. \tag{120}$$

The continuous representative of $V_{n+1}$ is bounded by $\mathfrak{E}_{n+1,m}$. Set $t=T_*$ in (120) to prove (115); spatial differentiation gives (117). The remaining size estimates follow from (110)–(111).

The divergence map $H^{m+1}\to H^m$ is continuous, so $\nabla\cdot u_*=0$. For each $a,m$, the curves $V_0,\dots,V_{a+1}$ are continuous in $H^m$, and (120) identifies the derivative of $V_j$ with $V_{j+1}$. Induction in $j$ proves (118). □

**Theorem 5.6** (Pressure-complete endpoint jet)**.** *Assume the hypotheses of Theorem 5.2. For $n\in\mathbb{N}_0$, define*

$$Q_n^* := R_iR_j\sum_{a=0}^{n}\binom{n}{a}V_{a,i}^*V_{n-a,j}^*. \tag{121}$$

*Then $Q_n^*\in H^\infty(\mathbb{R}^3)$, and, for every $s\in\mathbb{N}_0$,*

$$\|Q_n^*\|_{H^s} \le 9C_s^{\mathrm{alg}}2^n\mathfrak{E}^2_{n,s+1}, \tag{122}$$

$$\|Q_n(t)-Q_n^*\|_{H^s} \le 9C_s^{\mathrm{alg}}2^{n+1}(T_*-t)\mathfrak{E}^2_{n+1,s+1}. \tag{123}$$

*For $\alpha\in\mathbb{N}_0^3$,*

$$\Pi^*_{n,\alpha} := \partial_x^\alpha Q_n^* = \lim_{t\uparrow T_*}\Pi_{n,\alpha}(t) \quad \text{in } H^m(\mathbb{R}^3) \text{ for every } m\in\mathbb{N}_0. \tag{124}$$

*More precisely, for every $m\in\mathbb{N}_0$,*

$$\big\|\Pi^*_{n,\alpha}\big\|_{H^m} \le 9C^{\mathrm{alg}}_{m+|\alpha|}2^n\mathfrak{E}^2_{n,m+|\alpha|+1}, \tag{125}$$

$$\big\|\Pi_{n,\alpha}(t)-\Pi^*_{n,\alpha}\big\|_{H^m} \le 9C^{\mathrm{alg}}_{m+|\alpha|}2^{n+1}(T_*-t)\mathfrak{E}^2_{n+1,m+|\alpha|+1}. \tag{126}$$

*The endpoint rows satisfy, for every $n, q \in \mathbb{N}_0$,*

$$V_{n+1}^* = \nu\Delta V_n^* - \sum_{a=0}^{n} \binom{n}{a} (V_a^* \cdot \nabla) V_{n-a}^* - \nabla Q_n^* \quad \text{in } H^q(\mathbb{R}^3). \tag{127}$$

*Among families rooted at $u_*$ that satisfy* (121) *and* (127) *in every finite Sobolev space, $(V_n^*, Q_n^*)_{n\in\mathbb{N}_0}$ is the unique pressure-normalized endpoint jet. Moreover,*

$$p \in \bigcap_{a,m\in\mathbb{N}_0} C^a([0,T_*]; H^m(\mathbb{R}^3)), \qquad \partial_t^a p(T_*) = Q_a^*. \tag{128}$$

*Proof.* Lemmas 2.4 and 2.5, followed by (110), give

$$\begin{aligned} \|Q_n^*\|_{H^s} &\le 9C_s^{\mathrm{alg}} \sum_{a=0}^{n} \binom{n}{a} \|V_a^*\|_{H^{s+1}} \left\|V_{n-a}^*\right\|_{H^{s+1}} \\ &\le 9C_s^{\mathrm{alg}} 2^n \mathfrak{E}_{n,s+1}^2. \end{aligned}$$

For each product in $Q_n(t) - Q_n^*$, insert $V_a^* V_{n-a}(t)$, use (115), and bound every retained factor by $\mathfrak{E}_{n+1,s+1}$. The two product differences and $\sum_{a=0}^{n} \binom{n}{a} = 2^n$ prove (123). The Fourier multiplier inequality

$$\|\partial_x^\alpha f\|_{H^m} \le \|f\|_{H^{m+|\alpha|}}$$

applied to (122) and (123) proves (125) and (126). The latter tends to zero as $t \uparrow T_*$, which proves (124) in every $H^m$.

Fix $n, q$. The trace of $V_n$ in $H^{q+2}$ carries the diffusion term in Proposition 3.4 to $H^q$ at $T_*$. The product estimate at order $q+1$, using velocity convergence in $H^{q+2}$, carries every transport term to $H^q$. Equation (123) at order $q+1$ carries the pressure gradient to $H^q$. The left side converges to $V_{n+1}^*$ in $H^q$, which proves (127).

The value $V_0^* = u_*$ fixes $Q_0^*$, and the velocity recurrence fixes $V_1^*$. Inductively, $V_0^*, \dots, V_n^*$ fix $Q_n^*$, which fixes $V_{n+1}^*$. This proves uniqueness within the displayed recurrence class. Extend each $Q_n$ to $T_*$ by $Q_n(T_*) := Q_n^*$. On $(0, T_*)$, $Q_n = \partial_t^n p$, so $\partial_t Q_n = Q_{n+1}$ in distributions. Both rows are continuous in every $H^m$ by (123). Hence, for $0 \le s \le t \le T_*$,

$$Q_n(t) - Q_n(s) = \int_s^t Q_{n+1}(\tau)\, \mathrm{d}\tau \quad \text{in } H^m(\mathbb{R}^3). \tag{129}$$

Indeed, the identity holds for $t < T_*$, and the endpoint bounds permit passage to $t = T_*$. Thus $Q_n \in C^1([0,T_*]; H^m)$ with derivative $Q_{n+1}$. Induction gives (128). □

**Lemma 5.7** (Late-slice overlap uniqueness)**.** *Assume the hypotheses of Theorem 5.2. Define*

$$\delta_0 := \min\{1, c_\nu (1 + \mathfrak{E}_{0,3})^{-2}\} > 0, \tag{130}$$

*where $c_\nu$ is the constant in Theorem 2.11. For every $t_0 \in (\max\{0, T_* - \delta_0/2\}, T_*)$, let*

$$(w^{t_0}, q^{t_0}) \in \mathscr{M}_\nu^3(t_0, t_0 + \delta_0; u(t_0)), \qquad (w^*, q^*) \in \mathscr{M}_\nu^3(T_*, T_* + \delta_0; u_*)$$

*be the unique pairs supplied by Theorem 2.11. Then*

$$(w^{t_0}, q^{t_0}) = (u, p) \qquad \text{on } [t_0, T_*), \tag{131}$$

$$(w^{t_0}, q^{t_0}) = (w^*, q^*) \qquad \text{on } [T_*, t_0 + \delta_0]. \tag{132}$$

*Proof.* Equation (110) gives

$$\|u(t_0)\|_{H^3} \le \mathfrak{E}_{0,3}, \qquad \|u_*\|_{H^3} \le \mathfrak{E}_{0,3}.$$

Hence

$$\delta_0 \le \delta_\nu(u(t_0)), \qquad \delta_0 \le \delta_\nu(u_*),$$

with $\delta_\nu$ as in Theorem 2.11. Theorem 2.11 gives both pairs on the stated intervals. For each $t_0 < b < T_*$, Lemma 2.10 places $(u,p)|_{[t_0,b]}$ and $(w^{t_0}, q^{t_0})|_{[t_0,b]}$ in $\mathscr{M}^3_\nu(t_0, b; u(t_0))$. Uniqueness in that class gives (131). Continuity in $H^3$ gives

$$w^{t_0}(T_*) = \lim_{t \uparrow T_*} u(t) = u_*.$$

Lemma 2.10 places the restrictions of $(w^{t_0}, q^{t_0})$ and $(w^*, q^*)$ to $[T_*, t_0 + \delta_0]$ in $\mathscr{M}^3_\nu(T_*, t_0 + \delta_0; u_*)$. Uniqueness in that class gives (132). The right overlap has positive length because $t_0 + \delta_0 > T_* + \delta_0/2$. □

**Theorem 5.8** (Same-history endpoint restart)**.** *Assume the hypotheses of Theorem 5.2. There is a pressure-normalized classical solution $(\widetilde{u}, \widetilde{p})$ on $[0, T_* + \delta_0/2] \times \mathbb{R}^3$ such that*

$$(\widetilde{u}, \widetilde{p})|_{[0,T_*)\times\mathbb{R}^3} = (u,p), \tag{133}$$

$$\widetilde{u}, \widetilde{p} \in C^\infty([0, T_* + \delta_0/2] \times \mathbb{R}^3), \tag{134}$$

$$\partial_t^n \widetilde{u}(T_*) = V_n^*, \qquad \partial_t^n \widetilde{p}(T_*) = Q_n^* \quad (n \in \mathbb{N}_0). \tag{135}$$

*If $0 < \eta \le \delta_0/2$ and $(\widehat{u}, \widehat{p})$ is another pressure-normalized classical extension satisfying*

$$(\widehat{u}, \widehat{p})|_{[0,T_*)} = (u,p), \qquad (\widehat{u}, \widehat{p})|_{[T_*, T_*+\eta]} \in \mathscr{M}^3_\nu(T_*, T_* + \eta; u_*), \tag{136}$$

*then*

$$(\widehat{u}, \widehat{p}) = (\widetilde{u}, \widetilde{p}) \quad \text{on } [0, T_* + \eta]. \tag{137}$$

*Proof.* Let $(w^*, q^*)$ be the endpoint solution from Lemma 5.7, and set

$$W_n^+ := \partial_t^n w^*(T_*), \qquad P_n^+ := \partial_t^n q^*(T_*).$$

Theorem 2.11 makes $(w^*, q^*)$ smooth and pressure-normalized. Differentiating its equations exactly as in the proof of Proposition 3.4 gives

$$P_n^+ = R_i R_j \sum_{a=0}^{n} \binom{n}{a} W_{a,i}^+ W_{n-a,j}^+,$$

$$W_{n+1}^+ = \nu \Delta W_n^+ - \sum_{a=0}^{n} \binom{n}{a} (W_a^+ \cdot \nabla) W_{n-a}^+ - \nabla P_n^+.$$

Since $W_0^+ = u_* = V_0^*$, induction with (121) and (127) gives

$$W_n^+ = V_n^*, \qquad P_n^+ = Q_n^*, \qquad n \in \mathbb{N}_0. \tag{138}$$

Define

$$(\widetilde{u}, \widetilde{p})(t) := \begin{cases} (u,p)(t), & 0 \le t < T_*, \\ (w^*, q^*)(t), & T_* \le t \le T_* + \delta_0/2. \end{cases}$$

Equations (118), (128), and (138) match every one-sided time derivative in every finite Sobolev space. Sobolev embedding gives (134), and the pressure-complete equation holds at $T_*$ by (127). Lemma 5.7 identifies the joined pair near $T_*$ with the single local history $(w^{t_0}, q^{t_0})$, so the gluing is one local classical history across the endpoint; the piecewise definition gives (133). Any pair in (136) has that datum $u_*$ on the right of $T_*$ and belongs to the shifted mild class containing $(w^*, q^*)$. The at-most-one assertion in Theorem 2.11 proves (137). □

The finite rectangle bounds give compatible velocity traces through $T_*$ in Theorem 5.5. The pressure recurrence fixes their normalized endpoint jet in Theorem 5.6. With those endpoint values, Theorem 5.8 extends the original history beyond $T_*$. A proposed finite maximal time would admit a classical continuation. Corollary 5.9 applies this contradiction, and Theorem 5.10 collects the global solution, pressure normalization, and energy identity.

**Corollary 5.9** (Maximal-time contradiction)**.** *The maximal pressure-complete classical history generated by $(u_0, \nu)$ cannot have $T_* < \infty$.*

*Proof.* If $T_* < \infty$, the mixed-index and critical-height descriptions of the finite rectangles in Theorem 3.9 recover that all-orders intersection. Those rectangles give the trace norms, and the chain from Theorem 5.2 through Theorem 5.8 produces a pressure-normalized classical extension to $[0, T_* + \delta_0/2]$, where $\delta_0 > 0$. This contradicts maximality in Definition 3.2. □

**Theorem 5.10** (Global smoothness on $\mathbb{R}^3$)**.** *For every $\nu > 0$ and every $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, there exists a pressure-normalized pair*

$$(u, p) \in C^\infty([0,\infty) \times \mathbb{R}^3; \mathbb{R}^3 \times \mathbb{R}) \tag{139}$$

*satisfying*

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \tag{140}$$

$$\nabla \cdot u = 0, \qquad u(0) = u_0, \tag{141}$$

$$p(t) = R_i R_j (u_i(t) u_j(t)), \qquad \lim_{|x| \to \infty} p(t, x) = 0, \tag{142}$$

$$\frac{1}{2} \|u(t)\|^2_{L^2} + \nu \int_0^t \|\nabla u(s)\|^2_{L^2} \, \mathrm{d}s = \frac{1}{2} \|u_0\|^2_{L^2}, \qquad t \geq 0. \tag{143}$$

*In particular,*

$$\sup_{t \geq 0} \|u(t)\|^2_{L^2(\mathbb{R}^3)} \leq \|u_0\|^2_{L^2(\mathbb{R}^3)} < \infty. \tag{144}$$

*For every finite $T > 0$, its restriction is the unique member of $\mathscr{M}^3_\nu(0, T; u_0)$.*

*Proof.* Let

$$\mathscr{T} := \left\{ \tau > 0 : \begin{array}{c} \text{there exists a pressure-complete classical history} \\ (\tau, u^\tau, p^\tau) \text{ generated by } (u_0, \nu) \end{array} \right\},$$

and put

$$T_* := \sup \mathscr{T}.$$

Theorem 2.11, applied to $u_0 \in H^\infty_\sigma$, supplies a shifted mild solution whose persistence statement gives every finite velocity and pressure row. Sobolev embedding and the displayed recurrences make this pair a pressure-complete classical history, so $\mathscr{T}$ is nonempty. Lemma 2.10 admits each strict classical restriction to the shifted mild class, whose at-most-one property identifies any two such

solutions on their common interval; their union defines a unique solution on $[0,T_*)$. Since $u_0 \in H^\infty_\sigma$, the persistence conclusion of Theorem 2.11, applied on overlapping local intervals, places every strict restriction in the class of Definition 3.2. An extension beyond $T_*$ would restrict to an interval $[0,\tau]$ with $\tau > T_*$, contrary to the definition of the supremum. Thus $(T_*,u,p)$ is maximal. Corollary 5.9 gives $T_* = \infty$.

Fix $T < \infty$. Persistence in Theorem 2.11, iterated along the unique history, retains every finite spatial Sobolev order on $[0,T]$. Proposition 3.4 retains every finite temporal row. Lemma 2.2 gives (139); arbitrariness of $T$ gives the global class. The pressure belongs to $H^s(\mathbb{R}^3)$ for every $s$, and the Fourier inversion proof of Lemma 2.2 places its continuous representative in $C_0(\mathbb{R}^3)$, which proves the limit in (142). Lemma 2.10 places $(u,p)|_{[0,T]}$ in $\mathscr{M}^3_\nu(0,T;u_0)$. If $(\widetilde{u},\widetilde{p}) \in \mathscr{M}^3_\nu(0,T;u_0)$, the at-most-one assertion in Theorem 2.11 gives $(\widetilde{u},\widetilde{p}) = (u,p)$ on $[0,T]$. This proves the stated uniqueness.

Proposition 4.1 holds on every finite interval. Since $T$ was arbitrary, it gives (143) for every $t \geq 0$; (144) follows by discarding the nonnegative dissipation term. □

# 6 Matrix forms of the completed intersection

The compatible intersection, its endpoint jet, and the original history's restart have now been derived. The matrices in this section reorganize those completed rows. The mixed-index and completed-height exhaustions, together with their compatibility on $(0,T_*)$, have already been proved before any matrix is introduced.

**Proposition 6.1** (Mixed-index transport matrix and pressure completion)**.** *Let*

$$\Gamma := \mathbb{N}_0 \times \mathbb{N}_0^3, \qquad \kappa = (n,\alpha), \qquad \lambda = (a,\beta),$$

*and write*

$$\lambda \preceq \kappa \quad \Longleftrightarrow \quad a \leq n \ \text{ and } \ \beta \leq \alpha, \qquad \kappa - \lambda := (n-a, \alpha-\beta), \qquad \binom{\kappa}{\lambda} := \binom{n}{a}\binom{\alpha}{\beta}.$$

*Set*

$$J_\kappa := J_{n,\alpha}, \qquad \Pi_\kappa := \Pi_{n,\alpha}, \qquad \mathbf{J} := (J_\lambda)_{\lambda\in\Gamma}.$$

*The block operator*

$$\mathbb{L}(J)_{\kappa,\lambda} := \begin{cases} \binom{\kappa}{\lambda}\mathbb{P}\big[(J_{\kappa-\lambda}\cdot\nabla)\cdot\big], & \lambda \preceq \kappa, \\ 0, & \lambda \not\preceq \kappa, \end{cases} \tag{145}$$

*has the finite block-row action*

$$[\mathbb{L}(J)\mathbf{J}]_\kappa = \sum_{\lambda\preceq\kappa}\binom{\kappa}{\lambda}\mathbb{P}\big[(J_\lambda\cdot\nabla)J_{\kappa-\lambda}\big]. \tag{146}$$

*Consequently the complete mixed row is the pair*

$$J_{n+1,\alpha} + [\mathbb{L}(J)\mathbf{J}]_\kappa = \nu\Delta J_\kappa, \tag{147}$$

$$\Pi_\kappa = R_iR_j\sum_{\lambda\preceq\kappa}\binom{\kappa}{\lambda}J_{\lambda,i}J_{\kappa-\lambda,j}, \tag{148}$$

*while the unprojected velocity identity retains $\nabla\Pi_\kappa$ exactly as in* (47).

*Order $\Gamma$ by increasing $n+|\alpha|$, with any fixed ordering among indices of equal depth. Then $\mathbb{L}(J)$ is block lower triangular. For*

$$\kappa! := n!\,\alpha!, \qquad \alpha! := \alpha_1!\alpha_2!\alpha_3!, \qquad \widehat{J}_\kappa := \frac{J_\kappa}{\kappa!}, \qquad \widehat{\Pi}_\kappa := \frac{\Pi_\kappa}{\kappa!},$$

*let $D$ be the diagonal matrix with $D_{\kappa,\kappa} = \kappa!$. On every finite downward-closed truncation of $\Gamma$,*

$$D^{-1}\mathbb{L}(D\widehat{J})D = \widehat{\mathbb{L}}(\widehat{J}), \qquad \widehat{\mathbb{L}}(\widehat{J})_{\kappa,\lambda} := \begin{cases} \mathbb{P}\big[(\widehat{J}_{\kappa-\lambda}\cdot\nabla)\cdot\big], & \lambda \preceq \kappa, \\ 0, & \lambda \not\preceq \kappa. \end{cases} \tag{149}$$

*Thus the factorial-normalized pressure-complete recurrence is*

$$(n+1)\widehat{J}_{n+1,\alpha} + \sum_{\lambda\preceq\kappa}(\widehat{J}_\lambda\cdot\nabla)\widehat{J}_{\kappa-\lambda} + \nabla\widehat{\Pi}_\kappa = \nu\Delta\widehat{J}_\kappa, \tag{150}$$

$$\widehat{\Pi}_\kappa = R_iR_j\sum_{\lambda\preceq\kappa}\widehat{J}_{\lambda,i}\widehat{J}_{\kappa-\lambda,j}. \tag{151}$$

*The normalized transport blocks depend on the difference $\kappa-\lambda$; hence* (149) *is the block lower-triangular convolution form of the mixed transport array.*

*Proof.* For $\lambda \preceq \kappa$, change the summation index from $\lambda$ to $\kappa-\lambda$ and use $\binom{\kappa}{\kappa-\lambda} = \binom{\kappa}{\lambda}$. This gives (146). Apply $\mathbb{P}$ to (47). The identities

$$\mathbb{P}J_{n+1,\alpha} = J_{n+1,\alpha}, \qquad \mathbb{P}\Delta J_\kappa = \Delta J_\kappa, \qquad \mathbb{P}\nabla\Pi_\kappa = 0$$

follow from (48) and the Fourier symbols. They prove (147); equation (148) is (46) in mixed-index notation. The pressure formula also shows that projection of the velocity row loses no pressure coordinate.

If $\lambda \preceq \kappa$, then $a+|\beta| \le n+|\alpha|$, which proves lower triangularity in the stated ordering. Finally,

$$\binom{\kappa}{\lambda}\lambda!\,(\kappa-\lambda)! = \kappa!.$$

Dividing the mixed velocity and pressure rows by $\kappa!$ proves (150) and (151). This factorial cancellation in each operator block proves (149). Every row has $(n+1)\prod_{j=1}^{3}(\alpha_j+1)$ supported blocks, so all displayed row sums are finite. □

**Proposition 6.2** (Sobolev rectangle matrix and adjacent stencil)**.** *For $0<T<T_*$, $N, M \in \mathbb{N}_0$, $0 \le n \le N$, and $m \in \mathbb{Z}$, set*

$$G_{n,m}(T) := \|V_n\|_{L^2(0,T;H^m)},$$

*and set*

$$P_{n,M}(T) := \|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})}.$$

*The object matrix of the finite pressure-complete rectangle is*

$$\mathbf{R}_{N,M}[u_0;T] := \begin{pmatrix} V_0|_{(0,T)} & V_1|_{(0,T)} & Q_0|_{(0,T)} \\ V_1|_{(0,T)} & V_2|_{(0,T)} & Q_1|_{(0,T)} \\ \vdots & \vdots & \vdots \\ V_N|_{(0,T)} & V_{N+1}|_{(0,T)} & Q_N|_{(0,T)} \end{pmatrix}. \tag{152}$$

*The corresponding matrix of the three column norms is*

$$\mathbf{C}_{N,M}(T) := \begin{pmatrix} G_{0,M+1}(T) & G_{1,M-1}(T) & P_{0,M}(T) \\ G_{1,M+1}(T) & G_{2,M-1}(T) & P_{1,M}(T) \\ \vdots & \vdots & \vdots \\ G_{N,M+1}(T) & G_{N+1,M-1}(T) & P_{N,M}(T) \end{pmatrix}. \tag{153}$$

*If $\mathbf{G}_{N,M}(T)$ denotes its first two columns, then*

$$\|\mathbf{G}_{N,M}(T)\|_{\mathrm{F}}^2 = \mathfrak{R}_{N,M}(u;T), \qquad \sum_{n=0}^{N} P_{n,M}(T) = \mathfrak{Q}_{N,M}(p;T). \tag{154}$$

*For each $0 \le n \le N$, the first two entries in row $n$ form the adjacent Hilbert-scale stencil*

$$\left.\begin{array}{r} (n, M+1): \quad V_n \in L^2(0,T;H^{M+1}), \\ (n+1, M-1): \quad V_{n+1} = \partial_t V_n \in L^2(0,T;H^{M-1}) \end{array}\right\} \longrightarrow (n,M): \quad V_n \in C([0,T];H^M). \tag{155}$$

*For every $m \in \mathbb{N}_0$, the Sobolev matrix measures the mixed-index array through the exact identity*

$$G_{n,m}(T)^2 = \sum_{\substack{\alpha \in \mathbb{N}_0^3 \\ |\alpha| \le m}} \frac{m!}{(m-|\alpha|)!\,\alpha!} \, \|J_{n,\alpha}\|_{L^2((0,T)\times\mathbb{R}^3)}^2 . \tag{156}$$

*The pressure entry in row $n$ is $Q_n = \Pi_{n,0}$, and its spatial rows $\partial_x^\alpha Q_n = \Pi_{n,\alpha}$ are fixed by* (148).

*Proof.* The object matrix is the tuple in Definition 4.5, arranged by temporal row. The two identities in (154) are (80) and (81). Lemma 2.7 gives (155). Finally, Plancherel and

$$(1+|\xi|^2)^m = \sum_{\substack{\alpha \in \mathbb{N}_0^3 \\ |\alpha| \le m}} \frac{m!}{(m-|\alpha|)!\,\alpha!} \, \xi^{2\alpha}$$

give (156). The pressure claims follow from the definitions of $Q_n, \Pi_{n,\alpha}$ and (148). □

**Proposition 6.3** (Full-interval closed estimate matrix)**.** *Under the hypotheses of Theorem 4.6, fix $N, M \in \mathbb{N}_0$ and set*

$$\begin{aligned} a_{n,M} &:= \|V_n\|_{L^2(0,T_*;H^{M+1})}, \\ b_{n,M} &:= \|V_{n+1}\|_{L^2(0,T_*;H^{M-1})}, \qquad 0 \le n \le N. \end{aligned} \tag{157}$$

*Define entrywise order on nonnegative matrices by $A \preceq_{\mathrm{ent}} B$ when $A_{ij} \le B_{ij}$ for every entry. Then, for every $0 < T < T_*$,*

$$\mathbf{G}_{N,M}(T) \preceq_{\mathrm{ent}} \mathbf{G}_{N,M}^* := \begin{pmatrix} a_{0,M} & b_{0,M} \\ a_{1,M} & b_{1,M} \\ \vdots & \vdots \\ a_{N,M} & b_{N,M} \end{pmatrix}. \tag{158}$$

*Moreover,*

$$\begin{aligned} \mathfrak{R}_{N,M}(u;T) &= \|\mathbf{G}_{N,M}(T)\|_{\mathrm{F}}^2 \\ &\le \left\|\mathbf{G}_{N,M}^*\right\|_{\mathrm{F}}^2 = \sum_{n=0}^{N} \left(a_{n,M}^2 + b_{n,M}^2\right) = \mathfrak{R}_{N,M}^*[u_0,\nu,T_*] < \infty. \end{aligned} \tag{159}$$

*Proof.* Proposition 4.7 makes every $a_{n,M}$ and $b_{n,M}$ finite. Restriction from $(0,T_*)$ to $(0,T)$ decreases each nonnegative coordinate norm and gives (158). Taking Frobenius norms and using (154) together with (84) proves (159). □

# 7 Quantitative consequences of the compatible rectangles

## 7.1 Completion in the global $H^1$ mild class

The construction begins with every datum in $H^\infty_\sigma(\mathbb{R}^3)$, not only with Schwartz data. Parabolic smoothing now transfers that result to the full strong $H^1$ class.

**Theorem 7.1** (Global $H^1$ completion)**.** *Let $\nu > 0$ and $a \in H^1_\sigma(\mathbb{R}^3)$. There is a unique pressure-normalized mild solution*

$$u^a \in C([0,\infty); H^1_\sigma) \cap L^2_{\mathrm{loc}}([0,\infty); H^2_\sigma), \qquad u^a \in C^\infty((0,\infty)\times\mathbb{R}^3), \tag{160}$$

*of the three-dimensional Navier–Stokes equations with datum $a$ and viscosity $\nu$. Its pressure is*

$$p^a = R_i R_j (u^a_i u^a_j)$$

*in distributions. The solution obeys the kinetic-energy identity on every finite interval.*

*Proof.* Write

$$Y(t) := \|\nabla u(t)\|_2^2, \qquad Z(t) := \|\Delta u(t)\|_2^2, \qquad c_1 := \frac{27}{16}\mathsf{S}(6)^6.$$

For a smooth solenoidal approximation, the $L^2$ pairing of the projected equation with $-\Delta u$ gives

$$\frac{1}{2}Y'(t) + \nu Z(t) = \mathrm{Re}\,\langle (u\cdot\nabla)u, \Delta u\rangle_{L^2}\,. \tag{161}$$

The explicit three-dimensional Sobolev inequality and interpolation give

$$\|u\|_6 \le \mathsf{S}(6) Y^{1/2}, \qquad \|\nabla u\|_3 \le \mathsf{S}(6)^{1/2} Y^{1/4} Z^{1/4}.$$

Consequently,

$$|\langle (u\cdot\nabla)u, \Delta u\rangle_{L^2}| \le \mathsf{S}(6)^{3/2} Y^{3/4} Z^{3/4} \le \frac{\nu}{2} Z + \frac{27\mathsf{S}(6)^6}{32\nu^3} Y^3,$$

and hence

$$Y'(t) + \nu Z(t) \le c_1 \nu^{-3} Y(t)^3. \tag{162}$$

Comparison with $y' = c_1\nu^{-3}y^3$ shows that

$$Y(t) \le \frac{Y(0)}{\big(1 - 2c_1\nu^{-3}Y(0)^2 t\big)^{1/2}} \tag{163}$$

while the denominator is positive.

The estimate holds on the explicit lifespan

$$\tau_\nu(a) := \min\left\{1, \frac{\nu^3}{4c_1(1+\|\nabla a\|_2^2)^2}\right\},$$
$$\sup_{0\le t\le\tau_\nu(a)} Y(t) \le \sqrt{2}Y(0), \qquad \int_0^{\tau_\nu(a)} Z(t)\,\mathrm{d}t < \infty. \tag{164}$$

Let $\mathcal{J}_k$ be the orthogonal Fourier projection to $\{|\xi| \le k\}$. On the range of $\mathcal{J}_k$, the regularized problem

$$\partial_t u^{(k)} - \nu\Delta u^{(k)} = -\mathcal{J}_k\mathbb{P}\nabla\cdot(u^{(k)}\otimes u^{(k)}), \qquad u^{(k)}(0) = \mathcal{J}_k a, \tag{165}$$

is a Banach-space ordinary differential equation in $H^1_\sigma$. Indeed, if

$$\mathcal{N}_k(z) := \mathcal{J}_k\mathbb{P}\nabla\cdot(z\otimes z),$$

then the Bernstein inequalities on the range of $\mathcal{J}_k$ give

$$\|\mathcal{N}_k(z) - \mathcal{N}_k(y)\|_{H^1} \le C_k\big(\|z\|_{H^1} + \|y\|_{H^1}\big)\,\|z-y\|_{H^1}, \qquad C_k < \infty. \tag{166}$$

Picard's theorem supplies a maximal trajectory. The projection is self-adjoint and commutes with $\mathbb{P}$, $\nabla$, and $\Delta$, so the kinetic identity holds along that trajectory. Since

$$\|z\|_{H^1}^2 \le (1+k^2)\,\|z\|_2^2 \qquad (z = \mathcal{J}_k z), \tag{167}$$

the kinetic identity and the Banach-space continuation criterion extend the trajectory for all positive times. Estimate (162) holds for $u^{(k)}$, with constants independent of $k$. Equations (163)–(164) give

$$\sup_k\left(\left\|u^{(k)}\right\|_{L^\infty(0,\tau_\nu(a);H^1)} + \left\|u^{(k)}\right\|_{L^2(0,\tau_\nu(a);H^2)}\right) < \infty. \tag{168}$$

The nonlinear estimate

$$\|\mathbb{P}\nabla\cdot(u\otimes u)\|_2 \le \|u\|_6\,\|\nabla u\|_3 \le \mathsf{S}(6)^{3/2}Y^{3/4}Z^{1/4},$$

shows that the regularized nonlinear terms are bounded in $L^2(0,\tau_\nu(a);L^2)$, uniformly in $k$. The equations bound $\partial_t u^{(k)}$ in this space. Weak compactness and the Aubin–Lions–Simon lemma [1, 14] on every ball give, after diagonal extraction,

$$\begin{aligned} u^{(k)} &\overset{*}{\rightharpoonup} u && \text{in } L^\infty(0,\tau_\nu(a);H^1),\\ u^{(k)} &\rightharpoonup u && \text{in } L^2(0,\tau_\nu(a);H^2),\\ u^{(k)} &\longrightarrow u && \text{in } L^2(0,\tau_\nu(a);H^1(B_R)) \quad (R<\infty). \end{aligned}$$

The last convergence passes the quadratic term in distributions; lower semicontinuity preserves (168). The displayed nonlinear estimate now gives $\mathbb{P}\nabla\cdot(u\otimes u) \in L^2L^2$, and the equation gives $u_t \in L^2(0,\tau_\nu(a);L^2)$. For every $\phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$ with $\nabla\cdot\phi = 0$ and every $\eta \in C_c^\infty([0,\tau_\nu(a)))$, passage to the limit in the time-integrated regularized equation gives

$$\begin{aligned} -\int_0^{\tau_\nu(a)} \langle u,\phi\rangle_{L^2}\,\eta'\,\mathrm{d}t + \nu\int_0^{\tau_\nu(a)} \langle\nabla u,\nabla\phi\rangle_{L^2}\,\eta\,\mathrm{d}t - \int_0^{\tau_\nu(a)} \langle u\otimes u,\nabla\phi\rangle_{L^2}\,\eta\,\mathrm{d}t \\ = \langle a,\phi\rangle_{L^2}\,\eta(0). \end{aligned} \tag{169}$$

The right-hand side fixes the weak initial trace as $a$.

The Lions–Magenes theorem [12] for $H^2_\sigma \hookrightarrow H^1_\sigma \hookrightarrow L^2_\sigma$ gives $u \in C([0,\tau_\nu(a)];H^1_\sigma)$; its continuous trace at zero is the trace in (169), hence $u(0)=a$. Variation of constants gives the mild equation, and $p = R_iR_j(u_iu_j)$ supplies its normalized pressure. Since $u_t \in L^2L^2$ and $u \in L^2H^2$, pairing the limiting equation with $u$ is legitimate; incompressibility cancels the transport term and proves the kinetic-energy identity.

If $u, v$ are two such solutions and $w = u - v$, incompressibility cancels the $(u\cdot\nabla)w$ pairing and gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|w\|_2^2 + \nu\|\nabla w\|_2^2 = -\operatorname{Re}\langle (w\cdot\nabla)v, w\rangle_{L^2}.$$

This Sobolev constant yields

$$|\langle (w\cdot\nabla)v, w\rangle_{L^2}| \le \mathsf{S}(6)^{3/2}\|\nabla v\|_2\|w\|_2^{1/2}\|\nabla w\|_2^{3/2} \le \frac{\nu}{2}\|\nabla w\|_2^2 + \frac{27\mathsf{S}(6)^6}{32\nu^3}\|\nabla v\|_2^4\|w\|_2^2.$$

Gronwall's inequality proves uniqueness. This calculation one derivative higher also gives the finite-interval stability needed below. Indeed, pairing the difference equation with $-\Delta w$, integrating the $(v\cdot\nabla)w$ term once in space, and applying Young's inequality gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\nabla w\|_2^2 + \nu\|\Delta w\|_2^2 \le \frac{C_{\mathrm{st}}}{\nu}\left(\|\nabla u\|_3^2 + \|\nabla v\|_3^2\right)\|\nabla w\|_2^2. \tag{170}$$

The coefficient is integrable because

$$\|\nabla u\|_3^2 \le \mathsf{S}(6)\|\nabla u\|_2\|\Delta u\|_2,$$

and likewise for $v$. The $L^2$ difference estimate and (170) show continuous dependence in $C([0,S];H^1)$ on every finite interval $[0,S]$.

Let $c_0 > 0$ be the absolute smallness constant in the full-space $H^1$ local existence theorem, Theorem 5.4(ii) of [15], written at viscosity one as $\|b\|_{H^1}^4 h \le c_0$. After viscosity rescaling, its condition is

$$h\|b\|_{H^1}^4 \le c_0\nu^3. \tag{171}$$

Fix $0 < \varepsilon < S \le \tau_\nu(a)$, and put $K_S := 1 + \sup_{0\le t\le S}\|u(t)\|_{H^1}$. Choose

$$0 < h \le \min\left\{\varepsilon, c_0\nu^3 K_S^{-4}\right\}.$$

The intervals $[0,h]$ and $[t-h,t]$, $h \le t \le S$, obey (171). The quantitative positive-time regularity estimate in Lemma 5.5 of [15], after viscosity rescaling with interior time $h/2$, and the heat-semigroup/product iteration in its proof give, for every $m \in \mathbb{N}$,

$$\sup_{\varepsilon\le t\le S}\|u(t)\|_{H^m} \le C_{m,\varepsilon,S}(\nu, K_S) < \infty. \tag{172}$$

Apply the mild formula from time $\varepsilon/2$. The bounds (172), the heat multiplier, and Lemma 2.4 show that $u \in C([\varepsilon,S];H^m)$ for every $m$. The identities

$$p = R_iR_j(u_iu_j), \qquad u_t = \nu\Delta u - \mathbb{P}\nabla\cdot(u\otimes u) \tag{173}$$

then give $p \in C([\varepsilon,S];H^m)$ and $u_t \in C([\varepsilon,S];H^m)$ for every $m$. Differentiating (173) and the pressure identity, and applying this product estimate, proves inductively that all temporal derivatives of $u$ and $p$ belong to every spatial Sobolev space continuously on $[\varepsilon,S]$. Sobolev embedding proves $(u,p) \in C^\infty((0,\tau_\nu(a)]\times\mathbb{R}^3)$.

Choose $0 < t_0 < \tau_\nu(a)$. Equation (172) places $u(t_0)$ in $H^\infty_\sigma$. Theorem 5.10, with time origin $t_0$, supplies a global smooth history from that datum. The uniqueness estimate above identifies it with the local history on their overlap, so the two histories glue to (160). This estimate proves uniqueness in the full displayed class. The kinetic identity on the first interval and the classical identity after $t_0$ agree at the common trace and prove the final assertion. □

## 7.2 Finite norm and trace majorants

The compatible rectangles carry more information than the endpoint argument uses. We first name scalar majorants of their finite norms and close the estimates that will feed the mixed rows, frequency tails, and continuation bounds below.

**Definition 7.2** (Rectangle-norm and trace majorants)**.** Let $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, $\nu > 0$, and let $(u, p)$ be the datum-generated pressure-normalized solution. Let $0 < T < \infty$ satisfy either $T < T_*$ or $T = T_* < \infty$, and set $I_T = (0, T)$,

$$V_n := \partial_t^n u, \qquad Q_n := \partial_t^n p, \qquad n \in \mathbb{N}_0.$$

Define the monotone rectangle-norm majorant

$$\mathfrak{P}_{N,M}[u_0, \nu; T] := 1 + \sup_{0<S<T} \mathfrak{R}_{N,M}(u; S), \tag{174}$$

and the trace constant

$$\mathfrak{T}_{N,M}[u_0, \nu; T] := \left(1 + T^{-1}\right)^{1/2} \mathfrak{P}_{N,M}[u_0, \nu; T]^{1/2}. \tag{175}$$

All occurrences of $\mathfrak{P}_{N,M}$ and $\mathfrak{T}_{N,M}$ below carry the arguments $[u_0, \nu; T]$. The datum $u_0$, viscosity $\nu$, pressure-normalized solution $(u, p)$, horizon $T$, interval $I_T$, and row notation $V_n, Q_n$ fixed here remain in force throughout the rest of this section. When $T = T_* < \infty$, the full-interval representatives are those of Proposition 4.7, and every endpoint value used below is the compatible trace supplied by Theorem 5.5.

**Lemma 7.3** (Majorant finiteness)**.** *For every $N, M \in \mathbb{N}_0$,*

$$1 \le \mathfrak{P}_{N,M}[u_0, \nu; T] < \infty. \tag{176}$$

*Proof.* If $T = T_* < \infty$, equation (82) proves (176). If $T < T_*$, classical persistence on $[0, T]$ gives

$$\sup_{0\le t\le T} \|V_n(t)\|_{H^m} < \infty \qquad (n, m \in \mathbb{N}_0),$$

and (80) proves (176). □

**Lemma 7.4** (Fixed analytic constants)**.** *For $r \in \mathbb{N}_0$, set*

$$A_r := 2C_r^{\mathrm{alg}}.$$

*Then*

$$\|R_i R_j(f_i g_j)\|_{H^r} + \|\mathbb{P}\nabla\cdot(f \otimes g)\|_{H^{r-1}} \le A_r \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}. \tag{177}$$

*This constant satisfies*

$$\|R_i R_j(f_i g_j)\|_{H^r} + \|\nabla R_i R_j(f_i g_j)\|_{H^{r-1}} \le A_r \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}. \tag{178}$$

*For $s, k \in \mathbb{N}_0$, $2 \le b \le \infty$, and $s > k + \frac{3}{2} - \frac{3}{b}$, define*

$$\nabla^k f := (\partial^\alpha f)_{|\alpha|=k}, \qquad \left\|\nabla^k f\right\|_{L^b} := \left\| \left( \sum_{|\alpha|=k} |\partial^\alpha f|^2 \right)^{1/2} \right\|_{L^b},$$

*and set*

$$d_k := \#\{\alpha \in \mathbb{N}_0^3 : |\alpha| = k\}, \qquad \frac{1}{q_b} := \frac{1}{2} - \frac{1}{b},$$

*with $q_2 = \infty$, and*

$$S_{s,k,b} := d_k^{1/2} (2\pi)^{-3(1/2-1/b)} \left\| |\xi|^k \langle \xi \rangle^{-s} \right\|_{L^{q_b}_\xi}. \tag{179}$$

*Then $S_{s,k,b} < \infty$ and*

$$\left\|\nabla^k f\right\|_{L^b} \le S_{s,k,b} \|f\|_{H^s}. \tag{180}$$

*For $m \ge 2$, set $L_m := \max\{1, C_m^{\mathrm{alg}}\}$. Then*

$$\|fg\|_{H^m} \le L_m \|f\|_{H^m} \|g\|_{H^m}, \tag{181}$$

$$\left\|e^{\nu t\Delta} \mathbb{P}\nabla \cdot F\right\|_{H^m} \le L_m (\nu t)^{-1/2} \|F\|_{H^m}, \qquad t > 0. \tag{182}$$

*Proof.* Lemmas 2.4 and 2.5, applied componentwise before summing squares, give

$$\|R_i R_j (f_i g_j)\|_{H^r} \le \|f \otimes g\|_{H^r} \le C_r^{\mathrm{alg}} \|f\|_{H^{r+1}} \|g\|_{H^{r+1}},$$
$$\|\mathbb{P}\nabla \cdot (f \otimes g)\|_{H^{r-1}} \le \|f \otimes g\|_{H^r} \le C_r^{\mathrm{alg}} \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}.$$

Moreover,

$$\|\nabla R_i R_j (f_i g_j)\|_{H^{r-1}} \le \|R_i R_j (f_i g_j)\|_{H^r}.$$

This proves (177) and (178).

Let $b' = b/(b-1)$. Interpolation between Fourier inversion $L^1_\xi \to L^\infty_x$, with norm $(2\pi)^{-3/2}$, and Plancherel gives

$$\left\|\mathcal{F}^{-1} h\right\|_{L^b_x} \le (2\pi)^{-3(1/2-1/b)} \|h\|_{L^{b'}_\xi}.$$

For $|\alpha| = k$, Hölder's inequality with $1/b' = 1/2 + 1/q_b$ gives

$$\|\partial^\alpha f\|_{L^b} \le (2\pi)^{-3(1/2-1/b)} \left\| |\xi|^k \langle \xi \rangle^{-s} \right\|_{L^{q_b}} \|f\|_{H^s}.$$

The weight belongs to $L^{q_b}$ under the displayed strict inequality for $s$; for $b = 2$, its $L^\infty$-norm is finite because $s > k$. Summation over $|\alpha| = k$ proves (180).

The first estimate in (181) is Lemma 2.4. For the heat estimate, the symbol of $\mathbb{P}\nabla\cdot$, acting from tensors to vectors, has norm at most $|\xi|$, while

$$\sup_{\rho \ge 0} \rho e^{-\nu t \rho^2} = (2e\nu t)^{-1/2}.$$

Plancherel gives

$$\left\|e^{\nu t\Delta} \mathbb{P}\nabla \cdot F\right\|_{H^m} \le (2e\nu t)^{-1/2} \|F\|_{H^m} \le L_m (\nu t)^{-1/2} \|F\|_{H^m}.$$

□

**Definition 7.5** (Canonical rectangle majorants)**.** For $N, M \in \mathbb{N}_0$, define

$$\mathfrak{B}_{N,M}[u_0, \nu; T] := \inf \{B \geq 0 : \mathfrak{R}_{N,M}(u; S) \leq B \text{ for every } 0 < S < T\}. \tag{183}$$

Monotone convergence gives

$$\mathfrak{B}_{N,M}[u_0, \nu; T] = \begin{cases} \mathfrak{R}_{N,M}(u; T), & T < T_*, \\ \mathfrak{R}^*_{N,M}[u_0, \nu, T_*], & T = T_* < \infty. \end{cases} \tag{184}$$

The global theorem fixes one history for $(u_0, \nu)$. Thus $\mathfrak{B}_{N,M}[u_0, \nu; T]$ is the exact minimal rectangle bound of that datum-generated history, not a supremum over a family of solutions and not an independently chosen constant. Equations (154) and (183) give the exact matrix identities

$$\mathfrak{B}_{N,M}[u_0, \nu; T] = \sup_{0<S<T} \|\mathbf{G}_{N,M}(S)\|_{\mathrm{F}}^2, \tag{185}$$

$$\mathfrak{P}_{N,M}[u_0, \nu; T] = 1 + \sup_{0<S<T} \|\mathbf{G}_{N,M}(S)\|_{\mathrm{F}}^2. \tag{186}$$

Consequently,

$$\mathfrak{P}_{N,M}[u_0, \nu; T] = 1 + \mathfrak{B}_{N,M}[u_0, \nu; T]. \tag{187}$$

For $n, m \in \mathbb{N}_0$, define the initial-jet majorants $\mathfrak{D}_{n,m} = \mathfrak{D}_{n,m}[u_0, \nu]$ by

$$\mathfrak{D}_{0,m} := \|u_0\|_{H^m}, \tag{188}$$

$$\mathfrak{D}_{n+1,m} := \nu \mathfrak{D}_{n,m+2} + A_{m+1} \sum_{a=0}^{n} \binom{n}{a} \mathfrak{D}_{a,m+2} \mathfrak{D}_{n-a,m+2}. \tag{189}$$

Define the pointwise recursion $\mathfrak{Y}_{n,m} = \mathfrak{Y}_{n,m}[u_0, \nu; T]$ by

$$\mathfrak{Y}_{0,0} := \|u_0\|_2, \tag{190}$$

$$\mathfrak{Y}_{0,m} := \min \left\{ (1 + T^{-1})^{1/2} \mathfrak{B}_{0,m}^{1/2}, \|u_0\|_{H^m} + T^{1/2} \mathfrak{B}_{0,m+1}^{1/2} \right\}, \qquad m \geq 1, \tag{191}$$

$$\mathfrak{Y}_{n+1,m} := \nu \mathfrak{Y}_{n,m+2} + A_{m+1} \sum_{a=0}^{n} \binom{n}{a} \mathfrak{Y}_{a,m+2} \mathfrak{Y}_{n-a,m+2}. \tag{192}$$

For $m \in \{-1\} \cup \mathbb{N}_0$, put

$$m_+ := \max\{m, 0\}, \qquad \mu(m) := \max\{m - 1, 0\},$$

and define

$$\widetilde{\mathfrak{X}}_{n,m} := \min \left\{ T^{1/2} \mathfrak{Y}_{n,m_+}, \mathfrak{B}_{n,\mu(m)}^{1/2} \right\}. \tag{193}$$

Set

$$\mathfrak{X}_{0,0} := \min \left\{ \widetilde{\mathfrak{X}}_{0,0}, T^{1/2} \|u_0\|_2 \right\}, \tag{194}$$

$$\mathfrak{X}_{0,1} := \min \left\{ \widetilde{\mathfrak{X}}_{0,1}, \left( T + \frac{1}{2\nu} \right)^{1/2} \|u_0\|_2 \right\}, \tag{195}$$

$$\mathfrak{X}_{n,m} := \widetilde{\mathfrak{X}}_{n,m} \quad \text{for all other } (n, m). \tag{196}$$

For $n, m \in \mathbb{N}_0$, define

$$\mathfrak{Z}_{n,m}[u_0, \nu; T] := \min \left\{ \mathfrak{Y}_{n,m}, (1 + T^{-1})^{1/2} \mathfrak{B}_{n,m}^{1/2}, \mathfrak{D}_{n,m} + T^{1/2} \mathfrak{X}_{n+1,m} \right\}. \tag{197}$$

All $\mathfrak{B}, \mathfrak{D}, \mathfrak{X}, \mathfrak{Y}, \mathfrak{Z}$ in this section carry the arguments $[u_0, \nu; T]$, except that $\mathfrak{D}$ is independent of $T$.

**Proposition 7.6** (Exact viscosity and horizon scaling)**.** *Let*

$$a := \nu^{-1}u_0, \qquad v(s,x) := \nu^{-1}u(s/\nu, x), \qquad q(s,x) := \nu^{-2}p(s/\nu, x).$$

*Then $(v,q)$ is the unit-viscosity solution generated by $a$. If*

$$\widetilde{V}_n := \partial_s^n v, \qquad \widetilde{Q}_n := \partial_s^n q,$$

*then*

$$V_n(t) = \nu^{n+1}\widetilde{V}_n(\nu t), \qquad Q_n(t) = \nu^{n+2}\widetilde{Q}_n(\nu t). \tag{198}$$

*Substitution of* (198) *into the defining norms gives*

$$\mathfrak{B}_{N,M}[u_0,\nu;T] = \sum_{n=0}^{N}\Big[\nu^{2n+1}\left\|\widetilde{V}_n\right\|^2_{L^2(0,\nu T;H^{M+1})} + \nu^{2n+3}\left\|\widetilde{V}_{n+1}\right\|^2_{L^2(0,\nu T;H^{M-1})}\Big], \tag{199}$$

$$\|Q_n\|_{L^1(0,T;H^m)} + \|\nabla Q_n\|_{L^1(0,T;H^{m-1})} = \nu^{n+1}\Big(\left\|\widetilde{Q}_n\right\|_{L^1(0,\nu T;H^m)} + \left\|\nabla\widetilde{Q}_n\right\|_{L^1(0,\nu T;H^{m-1})}\Big), \tag{200}$$

$$\sup_{0\le t\le T}\|Q_n(t)\|_{H^m} = \nu^{n+2}\sup_{0\le s\le \nu T}\left\|\widetilde{Q}_n(s)\right\|_{H^m}. \tag{201}$$

*The initial-jet recursion obeys*

$$\mathfrak{D}_{n,m}[u_0,\nu] = \nu^{n+1}\mathfrak{D}_{n,m}[u_0/\nu, 1]. \tag{202}$$

*Write*

$$E^u_\sigma(t) := \frac{1}{2}\|\Lambda^\sigma u(t)\|_2^2, \qquad E^v_\sigma(s) := \frac{1}{2}\|\Lambda^\sigma v(s)\|_2^2.$$

*For every $j\in\mathbb{N}_0$, the critical-height rows obey*

$$E^u_{j+1/2}(t) = \nu^2 E^v_{j+1/2}(\nu t), \tag{203}$$

$$\int_0^T E^u_{j+1/2}(t)\,\mathrm{d}t = \nu\int_0^{\nu T} E^v_{j+1/2}(s)\,\mathrm{d}s. \tag{204}$$

*For every $b\in H^3_\sigma(\mathbb{R}^3)$, the maximal elapsed lifespan in the shifted mild class has the exact covariance*

$$\tau^{\max}_{\nu,3}(s,b) = \nu^{-1}\tau^{\max}_{1,3}(0,b/\nu). \tag{205}$$

*Proof.* The change of variables gives $u(t) = \nu v(\nu t)$ and $p(t) = \nu^2 q(\nu t)$. Differentiation proves (198); the substitution $s = \nu t$ proves (199)–(201). Equation (202) follows by induction in (188)–(189). The amplitude relation $u(t) = \nu v(\nu t)$ proves (203), and the substitution $s = \nu t$ proves (204). For $(w,\pi)\in\mathscr{M}^3_\nu(s,s+\delta;b)$, set

$$\mathcal{S}_\nu(w,\pi)(\sigma) := \left(\nu^{-1}w(s+\sigma/\nu), \nu^{-2}\pi(s+\sigma/\nu)\right), \qquad 0\le\sigma\le\nu\delta.$$

Time translation and this change of variables show that $\mathcal{S}_\nu$ is a bijection from $\mathscr{M}^3_\nu(s,s+\delta;b)$ onto $\mathscr{M}^3_1(0,\nu\delta;b/\nu)$. Its inverse is the corresponding amplitude and time rescaling, which proves (205). □

**Proposition 7.7** (Height-coordinate reconstruction of the finite rows)**.** *For $m \in \mathbb{N}_0$, define*

$$\mathfrak{H}_m[u_0,\nu;T] := \frac{1}{2}\left\|\Lambda^{m+1/2}u\right\|^2_{L^2(I_T;L^2)}, \tag{206}$$

$$\mathfrak{S}_m[u_0,\nu;T]^2 := 2^m\left(T\,\|u_0\|_2^2 + 2\mathfrak{H}_m[u_0,\nu;T]\right). \tag{207}$$

*Then*

$$2\mathfrak{H}_m[u_0,\nu;T] \le \mathfrak{B}_{0,m}[u_0,\nu;T], \qquad \|u\|_{L^2(I_T;H^m)} \le \mathfrak{S}_m[u_0,\nu;T]. \tag{208}$$

*Define recursively, for $n,m \in \mathbb{N}_0$,*

$$\mathfrak{A}^{\mathrm{v}}_{0,m} := T^{-1/2}\mathfrak{S}_m + \nu T^{1/2}\mathfrak{S}_{m+2} + A_{m+1}\mathfrak{S}^2_{m+2}, \tag{209}$$

$$\mathfrak{A}^{\mathrm{p}}_{n,m} := A_m\sum_{a=0}^{n}\binom{n}{a}\mathfrak{A}^{\mathrm{v}}_{a,m+1}\mathfrak{A}^{\mathrm{v}}_{n-a,m+1}, \tag{210}$$

$$\mathfrak{A}^{\mathrm{v}}_{n+1,m} := \nu\mathfrak{A}^{\mathrm{v}}_{n,m+2} + A_{m+1}\sum_{a=0}^{n}\binom{n}{a}\mathfrak{A}^{\mathrm{v}}_{a,m+2}\mathfrak{A}^{\mathrm{v}}_{n-a,m+2}. \tag{211}$$

*Every finite quantity in this recursion is finite, and*

$$\sup_{0\le t\le T}\|V_n(t)\|_{H^m} \le \mathfrak{A}^{\mathrm{v}}_{n,m}, \tag{212}$$

$$\sup_{0\le t\le T}\left(\|Q_n(t)\|_{H^m} + \|\nabla Q_n(t)\|_{H^{m-1}}\right) \le \mathfrak{A}^{\mathrm{p}}_{n,m}. \tag{213}$$

*Let $M_- := \max\{M-1,0\}$ and set*

$$\widehat{\mathfrak{B}}^{\mathrm{h}}_{N,M}[u_0,\nu;T] := T\sum_{n=0}^{N}\left((\mathfrak{A}^{\mathrm{v}}_{n,M+1})^2 + (\mathfrak{A}^{\mathrm{v}}_{n+1,M_-})^2\right), \tag{214}$$

$$\widehat{\mathfrak{Q}}^{\mathrm{h}}_{N,M}[u_0,\nu;T] := T\sum_{n=0}^{N}\mathfrak{A}^{\mathrm{p}}_{n,M}. \tag{215}$$

*Then the scalar norm and pressure bounds satisfy*

$$\mathfrak{B}_{N,M}[u_0,\nu;T] \le \widehat{\mathfrak{B}}^{\mathrm{h}}_{N,M}[u_0,\nu;T], \tag{216}$$

$$\mathfrak{Q}_{N,M}(p;T) \le \widehat{\mathfrak{Q}}^{\mathrm{h}}_{N,M}[u_0,\nu;T]. \tag{217}$$

*For fixed $N,M$, the right-hand sides use only*

$$\nu,\quad T,\quad \|u_0\|_2,\quad N,\quad M,\quad \{\mathfrak{H}_q[u_0,\nu;T] : 0\le q\le M+2N+4\},\quad \{A_q : 0\le q\le M+2N+4\}. \tag{218}$$

*The quantities $\mathfrak{H}_q$ are the exact completed-height coordinates of the fixed datum-generated history. Equation* (208) *converts every finite height into a spatial Sobolev row, while* (216) *uses the equation to bound the temporal and pressure rows of that prescribed finite rectangle. Over all finite indices this recovers that mixed intersection.*

*Proof.* The first estimate in (208) follows from the upper row of $\mathfrak{B}_{0,m}$. The Fourier inequality

$$(1+|\xi|^2)^m \le 2^m(1+|\xi|^{2m+1})$$

and (69) give the second estimate.

The projected equation and (177) give

$$\|u_t\|_{L^1(I_T;H^m)} \le \nu T^{1/2}\mathfrak{S}_{m+2} + A_{m+1}\mathfrak{S}_{m+2}^2.$$

Choose a Lebesgue time $t_m \in I_T$ satisfying

$$\|u(t_m)\|_{H^m} \le T^{-1/2}\,\|u\|_{L^2(I_T;H^m)}\,.$$

The Banach-valued fundamental identity gives (212) at $n = 0$. Assume that bound through row $n$. The pressure recurrence and (178) give (213). The projected temporal recurrence and (177) give the next velocity row through (211). Induction proves (212)–(213).

Integrating the pointwise bounds over $I_T$ gives (216); when $M = 0$, use $H^0 \hookrightarrow H^{-1}$ on the lower row. This integration in (213) gives (217). Induction in (211) gives the finite dependency range in (218). □

**Lemma 7.8** (Closed velocity majorants)**.** *For every $n, m \in \mathbb{N}_0$,*

$$\|V_n(0)\|_{H^m} \le \mathfrak{D}_{n,m}, \tag{219}$$

$$\|V_n\|_{L^2(I_T;H^m)} \le \mathfrak{X}_{n,m}, \tag{220}$$

$$\sup_{0\le t\le T} \|V_n(t)\|_{H^m} \le \mathfrak{Z}_{n,m}. \tag{221}$$

*For $m = -1$, equation* (220) *remains valid. Every right-hand side is finite and depends on $(u_0, \nu, n, m, T)$ through the finite rectangle-matrix suprema in* (185)*.*

*Proof.* Applying $\mathbb{P}$ to the temporal recurrence gives

$$V_{n+1} = \nu\Delta V_n - \mathbb{P}\nabla\cdot\sum_{a=0}^{n}\binom{n}{a}(V_a \otimes V_{n-a}). \tag{222}$$

Equation (177) at order $m + 1$, evaluated at $t = 0$, proves (219) by induction through (189). This estimate, taken pointwise on $[0, T]$, proves by induction that

$$\sup_{0\le t\le T} \|V_n(t)\|_{H^m} \le \mathfrak{Y}_{n,m}. \tag{223}$$

The base case $m = 0$ is (69). For $m \ge 1$, both terms in (191) are valid: the first is the adjacent trace bound, and the second follows from

$$u(t) = u_0 + \int_0^t V_1(s)\,\mathrm{d}s \quad \text{in } H^m.$$

The rectangle with middle order $\mu(m)$, Sobolev nesting, and (223) give respectively

$$\|V_n\|_{L^2(I_T;H^m)} \le \mathfrak{B}_{n,\mu(m)}^{1/2}, \qquad \|V_n\|_{L^2(I_T;H^m)} \le T^{1/2}\mathfrak{Y}_{n,m_+}.$$

This proves (220) through (193); the two special energy cases follow from (69) and (70). The adjacent trace lemma gives

$$\sup_{0\le t\le T} \|V_n(t)\|_{H^m} \le (1 + T^{-1})^{1/2}\mathfrak{B}_{n,m}^{1/2},$$

while the Banach-valued fundamental identity and (219) give

$$\sup_{0\le t\le T} \|V_n(t)\|_{H^m} \le \mathfrak{D}_{n,m} + T^{1/2}\mathfrak{X}_{n+1,m}.$$

Taking the minimum of these bounds and (223) proves (221). □

**Definition 7.9** (Closed pressure majorants)**.** For $n, m \in \mathbb{N}_0$, define

$$\mathfrak{Q}^{(1)}_{n,m} := A_m \sum_{a=0}^{n} \binom{n}{a} \mathfrak{X}_{a,m+1} \mathfrak{X}_{n-a,m+1}, \tag{224}$$

$$\mathfrak{Q}^{(2)}_{n,m} := A_m \sum_{a=0}^{n} \binom{n}{a} \min\left\{ \mathfrak{Z}_{a,m+1} \mathfrak{X}_{n-a,m+1}, \mathfrak{X}_{a,m+1} \mathfrak{Z}_{n-a,m+1} \right\}, \tag{225}$$

$$\mathfrak{Q}^{(\infty)}_{n,m} := A_m \sum_{a=0}^{n} \binom{n}{a} \mathfrak{Z}_{a,m+1} \mathfrak{Z}_{n-a,m+1}, \tag{226}$$

$$\mathfrak{Q}^{(\mathrm{mod})}_{n,m} := A_m \sum_{a=0}^{n} \binom{n}{a} \left( \mathfrak{X}_{a+1,m+1} \mathfrak{Z}_{n-a,m+1} + \mathfrak{Z}_{a,m+1} \mathfrak{X}_{n-a+1,m+1} \right). \tag{227}$$

These quantities carry the arguments $[u_0, \nu; T]$ through $\mathfrak{X}$ and $\mathfrak{Z}$.

## 7.3 Mixed velocity and pressure rows

The canonical majorants now turn each finite rectangle into bounds for the actual differentiated velocity and pressure rows, including their traces and time moduli.

**Theorem 7.10** (Quantitative mixed-jet rows and traces)**.** *Let $N, s \in \mathbb{N}_0$, let $0 \le n \le N$, and let $\alpha \in \mathbb{N}_0^3$. Put*

$$J_{n,\alpha} := \partial_x^\alpha V_n.$$

*Then*

$$\|J_{n,\alpha}\|^2_{L^2(I_T;H^{s+1})} + \|J_{n+1,\alpha}\|^2_{L^2(I_T;H^{s-1})} \le \mathfrak{P}_{N,s+|\alpha|}, \tag{228}$$

$$\sup_{0\le t\le T} \|J_{n,\alpha}(t)\|_{H^s} \le \mathfrak{T}_{N,s+|\alpha|}. \tag{229}$$

*There is a unique trace $J^T_{n,\alpha} \in H^s$ at $t = T$, and for $0 \le r \le t \le T$,*

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^{s-1}} \le (t-r)^{1/2} \mathfrak{P}^{1/2}_{N,s+|\alpha|}, \tag{230}$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^{s}} \le (t-r)^{1/2} \mathfrak{P}^{1/2}_{N,s+|\alpha|+1}, \tag{231}$$

$$\left\|J^T_{n,\alpha}\right\|_{H^s} \le \mathfrak{T}_{N,s+|\alpha|}. \tag{232}$$

*If $0 \le \theta \le 1$, then*

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^{s-\theta}} \le 2^{1-\theta} \mathfrak{T}^{1-\theta}_{N,s+|\alpha|} \mathfrak{P}^{\theta/2}_{N,s+|\alpha|} (t-r)^{\theta/2}. \tag{233}$$

*For $1 \le a \le \infty$, $2 \le b \le \infty$, $k, q \in \mathbb{N}_0$, and $q > k + \frac{3}{2} - \frac{3}{b}$,*

$$\left\|\nabla^k J_{n,\alpha}\right\|_{L^a(I_T;L^b)} \le S_{q,k,b} T^{1/a} \mathfrak{T}_{N,q+|\alpha|}, \tag{234}$$

$$\left\|\nabla^k J_{n,\alpha}\right\|_{L^2(I_T;L^b)} \le S_{q,k,b} \mathfrak{P}^{1/2}_{N,q+|\alpha|}, \tag{235}$$

*where $T^{1/\infty} = 1$.*

*Proof.* Take the rectangle with middle order $M = s + |\alpha|$. The Fourier multiplier inequality

$$\|\partial_x^\alpha f\|_{H^r} \le \|f\|_{H^{r+|\alpha|}}$$

and (174) give (228). The adjacent-row trace estimate gives

$$\begin{aligned}\sup_{0\le t\le T} \|J_{n,\alpha}(t)\|_{H^s}^2 &\le T^{-1}\,\|J_{n,\alpha}\|_{L^2(I_T;H^s)}^2 \\ &\quad + 2\,\|J_{n,\alpha}\|_{L^2(I_T;H^{s+1})}\,\|J_{n+1,\alpha}\|_{L^2(I_T;H^{s-1})} \\ &\le (T^{-1}+1)\mathfrak{P}_{N,s+|\alpha|},\end{aligned}$$

which proves (229) and (232).

The Bochner identity

$$J_{n,\alpha}(t) - J_{n,\alpha}(r) = \int_r^t J_{n+1,\alpha}(\tau)\,d\tau$$

in $H^{s-1}$, followed by Cauchy–Schwarz, proves (230). Applying this argument to the rectangle with middle order $s+|\alpha|+1$ proves (231). Interpolation between $H^{s-1}$ and $H^s$, together with $\|J(t)-J(r)\|_{H^s} \le 2\mathfrak{T}_{N,s+|\alpha|}$, proves (233). Sobolev embedding and (229) prove (234); Sobolev embedding and (228) prove (235). □

**Theorem 7.11** (Quantitative pressure rows)**.** *Let $N, s \in \mathbb{N}_0$, $0 \le n \le N$, and $\alpha \in \mathbb{N}_0^3$. Set*

$$\Pi_{n,\alpha} := \partial_x^\alpha Q_n, \qquad r := s + |\alpha|.$$

*Then*

$$\|\Pi_{n,\alpha}\|_{L^1(I_T;H^s)} + \|\nabla\Pi_{n,\alpha}\|_{L^1(I_T;H^{s-1})} \le 2A_r 2^n \mathfrak{P}_{N,r}, \tag{236}$$

$$\|\Pi_{n,\alpha}\|_{L^2(I_T;H^s)} \le A_r 2^n \mathfrak{T}_{N,r+1}\mathfrak{P}_{N,r}^{1/2}, \tag{237}$$

$$\sup_{0\le t\le T}\|\Pi_{n,\alpha}(t)\|_{H^s} \le A_r 2^n \mathfrak{T}_{N,r+1}^2. \tag{238}$$

*There is a unique endpoint pressure row $\Pi_{n,\alpha}^T \in H^s$, and*

$$\left\|\Pi_{n,\alpha}^T\right\|_{H^s} \le A_r 2^n \mathfrak{T}_{N,r+1}^2, \tag{239}$$

$$\|\Pi_{n,\alpha}(t) - \Pi_{n,\alpha}(r_0)\|_{H^s} \le 2^{n+1} A_r \mathfrak{T}_{N,r+1}\mathfrak{P}_{N,r+2}^{1/2}|t-r_0|^{1/2} \tag{240}$$

*for $0 \le r_0 \le t \le T$. If $1 \le a \le \infty$, $2 \le b \le \infty$, $k, q \in \mathbb{N}_0$, and $q > k + \frac{3}{2} - \frac{3}{b}$, then*

$$\left\|\nabla^k \Pi_{n,\alpha}\right\|_{L^a(I_T;L^b)} \le S_{q,k,b} T^{1/a} A_{q+|\alpha|} 2^n \mathfrak{T}_{N,q+|\alpha|+1}^2. \tag{241}$$

*Proof.* The normalized pressure recurrence is

$$Q_n = R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i} V_{n-a,j}. \tag{242}$$

Equation (177), Cauchy–Schwarz in time, and $\sum_{a=0}^n \binom{n}{a} = 2^n$ give

$$\|Q_n\|_{L^1(I_T;H^r)} \le A_r 2^n \mathfrak{P}_{N,r}. \tag{243}$$

The gradient estimate in (236) follows from $\|\nabla f\|_{H^{r-1}} \le \|f\|_{H^r}$. Using one factor in $L_t^\infty H_x^{r+1}$ and the other in $L_t^2 H_x^{r+1}$ proves (237). Using both factors in $L_t^\infty H_x^{r+1}$ proves (238) and (239).

Subtract (242) at times $t$ and $r_0$, split each product difference into two terms, apply (231) in $H^{r+1}$, and use (229). The two product differences and the binomial sum give (240). Equation (241) follows from (238) and Sobolev embedding. □

**Proposition 7.12** (Matrix form of the closed rectangle estimates)**.** *For nonnegative matrices of identical dimensions, write $A \preceq_{\mathrm{ent}} B$ when every entry of $A$ is at most the corresponding entry of $B$. When $T = T_* < \infty$, set the closed rectangle matrix by monotone exhaustion and use the terminal scalar norms from Proposition 4.7:*

$$\begin{aligned}
\mathbf{C}_{N,M}(T_*) &:= \lim_{S \uparrow T_*} \mathbf{C}_{N,M}(S), \\
\mathfrak{R}_{N,M}(u; T_*) &= \lim_{S \uparrow T_*} \mathfrak{R}_{N,M}(u; S), \\
\mathfrak{Q}_{N,M}(p; T_*) &:= \lim_{S \uparrow T_*} \mathfrak{Q}_{N,M}(p; S).
\end{aligned}$$

*All three limits exist in the extended nonnegative reals by monotonicity. Theorem 4.6 and Proposition 4.9 make the scalar limits finite, and the estimates below make every matrix entry finite. Then*

$$\mathbf{C}_{N,M}(T) \preceq_{\mathrm{ent}} \begin{pmatrix} \mathfrak{X}_{n,M+1} & \mathfrak{X}_{n+1,M-1} & \mathfrak{Q}^{(1)}_{n,M} \end{pmatrix}_{n=0}^{N}. \tag{244}$$

*Consequently,*

$$\mathfrak{R}_{N,M}(u; T) \le \sum_{n=0}^{N} \left( \mathfrak{X}^2_{n,M+1} + \mathfrak{X}^2_{n+1,M-1} \right), \tag{245}$$

$$\mathfrak{Q}_{N,M}(p; T) \le \sum_{n=0}^{N} \mathfrak{Q}^{(1)}_{n,M}. \tag{246}$$

*Proof.* The first two columns of (244) are (220). The pressure recurrence (242), the product estimate (178), and Cauchy–Schwarz in time give the third column through (224). Take the Frobenius norm of the first two columns and sum the third column, using (154). □

## 7.4 Frequency and critical-scale consequences

**Proposition 7.13** (Velocity and pressure frequency tails)**.** *Let $\mathsf{P}_{\ge K}$ be the Fourier multiplier of $\mathbf{1}_{\{|\xi| \ge K\}}$, where $K \ge 1$. Let $r_1 \in \mathbb{N}$, $s \in \mathbb{N}_0$, $0 \le n \le N$, and $\alpha \in \mathbb{N}_0^3$. Then*

$$\|\mathsf{P}_{\ge K} J_{n,\alpha}\|_{L^2(I_T; H^s)} \le K^{-r_1} \mathfrak{P}^{1/2}_{N, s+r_1+|\alpha|}, \tag{247}$$

$$\sup_{0 \le t \le T} \|\mathsf{P}_{\ge K} J_{n,\alpha}(t)\|_{H^s} \le K^{-r_1} \mathfrak{T}_{N, s+r_1+|\alpha|}, \tag{248}$$

$$\|\mathsf{P}_{\ge K} \Pi_{n,\alpha}\|_{L^1(I_T; H^s)} \le K^{-r_1} A_{s+r_1+|\alpha|} 2^n \mathfrak{P}_{N, s+r_1+|\alpha|}, \tag{249}$$

$$\sup_{0 \le t \le T} \|\mathsf{P}_{\ge K} \Pi_{n,\alpha}(t)\|_{H^s} \le K^{-r_1} A_{s+r_1+|\alpha|} 2^n \mathfrak{T}^2_{N, s+r_1+|\alpha|+1}. \tag{250}$$

*The pointwise estimates* (248) *and* (250) *hold at $t = T$.*

*Proof.* For every $f \in H^{s+r_1}$,

$$\|\mathsf{P}_{\geq K} f\|_{H^s} \leq K^{-r_1} \|f\|_{H^{s+r_1}} .$$

Apply this inequality to (228) and (229) at spatial order $s + r_1$ to obtain the two velocity bounds. Since Fourier multipliers commute,

$$\|\mathsf{P}_{\geq K} \Pi_{n,\alpha}\|_{H^s} \leq K^{-r_1} \|Q_n\|_{H^{s+r_1+|\alpha|}} .$$

Integration in time followed by (243) at order $s + r_1 + |\alpha|$ gives (249). Applying the tail inequality directly to $\Pi_{n,\alpha}$, followed by (238) with $s$ replaced by $s + r_1$, gives (250). □

**Proposition 7.14** (Fixed weighted dyadic shell sums)**.** *Let $(\Delta_j)_{j\geq -1}$ be a fixed inhomogeneous Littlewood–Paley decomposition on $\mathbb{R}^3$. Let $\sigma \in \mathbb{R}$, $k \in \mathbb{N}_0$, and choose $M \in \mathbb{N}_0$ such that*

$$M > \sigma + k + \frac{3}{2}. \tag{251}$$

*Then*

$$\int_0^T \sum_{j\geq -1} 2^{\sigma j} \left\| \Delta_j \nabla^k u(t) \right\|_{L^\infty} \, \mathrm{d}t \leq C_{\Delta,\sigma,k,M} T^{1/2} \mathfrak{X}_{0,M}. \tag{252}$$

*In particular, with $\omega = \nabla \times u$,*

$$\sum_{j\geq -1} \int_0^T \|\Delta_j \omega(t)\|_{L^\infty} \, \mathrm{d}t \leq C_\Delta T^{1/2} \mathfrak{X}_{0,3} < \infty. \tag{253}$$

*Each estimate uses one fixed finite Sobolev coordinate; no summation over Sobolev orders is asserted.*

*Proof.* Bernstein's inequality on a shell gives, for $j \geq 0$,

$$2^{\sigma j} \left\| \Delta_j \nabla^k f \right\|_{L^\infty} \leq C_\Delta 2^{(\sigma+k+3/2)j} \|\Delta_j f\|_2 .$$

Put $d := M - \sigma - k - 3/2 > 0$. Cauchy–Schwarz in $j$ yields

$$\begin{aligned} \sum_{j\geq 0} 2^{\sigma j} \left\| \Delta_j \nabla^k f \right\|_{L^\infty} &\leq C_\Delta \left( \sum_{j\geq 0} 2^{-2dj} \right)^{1/2} \left( \sum_{j\geq 0} 2^{2Mj} \|\Delta_j f\|_2^2 \right)^{1/2} \\ &\leq C_{\Delta,\sigma,k,M} \|f\|_{H^M} . \end{aligned}$$

The single low block $j = -1$ obeys this final bound. The last inequality follows from Plancherel's theorem and the finite overlap of the dyadic multipliers. Integrating in time, applying Cauchy–Schwarz, and using (220) prove (252). Since curl commutes with every $\Delta_j$, the choice $(\sigma, k, M) = (0, 1, 3)$ gives (253). □

**Proposition 7.15** (Critical-height bounds)**.** *Define*

$$E_\sigma(t) := \frac{1}{2} \|\Lambda^\sigma u(t)\|_2^2 , \qquad H(t) := E_{1/2}(t),$$

*and*

$$E_{j+1/2,\geq K}(t) := \frac{1}{2} \int_{|\xi|\geq K} |\xi|^{2j+1} |\widehat{u}(t,\xi)|^2 \, d\xi.$$

*For $j \in \mathbb{N}_0$,*

$$\sup_{0\le t\le T} E_{j+1/2}(t) \le \frac{1}{2}\mathfrak{T}_{0,j+1}^2, \tag{254}$$

$$\int_0^T E_{j+1/2}(t)\,dt \le \frac{1}{2}\mathfrak{P}_{0,j}. \tag{255}$$

*If $m > j + \frac{1}{2}$ is an integer and $K \ge 1$, then*

$$\sup_{0\le t\le T} E_{j+1/2,\ge K}(t) \le \frac{1}{2}K^{2j+1-2m}\mathfrak{T}_{0,m}^2, \tag{256}$$

$$\int_0^T E_{j+1/2,\ge K}(t)\,dt \le \frac{1}{2}K^{2j+1-2m}\mathfrak{P}_{0,m-1}. \tag{257}$$

*For $k \in \mathbb{N}$,*

$$\left\|H^{(k)}\right\|_{L^1(0,T)} \le 2^{k-1}\mathfrak{P}_{k,0}. \tag{258}$$

*In particular,*

$$\left\|H'\right\|_{L^1(0,T)} \le \frac{1}{2}\mathfrak{P}_{0,2}, \tag{259}$$

$$\int_0^T \left\|\Lambda^{3/2}u(t)\right\|_2^2\,dt \le \mathfrak{P}_{0,1}, \tag{260}$$

$$\left\|\mathcal{P}_{1/2}\right\|_{L^1(0,T)} \le \frac{1}{2}\mathfrak{P}_{0,2} + \nu\mathfrak{P}_{0,1}, \tag{261}$$

$$|H(T) - H(t)| \le \mathfrak{T}_{0,1}\mathfrak{P}_{0,2}^{1/2}(T-t)^{1/2}, \quad 0 \le t \le T, \tag{262}$$

*where $\mathcal{P}_{1/2} = H' + \nu\left\|\Lambda^{3/2}u\right\|_2^2$.*

*Proof.* Equations (254) and (255) follow from $\dot{H}^{j+1/2} \hookleftarrow H^{j+1}$, (229), and (228). On $|\xi| \ge K$,

$$|\xi|^{2j+1} \le K^{2j+1-2m}|\xi|^{2m},$$

which proves (256) and (257).

The identity

$$H^{(k)}(t) = \frac{1}{2}\sum_{a=0}^{k}\binom{k}{a}\operatorname{Re}\left\langle \Lambda^{1/2}V_a(t), \Lambda^{1/2}V_{k-a}(t)\right\rangle$$

and Cauchy–Schwarz in time give (258). At $k = 1$, the two equal terms give

$$\left\|H'\right\|_{L^1(0,T)} \le \|u_t\|_{L^2(I_T;\dot{H}^{1/2})}\,\|u\|_{L^2(I_T;\dot{H}^{1/2})} \le \frac{1}{2}\mathfrak{P}_{0,2}.$$

Equation (260) follows from the upper velocity row of the $(0,1)$ rectangle. The identity defining $\mathcal{P}_{1/2}$ gives (261). Finally,

$$\|u(T) - u(t)\|_{\dot{H}^{1/2}} \le \|u(T) - u(t)\|_{H^1} \le (T-t)^{1/2}\,\|u_t\|_{L^2(t,T;H^1)} \le (T-t)^{1/2}\mathfrak{P}_{0,2}^{1/2},$$

and

$$|H(T) - H(t)| \le \frac{1}{2}\left(\|u(T)\|_{\dot{H}^{1/2}} + \|u(t)\|_{\dot{H}^{1/2}}\right)\|u(T) - u(t)\|_{\dot{H}^{1/2}}.$$

Equation (229) bounds each retained velocity factor, so these two estimates prove (262). □

Taking $j = 0$ and any integer $m \ge 1$ in (256) gives the uniform tail limit

$$\lim_{K\to\infty}\sup_{0\le t\le T} E_{1/2,\ge K}(t) = 0. \tag{263}$$

### 7.5 Continuation and quantitative restart

**Proposition 7.16** (Continuation norms)**.** *The finite-horizon solution satisfies*

$$\|u\|_{L^2(I_T;L^\infty)} \le S_{2,0,\infty}\mathfrak{P}_{0,1}^{1/2}, \tag{264}$$

$$\int_0^T \|\nabla u(t)\|_{L^\infty}\, dt \le S_{3,1,\infty}T^{1/2}\mathfrak{P}_{0,2}^{1/2}, \tag{265}$$

$$\int_0^T \|\nabla \times u(t)\|_{L^\infty}\, dt \le 2S_{3,1,\infty}T^{1/2}\mathfrak{P}_{0,2}^{1/2}. \tag{266}$$

*For every integer $m \ge 3$, let $K_m$ be the explicit constant in* (14)*. Then*

$$|\mathrm{Re}\,\langle A^m\mathbb{P}\nabla\cdot(u\otimes u), A^m u\rangle_{L^2}| \le K_m \|\nabla u\|_{L^\infty}\|u\|_{H^m}^2. \tag{267}$$

$$\sup_{0\le t\le T}\|u(t)\|_{H^m}^2 \le \|u_0\|_{H^m}^2 \exp\Big(2K_m S_{3,1,\infty}T^{1/2}\mathfrak{P}_{0,2}^{1/2}\Big). \tag{268}$$

*Proof.* Sobolev embedding and the upper rows of the $(0,1)$ and $(0,2)$ rectangles give (264) and (265); the pointwise bound $\|\nabla\times u\|_\infty \le 2\|\nabla u\|_\infty$ gives (266). The $H^m$ energy identity and (267) give

$$\frac{1}{2}\frac{d}{dt}\|u\|_{H^m}^2 + \nu\|\nabla u\|_{H^m}^2 \le K_m\|\nabla u\|_{L^\infty}\|u\|_{H^m}^2,$$

and hence

$$\frac{d}{dt}\|u\|_{H^m}^2 \le 2K_m\|\nabla u\|_{L^\infty}\|u\|_{H^m}^2.$$

Gronwall's inequality and (265) prove (268). □

**Theorem 7.17** (Explicit endpoint-restart bound)**.** *Assume $T = T_* < \infty$, and set $u_T := u_*$. Define the maximal restart lifespan inside the shifted pressure-normalized $H^3$ mild class by*

$$\begin{aligned}\delta_T^{\max} &:= \tau_{\nu,3}^{\max}(T,u_T)\\ &= \sup\big\{\delta>0 : \mathscr{M}_\nu^3(T,T+\delta;u_T)\neq\varnothing\big\} \in (0,\infty].\end{aligned} \tag{269}$$

*Define*

$$\delta_{\mathrm{qc}}[u_0,\nu;T] := \min\Big\{1, c_\nu\big(1+\mathfrak{Z}_{0,3}[u_0,\nu;T]\big)^{-2}\Big\}. \tag{270}$$

*The class $\mathscr{M}_\nu^3(T,T+\delta_{\mathrm{qc}};u_T)$ contains exactly one pair $(w,q)$. It is the original history's continuation constructed in Theorem 5.8, and*

$$\delta_T^{\max} \ge \delta_{\mathrm{qc}}[u_0,\nu;T] > 0. \tag{271}$$

*At the base restart order,*

$$\begin{aligned}&\|w\|_{C([T,T+\delta_{\mathrm{qc}}];H^3)} + \|w\|_{L^2(T,T+\delta_{\mathrm{qc}};H^4)} + \|w_t\|_{L^2(T,T+\delta_{\mathrm{qc}};H^2)}\\ &\quad\le 2L_\nu\mathfrak{Z}_{0,3}[u_0,\nu;T].\end{aligned} \tag{272}$$

*Set*

$$\mathfrak{A}_T := 2S_{3,1,\infty}L_\nu\delta_{\mathrm{qc}}\mathfrak{Z}_{0,3}[u_0,\nu;T]. \tag{273}$$

*For every integer $m \ge 3$,*

$$\sup_{T\le t\le T+\delta_{\mathrm{qc}}}\|w(t)\|_{H^m}^2 \le \mathfrak{Z}_{0,m}[u_0,\nu;T]^2\exp(2K_m\mathfrak{A}_T). \tag{274}$$

*Proof.* Equations (114) and (221) give

$$\|u_T\|_{H^3} \le \mathfrak{Z}_{0,3}.$$

Consequently,

$$\delta_{\mathrm{qc}}[u_0,\nu;T] \le \min\{1, c_\nu(1+\|u_T\|_{H^3})^{-2}\}.$$

Theorem 2.11 gives the unique member of $\mathscr{M}^3_\nu(T,T+\delta_{\mathrm{qc}};u_T)$. The fixed-point ball in its proof gives (272). Hence

$$\int_T^{T+\delta_{\mathrm{qc}}} \|\nabla w(t)\|_\infty \,\mathrm{d}t \le S_{3,1,\infty}\delta_{\mathrm{qc}}\,\|w\|_{C([T,T+\delta_{\mathrm{qc}}];H^3)} \le \mathfrak{A}_T.$$

The $H^m$ energy estimate in Proposition 7.16, now on the restart interval, and $\|u_T\|_{H^m} \le \mathfrak{Z}_{0,m}$ prove (274). Theorem 5.8 identifies this unique local pair with the continuation of the preterminal history, and its membership on the displayed interval proves (271). □

# 8 Three-dimensional applications

## 8.1 Explicit closure for small critical data

The construction below gives an explicit three-dimensional $\dot H^{1/2}$ realization of the critical-data mechanism introduced by Fujita and Kato [6]; every constant used here is derived in the displayed estimates.

Use the explicit homogeneous Sobolev constants

$$\mathsf{S}_3 := \mathsf{S}(3), \qquad C_{\mathrm{src}} := \mathsf{S}_3^3 = \frac{1}{\sqrt{2}\,\pi}.$$

These constants hold for vector fields. Indeed, apply the scalar inequality to the Euclidean length $|F|$: one has $|\nabla|F|| \le |\nabla F|$, while the difference-quotient representation of $\dot H^{1/2}$ and $\big||F(x)|-|F(y)|\big| \le |F(x)-F(y)|$ give $\||F|\|_{\dot H^{1/2}} \le \|F\|_{\dot H^{1/2}}$.

**Lemma 8.1** (Critical action identity)**.** *Let $(u,p)$ be a smooth pressure-normalized solution on a time interval $I \subset [0,T_*)$, and set*

$$\mathcal{B}(u) := \mathbb{P}\nabla\cdot(u\otimes u).$$

*Then, pointwise on $I$,*

$$\nu\frac{\mathrm{d}}{\mathrm{d}t}\|u\|^2_{\dot H^{1/2}} + \nu^2\|u\|^2_{\dot H^{3/2}} + \|u_t\|^2_{\dot H^{-1/2}} = \|\mathcal{B}(u)\|^2_{\dot H^{-1/2}}\,. \tag{275}$$

*Proof.* The projected equation is

$$u_t = -\nu\Lambda^2 u - \mathcal{B}(u).$$

Put

$$g := \Lambda^{-1/2}\mathcal{B}(u), \qquad w := \Lambda^{-1/2}u_t = -\nu\Lambda^{3/2}u - g.$$

Pairing the projected equation with $\Lambda u$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u\|^2_{\dot H^{1/2}} = -2\nu\|u\|^2_{\dot H^{3/2}} - 2\,\mathrm{Re}\left\langle g, \Lambda^{3/2}u\right\rangle_2.$$

Multiply this equality by $\nu$ and expand $\|w\|_2^2 = \big\|\nu\Lambda^{3/2}u+g\big\|_2^2$. The cross terms cancel and give (275). □

For $m \in \mathbb{N}$, define the explicit homogeneous commutator constant

$$\mathsf{K}_m^{\mathrm{hom}} := 3^m \sum_{\ell=1}^{m} \binom{m}{\ell} \mathsf{S}(\mathfrak{p}_{m,\ell}) \mathsf{S}(\mathfrak{q}_{m,\ell}), \tag{276}$$

where $(\mathfrak{p}_{1,1}, \mathfrak{q}_{1,1}) = (3, 6)$, and, for $m \geq 2$,

$$\vartheta_{m,\ell} := \frac{\ell - 1}{m - 1}, \qquad \frac{1}{\mathfrak{p}_{m,\ell}} := \frac{1}{3} - \frac{\vartheta_{m,\ell}}{6}, \qquad \frac{1}{\mathfrak{q}_{m,\ell}} := \frac{1}{6} + \frac{\vartheta_{m,\ell}}{6}.$$

The corresponding estimate is

$$|\mathrm{Re}\, \langle \Lambda^m \mathbb{P} \nabla \cdot (z \otimes z), \Lambda^m z \rangle_{L^2}| \leq \mathsf{K}_m^{\mathrm{hom}} \left\| \Lambda^{3/2} z \right\|_2 \|\Lambda^m z\|_2 \left\| \Lambda^{m+1} z \right\|_2 \tag{277}$$

for every $z \in \mathcal{S}_\sigma(\mathbb{R}^3)$.

**Lemma 8.2** (Explicit homogeneous critical commutator)**.** *For every $m \in \mathbb{N}$, $\mathsf{K}_m^{\mathrm{hom}} < \infty$, and* (277) *holds with the displayed constant. Moreover,* (277) *extends to every divergence-free $z \in H^{m+1}(\mathbb{R}^3) \cap \dot{H}^{3/2}(\mathbb{R}^3)$.*

*Proof.* The multinomial identity gives the exact pairing

$$\langle \Lambda^m f, \Lambda^m g \rangle_{L^2} = \sum_{|\alpha| = m} \frac{m!}{\alpha!} \langle \partial^\alpha f, \partial^\alpha g \rangle_{L^2}.$$

After differentiating $(z \cdot \nabla) z$, the undifferentiated transport term has zero pairing by incompressibility. A remaining term has the form

$$\left\langle (\partial^\beta z \cdot \nabla) \partial^{\alpha - \beta} z, \partial^\alpha z \right\rangle_{L^2}, \qquad |\alpha| = m, \quad 0 < \beta \leq \alpha.$$

Put $\ell = |\beta|$. For $m \geq 2$, Hölder's inequality, Lemma 2.3, and Fourier log-convexity give

$$\begin{aligned} &\left\| \partial^\beta z \right\|_{\mathfrak{p}_{m,\ell}} \left\| \nabla \partial^{\alpha - \beta} z \right\|_{\mathfrak{q}_{m,\ell}} \\ &\quad \leq \mathsf{S}(\mathfrak{p}_{m,\ell}) \mathsf{S}(\mathfrak{q}_{m,\ell}) \|z\|_{\dot{H}^{3/2}} \|z\|_{\dot{H}^{m+1}}. \end{aligned}$$

Indeed, the two differentiated Sobolev orders are the complementary interpolants

$$\begin{aligned} \ell + s(\mathfrak{p}_{m,\ell}) &= (1 - \vartheta_{m,\ell}) \frac{3}{2} + \vartheta_{m,\ell}(m+1), \\ m - \ell + 1 + s(\mathfrak{q}_{m,\ell}) &= \vartheta_{m,\ell} \frac{3}{2} + (1 - \vartheta_{m,\ell})(m+1). \end{aligned}$$

For $m = 1$, this bound is the direct $L^3$-$L^6$ estimate. Finally,

$$\sum_{|\alpha| = m} \frac{m!}{\alpha!} = 3^m, \qquad \sum_{\substack{\beta \leq \alpha \\ |\beta| = \ell}} \binom{\alpha}{\beta} = \binom{m}{\ell}.$$

Summation gives exactly (276) and proves (277). Solenoidal Fourier truncation and convergence in $H^{m+1} \cap \dot{H}^{3/2}$ give the asserted extension. □

**Theorem 8.3** (Explicit small-critical-data global regularity)**.** *Let $\nu > 0$ and $a \in \mathcal{S}_\sigma(\mathbb{R}^3)$. Set*

$$a_{\mathrm{c}} := \left\|\Lambda^{1/2} a\right\|_2, \qquad \delta_0 := \nu^2 - C_{\mathrm{src}}^2 a_{\mathrm{c}}^2.$$

*If*

$$a_{\mathrm{c}} < \frac{\nu}{C_{\mathrm{src}}} = \sqrt{2}\,\pi\nu, \tag{278}$$

*then $a$ generates a unique global pressure-normalized smooth solution*

$$(u, p) \in C^\infty([0,\infty) \times \mathbb{R}^3; \mathbb{R}^3 \times \mathbb{R}).$$

*For every finite $T > 0$, the solution is the unique member of $\mathscr{M}_\nu^3(0, T; a)$. For every $t \geq 0$,*

$$\nu \left\|\Lambda^{1/2} u(t)\right\|_2^2 + \delta_0 \int_0^t \left\|\Lambda^{3/2} u(s)\right\|_2^2 \,\mathrm{d}s + \int_0^t \|u_t(s)\|_{\dot{H}^{-1/2}}^2 \,\mathrm{d}s \leq \nu a_{\mathrm{c}}^2, \qquad \delta_0 > 0. \tag{279}$$

*The critical nonlinear source and the first temporal response satisfy*

$$\int_0^\infty \|\mathbb{P}\nabla\cdot(u \otimes u)(t)\|_{\dot{H}^{-1/2}}^2 \,\mathrm{d}t \leq \frac{C_{\mathrm{src}}^2 \nu a_{\mathrm{c}}^4}{\delta_0} = \frac{\nu a_{\mathrm{c}}^4}{2\pi^2\nu^2 - a_{\mathrm{c}}^2}, \tag{280}$$

$$\nu^2 \int_0^\infty \left\|\Lambda^{3/2} u(t)\right\|_2^2 \,\mathrm{d}t + \int_0^\infty \|u_t(t)\|_{\dot{H}^{-1/2}}^2 \,\mathrm{d}t \leq \nu a_{\mathrm{c}}^2 + \frac{C_{\mathrm{src}}^2 \nu a_{\mathrm{c}}^4}{\delta_0}. \tag{281}$$

*For every $m \in \mathbb{N}$, define*

$$\Gamma_m^{\mathrm{sm}} := \frac{(\mathsf{K}_m^{\mathrm{hom}})^2 a_{\mathrm{c}}^2}{\delta_0}.$$

*Then*

$$\begin{aligned} &\|\Lambda^m u(t)\|_2^2 + \nu \int_0^t \left\|\Lambda^{m+1} u(s)\right\|_2^2 \,\mathrm{d}s \\ &\qquad \leq \|\Lambda^m a\|_2^2 \exp(\Gamma_m^{\mathrm{sm}}). \end{aligned} \tag{282}$$

*In particular,*

$$\sup_{t \geq 0} \|u(t)\|_{H^m}^2 \leq 2^{m-1} \left[\|a\|_2^2 + \|\Lambda^m a\|_2^2 \exp(\Gamma_m^{\mathrm{sm}})\right]. \tag{283}$$

*Proof.* Iterate Theorem 2.11 and join overlapping intervals by its at-most-one assertion for the shifted mild class. The union is the maximal shifted pressure-normalized $H^3$ mild history on $[0, T_*)$, and persistence retains $H^\infty$ on every compact subinterval. Put

$$A(t) := \left\|\Lambda^{1/2} u(t)\right\|_2, \qquad B(t) := \left\|\Lambda^{3/2} u(t)\right\|_2.$$

Dual Sobolev, Hölder, and the unit Fourier-multiplier norm of $\mathbb{P}$ give the complete source-constant chain

$$\begin{aligned} \|\mathbb{P}\nabla\cdot(u \otimes u)\|_{\dot{H}^{-1/2}} &\leq \mathsf{S}_3 \,\|(u\cdot\nabla)u\|_{L^{3/2}} \\ &\leq \mathsf{S}_3 \,\|u\|_{L^3} \,\|\nabla u\|_{L^3} \\ &\leq \mathsf{S}_3^3 \left\|\Lambda^{1/2} u\right\|_2 \left\|\Lambda^{3/2} u\right\|_2 = C_{\mathrm{src}} A(t) B(t). \end{aligned} \tag{284}$$

Combining (284) with the exact identity (275) gives, on every classical subinterval,

$$\nu\frac{\mathrm{d}}{\mathrm{d}t}A(t)^2 + \big(\nu^2 - C_{\mathrm{src}}^2 A(t)^2\big)B(t)^2 + \|u_t(t)\|_{\dot{H}^{-1/2}}^2 \le 0.$$

Continuity and a first-exit argument give $A(t) \le a_{\mathrm{c}}$ throughout $[0, T_*)$. Integration and the definition of $\delta_0$ prove (279). In particular,

$$\int_0^t B(s)^2\,\mathrm{d}s \le \frac{\nu a_{\mathrm{c}}^2}{\delta_0}.$$

For every $t < T_*$, this source estimate and the integrated identity (275) prove (280) and (281) with $\infty$ replaced by $t$. After the continuation argument below proves $T_* = \infty$, monotone convergence gives the stated forms.

At order $m$, Lemma 8.2 and Young's inequality yield

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\|\Lambda^m u\|_2^2 + \nu\,\big\|\Lambda^{m+1}u\big\|_2^2 \le \frac{(\mathsf{K}_m^{\mathrm{hom}})^2}{\nu}B(t)^2\,\|\Lambda^m u\|_2^2\,.$$

Multiplication by the corresponding integrating factor and

$$\int_0^t B(s)^2\,\mathrm{d}s \le \frac{\nu a_{\mathrm{c}}^2}{\delta_0}$$

prove (282). The Fourier inequality

$$(1+|\xi|^2)^m \le 2^{m-1}(1+|\xi|^{2m})$$

and the kinetic-energy identity prove (283).

Let $M_3$ be the square root of the right-hand side of (283) with $m = 3$, and set

$$\delta_{\mathrm{crit}} := \min\{1, c_\nu(1+M_3)^{-2}\} > 0.$$

Every datum $u(t_0)$, $t_0 < T_*$, has $H^3$-norm at most $M_3$. Theorem 2.11 restarts the solution for at least $\delta_{\mathrm{crit}}$ units of time. If $T_* < \infty$, choose $t_0 < T_*$ with $T_* - t_0 < \delta_{\mathrm{crit}}/2$. Both restrictions belong to $\mathscr{M}_\nu^3(t_0, b; u(t_0))$ for every $b < T_*$, so uniqueness on each such closed subinterval identifies them throughout $[t_0, T_*)$. The restarted member then extends the original history beyond $T_*$, contradicting maximality. Hence $T_* = \infty$, and persistence gives the asserted smoothness. This local uniqueness on each finite interval proves the stated global uniqueness class. □

For $m \ge 1$, define

$$U_m^{\mathrm{sm}} := 2^{(m-1)/2}\left[\|a\|_2^2 + \|\Lambda^m a\|_2^2\exp(\Gamma_m^{\mathrm{sm}})\right]^{1/2}, \qquad U_0^{\mathrm{sm}} := \|a\|_2\,, \qquad U_{-1}^{\mathrm{sm}} := U_0^{\mathrm{sm}}.$$

For $n \in \mathbb{N}_0$ and $m \ge -1$, define the finite-order recursion

$$\mathsf{Y}_{0,m}^{\mathrm{sm}} := U_m^{\mathrm{sm}}, \tag{285}$$

$$\mathsf{Y}_{n+1,m}^{\mathrm{sm}} := \nu\mathsf{Y}_{n,m+2}^{\mathrm{sm}} + A_{m+1}\sum_{a'=0}^{n}\binom{n}{a'}\mathsf{Y}_{a',m+2}^{\mathrm{sm}}\mathsf{Y}_{n-a',m+2}^{\mathrm{sm}}. \tag{286}$$

**Proposition 8.4** (Explicit small-data pressure-complete rectangles)**.** *Under the hypotheses of Theorem 8.3, for every $n \in \mathbb{N}_0$ and $m \geq -1$,*

$$\sup_{t\geq 0} \|V_n(t)\|_{H^m} \leq \mathsf{Y}^{\mathrm{sm}}_{n,m}. \tag{287}$$

*For every $T < \infty$ and $N, M \in \mathbb{N}_0$,*

$$\mathfrak{R}_{N,M}(u;T) \leq T \sum_{n=0}^{N} \left[ (\mathsf{Y}^{\mathrm{sm}}_{n,M+1})^2 + (\mathsf{Y}^{\mathrm{sm}}_{n+1,M-1})^2 \right], \tag{288}$$

$$\mathfrak{Q}_{N,M}(p;T) \leq T A_M \sum_{n=0}^{N} \sum_{a'=0}^{n} \binom{n}{a'} \mathsf{Y}^{\mathrm{sm}}_{a',M+1} \mathsf{Y}^{\mathrm{sm}}_{n-a',M+1}. \tag{289}$$

*In the entrywise order from Proposition 7.12, these bounds have the matrix form*

$$\mathbf{C}_{N,M}(T) \preceq_{\mathrm{ent}} \left( \begin{array}{ccc} T^{1/2}\mathsf{Y}^{\mathrm{sm}}_{n,M+1} & T^{1/2}\mathsf{Y}^{\mathrm{sm}}_{n+1,M-1} & T A_M \displaystyle\sum_{a'=0}^{n} \binom{n}{a'} \mathsf{Y}^{\mathrm{sm}}_{a',M+1} \mathsf{Y}^{\mathrm{sm}}_{n-a',M+1} \end{array} \right)_{n=0}^{N}. \tag{290}$$

*Every right-hand side is an explicit finite function of $\nu, T, N, M$ and the datum norms*

$$\|a\|_2, \qquad \left\|\Lambda^{1/2} a\right\|_2, \qquad \{\|\Lambda^r a\|_2 : 1 \leq r \leq M + 2N + 1\}. \tag{291}$$

*Indeed, induction in $n$ shows that $\mathsf{Y}^{\mathrm{sm}}_{n,m}$ uses only the entries $U^{\mathrm{sm}}_j$ with $-1 \leq j \leq m + 2n$; both velocity columns of (290) consequently stop at $j = M + 2N + 1$, and its pressure column stops no later.*

*Proof.* Equation (283) proves the base (287) for $m \geq 1$; the kinetic-energy identity proves it for $m = 0$, and $H^0 \hookrightarrow H^{-1}$ proves it for $m = -1$. Apply $\mathbb{P}$ to the temporal recurrence. This is the $\alpha = 0$ temporal block-row of (147); its factorial conjugate is (150). Equation (177) at order $m + 1$ and induction give (287) through (286); for $m = -1$, use the order-zero product estimate and $H^0 \hookrightarrow H^{-1}$. Integration in time gives (288). The pressure recurrence and (178) at order $M$, followed by integration and summation in $n$, give (289). These rowwise bounds give (290); taking the Frobenius norm of its first two columns and summing its pressure column recovers (288) and (289). □

## 8.2 Low-rectangle and scale-local bounds

The following results use distinct outputs of the compatible rectangles: mixed space–time embeddings, pressure completion, and integrable Lipschitz action.

**Proposition 8.5** (Three low-rectangle regularity classes)**.** *Let $u$ be the global solution generated by Theorem 5.10, and let $0 < T < \infty$. Then*

$$\|u\|_{L^\infty(0,T;L^3)} \leq S_{1,0,3}\mathfrak{Z}_{0,1}[u_0, \nu; T], \tag{292}$$

$$\|u\|_{L^2(0,T;L^\infty)} \leq S_{2,0,\infty}\mathfrak{X}_{0,2}[u_0, \nu; T], \tag{293}$$

$$\|\nabla u\|_{L^2(0,T;L^3)} \leq S_{2,1,3}\mathfrak{X}_{0,2}[u_0, \nu; T]. \tag{294}$$

*In particular, each of the three norms is finite on every finite horizon, with the stated datum, viscosity, and horizon dependence.*

*Proof.* Use (772) with $(n,\alpha,a,b,q,k)=(0,0,\infty,3,1,0)$ to obtain (292). Use (773) with $(n,\alpha,b,q,k)=(0,0,\infty,2,0)$ and $(0,0,3,2,1)$ to obtain (293) and (294). The strict embedding conditions are, respectively, $1>1/2$, $2>3/2$, and $2>3/2$. □

**Corollary 8.6** (Uniform extinction of parabolic scale content)**.** *Let $\nu>0$, let $u_0\in H^\infty_\sigma(\mathbb{R}^3)$, and let $(u,p)$ be the pressure-normalized global pair supplied by Theorem 5.10. This is a post-regularity consequence for the Caffarelli–Kohn–Nirenberg scale-normalized quantity [3]; it is not used as an $\varepsilon$-regularity input or continuation criterion. For $0<T<\infty$, define*

$$\begin{aligned} U_T &:= S_{2,0,\infty}\mathfrak{Z}_{0,2}[u_0,\nu;T],\\ P_T &:= S_{2,0,\infty}\mathfrak{Q}^{(\infty)}_{0,2}[u_0,\nu;T]. \end{aligned} \tag{295}$$

*Let $z_0=(t_0,x_0)$, $\rho>0$, and suppose that*

$$Q_\rho(z_0):=(t_0-\rho^2,t_0)\times B_\rho(x_0)\subset(0,T)\times\mathbb{R}^3.$$

*Then the scale-normalized velocity–pressure content satisfies*

$$\rho^{-2}\iint_{Q_\rho(z_0)}\left(|u|^3+|p|^{3/2}\right)\,\mathrm{d}x\,\mathrm{d}t\le\frac{4\pi}{3}\rho^3\left(U_T^3+P_T^{3/2}\right). \tag{296}$$

*Consequently,*

$$\lim_{\rho\downarrow0}\sup_{\{z_0:\,Q_\rho(z_0)\subset(0,T)\times\mathbb{R}^3\}}\rho^{-2}\iint_{Q_\rho(z_0)}\left(|u|^3+|p|^{3/2}\right)\,\mathrm{d}x\,\mathrm{d}t=0. \tag{297}$$

*Proof.* Equation (767), followed by (180) with $(s,k,b)=(2,0,\infty)$, gives

$$\|u\|_{L^\infty((0,T)\times\mathbb{R}^3)}\le U_T.$$

Equations (776) and (180) give

$$\|p\|_{L^\infty((0,T)\times\mathbb{R}^3)}\le P_T.$$

Since $|B_\rho|=\frac{4\pi}{3}\rho^3$ and the time length of $Q_\rho$ is $\rho^2$,

$$\iint_{Q_\rho(z_0)}\left(|u|^3+|p|^{3/2}\right)\,\mathrm{d}x\,\mathrm{d}t\le\frac{4\pi}{3}\rho^5\left(U_T^3+P_T^{3/2}\right).$$

Division by $\rho^2$ proves (296); its right-hand side is independent of the center and tends to zero with $\rho$, which proves (297). □

## 8.3 Vorticity and material flow

**Theorem 8.7** (Global vorticity amplification and dissipation)**.** *Let $\nu>0$, let $u_0\in H^\infty_\sigma(\mathbb{R}^3)$, and let $(u,p)$ be the pressure-normalized global pair supplied by Theorem 5.10. Put*

$$\omega:=\nabla\times u,\qquad \omega_0:=\nabla\times u_0,\qquad L^\omega_T:=S_{3,1,\infty}T^{1/2}\mathfrak{X}_{0,3}[u_0,\nu;T].$$

*For every $T<\infty$ and $2\le r\le\infty$,*

$$\sup_{0\le t\le T}\|\omega(t)\|_{L^r}\le e^{L^\omega_T}\,\|\omega_0\|_{L^r}\,, \tag{298}$$

$$\int_0^T\|(\omega\cdot\nabla)u(t)\|_{L^r}\,\mathrm{d}t\le L^\omega_T e^{L^\omega_T}\,\|\omega_0\|_{L^r}\,. \tag{299}$$

*If $2\le r<\infty$, then*

$$\int_0^T\left\|\nabla|\omega(t)|^{r/2}\right\|_2^2\,\mathrm{d}t\le\frac{r}{4\nu(r-1)}e^{rL^\omega_T}\,\|\omega_0\|_{L^r}^r\,. \tag{300}$$

*Proof.* Taking the curl of the momentum equation gives the three-dimensional vorticity equation

$$\partial_t \omega + (u \cdot \nabla)\omega = (\omega \cdot \nabla)u + \nu \Delta \omega. \tag{301}$$

Fix $2 \le r < \infty$. Pair (301) with $|\omega|^{r-2}\omega$. To justify this power test, choose $\chi \in C_c^\infty(\mathbb{R}^3)$, with $0 \le \chi \le 1$, $\chi = 1$ on $B_1$, and $\chi = 0$ outside $B_2$, put $\chi_R(x) = \chi(x/R)$, and first test against

$$\chi_R \Phi_\varepsilon(\omega), \qquad \Phi_\varepsilon(z) := (|z|^2 + \varepsilon)^{(r-2)/2} z.$$

Define the primitive

$$F_\varepsilon(z) := \frac{(|z|^2 + \varepsilon)^{r/2} - \varepsilon^{r/2}}{r}. \tag{302}$$

Then $\nabla_z F_\varepsilon = \Phi_\varepsilon$, and incompressibility gives

$$\int_{\mathbb{R}^3} \chi_R (u \cdot \nabla)\omega \cdot \Phi_\varepsilon(\omega)\, \mathrm{d}x = -\int_{\mathbb{R}^3} F_\varepsilon(\omega)\, u \cdot \nabla \chi_R \, \mathrm{d}x. \tag{303}$$

With $A_R := B_{2R} \setminus B_R$,

$$\left| \int F_\varepsilon(\omega)\, u \cdot \nabla \chi_R \, \mathrm{d}x \right| \le \frac{C_r}{R} \|u\|_\infty \left( \|\omega\|_{L^r(A_R)}^r + \varepsilon^{(r-2)/2} \|\omega\|_{L^2(A_R)}^2 \right), \tag{304}$$

$$\left| \int \nabla \chi_R \cdot \nabla \omega\, \Phi_\varepsilon(\omega)\, \mathrm{d}x \right| \le \frac{C_r}{R} \|\nabla \omega\|_{L^2(A_R)} \left( \|\omega\|_{L^{2r-2}(A_R)}^{r-1} + \varepsilon^{(r-2)/2} \|\omega\|_{L^2(A_R)} \right). \tag{305}$$

All norms on the right are finite because $u, \omega$ belong to every spatial Sobolev space. Their annular tails vanish as $R \to \infty$, so (303) converges to the transport cancellation and the diffusion cutoff term vanishes.

The remaining diffusion form is nonnegative:

$$\begin{aligned} \nabla\omega : D\Phi_\varepsilon(\omega)\nabla\omega &= (|\omega|^2 + \varepsilon)^{(r-2)/2} |\nabla\omega|^2 \\ &\quad + (r-2)(|\omega|^2 + \varepsilon)^{(r-4)/2} \sum_{j=1}^{3} (\omega \cdot \partial_j \omega)^2. \end{aligned} \tag{306}$$

After the limit $R \to \infty$, dominated convergence applies to $F_\varepsilon(\omega)$ and to the stretching term, while Fatou's lemma applies to (306) as $\varepsilon \downarrow 0$. This yields the unregularized chain rule and diffusion inequality used below. The stretching term satisfies

$$\left| \int_{\mathbb{R}^3} (\omega \cdot \nabla)u \cdot |\omega|^{r-2}\omega \, \mathrm{d}x \right| \le \|\nabla u\|_\infty \|\omega\|_r^r .$$

For the diffusion term, spatial integration by parts gives

$$\begin{aligned} -\int_{\mathbb{R}^3} \Delta\omega \cdot |\omega|^{r-2}\omega \, \mathrm{d}x &\ge (r-1) \int_{\mathbb{R}^3} |\omega|^{r-2} |\nabla|\omega||^2 \, \mathrm{d}x \\ &= \frac{4(r-1)}{r^2} \left\| \nabla |\omega|^{r/2} \right\|_2^2 . \end{aligned}$$

Hence

$$\frac{1}{r} \frac{\mathrm{d}}{\mathrm{d}t} \|\omega\|_r^r + \frac{4\nu(r-1)}{r^2} \left\| \nabla |\omega|^{r/2} \right\|_2^2 \le \|\nabla u\|_\infty \|\omega\|_r^r . \tag{307}$$

Combining (307) with the quantitative rectangle estimate (798) gives

$$\int_0^T \|\nabla u(t)\|_\infty \, \mathrm{d}t \le L_T^\omega.$$

Gronwall's inequality proves (298) for finite $r$. Multiplying (307) by $\exp(-r\int_0^t \|\nabla u(s)\|_\infty \, \mathrm{d}s)$ and integrating proves (300). Since $\omega_0, \omega(t) \in L^2 \cap L^\infty$, $\|f\|_r \to \|f\|_\infty$ as $r \to \infty$ for each such $f$. Consequently,

$$\sup_{0\le t\le T} \|\omega(t)\|_\infty \le \liminf_{r\to\infty} \sup_{0\le t\le T} \|\omega(t)\|_r \le e^{L_T^\omega} \lim_{r\to\infty} \|\omega_0\|_r = e^{L_T^\omega} \|\omega_0\|_\infty .$$

This is (298) at the endpoint. Finally,

$$\|(\omega\cdot\nabla)u(t)\|_r \le \|\nabla u(t)\|_\infty \|\omega(t)\|_r ;$$

integration and (298) give (299). □

**Theorem 8.8** (Quantitative volume-preserving material flow)**.** *Let $\nu > 0$, let $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and let $(u, p)$ be the pressure-normalized global pair supplied by Theorem 5.10. For $0 < T < \infty$, put*

$$\begin{aligned} D_T &:= S_{2,0,\infty} T^{1/2} \mathfrak{X}_{0,2}[u_0, \nu; T], \\ L_T &:= S_{3,1,\infty} T^{1/2} \mathfrak{X}_{0,3}[u_0, \nu; T]. \end{aligned} \tag{308}$$

*For every $a \in \mathbb{R}^3$, the problem*

$$\partial_t X(t,a) = u(t, X(t,a)), \qquad X(0,a) = a, \tag{309}$$

*has a unique solution on $[0,T]$. For every $t \in [0,T]$, the map $X(t,\cdot) : \mathbb{R}^3 \to \mathbb{R}^3$ is a smooth volume-preserving diffeomorphism and*

$$|X(t,a) - a| \le D_T, \tag{310}$$

$$e^{-L_T}|a-b| \le |X(t,a) - X(t,b)| \le e^{L_T}|a-b|, \tag{311}$$

$$\det D_a X(t,a) = 1. \tag{312}$$

*Proof.* Equations (797) and (798) give

$$\int_0^T \|u(t)\|_\infty \, \mathrm{d}t \le D_T,$$
$$\int_0^T \|\nabla u(t)\|_\infty \, \mathrm{d}t \le L_T.$$

Picard iteration on successive compact time intervals gives local existence and uniqueness for (309). The first integral bound prevents escape to spatial infinity and extends every trajectory to $[0,T]$; it also proves (310). For two trajectories, the second integral bound and Gronwall's inequality give the upper bound in (311). Solving this equation backward from time $t$ produces the inverse map and gives the lower bound. Spatial differentiation of (309) and induction give smoothness of the map and its inverse. Finally, Jacobi's formula and $\nabla\cdot u = 0$ give

$$\partial_t \det D_a X = (\nabla\cdot u)(t,X) \det D_a X = 0, \qquad \det D_a X(0,a) = 1,$$

which proves (312). □

## 8.4 Relative energy and the Leray–Hopf class

The weak class is stated in the whole-space form originating with Leray and Hopf [10, 7]. The argument is the classical weak–strong uniqueness mechanism associated with Serrin [13], written here against the pressure-complete smooth history and derived directly from the weak formulation and the energy inequality.

**Definition 8.9** (Three-dimensional Leray–Hopf solution)**.** Let $\nu > 0$, $T > 0$, and $v_0 \in H^0_\sigma(\mathbb{R}^3)$. A Leray–Hopf solution on $[0,T]$ is a pair $(v,q)$ such that

$$v \in C_w([0,T]; H^0_\sigma) \cap L^\infty(0,T; L^2_\sigma) \cap L^2(0,T; \dot{H}^1_\sigma), \qquad q \in \mathcal{D}'((0,T)\times\mathbb{R}^3),$$

the equations

$$\partial_t v + (v\cdot\nabla)v + \nabla q = \nu\Delta v, \qquad \nabla\cdot v = 0$$

hold in distributions, $v(t) \to v_0$ strongly in $L^2$ as $t \downarrow 0$, and

$$\frac{1}{2}\|v(t)\|_2^2 + \nu\int_0^t \|\nabla v(s)\|_2^2 \,\mathrm{d}s \le \frac{1}{2}\|v_0\|_2^2 \tag{313}$$

for every $t \in [0,T]$. Every divergence-free test field $\phi \in C^\infty_c([0,T)\times\mathbb{R}^3;\mathbb{R}^3)$ satisfies

$$\int_0^T \left[ -\langle v, \partial_t\phi\rangle - \int_{\mathbb{R}^3} (v\otimes v) : \nabla\phi \,\mathrm{d}x + \nu\langle \nabla v, \nabla\phi\rangle \right] \mathrm{d}t = \langle v_0, \phi(0)\rangle\,. \tag{314}$$

A global Leray–Hopf solution at viscosity $\nu$ is a pair on $[0,\infty)$ whose restriction to every finite $[0,T]$ is a Leray–Hopf solution in Definition 8.9, with identical initial data and viscosity.

**Lemma 8.10** (Solenoidal cutoff density and time commutation)**.** *Let $\mathscr{A}$ be a finite subset of $\{(r,p) : r \in \mathbb{N}_0,\ 2 \le p < \infty\}$. There are linear operators $\mathcal{T}_k$, independent of time, such that, for every $f \in H^\infty_\sigma(\mathbb{R}^3)$,*

$$\mathcal{T}_k f \in C^\infty_{c,\sigma}(\mathbb{R}^3)$$

*and, for every $(r,p) \in \mathscr{A}$,*

$$\sup_k \|\mathcal{T}_k f\|_{W^{r,p}} \le C_{\mathscr{A}} \|f\|_{W^{r,p}}\,, \tag{315}$$

$$\|\mathcal{T}_k f - f\|_{W^{r,p}} \longrightarrow 0. \tag{316}$$

*The convergence is uniform for $f$ in a compact subset of the retained intersection of $W^{r,p}$ spaces. If $F \in C^\infty([0,T]; H^\infty_\sigma)$, then*

$$\partial_t^a(\mathcal{T}_k F) = \mathcal{T}_k(\partial_t^a F) \qquad (a \in \mathbb{N}_0), \tag{317}$$

*and the convergence in* (316) *is uniform in $t$ for every fixed temporal derivative.*

*Proof.* Choose $\chi \in C^\infty_c(B_2)$ with $\chi = 1$ on $B_1$, and put $\chi_R(x) = \chi(x/R)$ and $A_R = B_{2R} \setminus \overline{B_R}$. For a smooth solenoidal $f$, set

$$g_R := \nabla\chi_R \cdot f.$$

It is supported in $A_R$, and

$$\int_{A_R} g_R \,\mathrm{d}x = \int_{\mathbb{R}^3} \nabla\cdot(\chi_R f)\,\mathrm{d}x = 0.$$

We now construct the required compact anti-divergence explicitly. Fix $\rho \in C_c^\infty((-3,3))$ with $\int_{\mathbb{R}} \rho = 1$. If $h \in C_c^\infty(B_2)$ and $\int h = 0$, set

$$m_1(y_2, y_3) := \int_{\mathbb{R}} h(s, y_2, y_3)\, \mathrm{d}s, \qquad m_2(y_3) := \int_{\mathbb{R}} m_1(s, y_3)\, \mathrm{d}s$$

and define $\mathcal{G}h = (G_1, G_2, G_3)$ by

$$\begin{aligned} G_1(y) &:= \int_{-\infty}^{y_1} \big(h(s, y_2, y_3) - \rho(s) m_1(y_2, y_3)\big)\, \mathrm{d}s, \\ G_2(y) &:= \rho(y_1) \int_{-\infty}^{y_2} \big(m_1(s, y_3) - \rho(s) m_2(y_3)\big)\, \mathrm{d}s, \\ G_3(y) &:= \rho(y_1)\rho(y_2) \int_{-\infty}^{y_3} m_2(s)\, \mathrm{d}s. \end{aligned}$$

The three integrands have zero total integral in their integration variables; for the last one this is $\int h = 0$. Thus $\mathcal{G}h \in C_c^\infty((-3,3)^3; \mathbb{R}^3)$, and direct differentiation gives

$$\nabla \cdot \mathcal{G}h = h - \rho(y_1) m_1 + \rho(y_1) m_1 - \rho(y_1)\rho(y_2) m_2 + \rho(y_1)\rho(y_2) m_2 = h.$$

Integration over intervals of length at most six, Hölder's inequality, Fubini's theorem, and differentiation under the integral give, for $a \in \mathbb{N}_0$ and $1 < p < \infty$,

$$\|\mathcal{G}h\|_{W^{a,p}((-3,3)^3)} \le C_{a,p,\rho} \|h\|_{W^{a,p}(B_2)}. \tag{318}$$

For $g \in C_c^\infty(B_{2R})$ of mean zero, put

$$(\mathcal{G}_R g)(x) := R\big[\mathcal{G}(g(R\,\cdot))\big](x/R).$$

Then $\nabla \cdot \mathcal{G}_R g = g$, its support lies in $(-3R, 3R)^3$, and scaling (318) gives

$$\begin{aligned} \|\mathcal{G}_R g\|_{L^p} &\le C_p R \|g\|_{L^p}, \\ \left\|\nabla^j \mathcal{G}_R g\right\|_{L^p} &\le C_{j,p} \sum_{b=0}^{j} R^{1-j+b} \left\|\nabla^b g\right\|_{L^p}, \qquad j \ge 1. \end{aligned} \tag{319}$$

Define

$$\mathcal{S}_R f := \chi_R f - \mathcal{G}_R g_R.$$

Then $\mathcal{S}_R f \in C_{c,\sigma}^\infty(\mathbb{R}^3)$, with support in $(-3R, 3R)^3$. Leibniz' rule gives, for $0 \le a \le r$,

$$\|\nabla^a g_R\|_{L^p(A_R)} \le C_a \sum_{b=0}^{a} R^{-1-a+b} \left\|\nabla^b f\right\|_{L^p(A_R)}.$$

Combining this estimate with (319) and the analogous cutoff estimate yields

$$\|\mathcal{S}_R f - f\|_{W^{r,p}} \le C_{r,p} \sum_{b=0}^{r} (1 + R^{b-r}) \left\|\nabla^b f\right\|_{L^p(\mathbb{R}^3 \setminus B_R)}. \tag{320}$$

For $r = 0$, this conclusion follows from (319) and $\|g_R\|_p \le C R^{-1} \|f\|_{L^p(A_R)}$. The right side tends to zero as $R \to \infty$, and these calculations give operator norms bounded independently of $R$.

Choose $R_k \uparrow \infty$ and set $\mathcal{T}_k := \mathcal{S}_{R_k}$. Estimate (320) holds simultaneously for the finite family $\mathscr{A}$, while the preceding operator estimates give (315) simultaneously for that family. Strong convergence plus the uniform operator bound makes convergence uniform on compact subsets: cover a compact set by finitely many norm balls and use the bound on each remainder. Each $\mathcal{T}_k$ is independent of time, so differentiation commutes with it. The ranges of $\partial_t^a F$ on $[0,T]$ are compact in every retained space, which proves the final assertion. □

**Lemma 8.11** (Admissible strong-history test)**.** *Let $u$ be the global solution from Theorem 5.10, and let $v$ be a Leray–Hopf solution on $[0,T]$ at this viscosity. Put $w := v - u$. Then, for every $t \in [0,T]$,*

$$\begin{aligned}\langle v(t), u(t)\rangle = \langle v_0, u_0\rangle + \int_0^t \int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot w \,\mathrm{d}x \,\mathrm{d}s \\ - 2\nu \int_0^t \langle \nabla v, \nabla u\rangle \,\mathrm{d}s.\end{aligned} \tag{321}$$

*Proof.* The energy class and spatial interpolation give

$$v \in L^4(0,T;L^3), \qquad \|v\|_{L^4L^3}^4 \le C\,\|v\|_{L^\infty L^2}^2\,\|\nabla v\|_{L^2L^2}^2\,,$$

so $v \otimes v \in L^2(0,T;L^{3/2})$.

Fix $0 < t < T$. Solenoidal density on $\mathbb{R}^3$, applied uniformly to the compact sets $u([0,T])$ and $u_t([0,T])$ in their finite Sobolev spaces, gives

$$u^{(k)} \in C^\infty([0,T]; C^\infty_{c,\sigma}(\mathbb{R}^3))$$

such that

$$u^{(k)} \longrightarrow u \quad \text{in} \quad W^{1,1}(0,T;L^2) \cap L^2(0,T;H^1) \cap L^2(0,T;W^{1,3}) \cap C([0,T];L^2). \tag{322}$$

Apply Lemma 8.10 with

$$\mathscr{A} = \{(0,2),(1,2),(1,3)\}$$

and set $u^{(k)} = \mathcal{T}_k u$. Its uniform compact-range conclusion for $u$ and $u_t$, together with time commutation (317), gives (322).

For $0 < h < T - t$, choose $\eta_h \in C_c^\infty([0,T))$ with $0 \le \eta_h \le 1$, $\eta_h = 1$ on $[0,t]$, $\eta_h = 0$ on $[t+h,T]$, and $\eta_h' \le 0$. The field $\eta_h u^{(k)}$ is an admissible test in (314). First let $k \to \infty$. The convergence in (322), together with $v \otimes v \in L^2L^{3/2}$, permits passage in every term.

Choose the time cutoffs so that $-\eta_h'$ is an approximate identity from the right at $t$, and let $h \downarrow 0$. Weak continuity of $v$, strong continuity of $u$, and the Lebesgue differentiation theorem give

$$\begin{aligned}\langle v(t), u(t)\rangle = \langle v_0, u_0\rangle &+ \int_0^t \langle v, u_t\rangle \, \mathrm{d}s + \int_0^t \int_{\mathbb{R}^3} (v \otimes v) : \nabla u \, \mathrm{d}x \, \mathrm{d}s \\ &- \nu \int_0^t \langle \nabla v, \nabla u\rangle \, \mathrm{d}s.\end{aligned}$$

This identity first holds for almost every $t$. Its right-hand side is continuous in $t$, while $t \mapsto \langle v(t), u(t)\rangle$ is continuous by weak continuity of $v$ and norm continuity of $u$. Hence it holds for every $0 \le t \le T$, including $t = T$ by passage from below. Using

$$u_t = -(u \cdot \nabla)u - \nabla p + \nu \Delta u$$

is admissible because $v$ is divergence free. It gives

$$\int_0^t \langle v, u_t\rangle \, \mathrm{d}s = -\int_0^t \int_{\mathbb{R}^3} v \cdot (u \cdot \nabla)u \, \mathrm{d}x \, \mathrm{d}s - \nu \int_0^t \langle \nabla v, \nabla u\rangle \, \mathrm{d}s.$$

Finally,

$$\int_{\mathbb{R}^3} [(v \otimes v) : \nabla u - v \cdot (u \cdot \nabla)u] \, \mathrm{d}x = \int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot w \, \mathrm{d}x,$$

because

$$\int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot u \, \mathrm{d}x = \frac{1}{2} \int_{\mathbb{R}^3} w \cdot \nabla |u|^2 \, \mathrm{d}x = 0.$$

This proves (321). □

**Theorem 8.12** (Quantitative Leray–Hopf stability)**.** *Let $\nu > 0$ and $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and let $u$ be the global strong history from Theorem 5.10. Fix $T > 0$, and let $v$ be any Leray–Hopf solution on $[0, T]$ at this viscosity $\nu$, with datum $v_0 \in H^0_\sigma(\mathbb{R}^3)$. Then, for $0 \le t \le T$,*

$$\begin{aligned}&\frac{1}{2} \|v(t) - u(t)\|_2^2 + \nu \int_0^t \|\nabla(v - u)(s)\|_2^2 \, \mathrm{d}s \\ &\quad \le \frac{1}{2} \|v_0 - u_0\|_2^2 \exp\left( 2 \int_0^t \|\nabla u(s)\|_\infty \, \mathrm{d}s \right) \qquad (323) \\ &\quad \le \frac{1}{2} \|v_0 - u_0\|_2^2 \exp\left( 2 S_{3,1,\infty} T^{1/2} \mathfrak{P}_{0,2}[u_0, \nu; T]^{1/2} \right). \qquad (324)\end{aligned}$$

*The lower spatial rectangle also gives the viscosity-weighted estimate*

$$\begin{aligned}&\frac{1}{2} \|v(t) - u(t)\|_2^2 + \frac{\nu}{2} \int_0^t \|\nabla(v - u)(s)\|_2^2 \, \mathrm{d}s \\ &\quad \le \frac{1}{2} \|v_0 - u_0\|_2^2 \exp\left( \frac{S_{2,0,\infty}^2}{\nu} \mathfrak{X}_{0,2}[u_0, \nu; T]^2 \right). \qquad (325)\end{aligned}$$

*Proof.* Combine (313), the strong energy equality (143), and (321). With $w = v - u$, this gives

$$\frac{1}{2} \|w(t)\|_2^2 + \nu \int_0^t \|\nabla w(s)\|_2^2 \, \mathrm{d}s \le \frac{1}{2} \|w(0)\|_2^2 - \int_0^t \int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot w \, \mathrm{d}x \, \mathrm{d}s. \qquad (326)$$

The integral term in (326) has absolute value at most

$$\|\nabla u(s)\|_\infty \|w(s)\|_2^2 .$$

Gronwall's inequality bounds $\|w(t)\|_2^2$. Substitution of that bound back into (326) proves (323). Proposition 7.16 gives

$$\int_0^T \|\nabla u(t)\|_\infty \, \mathrm{d}t \le S_{3,1,\infty} T^{1/2} \mathfrak{P}_{0,2}[u_0, \nu; T]^{1/2},$$

which proves (324).

For the lower-rectangle estimate, integration by parts and $\nabla \cdot w = 0$ give

$$\left| \int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot w \, \mathrm{d}x \right| = \left| \int_{\mathbb{R}^3} (w \cdot \nabla w) \cdot u \, \mathrm{d}x \right| \le \|u\|_\infty \|w\|_2 \|\nabla w\|_2 .$$

Young's inequality in (326) gives

$$\frac{1}{2}\|w(t)\|_2^2 + \frac{\nu}{2}\int_0^t \|\nabla w(s)\|_2^2 \, \mathrm{d}s \le \frac{1}{2}\|w(0)\|_2^2 + \frac{1}{2\nu}\int_0^t \|u(s)\|_\infty^2 \|w(s)\|_2^2 \, \mathrm{d}s.$$

Set

$$A(t) := \frac{1}{\nu}\int_0^t \|u(s)\|_\infty^2 \, \mathrm{d}s.$$

Gronwall's inequality gives $\|w(t)\|_2^2 \le \|w(0)\|_2^2 \, e^{A(t)}$. Substitution in the preceding inequality gives

$$\frac{1}{2}\|w(t)\|_2^2 + \frac{\nu}{2}\int_0^t \|\nabla w(s)\|_2^2 \, \mathrm{d}s \le \frac{1}{2}\|w(0)\|_2^2 \, e^{A(t)}.$$

Finally, (797) yields

$$A(t) \le \frac{S_{2,0,\infty}^2}{\nu} \mathfrak{X}_{0,2}[u_0, \nu; T]^2,$$

which proves (325). □

**Corollary 8.13** (Explicit weak-class stability for small critical data)**.** *Under the hypotheses of Theorem 8.3, let $v$ be a global Leray–Hopf solution at this viscosity $\nu$, with datum $v_0 \in H^0_\sigma(\mathbb{R}^3)$. Then*

$$\begin{aligned} &\frac{1}{2}\|v(t) - u(t)\|_2^2 + \nu \int_0^t \|\nabla(v-u)(s)\|_2^2 \, \mathrm{d}s \\ &\qquad \le \frac{1}{2}\|v_0 - a\|_2^2 \exp(2 S_{3,1,\infty} t U_3^{\mathrm{sm}}) \qquad (t \ge 0). \end{aligned} \tag{327}$$

*Proof.* Equation (283) gives $\sup_{s \ge 0} \|u(s)\|_{H^3} \le U_3^{\mathrm{sm}}$. Hence

$$\int_0^t \|\nabla u(s)\|_\infty \, \mathrm{d}s \le S_{3,1,\infty} t U_3^{\mathrm{sm}}.$$

The solution constructed in Theorem 8.3 and the global history from Theorem 5.10, with datum $a$, both restrict to members of $\mathscr{M}^3_\nu(0, T; a)$ on every finite interval. The at-most-one assertion in Theorem 2.11 identifies the two pressure-normalized histories. Apply (323). □

**Corollary 8.14** (Uniqueness and energy equality in the Leray–Hopf class)**.** *Let $\nu > 0$ and $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$. If $(v, q)$ is any global three-dimensional Leray–Hopf solution at viscosity $\nu$, with datum $u_0$, then $v$ agrees on every finite interval with the velocity of the global solution from Theorem 5.10. Thus $v$ has the smooth representative*

$$u \in C^\infty([0,\infty) \times \mathbb{R}^3; \mathbb{R}^3),$$

*and it satisfies*

$$\frac{1}{2}\|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \, \mathrm{d}s = \frac{1}{2}\|u_0\|_2^2 \qquad (t \geq 0).$$

*Its distributional pressure has the form*

$$q = p + c(t) \quad \text{in } \mathcal{D}'((0,\infty) \times \mathbb{R}^3), \tag{328}$$

*where $p$ is the pressure in* (142) *and $c$ is a scalar distribution in time. The canonical Riesz representative $R_i R_j(u_i u_j)$ equals $p$. If $q$ is represented by a locally integrable function with $q(t,\cdot) \in C_0(\mathbb{R}^3)$ for almost every $t$, then $q = p$ almost everywhere.*

*Proof.* Set $v_0 = u_0$ in Theorem 8.12. The right-hand side of (324) is zero, so $v = u$ in $C_w([0,T]; L^2) \cap L^2(0,T; \dot{H}^1)$. Since $T$ is arbitrary, the identity holds globally. Smoothness and energy equality follow from (139) and (143). If $q$ is a distributional pressure paired with this velocity and $p$ is the pressure in (142), subtraction of the two momentum equations gives

$$\nabla(q - p) = 0 \quad \text{in } \mathcal{D}'((0,\infty) \times \mathbb{R}^3).$$

A distribution on a product domain whose spatial gradient vanishes is the pullback of a scalar distribution in time: testing first against spatial test functions of integral zero and then against one fixed test function of integral one gives (328). The Riesz formula for $p$ proves the canonical-representative assertion. In the function-valued case, $q(t,x) - p(t,x) = c(t)$ for almost every $(t,x)$; both spatial slices tend to zero as $|x| \to \infty$, so $c(t) = 0$ for almost every $t$. □

# 9 Conclusion

The datum and viscosity determine one pressure-normalized velocity history and one compatible family of finite pressure-complete rectangles. Their mixed-index exhaustion gives every temporal–spatial Sobolev row. The completed critical-height identities are evaluated on those same rectangles; their height exhaustion gives every spatial Sobolev row, and the equation reconstructs those temporal and pressure rows. The scalar $\mathfrak{R}_{N,M}$ measures $\mathcal{R}_{N,M}$, Proposition 4.7 identifies its terminal value, and Theorem 5.2 proves that every larger rectangle restricts to identical rows in each smaller one and that the family exhausts all finite mixed indices.

The adjacent rows in each finite rectangle give continuous Sobolev traces through the proposed terminal time. Their compatible endpoint values determine $u_* \in H^\infty_\sigma$, and the pressure formula and velocity recurrence recover its complete jet. Local well-posedness restarts from that state, while uniqueness on the overlap joins the continuation to the incoming history. A finite maximal time would admit an extension, so the maximal interval is $[0,\infty)$.

These same rectangles also quantify the critical-height rows, frequency tails, Lipschitz action, and relative-energy stability, with constants depending on the datum, viscosity, and horizon. In the small-critical regime, the scale-invariant norm closes the critical stock/action estimate, and the

higher rectangles admit explicit formulas using the corresponding finite collection of higher datum norms.

The resulting global solution is smooth, satisfies the exact energy identity, and carries the decay-normalized pressure. Relative energy then identifies every Leray–Hopf solution with that datum with this smooth velocity, and the pressure is unique up to the unavoidable time-dependent gauge unless this decay normalization is imposed. The endpoint contradiction and these consequences all belong to the one fixed-viscosity evolution constructed from $u_0$.

# Part III
# The Forced Equations

## 10 Introduction

Fix one viscosity $\nu > 0$, one divergence-free datum $u_0 \in H^\infty(\mathbb{R}^3;\mathbb{R}^3)$, and one prescribed source

$$f \in C^\infty([0,\infty);\mathcal{S}(\mathbb{R}^3;\mathbb{R}^3)).$$

Forced local theory produces a unique pressure-normalized classical history $(u,p)$ on a maximal interval $[0,T_*)$. Part III follows that one history. Its question is whether a finite value of $T_*$ can terminate the solution generated by $(u_0,\nu,f)$.

**Theorem.** *For every $\nu > 0$, every $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and every $f \in C^\infty([0,\infty);\mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$, there is a unique pressure-normalized pair*

$$(u,p) \in C^\infty([0,\infty)\times\mathbb{R}^3;\mathbb{R}^3\times\mathbb{R})$$

*satisfying*

$$\partial_t u + (u\cdot\nabla)u + \nabla p = \nu\Delta u + f, \qquad \nabla\cdot u = 0, \qquad u(0) = u_0, \qquad \lim_{|x|\to\infty} p(t,x) = 0.$$

*It obeys the forced energy identity*

$$\frac{1}{2}\|u(t)\|_{L^2}^2 + \nu\int_0^t \|\nabla u(s)\|_{L^2}^2\,\mathrm{d}s = \frac{1}{2}\|u_0\|_{L^2}^2 + \int_0^t \langle f(s),u(s)\rangle_{L^2}\,\mathrm{d}s \qquad (t\ge 0).$$

**Corollary 10.1** (Weak–strong uniqueness and pressure normalization)**.** *For the solution of the preceding theorem, every forced Leray–Hopf pair $(v,q)$ with that datum, viscosity, and prescribed source has $v = u$. Distributionally, $q = p + c(t)$ for a scalar distribution $c$; if $q(t,\cdot) \in C_0(\mathbb{R}^3)$ almost everywhere, then $q = p$ almost everywhere.*

Leray's weak solutions establish global finite-energy existence [10]. The regularity results of Serrin [13], Caffarelli–Kohn–Nirenberg [3], and Escauriaza–Seregin–Šverák [4] identify increasingly sharp ways in which a singularity would have to evade classical control. Fujita–Kato and Kato provide the local strong theory used both at the initial time and at the terminal restart [6, 8]. The proof below works directly with the complete derivative structure of the prescribed forced history.

Forcing changes every level of that structure. Its solenoidal part $g = \mathbb{P}f$ enters the projected velocity equation. Its gradient part is absorbed by the normalized pressure,

$$-\Delta p = \partial_i\partial_j(u_iu_j) - \nabla\cdot f, \qquad p = R_iR_j(u_iu_j) - (-\Delta)^{-1}\nabla\cdot f.$$

Every temporal row contains the corresponding derivative $F_n = \partial_t^n f$; every energy row contains force work; and a restart at $T_*$ retains that absolute-time source. These are parts of a single forced velocity–pressure–source history.

The proof uses two constructions from the pressure-complete mixed jet. For every finite pair $(N,M)$, the datum construction gives the corresponding whole-terminal velocity–pressure–force record, and deleting its outer coordinates gives every smaller record. Ranging over all finite mixed indices exhausts every temporal and spatial order and hence gives the full space–time

Sobolev intersection. Independently, the critical-height and viscous identities are derived from this differentiated equation. Their completed coordinates recover every spatial Sobolev row, after which the forced recurrence reconstructs the temporal and pressure rows. Pairing, squaring, and integrating any selected rectangle estimates every face of that forced equation together. The resulting all-orders membership gives the adjacent rows one common smooth endpoint. The pressure formula and forced recurrences determine the complete endpoint jet, and local theory restarts the equation from $u_* = u(T_*)$ with that source. Complete endpoint-jet matching glues the restart to the incoming history, contradicting a finite maximal time and giving $T_* = \infty$.

Section 11 derives this architecture in one view. The analytic framework and the mixed-jet construction follow, then the compatible finite datum records and their mixed-index exhaustion. The critical-height and viscous identities are derived independently and then evaluated on those same finite rectangle coordinates before both routes enter the common endpoint continuation. The remaining sections record matrix formulations and quantitative consequences of these finite rectangles.

# 11 From a Prescribed Force to Terminal Continuation

Fix one divergence-free datum $u_0$, one viscosity $\nu > 0$, and one prescribed smooth force $f$. Let $(u, p)$ be the corresponding maximal classical solution on $[0, T_*)$. The proof never changes this history. Every derivative, finite rectangle, endpoint value, and restarted solution below belongs to the trajectory generated by the original triple $(u_0, \nu, f)$.

The force enters the proof in two linked forms. Its solenoidal part $g = \mathbb{P}f$ acts in the projected velocity equation, while its gradient part is fixed by the normalized pressure:

$$\partial_t u + \mathbb{P}\nabla\cdot(u\otimes u) - \nu\Delta u = g, \qquad p = R_i R_j(u_i u_j) - (-\Delta)^{-1}\nabla\cdot f.$$

The source enters the velocity and pressure at the initial construction and remains present in every differentiated row.

The Navier–Stokes scaling acts on the complete forced equation. If $(u, p, f)$ is a solution, then

$$u_\lambda(t,x) = \lambda u(\lambda^2 t, \lambda x), \qquad p_\lambda(t,x) = \lambda^2 p(\lambda^2 t, \lambda x), \qquad f_\lambda(t,x) = \lambda^3 f(\lambda^2 t, \lambda x)$$

solves the equation with transformed datum $u_{0,\lambda}(x) = \lambda u_0(\lambda x)$. For every real $s$,

$$\|\Lambda^s u_\lambda(t)\|_2^2 = \lambda^{2s-1}\|\Lambda^s u(\lambda^2 t)\|_2^2.$$

The invariant spatial height is $s = \frac{1}{2}$. This identifies the critical row of this forced differential structure; it does not replace the prescribed history by its rescaling.

Write

$$V_n = \partial_t^n u, \qquad Q_n = \partial_t^n p, \qquad F_n = \partial_t^n f,$$

and let

$$B_n = \sum_{a=0}^{n}\binom{n}{a}(V_a\cdot\nabla)V_{n-a}.$$

Repeated differentiation of the original forced equation gives the complete temporal row

$$V_{n+1} + B_n + \nabla Q_n = \nu\Delta V_n + F_n, \tag{329}$$

$$-\Delta Q_n = \partial_i\partial_j\sum_{a=0}^{n}\binom{n}{a}(V_a)_i(V_{n-a})_j - \nabla\cdot F_n. \tag{330}$$

Spatial differentiation distributes through both velocity factors and through the prescribed source. The fields

$$\partial_x^\alpha V_n, \qquad \partial_x^\alpha Q_n, \qquad \partial_x^\alpha F_n$$

form the pressure-complete forced mixed jet. They are coordinates of one solution, rather than a collection of independently chosen functions.

At a proposed finite maximal time $T_*$, the finite object through temporal depth $N$ and spatial height $M$ is the record

$$\mathcal{R}_{N,M}[u_0,\nu,f;T_*] := \big((V_n)_{n=0}^N,(V_{n+1})_{n=0}^N;(Q_n)_{n=0}^N;(F_n)_{n=0}^N\big).$$

Its scalar velocity norm on a strict horizon is

$$\mathfrak{R}_{N,M}(T) := \sum_{n=0}^{N}\left(\|V_n\|^2_{L^2(0,T;H^{M+1})}+\|V_{n+1}\|^2_{L^2(0,T;H^{M-1})}\right).$$

The datum construction places the record on $(0,T_*)$ and gives

$$\sup_{0<T<T_*}\mathfrak{R}_{N,M}(T)<\infty \qquad (N,M<\infty). \tag{A}$$

Deleting outer temporal or spatial coordinates recovers the corresponding smaller record. As $N$ and $M$ range over all finite values, these blocks exhaust every mixed derivative of the one forced history. Restriction in time gives the strict-horizon blocks, while monotone exhaustion identifies each finite norm on $(0,T_*)$.

Letting the temporal and spatial indices range over all finite values gives

$$u\in\bigcap_{r,m\ge0}H^r(I;H^m_\sigma(\mathbb{R}^3)), \qquad I=(0,T_*). \tag{A$'$}$$

Independently, the completed critical heights and the viscous diagonal relation are derived from this mixed jet. They retain transport, pressure, viscosity, and force work while exposing every finite spatial height. Evaluating those identities on the datum-generated blocks and intersecting the heights gives the complete spatial Sobolev column; the forced recurrence then recovers this intersection (A$'$). There is no second family of fields: both constructions use this datum-generated block $\mathcal{R}_{N,M}[u_0,\nu,f;T_*]$ at every finite pair $(N,M)$. The independently derived critical-height relationship rectangle is the finite system of coupled identities evaluated on those same coordinates.

The viscous relation is visible already in one row. Project (329), apply $\Lambda^s$, and use $\Delta=-\Lambda^2$:

$$\Lambda^sV_{n+1}+\nu\Lambda^{s+2}V_n=\Lambda^s\mathbb{P}(F_n-B_n). \tag{B}$$

Squaring this equality does not separate the nonlinearity from the fluid. The cross term is the derivative of the adjacent spatial norm, so

$$\begin{aligned}\|\Lambda^sV_{n+1}\|_2^2+\nu^2\|\Lambda^{s+2}V_n\|_2^2+\nu\frac{d}{dt}\|\Lambda^{s+1}V_n\|_2^2\\ =\|\Lambda^s\mathbb{P}(F_n-B_n)\|_2^2.\end{aligned} \tag{C}$$

The complementary projection gives the pressure allocation

$$\nabla Q_n=(I-\mathbb{P})(F_n-B_n),$$

and orthogonality recombines the complete source:

$$\|\Lambda^s\mathbb{P}(F_n - B_n)\|_2^2 + \|\Lambda^s\nabla Q_n\|_2^2 = \|\Lambda^s(F_n - B_n)\|_2^2. \tag{D}$$

Thus the rectangle keeps viscosity, transport, pressure, and forcing in one estimate.

The critical-height tower is another reading of this viscous tether. For

$$E_{r,s} = \frac{1}{2}\|\Lambda^s V_r\|_2^2,$$

the complete row is

$$\partial_t E_{r,s} + 2\nu E_{r,s+1} = \mathcal{P}_{r,s} + \mathcal{W}_{r,s}, \qquad \mathcal{W}_{r,s} = \mathrm{Re}\langle\Lambda^s V_r, \Lambda^s F_r\rangle. \tag{E}$$

At $r = 0$ and $s = \frac{1}{2}$, differentiating (E) repeatedly exposes $E_{3/2}, E_{5/2}, \ldots$, while the differentiated transport, pressure, and force work remains in that completed row. The tower then describes the geometry among mixed-jet coordinates; it does not replace the velocity field by a scalar height. The first two spatial heights are, explicitly,

$$H' + \nu\left\|\Lambda^{3/2}u\right\|_2^2 = \mathcal{P}_{1/2} + \mathcal{W}_{1/2},$$
$$E'_{3/2} + \nu\left\|\Lambda^{5/2}u\right\|_2^2 = \mathcal{P}_{3/2} + \mathcal{W}_{3/2},$$

and hence

$$H'' = 4\nu^2 E_{5/2} - 2\nu(\mathcal{P}_{3/2} + \mathcal{W}_{3/2}) + \partial_t(\mathcal{P}_{1/2} + \mathcal{W}_{1/2}).$$

These are the actual $s = \frac{1}{2}$ and $s = \frac{3}{2}$ rows. Their transport, pressure, viscosity, and force coordinates remain joined.

For finite $N, M$, use the corresponding datum-generated block $\mathcal{R}_{N,M}[u_0, \nu, f; T_*]$. At the spatial orders required by (B)–(E), its restriction to a strict horizon satisfies

$$\mathfrak{R}_{N,M}(T) \le \mathfrak{R}^*_{N,M} := \sup_{0<S<T_*} \mathfrak{R}_{N,M}(S) < \infty. \tag{F}$$

Restriction compatibility identifies every common row in two such rectangles, while monotone exhaustion gives

$$\lim_{T\uparrow T_*} \mathfrak{R}_{N,M}(T) = \mathfrak{R}^*_{N,M} < \infty. \tag{G}$$

The bound may depend on the selected rectangle. No estimate uniform in the infinitely many indices is used. The bounds in (G) are scalar norms of the finite blocks whose mixed-index exhaustion gives (A′). The independently derived identities in (E), evaluated on those same coordinates, give the completed heights. Exhausting their spatial heights and applying the Fourier Sobolev split gives the spatial column, from which (329)–(330) recover (A′) again. The positive-sign square (C), together with (D), relates the adjacent temporal, viscous, and pressure faces. Sobolev multiplication is applied within that same finite rectangle to its binomial transport face $B_n$; it does not replace the nonlinear term or introduce a second premise.

The mixed-index exhaustion and the critical-height relationships use these finite blocks. The former records every mixed derivative; the latter recover every spatial row and return through the forced recurrence. Pairing, squaring, and integrating (B)–(E) inside any selected rectangle gives its complete-equation estimates, with the nonlinearity retained as one face of that forced rectangle.

The datum rectangles now carry the complete equation to the terminal step. Their zeroth temporal faces give, for every finite $m$,

$$u \in L^2(0,T_*;H^{m+2}).$$

Sobolev multiplication and the prescribed force then give

$$\begin{aligned}\|u_t\|_{L^1(0,T_*;H^m)} &\le \nu\sqrt{T_*}\,\|u\|_{L^2(0,T_*;H^{m+2})}\\ &\quad + C_m\|u\|^2_{L^2(0,T_*;H^{m+2})} + \|g\|_{L^1(0,T_*;H^m)} < \infty.\end{aligned}$$

Hence

$$u \in W^{1,1}(0,T_*;H^m) \hookrightarrow C([0,T_*];H^m) \qquad (m \ge 0).$$

All these traces come from one velocity and determine one terminal state

$$u_* = u(T_*) \in \bigcap_{m\ge 0} H^m_\sigma(\mathbb{R}^3).$$

Passing (329)–(330) to $T_*$ determines every temporal and pressure coordinate from $u_*$ and the values $F_n(T_*)$. Local theory from $u_*$, driven by that prescribed force $f(t)$, continues that history past $T_*$. This contradicts maximality, so $T_* = \infty$.

# 12 Analytic framework

The argument has two analytic joints. Products and Fourier multipliers must keep every differentiated velocity–pressure row in a common Sobolev scale; at the terminal time, an $H^3$ local theory must restart that same pressure-normalized history. We record the estimates needed at those two joints, with the constants later used in the quantitative bounds.

**Definition 12.1.** For $f \in \mathcal{S}(\mathbb{R}^3)$, set

$$\widehat{f}(\xi) := (2\pi)^{-3/2}\int_{\mathbb{R}^3} e^{-ix\cdot\xi} f(x)\,\mathrm{d}x, \qquad \langle\xi\rangle := (1+|\xi|^2)^{1/2}.$$

This integral defines $\widehat{f}$ for $f \in L^1(\mathbb{R}^3)$. The Fourier transform on $\mathcal{S}'(\mathbb{R}^3)$ is its extension by duality. For $s \in \mathbb{R}$, define

$$H^s(\mathbb{R}^3) := \left\{ f \in \mathcal{S}'(\mathbb{R}^3) : \langle\xi\rangle^s \widehat{f} \in L^2(\mathbb{R}^3)\right\},$$

and define the finite homogeneous-norm domain by

$$\dot{H}^s(\mathbb{R}^3) := \left\{ f \in \mathcal{S}'(\mathbb{R}^3) : \begin{array}{l}\widehat{f} \text{ is represented by a measurable function on } \mathbb{R}^3,\\ |\xi|^s\,\widehat{f} \in L^2(\mathbb{R}^3)\end{array} \right\}.$$

On these domains set

$$\|f\|^2_{H^s} := \int_{\mathbb{R}^3}\langle\xi\rangle^{2s}\left|\widehat{f}(\xi)\right|^2\,\mathrm{d}\xi, \qquad \|f\|^2_{\dot{H}^s} := \int_{\mathbb{R}^3}|\xi|^{2s}\left|\widehat{f}(\xi)\right|^2\,\mathrm{d}\xi.$$

Set $A := (1-\Delta)^{1/2}$, $\Lambda := (-\Delta)^{1/2}$,

$$H^s_\sigma := \{v \in H^s(\mathbb{R}^3;\mathbb{R}^3) : \nabla\cdot v = 0\}, \qquad H^\infty := \bigcap_{m\in\mathbb{N}_0} H^m, \qquad H^\infty_\sigma := \bigcap_{m\in\mathbb{N}_0} H^m_\sigma.$$

$$\mathcal{S}_\sigma(\mathbb{R}^3) := \{v \in \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3) : \nabla \cdot v = 0\}.$$

For a vector or tensor $F = (F_I)_I$, all Sobolev norms are componentwise:

$$\|F\|_{H^s}^2 := \sum_I \|F_I\|_{H^s}^2 , \qquad \|F\|_{\dot{H}^s}^2 := \sum_I \|F_I\|_{\dot{H}^s}^2 .$$

Repeated Latin indices are summed from 1 to 3.

**Lemma 12.2** (Sobolev nesting and embedding)**.** *If $s_2 \geq s_1$, then*

$$\|f\|_{H^{s_1}} \leq \|f\|_{H^{s_2}} .$$

*If $k \in \mathbb{N}_0$ and $s > k + 3/2$, then*

$$\max_{|\alpha|\leq k} \|\partial^\alpha f\|_{L^\infty} \leq C_{s,k}^{\mathrm{emb}} \|f\|_{H^s} ,$$

*where*

$$C_{s,k}^{\mathrm{emb}} := (2\pi)^{-3/2} \max_{0\leq j\leq k} \left( \int_{\mathbb{R}^3} |\xi|^{2j} \langle \xi \rangle^{-2s} \, \mathrm{d}\xi \right)^{1/2} < \infty.$$

*In particular,*

$$H^\infty(\mathbb{R}^3) \hookrightarrow C_b^\infty(\mathbb{R}^3).$$

*Proof.* The first assertion follows from $\langle \xi \rangle^{2s_1} \leq \langle \xi \rangle^{2s_2}$. For $|\alpha| \leq k$, Fourier inversion and Cauchy–Schwarz give

$$|\partial^\alpha f(x)| \leq (2\pi)^{-3/2} \left( \int_{\mathbb{R}^3} |\xi|^{2|\alpha|} \langle \xi \rangle^{-2s} \, \mathrm{d}\xi \right)^{1/2} \|f\|_{H^s} .$$

The integral is finite exactly when $s > |\alpha| + 3/2$. Approximation by Schwartz functions and dominated convergence give continuous representatives. For each $k$, choose $s = k + 2$. □

**Lemma 12.3** (Explicit three-dimensional homogeneous Sobolev constants)**.** *For $3 \leq p \leq 6$, set*

$$s(p) := 3 \left( \frac{1}{2} - \frac{1}{p} \right)$$

*and*

$$\mathsf{S}(p) := 2^{-s(p)} \pi^{-s(p)/2} \left( \frac{\Gamma((3 - 2s(p))/2)}{\Gamma((3 + 2s(p))/2)} \right)^{1/2} \left( \frac{\Gamma(3)}{\Gamma(3/2)} \right)^{s(p)/3} . \tag{331}$$

*Then every scalar, vector, or finite tensor field $F$ in $\dot{H}^{s(p)}(\mathbb{R}^3)$ satisfies*

$$\|F\|_{L^p(\mathbb{R}^3)} \leq \mathsf{S}(p) \, \|F\|_{\dot{H}^{s(p)}(\mathbb{R}^3)} . \tag{332}$$

*In particular,*

$$\mathsf{S}(3) = 2^{-1/6} \pi^{-1/3}, \qquad \mathsf{S}(6) = \frac{2^{2/3}}{\sqrt{3}\, \pi^{2/3}}, \qquad \mathsf{S}(6)\mathsf{S}(3) = \frac{1}{\pi} \sqrt{\frac{2}{3}}.$$

*Proof.* For scalar fields, (332) is the sharp three-dimensional homogeneous Sobolev inequality obtained from the sharp Hardy–Littlewood–Sobolev inequality [11]. After duality, insertion of the Riesz-potential kernel constant, and conversion to the unitary Fourier transform fixed above, the squared sharp Sobolev constant for the multiplier $\widehat{\Lambda^s f}(\xi) = |\xi|^s \widehat{f}(\xi)$ is

$$2^{-2s}\pi^{-s}\frac{\Gamma((3-2s)/2)}{\Gamma((3+2s)/2)}\left(\frac{\Gamma(3)}{\Gamma(3/2)}\right)^{2s/3}. \tag{333}$$

Taking the square root in (333) and putting $p = 6/(3-2s)$ gives (331)– (332). For a finite component array $F = (F_I)_I$, Minkowski's inequality in $L^{p/2}$ gives

$$\|F\|_p^2 = \left\| \sum_I |F_I|^2 \right\|_{p/2} \le \sum_I \|F_I\|_p^2 \le \mathsf{S}(p)^2 \sum_I \|F_I\|_{\dot{H}^{s(p)}}^2 .$$

The two endpoint formulas follow by evaluating the gamma factors at $s = 1/2$ and $s = 1$. □

**Lemma 12.4** (Sobolev multiplication)**.** *Set*

$$\mathsf{H}_4 := (2\pi)^{-3/4}\left(\int_{\mathbb{R}^3}\langle\xi\rangle^{-4}\,\mathrm{d}\xi\right)^{1/4}, \qquad C_0^{\mathrm{alg}} := \mathsf{H}_4^2.$$

*For $m \geq 1$, define*

$$\mathsf{E}_m := \max_{0\leq j\leq \lfloor m/2\rfloor} C^{\mathrm{emb}}_{j+2,j}, \qquad \mathsf{C}_m^{\mathrm{mix}} := 13^{m/2}\mathsf{E}_m,$$

$$\mathsf{C}_m^{\mathrm{F}} := 2^m(2\pi)^{-3/2}\left(\int_{\mathbb{R}^3}\langle\xi\rangle^{-2m}\,\mathrm{d}\xi\right)^{1/2} \qquad (m\geq 2),$$

*and set*

$$C_1^{\mathrm{alg}} := \mathsf{C}_1^{\mathrm{mix}}, \qquad C_m^{\mathrm{alg}} := \max\{\mathsf{C}_m^{\mathrm{mix}}, \mathsf{C}_m^{\mathrm{F}}\} \quad (m\geq 2).$$

*All these constants are finite and explicit. For every $m \in \mathbb{N}_0$,*

$$\|fg\|_{H^m} \leq C_m^{\mathrm{alg}}\,\|f\|_{H^{m+1}}\,\|g\|_{H^{m+1}}.$$

*For $m \geq 2$,*

$$\|fg\|_{H^m} \leq C_m^{\mathrm{alg}}\,\|f\|_{H^m}\,\|g\|_{H^m}.$$

*Proof.* Hausdorff–Young and Hölder give

$$\|f\|_4 \leq (2\pi)^{-3/4}\left\|\widehat{f}\right\|_{4/3} \leq \mathsf{H}_4\,\|f\|_{H^1}.$$

Thus $\|fg\|_2 \leq C_0^{\mathrm{alg}}\,\|f\|_{H^1}\,\|g\|_{H^1}$.

Let $m \geq 1$. The exact integer-order identity

$$\|h\|_{H^m}^2 = \sum_{|\alpha|\leq m}\frac{m!}{(m-|\alpha|)!\,\alpha!}\,\|\partial^\alpha h\|_2^2$$

follows by expanding $(1+|\xi|^2)^m$. In each Leibniz term for $\partial^\alpha(fg)$, place the factor carrying at most $\lfloor|\alpha|/2\rfloor$ derivatives in $L^\infty$. The definition of $\mathsf{E}_m$, Sobolev nesting, and $\sum_{\beta\leq\alpha}\binom{\alpha}{\beta} = 2^{|\alpha|}$ give

$$\|\partial^\alpha(fg)\|_2 \leq \mathsf{E}_m 2^{|\alpha|}\,\|f\|_{H^{m+1}}\,\|g\|_{H^{m+1}}.$$

The multinomial identity

$$\sum_{|\alpha|\leq m}\frac{m!}{(m-|\alpha|)!\,\alpha!}4^{|\alpha|} = 13^m$$

proves the first estimate with $\mathsf{C}_m^{\mathrm{mix}}$.

For $m \geq 2$, the inequalities

$$\langle\eta+\zeta\rangle^m \leq 2^{m-1}\big(\langle\eta\rangle^m + \langle\zeta\rangle^m\big), \qquad \left\|\widehat{f}\right\|_1 \leq \left(\int_{\mathbb{R}^3}\langle\xi\rangle^{-2m}\,\mathrm{d}\xi\right)^{1/2}\|f\|_{H^m},$$

combined with $\widehat{fg} = (2\pi)^{-3/2}\widehat{f}*\widehat{g}$ and Young's convolution inequality give

$$\|fg\|_{H^m} \leq \mathsf{C}_m^{\mathrm{F}}\,\|f\|_{H^m}\,\|g\|_{H^m}.$$

The chosen maximum proves both asserted estimates with this constant. □

**Lemma 12.5** (Riesz and Leray multipliers)**.** *Let*

$$r_j(\xi) := \begin{cases} i\xi_j/\left|\xi\right|, & \xi \neq 0, \\ 0, & \xi = 0, \end{cases} \qquad P(\xi) := \begin{cases} I - \dfrac{\xi \otimes \xi}{\left|\xi\right|^2}, & \xi \neq 0, \\ I, & \xi = 0, \end{cases}$$

*and*

$$\widehat{R_j f}(\xi) = r_j(\xi)\widehat{f}(\xi), \qquad \widehat{\mathbb{P}f}(\xi) = P(\xi)\widehat{f}(\xi).$$

*For all* $s \in \mathbb{R}$*,*

$$\left\|R_j f\right\|_{H^s} \leq \left\|f\right\|_{H^s}, \qquad \left\|\mathbb{P}f\right\|_{H^s} \leq \left\|f\right\|_{H^s}.$$

*If* $m \in \mathbb{N}_0$ *and* $h \in L^1(\mathbb{R}^3) \cap H^m(\mathbb{R}^3;\mathbb{R}^3)$*, then*

$$\left\|(-\Delta)^{-1}\nabla \cdot h\right\|_{H^m} \leq C_m^{\mathrm{fp}}\big(\left\|h\right\|_{L^1} + \left\|h\right\|_{H^m}\big). \tag{334}$$

*If* $m \in \mathbb{N}_0$ *and* $v \in H^{m+1}_\sigma$*, then*

$$q := \sum_{i,j=1}^{3} R_i R_j(v_i v_j) \in H^m, \qquad -\Delta q = \sum_{i,j=1}^{3} \partial_i \partial_j (v_i v_j),$$

*and*

$$\mathbb{P}\nabla \cdot (v \otimes v) = (v \cdot \nabla)v + \nabla q.$$

*Here*

$$(v \otimes v)_{ij} := v_i v_j, \qquad (\nabla \cdot (v \otimes v))_i := \sum_{j=1}^{3} \partial_j (v_i v_j), \qquad ((v \cdot \nabla)v)_i := \sum_{j=1}^{3} v_j \partial_j v_i.$$

*Proof.* The multiplier symbols have operator norm at most one. Plancherel proves the first two estimates. For the force potential, split Fourier space at $|\xi| = 1$. On the unit ball, $\left|\widehat{h}(\xi)\right| \leq (2\pi)^{-3/2}\left\|h\right\|_{L^1}$ and $\int_{|\xi|\leq 1}\left|\xi\right|^{-2}\,\mathrm{d}\xi < \infty$ in three dimensions. Outside the unit ball, $\left|\xi\right|^{-1} \leq 1$, so the $H^m$ norm of the multiplier is bounded by the $H^m$ norm of $h$. This proves (334). Lemma 12.4 gives $v_i v_j \in H^m$. The remaining identities follow after multiplication by $\left|\xi\right|^2$ and by the symbol of $\mathbb{P}$. □

**Lemma 12.6** (Mixed Leibniz identities)**.** *For* $N \in \mathbb{N}_0$ *and* $\alpha \in \mathbb{N}_0^3$*,*

$$\partial_t^N \partial_x^\alpha (fg) = \sum_{a=0}^{N} \sum_{\beta \leq \alpha} \binom{N}{a} \binom{\alpha}{\beta} (\partial_t^a \partial_x^\beta f)(\partial_t^{N-a} \partial_x^{\alpha-\beta} g).$$

*Proof.* Apply the one-variable Leibniz identity successively in $t, x_1, x_2, x_3$. □

**Lemma 12.7** (Adjacent Hilbert-scale trace)**.** *Let $0 < T < \infty$, $m \in \mathbb{R}$, and*

$$v \in L^2(0,T;H^{m+1}), \qquad \partial_t v \in L^2(0,T;H^{m-1}).$$

*Then $v$ has a unique representative in $C([0,T];H^m)$, and, for $0 \le r \le t \le T$,*

$$\|v(t)\|_{H^m}^2 - \|v(r)\|_{H^m}^2 = 2\,\mathrm{Re}\int_r^t \left\langle A^{m-1}\partial_t v(s), A^{m+1}v(s)\right\rangle_{L^2}\,\mathrm{d}s, \tag{335}$$

$$\left|\|v(t)\|_{H^m}^2 - \|v(r)\|_{H^m}^2\right| \le 2\,\|\partial_t v\|_{L^2(r,t;H^{m-1})}\,\|v\|_{L^2(r,t;H^{m+1})}\,, \tag{336}$$

$$\sup_{0\le t\le T}\|v(t)\|_{H^m}^2 \le T^{-1}\,\|v\|_{L^2(0,T;H^m)}^2 + 2\,\|\partial_t v\|_{L^2(0,T;H^{m-1})}\,\|v\|_{L^2(0,T;H^{m+1})}\,. \tag{337}$$

*Proof.* Let $S_R$ denote Fourier restriction to $\{|\xi| \le R\}$. For $v_R = S_R v$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\|v_R(t)\|_{H^m}^2 = 2\,\mathrm{Re}\left\langle A^{m-1}\partial_t v_R(t), A^{m+1}v_R(t)\right\rangle_{L^2}.$$

Integration and Cauchy–Schwarz prove (335)–(336) for $v_R$. Applied to $S_R v - S_S v$, this estimate shows that $(v_R)_R$ is Cauchy in $C([0,T];H^m)$; choose one Lebesgue time for the initial Cauchy term. Its limit equals $v$ in $L^2(0,T;H^m)$. Passage to the limit proves the first two formulas. Choose $r \in (0,T)$ with $\|v(r)\|_{H^m}^2 \le T^{-1}\,\|v\|_{L^2H^m}^2$ and apply (336). □

**Lemma 12.8** (Explicit integer transport commutator)**.** *Set*

$$\Gamma_{1,1} := 1,$$

$$\Gamma_{\ell,k} := (2\ell - 4 + \sqrt{3})^{(k-1)(\ell-k)} \qquad (\ell \ge 2,\ 1 \le k \le \ell), \tag{338}$$

$$K_m := \sum_{\ell=1}^{m} 3^\ell \binom{m}{\ell} \sum_{k=1}^{\ell} \binom{\ell}{k} \Gamma_{\ell,k} \qquad (m \ge 3). \tag{339}$$

*Every $v \in H^{m+1}_\sigma(\mathbb{R}^3)$ satisfies*

$$|\mathrm{Re}\,\langle A^m \mathbb{P}\nabla\cdot(v\otimes v), A^m v\rangle_{L^2}| \le K_m\,\|\nabla v\|_{L^\infty}\,\|v\|_{H^m}^2\,. \tag{340}$$

*In particular,*

$$K_m \le (7^m - 1)(2m - 4 + \sqrt{3})^{(m-1)^2/4}. \tag{341}$$

*Proof.* We first prove the interpolation inequality used in the commutator. For the full ordered derivative tensor $\nabla^j F$, fix $M \ge 1$ and set

$$p_{M,0} := \infty, \qquad p_{M,j} := \frac{2M}{j} \quad (1 \le j \le M), \qquad A_M := 2M - 2 + \sqrt{3}.$$

Then

$$\left\|\nabla^j F\right\|_{L^{p_{M,j}}} \le A_M^{j(M-j)/2}\,\|F\|_\infty^{1-j/M}\,\left\|\nabla^M F\right\|_2^{j/M} \qquad (0 \le j \le M). \tag{342}$$

The endpoint cases are identities. For $1 \le j \le M-1$, put

$$G := \nabla^{j-1}F, \qquad W := \nabla G, \qquad p := p_{M,j}.$$

All contractions below are the Euclidean contractions of the corresponding ordered tensors. Integration by parts gives

$$\int_{\mathbb{R}^3} |W|^p = -\int_{\mathbb{R}^3} |W|^{p-2} G : \Delta G \\ -(p-2)\int_{\mathbb{R}^3} |W|^{p-4} \sum_{q=1}^{3} (G : W_{\cdot q})(W : \partial_q W).$$

The pointwise bounds

$$|\Delta G| \le \sqrt{3}\,|\nabla^2 G|, \qquad \left|\sum_{q=1}^{3} (G : W_{\cdot q})(W : \partial_q W)\right| \le |G|\,|W|^2 |\nabla^2 G|$$

give

$$\int_{\mathbb{R}^3} |W|^p \le (p - 2 + \sqrt{3}) \int_{\mathbb{R}^3} |G|\,|W|^{p-2} |\nabla^2 G|.$$

Since

$$\frac{1}{p_{M,j-1}} + \frac{p-2}{p} + \frac{1}{p_{M,j+1}} = 1,$$

Hölder's inequality, with $X_j := \left\|\nabla^j F\right\|_{L^{p_{M,j}}}$, gives

$$X_j^2 \le \left(\frac{2M}{j} - 2 + \sqrt{3}\right) X_{j-1} X_{j+1} \le A_M X_{j-1} X_{j+1}. \tag{343}$$

The numbers $d_j := j(M-j)/2$ obey

$$2d_j - d_{j-1} - d_{j+1} = 1, \qquad d_0 = d_M = 0.$$

The discrete maximum principle applied to (343) gives

$$X_j \le A_M^{d_j} X_0^{1-j/M} X_M^{j/M},$$

which is (342). The integration by parts is justified first with $(|W|^2 + \varepsilon)^{(p-2)/2}$, followed by $\varepsilon \downarrow 0$; smooth approximation handles zero tensor norms.

For $|\alpha| \le m$, set

$$\alpha! := \alpha_1! \alpha_2! \alpha_3!, \qquad c_{m,\alpha} := \frac{m!}{(m - |\alpha|)!\,\alpha!}.$$

The multiplier $A^{2m} = (1-\Delta)^m$ and the multinomial identity give the exact pairing formula

$$\langle A^m f, A^m g \rangle_{L^2} = \sum_{|\alpha| \le m} c_{m,\alpha} \langle \partial^\alpha f, \partial^\alpha g \rangle_{L^2}. \tag{344}$$

The Leray projection is self-adjoint, commutes with $A^m$, and fixes $A^m v$. Since $\nabla \cdot (v \otimes v) = (v \cdot \nabla) v$, it follows that

$$\langle A^m \mathbb{P} \nabla \cdot (v \otimes v), A^m v \rangle_{L^2} = \sum_{|\alpha| \le m} c_{m,\alpha} \langle \partial^\alpha ((v \cdot \nabla) v), \partial^\alpha v \rangle_{L^2}. \tag{345}$$

For each $\alpha$,

$$\partial^\alpha ((v \cdot \nabla) v) = (v \cdot \nabla) \partial^\alpha v + \sum_{0 < \beta \le \alpha} \binom{\alpha}{\beta} (\partial^\beta v \cdot \nabla) \partial^{\alpha - \beta} v.$$

The divergence-free transport term has zero real pairing with $\partial^\alpha v$, because

$$\operatorname{Re}\langle (v\cdot\nabla)\partial^\alpha v, \partial^\alpha v\rangle_{L^2} = \frac{1}{2}\int_{\mathbb{R}^3} v\cdot\nabla\,|\partial^\alpha v|^2\,\mathrm{d}x = 0.$$

For a remaining term, put

$$\ell := |\alpha|\,, \qquad k := |\beta|\,.$$

If $\ell \geq 2$, apply (342) to $F = \nabla v$, with

$$M = \ell - 1, \qquad j = k - 1.$$

The two Hölder exponents are $p_{M,j}$ and $p_{M,M-j}$, whose reciprocal sum is $1/2$. The two factors are subarrays of the corresponding ordered derivative tensors, so

$$\begin{aligned}\left\|(\partial^\beta v\cdot\nabla)\partial^{\alpha-\beta}v\right\|_2 &\leq \left\|\partial^\beta v\right\|_{L^{p_{M,j}}}\left\|\nabla\partial^{\alpha-\beta}v\right\|_{L^{p_{M,M-j}}}\\ &\leq \Gamma_{\ell,k}\,\|\nabla v\|_\infty\left\|\nabla^\ell v\right\|_2.\end{aligned} \tag{346}$$

For $\ell = 1$, this formula follows from the direct $L^\infty$-$L^2$ estimate and $\Gamma_{1,1} = 1$. Since

$$\left\|\nabla^\ell v\right\|_2 = \|v\|_{\dot H^\ell} \leq \|v\|_{H^m}\,, \qquad \|\partial^\alpha v\|_2 \leq \|v\|_{H^m}\,,$$

each remaining pairing is bounded by

$$\Gamma_{\ell,k}\,\|\nabla v\|_\infty\,\|v\|_{H^m}^2\,.$$

The multinomial identities

$$\sum_{|\alpha|=\ell} c_{m,\alpha} = 3^\ell\binom{m}{\ell}, \qquad \sum_{\substack{\beta\leq\alpha\\ |\beta|=k}}\binom{\alpha}{\beta} = \binom{\ell}{k}$$

now give (340) with the constant (339). Finally,

$$\sum_{k=1}^{\ell}\binom{\ell}{k} \leq 2^\ell, \qquad (k-1)(\ell-k) \leq \frac{(\ell-1)^2}{4},$$

and summation of $\binom{m}{\ell}3^\ell 2^\ell$ over $1\leq\ell\leq m$ proves (341). Approximation by divergence-free Schwartz fields justifies all integrations by parts for $v\in H^{m+1}_\sigma$. □

The preceding estimates control the differentiated rows as they approach an endpoint. To continue their common limit as the original solution, we now fix a time-translation-invariant local class that admits both late preterminal slices and the endpoint restart, and in which uniqueness holds on every overlap.

**Definition 12.9** (Shifted pressure-normalized $H^3$ mild class)**.** Fix the prescribed force $f\in C^\infty([0,\infty);\mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$ and set $g=\mathbb{P}f$. Let $0\leq s<b<\infty$ and $a\in H^3_\sigma(\mathbb{R}^3)$. Define $\mathscr{M}^3_{\nu,f}(s,b;a)$ to be the set of pairs $(w,q)$ such that

$$w\in C([s,b];H^3_\sigma(\mathbb{R}^3)), \qquad q\in C([s,b];H^3(\mathbb{R}^3)), \tag{347}$$

$$w(s) = a, \qquad q(t) = R_iR_j\big(w_i(t)w_j(t)\big) - (-\Delta)^{-1}\nabla\cdot f(t), \tag{348}$$

The remaining defining condition is the Duhamel identity: for every $t \in [s,b]$,

$$w(t) = e^{\nu(t-s)\Delta}a + \int_s^t e^{\nu(t-r)\Delta}g(r)\,\mathrm{d}r - \int_s^t e^{\nu(t-r)\Delta}\mathbb{P}\nabla\cdot(w\otimes w)(r)\,\mathrm{d}r \quad \text{in } H^3_\sigma(\mathbb{R}^3). \tag{349}$$

The integral is an $H^3_\sigma$-valued Bochner integral: multiplication in $H^3(\mathbb{R}^3)$ places $\mathbb{P}\nabla\cdot(w\otimes w)$ in $C([s,b];H^2_\sigma)$, and the heat multiplier from $H^2$ to $H^3$ has an integrable $(t-r)^{-1/2}$ singularity. The maximal lifespan in this class is

$$\tau^{\max}_{\nu,f,3}(s,a) := \sup\big\{b-s:\ b>s,\ \mathscr{M}^3_{\nu,f}(s,b;a)\neq\varnothing\big\} \in [0,\infty]. \tag{350}$$

**Lemma 12.10** (Restriction and classical admission)**.** *If $(w,q)\in\mathscr{M}^3_{\nu,f}(s,b;a)$ and $s\le r<c\le b$, then*

$$(w,q)|_{[r,c]} \in \mathscr{M}^3_{\nu,f}(r,c;w(r)). \tag{351}$$

*If $(u,p)$ is a pressure-complete classical history in the sense of Definition 13.2 and $[s,b]$ is contained in its interval of existence, then*

$$(u,p)|_{[s,b]} \in \mathscr{M}^3_{\nu,f}(s,b;u(s)). \tag{352}$$

*Proof.* Split (349) at $r$ and use the heat semigroup identity. This gives, for $r\le t\le c$,

$$w(t) = e^{\nu(t-r)\Delta}w(r) + \int_r^t e^{\nu(t-\rho)\Delta}g(\rho)\,\mathrm{d}\rho - \int_r^t e^{\nu(t-\rho)\Delta}\mathbb{P}\nabla\cdot(w\otimes w)(\rho)\,\mathrm{d}\rho.$$

The pressure formula is unchanged under restriction. For a classical history, apply $\mathbb{P}$ to the momentum equation and use variation of constants; the decaying pressure formula gives (348). □

**Theorem 12.11** (Quantitative local restart)**.** *Set $C_{\mathrm{q}} := \sqrt{3}\,C_3^{\mathrm{alg}}$. For $\nu>0$, set*

$$L_\nu := 4(1+\nu+\nu^{-1})\exp\!\left(\frac{\nu+\nu^{-1}}{2}\right), \qquad c_\nu := (8C_{\mathrm{q}}L_\nu^2)^{-2}.$$

*For every $s\ge 0$, put*

$$\Gamma_s := \|\mathbb{P}f\|_{L^2(s,s+1;H^2)}.$$

*For every $a\in H^3_\sigma(\mathbb{R}^3)$ and every*

$$0<\delta\le\delta_{\nu,f}(s,a) := \min\{1, c_\nu(1+\|a\|_{H^3}+\Gamma_s)^{-2}\},$$

*the class $\mathscr{M}^3_{\nu,f}(s,s+\delta;a)$ contains exactly one pair $(w,q)$. Its velocity satisfies*

$$w\in C([s,s+\delta];H^3_\sigma)\cap L^2(s,s+\delta;H^4), \qquad w_t\in L^2(s,s+\delta;H^2).$$

*If $a\in H^\infty_\sigma$, then*

$$\partial_t^n w,\ \partial_t^n q \in C([s,s+\delta];H^m) \qquad (n,m\in\mathbb{N}_0).$$

*For every finite $b>s$, the class $\mathscr{M}^3_{\nu,f}(s,b;a)$ contains at most one pair. Its compatible members on strict subintervals of $[s,s+\tau^{\max}_{\nu,f,3}(s,a))$ define the maximal shifted pressure-normalized $H^3$ mild history from $a$.*

*Proof.* Translate the initial time to zero while retaining the shifted prescribed source $r \mapsto f(s+r)$. It is enough to write the proof for $s = 0$. Let $0 < \delta \le 1$, let

$$z_t - \nu\Delta z = F, \qquad z(0) = a, \qquad a \in H^3_\sigma, \quad F \in L^2(0,\delta; H^2_\sigma),$$

and set

$$A_0 := \|a\|_{H^3}, \qquad G := \|F\|_{L^2(0,\delta;H^2)}.$$

The $H^3$ energy identity is

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|z\|^2_{H^3} + \nu\|\nabla z\|^2_{H^3} = \mathrm{Re}\left\langle A^2F, A^4z\right\rangle_{L^2}.$$

Since $\|z\|^2_{H^4} = \|z\|^2_{H^3} + \|\nabla z\|^2_{H^3}$, Young's inequality gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\|z\|^2_{H^3} + \nu\|\nabla z\|^2_{H^3} \le \nu\|z\|^2_{H^3} + \nu^{-1}\|F\|^2_{H^2}.$$

Consequently,

$$\|z\|_{C([0,\delta];H^3)} \le e^{\nu/2}\big(A_0 + \nu^{-1/2}G\big). \tag{353}$$

Pairing $A^2z_t - \nu\Delta A^2 z = A^2F$ with $-\Delta A^2 z$ gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left\|\nabla A^2 z\right\|_2^2 + \nu\left\|\Delta A^2 z\right\|_2^2 = \mathrm{Re}\left\langle A^2F, -\Delta A^2 z\right\rangle_{L^2}.$$

Integration and Young's inequality yield

$$\|\Delta z\|_{L^2(0,\delta;H^2)} \le \nu^{-1/2}A_0 + \nu^{-1}G. \tag{354}$$

The pointwise multiplier inequality

$$\langle\xi\rangle^8 \le 2\big(\langle\xi\rangle^6 + |\xi|^4\langle\xi\rangle^4\big)$$

and $\delta \le 1$ imply

$$\|z\|_{L^2(0,\delta;H^4)} \le \sqrt{2}\left(\|z\|_{C([0,\delta];H^3)} + \|\Delta z\|_{L^2(0,\delta;H^2)}\right). \tag{355}$$

The equation and (354) give

$$\|z_t\|_{L^2(0,\delta;H^2)} \le \nu^{1/2}A_0 + 2G. \tag{356}$$

Equations (353)–(356) give

$$\begin{aligned}
&\|z\|_{CH^3} + \|z\|_{L^2H^4} + \|z_t\|_{L^2H^2} \\
&\quad\le e^{\nu/2}(A_0 + \nu^{-1/2}G) + \sqrt{2}\left(e^{\nu/2}(A_0 + \nu^{-1/2}G) + \nu^{-1/2}A_0 + \nu^{-1}G\right) + \nu^{1/2}A_0 + 2G \\
&\quad\le L_\nu(A_0 + G).
\end{aligned}$$

The last inequality follows from

$$\nu^{1/2} \le \frac{1+\nu}{2}, \qquad \nu^{-1/2} \le \frac{1+\nu^{-1}}{2}, \qquad e^{\nu/2} \le e^{(\nu+\nu^{-1})/2}.$$

Fourier truncation and passage to the limit justify the preceding linear estimates for the stated data class.

For $v, z \in H^3_\sigma$, Lemmas 12.4 and 12.5 give

$$\|\mathbb{P}\nabla\cdot(v\otimes z)\|_{H^2} \le C_{\mathrm{q}}\,\|v\|_{H^3}\,\|z\|_{H^3}\,, \tag{357}$$

$$\|\mathbb{P}\nabla\cdot(v\otimes v - z\otimes z)\|_{H^2} \le C_{\mathrm{q}}(\|v\|_{H^3}+\|z\|_{H^3})\,\|v-z\|_{H^3}\,. \tag{358}$$

Indeed,

$$\|\nabla\cdot(v\otimes z)\|^2_{H^2} \le 3\sum_{i,j=1}^{3}\|v_i z_j\|^2_{H^3} \le 3(C_3^{\mathrm{alg}})^2\,\|v\|^2_{H^3}\,\|z\|^2_{H^3}\,.$$

Define

$$X_\delta := \left\{v \in C([0,\delta];H^3_\sigma)\cap L^2(0,\delta;H^4_\sigma) : v_t \in L^2(0,\delta;H^2_\sigma)\right\},$$

$$\|v\|_{X_\delta} := \|v\|_{CH^3} + \|v\|_{L^2H^4} + \|v_t\|_{L^2H^2}\,, \qquad X_\delta(a) := \{v\in X_\delta : v(0)=a\}.$$

$X_\delta$ is a Banach space with this norm, and $X_\delta(a)$ is closed. Put

$$\Gamma_0 := \|g\|_{L^2(0,1;H^2)}\,, \qquad B_0 := A_0+\Gamma_0, \qquad R := 2L_\nu B_0,$$

and define

$$\mathbb{B}_R := \{v\in X_\delta(a) : \|v\|_{X_\delta}\le R\}.$$

The set $\mathbb{B}_R$ is nonempty: the forced linear trajectory

$$v^{\mathrm{lin}}(t) = e^{\nu t\Delta}a + \int_0^t e^{\nu(t-r)\Delta}g(r)\,\mathrm{d}r$$

belongs to $X_\delta(a)$, and the linear estimate gives $\|v^{\mathrm{lin}}\|_{X_\delta}\le L_\nu B_0 \le R$. For $v\in\mathbb{B}_R$, define

$$\Phi(v)(t) := e^{\nu t\Delta}a + \int_0^t e^{\nu(t-s)\Delta}g(s)\,\mathrm{d}s - \int_0^t e^{\nu(t-s)\Delta}\mathbb{P}\nabla\cdot(v\otimes v)(s)\,\mathrm{d}s.$$

If $\delta\le\delta_{\nu,f}(0,a)$, then

$$\delta^{1/2} \le \frac{1}{8C_{\mathrm{q}}L_\nu^2(1+B_0)}.$$

The linear estimate and (357) imply

$$\begin{aligned}\|\Phi(v)\|_{X_\delta} &\le L_\nu\big(B_0 + C_{\mathrm{q}}\delta^{1/2}R^2\big)\\ &\le L_\nu B_0 + \frac{L_\nu B_0^2}{2(1+B_0)} \le R.\end{aligned}$$

For $v, z\in\mathbb{B}_R$,

$$\begin{aligned}\|\Phi(v)-\Phi(z)\|_{X_\delta} &\le L_\nu C_{\mathrm{q}}\delta^{1/2}(\|v\|_{CH^3}+\|z\|_{CH^3})\,\|v-z\|_{CH^3}\\ &\le \frac{B_0}{2(1+B_0)}\,\|v-z\|_{X_\delta} \le \frac{1}{2}\,\|v-z\|_{X_\delta}\,.\end{aligned}$$

Thus $\Phi$ has a unique fixed point in $\mathbb{B}_R$. If $B_0=0$, then both the datum and projected force vanish on the interval and the zero function is the fixed point. Let $b>0$ and let

$$(v,q_v),(z,q_z)\in\mathscr{M}^3_{\nu,f}(0,b;a).$$

Set

$$M_b := \|v\|_{C([0,b];H^3)} + \|z\|_{C([0,b];H^3)}.$$

On every subinterval $I \subset [0,b]$ of length $h \le 1$ on which the two velocities have a common left trace, their shifted Duhamel identities and the zero-data linear estimate give

$$\|v - z\|_{C(I;H^3)} \le L_\nu C_{\mathrm{q}} h^{1/2} M_b \|v - z\|_{C(I;H^3)}.$$

A finite partition with $L_\nu C_{\mathrm{q}} h^{1/2} M_b < 1$ proves $v = z$ on $[0,b]$. The common force cancels in the difference equation. The pressure formula, including its common force potential, gives $q_v = q_z$. This proves the at-most-one assertion for the full shifted mild class. For the fixed point, (357) places the quadratic momentum term in $L^2(0,\delta;H^2)$; the linear estimates give

$$w \in L^2(0,\delta;H^4), \qquad w_t \in L^2(0,\delta;H^2).$$

Lemma 12.5 gives the normalized pressure

$$q = R_i R_j (w_i w_j) - (-\Delta)^{-1} \nabla \cdot f.$$

Fix $m \ge 3$ and assume $a \in H^m_\sigma$. Let $S_k$ be Fourier restriction to $\{|\xi| \le k\}$, set $a_k = S_k a$, and replace the fixed-point map above by

$$\Phi_k(v)(t) := e^{\nu t\Delta} a_k + \int_0^t e^{\nu(t-s)\Delta} S_k g(s)\,\mathrm{d}s - \int_0^t e^{\nu(t-s)\Delta} S_k \mathbb{P}\nabla\cdot (v\otimes v)(s)\,\mathrm{d}s.$$

The projection $S_k$ is self-adjoint, contractive on every Sobolev space, and commutes with the heat flow, $\mathbb{P}$, and spatial derivatives. The estimates that gave $\Phi$ its contraction factor at most $1/2$ apply to $\Phi_k$ on the closed radius-$R$ ball in $X_\delta(a_k)$. Its fixed point $w^{(k)} = S_k w^{(k)}$ consequently exists on $[0,\delta]$ and satisfies

$$\sup_k \left\| w^{(k)} \right\|_{X_\delta} \le R. \tag{359}$$

Because its Fourier support is bounded, $w^{(k)}$ belongs to every spatial Sobolev space. Self-adjointness of $S_k$ removes the truncation from the energy pairing, so Lemma 12.8 gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\left\|w^{(k)}\right\|^2_{H^m} + \nu \left\|\nabla w^{(k)}\right\|^2_{H^m} \le K_m \left\|\nabla w^{(k)}\right\|_{L^\infty} \left\|w^{(k)}\right\|^2_{H^m} + \|g\|_{H^m} \left\|w^{(k)}\right\|_{H^m}.$$

Set

$$C_{\nabla,3} := (2\pi)^{-3/2} \left( \int_{\mathbb{R}^3} |\xi|^2 \langle \xi \rangle^{-6}\,\mathrm{d}\xi \right)^{1/2}.$$

Fourier inversion and Cauchy–Schwarz give $\|\nabla v\|_\infty \le C_{\nabla,3} \|v\|_{H^3}$. From (359),

$$\int_0^\delta \left\|\nabla w^{(k)}(s)\right\|_\infty \,\mathrm{d}s \le C_{\nabla,3}\delta R.$$

Put

$$F_k(t) := \left\|w^{(k)}(t)\right\|^2_{H^m} + 2\nu \int_0^t \left\|\nabla w^{(k)}(s)\right\|^2_{H^m}\,\mathrm{d}s.$$

The differential inequality above gives

$$F_k(t) \le \|a\|^2_{H^m} + \int_0^t \|g(s)\|^2_{H^m}\,\mathrm{d}s + \int_0^t \left(1 + 2K_m \left\|\nabla w^{(k)}(s)\right\|_\infty\right) F_k(s)\,\mathrm{d}s.$$

Gronwall's inequality now gives the uniform bound

$$\sup_{0\le t\le\delta}\left(\left\|w^{(k)}(t)\right\|_{H^m}^2+2\nu\int_0^t\left\|\nabla w^{(k)}(s)\right\|_{H^m}^2\,\mathrm{d}s\right)\le e^{\delta+2K_mC_{\nabla,3}\delta R}\left(\|a\|_{H^m}^2+\|g\|_{L^2(0,\delta;H^m)}^2\right). \tag{360}$$

Since $\left\|w^{(k)}\right\|_{H^{m+1}}^2=\left\|w^{(k)}\right\|_{H^m}^2+\left\|\nabla w^{(k)}\right\|_{H^m}^2$, this also bounds $w^{(k)}$ in $L^2H^{m+1}$. The truncated equations and Lemma 12.4 give, with a constant depending only on $m$,

$$\left\|\partial_t w^{(k)}\right\|_{L^2H^{m-1}}\le\nu\left\|w^{(k)}\right\|_{L^2H^{m+1}}+\sqrt{3}\,C_m^{\mathrm{alg}}\delta^{1/2}\left\|w^{(k)}\right\|_{L^\infty H^m}^2+\|g\|_{L^2H^{m-1}},$$

so the time derivatives are uniformly bounded in $L^2H^{m-1}$.

It remains to identify the higher-order weak limit with the fixed point already constructed in $X_\delta(a)$. The contraction estimate gives

$$\left\|w^{(k)}-w\right\|_{X_\delta}\le 2\left\|\Phi_k(w)-\Phi(w)\right\|_{X_\delta}.$$

The linear bound proved above controls the right-hand side by a constant times

$$\|(S_k-I)a\|_{H^3}+\|(S_k-I)g\|_{L^2H^2}+\|(S_k-I)\mathbb{P}\nabla\cdot(w\otimes w)\|_{L^2H^2},$$

which tends to zero. Hence $w^{(k)}\to w$ in $X_\delta$. Weak compactness in the three uniformly bounded higher-order spaces and uniqueness of the distributional limit place

$$w\in L^\infty(0,\delta;H^m_\sigma)\cap L^2(0,\delta;H^{m+1}_\sigma),\qquad w_t\in L^2(0,\delta;H^{m-1}_\sigma).$$

Lemma 12.7 supplies the representative $w\in C([0,\delta];H^m_\sigma)$; its trace at zero is $a$, since it is the distributional trace of the already constructed $CH^3$ solution. Assume now that $a\in H^\infty_\sigma$. Set $W_0=w$ and define recursively

$$F_n=\partial_t^n f,\qquad Q_n=R_iR_j\sum_{a=0}^n\binom{n}{a}W_{a,i}W_{n-a,j}-(-\Delta)^{-1}\nabla\cdot F_n,$$

$$W_{n+1}=\nu\Delta W_n-\sum_{a=0}^n\binom{n}{a}(W_a\cdot\nabla)W_{n-a}-\nabla Q_n+F_n.$$

If $W_0,\dots,W_n\in CH^{m+2}$, then Lemmas 12.4 and 12.5 give

$$Q_n\in CH^{m+1},\qquad W_{n+1}\in CH^m.$$

Since $m$ is arbitrary, induction gives $W_n,Q_n\in CH^m$ for every $n,m\in\mathbb{N}_0$. The equation first gives $W_0\in C^1H^m$ and $\partial_tW_0=W_1$. If $W_a=\partial_t^a w$ for $0\le a\le n$, distributional differentiation of $q=R_iR_j(w_iw_j)-(-\Delta)^{-1}\nabla\cdot f$ and Lemma 12.6 give $\partial_t^n q=Q_n$. Differentiation of the momentum equation then gives $\partial_tW_n=W_{n+1}$. Induction gives

$$W_n\in C^1H^m,\qquad\partial_tW_n=W_{n+1},\qquad W_n=\partial_t^n w,\qquad Q_n=\partial_t^n q.$$

This proof with the translated prescribed source $r\mapsto f(s+r)$ proves the result for every $s\ge0$. Lemma 12.10 and the at-most-one assertion make all nonempty classes below $\tau^{\max}_{\nu,f,3}(s,a)$ compatible; their union is the maximal shifted pressure-normalized $H^3$ mild history. □

# 13 The pressure-complete setting and mixed jet

**Proposition 13.1** (Pressure normalization)**.** *Let $u, p, f$ be smooth on $I \times \mathbb{R}^3$, let $\nabla \cdot u = 0$, and suppose*

$$u(t) \in H^1_\sigma(\mathbb{R}^3), \qquad p(t) \in H^0(\mathbb{R}^3), \qquad f(t) \in L^1(\mathbb{R}^3) \cap H^1(\mathbb{R}^3) \qquad (t \in I),$$

*and*

$$u_t + (u \cdot \nabla)u + \nabla p = \nu \Delta u + f. \tag{361}$$

*Then*

$$-\Delta p = \partial_i \partial_j (u_i u_j) - \nabla \cdot f, \tag{362}$$

$$p = R_i R_j (u_i u_j) - (-\Delta)^{-1} \nabla \cdot f, \tag{363}$$

$$(u \cdot \nabla)u + \nabla p = \mathbb{P} \nabla \cdot (u \otimes u) + (I - \mathbb{P}) f. \tag{364}$$

*The condition $p(t) \in H^0(\mathbb{R}^3)$ fixes the pressure uniquely.*

*Proof.* Incompressibility gives

$$(u \cdot \nabla) u_i = \partial_j (u_j u_i).$$

Taking the divergence of (361) cancels $\nabla \cdot u_t$ and $\nabla \cdot \Delta u$, which yields (362). Lemma 12.5, applied with $m = 0$, gives the force potential in $H^0$ and

$$q := R_i R_j (u_i u_j) - (-\Delta)^{-1} \nabla \cdot f \in H^0.$$

It satisfies this Poisson identity and

$$(u \cdot \nabla)u + \nabla q = \mathbb{P} \nabla \cdot (u \otimes u) + (I - \mathbb{P}) f.$$

If $p_1, p_2 \in H^0$ satisfy (362), then $h = p_1 - p_2$ satisfies $|\xi|^2 \widehat{h} = 0$. Since $\widehat{h} \in L^2$ and is supported at $\{0\}$, it vanishes. Hence $p = q$, which proves (363) and (364). □

**Definition 13.2** (Maximal pressure-complete classical history)**.** Fix $\nu > 0$, $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and $f \in C^\infty([0, \infty); \mathcal{S}(\mathbb{R}^3; \mathbb{R}^3))$. A pressure-complete classical history generated by $(u_0, \nu, f)$ is a triple

$$(T_*, u, p), \qquad 0 < T_* \le \infty,$$

such that, for every $0 < T < T_*$ and $n, m \in \mathbb{N}_0$,

$$\partial_t^n u \in C([0, T]; H^m_\sigma(\mathbb{R}^3)),$$
$$\partial_t^n p \in C([0, T]; H^m(\mathbb{R}^3)),$$

and

$$u_t + (u \cdot \nabla)u + \nabla p = \nu \Delta u + f, \tag{365}$$

$$\nabla \cdot u = 0, \tag{366}$$

$$u(0) = u_0, \tag{367}$$

$$p(t) = R_i R_j (u_i(t) u_j(t)) - (-\Delta)^{-1} \nabla \cdot f(t). \tag{368}$$

The history is maximal if no pressure-complete classical history generated by that $(u_0, \nu, f)$ continues $(u, p)$ past $T_*$.

**Definition 13.3** (Temporal and mixed velocity–pressure jets)**.** Let $(T_*, u, p)$ be a maximal pressure-complete classical history. For $n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$, define

$$V_n := \partial_t^n u, \qquad Q_n := \partial_t^n p, \qquad F_n := \partial_t^n f, \qquad J_{n,\alpha} := \partial_t^n \partial_x^\alpha u, \qquad \Pi_{n,\alpha} := \partial_t^n \partial_x^\alpha p.$$

Set also $F_{n,\alpha} := \partial_t^n \partial_x^\alpha f$. For $\beta \le \alpha$, define

$$\binom{\alpha}{\beta} := \prod_{\ell=1}^{3} \binom{\alpha_\ell}{\beta_\ell}, \qquad |\alpha| := \alpha_1 + \alpha_2 + \alpha_3.$$

**Proposition 13.4** (Pressure-complete binomial recurrences)**.** *For every $n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,*

$$Q_n = R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i} V_{n-a,j} - (-\Delta)^{-1} \nabla \cdot F_n, \tag{369}$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^{n} \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n + F_n, \tag{370}$$

$$\Pi_{n,\alpha} = R_i R_j \sum_{a=0}^{n} \sum_{\beta \le \alpha} \binom{n}{a} \binom{\alpha}{\beta} J_{a,\beta,i} J_{n-a,\alpha-\beta,j} - \partial_x^\alpha (-\Delta)^{-1} \nabla \cdot F_n, \tag{371}$$

$$J_{n+1,\alpha} + \sum_{a=0}^{n} \sum_{\beta \le \alpha} \binom{n}{a} \binom{\alpha}{\beta} (J_{a,\beta} \cdot \nabla) J_{n-a,\alpha-\beta} + \nabla \Pi_{n,\alpha} = \nu \Delta J_{n,\alpha} + F_{n,\alpha}, \tag{372}$$

$$\nabla \cdot J_{n,\alpha} = 0. \tag{373}$$

*Proof.* Apply $\partial_t^n$ to (368) and use Lemma 12.6 with $\alpha = 0$ to obtain (369). Apply $\partial_t^n$ to (365) to obtain (370). Apply $\partial_x^\alpha$ to both identities, use Lemma 12.6, and commute the derivatives with $R_i R_j$ to obtain (371) and (372). Differentiation of (366) gives (373). □

**The forced datum and its derivative rows.** The initial velocity, viscosity, and complete prescribed force specify all inputs to these recurrences. Pressure is determined by the velocity products and the gradient part of that same force. Every temporal and spatial force derivative remains in its corresponding row. In the kinetic-energy calculation of Proposition 14.1, transport and pressure have zero global pairing with the velocity while the external work remains. At higher Sobolev orders, the full transport–pressure contribution and force work are retained in Proposition 14.3. These are the governing rows and energy identities used below.

**Definition 13.5** (Canonical pressure constant)**.** For vector fields $v, w$, set

$$\mathcal{Q}(v, w) := \sum_{i,j=1}^{3} R_i R_j (v_i w_j).$$

For $m \in \mathbb{N}_0$, define

$$C_m^{\mathrm{p}} := \sup_{v,w \in H^{m+1}(\mathbb{R}^3;\mathbb{R}^3)\setminus\{0\}} \frac{\|\mathcal{Q}(v,w)\|_{H^m} + \|\nabla \mathcal{Q}(v,w)\|_{H^{m-1}}}{\|v\|_{H^{m+1}} \, \|w\|_{H^{m+1}}}.$$

**Lemma 13.6.** *For every $m \in \mathbb{N}_0$,*

$$C_m^{\mathrm{p}} \le 2C_m^{\mathrm{alg}}.$$

*Proof.* Regard $(v_i w_j)_{i,j}$ as one tensor. The contraction $F \mapsto R_i R_j F_{ij}$ has Fourier-multiplier norm one from the tensor $H^m$ norm to scalar $H^m$. Lemma 12.4 and $\|\nabla h\|_{H^{m-1}} \le \|h\|_{H^m}$ give

$$\|\mathcal{Q}(v,w)\|_{H^m} + \|\nabla \mathcal{Q}(v,w)\|_{H^{m-1}} \le 2C_m^{\mathrm{alg}} \|v\|_{H^{m+1}} \|w\|_{H^{m+1}}.$$

□

**Lemma 13.7** (Initial pressure-complete jet)**.** *Set*

$$V_n^0 := V_n(0), \qquad Q_n^0 := Q_n(0), \qquad J_{n,\alpha}^0 := J_{n,\alpha}(0), \qquad \Pi_{n,\alpha}^0 := \Pi_{n,\alpha}(0).$$

*Write $F_n^0 := F_n(0)$. Then*

$$V_0^0 = u_0, \tag{374}$$

$$Q_n^0 = R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i}^0 V_{n-a,j}^0 - (-\Delta)^{-1} \nabla \cdot F_n^0, \tag{375}$$

$$V_{n+1}^0 = \nu \Delta V_n^0 - \sum_{a=0}^{n} \binom{n}{a} (V_a^0 \cdot \nabla) V_{n-a}^0 - \nabla Q_n^0 + F_n^0. \tag{376}$$

*For every $n, m \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,*

$$V_n^0, Q_n^0, J_{n,\alpha}^0, \Pi_{n,\alpha}^0 \in H^m(\mathbb{R}^3).$$

*Proof.* Equations (374)–(376) are (369) and (370) at $t = 0$. Assume $V_0^0, \dots, V_n^0 \in \bigcap_{m \ge 0} H^m$. Lemmas 12.4 and 12.5 give

$$Q_n^0 \in \bigcap_{m \ge 0} H^m, \qquad V_{n+1}^0 \in \bigcap_{m \ge 0} H^m.$$

Induction starts from $u_0 \in H^\infty_\sigma$, while every $F_n^0$ is Schwartz by the prescribed force class. Spatial differentiation and (371) give the claims for $J_{n,\alpha}^0$ and $\Pi_{n,\alpha}^0$. □

**Theorem 13.8** (Finite-row pressure estimates)**.** *Let $I = (0,T) \subset (0,T_*)$, let $n, M \in \mathbb{N}_0$, and suppose*

$$V_a \in L^2(I; H^{M+1}(\mathbb{R}^3)) \qquad (0 \le a \le n).$$

*For $\alpha \in \mathbb{N}_0^3$, define the prescribed pressure-source budget*

$$\mathfrak{F}_{n,\alpha,M}^{\mathrm{p}}(I) := \left\| \partial_x^\alpha (-\Delta)^{-1} \nabla \cdot F_n \right\|_{L^1(I;H^M)} + \left\| \nabla \partial_x^\alpha (-\Delta)^{-1} \nabla \cdot F_n \right\|_{L^1(I;H^{M-1})}. \tag{377}$$

*It is finite by* (334) *and the prescribed force class. Then*

$$\begin{aligned} &\|Q_n\|_{L^1(I;H^M)} + \|\nabla Q_n\|_{L^1(I;H^{M-1})} \\ &\quad \le C_M^{\mathrm{p}} \sum_{a=0}^{n} \binom{n}{a} \|V_a\|_{L^2(I;H^{M+1})} \|V_{n-a}\|_{L^2(I;H^{M+1})} + \mathfrak{F}_{n,0,M}^{\mathrm{p}}(I). \end{aligned} \tag{378}$$

*If*

$$\mathcal{V}_{n,M}(I)^2 := \sum_{a=0}^{n} \|V_a\|^2_{L^2(I;H^{M+1})},$$

*then*

$$\|Q_n\|_{L^1(I;H^M)} + \|\nabla Q_n\|_{L^1(I;H^{M-1})} \le C_M^{\mathrm{p}} 2^n \mathcal{V}_{n,M}(I)^2 + \mathfrak{F}^{\mathrm{p}}_{n,0,M}(I). \tag{379}$$

*For $\alpha \in \mathbb{N}_0^3$, assume*

$$J_{a,\beta} \in L^2(I; H^{M+1}(\mathbb{R}^3)) \qquad (0 \le a \le n,\ \beta \le \alpha).$$

*Then*

$$\begin{aligned}&\|\Pi_{n,\alpha}\|_{L^1(I;H^M)} + \|\nabla \Pi_{n,\alpha}\|_{L^1(I;H^{M-1})} \\ &\quad \le C_M^{\mathrm{p}} \sum_{a=0}^{n} \sum_{\beta \le \alpha} \binom{n}{a}\binom{\alpha}{\beta} \|J_{a,\beta}\|_{L^2(I;H^{M+1})} \|J_{n-a,\alpha-\beta}\|_{L^2(I;H^{M+1})} + \mathfrak{F}^{\mathrm{p}}_{n,\alpha,M}(I).\end{aligned} \tag{380}$$

*If*

$$\mathcal{J}_{n,\alpha,M}(I)^2 := \sum_{a=0}^{n} \sum_{\beta \le \alpha} \|J_{a,\beta}\|^2_{L^2(I;H^{M+1})},$$

*then*

$$\|\Pi_{n,\alpha}\|_{L^1(I;H^M)} + \|\nabla \Pi_{n,\alpha}\|_{L^1(I;H^{M-1})} \le C_M^{\mathrm{p}} 2^{n+|\alpha|} \mathcal{J}_{n,\alpha,M}(I)^2 + \mathfrak{F}^{\mathrm{p}}_{n,\alpha,M}(I). \tag{381}$$

*Proof.* Lemmas 12.4 and 12.5 give

$$\|R_i R_j (v_i w_j)\|_{H^M} + \|\nabla R_i R_j (v_i w_j)\|_{H^{M-1}} \le C_M^{\mathrm{p}} \|v\|_{H^{M+1}} \|w\|_{H^{M+1}}. \tag{382}$$

Equations (369) and (382) give, for almost every $t \in I$,

$$\begin{aligned}\|Q_n(t)\|_{H^M} + \|\nabla Q_n(t)\|_{H^{M-1}} \le{}& C_M^{\mathrm{p}} \sum_{a=0}^{n} \binom{n}{a} \|V_a(t)\|_{H^{M+1}} \|V_{n-a}(t)\|_{H^{M+1}} \\ &+ \left\|(-\Delta)^{-1}\nabla \cdot F_n(t)\right\|_{H^M} \\ &+ \left\|\nabla(-\Delta)^{-1}\nabla \cdot F_n(t)\right\|_{H^{M-1}}.\end{aligned}$$

Integration and Cauchy–Schwarz prove (378), with the two force-potential terms collected exactly in $\mathfrak{F}^{\mathrm{p}}_{n,0,M}(I)$. For nonnegative $x_a$,

$$\sum_{a=0}^{n} \binom{n}{a} x_a x_{n-a} \le \sum_{a=0}^{n} \binom{n}{a} x_a^2 \le 2^n \sum_{a=0}^{n} x_a^2.$$

This proves (379).

Apply (382) to each summand in (371). Integration and Cauchy–Schwarz give (380); the differentiated prescribed pressure source is $\mathfrak{F}^{\mathrm{p}}_{n,\alpha,M}(I)$. The involution

$$(a,\beta) \longmapsto (n-a, \alpha-\beta)$$

preserves the weights $\binom{n}{a}\binom{\alpha}{\beta}$, and

$$\sum_{a=0}^{n} \sum_{\beta \le \alpha} \binom{n}{a}\binom{\alpha}{\beta} = 2^{n+|\alpha|}.$$

In particular, every individual weight is at most $2^{n+|\alpha|}$. The inequality $2xy \le x^2 + y^2$ proves (381). □

## 13.1 One compatible forced rectangle family and its two descriptions

Start with $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, $\nu > 0$, and the prescribed $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$. Theorem 12.11 constructs a pressure-normalized classical history on a positive interval. Uniqueness identifies the local solutions on their overlaps; their compatible union gives the maximal history of Definition 13.2.

The datum fixes the initial velocity row. The pressure formula in Proposition 13.4 fixes its normalized pressure, and the velocity recurrence fixes the next temporal row. Repetition and spatial differentiation produce the complete initial mixed jet. Along the history these are the actual derivatives of $u, p, f$, with the complete transport sum, pressure, viscosity, and force row retained. These finite jet realizations give every finite mixed Sobolev row on every strict horizon.

At a proposed finite maximal time, fix finite temporal and spatial orders and close that index set under the adjacent temporal row, the viscous derivatives, every Leibniz allocation, the normalized pressure coordinate, and the prescribed force row. The datum construction produces that finite pressure-complete record on $I = (0, T_*)$, and coordinate deletion recovers every smaller record. Ranging through the mixed indices exhausts every finite temporal and spatial order and gives the direct all-orders mixed Sobolev membership. The next section independently derives the completed-height and viscous identities and evaluates them on these same coordinates.

**Theorem 13.9** (Datum-generated strict-horizon mixed-jet intersection)**.** *Let $\nu > 0$, $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$. Let $(T_*, u, p)$ be the maximal pressure-complete classical history generated by $(u_0, \nu, f)$. For every $0 < T < T_*$,*

$$u \in \bigcap_{r,m\in\mathbb{N}_0} H^r((0,T); H^m_\sigma(\mathbb{R}^3)). \tag{383}$$

*All velocity, pressure, and force coordinates are the actual mixed derivatives of this one history, obey Proposition 13.4, and agree under restriction.*

*Proof.* Write $A_n = V_n(0)$ and $P_n = Q_n(0)$. The complete initial recurrence is

$$\begin{aligned} A_0 &= u_0, \\ P_n &= R_i R_j \sum_{a=0}^{n} \binom{n}{a} (A_a)_i (A_{n-a})_j - (-\Delta)^{-1} \nabla \cdot F_n(0), \\ A_{n+1} &= \nu \Delta A_n - \sum_{a=0}^{n} \binom{n}{a} (A_a \cdot \nabla) A_{n-a} - \nabla P_n + F_n(0). \end{aligned}$$

Lemma 13.7 shows inductively that every selected velocity and pressure row lies in every finite Sobolev space.

Fix $T < T_*$ and finite $r, m$. Apply the forced local construction simultaneously to the finite family

$$\{\partial_t^n \partial_x^\alpha u : 0 \le n \le r,\ |\alpha| \le m\},$$

using the displayed recurrence at the initial face. Higher-order persistence realizes these fields as the actual derivatives of one velocity. Whenever two finite constructions overlap in indices or time, local uniqueness identifies their common coordinates. On the compact interval $[0, T]$, persistence gives

$$\sum_{n=0}^{r} \|V_n\|^2_{L^2(0,T;H^m)} < \infty.$$

Taking every finite $(r, m)$ proves (383). The pressure formula and Theorem 13.8 place each selected pressure row in its stated strict-horizon Sobolev space. Restriction compatibility makes the resulting family one mixed jet of one velocity. □

At a proposed finite maximal time, put $I = (0, T_*)$ and fix finite $N, M$. The prescribed triple $(u_0, \nu, f)$, its initial mixed jet, and the complete velocity–pressure–force equations determine the record below.

**Theorem 13.10** (Datum-generated compatible forced finite rectangles)**.** *Let $\nu > 0$, $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$. Let $(T_*, u, p)$ be the maximal pressure-complete classical history generated by the single forced datum $(u_0, \nu, f)$. Assume, toward contradiction, that $T_* < \infty$, and put $I = (0, T_*)$. For finite $N, M$, the corresponding pressure-complete record is*

$$\mathcal{R}_{N,M}[u_0, \nu, f; T_*] := \left((V_n)_{n=0}^N, (V_{n+1})_{n=0}^N; (Q_n)_{n=0}^N; (F_n)_{n=0}^N\right). \tag{384}$$

*The forced construction produces, for $0 \le n \le N$,*

$$V_n \in L^2(I; H^{M+1}), \qquad V_{n+1} \in L^2(I; H^{M-1}). \tag{385}$$

*Define its velocity norm on $I$ by*

$$\mathfrak{R}_{N,M}(u; T_*) := \sum_{n=0}^{N} \left( \|V_n\|^2_{L^2(I;H^{M+1})} + \|V_{n+1}\|^2_{L^2(I;H^{M-1})} \right). \tag{386}$$

*The forced whole-terminal output is*

$$\mathfrak{R}_{N,M}(u; T_*) < \infty \qquad (N, M < \infty). \tag{387}$$

*Taking $N \ge n$ and $M = m$ gives the corresponding statement for every finite pair $(n, m)$. For $N' \le N$ and $M' \le M$, the larger block restricts to the velocity, pressure, and force coordinates of the smaller block:*

$$\mathcal{R}_{N,M}[u_0, \nu, f; T_*]|_{(N',M')} = \mathcal{R}_{N',M'}[u_0, \nu, f; T_*]. \tag{388}$$

*Here restriction deletes the outer temporal rows and reads every retained distribution in the lower Sobolev order. The mixed coordinates are*

$$J_{n,\alpha} = \partial_x^\alpha \partial_t^n u, \qquad \Pi_{n,\alpha} = \partial_x^\alpha \partial_t^n p, \qquad F_{n,\alpha} = \partial_x^\alpha \partial_t^n f.$$

*For every finite $(n, m, \alpha)$,*

$$J_{n,\alpha} \in L^2(I; H^m), \qquad \Pi_{n,\alpha} \in L^1(I; H^m), \qquad F_{n,\alpha} \in C([0, T_*]; H^m).$$

*Datum construction on the complete forced graph.* Put $I = (0, T_*)$, and fix finite $N, M$. The prescribed data first generate the complete initial jet:

$$\begin{aligned} A_0 &= u_0, \\ P_n &= R_i R_j \sum_{a=0}^{n} \binom{n}{a} (A_a)_i (A_{n-a})_j - (-\Delta)^{-1} \nabla \cdot F_n(0), \\ A_{n+1} &= \nu \Delta A_n - \sum_{a=0}^{n} \binom{n}{a} (A_a \cdot \nabla) A_{n-a} - \nabla P_n + F_n(0). \end{aligned}$$

At step $n$, this recurrence determines $A_{n+1}$ from the preceding velocity rows and the prescribed force row $F_n(0)$. Lemma 13.7 places every required initial row in every finite spatial Sobolev order, and spatial differentiation supplies the initial mixed coordinates.

For an equation indexed by $(n,\alpha)$, retain the adjacent temporal coordinate $J_{n+1,\alpha}$, the viscous coordinates $J_{n,\alpha+2e_\ell}$, and the pressure coordinates $\Pi_{n,\alpha+e_\ell}$, for $1\le \ell\le 3$, together with the direct source coordinate $F_{n,\alpha}$. For every $a\le n$, $\beta\le\alpha$, and velocity component $i$, retain both transport factors

$$J_{a,\beta,i}, \qquad J_{n-a,\alpha-\beta+e_i}.$$

Here $e_i$ is the $i$-th spatial unit multi-index. Set $\Gamma_\alpha := \{\alpha\}\cup\{\alpha+e_\ell : 1\le\ell\le 3\}$. For every $\gamma\in\Gamma_\alpha$, every $a\le n$, and every $\beta\le\gamma$, also retain the pressure-product factors $J_{a,\beta,i}$, $J_{n-a,\gamma-\beta,j}$, and the pressure-source coordinate $F_{n,\gamma}$. Each retained normalized pressure coordinate then carries both its full velocity product and its prescribed source potential:

$$\Pi_{n,\gamma} = R_iR_j\sum_{a=0}^{n}\sum_{\beta\le\gamma}\binom{n}{a}\binom{\gamma}{\beta}J_{a,\beta,i}J_{n-a,\gamma-\beta,j} - (-\Delta)^{-1}\nabla\cdot F_{n,\gamma} \qquad (\gamma\in\Gamma_\alpha).$$

The retained equation is

$$\mathscr{E}_{n,\alpha}(J,\Pi;F) := J_{n+1,\alpha} + \sum_{a=0}^{n}\sum_{\beta\le\alpha}\binom{n}{a}\binom{\alpha}{\beta}(J_{a,\beta}\cdot\nabla)J_{n-a,\alpha-\beta} + \nabla\Pi_{n,\alpha} - \nu\Delta J_{n,\alpha} - F_{n,\alpha} = 0. \tag{389}$$

There are finitely many retained coordinates for the fixed selected equations. This dependency cover is not a second or truncated flow: every velocity and pressure entry is a derivative of the one prescribed forced history, and every force entry is a derivative of the one prescribed source. Applying $\mathbb{P}$ reads the solenoidal evolution; applying $I-\mathbb{P}$ reads the normalized pressure equation.

The prescribed source adds one known affine coordinate to each coupled row. At the block's velocity-row order, for every $0\le n\le N$, Fourier Cauchy–Schwarz and Young's inequality give the source payment

$$2\,|\langle F_n, V_n\rangle_{H^M}| \le \nu\,\|V_n\|^2_{H^{M+1}} + \nu^{-1}\,\|F_n\|^2_{H^{M-1}}\,. \tag{390}$$

The first right-hand term is absorbed by the viscous $H^{M+1}$-dissipation in that row. The second is finite on $[0,T_*]$ because $F_n$ is smooth in time with Schwartz spatial values. For each retained mixed pressure index $\gamma\in\Gamma_\alpha$, the gradient component has the finite budget $\mathfrak{F}^{\mathrm{p}}_{n,\gamma,M}(I)$ from (377). Thus the affine force and every force-induced pressure coordinate are paid inside this finite forced record.

The initial face, complete dependency cover, two projections, and prescribed affine source now form the input to the forced datum construction. Applied to this one graph and this one history, it supplies, for every $n=0,\dots,N$,

$$V_n\in L^2(I;H^{M+1}), \qquad V_{n+1}\in L^2(I;H^{M-1}).$$

These are the memberships in (385).

For this fixed finite pair, form the sum of their defining squared norms:

$$C^f_{N,M}[u_0,\nu,f;I] := \sum_{n=0}^{N}\Big(\|V_n\|^2_{L^2(I;H^{M+1})} + \|V_{n+1}\|^2_{L^2(I;H^{M-1})}\Big) < \infty.$$

These norms give

$$\mathfrak{R}_{N,M}(u;T_*) = C^f_{N,M}[u_0,\nu,f;I] < \infty. \tag{391}$$

This proves (387). The record also retains the pressure and prescribed-force coordinates constrained by (389).

Fix $n,m,\alpha$. Choosing $N\ge n$ and $M\ge m+|\alpha|$ gives the stated velocity and mixed-velocity memberships. The normalized Riesz formula and Theorem 13.8 give the pressure membership,

including its prescribed source potential, while the prescribed force class gives $F_{n,\alpha} \in C([0,T_*];H^m)$. The temporal recurrence, spatial Leibniz rule, pressure normalization, and force jet commute with deletion of the outer indices. Overlap uniqueness identifies every retained coordinate with the corresponding derivative of the original maximal history. Hence each enlarged rectangle restricts to the smaller one, so the finite records satisfy the literal restriction identity (388) on $I$. □

The datum-generated blocks admit two all-orders readings. Ranging over their mixed indices gives the following intersection. The next section independently derives the completed-height identities and uses their coordinates to recover the spatial column before the forced recurrence reconstructs those mixed rows.

**Corollary 13.11** (Mixed-index exhaustion of the forced finite rectangles)**.** *Under the hypotheses of Theorem 13.10,*

$$u \in \bigcap_{r,m\in\mathbb{N}_0} H^r(I;H^m_\sigma(\mathbb{R}^3)), \qquad I=(0,T_*). \tag{392}$$

*This is the direct mixed-index exhaustion of the datum-generated family* $\big(\mathcal{R}_{N,M}[u_0,\nu,f;T_*]\big)_{N,M\in\mathbb{N}_0}$.

*Proof.* Fix finite $r,m$. A rectangle with temporal depth at least $r$ and spatial height at least $m$ contains

$$\sum_{n=0}^{r} \|V_n\|^2_{L^2(I;H^m)} < \infty.$$

The recurrence identifies $V_n$ with the weak derivative $\partial_t^n u$, which proves (392). Choosing the spatial height higher by $|\alpha|$ gives every $J_{n,\alpha}$. The finite-row pressure estimate gives every $\Pi_{n,\alpha}$, and the prescribed force class gives every $F_{n,\alpha}$. Equation (388) identifies these rows as literal common coordinates of every sufficiently large block on this one forced history. □

**Definition 13.12** (Finite pressure-complete forced rectangle)**.** Let $0<T<T_*$ and $N,M\in\mathbb{N}_0$. Define

$$\mathcal{R}_{N,M}[u_0,\nu,f;T] := \big((V_n|_{(0,T)})_{n=0}^N, (V_{n+1}|_{(0,T)})_{n=0}^N; (Q_n|_{(0,T)})_{n=0}^N; (F_n|_{(0,T)})_{n=0}^N\big)\,,$$

with every row constrained by Proposition 13.4. Its associated scalar norms are

$$\mathfrak{R}_{N,M}(u;T) := \sum_{n=0}^{N} \|V_n\|^2_{L^2(0,T;H^{M+1})} + \sum_{n=0}^{N} \|V_{n+1}\|^2_{L^2(0,T;H^{M-1})}\,, \tag{393}$$

$$\mathfrak{Q}_{N,M}(p;T) := \sum_{n=0}^{N} \Big(\|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})}\Big)\,, \tag{394}$$

$$\mathfrak{F}_{N,M}(f;T) := \sum_{n=0}^{N} \Big(\|F_n\|_{L^1(0,T;H^{M-1})} + \mathfrak{F}^{\mathrm{p}}_{n,0,M}(0,T)\Big)\,. \tag{395}$$

**Theorem 13.13** (Compatible forced rectangle norms)**.** *Under the hypotheses of Theorem 13.10, every finite pair* $(N,M)$ *satisfies*

$$\mathfrak{R}^*_{N,M}[u_0,\nu,f,T_*] := \sup_{0<T<T_*} \mathfrak{R}_{N,M}(u;T) < \infty. \tag{396}$$

*For every finite $n, m$, the adjacent coordinates obey*

$$V_n \in L^2(0,T_*;H^{m+1}), \qquad V_{n+1} \in L^2(0,T_*;H^{m-1}). \tag{397}$$

*The constant in* (396) *may depend on the selected finite orders. Every entry is the corresponding derivative of the original forced history.*

*Proof.* Put $I = (0,T_*)$. By Theorem 13.10, for $0 \le n \le N$,

$$V_n \in L^2(I;H^{M+1}), \qquad V_{n+1} \in L^2(I;H^{M-1}).$$

Their finite squared sum

$$C^f_{N,M} := \sum_{n=0}^{N} \left( \|V_n\|^2_{L^2(I;H^{M+1})} + \|V_{n+1}\|^2_{L^2(I;H^{M-1})} \right)$$

is finite. Restriction of each nonnegative integral gives

$$\mathfrak{R}_{N,M}(u;T) \le C^f_{N,M} \qquad (0 < T < T_*),$$

and proves (396). Selecting $(N,M) = (n,m)$ gives (397). Enlarging the indices and deleting the outer coordinates recovers the smaller block because every retained entry is that derivative of the prescribed forced history. □

The next section derives the critical-height and viscous relationships from this mixed jet. It evaluates those identities on these finite blocks $\mathcal{R}_{N,M}[u_0,\nu,f;T_*]$, exhausts their completed heights through the spatial Sobolev column, and returns through the forced recurrence to that mixed intersection.

# 14 Critical-height relationships on these rectangles

A finite rectangle is the velocity–pressure–force record selected at fixed temporal and spatial orders from the one forced history. Ranging over the mixed indices exhausts all of these finite records. Independently, differentiating the critical heights and squaring the viscous cell produces relationships on those same rectangle coordinates. Their completed heights recover every spatial Sobolev row, and the forced recurrence reconstructs that temporal and pressure intersection, with the nonlinear, pressure, and force faces retained.

**Proposition 14.1** (Kinetic-energy identity)**.** *For every $0 \le t < T_*$,*

$$\frac{1}{2}\|u(t)\|_2^2 + \nu \int_0^t \|\nabla u(s)\|_2^2 \, \mathrm{d}s = \frac{1}{2}\|u_0\|_2^2 + \int_0^t \langle f(s), u(s)\rangle_{L^2} \, \mathrm{d}s. \tag{398}$$

*Put*

$$K_f(T) := \|u_0\|_2 + \int_0^T \|\mathbb{P} f(s)\|_2 \, \mathrm{d}s.$$

*Consequently, for $0 < T < T_*$,*

$$\|u\|_{L^\infty(0,T;L^2)} \le K_f(T), \tag{399}$$

$$\|u\|^2_{L^2(0,T;H^1)} \le \left(T + \frac{1}{2\nu}\right) K_f(T)^2. \tag{400}$$

*Proof.* Take the $L^2$ pairing of the momentum equation with $u$. Integration by parts and $\nabla \cdot u = 0$ give

$$\langle (u \cdot \nabla)u, u \rangle_{L^2} = 0, \qquad \langle \nabla p, u \rangle_{L^2} = 0, \qquad \langle \Delta u, u \rangle_{L^2} = - \|\nabla u\|_2^2 .$$

Since $u$ is divergence free, $\langle f, u \rangle = \langle \mathbb{P} f, u \rangle$. Integration in time proves (398). Applying the identity first to $(\|u\|_2^2 + \varepsilon)^{1/2}$ and then letting $\varepsilon \downarrow 0$ gives $\|u(t)\|_2 \le K_f(t)$. Moreover,

$$\int_0^T K_f(t) \, \|\mathbb{P} f(t)\|_2 \, \mathrm{d}t = \frac{1}{2}\big(K_f(T)^2 - \|u_0\|_2^2\big),$$

so (398) gives $\int_0^T \|\nabla u\|_2^2 \, \mathrm{d}t \le K_f(T)^2/(2\nu)$. Adding $\int_0^T \|u(s)\|_2^2 \, \mathrm{d}s$ proves (400). □

Pairing a differentiated velocity equation with a spatial derivative of that row expresses the change of its Sobolev energy through viscous dissipation, the full transport–pressure contribution, and the prescribed force work. Repeated time differentiation then exposes higher spatial energies, with every differentiated nonlinear and force term retained in each identity.

**Definition 14.2** (Completed height rows)**.** For $r \in \mathbb{N}_0$ and $s \ge 0$, set

$$E_{r,s}(t) := \frac{1}{2} \, \|\Lambda^s V_r(t)\|_2^2 , \tag{401}$$

$$\mathcal{P}_{r,s}(t) := - \operatorname{Re} \left\langle \Lambda^s V_r(t), \Lambda^s \left( \sum_{a=0}^{r} \binom{r}{a} (V_a \cdot \nabla) V_{r-a} + \nabla Q_r \right)(t) \right\rangle_{L^2} . \tag{402}$$

$$\mathcal{W}_{r,s}(t) := \operatorname{Re} \langle \Lambda^s V_r(t), \Lambda^s F_r(t) \rangle_{L^2} . \tag{403}$$

For the base row write $E_s := E_{0,s}$, $\mathcal{P}_s := \mathcal{P}_{0,s}$, $\mathcal{W}_s := \mathcal{W}_{0,s}$, and $H := E_{1/2}$.

Differentiating this quadratic height along the classical history gives

$$H^{(r)}(t) = \frac{1}{2} \sum_{a=0}^{r} \binom{r}{a} \left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{r-a}(t) \right\rangle_{L^2} .$$

On every strict interval the mixed jet makes this contraction an identity of this forced history. The following recurrence exposes its spatial face without discarding the transfer or force work. For each finite $k$, choose a coordinate rectangle high enough to contain the required differentiated rows. Deriving the height identities on those rows gives a finite relationship rectangle containing $H^{(k)}$, every differentiated transfer and force-work term, and the spatial energy $E_{k+1/2}$. Compatibility under enlargement turns these finite identities into the completed-height exhaustion below.

**Proposition 14.3** (Completed height recurrence in every temporal row)**.** *For every strict classical interval $(0, T)$, every $r \in \mathbb{N}_0$, and every $s \ge 0$,*

$$\partial_t E_{r,s} = \mathcal{P}_{r,s} + \mathcal{W}_{r,s} - 2\nu E_{r,s+1}. \tag{404}$$

*For every $k \in \mathbb{N}$,*

$$\partial_t^k E_{r,1/2} = (-2\nu)^k E_{r,k+1/2} + \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} (\mathcal{P}_{r,j+1/2} + \mathcal{W}_{r,j+1/2}). \tag{405}$$

*In the base row $r = 0$,*

$$H^{(k)} = (-2\nu)^k E_{k+1/2} + \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} (\mathcal{P}_{j+1/2} + \mathcal{W}_{j+1/2}), \tag{406}$$

*and the highest spatial energy is the exact coordinate*

$$E_{k+1/2} = \frac{H^{(k)} - \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} (\mathcal{P}_{j+1/2} + \mathcal{W}_{j+1/2})}{(-2\nu)^k}. \tag{407}$$

*Proof.* Pair the temporal recurrence for $V_r$ with $\Lambda^{2s} V_r$. The pressure-complete nonlinear sum is exactly $\mathcal{P}_{r,s}$, the force pairing is exactly $\mathcal{W}_{r,s}$, and

$$\nu \operatorname{Re} \langle \Lambda^s V_r, \Lambda^s \Delta V_r \rangle_{L^2} = -\nu \left\| \Lambda^{s+1} V_r \right\|_2^2 = -2\nu E_{r,s+1}.$$

This proves (404). Induction on $k$ gives the stronger identity

$$\partial_t^k E_{r,s} = (-2\nu)^k E_{r,s+k} + \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} (\mathcal{P}_{r,s+j} + \mathcal{W}_{r,s+j}). \tag{408}$$

Indeed, differentiation of the order-$k$ identity followed by $\partial_t E_{r,s+k} = \mathcal{P}_{r,s+k} + \mathcal{W}_{r,s+k} - 2\nu E_{r,s+k+1}$ produces the order-$k+1$ identity. Taking $s = 1/2$ proves (405). Taking $r = 0$ gives (406); since $\nu > 0$, division by $(-2\nu)^k$ gives (407). □

**Definition 14.4** (Completed critical-height coordinates)**.** Set

$$R_0 := H = E_{1/2}, \qquad R_k := \frac{H^{(k)} - \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} (\mathcal{P}_{j+1/2} + \mathcal{W}_{j+1/2})}{(-2\nu)^k} \quad (k \in \mathbb{N}). \tag{409}$$

Proposition 14.3 gives the exact identity

$$R_k = E_{k+1/2} = \frac{1}{2} \left\| \Lambda^{k+1/2} u \right\|_2^2 \qquad (k \in \mathbb{N}_0). \tag{410}$$

**Proposition 14.5** (Complete forced rectangle cell)**.** *For every temporal row $n \in \mathbb{N}_0$, every $s \geq 0$, and every strict classical interval, the projected differentiated equation is*

$$\Lambda^s V_{n+1} + \nu \Lambda^{s+2} V_n = \Lambda^s \mathbb{P}(F_n - B_n), \qquad B_n := \sum_{a=0}^{n} \binom{n}{a} (V_a \cdot \nabla) V_{n-a}. \tag{411}$$

*Its square is the viscous diagonal identity*

$$\|\Lambda^s V_{n+1}\|_2^2 + \nu^2 \left\| \Lambda^{s+2} V_n \right\|_2^2 + \nu \frac{\mathrm{d}}{\mathrm{d}t} \left\| \Lambda^{s+1} V_n \right\|_2^2 = \|\Lambda^s \mathbb{P}(F_n - B_n)\|_2^2 . \tag{412}$$

*Before projection, the gradient face is orthogonal to both divergence-free faces. Consequently,*

$$\|V_{n+1}\|_{\dot{H}^s}^2 + \nu^2 \|\Delta V_n\|_{\dot{H}^s}^2 + \|\nabla Q_n\|_{\dot{H}^s}^2 + \nu \frac{\mathrm{d}}{\mathrm{d}t} \|\nabla V_n\|_{\dot{H}^s}^2 = \|F_n - B_n\|_{\dot{H}^s}^2 . \tag{413}$$

*Thus, for* $0 < S < T_*$,

$$\nu \left\|\Lambda^{s+1} V_n(S)\right\|_2^2 + \int_0^S \left( \|\Lambda^s V_{n+1}\|_2^2 + \nu^2 \left\|\Lambda^{s+2} V_n\right\|_2^2 + \|\Lambda^s \nabla Q_n\|_2^2 \right) \mathrm{d}t = \nu \left\|\Lambda^{s+1} V_n^0\right\|_2^2 + \int_0^S \|\Lambda^s (F_n - B_n)\|_2^2 \, \mathrm{d}t. \tag{414}$$

*These identities retain the temporal, viscous, pressure, transport, and force faces of the differentiated momentum row.*

*Proof.* Apply $\mathbb{P}\Lambda^s$ to the temporal row in Proposition 13.4. Since $-\Delta = \Lambda^2$, moving the viscous term to the left gives (411) with a positive sign. On squaring, the cross term is

$$2\nu \operatorname{Re} \left\langle \Lambda^s V_{n+1}, \Lambda^{s+2} V_n \right\rangle_{L^2} = \nu \frac{\mathrm{d}}{\mathrm{d}t} \left\|\Lambda^{s+1} V_n\right\|_2^2 ,$$

which proves (412). The complementary projection gives $\nabla Q_n = (I - \mathbb{P})(F_n - B_n)$. Orthogonality of $\mathbb{P}$ and $I - \mathbb{P}$ proves (413); integration in time gives (414). □

**Proposition 14.6** (Complete-equation estimates on finite rectangles)**.** *Assume the hypotheses of Theorem 13.13. The rectangle identities estimate the complete equation inside each selected finite rectangle. For every* $n \in \mathbb{N}_0$ *and* $s \geq 0$*, put*

$$\mathfrak{D}_{n,s}(S) := \nu \left\|\Lambda^{s+1} V_n(S)\right\|_2^2 + \int_0^S \left( \|\Lambda^s V_{n+1}\|_2^2 + \nu^2 \left\|\Lambda^{s+2} V_n\right\|_2^2 + \|\Lambda^s \nabla Q_n\|_2^2 \right) \mathrm{d}t. \tag{415}$$

*Then*

$$\sup_{0<S<T_*} \mathfrak{D}_{n,s}(S) \leq \nu \left\|\Lambda^{s+1} V_n^0\right\|_2^2 + \|\Lambda^s (F_n - B_n)\|_{L^2(0,T_*;L^2)}^2 < \infty. \tag{416}$$

*Thus the selected rectangle bounds, on that terminal interval, its adjacent temporal face, viscous face, full binomial nonlinear row, normalized pressure-gradient face, and prescribed-force face.*

*Proof.* Let $I = (0, T_*)$. Fix $(n, s)$. Choose a datum-generated finite rectangle high enough to contain every velocity factor in $B_n$ at spatial order $s + 2$. The adjacent trace lemma gives the required $L_t^\infty H_x^{s+2}$ factor, while the higher spatial face gives the other factor in $L_t^2 H_x^{s+2}$. Sobolev multiplication, summed over the finite binomial row, gives the explicit same-history estimate

$$\|B_n\|_{L^2(I;\dot{H}^s)} \leq C_s \sum_{a=0}^{n} \binom{n}{a} \|V_a\|_{L^\infty(I;H^{s+2})} \|V_{n-a}\|_{L^2(I;H^{s+2})} < \infty. \tag{417}$$

The prescribed Schwartz force gives $F_n \in L^2(I; \dot{H}^s)$. Now evaluate the independently derived positive-sign identity (414) on these selected coordinates. Its right-hand side is finite, and taking the supremum over $S < T_*$ proves (416). The pressure term is present because (413) is the unprojected orthogonal completion of that row. Every quantity in the resulting estimate is a coordinate of that forced equation row. □

**Proposition 14.7** (Complete momentum rows inside one finite rectangle)**.** *Assume the hypotheses of Theorem 13.13. Fix $N \in \mathbb{N}_0$, $M \geq 2$, and $0 < T \leq T_*$, and put*

$$R_v := \sum_{n=0}^{N} \left( \|V_n\|^2_{L^2(0,T;H^{M+1})} + \|V_{n+1}\|^2_{L^2(0,T;H^{M-1})} \right),$$
$$I_{N,M} := \sum_{n=0}^{N} \left\|V_n^0\right\|^2_{H^M}.$$

*Then the adjacent faces of this same rectangle give*

$$\sum_{n=0}^{N} \sup_{0\leq t\leq T} \|V_n(t)\|^2_{H^M} \leq I_{N,M} + R_v, \tag{418}$$

*and the complete binomial nonlinear rows obey*

$$\sum_{n=0}^{N} \|B_n\|^2_{L^2(0,T;H^{M-1})} \leq C_{N,M}(I_{N,M} + R_v)R_v. \tag{419}$$

*For $0 < S \leq T$, define the pressure-complete dissipation record*

$$\mathcal{D}_{N,M}(S) := \sum_{n=0}^{N} \Big[ \nu \|V_n(S)\|^2_{\dot{H}^M} + \|V_{n+1}\|^2_{L^2(0,S;\dot{H}^{M-1})} + \nu^2 \|V_n\|^2_{L^2(0,S;\dot{H}^{M+1})} + \|\nabla Q_n\|^2_{L^2(0,S;\dot{H}^{M-1})} \Big].$$

*The integrated complete rows satisfy*

$$\mathcal{D}_{N,M}(S) \leq \nu I_{N,M} + 2\sum_{n=0}^{N} \|F_n\|^2_{L^2(0,T;H^{M-1})} + 2C_{N,M}(I_{N,M} + R_v)R_v. \tag{420}$$

*Thus one finite velocity rectangle, together with the displayed prescribed force coordinates, bounds on that interval and in those equations its adjacent temporal rows, viscous Laplacians, complete nonlinear rows, and normalized pressure gradients. Lower spatial orders follow by Sobolev nesting from a rectangle with $M \geq 2$.*

*Proof.* For $0 \leq n \leq N$, the Hilbert triple $H^{M+1} \subset H^M \subset H^{M-1}$ and $\partial_t V_n = V_{n+1}$ give

$$\|V_n(t)\|^2_{H^M} - \left\|V_n^0\right\|^2_{H^M} = 2\,\mathrm{Re}\int_0^t \left\langle (1-\Delta)^{(M+1)/2} V_n, (1-\Delta)^{(M-1)/2} V_{n+1} \right\rangle_{L^2} \mathrm{d}s.$$

Cauchy–Schwarz in time and $2ab \leq a^2 + b^2$, followed by summation in $n$, prove (418). Sobolev multiplication applied to every allocation in $B_n$, using one factor in $L^\infty(0,T;H^M)$ and the other in $L^2(0,T;H^{M+1})$, proves (419). Finally, sum (414) over $0 \leq n \leq N$ at $s = M-1$, use $\|F_n - B_n\|^2 \leq 2\|F_n\|^2 + 2\|B_n\|^2$, and substitute (419). This gives (420). The identities

$$\nabla Q_n = (I - \mathbb{P})(F_n - B_n), \qquad V_{n+1} = \nu\Delta V_n - \mathbb{P}B_n + \mathbb{P}F_n$$

show directly that these are the terms of that pressure-complete row. □

The finite sum above evaluates the coordinates selected by Definition 13.12. Theorem 13.13 records their finite norm, while the complete-equation estimate retains and relates every face of those coordinates. The next proposition records both directions between adjacent-row membership and its finite rectangle norm, together with monotone exhaustion of the interval.

**Proposition 14.8** (Norm equivalence and full-interval realization)**.**

*For the fixed maximal history, the family of bounds* (396) *is equivalent to the adjacent-row statement* (397)*. Whenever either formulation holds, for every $N, M \in \mathbb{N}_0$,*

$$\sum_{n=0}^{N} \left( \|V_n\|^2_{L^2(0,T_*;H^{M+1})} + \|V_{n+1}\|^2_{L^2(0,T_*;H^{M-1})} \right) = \lim_{T \uparrow T_*} \mathfrak{R}_{N,M}(u;T) = \mathfrak{R}^*_{N,M}[u_0, \nu, f, T_*]. \tag{421}$$

*Every row is the corresponding distributional derivative of the single pressure-normalized maximal history.*

*Proof.* Assume first (397). For fixed $N, M$, set

$$C_{N,M} := \sum_{n=0}^{N} \left( \|V_n\|^2_{L^2(0,T_*;H^{M+1})} + \|V_{n+1}\|^2_{L^2(0,T_*;H^{M-1})} \right).$$

The sum is finite because it contains finitely many instances of (397). Restriction from $(0, T_*)$ to $(0, T)$ gives

$$\mathfrak{R}_{N,M}(u;T) \le C_{N,M} \qquad (0 < T < T_*),$$

and hence (396). Every term in $\mathfrak{R}_{N,M}(u;T)$ is the integral of a nonnegative function, so monotone exhaustion of $(0, T_*)$ gives (421).

Conversely, assume (396). This monotone exhaustion gives the finite full-interval sum in (421). Taking $N = n$ and $M = m$ yields (397) for each finite $(n, m)$. Proposition 13.4 identifies every entry with the corresponding distributional derivative of the fixed pressure-normalized history. □

**Corollary 14.9** (Critical-height exhaustion and Sobolev recovery)**.** *For the datum-generated compatible finite rectangles of Theorem 13.10, every completed critical coordinate belongs to $L^1(0, T_*)$, and*

$$\mathscr{H}_k[u_0, \nu, f, T_*] := \|R_k\|_{L^1(0,T_*)} = \frac{1}{2} \left\| \Lambda^{k+1/2} u \right\|^2_{L^2(0,T_*;L^2)} \le \frac{1}{2} \mathfrak{R}^*_{0,k}[u_0, \nu, f, T_*] \qquad (k \in \mathbb{N}_0). \tag{422}$$

*For every $m \in \mathbb{N}_0$, the completed heights recover the spatial Sobolev row through*

$$\|u\|^2_{L^2(0,T_*;H^m)} \le 2^m \left( T_* K_f(T_*)^2 + 2 \|R_m\|_{L^1(0,T_*)} \right) \le 2^m \left( T_* K_f(T_*)^2 + \mathfrak{R}^*_{0,m}[u_0, \nu, f, T_*] \right). \tag{423}$$

*Thus the compatible critical-height exhaustion gives*

$$(R_k)_{k \in \mathbb{N}_0} \in \prod_{k \in \mathbb{N}_0} L^1(0, T_*), \qquad u \in \bigcap_{m \in \mathbb{N}_0} L^2(0, T_*; H^m(\mathbb{R}^3)). \tag{424}$$

*Proof.* Take the $n = 0, m = k$ finite rectangle in (385); equivalently, take $N = 0$ and $M = k$ in (421). Since $|\xi|^{2k+1} \le \langle \xi \rangle^{2k+2}$,

$$\begin{aligned} 2 \|R_k\|_{L^1(0,T_*)} &= \left\| \Lambda^{k+1/2} u \right\|^2_{L^2(0,T_*;L^2)} \\ &\le \|u\|^2_{L^2(0,T_*;H^{k+1})} \le \mathfrak{R}^*_{0,k}[u_0, \nu, f, T_*], \end{aligned}$$

which proves (422). For $|\xi| \le 1$, $\langle \xi \rangle^{2m} \le 2^m$; for $|\xi| > 1$, $\langle \xi \rangle^{2m} \le 2^m |\xi|^{2m+1}$. Hence

$$\|v\|^2_{H^m} \le 2^m \left( \|v\|^2_2 + \left\| \Lambda^{m+1/2} v \right\|^2_2 \right).$$

Integrating this inequality, using (399) and (422), proves (423). Since $k$ and $m$ are arbitrary finite indices, (424) follows. □

**Proposition 14.10** (Forced spatial-column reconstruction of the mixed intersection)**.** *Let $(T_*, u, p)$ be the pressure-normalized forced history generated by $(u_0, \nu, f)$, with $T_* < \infty$, and put $I = (0, T_*)$. If*

$$u \in \bigcap_{m \in \mathbb{N}_0} L^2(I; H^m_\sigma(\mathbb{R}^3)), \tag{425}$$

*then $V_n, Q_n \in C([0, T_*]; H^m)$ for every finite $n, m$, and (392) follows for this same forced history.*

*Proof.* Set $S_m := \|u\|_{L^2(I;H^m)}$. Sobolev multiplication and the forced velocity equation give

$$\|u_t\|_{L^1(I;H^m)} \le \nu\sqrt{T_*}\, S_{m+2} + C_m S^2_{m+2} + \|\mathbb{P}f\|_{L^1(I;H^m)} < \infty. \tag{426}$$

Thus $u \in W^{1,1}(I; H^m) \hookrightarrow C([0, T_*]; H^m)$, with

$$\sup_{0\le t\le T_*} \|u(t)\|_{H^m} \le M_{0,m} := \|u_0\|_{H^m} + \nu\sqrt{T_*}\, S_{m+2} + C_m S^2_{m+2} + \|\mathbb{P}f\|_{L^1(I;H^m)}\,. \tag{427}$$

Suppose $V_0, \dots, V_n$ are continuous on $[0, T_*]$ in every finite Sobolev order. The complete velocity row gives the finite recursive bound

$$M_{n+1,m} := \nu M_{n,m+2} + C_m \sum_{a=0}^{n} \binom{n}{a} M_{a,m+2} M_{n-a,m+2} + \sup_{[0,T_*]} \|\mathbb{P}F_n\|_{H^m} < \infty. \tag{428}$$

On the interior this continuous expression is $\partial_t V_n$; integration identifies its one-sided endpoint derivatives. Induction gives

$$V_n \in C([0, T_*]; H^m), \qquad \sup_{[0,T_*]} \|V_n\|_{H^m} \le M_{n,m} \quad (n, m < \infty). \tag{429}$$

Simultaneously, the normalized pressure formula

$$Q_n = R_i R_j \sum_{a=0}^{n} \binom{n}{a} (V_a)_i (V_{n-a})_j - (-\Delta)^{-1} \nabla \cdot F_n$$

and (334) give $Q_n \in C([0, T_*]; H^m)$ at every finite order. Since the rows are the actual temporal derivatives of $u$,

$$\|u\|^2_{H^r(I;H^m)} \le T_* \sum_{n=0}^{r} M^2_{n,m} < \infty. \tag{430}$$

Taking all finite $(r, m)$ proves (392). □

The preceding calculation recovers that all-orders membership through the completed heights. For each finite $k$, choose a sufficiently high coordinate rectangle and derive the corresponding critical-height relationship rectangle on its rows. This computes $R_k = E_{k+1/2}$, with every differentiated force-work term retained in its recurrence. Compatibility under enlargement identifies the family $(R_k)_{k\ge 0}$ on the original forced history. Taking every finite $k$ gives (424). Proposition 14.10 then reconstructs every temporal and pressure row and recovers (392). Conversely, the mixed-index exhaustion contains the velocity rows from which every finite completed height is computed. Both descriptions use this datum-generated block $\mathcal{R}_{N,M}[u_0, \nu, f; T_*]$ at each finite pair and identify identical velocity, pressure, and force derivatives.

**Proposition 14.11** (Pressure completion)**.** *Under the hypotheses of Theorem 13.13, for every* $N, M \in \mathbb{N}_0$*,*

$$\mathfrak{Q}^*_{N,M}[u_0,\nu,f,T_*] := \sup_{0<T<T_*} \mathfrak{Q}_{N,M}(p;T) \le C^{\mathrm{p}}_M(2^{N+1}-1)\mathfrak{R}^*_{N,M}[u_0,\nu,f,T_*] + \sum_{n=0}^{N} \mathfrak{F}^{\mathrm{p}}_{n,0,M}(0,T_*). \tag{431}$$

*For every* $0 \le n \le N$*, the identities*

$$Q_n = R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i} V_{n-a,j} - (-\Delta)^{-1} \nabla \cdot F_n, \tag{432}$$

$$V_{n+1} = \nu \Delta V_n - \sum_{a=0}^{n} \binom{n}{a} (V_a \cdot \nabla) V_{n-a} - \nabla Q_n + F_n \tag{433}$$

*hold respectively in* $L^1(0,T_*;H^M)$ *and* $L^1(0,T_*;H^{M-1})$*, including when* $M = 0$*.*

*Proof.* For every $0 < T < T_*$, Theorem 13.8 gives

$$\|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})} \le C^{\mathrm{p}}_M 2^n \mathfrak{R}^*_{N,M}[u_0,\nu,f,T_*] + \mathfrak{F}^{\mathrm{p}}_{n,0,M}(0,T).$$

Sum over $0 \le n \le N$, then let $T \uparrow T_*$. Monotone convergence in the nonnegative norm integrals proves (431). Since $\nabla \cdot V_a = 0$,

$$(V_a \cdot \nabla) V_{n-a} = \nabla \cdot (V_{n-a} \otimes V_a).$$

Lemma 12.4 and Cauchy–Schwarz in time give

$$\|V_{n-a} \otimes V_a\|_{L^1(0,T_*;H^M)} \le C^{\mathrm{alg}}_M \|V_a\|_{L^2(0,T_*;H^{M+1})} \|V_{n-a}\|_{L^2(0,T_*;H^{M+1})}.$$

Thus every transport term belongs to $L^1(0,T_*;H^{M-1})$, also for $M = 0$. Moreover,

$$\|\Delta V_n\|_{L^1 H^{M-1}} \le T_*^{1/2} \|V_n\|_{L^2 H^{M+1}}.$$

Theorem 13.13 places $V_{n+1}$ in $L^2(0,T_*;H^{M-1})$, hence in $L^1(0,T_*;H^{M-1})$. The prescribed row $F_n$ belongs to $L^1(0,T_*;H^{M-1})$, and its pressure potential belongs to the stated spaces by (334). The strict-interval identities are identities of distributions; their full-interval representatives belong to the stated spaces. This proves the two identities on $(0,T_*)$. □

**Definition 14.12** (Finite whole-terminal blocks)**.** For $N, M \in \mathbb{N}_0$, the full-interval block already defined in (384) has, for $0 \le n \le N$,

$$\begin{aligned} &V_n \in L^2(0,T_*;H^{M+1}), && V_{n+1} \in L^2(0,T_*;H^{M-1}),\\ &Q_n \in L^1(0,T_*;H^M), && \nabla Q_n \in L^1(0,T_*;H^{M-1}),\\ &F_n \in C([0,T_*];H^{M-1}) \subset L^1(0,T_*;H^{M-1}). \end{aligned} \tag{434}$$

Its finite velocity norm is given by Theorem 13.13, its pressure rows by Proposition 14.11, and its force rows by the prescribed force class. Every row is an actual derivative of the one forced history generated by $(u_0,\nu,f)$.

**Proposition 14.13** (Restriction compatibility)**.** *For $N' \le N$ and $M' \le M$,*

$$\mathcal{R}_{N,M}[u_0,\nu,f;T_*]|_{(N',M')} = \mathcal{R}_{N',M'}[u_0,\nu,f;T_*]. \tag{435}$$

*Here restriction deletes the temporal rows with index larger than $N'$ and reads each retained distribution in the lower Sobolev order $M'$. If $(N'',M'') \le (N',M') \le (N,M)$, two successive restrictions retain exactly the coordinates obtained by restricting directly to $(N'',M'')$. Thus the datum-generated blocks are a compatible family of literal coordinates of one forced history.*

*Proof.* All retained entries are the fixed distributions $\partial_t^n u$, $\partial_t^{n+1} u$, $\partial_t^n p$, and $\partial_t^n f$. The Sobolev embeddings preserve those distributions. Deleting rows in one or two stages leaves an identical tuple. □

# 15 All-orders regularity, endpoint jet, and restart

The mixed-index and critical-height exhaustions have now identified that all-orders membership of the forced maximal history. The endpoint argument no longer distinguishes their coordinates. For every finite spatial order, the common family contains the adjacent velocity, pressure, and force rows needed to reach $T_*$, recover one endpoint jet, and restart the history with that prescribed force.

**Definition 15.1** (Direct all-orders reading of the finite blocks)**.** For a proposed finite maximal time, consider the datum-generated family

$$\big(\mathcal{R}_{N,M}[u_0,\nu,f;T_*]\big)_{N,M\in\mathbb{N}_0}. \tag{436}$$

For $N' \le N$ and $M' \le M$, these blocks satisfy

$$\mathcal{R}_{N,M}[u_0,\nu,f;T_*]|_{(N',M')} = \mathcal{R}_{N',M'}[u_0,\nu,f;T_*]. \tag{437}$$

Every mixed coordinate occurs in a sufficiently large finite block, and every common coordinate is that derivative of the one history generated by $(u_0,\nu,f)$.

**Theorem 15.2** (Compatible all-orders realization)**.**
*Let $\nu > 0$, $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, prescribe $f \in C^\infty([0,\infty);\mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$, and let $(T_*,u,p)$ be the maximal pressure-complete classical history generated by $(u_0,\nu,f)$. If $T_* < \infty$, then*

$$\mathcal{R}_{N,M}[u_0,\nu,f;T_*] = \big((V_n)_{n=0}^N,(V_{n+1})_{n=0}^N;(Q_n)_{n=0}^N;(F_n)_{n=0}^N\big) \tag{438}$$

*at every finite pair $(N,M)$, and larger blocks restrict as in* (437)*. All rows are restrictions of the original pressure-complete history. The velocity, pressure, and force rows retain the bounds obtained by either all-orders exhaustion:*

$$V_n \in \bigcap_{m\in\mathbb{N}_0} L^2(0,T_*;H^m_\sigma(\mathbb{R}^3)), \tag{439}$$

$$Q_n \in \bigcap_{m\in\mathbb{N}_0} L^1(0,T_*;H^m(\mathbb{R}^3)), \tag{440}$$

$$F_n \in \bigcap_{m\in\mathbb{N}_0} C([0,T_*];H^m(\mathbb{R}^3)). \tag{441}$$

*Every finite pair selects the corresponding block $\mathcal{R}_{N,M}[u_0,\nu,f;T_*]$. As the finite indices range, these blocks exhaust*

$$u \in \bigcap_{r,m\in\mathbb{N}_0} H^r(0,T_*;H^m(\mathbb{R}^3)), \tag{442}$$

*and, for every $r,m\in\mathbb{N}_0$,*

$$\sum_{n=0}^{r} \|\partial_t^n u\|^2_{L^2(0,T_*;H^m)} \le \mathfrak{R}^*_{r,m}[u_0,\nu,f,T_*]. \tag{443}$$

*Proof.* The mixed-index exhaustion in Theorem 13.11 and the critical-height exhaustion in Corollary 14.9, followed by Proposition 14.10, give those velocity memberships in (439) and (442). The forced pressure recurrence gives (440), while Proposition 14.11 gives its finite quantitative bound. The membership (441) follows directly from the prescribed class $f\in C^\infty([0,\infty);\mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$. For each finite $N,M$, take the corresponding datum-generated block. Theorem 13.13 and Proposition 14.11 give its finite norm bounds. Proposition 14.13 says that every common row is literally one distribution in every larger block. Every $V_n,Q_n,F_n$ appears in a sufficiently deep block, and every spatial derivative is fixed by its distributional velocity, pressure, or force row. The full family consequently gives identical rows in either all-orders exhaustion.

Fix finite $r,m$, put $I=(0,T_*)$, and choose $N\ge r$, $M\ge m$. The upper rows of that finite record belong to $L^2(I;H^{M+1})\hookrightarrow L^2(I;H^m)$ for $0\le n\le r$. Proposition 13.4 identifies them as the distributional derivatives $V_n=\partial_t^n u$. Thus the finite record contains every term in the derivative-sum norm of $H^r(I;H^m)$. Taking $N=r$, $M=m$ gives

$$\begin{aligned}\|u\|^2_{H^r(I;H^m)} &= \sum_{n=0}^{r} \|V_n\|^2_{L^2(I;H^m)} \\ &\le \sum_{n=0}^{r} \|V_n\|^2_{L^2(I;H^{m+1})} \le \mathfrak{R}^*_{r,m}[u_0,\nu,f,T_*].\end{aligned}$$

This proves (443); each mixed Sobolev norm is recovered from a finite rectangle. The completed heights recover those spatial rows, and the viscous identities retain the complete forced equation on every such rectangle. □

**Proposition 15.3** (Complete spatial row and common endpoint)**.** *Let the prescribed datum and finite maximal history be as in Theorem 15.2. Then, for every $m\in\mathbb{N}_0$,*

$$u\in W^{1,1}(0,T_*;H^m_\sigma(\mathbb{R}^3)) \hookrightarrow C([0,T_*];H^m_\sigma(\mathbb{R}^3)). \tag{444}$$

*The endpoint values supplied at the different Sobolev orders are the images of one common state*

$$u_* := u(T_*) \in \bigcap_{m\in\mathbb{N}_0} H^m_\sigma(\mathbb{R}^3) = H^\infty_\sigma(\mathbb{R}^3). \tag{445}$$

*Proof.* Fix $m$. The zeroth temporal coordinate of (442) gives

$$u\in L^2(0,T_*;H^{m+2}).$$

The pressure-completed nonlinearity satisfies

$$\|\mathbb{P}\nabla\cdot(u\otimes u)\|_{H^m} \le C_m \|u\|^2_{H^{m+2}}$$

by Lemmas 12.4 and 12.5. It belongs to $L^1(0,T_*;H^m)$. This spatial row and $T_* < \infty$ give $\Delta u \in L^1(0,T_*;H^m)$, and the projected equation

$$u_t = \nu\Delta u - \mathbb{P}\nabla\cdot(u\otimes u) + \mathbb{P}f$$

places $u_t$ in $L^1(0,T_*;H^m)$, because the prescribed source is integrable there in every finite Sobolev order. Also $u \in L^1(0,T_*;H^m)$, so the Banach-valued fundamental theorem gives (444).

If $k \geq m$, the $H^k$- and $H^m$-continuous representatives are the original preterminal function after the embedding $H^k \hookrightarrow H^m$. Uniqueness of continuous representatives makes their endpoint values agree under that embedding. These compatible values define the single state in (445). Continuity of divergence from $H^{m+1}$ to $H^m$ gives $\nabla\cdot u_* = 0$. □

**Corollary 15.4** (Full-interval pressure-complete mixed jet)**.** *Let the prescribed datum and finite maximal history be as in Theorem 15.2. Then, for every $r,m,n \in \mathbb{N}_0$ and $\alpha \in \mathbb{N}_0^3$,*

$$u \in H^r(0,T_*;H^m_\sigma(\mathbb{R}^3)), \tag{446}$$

$$J_{n,\alpha} := \partial_x^\alpha V_n \in L^2(0,T_*;H^{m+1}(\mathbb{R}^3)), \tag{447}$$

$$J_{n+1,\alpha} \in L^2(0,T_*;H^{m-1}(\mathbb{R}^3)), \tag{448}$$

$$\Pi_{n,\alpha} := \partial_x^\alpha Q_n \in L^1(0,T_*;H^m(\mathbb{R}^3)). \tag{449}$$

*The following finite bounds hold:*

$$\sum_{j=0}^{r} \|V_j\|^2_{L^2(0,T_*;H^m)} \leq \mathfrak{R}^*_{r,m}[u_0,\nu,f,T_*], \tag{450}$$

$$\begin{aligned}\|J_{n,\alpha}\|^2_{L^2(0,T_*;H^{m+1})} &\quad + \|J_{n+1,\alpha}\|^2_{L^2(0,T_*;H^{m-1})}\\ &\leq \mathfrak{R}^*_{n,m+|\alpha|}[u_0,\nu,f,T_*],\end{aligned} \tag{451}$$

$$\|\Pi_{n,\alpha}\|_{L^1(0,T_*;H^m)} \leq C^{\mathrm{p}}_{m+|\alpha|}(2^{n+1}-1)\mathfrak{R}^*_{n,m+|\alpha|}[u_0,\nu,f,T_*] + \mathfrak{F}^{\mathrm{p}}_{n,\alpha,m}(0,T_*). \tag{452}$$

*Every mixed row satisfies the recurrences in Proposition 13.4 on $(0,T_*)$.*

*Proof.* Equations (446) and (450) follow from Theorem 15.2. The Fourier multiplier bound

$$\|\partial_x^\alpha h\|_{H^s} \leq \|h\|_{H^{s+|\alpha|}}$$

and the rectangle $(n, m+|\alpha|)$ give (447)–(451). Apply this multiplier bound to the pressure completion in Proposition 14.11 to obtain (449) and (452). Proposition 13.4 supplies the stated identities. □

**Theorem 15.5** (Compatible full-interval Sobolev traces)**.** *Let the prescribed datum and finite maximal history be as in Theorem 15.2. For $N, m \in \mathbb{N}_0$, define*

$$\mathfrak{E}_{N,m} := (1 + T_*^{-1})^{1/2} \big(\mathfrak{R}^*_{N,m}[u_0, \nu, f, T_*]\big)^{1/2}. \tag{453}$$

*If $0 \le n \le N$ and $\alpha \in \mathbb{N}_0^3$, then*

$$V_n \in C([0, T_*]; H^m_\sigma(\mathbb{R}^3)), \qquad \sup_{0 \le t \le T_*} \|V_n(t)\|_{H^m} \le \mathfrak{E}_{N,m}, \tag{454}$$

$$J_{n,\alpha} \in C([0, T_*]; H^m(\mathbb{R}^3)), \qquad \sup_{0 \le t \le T_*} \|J_{n,\alpha}(t)\|_{H^m} \le \mathfrak{E}_{N,m+|\alpha|}. \tag{455}$$

*There are unique values*

$$V_n^* := \lim_{t \uparrow T_*} V_n(t) \in H^\infty(\mathbb{R}^3), \tag{456}$$

$$J^*_{n,\alpha} := \lim_{t \uparrow T_*} J_{n,\alpha}(t) = \partial_x^\alpha V_n^* \in H^\infty(\mathbb{R}^3), \tag{457}$$

*such that, for $k \ge m$, the image of the $H^k$ trace under $H^k \hookrightarrow H^m$ is the $H^m$ trace. Quantitatively,*

$$\|V_n^*\|_{H^m} \le \mathfrak{E}_{n,m}, \tag{458}$$

$$\|V_n(t) - V_n^*\|_{H^m} \le (T_* - t)\mathfrak{E}_{n+1,m}, \qquad 0 \le t \le T_*, \tag{459}$$

$$\big\|J^*_{n,\alpha}\big\|_{H^m} \le \mathfrak{E}_{n,m+|\alpha|}, \tag{460}$$

$$\big\|J_{n,\alpha}(t) - J^*_{n,\alpha}\big\|_{H^m} \le (T_* - t)\mathfrak{E}_{n+1,m+|\alpha|}. \tag{461}$$

*In particular, the zeroth trace agrees with the common endpoint in Proposition 15.3, and*

$$u_* = V_0^* \in H^\infty_\sigma(\mathbb{R}^3), \qquad u \in \bigcap_{a,m \in \mathbb{N}_0} C^a([0, T_*]; H^m_\sigma(\mathbb{R}^3)), \qquad \partial_t^a u(T_*) = V_a^*. \tag{462}$$

*Proof.* For $0 \le n \le N$, choose the datum rectangle supplied by Theorem 13.13 that contains the two coordinates

$$V_n \in L^2(0, T_*; H^{m+1}), \qquad \partial_t V_n = V_{n+1} \in L^2(0, T_*; H^{m-1}).$$

That finite rectangle supplies both norms. With $A = (1 - \Delta)^{1/2}$, the adjacent-row identity is

$$\frac{\mathrm{d}}{\mathrm{d}t} \|V_n(t)\|^2_{H^m} = 2 \operatorname{Re} \big\langle A^{m-1} V_{n+1}(t), A^{m+1} V_n(t) \big\rangle_{L^2}.$$

Integrating from the initial time and applying Cauchy–Schwarz gives

$$\sup_{0 \le t \le T_*} \|V_n(t)\|^2_{H^m} \le \|V_n(0)\|^2_{H^m} + 2 \|V_{n+1}\|_{L^2(I;H^{m-1})} \|V_n\|_{L^2(I;H^{m+1})}, \qquad I = (0, T_*). \tag{463}$$

The initial norm is finite by Lemma 13.7. Lemma 12.7 supplies the continuous representative on $[0, T_*]$, justifies the identity by Fourier truncation, and gives the first membership in (454) and

$$\begin{aligned} \sup_{0 \le t \le T_*} \|V_n(t)\|^2_{H^m} &\le T_*^{-1} \|V_n\|^2_{L^2 H^m} + 2 \|V_n\|_{L^2 H^{m+1}} \|V_{n+1}\|_{L^2 H^{m-1}} \\ &\le (1 + T_*^{-1}) \mathfrak{R}^*_{N,m}[u_0, \nu, f, T_*]. \end{aligned}$$

This proves the bound in (454). Apply this lemma to $J_{n,\alpha}$, using (451), to prove (455).

For $k \geq m$, the $H^k$-continuous and $H^m$-continuous representatives equal the original preterminal distribution. After applying $H^k \hookrightarrow H^m$, uniqueness of continuous representatives makes them equal on $[0, T_*]$. Their endpoint values are compatible and define (456)–(457). Spatial differentiation is continuous between the corresponding finite Sobolev spaces, so $J^*_{n,\alpha} = \partial_x^\alpha V_n^*$.

Corollary 15.4 gives $V_n, V_{n+1} \in L^1(0, T_*; H^m)$ and $\partial_t V_n = V_{n+1}$ in distributions. The Banach-valued fundamental identity gives

$$V_n(t) - V_n(s) = \int_s^t V_{n+1}(\tau)\,\mathrm{d}\tau \quad \text{in } H^m, \qquad 0 \leq s \leq t \leq T_*. \tag{464}$$

The continuous representative of $V_{n+1}$ is bounded by $\mathfrak{E}_{n+1,m}$. Set $t = T_*$ in (464) to prove (459); spatial differentiation gives (461). The remaining size estimates follow from (454)–(455).

The divergence map $H^{m+1} \to H^m$ is continuous, so $\nabla \cdot u_* = 0$. For each $a, m$, the curves $V_0, \dots, V_{a+1}$ are continuous in $H^m$, and (464) identifies the derivative of $V_j$ with $V_{j+1}$. Induction in $j$ proves (462). □

**Theorem 15.6** (Pressure-complete endpoint jet)**.** *Let the prescribed datum and finite maximal history be as in Theorem 15.2. For $n \in \mathbb{N}_0$, define*

$$\Phi_n(t) := (-\Delta)^{-1} \nabla \cdot F_n(t).$$

*The prescribed force class gives $\Phi_n \in C^\infty([0,\infty); H^s)$ for every finite $n, s$. Define*

$$Q_n^* := R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i}^* V_{n-a,j}^* - \Phi_n(T_*). \tag{465}$$

*Then $Q_n^* \in H^\infty(\mathbb{R}^3)$, and, for every $s \in \mathbb{N}_0$,*

$$\|Q_n^*\|_{H^s} \leq 9 C_s^{\mathrm{alg}} 2^n \mathfrak{E}_{n,s+1}^2 + \|\Phi_n(T_*)\|_{H^s}\,, \tag{466}$$

$$\|Q_n(t) - Q_n^*\|_{H^s} \leq 9 C_s^{\mathrm{alg}} 2^{n+1} (T_* - t) \mathfrak{E}_{n+1,s+1}^2 \tag{467}$$

$$+ (T_* - t) \sup_{0 \leq r \leq T_*} \|\Phi_{n+1}(r)\|_{H^s}\,. \tag{468}$$

*For $\alpha \in \mathbb{N}_0^3$,*

$$\Pi_{n,\alpha}^* := \partial_x^\alpha Q_n^* = \lim_{t \uparrow T_*} \Pi_{n,\alpha}(t) \quad \text{in } H^m(\mathbb{R}^3) \text{ for every } m \in \mathbb{N}_0. \tag{469}$$

*More precisely, for every $m \in \mathbb{N}_0$,*

$$\big\|\Pi_{n,\alpha}^*\big\|_{H^m} \leq 9 C_{m+|\alpha|}^{\mathrm{alg}} 2^n \mathfrak{E}_{n,m+|\alpha|+1}^2 + \|\partial_x^\alpha \Phi_n(T_*)\|_{H^m}\,, \tag{470}$$

$$\big\|\Pi_{n,\alpha}(t) - \Pi_{n,\alpha}^*\big\|_{H^m} \leq 9 C_{m+|\alpha|}^{\mathrm{alg}} 2^{n+1} (T_* - t) \mathfrak{E}_{n+1,m+|\alpha|+1}^2 \tag{471}$$

$$+ (T_* - t) \sup_{0 \leq r \leq T_*} \|\partial_x^\alpha \Phi_{n+1}(r)\|_{H^m}\,. \tag{472}$$

*The endpoint rows satisfy, for every $n, q \in \mathbb{N}_0$,*

$$V_{n+1}^* = \nu \Delta V_n^* - \sum_{a=0}^{n} \binom{n}{a} (V_a^* \cdot \nabla) V_{n-a}^* - \nabla Q_n^* + F_n(T_*) \quad \text{in } H^q(\mathbb{R}^3). \tag{473}$$

*Among families rooted at $u_*$, carrying the prescribed values $(F_n(T_*))_{n\in\mathbb{N}_0}$, and satisfying* (465) *and* (473) *in every finite Sobolev space, $(V_n^*, Q_n^*)_{n\in\mathbb{N}_0}$ is the unique pressure-normalized endpoint jet. Moreover,*

$$p \in \bigcap_{a,m\in\mathbb{N}_0} C^a([0,T_*];H^m(\mathbb{R}^3)), \qquad \partial_t^a p(T_*) = Q_a^*. \tag{474}$$

*Proof.* Lemmas 12.4 and 12.5, followed by (454), give

$$\begin{aligned}\|Q_n^*\|_{H^s} &\le 9C_s^{\mathrm{alg}} \sum_{a=0}^{n} \binom{n}{a} \|V_a^*\|_{H^{s+1}} \left\|V_{n-a}^*\right\|_{H^{s+1}} + \|\Phi_n(T_*)\|_{H^s} \\ &\le 9C_s^{\mathrm{alg}} 2^n \mathfrak{E}_{n,s+1}^2 + \|\Phi_n(T_*)\|_{H^s}\,.\end{aligned}$$

This is (466). For each product in $Q_n(t) - Q_n^*$, insert $V_a^* V_{n-a}(t)$, use (459), and bound every retained factor by $\mathfrak{E}_{n+1,s+1}$. The two product differences and $\sum_{a=0}^n \binom{n}{a} = 2^n$ prove the product part of (468). Since $\partial_t \Phi_n = \Phi_{n+1}$,

$$\|\Phi_n(t) - \Phi_n(T_*)\|_{H^s} \le (T_* - t) \sup_{0\le r\le T_*} \|\Phi_{n+1}(r)\|_{H^s}\,,$$

which supplies the remaining term. The Fourier multiplier inequality

$$\|\partial_x^\alpha h\|_{H^m} \le \|h\|_{H^{m+|\alpha|}}$$

applied to the nonlinear Riesz-product component, together with the exact derivative $\partial_x^\alpha \Phi_n$ of the force potential, proves (470) and (472). The latter tends to zero as $t \uparrow T_*$, which proves (469) in every $H^m$.

Fix $n, q$. The trace of $V_n$ in $H^{q+2}$ carries the diffusion term in Proposition 13.4 to $H^q$ at $T_*$. The product estimate at order $q+1$, using velocity convergence in $H^{q+2}$, carries every transport term to $H^q$. Equation (468) at order $q+1$ carries the pressure gradient to $H^q$. The left side converges to $V_{n+1}^*$ in $H^q$, and $F_n(t) \to F_n(T_*)$ in $H^q$, which proves (473).

The value $V_0^* = u_*$ fixes $Q_0^*$, and the velocity recurrence fixes $V_1^*$. Inductively, $V_0^*, \dots, V_n^*$ fix $Q_n^*$, which fixes $V_{n+1}^*$. This proves uniqueness within the displayed recurrence class. Extend each $Q_n$ to $T_*$ by $Q_n(T_*) := Q_n^*$. On $(0, T_*)$, $Q_n = \partial_t^n p$, so $\partial_t Q_n = Q_{n+1}$ in distributions. Both rows are continuous in every $H^m$ by (468). Hence, for $0 \le s \le t \le T_*$,

$$Q_n(t) - Q_n(s) = \int_s^t Q_{n+1}(\tau)\,\mathrm{d}\tau \quad \text{in } H^m(\mathbb{R}^3). \tag{475}$$

Indeed, the identity holds for $t < T_*$, and the endpoint bounds permit passage to $t = T_*$. Thus $Q_n \in C^1([0,T_*];H^m)$ with derivative $Q_{n+1}$. Induction gives (474). □

**Theorem 15.7** (Same-forced-history global continuation)**.** *Let $\nu > 0$, $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, prescribe $f \in C^\infty([0,\infty);\mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$, and let $(T_*, u, p)$ be the maximal pressure-complete classical history generated by $(u_0, \nu, f)$. Then*

$$T_* = \infty. \tag{476}$$

*Proof.* Suppose $T_* < \infty$. Theorem 15.5 gives $u_* \in H^\infty_\sigma$. Theorem 12.11 then gives a unique smooth pressure-normalized pair $(w^*, q^*)$ on a nontrivial right-hand time neighborhood of $T_*$, driven by the original prescribed force $f(t)$, with $w^*(T_*) = u_*$. Define

$$W_n^+ := \partial_t^n w^*(T_*), \qquad P_n^+ := \partial_t^n q^*(T_*).$$

The restart datum is exactly

$$w^*(T_*) = u_* = u(T_*).$$

Differentiating the restarted equations exactly as in the proof of Proposition 13.4 gives

$$P_n^+ = R_i R_j \sum_{a=0}^{n} \binom{n}{a} W_{a,i}^+ W_{n-a,j}^+ - (-\Delta)^{-1} \nabla \cdot F_n(T_*),$$

$$W_{n+1}^+ = \nu \Delta W_n^+ - \sum_{a=0}^{n} \binom{n}{a} (W_a^+ \cdot \nabla) W_{n-a}^+ - \nabla P_n^+ + F_n(T_*).$$

Since $W_0^+ = u_* = V_0^*$, induction with (465) and (473) gives

$$W_n^+ = V_n^*, \qquad P_n^+ = Q_n^*, \qquad n \in \mathbb{N}_0. \tag{477}$$

Equations (462), (474), and (477) match every one-sided time derivative in every finite Sobolev space. Sobolev embedding then glues the incoming pair $(u, p)$ to $(w^*, q^*)$ through $T_*$ as one smooth pressure-complete history generated by that $(u_0, \nu, f)$. The pressure-complete equation holds at $T_*$ by (473), and the glued history continues $(u, p)$ past $T_*$. This contradicts maximality. Hence $T_* = \infty$. □

The mixed-index exhaustion of the compatible coordinate rectangles in Theorem 13.11 gives the full-interval intersection. Independently, the critical-height relationship rectangles derived on those same coordinates recover that intersection through Proposition 14.10. Their adjacent rows form the compatible velocity traces through $T_*$ in Theorem 15.5, while each finite rectangle carries the complete-equation identities and finite norm estimates. The pressure recurrence fixes their normalized endpoint jet in Theorem 15.6. With those endpoint values, Theorem 15.7 extends the original history past $T_*$, so $T_* = \infty$. Theorem 15.8 collects the global solution, pressure normalization, and energy identity.

**Theorem 15.8** (Global smoothness and uniqueness for prescribed smooth data)**.** *For every $\nu > 0$, every $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and every prescribed $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$, there exists a pressure-normalized pair*

$$(u, p) \in C^\infty([0,\infty) \times \mathbb{R}^3; \mathbb{R}^3 \times \mathbb{R}) \tag{478}$$

*satisfying*

$$\partial_t u + (u \cdot \nabla) u + \nabla p = \nu \Delta u + f, \tag{479}$$

$$\nabla \cdot u = 0, \qquad u(0) = u_0, \tag{480}$$

$$p(t) = R_i R_j (u_i(t) u_j(t)) - (-\Delta)^{-1} \nabla \cdot f(t), \qquad \lim_{|x| \to \infty} p(t,x) = 0, \tag{481}$$

$$\frac{1}{2} \|u(t)\|_{L^2}^2 + \nu \int_0^t \|\nabla u(s)\|_{L^2}^2 \, \mathrm{d}s = \frac{1}{2} \|u_0\|_{L^2}^2 + \int_0^t \langle f(s), u(s) \rangle_{L^2} \, \mathrm{d}s, \qquad t \geq 0. \tag{482}$$

*In particular, for every finite $t$,*

$$\|u(t)\|_{L^2(\mathbb{R}^3)} \leq \|u_0\|_{L^2} + \int_0^t \|\mathbb{P} f(s)\|_{L^2} \, \mathrm{d}s < \infty. \tag{483}$$

*For every finite $T > 0$, its restriction is the unique member of $\mathscr{M}^3_{\nu,f}(0, T; u_0)$.*

*Proof.* Let

$$\mathscr{T} := \left\{ \tau > 0 : \begin{array}{c} \text{there exists a pressure-complete classical history} \\ (\tau, u^\tau, p^\tau) \text{ generated by } (u_0, \nu, f) \end{array} \right\},$$

and put

$$T_* := \sup \mathscr{T}.$$

Theorem 12.11, applied to $u_0 \in H^\infty_\sigma$, supplies a shifted mild solution whose persistence statement gives every finite velocity and pressure row. Sobolev embedding and the displayed recurrences make this pair a pressure-complete classical history, so $\mathscr{T}$ is nonempty. Lemma 12.10 admits each strict classical restriction to the shifted mild class, whose at-most-one property identifies any two such solutions on their common interval; their union defines a unique solution on $[0, T_*)$. Since $u_0 \in H^\infty_\sigma$, the persistence conclusion of Theorem 12.11, applied on overlapping local intervals, places every strict restriction in the class of Definition 13.2. An extension beyond $T_*$ would restrict to an interval $[0, \tau]$ with $\tau > T_*$, contrary to the definition of the supremum. Thus $(T_*, u, p)$ is maximal. Theorem 15.7 gives $T_* = \infty$.

Fix $T < \infty$. Persistence in Theorem 12.11, iterated along the unique history, retains every finite spatial Sobolev order on $[0, T]$. Proposition 13.4 retains every finite temporal row. Lemma 12.2 gives (478); arbitrariness of $T$ gives the global class. The pressure belongs to $H^s(\mathbb{R}^3)$ for every $s$, and the Fourier inversion proof of Lemma 12.2 places its continuous representative in $C_0(\mathbb{R}^3)$, which proves the limit in (481). Lemma 12.10 places $(u, p)|_{[0,T]}$ in $\mathscr{M}^3_{\nu,f}(0, T; u_0)$. If $(\widetilde{u}, \widetilde{p}) \in \mathscr{M}^3_{\nu,f}(0, T; u_0)$, the at-most-one assertion in Theorem 12.11 gives $(\widetilde{u}, \widetilde{p}) = (u, p)$ on $[0, T]$. This proves the stated uniqueness.

Proposition 14.1 holds on every finite interval. Since $T$ was arbitrary, it gives (482) for every $t \geq 0$; its regularized square-root estimate gives (483). □

# 16 Matrix forms of the completed intersection

The mixed indices exhaust the compatible coordinate rectangles of the single forced mixed jet. Independently, the completed critical heights exhaust the relationship rectangles derived on those coordinates and recover that all-orders intersection. The positive-sign viscous cell relates the faces of every finite coordinate rectangle in that family. The matrices in this section display how the velocity, nonlinear, pressure, viscous, and prescribed-force coordinates remain coupled inside each such rectangle.

**Proposition 16.1** (Mixed-index transport matrix and pressure completion)**.** *Let*

$$\Gamma := \mathbb{N}_0 \times \mathbb{N}_0^3, \qquad \kappa = (n, \alpha), \qquad \lambda = (a, \beta),$$

*and write*

$$\lambda \preceq \kappa \quad \Longleftrightarrow \quad a \leq n \ \text{ and } \ \beta \leq \alpha, \qquad \kappa - \lambda := (n - a, \alpha - \beta), \qquad \binom{\kappa}{\lambda} := \binom{n}{a}\binom{\alpha}{\beta}.$$

*Set*

$$J_\kappa := J_{n,\alpha}, \qquad \Pi_\kappa := \Pi_{n,\alpha}, \qquad F_\kappa := F_{n,\alpha}, \qquad \Psi_\kappa := (-\Delta)^{-1} \nabla \cdot F_\kappa, \qquad \mathbf{J} := (J_\lambda)_{\lambda \in \Gamma}.$$

*The block operator*

$$\mathbb{L}(J)_{\kappa,\lambda} := \begin{cases} \binom{\kappa}{\lambda}\mathbb{P}\big[(J_{\kappa-\lambda}\cdot\nabla)\cdot\big], & \lambda \preceq \kappa, \\ 0, & \lambda \not\preceq \kappa, \end{cases} \tag{484}$$

*has the finite block-row action*

$$[\mathbb{L}(J)\mathbf{J}]_\kappa = \sum_{\lambda\preceq\kappa}\binom{\kappa}{\lambda}\mathbb{P}\big[(J_\lambda\cdot\nabla)J_{\kappa-\lambda}\big]. \tag{485}$$

*Consequently the complete mixed row is the pair*

$$J_{n+1,\alpha} + [\mathbb{L}(J)\mathbf{J}]_\kappa = \nu\Delta J_\kappa + \mathbb{P}F_\kappa, \tag{486}$$

$$\Pi_\kappa = R_iR_j\sum_{\lambda\preceq\kappa}\binom{\kappa}{\lambda}J_{\lambda,i}J_{\kappa-\lambda,j} - \Psi_\kappa, \tag{487}$$

*while the unprojected velocity identity retains $\nabla\Pi_\kappa$ and the full force coordinate $F_\kappa$ exactly as in* (372).

*Order $\Gamma$ by increasing $n+|\alpha|$, with any fixed ordering among indices of equal depth. Then $\mathbb{L}(J)$ is block lower triangular. For*

$$\begin{aligned} \kappa! &:= n!\,\alpha!, & \alpha! &:= \alpha_1!\alpha_2!\alpha_3!, \\ \widehat{J}_\kappa &:= \frac{J_\kappa}{\kappa!}, & \widehat{\Pi}_\kappa &:= \frac{\Pi_\kappa}{\kappa!}, \\ \widehat{F}_\kappa &:= \frac{F_\kappa}{\kappa!}, & \widehat{\Psi}_\kappa &:= \frac{\Psi_\kappa}{\kappa!} = (-\Delta)^{-1}\nabla\cdot\widehat{F}_\kappa. \end{aligned}$$

*let $D$ be the diagonal matrix with $D_{\kappa,\kappa} = \kappa!$. On every finite downward-closed truncation of $\Gamma$,*

$$D^{-1}\mathbb{L}(D\widehat{J})D = \widehat{\mathbb{L}}(\widehat{J}), \qquad \widehat{\mathbb{L}}(\widehat{J})_{\kappa,\lambda} := \begin{cases} \mathbb{P}\big[(\widehat{J}_{\kappa-\lambda}\cdot\nabla)\cdot\big], & \lambda \preceq \kappa, \\ 0, & \lambda \not\preceq \kappa. \end{cases} \tag{488}$$

*Thus the factorial-normalized pressure-complete recurrence is*

$$(n+1)\widehat{J}_{n+1,\alpha} + \sum_{\lambda\preceq\kappa}(\widehat{J}_\lambda\cdot\nabla)\widehat{J}_{\kappa-\lambda} + \nabla\widehat{\Pi}_\kappa = \nu\Delta\widehat{J}_\kappa + \widehat{F}_\kappa, \tag{489}$$

$$\widehat{\Pi}_\kappa = R_iR_j\sum_{\lambda\preceq\kappa}\widehat{J}_{\lambda,i}\widehat{J}_{\kappa-\lambda,j} - \widehat{\Psi}_\kappa. \tag{490}$$

*The normalized transport blocks depend on the difference $\kappa-\lambda$; hence* (488) *is the block lower-triangular convolution form of the mixed transport array. The prescribed force is the affine row source, and its gradient component remains in the pressure coordinate through $\widehat{\Psi}_\kappa$.*

*Proof.* For $\lambda \preceq \kappa$, change the summation index from $\lambda$ to $\kappa-\lambda$ and use $\binom{\kappa}{\kappa-\lambda} = \binom{\kappa}{\lambda}$. This gives (485). Apply $\mathbb{P}$ to (372). The identities

$$\mathbb{P}J_{n+1,\alpha} = J_{n+1,\alpha}, \qquad \mathbb{P}\Delta J_\kappa = \Delta J_\kappa, \qquad \mathbb{P}\nabla\Pi_\kappa = 0, \qquad \mathbb{P}F_{n,\alpha} = \mathbb{P}F_\kappa$$

follow from (373) and the Fourier symbols. They prove (486); equation (487) is (371) in mixed-index notation. The paired projected and pressure formulas, including $-\Psi_\kappa$, retain every prescribed force

component: the solenoidal component remains in $\mathbb{P}F_\kappa$, while the gradient component is retained by $\Pi_\kappa$.

If $\lambda \preceq \kappa$, then $a + |\beta| \le n + |\alpha|$, which proves lower triangularity in the stated ordering. Finally,

$$\binom{\kappa}{\lambda} \lambda!\,(\kappa-\lambda)! = \kappa!.$$

Dividing the mixed velocity, pressure, and prescribed force rows by $\kappa!$ proves (489) and (490). This factorial cancellation in each operator block proves (488). Every row has $(n+1)\prod_{j=1}^{3}(\alpha_j+1)$ supported blocks, so all displayed row sums are finite. □

**Proposition 16.2** (Sobolev rectangle matrix and adjacent stencil)**.** *For $0 < T < T_*$, $N, M \in \mathbb{N}_0$, $0 \le n \le N$, and $m \in \mathbb{Z}$, set*

$$G_{n,m}(T) := \|V_n\|_{L^2(0,T;H^m)},$$

*and set*

$$\begin{aligned} P_{n,M}(T) &:= \|Q_n\|_{L^1(0,T;H^M)} + \|\nabla Q_n\|_{L^1(0,T;H^{M-1})}, \\ D_{n,M}(T) &:= \|F_n\|_{L^1(0,T;H^{M-1})} + \mathfrak{F}^{\mathrm{p}}_{n,0,M}(0,T). \end{aligned}$$

*The object matrix of the finite pressure-complete forced rectangle is*

$$\mathbf{R}_{N,M}[u_0,\nu,f;T] := \begin{pmatrix} V_0|_{(0,T)} & V_1|_{(0,T)} & Q_0|_{(0,T)} & F_0|_{(0,T)} \\ V_1|_{(0,T)} & V_2|_{(0,T)} & Q_1|_{(0,T)} & F_1|_{(0,T)} \\ \vdots & \vdots & \vdots & \vdots \\ V_N|_{(0,T)} & V_{N+1}|_{(0,T)} & Q_N|_{(0,T)} & F_N|_{(0,T)} \end{pmatrix}. \tag{491}$$

*The corresponding matrix of the two velocity norms, the pressure budget, and the prescribed-force budget is*

$$\mathbf{C}_{N,M}(T) := \begin{pmatrix} G_{0,M+1}(T) & G_{1,M-1}(T) & P_{0,M}(T) & D_{0,M}(T) \\ G_{1,M+1}(T) & G_{2,M-1}(T) & P_{1,M}(T) & D_{1,M}(T) \\ \vdots & \vdots & \vdots & \vdots \\ G_{N,M+1}(T) & G_{N+1,M-1}(T) & P_{N,M}(T) & D_{N,M}(T) \end{pmatrix}. \tag{492}$$

*If $\mathbf{G}_{N,M}(T)$ denotes its first two columns, then*

$$\|\mathbf{G}_{N,M}(T)\|_{\mathrm{F}}^2 = \mathfrak{R}_{N,M}(u;T), \qquad \sum_{n=0}^{N} P_{n,M}(T) = \mathfrak{Q}_{N,M}(p;T), \qquad \sum_{n=0}^{N} D_{n,M}(T) = \mathfrak{F}_{N,M}(f;T). \tag{493}$$

*For each $0 \le n \le N$, the first two entries in row $n$ form the adjacent Hilbert-scale stencil*

$$\left.\begin{aligned} (n, M+1):&\quad V_n \in L^2(0,T;H^{M+1}), \\ (n+1, M-1):&\quad V_{n+1} = \partial_t V_n \in L^2(0,T;H^{M-1}) \end{aligned}\right\} \longrightarrow (n,M):\quad V_n \in C([0,T];H^M). \tag{494}$$

*For every $m \in \mathbb{N}_0$, the Sobolev matrix measures the mixed-index array through the exact identity*

$$G_{n,m}(T)^2 = \sum_{\substack{\alpha \in \mathbb{N}_0^3 \\ |\alpha| \le m}} \frac{m!}{(m-|\alpha|)!\,\alpha!} \|J_{n,\alpha}\|^2_{L^2((0,T)\times\mathbb{R}^3)}. \tag{495}$$

*The pressure entry in row $n$ is $Q_n = \Pi_{n,0}$, and its spatial rows $\partial_x^\alpha Q_n = \Pi_{n,\alpha}$ are fixed by (487), including the force-pressure potential. The force entry is the prescribed row $F_n = \partial_t^n f$, whose spatial rows are $F_{n,\alpha} = \partial_x^\alpha F_n$; these are those force coordinates that occur in the mixed velocity and pressure rows.*

*Proof.* The object matrix is the tuple in Definition 13.12, arranged by temporal row. The three identities in (493) are (393) and (394)–(395). Lemma 12.7 gives (494). Finally, Plancherel and

$$(1+|\xi|^2)^m = \sum_{\substack{\alpha\in\mathbb{N}_0^3\\ |\alpha|\le m}} \frac{m!}{(m-|\alpha|)!\,\alpha!}\,\xi^{2\alpha}$$

give (495). The pressure and force claims follow from the definitions of $Q_n, \Pi_{n,\alpha}, F_n, F_{n,\alpha}$ and (487). □

**Proposition 16.3** (Full-interval closed estimate matrix)**.** *Under the hypotheses of Theorem 13.13, fix $N, M \in \mathbb{N}_0$ and set*

$$\begin{aligned} a_{n,M} &:= \|V_n\|_{L^2(0,T_*;H^{M+1})}\,,\\ b_{n,M} &:= \|V_{n+1}\|_{L^2(0,T_*;H^{M-1})}\,,\\ q_{n,M} &:= \|Q_n\|_{L^1(0,T_*;H^M)} + \|\nabla Q_n\|_{L^1(0,T_*;H^{M-1})}\,,\\ d_{n,M} &:= \|F_n\|_{L^1(0,T_*;H^{M-1})} + \mathfrak{F}^{\mathrm{p}}_{n,0,M}(0,T_*), \qquad 0\le n\le N. \end{aligned} \tag{496}$$

*Define entrywise order on nonnegative matrices by $A \preceq_{\mathrm{ent}} B$ when $A_{ij} \le B_{ij}$ for every entry. Then, for every $0 < T < T_*$,*

$$\mathbf{C}_{N,M}(T) \preceq_{\mathrm{ent}} \mathbf{C}^*_{N,M} := \begin{pmatrix} a_{0,M} & b_{0,M} & q_{0,M} & d_{0,M}\\ a_{1,M} & b_{1,M} & q_{1,M} & d_{1,M}\\ \vdots & \vdots & \vdots & \vdots\\ a_{N,M} & b_{N,M} & q_{N,M} & d_{N,M} \end{pmatrix}. \tag{497}$$

*Let $\mathbf{G}^*_{N,M}$ denote the first two columns of $\mathbf{C}^*_{N,M}$. Then*

$$\begin{aligned} \mathfrak{R}_{N,M}(u;T) &= \|\mathbf{G}_{N,M}(T)\|_{\mathrm{F}}^2\\ &\le \|\mathbf{G}^*_{N,M}\|_{\mathrm{F}}^2 = \sum_{n=0}^{N}\left(a_{n,M}^2 + b_{n,M}^2\right) = \mathfrak{R}^*_{N,M}[u_0,\nu,f,T_*] < \infty. \end{aligned} \tag{498}$$

*The other two columns satisfy the exact full-interval identities*

$$\sum_{n=0}^{N} q_{n,M} = \mathfrak{Q}^*_{N,M}[u_0,\nu,f,T_*] < \infty, \tag{499}$$

$$\sum_{n=0}^{N} d_{n,M} =: \mathfrak{F}^*_{N,M}[f,T_*] < \infty. \tag{500}$$

*Proof.* Proposition 14.8 makes every $a_{n,M}$ and $b_{n,M}$ finite. Proposition 14.11 makes every $q_{n,M}$ finite. The prescribed force class, together with (334), makes every $d_{n,M}$ finite. Restriction from $(0,T_*)$ to $(0,T)$ decreases each nonnegative coordinate norm and gives (497). Taking Frobenius norms and using (493) together with (421) proves (498). Monotone exhaustion of the pressure and prescribed-force columns gives (499) and (500). □

# 17 Quantitative consequences of the compatible rectangles

## 17.1 Completion in the global $H^1$ mild class

The forced construction begins with velocity data in $H^\infty_\sigma(\mathbb{R}^3)$, not only with Schwartz data, while retaining the one prescribed force throughout. The datum-generated full-interval intersection, its finite rectangle projections, and same-force continuation transfer, after positive-time smoothing, to the strong $H^1$ class for that force.

**Theorem 17.1** (Global $H^1$ completion)**.** *Let $\nu > 0$, let $a \in H^1_\sigma(\mathbb{R}^3)$, and prescribe $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$. Then there is a unique pressure-normalized mild solution*

$$u^a \in C([0,\infty); H^1_\sigma) \cap L^2_{\mathrm{loc}}([0,\infty); H^2_\sigma), \qquad u^a \in C^\infty((0,\infty)\times\mathbb{R}^3), \tag{501}$$

*of the forced three-dimensional Navier–Stokes equations with datum $a$, viscosity $\nu$, and that prescribed force $f$. Its pressure is*

$$p^a = R_i R_j (u^a_i u^a_j) - (-\Delta)^{-1} \nabla \cdot f$$

*in distributions. The solution obeys the forced kinetic-energy identity on every finite interval.*

*Proof.* Set $g := \mathbb{P} f$, and write

$$G_0 := \|g\|_{L^2(0,1;L^2)}, \qquad R_{a,f} := 1 + \|\nabla a\|_2^2 + 2\nu^{-1} G_0^2.$$

Write

$$Y(t) := \|\nabla u(t)\|_2^2, \qquad Z(t) := \|\Delta u(t)\|_2^2, \qquad c_1 := \frac{27}{16}\mathsf{S}(6)^6.$$

For a smooth solenoidal approximation, the $L^2$ pairing of the projected equation with $-\Delta u$ gives

$$\frac{1}{2} Y'(t) + \nu Z(t) = \mathrm{Re}\,\langle (u\cdot\nabla)u, \Delta u\rangle_{L^2} - \mathrm{Re}\,\langle g, \Delta u\rangle_{L^2}. \tag{502}$$

The explicit three-dimensional Sobolev inequality and interpolation give

$$\|u\|_6 \le \mathsf{S}(6) Y^{1/2}, \qquad \|\nabla u\|_3 \le \mathsf{S}(6)^{1/2} Y^{1/4} Z^{1/4}.$$

Consequently,

$$|\langle (u\cdot\nabla)u, \Delta u\rangle_{L^2}| \le \mathsf{S}(6)^{3/2} Y^{3/4} Z^{3/4} \le \frac{\nu}{2} Z + \frac{27\mathsf{S}(6)^6}{32\nu^3} Y^3,$$

while

$$|\langle g, \Delta u\rangle_{L^2}| \le \frac{\nu}{4} Z + \nu^{-1} \|g\|_2^2.$$

Consequently,

$$Y'(t) + \frac{\nu}{2} Z(t) \le c_1 \nu^{-3} Y(t)^3 + 2\nu^{-1} \|g(t)\|_2^2. \tag{503}$$

The integral inequality in (503), with the force contribution kept in this bootstrap, shows that

$$Y(t) \le 2 R_{a,f} \tag{504}$$

whenever $0 \le t \le 1$ and $8 c_1 \nu^{-3} R_{a,f}^2 t \le 1$.

The estimate holds on the explicit lifespan

$$\tau_{\nu,f}(a) := \min\left\{1, \frac{\nu^3}{8c_1 R_{a,f}^2}\right\},$$
$$\sup_{0\le t\le \tau_{\nu,f}(a)} Y(t) \le 2R_{a,f}, \qquad \int_0^{\tau_{\nu,f}(a)} Z(t)\,\mathrm{d}t < \infty. \tag{505}$$

Let $\mathcal{J}_k$ be the orthogonal Fourier projection to $\{|\xi| \le k\}$. On the range of $\mathcal{J}_k$, the regularized problem

$$\partial_t u^{(k)} - \nu\Delta u^{(k)} = -\mathcal{J}_k \mathbb{P}\nabla\cdot (u^{(k)} \otimes u^{(k)}) + \mathcal{J}_k g, \qquad u^{(k)}(0) = \mathcal{J}_k a, \tag{506}$$

is a Banach-space ordinary differential equation in $H^1_\sigma$. Indeed, if

$$\mathcal{N}_k(z) := \mathcal{J}_k \mathbb{P}\nabla\cdot (z\otimes z),$$

then the Bernstein inequalities on the range of $\mathcal{J}_k$ give

$$\|\mathcal{N}_k(z) - \mathcal{N}_k(y)\|_{H^1} \le C_k\big(\|z\|_{H^1} + \|y\|_{H^1}\big)\,\|z-y\|_{H^1}, \qquad C_k < \infty. \tag{507}$$

Picard's theorem supplies a maximal trajectory. The projection is self-adjoint and commutes with $\mathbb{P}$, $\nabla$, and $\Delta$, so the forced kinetic identity holds along that trajectory. Since

$$\|z\|_{H^1}^2 \le (1+k^2)\,\|z\|_2^2 \qquad (z = \mathcal{J}_k z), \tag{508}$$

the forced kinetic bound $\|u^{(k)}(t)\|_2 \le \|a\|_2 + \int_0^t \|g(s)\|_2\,\mathrm{d}s$ and the Banach-space continuation criterion extend the trajectory for all positive times. Estimate (503) holds for $u^{(k)}$, with constants independent of $k$. Equations (504)–(505) give

$$\sup_k \left( \left\| u^{(k)} \right\|_{L^\infty(0,\tau_{\nu,f}(a);H^1)} + \left\| u^{(k)} \right\|_{L^2(0,\tau_{\nu,f}(a);H^2)} \right) < \infty. \tag{509}$$

The nonlinear estimate

$$\|\mathbb{P}\nabla\cdot(u\otimes u)\|_2 \le \|u\|_6\,\|\nabla u\|_3 \le \mathsf{S}(6)^{3/2} Y^{3/4} Z^{1/4},$$

shows that the regularized nonlinear terms are bounded in $L^2(0,\tau_{\nu,f}(a);L^2)$, uniformly in $k$. The equations and $\mathcal{J}_k g \in L^2(0,\tau_{\nu,f}(a);L^2)$ bound $\partial_t u^{(k)}$ in this space. Weak compactness and the Aubin–Lions–Simon lemma [1, 14] on every ball give, after diagonal extraction,

$$\begin{aligned} u^{(k)} &\overset{*}{\rightharpoonup} u && \text{in } L^\infty(0,\tau_{\nu,f}(a);H^1),\\ u^{(k)} &\rightharpoonup u && \text{in } L^2(0,\tau_{\nu,f}(a);H^2),\\ u^{(k)} &\longrightarrow u && \text{in } L^2(0,\tau_{\nu,f}(a);H^1(B_R)) \quad (R<\infty). \end{aligned}$$

The last convergence passes the quadratic term in distributions; lower semicontinuity preserves (509). The displayed nonlinear estimate now gives $\mathbb{P}\nabla\cdot(u\otimes u) \in L^2L^2$, and the equation gives $u_t \in L^2(0,\tau_{\nu,f}(a);L^2)$. For every $\phi \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$ with $\nabla\cdot\phi = 0$ and every $\eta \in C_c^\infty([0,\tau_{\nu,f}(a)))$, passage to the limit in the time-integrated regularized equation gives

$$\begin{aligned} -\int_0^{\tau_{\nu,f}(a)} \langle u,\phi\rangle_{L^2}\,\eta'\,\mathrm{d}t + \nu\int_0^{\tau_{\nu,f}(a)} \langle \nabla u, \nabla\phi\rangle_{L^2}\,\eta\,\mathrm{d}t - \int_0^{\tau_{\nu,f}(a)} \langle u\otimes u, \nabla\phi\rangle_{L^2}\,\eta\,\mathrm{d}t \\ = \langle a,\phi\rangle_{L^2}\,\eta(0) + \int_0^{\tau_{\nu,f}(a)} \langle g,\phi\rangle_{L^2}\,\eta\,\mathrm{d}t. \end{aligned} \tag{510}$$

The right-hand side fixes the weak initial trace as $a$.

The Lions–Magenes theorem [12] for $H^2_\sigma \hookrightarrow H^1_\sigma \hookrightarrow L^2_\sigma$ gives $u \in C([0,\tau_{\nu,f}(a)];H^1_\sigma)$; its continuous trace at zero is the trace in (510), hence $u(0)=a$. Variation of constants gives the mild equation, and $p = R_iR_j(u_iu_j) - (-\Delta)^{-1}\nabla\cdot f$ supplies its normalized pressure. Since $u_t \in L^2L^2$ and $u \in L^2H^2$, pairing the limiting equation with $u$ is legitimate; incompressibility cancels the transport term and proves the forced kinetic-energy identity.

If $u, v$ are two such solutions and $w = u - v$, incompressibility cancels the $(u\cdot\nabla)w$ pairing and gives

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|w\|_2^2 + \nu\|\nabla w\|_2^2 = -\operatorname{Re}\langle (w\cdot\nabla)v, w\rangle_{L^2}.$$

This Sobolev constant yields

$$|\langle (w\cdot\nabla)v, w\rangle_{L^2}| \le \mathsf{S}(6)^{3/2}\|\nabla v\|_2\|w\|_2^{1/2}\|\nabla w\|_2^{3/2} \le \frac{\nu}{2}\|\nabla w\|_2^2 + \frac{27\mathsf{S}(6)^6}{32\nu^3}\|\nabla v\|_2^4\|w\|_2^2.$$

Gronwall's inequality proves uniqueness. This calculation one derivative higher also gives the finite-interval stability needed below. Indeed, pairing the difference equation with $-\Delta w$, integrating the $(v\cdot\nabla)w$ term once in space, and applying Young's inequality gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\nabla w\|_2^2 + \nu\|\Delta w\|_2^2 \le \frac{C_{\mathrm{st}}}{\nu}\left(\|\nabla u\|_3^2 + \|\nabla v\|_3^2\right)\|\nabla w\|_2^2. \tag{511}$$

The coefficient is integrable because

$$\|\nabla u\|_3^2 \le \mathsf{S}(6)\|\nabla u\|_2\|\Delta u\|_2,$$

and likewise for $v$. The $L^2$ difference estimate and (511) show continuous dependence in $C([0,S];H^1)$ on every finite interval $[0,S]$.

Let $c_0 > 0$ be the absolute smallness constant in the full-space $H^1$ local existence theorem, Theorem 5.4(ii) of [15]. Applied on an interval $I$ of length $h$, with left trace $b$ and projected source $g$, its viscosity-one hypothesis is

$$h\left(\|b\|_{H^1} + \|g\|_{L^1(I;H^1)}\right)^4 \le c_0.$$

After viscosity rescaling, the hypothesis becomes

$$h\left(\|b\|_{H^1} + \|g\|_{L^1(I;H^1)}\right)^4 \le c_0\nu^3. \tag{512}$$

Fix $0 < \varepsilon < S \le \tau_{\nu,f}(a)$, and put

$$K_{S,f} := 1 + \sup_{0\le t\le S}\|u(t)\|_{H^1} + \sup_{0\le t\le S}\|f(t)\|_{H^2}.$$

Choose

$$0 < h \le \min\left\{\varepsilon, c_0\nu^3K_{S,f}^{-4}\right\}.$$

For $I = [0,h]$ or $I = [t-h,t]$, $h \le t \le S$, and left endpoint $s$, the facts that $S \le \tau_{\nu,f}(a) \le 1$ and $g = \mathbb{P}f$ give

$$\|u(s)\|_{H^1} + \|g\|_{L^1(I;H^1)} \le \sup_{0\le r\le S}\|u(r)\|_{H^1} + h\sup_{0\le r\le S}\|f(r)\|_{H^2} \le K_{S,f}.$$

Thus these intervals obey (512). The quantitative positive-time regularity estimate in Lemma 5.5 of [15], after viscosity rescaling with interior time $h/2$, applies to this inhomogeneous mild formula.

Its source terms are bounded by the displayed smooth-force norms, so the heat-semigroup/product iteration gives, for every $m \in \mathbb{N}$, with

$$\Gamma_{m,S}(f) := \sum_{j=0}^{m} \sup_{0\le t\le S} \left\| \partial_t^j f(t) \right\|_{H^{m+2}},$$

$$\sup_{\varepsilon\le t\le S} \|u(t)\|_{H^m} \le C_{m,\varepsilon,S}\big(\nu, K_{S,f}, \Gamma_{m,S}(f)\big) < \infty. \tag{513}$$

Apply the mild formula from time $\varepsilon/2$. The bounds (513), the heat multiplier, and Lemma 12.4 show that $u \in C([\varepsilon, S]; H^m)$ for every $m$. The identities

$$p = R_i R_j (u_i u_j) - (-\Delta)^{-1} \nabla \cdot f, \qquad u_t = \nu \Delta u - \mathbb{P} \nabla \cdot (u \otimes u) + g \tag{514}$$

then give $p \in C([\varepsilon, S]; H^m)$ and $u_t \in C([\varepsilon, S]; H^m)$ for every $m$. Differentiating (514) and the pressure identity, and applying this product estimate, proves inductively that all temporal derivatives of $u$ and $p$ belong to every spatial Sobolev space continuously on $[\varepsilon, S]$. Sobolev embedding proves $(u, p) \in C^\infty((0, \tau_{\nu,f}(a)] \times \mathbb{R}^3)$.

Choose $0 < t_0 < \tau_{\nu,f}(a)$. Equation (513) places $u(t_0)$ in $H^\infty_\sigma$. The absolute-time form of Theorem 15.8, begun from $u(t_0)$ at $t_0$, supplies a global smooth history driven for $t \ge t_0$ by the original prescribed force $f(t)$. The uniqueness estimate above identifies it with the local history on their overlap, so the two histories glue to (501). This estimate proves uniqueness in the full displayed class. The forced kinetic identity on the first interval and the classical forced identity after $t_0$ agree at the common trace and prove the final assertion. □

## 17.2 Finite norm and trace majorants

The compatible rectangles carry more information than the endpoint argument uses. We first name scalar majorants of their finite norms and close the estimates that will feed the mixed rows, frequency tails, and continuation bounds below.

**Definition 17.2** (Rectangle-norm and trace majorants)**.** Let $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, $\nu > 0$, prescribe $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3; \mathbb{R}^3))$, and let $(u, p)$ be the pressure-normalized solution generated by $(u_0, \nu, f)$. Let $0 < T < \infty$ satisfy either $T < T_*$ or $T = T_* < \infty$. Set $I_T = (0, T)$,

$$V_n := \partial_t^n u, \qquad Q_n := \partial_t^n p, \qquad F_n := \partial_t^n f, \qquad \Phi_n := (-\Delta)^{-1} \nabla \cdot F_n, \qquad n \in \mathbb{N}_0.$$

Define the monotone rectangle-norm majorant

$$\mathfrak{P}_{N,M}[u_0, \nu, f; T] := 1 + \sup_{0<S<T} \mathfrak{R}_{N,M}(u; S), \tag{515}$$

and the trace constant

$$\mathfrak{T}_{N,M}[u_0, \nu, f; T] := \big(1 + T^{-1}\big)^{1/2} \mathfrak{P}_{N,M}[u_0, \nu, f; T]^{1/2}. \tag{516}$$

All occurrences of $\mathfrak{P}_{N,M}$ and $\mathfrak{T}_{N,M}$ below carry the arguments $[u_0, \nu, f; T]$. The velocity datum $u_0$, viscosity $\nu$, prescribed force $f$, pressure-normalized solution $(u, p)$, horizon $T$, interval $I_T$, and row notation $V_n, Q_n, F_n, \Phi_n$ fixed here remain in force throughout the rest of this section. In the terminal alternative $T = T_* < \infty$, the full-interval representatives are those of Proposition 14.8, and every endpoint value used below is the compatible trace supplied by Theorem 15.5.

**Lemma 17.3** (Majorant finiteness)**.** *For every $N, M \in \mathbb{N}_0$,*

$$1 \le \mathfrak{P}_{N,M}[u_0, \nu, f; T] < \infty. \tag{517}$$

*Proof.* If $T = T_* < \infty$, equation (396) proves (517). If $T < T_*$, classical persistence on $[0, T]$ gives

$$\sup_{0 \le t \le T} \|V_n(t)\|_{H^m} < \infty \qquad (n, m \in \mathbb{N}_0),$$

and (393) proves (517). □

**Lemma 17.4** (Fixed analytic constants)**.** *For $r \in \mathbb{N}_0$, set*

$$A_r := 2C_r^{\mathrm{alg}}.$$

*Then*

$$\|R_i R_j(f_i g_j)\|_{H^r} + \|\mathbb{P}\nabla\cdot(f \otimes g)\|_{H^{r-1}} \le A_r \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}. \tag{518}$$

*This constant satisfies*

$$\|R_i R_j(f_i g_j)\|_{H^r} + \|\nabla R_i R_j(f_i g_j)\|_{H^{r-1}} \le A_r \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}. \tag{519}$$

*For $s, k \in \mathbb{N}_0$, $2 \le b \le \infty$, and $s > k + \frac{3}{2} - \frac{3}{b}$, define*

$$\nabla^k f := (\partial^\alpha f)_{|\alpha|=k}, \qquad \left\|\nabla^k f\right\|_{L^b} := \left\| \left( \sum_{|\alpha|=k} |\partial^\alpha f|^2 \right)^{1/2} \right\|_{L^b},$$

*and set*

$$d_k := \#\{\alpha \in \mathbb{N}_0^3 : |\alpha| = k\}, \qquad \frac{1}{q_b} := \frac{1}{2} - \frac{1}{b},$$

*with $q_2 = \infty$, and*

$$S_{s,k,b} := d_k^{1/2} (2\pi)^{-3(1/2-1/b)} \left\| |\xi|^k \langle \xi \rangle^{-s} \right\|_{L^{q_b}_\xi}. \tag{520}$$

*Then $S_{s,k,b} < \infty$ and*

$$\left\|\nabla^k f\right\|_{L^b} \le S_{s,k,b} \|f\|_{H^s}. \tag{521}$$

*For $m \ge 2$, set $L_m := \max\{1, C_m^{\mathrm{alg}}\}$. Then*

$$\|fg\|_{H^m} \le L_m \|f\|_{H^m} \|g\|_{H^m}, \tag{522}$$

$$\left\|e^{\nu t\Delta}\mathbb{P}\nabla\cdot F\right\|_{H^m} \le L_m (\nu t)^{-1/2} \|F\|_{H^m}, \qquad t > 0. \tag{523}$$

*Proof.* Lemmas 12.4 and 12.5, applied componentwise before summing squares, give

$$\|R_i R_j(f_i g_j)\|_{H^r} \le \|f \otimes g\|_{H^r} \le C_r^{\mathrm{alg}} \|f\|_{H^{r+1}} \|g\|_{H^{r+1}},$$
$$\|\mathbb{P}\nabla\cdot(f \otimes g)\|_{H^{r-1}} \le \|f \otimes g\|_{H^r} \le C_r^{\mathrm{alg}} \|f\|_{H^{r+1}} \|g\|_{H^{r+1}}.$$

Moreover,

$$\|\nabla R_i R_j(f_i g_j)\|_{H^{r-1}} \le \|R_i R_j(f_i g_j)\|_{H^r}.$$

This proves (518) and (519).

Let $b' = b/(b-1)$. Interpolation between Fourier inversion $L^1_\xi \to L^\infty_x$, with norm $(2\pi)^{-3/2}$, and Plancherel gives

$$\left\|\mathcal{F}^{-1}h\right\|_{L^b_x} \le (2\pi)^{-3(1/2-1/b)} \|h\|_{L^{b'}_\xi}.$$

For $|\alpha| = k$, Hölder's inequality with $1/b' = 1/2 + 1/q_b$ gives

$$\|\partial^\alpha f\|_{L^b} \le (2\pi)^{-3(1/2-1/b)} \left\||\xi|^k \langle\xi\rangle^{-s}\right\|_{L^{q_b}} \|f\|_{H^s}.$$

The weight belongs to $L^{q_b}$ under the displayed strict inequality for $s$; for $b = 2$, its $L^\infty$-norm is finite because $s > k$. Summation over $|\alpha| = k$ proves (521).

The first estimate in (522) is Lemma 12.4. For the heat estimate, the symbol of $\mathbb{P}\nabla\cdot$, acting from tensors to vectors, has norm at most $|\xi|$, while

$$\sup_{\rho\ge 0} \rho e^{-\nu t\rho^2} = (2e\nu t)^{-1/2}.$$

Plancherel gives

$$\left\|e^{\nu t\Delta}\mathbb{P}\nabla\cdot F\right\|_{H^m} \le (2e\nu t)^{-1/2} \|F\|_{H^m} \le L_m(\nu t)^{-1/2} \|F\|_{H^m}.$$

□

**Definition 17.5** (Canonical rectangle majorants)**.** For $N, M \in \mathbb{N}_0$, define

$$\mathfrak{B}_{N,M}[u_0,\nu,f;T] := \inf\{B \ge 0 : \mathfrak{R}_{N,M}(u;S) \le B \text{ for every } 0 < S < T\}. \tag{524}$$

Monotone convergence gives

$$\mathfrak{B}_{N,M}[u_0,\nu,f;T] = \begin{cases} \mathfrak{R}_{N,M}(u;T), & T < T_*, \\ \mathfrak{R}^*_{N,M}[u_0,\nu,f,T_*], & T = T_* < \infty. \end{cases} \tag{525}$$

The pressure-normalized history in Definition 17.2 is fixed for $(u_0,\nu,f)$. Thus $\mathfrak{B}_{N,M}[u_0,\nu,f;T]$ is the exact minimal rectangle bound of that datum-generated history, not a supremum over a family of solutions and not an independently chosen constant. Equations (493) and (524) give the exact matrix identities

$$\mathfrak{B}_{N,M}[u_0,\nu,f;T] = \sup_{0<S<T} \|\mathbf{G}_{N,M}(S)\|^2_{\mathrm{F}}, \tag{526}$$

$$\mathfrak{P}_{N,M}[u_0,\nu,f;T] = 1 + \sup_{0<S<T} \|\mathbf{G}_{N,M}(S)\|^2_{\mathrm{F}}. \tag{527}$$

Consequently,

$$\mathfrak{P}_{N,M}[u_0,\nu,f;T] = 1 + \mathfrak{B}_{N,M}[u_0,\nu,f;T]. \tag{528}$$

For $n \in \mathbb{N}_0$ and $m \in \mathbb{Z}$, define

$$\begin{aligned} \mathfrak{G}^{(0)}_{n,m} &:= \|F_n(0)\|_{H^m}, & \mathfrak{G}^{(1)}_{n,m} &:= \|F_n\|_{L^1(I_T;H^m)}, \\ \mathfrak{G}^{(2)}_{n,m} &:= \|F_n\|_{L^2(I_T;H^m)}, & \mathfrak{G}^{(\infty)}_{n,m} &:= \sup_{0\le t\le T} \|F_n(t)\|_{H^m}, \end{aligned} \tag{529}$$

For $n, m \in \mathbb{N}_0$, define the force-pressure coordinates

$$\begin{aligned} \mathfrak{L}^{(1)}_{n,m} &:= \|\Phi_n\|_{L^1(I_T;H^m)} + \|\nabla\Phi_n\|_{L^1(I_T;H^{m-1})}, & \mathfrak{L}^{(2)}_{n,m} &:= \|\Phi_n\|_{L^2(I_T;H^m)}, \\ \mathfrak{L}^{(\infty)}_{n,m} &:= \sup_{0\le t\le T} \left(\|\Phi_n(t)\|_{H^m} + \|\nabla\Phi_n(t)\|_{H^{m-1}}\right). \end{aligned} \tag{530}$$

Every one of these coordinates is finite by the prescribed force class and (334). They belong to this forced history and are not independently chosen sources.

For $n, m \in \mathbb{N}_0$, define the initial-jet majorants $\mathfrak{D}_{n,m} = \mathfrak{D}_{n,m}[u_0, \nu, f]$ by

$$\mathfrak{D}_{0,m} := \|u_0\|_{H^m}\,, \tag{531}$$

$$\mathfrak{D}_{n+1,m} := \nu \mathfrak{D}_{n,m+2} + A_{m+1} \sum_{a=0}^{n} \binom{n}{a} \mathfrak{D}_{a,m+2} \mathfrak{D}_{n-a,m+2} + \mathfrak{G}^{(0)}_{n,m}. \tag{532}$$

Define the pointwise recursion $\mathfrak{Y}_{n,m} = \mathfrak{Y}_{n,m}[u_0, \nu, f; T]$ by

$$\mathfrak{Y}_{0,0} := K_f(T), \tag{533}$$

$$\mathfrak{Y}_{0,m} := \min\left\{ (1+T^{-1})^{1/2} \mathfrak{B}^{1/2}_{0,m}, \|u_0\|_{H^m} + T^{1/2} \mathfrak{B}^{1/2}_{0,m+1} \right\}, \qquad m \geq 1, \tag{534}$$

$$\mathfrak{Y}_{n+1,m} := \nu \mathfrak{Y}_{n,m+2} + A_{m+1} \sum_{a=0}^{n} \binom{n}{a} \mathfrak{Y}_{a,m+2} \mathfrak{Y}_{n-a,m+2} + \mathfrak{G}^{(\infty)}_{n,m}. \tag{535}$$

For $m \in \{-1\} \cup \mathbb{N}_0$, put

$$m_+ := \max\{m, 0\}, \qquad \mu(m) := \max\{m-1, 0\},$$

and define

$$\widetilde{\mathfrak{X}}_{n,m} := \min\left\{ T^{1/2} \mathfrak{Y}_{n,m_+}, \mathfrak{B}^{1/2}_{n,\mu(m)} \right\}. \tag{536}$$

Set

$$\mathfrak{X}_{0,0} := \min\left\{ \widetilde{\mathfrak{X}}_{0,0}, T^{1/2} K_f(T) \right\}, \tag{537}$$

$$\mathfrak{X}_{0,1} := \min\left\{ \widetilde{\mathfrak{X}}_{0,1}, \left( T + \frac{1}{2\nu} \right)^{1/2} K_f(T) \right\}, \tag{538}$$

$$\mathfrak{X}_{n,m} := \widetilde{\mathfrak{X}}_{n,m} \quad \text{for all other } (n,m). \tag{539}$$

For $n, m \in \mathbb{N}_0$, define

$$\mathfrak{Z}_{n,m}[u_0, \nu, f; T] := \min\left\{ \mathfrak{Y}_{n,m}, (1+T^{-1})^{1/2} \mathfrak{B}^{1/2}_{n,m}, \mathfrak{D}_{n,m} + T^{1/2} \mathfrak{X}_{n+1,m} \right\}. \tag{540}$$

All $\mathfrak{B}, \mathfrak{D}, \mathfrak{X}, \mathfrak{Y}, \mathfrak{Z}$, $\mathfrak{G}$, and $\mathfrak{L}$ in this section carry the arguments $[u_0, \nu, f; T]$, except that $\mathfrak{D}$ is independent of $T$.

**Proposition 17.6** (Exact viscosity and horizon scaling)**.** *Let*

$$a := \nu^{-1} u_0, \qquad v(s,x) := \nu^{-1} u(s/\nu, x), \qquad q(s,x) := \nu^{-2} p(s/\nu, x), \qquad \widetilde{f}(s,x) := \nu^{-2} f(s/\nu, x).$$

*Then $(v, q)$ is the unit-viscosity solution generated by $(a, 1, \widetilde{f})$. If*

$$\widetilde{V}_n := \partial_s^n v, \qquad \widetilde{Q}_n := \partial_s^n q, \qquad \widetilde{F}_n := \partial_s^n \widetilde{f},$$

*then*

$$V_n(t) = \nu^{n+1} \widetilde{V}_n(\nu t), \qquad Q_n(t) = \nu^{n+2} \widetilde{Q}_n(\nu t), \qquad F_n(t) = \nu^{n+2} \widetilde{F}_n(\nu t). \tag{541}$$

*Substitution of* (541) *into the defining norms gives*

$$\mathfrak{B}_{N,M}[u_0,\nu,f;T]=\sum_{n=0}^{N}\Big[\nu^{2n+1}\left\|\widetilde{V}_n\right\|^2_{L^2(0,\nu T;H^{M+1})}+\nu^{2n+3}\left\|\widetilde{V}_{n+1}\right\|^2_{L^2(0,\nu T;H^{M-1})}\Big], \tag{542}$$

$$\|Q_n\|_{L^1(0,T;H^m)}+\|\nabla Q_n\|_{L^1(0,T;H^{m-1})}=\nu^{n+1}\Big(\left\|\widetilde{Q}_n\right\|_{L^1(0,\nu T;H^m)}+\left\|\nabla\widetilde{Q}_n\right\|_{L^1(0,\nu T;H^{m-1})}\Big), \tag{543}$$

$$\sup_{0\le t\le T}\|Q_n(t)\|_{H^m}=\nu^{n+2}\sup_{0\le s\le\nu T}\left\|\widetilde{Q}_n(s)\right\|_{H^m}. \tag{544}$$

*The initial-jet recursion obeys*

$$\mathfrak{D}_{n,m}[u_0,\nu,f]=\nu^{n+1}\mathfrak{D}_{n,m}[u_0/\nu,1,\widetilde{f}]. \tag{545}$$

*Write*

$$E^u_\sigma(t):=\frac{1}{2}\|\Lambda^\sigma u(t)\|_2^2,\qquad E^v_\sigma(s):=\frac{1}{2}\|\Lambda^\sigma v(s)\|_2^2.$$

*For every $j\in\mathbb{N}_0$, the critical-height rows obey*

$$E^u_{j+1/2}(t)=\nu^2E^v_{j+1/2}(\nu t), \tag{546}$$

$$\int_0^T E^u_{j+1/2}(t)\,\mathrm{d}t=\nu\int_0^{\nu T}E^v_{j+1/2}(s)\,\mathrm{d}s. \tag{547}$$

*For every $t_0\ge0$ and $b\in H^3_\sigma(\mathbb{R}^3)$, put*

$$f_{\nu,t_0}(\sigma,x):=\nu^{-2}f(t_0+\sigma/\nu,x).$$

*The maximal elapsed lifespan in the shifted mild class has the exact covariance*

$$\tau^{\max}_{\nu,f,3}(t_0,b)=\nu^{-1}\tau^{\max}_{1,f_{\nu,t_0},3}(0,b/\nu). \tag{548}$$

*Proof.* The change of variables gives $u(t)=\nu v(\nu t)$ and $p(t)=\nu^2q(\nu t)$. Differentiation proves (541); the substitution $s=\nu t$ proves (542)–(544). The force-row identity in (541) makes the force term in (532) homogeneous of degree equal to that of $V_{n+1}$. Equation (545) then follows by induction in (531)–(532). The amplitude relation $u(t)=\nu v(\nu t)$ proves (546), and the substitution $s=\nu t$ proves (547). For $(w,\pi)\in\mathscr{M}^3_{\nu,f}(t_0,t_0+\delta;b)$, set

$$\mathcal{S}_\nu(w,\pi)(\sigma):=\big(\nu^{-1}w(t_0+\sigma/\nu),\nu^{-2}\pi(t_0+\sigma/\nu)\big),\qquad 0\le\sigma\le\nu\delta.$$

Time translation and this change of variables show that $\mathcal{S}_\nu$ is a bijection from $\mathscr{M}^3_{\nu,f}(t_0,t_0+\delta;b)$ onto $\mathscr{M}^3_{1,f_{\nu,t_0}}(0,\nu\delta;b/\nu)$. Its inverse is the corresponding amplitude and time rescaling, which proves (548). □

**Proposition 17.7** (Height-coordinate reconstruction of the finite rows)**.** *For $m\in\mathbb{N}_0$, define*

$$\mathfrak{H}_m[u_0,\nu,f;T]:=\frac{1}{2}\left\|\Lambda^{m+1/2}u\right\|^2_{L^2(I_T;L^2)}, \tag{549}$$

$$\mathfrak{S}_m[u_0,\nu,f;T]^2:=2^m\left(TK_f(T)^2+2\mathfrak{H}_m[u_0,\nu,f;T]\right). \tag{550}$$

*Then*

$$2\mathfrak{H}_m[u_0,\nu,f;T]\le\mathfrak{B}_{0,m}[u_0,\nu,f;T],\qquad \|u\|_{L^2(I_T;H^m)}\le\mathfrak{S}_m[u_0,\nu,f;T]. \tag{551}$$

*Define recursively, for* $n,m\in\mathbb{N}_0$,

$$\mathfrak{A}^{\mathrm{v}}_{0,m}:=T^{-1/2}\mathfrak{S}_m+\nu T^{1/2}\mathfrak{S}_{m+2}+A_{m+1}\mathfrak{S}^2_{m+2}+\mathfrak{G}^{(1)}_{0,m}, \tag{552}$$

$$\mathfrak{A}^{\mathrm{p}}_{n,m}:=A_m\sum_{a=0}^{n}\binom{n}{a}\mathfrak{A}^{\mathrm{v}}_{a,m+1}\mathfrak{A}^{\mathrm{v}}_{n-a,m+1}+\mathfrak{L}^{(\infty)}_{n,m}, \tag{553}$$

$$\mathfrak{A}^{\mathrm{v}}_{n+1,m}:=\nu\mathfrak{A}^{\mathrm{v}}_{n,m+2}+A_{m+1}\sum_{a=0}^{n}\binom{n}{a}\mathfrak{A}^{\mathrm{v}}_{a,m+2}\mathfrak{A}^{\mathrm{v}}_{n-a,m+2}+\mathfrak{G}^{(\infty)}_{n,m}. \tag{554}$$

*Every finite quantity in this recursion is finite, and*

$$\sup_{0\le t\le T}\|V_n(t)\|_{H^m}\le\mathfrak{A}^{\mathrm{v}}_{n,m}, \tag{555}$$

$$\sup_{0\le t\le T}\left(\|Q_n(t)\|_{H^m}+\|\nabla Q_n(t)\|_{H^{m-1}}\right)\le\mathfrak{A}^{\mathrm{p}}_{n,m}. \tag{556}$$

*Let* $M_-:=\max\{M-1,0\}$ *and set*

$$\widehat{\mathfrak{B}}^{\mathrm{h}}_{N,M}[u_0,\nu,f;T]:=T\sum_{n=0}^{N}\left((\mathfrak{A}^{\mathrm{v}}_{n,M+1})^2+(\mathfrak{A}^{\mathrm{v}}_{n+1,M_-})^2\right), \tag{557}$$

$$\widehat{\mathfrak{Q}}^{\mathrm{h}}_{N,M}[u_0,\nu,f;T]:=T\sum_{n=0}^{N}\mathfrak{A}^{\mathrm{p}}_{n,M}. \tag{558}$$

*Then*

$$\mathfrak{B}_{N,M}[u_0,\nu,f;T]\le\widehat{\mathfrak{B}}^{\mathrm{h}}_{N,M}[u_0,\nu,f;T], \tag{559}$$

$$\mathfrak{Q}_{N,M}(p;T)\le\widehat{\mathfrak{Q}}^{\mathrm{h}}_{N,M}[u_0,\nu,f;T]. \tag{560}$$

*For fixed* $N,M$*, the right-hand sides use only*

$$\begin{gathered}\nu,\quad T,\quad K_f(T),\quad N,\quad M,\\ \{\mathfrak{H}_q[u_0,\nu,f;T]:0\le q\le M+2N+4\},\quad \{A_q:0\le q\le M+2N+4\},\\ \{\mathfrak{G}^{(1)}_{0,b}:0\le b\le M+2N+2\},\\ \{\mathfrak{G}^{(\infty)}_{a,b}:0\le a\le N,\ 0\le b\le M+2N+2\},\quad \{\mathfrak{L}^{(\infty)}_{a,M}:0\le a\le N\}.\end{gathered} \tag{561}$$

*The quantities* $\mathfrak{H}_q$ *are the exact completed-height coordinates of the fixed datum-generated history. Equation* (551) *converts every finite height into a spatial Sobolev row, while* (559) *uses the forced equation to recover the temporal and pressure rows of any prescribed finite rectangle. Over all finite indices this is the critical-height exhaustion of that mixed intersection.*

*Proof.* The first estimate in (551) follows from the upper row of $\mathfrak{B}_{0,m}$. The Fourier inequality

$$(1+|\xi|^2)^m\le 2^m(1+|\xi|^{2m+1})$$

and (399) give the second estimate.

The projected equation and (518) give

$$\|u_t\|_{L^1(I_T;H^m)} \le \nu T^{1/2}\mathfrak{S}_{m+2} + A_{m+1}\mathfrak{S}_{m+2}^2 + \mathfrak{G}_{0,m}^{(1)}.$$

Choose a Lebesgue time $t_m \in I_T$ satisfying

$$\|u(t_m)\|_{H^m} \le T^{-1/2}\,\|u\|_{L^2(I_T;H^m)}\,.$$

The Banach-valued fundamental identity gives (555) at $n = 0$. Assume that bound through row $n$. The forced pressure recurrence and (519) give (556). The projected temporal recurrence with $\mathbb{P}F_n$ retained and (518) give the next velocity row through (554). Induction proves (555)–(556).

Integrating the pointwise bounds over $I_T$ gives (559); when $M = 0$, use $H^0 \hookrightarrow H^{-1}$ on the lower row. This integration in (556) gives (560). Induction in (554) gives the finite dependency range in (561). □

**Lemma 17.8** (Closed velocity majorants)**.** *For every $n, m \in \mathbb{N}_0$,*

$$\|V_n(0)\|_{H^m} \le \mathfrak{D}_{n,m}, \tag{562}$$

$$\|V_n\|_{L^2(I_T;H^m)} \le \mathfrak{X}_{n,m}, \tag{563}$$

$$\sup_{0\le t\le T}\|V_n(t)\|_{H^m} \le \mathfrak{Z}_{n,m}. \tag{564}$$

*For $m = -1$, equation* (563) *remains valid. Every right-hand side is finite and depends on $(u_0, \nu, f, n, m, T)$ through the finite rectangle-matrix suprema in* (526) *and the corresponding finite prescribed-force coordinates.*

*Proof.* Applying $\mathbb{P}$ to the temporal recurrence gives

$$V_{n+1} = \nu\Delta V_n - \mathbb{P}\nabla\cdot\sum_{a=0}^{n}\binom{n}{a}(V_a \otimes V_{n-a}) + \mathbb{P}F_n. \tag{565}$$

Equation (518) at order $m+1$, with $\mathbb{P}F_n$ retained and evaluated at $t = 0$, proves (562) by induction through (532). This estimate, taken pointwise on $[0, T]$, proves by induction that

$$\sup_{0\le t\le T}\|V_n(t)\|_{H^m} \le \mathfrak{Y}_{n,m}. \tag{566}$$

The base case $m = 0$ is the forced estimate (399). For $m \ge 1$, both terms in (534) are valid: the first is the adjacent trace bound, and the second follows from

$$u(t) = u_0 + \int_0^t V_1(s)\,\mathrm{d}s \quad \text{in } H^m.$$

The rectangle with middle order $\mu(m)$, Sobolev nesting, and (566) give respectively

$$\|V_n\|_{L^2(I_T;H^m)} \le \mathfrak{B}_{n,\mu(m)}^{1/2}, \qquad \|V_n\|_{L^2(I_T;H^m)} \le T^{1/2}\mathfrak{Y}_{n,m_+}.$$

This proves (563) through (536); the two special forced-energy cases follow from (399) and (400). The adjacent trace lemma gives

$$\sup_{0\le t\le T}\|V_n(t)\|_{H^m} \le (1 + T^{-1})^{1/2}\mathfrak{B}_{n,m}^{1/2},$$

while the Banach-valued fundamental identity and (562) give

$$\sup_{0\le t\le T}\|V_n(t)\|_{H^m} \le \mathfrak{D}_{n,m} + T^{1/2}\mathfrak{X}_{n+1,m}.$$

Taking the minimum of these bounds and (566) proves (564). □

**Definition 17.9** (Closed pressure majorants)**.** For $n, m \in \mathbb{N}_0$, define

$$\mathfrak{O}^{(1)}_{n,m} := A_m \sum_{a=0}^{n} \binom{n}{a} \mathfrak{X}_{a,m+1}\mathfrak{X}_{n-a,m+1} + \mathfrak{L}^{(1)}_{n,m}, \tag{567}$$

$$\mathfrak{O}^{(2)}_{n,m} := A_m \sum_{a=0}^{n} \binom{n}{a} \min\{\mathfrak{Z}_{a,m+1}\mathfrak{X}_{n-a,m+1}, \mathfrak{X}_{a,m+1}\mathfrak{Z}_{n-a,m+1}\} + \mathfrak{L}^{(2)}_{n,m}, \tag{568}$$

$$\mathfrak{O}^{(\infty)}_{n,m} := A_m \sum_{a=0}^{n} \binom{n}{a} \mathfrak{Z}_{a,m+1}\mathfrak{Z}_{n-a,m+1} + \mathfrak{L}^{(\infty)}_{n,m}, \tag{569}$$

$$\mathfrak{O}^{(\mathrm{mod})}_{n,m} := A_m \sum_{a=0}^{n} \binom{n}{a} (\mathfrak{X}_{a+1,m+1}\mathfrak{Z}_{n-a,m+1} + \mathfrak{Z}_{a,m+1}\mathfrak{X}_{n-a+1,m+1}) + \mathfrak{L}^{(2)}_{n+1,m}. \tag{570}$$

These quantities carry the arguments $[u_0, \nu, f; T]$ through $\mathfrak{X}$ and $\mathfrak{Z}$.

## 17.3 Mixed velocity and pressure rows

The canonical majorants now turn each finite rectangle into bounds for the actual differentiated velocity and pressure rows, including their traces and time moduli.

**Theorem 17.10** (Quantitative mixed-jet rows and traces)**.** *Let $N, s \in \mathbb{N}_0$, let $0 \le n \le N$, and let $\alpha \in \mathbb{N}_0^3$. Put*

$$J_{n,\alpha} := \partial_x^\alpha V_n.$$

*Then*

$$\|J_{n,\alpha}\|^2_{L^2(I_T;H^{s+1})} + \|J_{n+1,\alpha}\|^2_{L^2(I_T;H^{s-1})} \le \mathfrak{P}_{N,s+|\alpha|}, \tag{571}$$

$$\sup_{0\le t\le T} \|J_{n,\alpha}(t)\|_{H^s} \le \mathfrak{T}_{N,s+|\alpha|}. \tag{572}$$

*There is a unique trace $J^T_{n,\alpha} \in H^s$ at $t = T$, and for $0 \le r \le t \le T$,*

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^{s-1}} \le (t-r)^{1/2}\mathfrak{P}^{1/2}_{N,s+|\alpha|}, \tag{573}$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^{s}} \le (t-r)^{1/2}\mathfrak{P}^{1/2}_{N,s+|\alpha|+1}, \tag{574}$$

$$\left\|J^T_{n,\alpha}\right\|_{H^s} \le \mathfrak{T}_{N,s+|\alpha|}. \tag{575}$$

*If $0 \le \theta \le 1$, then*

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r)\|_{H^{s-\theta}} \le 2^{1-\theta}\mathfrak{T}^{1-\theta}_{N,s+|\alpha|}\mathfrak{P}^{\theta/2}_{N,s+|\alpha|}(t-r)^{\theta/2}. \tag{576}$$

*For $1 \le a \le \infty$, $2 \le b \le \infty$, $k, q \in \mathbb{N}_0$, and $q > k + \frac{3}{2} - \frac{3}{b}$,*

$$\left\|\nabla^k J_{n,\alpha}\right\|_{L^a(I_T;L^b)} \le S_{q,k,b}T^{1/a}\mathfrak{T}_{N,q+|\alpha|}, \tag{577}$$

$$\left\|\nabla^k J_{n,\alpha}\right\|_{L^2(I_T;L^b)} \le S_{q,k,b}\mathfrak{P}^{1/2}_{N,q+|\alpha|}, \tag{578}$$

*where $T^{1/\infty} = 1$.*

*Proof.* Take the rectangle with middle order $M = s + |\alpha|$. The Fourier multiplier inequality

$$\|\partial_x^\alpha f\|_{H^r} \le \|f\|_{H^{r+|\alpha|}}$$

and (515) give (571). The adjacent-row trace estimate gives

$$\begin{aligned}\sup_{0\le t\le T} \|J_{n,\alpha}(t)\|_{H^s}^2 &\le T^{-1}\,\|J_{n,\alpha}\|_{L^2(I_T;H^s)}^2 \\ &\quad + 2\,\|J_{n,\alpha}\|_{L^2(I_T;H^{s+1})}\,\|J_{n+1,\alpha}\|_{L^2(I_T;H^{s-1})} \\ &\le (T^{-1}+1)\mathfrak{P}_{N,s+|\alpha|},\end{aligned}$$

which proves (572) and (575).

The Bochner identity

$$J_{n,\alpha}(t) - J_{n,\alpha}(r) = \int_r^t J_{n+1,\alpha}(\tau)\,d\tau$$

in $H^{s-1}$, followed by Cauchy–Schwarz, proves (573). Applying this argument to the rectangle with middle order $s+|\alpha|+1$ proves (574). Interpolation between $H^{s-1}$ and $H^s$, together with $\|J(t)-J(r)\|_{H^s} \le 2\mathfrak{T}_{N,s+|\alpha|}$, proves (576). Sobolev embedding and (572) prove (577); Sobolev embedding and (571) prove (578). □

**Theorem 17.11** (Quantitative pressure rows)**.** *Let $N, s \in \mathbb{N}_0$, $0 \le n \le N$, and $\alpha \in \mathbb{N}_0^3$. Set*

$$\Pi_{n,\alpha} := \partial_x^\alpha Q_n, \qquad r := s + |\alpha|.$$

*Then*

$$\|\Pi_{n,\alpha}\|_{L^1(I_T;H^s)} + \|\nabla\Pi_{n,\alpha}\|_{L^1(I_T;H^{s-1})} \le 2A_r 2^n \mathfrak{P}_{N,r} + \mathfrak{L}_{n,r}^{(1)}, \tag{579}$$

$$\|\Pi_{n,\alpha}\|_{L^2(I_T;H^s)} \le A_r 2^n \mathfrak{T}_{N,r+1}\mathfrak{P}_{N,r}^{1/2} + \mathfrak{L}_{n,r}^{(2)}, \tag{580}$$

$$\sup_{0\le t\le T}\|\Pi_{n,\alpha}(t)\|_{H^s} \le A_r 2^n \mathfrak{T}_{N,r+1}^2 + \mathfrak{L}_{n,r}^{(\infty)}. \tag{581}$$

*There is a unique endpoint pressure row $\Pi_{n,\alpha}^T \in H^s$, and*

$$\left\|\Pi_{n,\alpha}^T\right\|_{H^s} \le A_r 2^n \mathfrak{T}_{N,r+1}^2 + \mathfrak{L}_{n,r}^{(\infty)}, \tag{582}$$

$$\|\Pi_{n,\alpha}(t) - \Pi_{n,\alpha}(r_0)\|_{H^s} \le 2^{n+1} A_r \mathfrak{T}_{N,r+1}\mathfrak{P}_{N,r+2}^{1/2}|t-r_0|^{1/2} + \mathfrak{L}_{n+1,r}^{(2)}|t-r_0|^{1/2} \tag{583}$$

*for $0 \le r_0 \le t \le T$. If $1 \le a \le \infty$, $2 \le b \le \infty$, $k, q \in \mathbb{N}_0$, and $q > k + \frac{3}{2} - \frac{3}{b}$, then*

$$\left\|\nabla^k \Pi_{n,\alpha}\right\|_{L^a(I_T;L^b)} \le S_{q,k,b} T^{1/a} A_{q+|\alpha|} 2^n \mathfrak{T}_{N,q+|\alpha|+1}^2 + S_{q,k,b} T^{1/a} \mathfrak{L}_{n,q+|\alpha|}^{(\infty)}. \tag{584}$$

*Proof.* The normalized pressure recurrence is

$$Q_n = R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i} V_{n-a,j} - \Phi_n. \tag{585}$$

Equation (518), Cauchy–Schwarz in time, and $\sum_{a=0}^n \binom{n}{a} = 2^n$ give

$$\|Q_n\|_{L^1(I_T;H^r)} \le A_r 2^n \mathfrak{P}_{N,r} + \|\Phi_n\|_{L^1(I_T;H^r)}. \tag{586}$$

The gradient estimate in (579) follows from $\|\nabla h\|_{H^{r-1}} \le \|h\|_{H^r}$, with the two prescribed force-potential norms retained in $\mathfrak{L}^{(1)}_{n,r}$. Using one factor in $L^\infty_t H^{r+1}_x$ and the other in $L^2_t H^{r+1}_x$ proves (580). Using both factors in $L^\infty_t H^{r+1}_x$ proves (581) and (582).

Subtract (585) at times $t$ and $r_0$, split each product difference into two terms, apply (574) in $H^{r+1}$, and use (572). The two product differences and the binomial sum, together with

$$\Phi_n(t) - \Phi_n(r_0) = \int_{r_0}^{t} \Phi_{n+1}(\tau)\,\mathrm{d}\tau,$$

give (583). Equation (584) follows from (581) and Sobolev embedding, including the force-potential coordinate. □

**Proposition 17.12** (Matrix form of the closed rectangle estimates)**.** *For nonnegative matrices of identical dimensions, write $A \preceq_{\mathrm{ent}} B$ when every entry of $A$ is at most the corresponding entry of $B$. When $T = T_* < \infty$, set the closed rectangle matrix by monotone exhaustion and use the three terminal scalar norms by*

$$\begin{aligned}
\mathbf{C}_{N,M}(T_*) &:= \lim_{S\uparrow T_*} \mathbf{C}_{N,M}(S),\\
\mathfrak{R}_{N,M}(u;T_*) &= \lim_{S\uparrow T_*} \mathfrak{R}_{N,M}(u;S),\\
\mathfrak{Q}_{N,M}(p;T_*) &:= \lim_{S\uparrow T_*} \mathfrak{Q}_{N,M}(p;S),\\
\mathfrak{F}_{N,M}(f;T_*) &:= \lim_{S\uparrow T_*} \mathfrak{F}_{N,M}(f;S).
\end{aligned}$$

*All four limits exist in the extended nonnegative reals by monotonicity. Theorem 13.13 and Proposition 14.11 make the scalar limits for velocity and pressure finite; the prescribed force class makes the force limit finite, and the estimates below make every matrix entry finite. Then*

$$\mathbf{C}_{N,M}(T) \preceq_{\mathrm{ent}} \begin{pmatrix} \mathfrak{X}_{n,M+1} & \mathfrak{X}_{n+1,M-1} & \mathfrak{Q}^{(1)}_{n,M} & \mathfrak{G}^{(1)}_{n,M-1} + \mathfrak{L}^{(1)}_{n,M} \end{pmatrix}_{n=0}^{N}. \tag{587}$$

*Consequently,*

$$\mathfrak{R}_{N,M}(u;T) \le \sum_{n=0}^{N} \left(\mathfrak{X}^2_{n,M+1} + \mathfrak{X}^2_{n+1,M-1}\right), \tag{588}$$

$$\mathfrak{Q}_{N,M}(p;T) \le \sum_{n=0}^{N} \mathfrak{Q}^{(1)}_{n,M}. \tag{589}$$

$$\mathfrak{F}_{N,M}(f;T) \le \sum_{n=0}^{N} \left(\mathfrak{G}^{(1)}_{n,M-1} + \mathfrak{L}^{(1)}_{n,M}\right). \tag{590}$$

*Proof.* The first two columns of (587) are (563). The pressure recurrence (585), the product estimate (519), and Cauchy–Schwarz in time give the third column through (567). The fourth column is the prescribed-force budget itself, bounded by (529)–(530). Take the Frobenius norm of the first two columns and sum the last two columns, using (493). □

### 17.4 Frequency and critical-scale consequences

**Proposition 17.13** (Velocity and pressure frequency tails)**.** *Let $\mathsf{P}_{\geq K}$ be the Fourier multiplier of $\mathbf{1}_{\{|\xi|\geq K\}}$, where $K \geq 1$. Let $r_1 \in \mathbb{N}$, $s \in \mathbb{N}_0$, $0 \leq n \leq N$, and $\alpha \in \mathbb{N}_0^3$. Then*

$$\|\mathsf{P}_{\geq K} J_{n,\alpha}\|_{L^2(I_T;H^s)} \leq K^{-r_1} \mathfrak{P}^{1/2}_{N,s+r_1+|\alpha|}, \tag{591}$$

$$\sup_{0\leq t\leq T} \|\mathsf{P}_{\geq K} J_{n,\alpha}(t)\|_{H^s} \leq K^{-r_1} \mathfrak{T}_{N,s+r_1+|\alpha|}, \tag{592}$$

$$\|\mathsf{P}_{\geq K} \Pi_{n,\alpha}\|_{L^1(I_T;H^s)} \leq K^{-r_1} A_{s+r_1+|\alpha|} 2^n \mathfrak{P}_{N,s+r_1+|\alpha|} + K^{-r_1} \mathfrak{L}^{(1)}_{n,s+r_1+|\alpha|}, \tag{593}$$

$$\sup_{0\leq t\leq T} \|\mathsf{P}_{\geq K} \Pi_{n,\alpha}(t)\|_{H^s} \leq K^{-r_1} A_{s+r_1+|\alpha|} 2^n \mathfrak{T}^2_{N,s+r_1+|\alpha|+1} + K^{-r_1} \mathfrak{L}^{(\infty)}_{n,s+r_1+|\alpha|}. \tag{594}$$

*The pointwise estimates* (592) *and* (594) *hold at $t = T$.*

*Proof.* For every $f \in H^{s+r_1}$,

$$\|\mathsf{P}_{\geq K} f\|_{H^s} \leq K^{-r_1} \|f\|_{H^{s+r_1}}.$$

Apply this inequality to (571) and (572) at spatial order $s + r_1$ to obtain the two velocity bounds. Since Fourier multipliers commute,

$$\|\mathsf{P}_{\geq K} \Pi_{n,\alpha}\|_{H^s} \leq K^{-r_1} \|Q_n\|_{H^{s+r_1+|\alpha|}}.$$

Integration in time followed by the forced pressure bound (586) at order $s + r_1 + |\alpha|$ gives (593). Applying the tail inequality directly to $\Pi_{n,\alpha}$, followed by (581) with $s$ replaced by $s + r_1$, gives (594). □

**Proposition 17.14** (Fixed weighted dyadic shell sums)**.** *Let $(\Delta_j)_{j\geq -1}$ be a fixed inhomogeneous Littlewood–Paley decomposition on $\mathbb{R}^3$. Let $\sigma \in \mathbb{R}$, $k \in \mathbb{N}_0$, and choose $M \in \mathbb{N}_0$ such that*

$$M > \sigma + k + \frac{3}{2}. \tag{595}$$

*Then*

$$\int_0^T \sum_{j\geq -1} 2^{\sigma j} \left\|\Delta_j \nabla^k u(t)\right\|_{L^\infty} \mathrm{d}t \leq C_{\Delta,\sigma,k,M} T^{1/2} \mathfrak{X}_{0,M}. \tag{596}$$

*In particular, with $\omega = \nabla \times u$,*

$$\sum_{j\geq -1} \int_0^T \|\Delta_j \omega(t)\|_{L^\infty} \mathrm{d}t \leq C_\Delta T^{1/2} \mathfrak{X}_{0,3} < \infty. \tag{597}$$

*Each estimate uses one fixed finite Sobolev coordinate; no summation over Sobolev orders is asserted.*

*Proof.* Bernstein's inequality on a shell gives, for $j \geq 0$,

$$2^{\sigma j} \left\|\Delta_j \nabla^k f\right\|_{L^\infty} \leq C_\Delta 2^{(\sigma+k+3/2)j} \|\Delta_j f\|_2.$$

Put $d := M - \sigma - k - 3/2 > 0$. Cauchy–Schwarz in $j$ yields

$$\begin{aligned}\sum_{j\geq 0} 2^{\sigma j} \left\|\Delta_j \nabla^k f\right\|_{L^\infty} &\leq C_\Delta \left(\sum_{j\geq 0} 2^{-2dj}\right)^{1/2} \left(\sum_{j\geq 0} 2^{2Mj} \|\Delta_j f\|_2^2\right)^{1/2} \\ &\leq C_{\Delta,\sigma,k,M} \|f\|_{H^M}.\end{aligned}$$

The single low block $j=-1$ obeys this final bound. The last inequality follows from Plancherel's theorem and the finite overlap of the dyadic multipliers. Integrating in time, applying Cauchy–Schwarz, and using (563) prove (596). Since curl commutes with every $\Delta_j$, the choice $(\sigma,k,M)=(0,1,3)$ gives (597). □

**Proposition 17.15** (Critical-height bounds)**.** *Define*

$$E_\sigma(t):=\frac{1}{2}\left\|\Lambda^\sigma u(t)\right\|_2^2,\qquad H(t):=E_{1/2}(t),$$

*and*

$$E_{j+1/2,\geq K}(t):=\frac{1}{2}\int_{|\xi|\geq K}|\xi|^{2j+1}|\widehat{u}(t,\xi)|^2\,d\xi.$$

*For $j\in\mathbb{N}_0$,*

$$\sup_{0\leq t\leq T}E_{j+1/2}(t)\leq\frac{1}{2}\mathfrak{T}_{0,j+1}^2, \tag{598}$$

$$\int_0^T E_{j+1/2}(t)\,dt\leq\frac{1}{2}\mathfrak{P}_{0,j}. \tag{599}$$

*If $m>j+\frac{1}{2}$ is an integer and $K\geq 1$, then*

$$\sup_{0\leq t\leq T}E_{j+1/2,\geq K}(t)\leq\frac{1}{2}K^{2j+1-2m}\mathfrak{T}_{0,m}^2, \tag{600}$$

$$\int_0^T E_{j+1/2,\geq K}(t)\,dt\leq\frac{1}{2}K^{2j+1-2m}\mathfrak{P}_{0,m-1}. \tag{601}$$

*For $k\in\mathbb{N}$,*

$$\left\|H^{(k)}\right\|_{L^1(0,T)}\leq 2^{k-1}\mathfrak{P}_{k,0}. \tag{602}$$

*In particular,*

$$\left\|H'\right\|_{L^1(0,T)}\leq\frac{1}{2}\mathfrak{P}_{0,2}, \tag{603}$$

$$\int_0^T\left\|\Lambda^{3/2}u(t)\right\|_2^2\,dt\leq\mathfrak{P}_{0,1}, \tag{604}$$

$$\left\|\mathcal{P}_{1/2}\right\|_{L^1(0,T)}\leq\frac{1}{2}\mathfrak{P}_{0,2}+\nu\mathfrak{P}_{0,1}+\mathfrak{P}_{0,0}^{1/2}\mathfrak{G}_{0,1}^{(2)}, \tag{605}$$

$$|H(T)-H(t)|\leq\mathfrak{T}_{0,1}\mathfrak{P}_{0,2}^{1/2}(T-t)^{1/2},\quad 0\leq t\leq T, \tag{606}$$

*where*

$$\mathcal{P}_{1/2}=H'+\nu\left\|\Lambda^{3/2}u\right\|_2^2-\mathcal{W}_{1/2}.$$

*Proof.* Equations (598) and (599) follow from $\dot{H}^{j+1/2}\hookleftarrow H^{j+1}$, (572), and (571). On $|\xi|\geq K$,

$$|\xi|^{2j+1}\leq K^{2j+1-2m}|\xi|^{2m},$$

which proves (600) and (601).

The identity

$$H^{(k)}(t)=\frac{1}{2}\sum_{a=0}^{k}\binom{k}{a}\operatorname{Re}\left\langle \Lambda^{1/2}V_a(t),\Lambda^{1/2}V_{k-a}(t)\right\rangle$$

and Cauchy–Schwarz in time give (602). At $k=1$, the two equal terms give

$$\left\|H'\right\|_{L^1(0,T)}\le \|u_t\|_{L^2(I_T;\dot H^{1/2})}\,\|u\|_{L^2(I_T;\dot H^{1/2})}\le \frac{1}{2}\mathfrak{P}_{0,2}.$$

Equation (604) follows from the upper velocity row of the $(0,1)$ rectangle. Cauchy–Schwarz gives

$$\left\|\mathcal{W}_{1/2}\right\|_{L^1(0,T)}\le \|u\|_{L^2(I_T;H^1)}\,\|\mathbb{P}f\|_{L^2(I_T;H^1)}\le \mathfrak{P}_{0,0}^{1/2}\mathfrak{G}_{0,1}^{(2)}.$$

The forced height identity defining $\mathcal{P}_{1/2}$ then gives (605). Finally,

$$\|u(T)-u(t)\|_{\dot H^{1/2}}\le \|u(T)-u(t)\|_{H^1}\le (T-t)^{1/2}\,\|u_t\|_{L^2(t,T;H^1)}\le (T-t)^{1/2}\mathfrak{P}_{0,2}^{1/2},$$

and

$$|H(T)-H(t)|\le \frac{1}{2}\big(\|u(T)\|_{\dot H^{1/2}}+\|u(t)\|_{\dot H^{1/2}}\big)\,\|u(T)-u(t)\|_{\dot H^{1/2}}\,.$$

Equation (572) bounds each retained velocity factor, so these two estimates prove (606). □

Taking $j=0$ and any integer $m\ge 1$ in (600) gives the uniform tail limit

$$\lim_{K\to\infty}\sup_{0\le t\le T}E_{1/2,\ge K}(t)=0. \tag{607}$$

## 17.5 Continuation norms

**Proposition 17.16** (Continuation norms)**.** *The finite-horizon solution satisfies*

$$\|u\|_{L^2(I_T;L^\infty)}\le S_{2,0,\infty}\mathfrak{P}_{0,1}^{1/2}, \tag{608}$$

$$\int_0^T\|\nabla u(t)\|_{L^\infty}\,dt\le S_{3,1,\infty}T^{1/2}\mathfrak{P}_{0,2}^{1/2}, \tag{609}$$

$$\int_0^T\|\nabla\times u(t)\|_{L^\infty}\,dt\le 2S_{3,1,\infty}T^{1/2}\mathfrak{P}_{0,2}^{1/2}. \tag{610}$$

*For every integer $m\ge 3$, let $K_m$ be the explicit constant in* (339)*. Then*

$$|\operatorname{Re}\langle A^m\mathbb{P}\nabla\cdot(u\otimes u),A^m u\rangle_{L^2}|\le K_m\,\|\nabla u\|_{L^\infty}\,\|u\|_{H^m}^2\,. \tag{611}$$

$$\sup_{0\le t\le T}\|u(t)\|_{H^m}^2\le\Big(\|u_0\|_{H^m}+\mathfrak{G}_{0,m}^{(1)}\Big)^2\exp\Big(2K_mS_{3,1,\infty}T^{1/2}\mathfrak{P}_{0,2}^{1/2}\Big)\,. \tag{612}$$

*Proof.* Sobolev embedding and the upper rows of the $(0,1)$ and $(0,2)$ rectangles give (608) and (609); the pointwise bound $\|\nabla\times u\|_\infty\le 2\,\|\nabla u\|_\infty$ gives (610). The forced $H^m$ energy identity and (611) give

$$\frac{1}{2}\frac{d}{dt}\,\|u\|_{H^m}^2+\nu\,\|\nabla u\|_{H^m}^2\le K_m\,\|\nabla u\|_{L^\infty}\,\|u\|_{H^m}^2+\|\mathbb{P}f\|_{H^m}\,\|u\|_{H^m}\,.$$

Regularizing the norm by $(\|u\|_{H^m}^2+\varepsilon)^{1/2}$, applying Gronwall, and then letting $\varepsilon\downarrow 0$ gives

$$\|u(t)\|_{H^m}\le\left(\|u_0\|_{H^m}+\int_0^t\|\mathbb{P}f(s)\|_{H^m}\,\mathrm{d}s\right)\exp\left(K_m\int_0^t\|\nabla u(s)\|_\infty\,\mathrm{d}s\right).$$

Gronwall's inequality and (609) prove (612). □

# 18 Three-dimensional applications

## 18.1 Explicit closure for small critical forced data

The construction below gives an explicit three-dimensional forced $\dot{H}^{1/2}$ realization of the critical-data mechanism introduced by Fujita and Kato [6]. The projected-force stock is retained in this critical action identity, and every constant used here is derived in the displayed estimates.

Use the explicit homogeneous Sobolev constants

$$\mathsf{S}_3 := \mathsf{S}(3), \qquad C_{\mathrm{src}} := \mathsf{S}_3^3 = \frac{1}{\sqrt{2}\,\pi}.$$

These constants hold for vector fields. Indeed, apply the scalar inequality to the Euclidean length $|F|$: one has $|\nabla|F|| \le |\nabla F|$, while the difference-quotient representation of $\dot{H}^{1/2}$ and $\big||F(x)| - |F(y)|\big| \le |F(x) - F(y)|$ give $\||F|\|_{\dot{H}^{1/2}} \le \|F\|_{\dot{H}^{1/2}}$.

**Lemma 18.1** (Critical action identity)**.** *Let $(u,p)$ be a smooth pressure-normalized solution on a time interval $I \subset [0,T_*)$ of the forced equation with prescribed force $f$, and set*

$$\mathcal{B}(u) := \mathbb{P}\nabla\cdot(u\otimes u), \qquad g := \mathbb{P}f.$$

*Then, pointwise on $I$,*

$$\nu\frac{\mathrm{d}}{\mathrm{d}t}\|u\|_{\dot{H}^{1/2}}^2 + \nu^2\|u\|_{\dot{H}^{3/2}}^2 + \|u_t\|_{\dot{H}^{-1/2}}^2 = \|\mathcal{B}(u) - g\|_{\dot{H}^{-1/2}}^2 . \tag{613}$$

*Proof.* The projected equation is

$$u_t = -\nu\Lambda^2 u - \mathcal{B}(u) + g.$$

Put

$$h := \Lambda^{-1/2}(\mathcal{B}(u) - g), \qquad w := \Lambda^{-1/2}u_t = -\nu\Lambda^{3/2}u - h.$$

Pairing the projected equation with $\Lambda u$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\|u\|_{\dot{H}^{1/2}}^2 = -2\nu\|u\|_{\dot{H}^{3/2}}^2 - 2\,\mathrm{Re}\left\langle h, \Lambda^{3/2}u\right\rangle_2 .$$

Multiply this equality by $\nu$ and expand $\|w\|_2^2 = \big\|\nu\Lambda^{3/2}u + h\big\|_2^2$. The cross terms cancel and give (613). □

For $m \in \mathbb{N}$, define the explicit homogeneous commutator constant

$$\mathsf{K}_m^{\mathrm{hom}} := 3^m \sum_{\ell=1}^{m} \binom{m}{\ell} \mathsf{S}(\mathfrak{p}_{m,\ell})\mathsf{S}(\mathfrak{q}_{m,\ell}), \tag{614}$$

where $(\mathfrak{p}_{1,1}, \mathfrak{q}_{1,1}) = (3,6)$, and, for $m \ge 2$,

$$\vartheta_{m,\ell} := \frac{\ell - 1}{m-1}, \qquad \frac{1}{\mathfrak{p}_{m,\ell}} := \frac{1}{3} - \frac{\vartheta_{m,\ell}}{6}, \qquad \frac{1}{\mathfrak{q}_{m,\ell}} := \frac{1}{6} + \frac{\vartheta_{m,\ell}}{6}.$$

The corresponding estimate is

$$|\mathrm{Re}\,\langle \Lambda^m\mathbb{P}\nabla\cdot(z\otimes z), \Lambda^m z\rangle_{L^2}| \le \mathsf{K}_m^{\mathrm{hom}} \left\|\Lambda^{3/2}z\right\|_2 \|\Lambda^m z\|_2 \left\|\Lambda^{m+1}z\right\|_2 \tag{615}$$

for every $z \in \mathcal{S}_\sigma(\mathbb{R}^3)$.

**Lemma 18.2** (Explicit homogeneous critical commutator)**.** *For every $m \in \mathbb{N}$, $\mathsf{K}_m^{\mathrm{hom}} < \infty$, and* (615) *holds with the displayed constant. Moreover,* (615) *extends to every divergence-free $z \in H^{m+1}(\mathbb{R}^3) \cap \dot{H}^{3/2}(\mathbb{R}^3)$.*

*Proof.* The multinomial identity gives the exact pairing

$$\langle \Lambda^m f, \Lambda^m g \rangle_{L^2} = \sum_{|\alpha|=m} \frac{m!}{\alpha!} \langle \partial^\alpha f, \partial^\alpha g \rangle_{L^2} .$$

After differentiating $(z\cdot\nabla)z$, the undifferentiated transport term has zero pairing by incompressibility. A remaining term has the form

$$\left\langle (\partial^\beta z \cdot \nabla)\partial^{\alpha-\beta} z, \partial^\alpha z \right\rangle_{L^2}, \qquad |\alpha| = m, \quad 0 < \beta \le \alpha.$$

Put $\ell = |\beta|$. For $m \ge 2$, Hölder's inequality, Lemma 12.3, and Fourier log-convexity give

$$\begin{aligned} &\left\| \partial^\beta z \right\|_{\mathfrak{p}_{m,\ell}} \left\| \nabla \partial^{\alpha-\beta} z \right\|_{\mathfrak{q}_{m,\ell}} \\ &\quad \le \mathsf{S}(\mathfrak{p}_{m,\ell}) \mathsf{S}(\mathfrak{q}_{m,\ell}) \, \|z\|_{\dot{H}^{3/2}} \, \|z\|_{\dot{H}^{m+1}} . \end{aligned}$$

Indeed, the two differentiated Sobolev orders are the complementary interpolants

$$\begin{aligned} \ell + s(\mathfrak{p}_{m,\ell}) &= (1-\vartheta_{m,\ell})\frac{3}{2} + \vartheta_{m,\ell}(m+1), \\ m - \ell + 1 + s(\mathfrak{q}_{m,\ell}) &= \vartheta_{m,\ell}\frac{3}{2} + (1-\vartheta_{m,\ell})(m+1). \end{aligned}$$

For $m = 1$, this bound is the direct $L^3$-$L^6$ estimate. Finally,

$$\sum_{|\alpha|=m} \frac{m!}{\alpha!} = 3^m, \qquad \sum_{\substack{\beta \le \alpha \\ |\beta|=\ell}} \binom{\alpha}{\beta} = \binom{m}{\ell}.$$

Summation gives exactly (614) and proves (615). Solenoidal Fourier truncation and convergence in $H^{m+1} \cap \dot{H}^{3/2}$ give the asserted extension. □

**Theorem 18.3** (Explicit small-critical forced-data global regularity)**.** *Let $\nu > 0$, $a \in \mathcal{S}_\sigma(\mathbb{R}^3)$, and let $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$ be the prescribed force. Write $g := \mathbb{P}f$, and assume*

$$g \in L^1(0,\infty;L^2) \cap L^2(0,\infty;\dot{H}^{-1/2}) \cap \bigcap_{j\in\mathbb{N}_0} L^2(0,\infty;\dot{H}^j). \tag{616}$$

*Set*

$$a_{\mathrm{c}} := \left\| \Lambda^{1/2} a \right\|_2, \qquad \mathcal{G}_{-1/2} := \|g\|_{L^2(0,\infty;\dot{H}^{-1/2})} .$$

*If there is an $\eta > 0$ such that, with*

$$A_f^2 := a_{\mathrm{c}}^2 + \frac{1+\eta^{-1}}{\nu}\mathcal{G}_{-1/2}^2, \qquad \delta_f := \nu^2 - (1+\eta)C_{\mathrm{src}}^2 A_f^2,$$

$$A_f < \frac{\nu}{\sqrt{1+\eta}\, C_{\mathrm{src}}} = \sqrt{\frac{2}{1+\eta}}\,\pi\nu, \tag{617}$$

*then the forced datum $(a,\nu,f)$ generates a unique global pressure-normalized smooth solution*

$$(u,p)\in C^\infty([0,\infty)\times\mathbb{R}^3;\mathbb{R}^3\times\mathbb{R}).$$

*For every finite $T>0$, the solution is the unique member of $\mathscr{M}^3_{\nu,f}(0,T;a)$. For every $t\ge 0$,*

$$\nu\left\|\Lambda^{1/2}u(t)\right\|_2^2+\delta_f\int_0^t\left\|\Lambda^{3/2}u(s)\right\|_2^2\,\mathrm{d}s+\int_0^t\|u_t(s)\|_{\dot H^{-1/2}}^2\,\mathrm{d}s\le\nu A_f^2,\qquad\delta_f>0.\tag{618}$$

*The critical nonlinear source and the first temporal response satisfy*

$$\int_0^\infty\|\mathbb{P}\nabla\cdot(u\otimes u)(t)\|_{\dot H^{-1/2}}^2\,\mathrm{d}t\le\frac{C_{\mathrm{src}}^2\nu A_f^4}{\delta_f},\tag{619}$$

$$\nu^2\int_0^\infty\left\|\Lambda^{3/2}u(t)\right\|_2^2\,\mathrm{d}t+\int_0^\infty\|u_t(t)\|_{\dot H^{-1/2}}^2\,\mathrm{d}t\le\nu A_f^2+\frac{(1+\eta)C_{\mathrm{src}}^2\nu A_f^4}{\delta_f}.\tag{620}$$

*For every $m\in\mathbb{N}$, define*

$$\begin{aligned}\mathcal{D}_m^f&:=\|\Lambda^m a\|_2^2+\frac{1+\eta^{-1}}{\nu}\left\|\Lambda^{m-1}g\right\|_{L^2(0,\infty;L^2)}^2,\\ \Gamma_m^{\mathrm{sm}}&:=\frac{(1+\eta)(\mathsf{K}_m^{\mathrm{hom}})^2A_f^2}{\delta_f}.\end{aligned}\tag{621}$$

*Then*

$$\begin{aligned}&\|\Lambda^m u(t)\|_2^2+\nu\int_0^t\left\|\Lambda^{m+1}u(s)\right\|_2^2\,\mathrm{d}s\\&\qquad\le\mathcal{D}_m^f\exp(\Gamma_m^{\mathrm{sm}}).\end{aligned}\tag{622}$$

*Put*

$$K_{f,\infty}:=\|a\|_2+\|g\|_{L^1(0,\infty;L^2)}.$$

*In particular,*

$$\sup_{t\ge0}\|u(t)\|_{H^m}^2\le 2^{m-1}\left[K_{f,\infty}^2+\mathcal{D}_m^f\exp(\Gamma_m^{\mathrm{sm}})\right].\tag{623}$$

*Proof.* Let $(u,p)$ be the global pressure-normalized smooth history supplied by Theorem 15.8 for $(a,\nu,f)$. Put

$$A(t):=\left\|\Lambda^{1/2}u(t)\right\|_2,\qquad B(t):=\left\|\Lambda^{3/2}u(t)\right\|_2.$$

Dual Sobolev, Hölder, and the unit Fourier-multiplier norm of $\mathbb{P}$ give the complete source-constant chain

$$\begin{aligned}\|\mathbb{P}\nabla\cdot(u\otimes u)\|_{\dot H^{-1/2}}&\le\mathsf{S}_3\,\|(u\cdot\nabla)u\|_{L^{3/2}}\\&\le\mathsf{S}_3\,\|u\|_{L^3}\,\|\nabla u\|_{L^3}\\&\le\mathsf{S}_3^3\left\|\Lambda^{1/2}u\right\|_2\left\|\Lambda^{3/2}u\right\|_2=C_{\mathrm{src}}A(t)B(t).\end{aligned}\tag{624}$$

Combining (624) with the exact identity (613) gives, on every classical subinterval,

$$\nu\frac{\mathrm{d}}{\mathrm{d}t}A(t)^2+\left(\nu^2-(1+\eta)C_{\mathrm{src}}^2A(t)^2\right)B(t)^2+\|u_t(t)\|_{\dot H^{-1/2}}^2\le(1+\eta^{-1})\,\|g(t)\|_{\dot H^{-1/2}}^2.$$

Indeed, this is $\|\mathcal{B}(u)-g\|^2 \le (1+\eta)\,\|\mathcal{B}(u)\|^2 + (1+\eta^{-1})\,\|g\|^2$. Continuity and a first-exit argument give $A(t) \le A_f$ throughout $[0,\infty)$. Integration and the definition of $\delta_f$ prove (618). In particular,

$$\int_0^t B(s)^2\,\mathrm{d}s \le \frac{\nu A_f^2}{\delta_f}.$$

For every finite $t$, this source estimate and the integrated identity (613) prove (619) and (620) with $\infty$ replaced by $t$. Monotone convergence gives the stated forms.

For the first temporal response, integrate (613), use $\|\mathcal{B}(u)-g\|^2 \le (1+\eta)\,\|\mathcal{B}(u)\|^2 + (1+\eta^{-1})\,\|g\|^2$, and substitute the definition of $A_f$.

At order $m$, the forced projected equation, Lemma 18.2, and Young's inequality yield

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\|\Lambda^m u\|_2^2 + \nu\,\big\|\Lambda^{m+1}u\big\|_2^2 \le \frac{(1+\eta)(\mathsf{K}_m^{\mathrm{hom}})^2}{\nu} B(t)^2\,\|\Lambda^m u\|_2^2 + \frac{1+\eta^{-1}}{\nu}\,\big\|\Lambda^{m-1}g(t)\big\|_2^2.$$

Multiplication by the corresponding integrating factor and

$$\int_0^t B(s)^2\,\mathrm{d}s \le \frac{\nu A_f^2}{\delta_f}$$

prove (622). The Fourier inequality

$$(1+|\xi|^2)^m \le 2^{m-1}(1+|\xi|^{2m})$$

and the forced energy estimate $\sup_{t\ge 0}\|u(t)\|_2 \le K_{f,\infty}$ prove (623).

The global smoothness and uniqueness assertions, including uniqueness in $\mathscr{M}^3_{\nu,f}(0,T;a)$ on every finite interval, are those of Theorem 15.8. □

For $m \ge 1$, define

$$U_m^{\mathrm{sm}} := 2^{(m-1)/2}\left[K_{f,\infty}^2 + \mathcal{D}_m^f \exp(\Gamma_m^{\mathrm{sm}})\right]^{1/2}, \qquad U_0^{\mathrm{sm}} := K_{f,\infty}, \qquad U_{-1}^{\mathrm{sm}} := U_0^{\mathrm{sm}}.$$

For $T<\infty$, $n\in\mathbb{N}_0$, and $m\ge -1$, define

$$\mathsf{G}_{n,m}(T) := \sup_{0\le t\le T}\|F_n(t)\|_{H^m}$$

and the finite-order recursion

$$\mathsf{Y}_{0,m}^{\mathrm{sm}}(T) := U_m^{\mathrm{sm}}, \tag{625}$$

$$\mathsf{Y}_{n+1,m}^{\mathrm{sm}}(T) := \nu\mathsf{Y}_{n,m+2}^{\mathrm{sm}}(T) + A_{m+1}\sum_{a'=0}^{n}\binom{n}{a'}\mathsf{Y}_{a',m+2}^{\mathrm{sm}}(T)\mathsf{Y}_{n-a',m+2}^{\mathrm{sm}}(T) + \mathsf{G}_{n,m}(T). \tag{626}$$

**Proposition 18.4** (Explicit small-forced-data pressure-complete rectangles)**.** *Under the hypotheses of Theorem 18.3, for every $T<\infty$, $n\in\mathbb{N}_0$, and $m\ge -1$,*

$$\sup_{0\le t\le T}\|V_n(t)\|_{H^m} \le \mathsf{Y}_{n,m}^{\mathrm{sm}}(T). \tag{627}$$

*For every $T < \infty$ and $N, M \in \mathbb{N}_0$,*

$$\mathfrak{R}_{N,M}(u;T) \le T \sum_{n=0}^{N} \left[ (\mathsf{Y}^{\mathrm{sm}}_{n,M+1}(T))^2 + (\mathsf{Y}^{\mathrm{sm}}_{n+1,M-1}(T))^2 \right], \tag{628}$$

$$\mathfrak{Q}_{N,M}(p;T) \le T A_M \sum_{n=0}^{N} \sum_{a'=0}^{n} \binom{n}{a'} \mathsf{Y}^{\mathrm{sm}}_{a',M+1}(T) \mathsf{Y}^{\mathrm{sm}}_{n-a',M+1}(T) + \sum_{n=0}^{N} \mathfrak{F}^{\mathrm{p}}_{n,0,M}(0,T). \tag{629}$$

*In the entrywise order from Proposition 17.12, these bounds have the matrix form below, where*

$$\begin{aligned}
\mathsf{C}^{(1)}_{n,M}(T) &:= T^{1/2} \mathsf{Y}^{\mathrm{sm}}_{n,M+1}(T), \\
\mathsf{C}^{(2)}_{n,M}(T) &:= T^{1/2} \mathsf{Y}^{\mathrm{sm}}_{n+1,M-1}(T), \\
\mathsf{C}^{(3)}_{n,M}(T) &:= T A_M \sum_{a'=0}^{n} \binom{n}{a'} \mathsf{Y}^{\mathrm{sm}}_{a',M+1}(T) \mathsf{Y}^{\mathrm{sm}}_{n-a',M+1}(T) + \mathfrak{F}^{\mathrm{p}}_{n,0,M}(0,T), \\
\mathsf{C}^{(4)}_{n,M}(T) &:= T \mathsf{G}_{n,M-1}(T) + \mathfrak{F}^{\mathrm{p}}_{n,0,M}(0,T).
\end{aligned}$$

*Thus*

$$\mathbf{C}_{N,M}(T) \preceq_{\mathrm{ent}} \left( \mathsf{C}^{(1)}_{n,M}(T),\, \mathsf{C}^{(2)}_{n,M}(T),\, \mathsf{C}^{(3)}_{n,M}(T),\, \mathsf{C}^{(4)}_{n,M}(T) \right)_{n=0}^{N}. \tag{630}$$

*Every right-hand side is an explicit finite function of $\nu, T, N, M$, the datum norms*

$$\|a\|_2, \qquad \left\| \Lambda^{1/2} a \right\|_2, \qquad \{ \|\Lambda^r a\|_2 : 1 \le r \le M + 2N + 1 \}. \tag{631}$$

*Together with these datum norms, the bound uses the force norms in* (616) *and* (621)*, and the finite family*

$$\left\{ \mathsf{G}_{n,m}(T),\, \mathfrak{F}^{\mathrm{p}}_{n,0,M}(0,T) : 0 \le n \le N,\ -1 \le m \le M + 2N + 1 \right\}.$$

*Indeed, induction in $n$ shows that $\mathsf{Y}^{\mathrm{sm}}_{n,m}(T)$ uses only the entries $U^{\mathrm{sm}}_j$ with $-1 \le j \le m + 2n$; both velocity columns of* (630) *consequently stop at $j = M + 2N + 1$, and its pressure column stops no later. The fourth column is the prescribed force coordinate of that mixed rectangle; it is bounded directly by its displayed force norms.*

*Proof.* Equation (623) proves the base (627) for $m \ge 1$; the forced energy estimate proves it for $m = 0$, and $H^0 \hookrightarrow H^{-1}$ proves it for $m = -1$. Apply $\mathbb{P}$ to the temporal recurrence. This is the $\alpha = 0$ temporal block-row of (486); its factorial conjugate is (489). Equation (518) at order $m + 1$ and induction give (627) through (626), with $\|\mathbb{P} F_n\|_{H^m} \le \|F_n\|_{H^m}$; for $m = -1$, use the order-zero product estimate and $H^0 \hookrightarrow H^{-1}$. Integration in time gives (628). The pressure recurrence and (519) at order $M$, followed by integration and summation in $n$, give (629), with the force-pressure potential retained in $\mathfrak{F}^{\mathrm{p}}_{n,0,M}(0,T)$. Direct integration gives the fourth, prescribed-force column. These rowwise bounds give (630); taking the Frobenius norm of its first two columns and summing its pressure column recovers (628) and (629). □

## 18.2 Low-rectangle and scale-local bounds

The following results use distinct outputs of the compatible rectangles: mixed space–time embeddings, pressure completion, and integrable Lipschitz action.

**Proposition 18.5** (Three low-rectangle regularity classes)**.** *Let $\nu > 0$, let $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, prescribe $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$, and let $u$ be the global solution supplied by Theorem 15.8 from $(u_0,\nu,f)$, and let $0 < T < \infty$. Then*

$$\|u\|_{L^\infty(0,T;L^3)} \le S_{1,0,3}\mathfrak{Z}_{0,1}[u_0,\nu,f;T], \tag{632}$$

$$\|u\|_{L^2(0,T;L^\infty)} \le S_{2,0,\infty}\mathfrak{X}_{0,2}[u_0,\nu,f;T], \tag{633}$$

$$\|\nabla u\|_{L^2(0,T;L^3)} \le S_{2,1,3}\mathfrak{X}_{0,2}[u_0,\nu,f;T]. \tag{634}$$

*In particular, each of the three norms is finite on every finite horizon, with the stated datum, viscosity, force, and horizon dependence.*

*Proof.* Use (808) with $(n,\alpha,a,b,q,k) = (0,0,\infty,3,1,0)$ to obtain (632). Use (809) with $(n,\alpha,b,q,k) = (0,0,\infty,2,0)$ and $(0,0,3,2,1)$ to obtain (633) and (634). The strict embedding conditions are, respectively, $1 > 1/2$, $2 > 3/2$, and $2 > 3/2$. □

**Corollary 18.6** (Uniform extinction of parabolic scale content)**.** *Let $\nu > 0$, let $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, prescribe $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$, and let $(u,p)$ be the pressure-normalized global pair supplied by Theorem 15.8 for $(u_0,\nu,f)$. This is a post-regularity consequence for the Caffarelli–Kohn–Nirenberg scale-normalized quantity [3]; it is not used as an $\varepsilon$-regularity input or continuation criterion. For $0 < T < \infty$, define*

$$\begin{aligned} U_T &:= S_{2,0,\infty}\mathfrak{Z}_{0,2}[u_0,\nu,f;T], \\ P_T &:= S_{2,0,\infty}\mathfrak{Q}^{(\infty)}_{0,2}[u_0,\nu,f;T]. \end{aligned} \tag{635}$$

*Let $z_0 = (t_0,x_0)$, $\rho > 0$, and suppose that*

$$Q_\rho(z_0) := (t_0-\rho^2, t_0) \times B_\rho(x_0) \subset (0,T)\times\mathbb{R}^3.$$

*Then the scale-normalized velocity–pressure content satisfies*

$$\rho^{-2}\iint_{Q_\rho(z_0)} \left(|u|^3 + |p|^{3/2}\right)\,\mathrm{d}x\,\mathrm{d}t \le \frac{4\pi}{3}\rho^3\left(U_T^3 + P_T^{3/2}\right). \tag{636}$$

*Consequently,*

$$\lim_{\rho\downarrow 0}\sup_{\{z_0:\, Q_\rho(z_0)\subset(0,T)\times\mathbb{R}^3\}} \rho^{-2}\iint_{Q_\rho(z_0)} \left(|u|^3 + |p|^{3/2}\right)\,\mathrm{d}x\,\mathrm{d}t = 0. \tag{637}$$

*Proof.* Equation (803), followed by (521) with $(s,k,b) = (2,0,\infty)$, gives

$$\|u\|_{L^\infty((0,T)\times\mathbb{R}^3)} \le U_T.$$

Equations (812) and (521) give

$$\|p\|_{L^\infty((0,T)\times\mathbb{R}^3)} \le P_T.$$

Since $|B_\rho| = \frac{4\pi}{3}\rho^3$ and the time length of $Q_\rho$ is $\rho^2$,

$$\iint_{Q_\rho(z_0)} \left(|u|^3 + |p|^{3/2}\right)\,\mathrm{d}x\,\mathrm{d}t \le \frac{4\pi}{3}\rho^5\left(U_T^3 + P_T^{3/2}\right).$$

Division by $\rho^2$ proves (636); its right-hand side is independent of the center and tends to zero with $\rho$, which proves (637). □

## 18.3 Vorticity and material flow

**Theorem 18.7** (Global vorticity amplification and dissipation)**.** *Let $\nu > 0$, let $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, prescribe $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$, and let $(u,p)$ be the pressure-normalized global pair supplied by Theorem 15.8 for $(u_0,\nu,f)$. Put*

$$\omega := \nabla \times u, \qquad \omega_0 := \nabla \times u_0, \qquad \zeta := \nabla \times f, \qquad L^\omega_T := S_{3,1,\infty} T^{1/2} \mathfrak{X}_{0,3}[u_0,\nu,f;T].$$

*For every $T < \infty$ and $2 \le r \le \infty$, set*

$$Z_{r,T} := \int_0^T \|\zeta(t)\|_{L^r}\, \mathrm{d}t, \qquad \Omega_{r,T} := e^{L^\omega_T}\big(\|\omega_0\|_{L^r} + Z_{r,T}\big).$$

*Then*

$$\sup_{0\le t\le T} \|\omega(t)\|_{L^r} \le \Omega_{r,T}, \tag{638}$$

$$\int_0^T \|(\omega\cdot\nabla)u(t)\|_{L^r}\, \mathrm{d}t \le L^\omega_T \Omega_{r,T}. \tag{639}$$

*If $2 \le r < \infty$, then*

$$\int_0^T \Big\|\nabla|\omega(t)|^{r/2}\Big\|_2^2\, \mathrm{d}t \le \frac{r}{4\nu(r-1)} e^{rL^\omega_T}\Big[\|\omega_0\|^r_{L^r} + rZ_{r,T}\Omega^{r-1}_{r,T}\Big]. \tag{640}$$

*Proof.* Taking the curl of the momentum equation gives the three-dimensional vorticity equation

$$\partial_t\omega + (u\cdot\nabla)\omega = (\omega\cdot\nabla)u + \nu\Delta\omega + \zeta. \tag{641}$$

Fix $2 \le r < \infty$. Pair (641) with $|\omega|^{r-2}\omega$. To justify this power test, choose $\chi \in C^\infty_c(\mathbb{R}^3)$, with $0 \le \chi \le 1$, $\chi = 1$ on $B_1$, and $\chi = 0$ outside $B_2$, put $\chi_R(x) = \chi(x/R)$, and first test against

$$\chi_R \Phi_\varepsilon(\omega), \qquad \Phi_\varepsilon(z) := (|z|^2+\varepsilon)^{(r-2)/2} z.$$

Define the primitive

$$F_\varepsilon(z) := \frac{(|z|^2+\varepsilon)^{r/2} - \varepsilon^{r/2}}{r}. \tag{642}$$

Then $\nabla_z F_\varepsilon = \Phi_\varepsilon$, and incompressibility gives

$$\int_{\mathbb{R}^3} \chi_R (u\cdot\nabla)\omega\cdot\Phi_\varepsilon(\omega)\,\mathrm{d}x = -\int_{\mathbb{R}^3} F_\varepsilon(\omega)\, u\cdot\nabla\chi_R\,\mathrm{d}x. \tag{643}$$

With $A_R := B_{2R}\setminus B_R$,

$$\left|\int F_\varepsilon(\omega)\, u\cdot\nabla\chi_R\,\mathrm{d}x\right| \le \frac{C_r}{R}\|u\|_\infty\left(\|\omega\|^r_{L^r(A_R)} + \varepsilon^{(r-2)/2}\|\omega\|^2_{L^2(A_R)}\right), \tag{644}$$

$$\left|\int \nabla\chi_R\cdot\nabla\omega\,\Phi_\varepsilon(\omega)\,\mathrm{d}x\right| \le \frac{C_r}{R}\|\nabla\omega\|_{L^2(A_R)}\left(\|\omega\|^{r-1}_{L^{2r-2}(A_R)} + \varepsilon^{(r-2)/2}\|\omega\|_{L^2(A_R)}\right). \tag{645}$$

All norms on the right are finite because $u,\omega$ belong to every spatial Sobolev space. Their annular tails vanish as $R\to\infty$, so (643) converges to the transport cancellation and the diffusion cutoff term vanishes.

The remaining diffusion form is nonnegative:

$$\nabla\omega : D\Phi_\varepsilon(\omega)\nabla\omega = (|\omega|^2+\varepsilon)^{(r-2)/2}|\nabla\omega|^2 + (r-2)(|\omega|^2+\varepsilon)^{(r-4)/2}\sum_{j=1}^{3}(\omega\cdot\partial_j\omega)^2. \tag{646}$$

After the limit $R\to\infty$, dominated convergence applies to $F_\varepsilon(\omega)$, the stretching term, and the force term, while Fatou's lemma applies to (646) as $\varepsilon\downarrow 0$. This yields the unregularized chain rule and diffusion inequality used below. The stretching term satisfies

$$\left|\int_{\mathbb{R}^3}(\omega\cdot\nabla)u\cdot|\omega|^{r-2}\omega\,\mathrm{d}x\right| \le \|\nabla u\|_\infty\,\|\omega\|_r^r\,.$$

For the diffusion term, spatial integration by parts gives

$$-\int_{\mathbb{R}^3}\Delta\omega\cdot|\omega|^{r-2}\omega\,\mathrm{d}x \ge (r-1)\int_{\mathbb{R}^3}|\omega|^{r-2}|\nabla|\omega||^2\,\mathrm{d}x = \frac{4(r-1)}{r^2}\left\|\nabla|\omega|^{r/2}\right\|_2^2.$$

Hence

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}t}\|\omega\|_r^r + \frac{4\nu(r-1)}{r^2}\left\|\nabla|\omega|^{r/2}\right\|_2^2 \le \|\nabla u\|_\infty\,\|\omega\|_r^r + \|\zeta\|_r\,\|\omega\|_r^{r-1}\,. \tag{647}$$

Combining (647) with the quantitative rectangle estimate (836) gives

$$\int_0^T \|\nabla u(t)\|_\infty\,\mathrm{d}t \le L_T^\omega.$$

The scalar inequality

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\omega(t)\|_r \le \|\nabla u(t)\|_\infty\,\|\omega(t)\|_r + \|\zeta(t)\|_r$$

and Gronwall's inequality prove (638) for finite $r$. Multiplying (647) by $\exp(-r\int_0^t\|\nabla u(s)\|_\infty\,\mathrm{d}s)$ and integrating, while using $\|\omega(t)\|_r\le\Omega_{r,T}$, proves (640). Since $\omega_0,\omega(t),\zeta(t)\in L^2\cap L^\infty$, with uniform Schwartz bounds for $\zeta$ on $[0,T]$, $\|h\|_r\to\|h\|_\infty$ as $r\to\infty$ for each spatial slice and $Z_{r,T}\to Z_{\infty,T}$. Consequently,

$$\sup_{0\le t\le T}\|\omega(t)\|_\infty \le \liminf_{r\to\infty}\sup_{0\le t\le T}\|\omega(t)\|_r \le \Omega_{\infty,T}.$$

This is (638) at the endpoint. Finally,

$$\|(\omega\cdot\nabla)u(t)\|_r \le \|\nabla u(t)\|_\infty\,\|\omega(t)\|_r\,;$$

integration and (638) give (639). □

**Theorem 18.8** (Quantitative volume-preserving material flow)**.** *Let $\nu>0$, let $u_0\in H^\infty_\sigma(\mathbb{R}^3)$, prescribe $f\in C^\infty([0,\infty);\mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$, and let $(u,p)$ be the pressure-normalized global pair supplied by Theorem 15.8 for $(u_0,\nu,f)$. For $0<T<\infty$, put*

$$D_T := S_{2,0,\infty}T^{1/2}\mathfrak{X}_{0,2}[u_0,\nu,f;T],\qquad L_T := S_{3,1,\infty}T^{1/2}\mathfrak{X}_{0,3}[u_0,\nu,f;T]. \tag{648}$$

*For every $a \in \mathbb{R}^3$, the problem*

$$\partial_t X(t,a) = u(t, X(t,a)), \qquad X(0,a) = a, \tag{649}$$

*has a unique solution on $[0,T]$. For every $t \in [0,T]$, the map $X(t,\cdot) : \mathbb{R}^3 \to \mathbb{R}^3$ is a smooth volume-preserving diffeomorphism and*

$$|X(t,a) - a| \le D_T, \tag{650}$$

$$e^{-L_T}|a-b| \le |X(t,a) - X(t,b)| \le e^{L_T}|a-b|, \tag{651}$$

$$\det D_a X(t,a) = 1. \tag{652}$$

*Proof.* Equations (835) and (836) give

$$\int_0^T \|u(t)\|_\infty \, \mathrm{d}t \le D_T,$$

$$\int_0^T \|\nabla u(t)\|_\infty \, \mathrm{d}t \le L_T.$$

Picard iteration on successive compact time intervals gives local existence and uniqueness for (649). The first integral bound prevents escape to spatial infinity and extends every trajectory to $[0,T]$; it also proves (650). For two trajectories, the second integral bound and Gronwall's inequality give the upper bound in (651). Solving this equation backward from time $t$ produces the inverse map and gives the lower bound. Spatial differentiation of (649) and induction give smoothness of the map and its inverse. Finally, Jacobi's formula and $\nabla \cdot u = 0$ give

$$\partial_t \det D_a X = (\nabla \cdot u)(t, X) \det D_a X = 0, \qquad \det D_a X(0,a) = 1,$$

which proves (652). □

## 18.4 Relative energy and the Leray–Hopf class

The weak class is stated in the whole-space form originating with Leray and Hopf [10, 7]. The argument is the classical weak–strong uniqueness mechanism associated with Serrin [13], written here against the pressure-complete smooth history and derived directly from the weak formulation and the energy inequality.

**Definition 18.9** (Three-dimensional Leray–Hopf solution)**.** Let $\nu > 0$, $T > 0$, $v_0 \in H^0_\sigma(\mathbb{R}^3)$, and prescribe $f \in C^\infty([0,T]; \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$. A forced Leray–Hopf solution on $[0,T]$ for $f$ is a pair $(v,q)$ such that

$$v \in C_w([0,T]; H^0_\sigma) \cap L^\infty(0,T; L^2_\sigma) \cap L^2(0,T; \dot{H}^1_\sigma), \qquad q \in \mathcal{D}'((0,T)\times\mathbb{R}^3),$$

the equations

$$\partial_t v + (v\cdot\nabla)v + \nabla q = \nu\Delta v + f, \qquad \nabla \cdot v = 0$$

hold in distributions, $v(t) \to v_0$ strongly in $L^2$ as $t \downarrow 0$, and

$$\frac{1}{2}\|v(t)\|_2^2 + \nu \int_0^t \|\nabla v(s)\|_2^2 \, \mathrm{d}s \le \frac{1}{2}\|v_0\|_2^2 + \int_0^t \langle f(s), v(s)\rangle \, \mathrm{d}s \tag{653}$$

for every $t \in [0,T]$. Every divergence-free test field $\phi \in C^\infty_c([0,T)\times\mathbb{R}^3;\mathbb{R}^3)$ satisfies

$$\int_0^T \Big[ -\langle v, \partial_t\phi\rangle - \int_{\mathbb{R}^3} (v\otimes v) : \nabla\phi \, \mathrm{d}x + \nu \langle \nabla v, \nabla\phi\rangle \Big] \, \mathrm{d}t = \langle v_0, \phi(0)\rangle + \int_0^T \langle f, \phi\rangle \, \mathrm{d}t. \tag{654}$$

A global forced Leray–Hopf solution at viscosity $\nu$ for the prescribed force $f$ is a pair on $[0,\infty)$ whose restriction to every finite $[0,T]$ is a forced Leray–Hopf solution in Definition 18.9, with identical initial data, viscosity, and prescribed force.

**Lemma 18.10** (Solenoidal cutoff density and time commutation)**.** *Let $\mathscr{A}$ be a finite subset of $\{(r,p) : r \in \mathbb{N}_0,\ 2 \le p < \infty\}$. There are linear operators $\mathcal{T}_k$, independent of time, such that, for every $f \in H^\infty_\sigma(\mathbb{R}^3)$,*

$$\mathcal{T}_k f \in C^\infty_{c,\sigma}(\mathbb{R}^3)$$

*and, for every $(r,p) \in \mathscr{A}$,*

$$\sup_k \|\mathcal{T}_k f\|_{W^{r,p}} \le C_{\mathscr{A}} \|f\|_{W^{r,p}}\,, \tag{655}$$

$$\|\mathcal{T}_k f - f\|_{W^{r,p}} \longrightarrow 0. \tag{656}$$

*The convergence is uniform for $f$ in a compact subset of the retained intersection of $W^{r,p}$ spaces. If $F \in C^\infty([0,T]; H^\infty_\sigma)$, then*

$$\partial_t^a(\mathcal{T}_k F) = \mathcal{T}_k(\partial_t^a F) \qquad (a \in \mathbb{N}_0), \tag{657}$$

*and the convergence in* (656) *is uniform in $t$ for every fixed temporal derivative.*

*Proof.* Choose $\chi \in C^\infty_c(B_2)$ with $\chi = 1$ on $B_1$, and put $\chi_R(x) = \chi(x/R)$ and $A_R = B_{2R} \setminus \overline{B_R}$. For a smooth solenoidal $f$, set

$$g_R := \nabla\chi_R \cdot f.$$

It is supported in $A_R$, and

$$\int_{A_R} g_R \,\mathrm{d}x = \int_{\mathbb{R}^3} \nabla\cdot(\chi_R f)\,\mathrm{d}x = 0.$$

We now construct the required compact anti-divergence explicitly. Fix $\rho \in C^\infty_c((-3,3))$ with $\int_{\mathbb{R}} \rho = 1$. If $h \in C^\infty_c(B_2)$ and $\int h = 0$, set

$$m_1(y_2,y_3) := \int_{\mathbb{R}} h(s,y_2,y_3)\,\mathrm{d}s, \qquad m_2(y_3) := \int_{\mathbb{R}} m_1(s,y_3)\,\mathrm{d}s$$

and define $\mathcal{G}h = (G_1,G_2,G_3)$ by

$$\begin{aligned}
G_1(y) &:= \int_{-\infty}^{y_1} \big(h(s,y_2,y_3) - \rho(s) m_1(y_2,y_3)\big)\,\mathrm{d}s,\\
G_2(y) &:= \rho(y_1)\int_{-\infty}^{y_2} \big(m_1(s,y_3) - \rho(s) m_2(y_3)\big)\,\mathrm{d}s,\\
G_3(y) &:= \rho(y_1)\rho(y_2)\int_{-\infty}^{y_3} m_2(s)\,\mathrm{d}s.
\end{aligned}$$

The three integrands have zero total integral in their integration variables; for the last one this is $\int h = 0$. Thus $\mathcal{G}h \in C^\infty_c((-3,3)^3;\mathbb{R}^3)$, and direct differentiation gives

$$\nabla\cdot\mathcal{G}h = h - \rho(y_1)m_1 + \rho(y_1)m_1 - \rho(y_1)\rho(y_2)m_2 + \rho(y_1)\rho(y_2)m_2 = h.$$

Integration over intervals of length at most six, Hölder's inequality, Fubini's theorem, and differentiation under the integral give, for $a \in \mathbb{N}_0$ and $1 < p < \infty$,

$$\|\mathcal{G}h\|_{W^{a,p}((-3,3)^3)} \le C_{a,p,\rho}\,\|h\|_{W^{a,p}(B_2)}\,. \tag{658}$$

For $g \in C_c^\infty(B_{2R})$ of mean zero, put

$$(\mathcal{G}_R g)(x) := R\big[\mathcal{G}(g(R\,\cdot))\big](x/R).$$

Then $\nabla \cdot \mathcal{G}_R g = g$, its support lies in $(-3R, 3R)^3$, and scaling (658) gives

$$\begin{aligned} \|\mathcal{G}_R g\|_{L^p} &\le C_p R \, \|g\|_{L^p}\,, \\ \left\|\nabla^j \mathcal{G}_R g\right\|_{L^p} &\le C_{j,p} \sum_{b=0}^{j} R^{1-j+b} \left\|\nabla^b g\right\|_{L^p}\,, \qquad j \ge 1. \end{aligned} \tag{659}$$

Define

$$\mathcal{S}_R f := \chi_R f - \mathcal{G}_R g_R.$$

Then $\mathcal{S}_R f \in C_{c,\sigma}^\infty(\mathbb{R}^3)$, with support in $(-3R, 3R)^3$. Leibniz' rule gives, for $0 \le a \le r$,

$$\|\nabla^a g_R\|_{L^p(A_R)} \le C_a \sum_{b=0}^{a} R^{-1-a+b} \left\|\nabla^b f\right\|_{L^p(A_R)}.$$

Combining this estimate with (659) and the analogous cutoff estimate yields

$$\|\mathcal{S}_R f - f\|_{W^{r,p}} \le C_{r,p} \sum_{b=0}^{r} (1 + R^{b-r}) \left\|\nabla^b f\right\|_{L^p(\mathbb{R}^3 \setminus B_R)}. \tag{660}$$

For $r = 0$, this conclusion follows from (659) and $\|g_R\|_p \le C R^{-1} \|f\|_{L^p(A_R)}$. The right side tends to zero as $R \to \infty$, and these calculations give operator norms bounded independently of $R$.

Choose $R_k \uparrow \infty$ and set $\mathcal{T}_k := \mathcal{S}_{R_k}$. Estimate (660) holds simultaneously for the finite family $\mathscr{A}$, while the preceding operator estimates give (655) simultaneously for that family. Strong convergence plus the uniform operator bound makes convergence uniform on compact subsets: cover a compact set by finitely many norm balls and use the bound on each remainder. Each $\mathcal{T}_k$ is independent of time, so differentiation commutes with it. The ranges of $\partial_t^a F$ on $[0, T]$ are compact in every retained space, which proves the final assertion. □

**Lemma 18.11** (Admissible strong-history test)**.** *Let $\nu > 0$, let $u_0 \in H_\sigma^\infty(\mathbb{R}^3)$, prescribe $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3; \mathbb{R}^3))$, and let $u$ be the global solution supplied by Theorem 15.8 for $(u_0, \nu, f)$, and let $v$ be a forced Leray–Hopf solution on $[0, T]$ at this viscosity with that prescribed force $f$. Put $w := v - u$. Then, for every $t \in [0, T]$,*

$$\begin{aligned} \langle v(t), u(t)\rangle = \langle v_0, u_0\rangle &+ \int_0^t \int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot w \, \mathrm{d}x \, \mathrm{d}s \\ &- 2\nu \int_0^t \langle \nabla v, \nabla u\rangle \, \mathrm{d}s + \int_0^t \langle f, u + v\rangle \, \mathrm{d}s. \end{aligned} \tag{661}$$

*Proof.* The energy class and spatial interpolation give

$$v \in L^4(0, T; L^3), \qquad \|v\|_{L^4 L^3}^4 \le C \, \|v\|_{L^\infty L^2}^2 \, \|\nabla v\|_{L^2 L^2}^2\,,$$

so $v \otimes v \in L^2(0, T; L^{3/2})$.

Fix $0 < t < T$. Solenoidal density on $\mathbb{R}^3$, applied uniformly to the compact sets $u([0, T])$ and $u_t([0, T])$ in their finite Sobolev spaces, gives

$$u^{(k)} \in C^\infty([0, T]; C_{c,\sigma}^\infty(\mathbb{R}^3))$$

such that

$$u^{(k)} \longrightarrow u \quad \text{in} \quad W^{1,1}(0,T;L^2) \cap L^2(0,T;H^1) \cap L^2(0,T;W^{1,3}) \cap C([0,T];L^2). \tag{662}$$

Apply Lemma 18.10 with

$$\mathscr{A} = \{(0,2),(1,2),(1,3)\}$$

and set $u^{(k)} = \mathcal{T}_k u$. Its uniform compact-range conclusion for $u$ and $u_t$, together with time commutation (657), gives (662).

For $0 < h < T - t$, choose $\eta_h \in C_c^\infty([0,T))$ with $0 \le \eta_h \le 1$, $\eta_h = 1$ on $[0,t]$, $\eta_h = 0$ on $[t+h,T]$, and $\eta_h' \le 0$. The field $\eta_h u^{(k)}$ is an admissible test in (654). First let $k \to \infty$. The convergence in (662), together with $v \otimes v \in L^2 L^{3/2}$, permits passage in every term.

Choose the time cutoffs so that $-\eta_h'$ is an approximate identity from the right at $t$, and let $h \downarrow 0$. Weak continuity of $v$, strong continuity of $u$, and the Lebesgue differentiation theorem give

$$\begin{aligned}\langle v(t), u(t)\rangle = \langle v_0, u_0\rangle &+ \int_0^t \langle v, u_t\rangle \, \mathrm{d}s + \int_0^t \int_{\mathbb{R}^3} (v \otimes v) : \nabla u \, \mathrm{d}x \, \mathrm{d}s \\ &- \nu \int_0^t \langle \nabla v, \nabla u\rangle \, \mathrm{d}s + \int_0^t \langle f, u\rangle \, \mathrm{d}s.\end{aligned}$$

This identity first holds for almost every $t$. Its right-hand side is continuous in $t$, while $t \mapsto \langle v(t), u(t)\rangle$ is continuous by weak continuity of $v$ and norm continuity of $u$. Hence it holds for every $0 \le t \le T$, including $t = T$ by passage from below. Using

$$u_t = -(u \cdot \nabla)u - \nabla p + \nu \Delta u + f$$

is admissible because $v$ is divergence free. It gives

$$\int_0^t \langle v, u_t\rangle \, \mathrm{d}s = -\int_0^t \int_{\mathbb{R}^3} v \cdot (u \cdot \nabla)u \, \mathrm{d}x \, \mathrm{d}s - \nu \int_0^t \langle \nabla v, \nabla u\rangle \, \mathrm{d}s + \int_0^t \langle v, f\rangle \, \mathrm{d}s.$$

Finally,

$$\int_{\mathbb{R}^3} [(v \otimes v) : \nabla u - v \cdot (u \cdot \nabla)u] \, \mathrm{d}x = \int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot w \, \mathrm{d}x,$$

because

$$\int_{\mathbb{R}^3} (w \cdot \nabla u) \cdot u \, \mathrm{d}x = \frac{1}{2} \int_{\mathbb{R}^3} w \cdot \nabla |u|^2 \, \mathrm{d}x = 0.$$

This proves (661). □

**Theorem 18.12** (Quantitative Leray–Hopf stability)**.** *Let $\nu > 0$, $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and prescribe $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$. Let $u$ be the global strong history from Theorem 15.8 for $(u_0, \nu, f)$. Fix $T > 0$, and let $v$ be any forced Leray–Hopf solution on $[0,T]$ at this viscosity, with datum $v_0 \in H^0_\sigma(\mathbb{R}^3)$ and that prescribed force $f$. Then, for $0 \le t \le T$,*

$$\begin{aligned}&\frac{1}{2}\|v(t) - u(t)\|_2^2 + \nu \int_0^t \|\nabla(v-u)(s)\|_2^2 \, \mathrm{d}s \\ &\quad \le \frac{1}{2}\|v_0 - u_0\|_2^2 \exp\Big(2\int_0^t \|\nabla u(s)\|_\infty \, \mathrm{d}s\Big) \qquad (663) \\ &\quad \le \frac{1}{2}\|v_0 - u_0\|_2^2 \exp\Big(2S_{3,1,\infty} T^{1/2} \mathfrak{P}_{0,2}[u_0,\nu,f;T]^{1/2}\Big). \qquad (664)\end{aligned}$$

*The lower spatial rectangle also gives the viscosity-weighted estimate*

$$\frac{1}{2}\|v(t)-u(t)\|_2^2+\frac{\nu}{2}\int_0^t\|\nabla(v-u)(s)\|_2^2\,\mathrm{d}s \\ \le\frac{1}{2}\|v_0-u_0\|_2^2\exp\left(\frac{S_{2,0,\infty}^2}{\nu}\mathfrak{X}_{0,2}[u_0,\nu,f;T]^2\right). \tag{665}$$

*Proof.* Combine (653), the forced strong energy equality (482), and (661). The two energy relations contribute $\int_0^t\langle f,u+v\rangle\,\mathrm{d}s$, exactly the force work in the cross identity, so the common prescribed force cancels without changing histories. With $w=v-u$, this gives

$$\frac{1}{2}\|w(t)\|_2^2+\nu\int_0^t\|\nabla w(s)\|_2^2\,\mathrm{d}s\le\frac{1}{2}\|w(0)\|_2^2-\int_0^t\int_{\mathbb{R}^3}(w\cdot\nabla u)\cdot w\,\mathrm{d}x\,\mathrm{d}s. \tag{666}$$

The integral term in (666) has absolute value at most

$$\|\nabla u(s)\|_\infty\|w(s)\|_2^2.$$

Gronwall's inequality bounds $\|w(t)\|_2^2$. Substitution of that bound back into (666) proves (663). Proposition 17.16 gives

$$\int_0^T\|\nabla u(t)\|_\infty\,\mathrm{d}t\le S_{3,1,\infty}T^{1/2}\mathfrak{P}_{0,2}[u_0,\nu,f;T]^{1/2},$$

which proves (664).

For the lower-rectangle estimate, integration by parts and $\nabla\cdot w=0$ give

$$\left|\int_{\mathbb{R}^3}(w\cdot\nabla u)\cdot w\,\mathrm{d}x\right|=\left|\int_{\mathbb{R}^3}(w\cdot\nabla w)\cdot u\,\mathrm{d}x\right|\le\|u\|_\infty\|w\|_2\|\nabla w\|_2.$$

Young's inequality in (666) gives

$$\frac{1}{2}\|w(t)\|_2^2+\frac{\nu}{2}\int_0^t\|\nabla w(s)\|_2^2\,\mathrm{d}s\le\frac{1}{2}\|w(0)\|_2^2+\frac{1}{2\nu}\int_0^t\|u(s)\|_\infty^2\|w(s)\|_2^2\,\mathrm{d}s.$$

Set

$$A(t):=\frac{1}{\nu}\int_0^t\|u(s)\|_\infty^2\,\mathrm{d}s.$$

Gronwall's inequality gives $\|w(t)\|_2^2\le\|w(0)\|_2^2e^{A(t)}$. Substitution in the preceding inequality gives

$$\frac{1}{2}\|w(t)\|_2^2+\frac{\nu}{2}\int_0^t\|\nabla w(s)\|_2^2\,\mathrm{d}s\le\frac{1}{2}\|w(0)\|_2^2e^{A(t)}.$$

Finally, (835) yields

$$A(t)\le\frac{S_{2,0,\infty}^2}{\nu}\mathfrak{X}_{0,2}[u_0,\nu,f;T]^2,$$

which proves (665). □

**Corollary 18.13** (Explicit weak-class stability for small forced critical data)**.** *Under the hypotheses of Theorem 18.3, let $v$ be a global forced Leray–Hopf solution at this viscosity $\nu$, with datum $v_0 \in H^0_\sigma(\mathbb{R}^3)$ and that prescribed force $f$. Then*

$$\begin{aligned}\frac{1}{2}\|v(t)-u(t)\|_2^2+\nu\int_0^t\|\nabla(v-u)(s)\|_2^2\,\mathrm{d}s\\ \le\frac{1}{2}\|v_0-a\|_2^2\exp(2S_{3,1,\infty}tU_3^{\mathrm{sm}})\qquad(t\ge0).\end{aligned}\tag{667}$$

*Proof.* Equation (623) gives $\sup_{s\ge0}\|u(s)\|_{H^3}\le U_3^{\mathrm{sm}}$. Hence

$$\int_0^t\|\nabla u(s)\|_\infty\,\mathrm{d}s\le S_{3,1,\infty}tU_3^{\mathrm{sm}}.$$

The derivation of (663) uses only the smooth strong history, the Leray–Hopf energy inequality, and the common prescribed force. Applying that derivation directly to the global small-critical history from Theorem 18.3 gives the stated explicit small-data refinement. □

**Corollary 18.14** (Uniqueness and energy equality in the Leray–Hopf class)**.** *Let $\nu>0$, $u_0\in H^\infty_\sigma(\mathbb{R}^3)$, and prescribe $f\in C^\infty([0,\infty);\mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$. If $(v,q)$ is any global three-dimensional forced Leray–Hopf solution at viscosity $\nu$, with datum $u_0$ and prescribed force $f$, then $v$ agrees on every finite interval with the velocity of the global forced solution from Theorem 15.8. Thus $v$ has the smooth representative*

$$u\in C^\infty([0,\infty)\times\mathbb{R}^3;\mathbb{R}^3),$$

*and it satisfies*

$$\frac{1}{2}\|u(t)\|_2^2+\nu\int_0^t\|\nabla u(s)\|_2^2\,\mathrm{d}s=\frac{1}{2}\|u_0\|_2^2+\int_0^t\langle f(s),u(s)\rangle\,\mathrm{d}s\qquad(t\ge0).$$

*Its distributional pressure has the form*

$$q=p+c(t)\quad\text{in }\mathcal{D}'((0,\infty)\times\mathbb{R}^3),\tag{668}$$

*where $p$ is the pressure in* (481) *and $c$ is a scalar distribution in time. The canonical Riesz representative*

$$R_iR_j(u_iu_j)-(-\Delta)^{-1}\nabla\cdot f$$

*equals $p$. If $q$ is represented by a locally integrable function with $q(t,\cdot)\in C_0(\mathbb{R}^3)$ for almost every $t$, then $q=p$ almost everywhere.*

*Proof.* Set $v_0=u_0$ in Theorem 18.12. The right-hand side of (664) is zero, so $v=u$ in $C_w([0,T];L^2)\cap L^2(0,T;\dot H^1)$. Since $T$ is arbitrary, the identity holds globally. Smoothness and the forced energy equality follow from (478) and (482). If $q$ is a distributional pressure paired with this velocity and $p$ is the pressure in (481), subtraction of the two momentum equations gives

$$\nabla(q-p)=0\quad\text{in }\mathcal{D}'((0,\infty)\times\mathbb{R}^3).$$

A distribution on a product domain whose spatial gradient vanishes is the pullback of a scalar distribution in time: testing first against spatial test functions of integral zero and then against one fixed test function of integral one gives (668). The Riesz formula for $p$ proves the canonical-representative assertion. In the function-valued case, $q(t,x)-p(t,x)=c(t)$ for almost every $(t,x)$; both spatial slices tend to zero as $|x|\to\infty$, so $c(t)=0$ for almost every $t$. □

# 19 Conclusion

The argument follows one forced velocity–pressure history from its prescribed datum $(u_0, \nu, f)$. Differentiating the complete equation generates its velocity, pressure, and force mixed jet. The datum-generated finite records give every temporal–spatial Sobolev row. Independently, the completed critical-height identities, evaluated on this mixed jet with force work retained, recover every spatial Sobolev row, and the forced equation reconstructs the temporal and pressure rows. A finite rectangle selects the rows used at any prescribed orders, and its norm remains finite through the endpoint. The viscous diagonal relation keeps the adjacent faces coupled. Integrating the complete rectangle relations keeps the transport, pressure, viscosity, prescribed force, and adjacent temporal response inside those estimates as parts of one equation.

The adjacent rows then have continuous Sobolev representatives through the proposed terminal time. Their compatible values determine $u_* = u(T_*) \in H^\infty_\sigma$, while the forced pressure formula and the velocity recurrence determine the complete endpoint jet from $u_*$ and $(\partial_t^n f(T_*))_{n\geq 0}$. Local well-posedness restarts from the single condition $w(T_*) = u_*$ with that prescribed absolute-time force $f(t)$. Complete endpoint-jet matching glues this restart to the incoming history. Every proposed finite maximal time admits a classical continuation, contradicting maximality. Hence the maximal time is infinite.

The resulting global forced solution is smooth, obeys the exact energy identity with force work, and carries the decay-normalized pressure

$$p = R_i R_j (u_i u_j) - (-\Delta)^{-1} \nabla \cdot f.$$

Its compatible finite rectangles also quantify the critical-height rows, frequency tails, Lipschitz action, vorticity amplification and dissipation, material flow, and same-force relative-energy stability. The small-critical theory is an explicit quantitative refinement of this general construction: its scale-invariant velocity stock and projected-force stock give closed formulas for the corresponding finite rectangle bounds.

Finally, relative energy identifies every forced Leray–Hopf solution with that initial datum, viscosity, and prescribed force with this smooth velocity; the common force work cancels in the comparison. The pressure is unique up to the unavoidable time-dependent gauge unless the decay normalization is imposed. Every step belongs to the one forced evolution generated by $(u_0, \nu, f)$.

# Part IV
# Refutation of OpenAI's *Finite Time Blowup for Navier–Stokes*

## 20 Introduction

OpenAI's *Finite Time Blowup for Navier–Stokes* proposes a three-dimensional flow that starts from rest and develops unbounded velocity under a smooth compactly supported force [18, Theorem 1.1]. Its kinetic energy remains finite because the large velocities occupy shrinking spatial regions. A comparison based only on the size of the external force or the kinetic energy cannot distinguish this concentration from a regular flow. The relevant question is whether the proposed solution can satisfy the stronger space–time bounds imposed by a regularity theorem for its prescribed data.

Part II relates the temporal derivatives of one unforced velocity field to its spatial Sobolev energies. Part III carries that construction into a new forced history and retains every prescribed source row. Viscosity couples the energy at order $1/2$ to dissipation at order $3/2$, while temporal differentiation produces higher spatial orders together with differentiated nonlinear and source terms. Both the explicit force work and the velocity's nonlinear response remain in the comparison below.

The forced theorem proved in Part III gives the full-interval solution used here. By Corollary 20.1, finite collections of its mixed Sobolev norms bound the velocity and all the terms in its momentum equation. The proposed exterior has divergent positive contributions to precisely these norms. Uniqueness identifies a proposed classical solution with the regular solution for its data, so the bounds apply before any regularity at the candidate's terminal time is assumed.

The source estimates distinguish contributions controlled by the initial data and prescribed force from those that require a velocity norm. This distinction yields quantitative sufficient conditions for continuation, proved in the appendices together with the general-order source expansions and product estimates.

### 20.1 Force, pressure, and material acceleration

We work on $\mathbb{R}^3$, at fixed viscosity $\nu > 0$, with unit density and no material boundary. Let $u_0$ be divergence free and belong to $H^\infty = \bigcap_{m\geq 0} H^m$. Fix $0 < T < \infty$. The force $f(x,t)$, measured per unit mass, is smooth through $[0,T]$, and all its mixed derivatives decay rapidly in space, uniformly on this interval. A subscript $\sigma$ denotes the divergence-free subspace. The whole-space Navier–Stokes system is [5, equations (1)–(3)]:

$$u_t + (u\cdot\nabla)u + \nabla p - \nu\Delta u = f, \qquad \nabla\cdot u = 0, \qquad u(0) = u_0. \tag{669}$$

At a fixed location, $u_t$ records how the velocity changes with time. A moving parcel also encounters spatial changes in velocity. Along $\dot X = u(X,t)$, its acceleration is

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}u(X(t),t) = D_t u(X(t),t), \qquad D_t = \partial_t + u\cdot\nabla, \\ D_t u = -\nabla p + \nu\Delta u + f. \end{aligned} \tag{670}$$

Thus the prescribed force contributes to the acceleration alongside pressure and viscous stress. Pressure depends on the surrounding velocity through incompressibility. Viscosity depends on its spatial curvature. These two responses must be retained when asking what a finite external force can cause.

Eulerian and material derivatives describe different derivatives of the velocity field along fixed points and moving trajectories. With $V_n = \partial_t^n u$, acceleration is $D_t u = V_1 + (V_0 \cdot \nabla)V_0$. Every higher parcel derivative is also a finite combination of the velocity's mixed derivatives. At all finite orders the two derivative families determine each other through product and commutator identities. Appendix J proves equality of the two complete Sobolev intersections. Finite rectangles of that intersection bound both the fixed-location derivatives and the parcel responses.

Differentiating the force along a parcel already exhibits this coupling:

$$D_t f = \partial_t f + (u \cdot \nabla) f.$$

The spatial variation of a smooth prescribed force is sampled at a rate set by the velocity. Higher material derivatives also differentiate that velocity. The parcel's acceleration, jerk, and subsequent derivatives are calculated within the mixed jet by exact product and commutator identities. This relation also appears in the Euler formulation [16, Section 1; Proposition 6.1].

Incompressibility determines the pressure contribution explicitly. Let $\mathbb{P}$ denote the orthogonal Leray projection, with Fourier symbol $I - \xi \otimes \xi/|\xi|^2$ for $\xi \neq 0$. Its value at zero is immaterial to an $L^2$-based norm. Write $\mathbb{Q} = I - \mathbb{P}$ for the gradient projection. Then

$$\begin{aligned} u_t - \nu\Delta u + \mathbb{P}\nabla \cdot (u \otimes u) &= \mathbb{P} f, \\ \nabla p &= (I - \mathbb{P})\big[f - (u \cdot \nabla)u\big]. \end{aligned} \tag{671}$$

Pressure is determined up to a function of time; its gradient is the physical quantity in momentum. A conservative force $f = \nabla\phi$ can be absorbed by setting $q = p - \phi$. The velocity then obeys the unforced equation exactly. In particular, rest remains a solution, with $p = \phi$. Boundary-driven flows require additional boundary conditions and are outside the whole-space setting of this paper.

## 20.2 The forced theorem and the comparison principle

Part III has already proved global smoothness and uniqueness for every datum and force in the class used below. Its theorem gives the following closed-interval consequence in the notation of this part.

**Corollary 20.1** (Finite-horizon forced intersection)**.** *Let $\nu > 0$, $u_0 \in H^\infty_\sigma(\mathbb{R}^3)$, and $f \in C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3;\mathbb{R}^3))$. For every finite $T > 0$, the unique global pair $(U, P)$ supplied by Theorem 15.8 satisfies*

$$U \in \bigcap_{r,m \geq 0} H^r(0, T; H^m(\mathbb{R}^3)). \tag{672}$$

*The mixed derivatives satisfy the differentiated momentum and incompressibility equations.*

*Proof.* Part III gives the global pressure-normalized solution. On $[0, T]$, its persistence argument places each spatial row in $C([0,T]; H^m)$, and the complete forced recurrence produces every temporal row. Hence, for every finite $r, m$,

$$\sum_{n=0}^{r} \|\partial_t^n U\|^2_{L^2(0,T;H^m)} \leq T \sum_{n=0}^{r} \sup_{0 \leq t \leq T} \|\partial_t^n U(t)\|^2_{H^m} < \infty.$$

Taking the intersection over the finite indices gives (672); the differentiated momentum and incompressibility rows are restrictions of the complete forced mixed jet proved in Part III. □

To see the comparison principle, let $v$ be another classical solution for these data on $[0,T)$, with finite spatial Sobolev norms on each strict subinterval. Uniqueness gives $v=U$ there. The adjacent norms $U\in L^2(0,T;H^3)$ and $U_t\in L^2(0,T;H^1)$ imply

$$\sup_{0\le t<T}\|v(t)\|_\infty^2\le\frac{\|u_0\|_{H^2}^2+\|U\|_{L^2H^3}^2+\|U_t\|_{L^2H^1}^2}{8\pi}<\infty. \tag{673}$$

The trace and embedding calculations are proved in Section 23.1; the uniqueness argument is given in Section 27. Before that argument is applied to the candidate of [18], Section 26 proves that its full localized residual belongs to the prescribed force class of Part III. Once that source payment has been made, the estimate excludes unbounded velocity. The critical-height and material calculations below identify further norms in which the proposed concentration fails the forced theorem.

## 21 Forced mixed derivatives and Sobolev energies

On each compact interval of local smooth existence, define $V_n=\partial_t^n u$ at fixed spatial coordinates. All spatial norms use Euclidean vector and Frobenius tensor norms. Throughout the derivative formulas, $f$ is the prescribed force in (669).

Put $F_n=\partial_t^n f$, $Q_n=\partial_t^n p$, and

$$B_n=\sum_{a=0}^{n}\binom{n}{a}(V_a\cdot\nabla)V_{n-a}. \tag{674}$$

The product rule applied to (669) gives the forced recurrence already derived in Proposition 13.4:

$$V_{n+1}+B_n+\nabla Q_n-\nu\Delta V_n=F_n. \tag{675}$$

Applying $\partial_x^\alpha$ to transport gives the full mixed expansion

$$\partial_x^\alpha B_n=\sum_{a=0}^{n}\sum_{\beta\le\alpha}\binom{n}{a}\binom{\alpha}{\beta}(\partial_x^\beta V_a\cdot\nabla)\partial_x^{\alpha-\beta}V_{n-a}. \tag{676}$$

Every force coordinate in this equation is the prescribed derivative $\partial_x^\alpha F_n$.

Taking divergence of (675) gives the pressure Poisson equation. In the Sobolev class used below, every nonpressure term of that row belongs to $H^m$, so $\nabla Q_n$ does too. Applying the orthogonal projection $I-\mathbb{P}$ to the full row then fixes its gradient:

$$-\Delta Q_n=\nabla\cdot B_n-\nabla\cdot F_n,\qquad \nabla Q_n=(I-\mathbb{P})(F_n-B_n). \tag{677}$$

The gradient part of the force contributes to pressure, while its solenoidal part remains in the projected velocity equation. Formula (677) will bound pressure through its gradient, the quantity present in momentum.

The input regularity already gives a compatible source intersection:

$$f\in\bigcap_{r,m\ge0}H^r(0,T;H^m(\mathbb{R}^3)). \tag{678}$$

Indeed, each mixed derivative has finite spatial Sobolev norm uniformly on the compact time interval, and integration in time preserves finiteness. The initial velocity jet is generated recursively by (675), in the form proved in Lemma 13.7:

$$a_0=u_0,\qquad a_{n+1}=\nu\Delta a_n-\mathbb{P}\sum_{b=0}^{n}\binom{n}{b}(a_b\cdot\nabla)a_{n-b}+\mathbb{P}F_n(0),\qquad a_n=V_n(0). \tag{679}$$

The product bound proved below shows by induction that $a_n \in H^m$ for every finite $n, m$, when $u_0 \in H^\infty_\sigma$.

## 21.1 Viscous coupling of Sobolev orders

Set $\Lambda = (-\Delta)^{1/2}$. Its power $\Lambda^s$ weights a Fourier component of wave number $\xi$ by $|\xi|^s$. Higher Sobolev energies thus measure progressively finer spatial variation; only height zero is the ordinary kinetic energy. For a temporal row $r$ and real height $s \geq 0$, define

$$E_{r,s} = \tfrac{1}{2} \|\Lambda^s V_r\|_2^2 , \qquad \mathcal{P}_{r,s} = -\operatorname{Re} \langle \Lambda^s V_r, \Lambda^s B_r \rangle , \qquad W_{r,s} = \operatorname{Re} \langle \Lambda^s V_r, \Lambda^s F_r \rangle .$$

Pairing (675) with $\Lambda^{2s} V_r$ eliminates the pressure gradient by incompressibility. The viscous pairing is $-\nu \left\|\Lambda^{s+1} V_r\right\|_2^2$, giving the exact row balance

$$\partial_t E_{r,s} = -2\nu E_{r,s+1} + \mathcal{P}_{r,s} + W_{r,s}. \tag{680}$$

Differentiating repeatedly exposes the higher spatial heights:

$$\partial_t^k E_{r,s} = (-2\nu)^k E_{r,s+k} + \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} (\mathcal{P}_{r,s+j} + W_{r,s+j}), \qquad k \geq 1. \tag{681}$$

For $k = 1$ this is (680). Differentiating the formula for $k$, substituting (680) at height $s + k$, and collecting the added term proves the formula for $k + 1$. Force leaves the viscous coefficient of the exposed energy unchanged. At $r = 0, s = 1/2$, the successive spatial heights are $1/2, 3/2, 5/2, \ldots$. Here $\partial_t^k E_{r,s}$ differentiates a scalar energy. It is not $E_{r+k,s}$, the energy of a higher velocity derivative. For example,

$$\partial_t^2 E_{r,s} = \|\Lambda^s V_{r+1}\|_2^2 + \operatorname{Re} \langle \Lambda^s V_r, \Lambda^s V_{r+2} \rangle .$$

The mixed velocity rows organize both factors in this expression, and all factors in the differentiated nonlinear and force terms.

The work derivatives retain both the response and source jets:

$$\partial_t^\ell W_{r,s} = \sum_{b=0}^{\ell} \binom{\ell}{b} \operatorname{Re} \langle \Lambda^s V_{r+b}, \Lambda^s F_{r+\ell-b} \rangle . \tag{682}$$

The nonlinear current has the full product rule

$$\partial_t^\ell \mathcal{P}_{r,s} = -\sum_{a=0}^{r} \binom{r}{a} \sum_{b+c+d=\ell} \frac{\ell!}{b!c!d!} \operatorname{Re} \left\langle \Lambda^s V_{r+b}, \Lambda^s \big( (V_{a+c} \cdot \nabla) V_{r-a+d} \big) \right\rangle . \tag{683}$$

Spatial differentiation uses (676) and the multi-index product rule. Every source coordinate remains a derivative of the prescribed field $f$; the velocity factor in (682) remains part of the response. The source intersection (678) controls each source factor. Bounds on the velocity factors require coordinates of the velocity history.

These identities hold on every $[0, S]$ within smooth existence. If the velocity belongs to the full-interval intersection, the continuous limits in Appendix J extend them to $T$. All differentiated nonlinear and force terms are retained before taking that limit.

## 21.2 Forcing corrections at the critical heights

At the critical starting height, put $H_r = E_{r,1/2}$. Rearranging the forced height identity gives the completed coordinate

$$\begin{aligned} C^f_{r,k} &:= \frac{\partial_t^k H_r}{(-2\nu)^k} \\ &\quad - \sum_{j=0}^{k-1} (-2\nu)^{j-k} \partial_t^{k-1-j} (\mathcal{P}_{r,j+1/2} + W_{r,j+1/2}), \\ C^f_{r,k} &= E_{r,k+1/2}. \end{aligned} \tag{684}$$

The factor $(-2\nu)^k$ is unchanged by forcing and is nonzero at every finite order. Each next temporal differentiation thus exposes the spatial height $k + 1/2$, with its complete nonlinear and source contributions. The identity is exact on every interval of smooth existence.

At the first two exposed heights, the source correction is bounded using kinetic energy and source regularity. Put $W_s = W_{0,s}$ and $h_s = \Lambda^{2s}\mathbb{P}f$. Moving derivatives onto $h_s$ writes the work as $W_s = \langle u, h_s \rangle$. The energy inequality bounds $\|u\|_{L^\infty L^2}$; differentiating this pairing moves the transport and diffusion derivatives onto the smooth prescribed field. Consequently $W_s$ and $W_s'$ have finite bounds $\mathsf{B}_s$ and $\mathsf{L}_s$, defined in (840) and proved in Appendix C. Evaluate the completed-height formulas of Proposition 14.3 on the forced velocity itself, and denote these expressions by

$$\begin{aligned} \widehat{C}_1[u] &:= -\frac{H' - \mathcal{P}_{1/2}}{2\nu}, \\ \widehat{C}_2[u] &:= \frac{H'' - \partial_t \mathcal{P}_{1/2} + 2\nu \mathcal{P}_{3/2}}{4\nu^2}, \qquad H = E_{0,1/2}. \end{aligned} \tag{685}$$

Here $\mathcal{P}_s = \mathcal{P}_{0,s}$ and $W_s = W_{0,s}$. Substituting the forced balance, with both time derivatives expanded, gives

$$\begin{aligned} \widehat{C}_1[u] &= E_{0,3/2} - \frac{W_{1/2}}{2\nu}, \\ \widehat{C}_2[u] &= E_{0,5/2} + \frac{W'_{1/2} - 2\nu W_{3/2}}{4\nu^2}. \end{aligned} \tag{686}$$

The estimates (840) control every source expression on the right. With

$$\delta_1 := \frac{\mathsf{B}_{1/2}}{2\nu}, \qquad \delta_2 := \frac{\mathsf{L}_{1/2}}{4\nu^2} + \frac{\mathsf{B}_{3/2}}{2\nu},$$

we obtain, for $j = 1, 2$,

$$\left\| C^f_{0,j} - \widehat{C}_j[u] \right\|_{L^\infty(0,T)} \le \delta_j < \infty. \tag{687}$$

These constants depend on the prescribed force and the kinetic-energy bound; no smallness condition is imposed. On a finite interval a bounded additive correction belongs to every $L^p$, $1 \le p \le \infty$. Consequently, the exact membership relation is

$$\begin{aligned} C^f_{0,j} \in L^p(0,T) \quad &\Longleftrightarrow \quad \widehat{C}_j[u] \in L^p(0,T), \\ \left\| C^f_{0,j} \right\|_{L^p(0,T)} &\le \left\| \widehat{C}_j[u] \right\|_{L^p(0,T)} + T^{1/p} \delta_j, \end{aligned} \tag{688}$$

where $T^{1/\infty} = 1$. Thus the explicit force correction preserves these completed-coordinate spaces, including the unweighted time integral at the critical viscous height $3/2$. Every expression here is evaluated on the forced history. The notation $\widehat{C}_j[u]$ retains the earlier completion formula; it does not replace its argument by an unforced solution.

## 22 Finite rectangles and uniform Sobolev bounds

Assume that the forced solution belongs to the class

$$u \in \mathscr{I}_T := \bigcap_{r,m\ge 0} H^r(0,T;H^m(\mathbb{R}^3)). \tag{I}$$

For integers $N, M \ge 0$, collect two adjacent space–time norms in

$$R_{N,M} := \sum_{n=0}^{N} \left( \|V_n\|^2_{L^2H^{M+1}} + \|V_{n+1}\|^2_{L^2H^{M-1}} \right) < \infty. \tag{689}$$

This is the velocity part of the finite rectangle in Definition 13.12; its organization by restriction is given by Definition 15.1 and Theorem 15.2. Time norms in this paper are over $(0,T)$. Different finite selections agree under restriction because their entries are derivatives of $u$. There is no bound common to all $N, M$ in the hypothesis. For $M = 0$, $H^{M-1} = H^{-1}$ is the usual dual Sobolev space.

We first convert these integral bounds into uniform bounds on a row. For $v \in L^2H^{m+1}$ with $v_t \in L^2H^{m-1}$, the Hilbert-pair identity is

$$\frac{\mathrm{d}}{\mathrm{d}t} \|v\|^2_{H^m} = 2\,\mathrm{Re}\left\langle (1-\Delta)^{(m+1)/2} v, (1-\Delta)^{(m-1)/2} v_t \right\rangle.$$

For classical rows it follows by differentiation and Plancherel. The identity extends to the displayed Hilbert spaces by spatial Fourier truncation and time smoothing. To justify the continuous representative, apply the resulting estimate to the difference of two truncated fields: their spatial $L^2H^{m+1}$ tails and temporal $L^2H^{m-1}$ tails tend to zero. The truncations are consequently Cauchy in $C([0,T];H^m)$. Integrate the identity in either direction from a time whose squared norm is at most its average. Cauchy–Schwarz gives

$$\sup_{0\le t\le T} \|v\|^2_{H^m} \le T^{-1} \|v\|^2_{L^2H^m} + 2\,\|v\|_{L^2H^{m+1}}\,\|v_t\|_{L^2H^{m-1}}. \tag{690}$$

Define the finite row constants

$$A_{n,m} := \sqrt{(1+T^{-1})R_{n,m}}, \qquad L_{n,m} := \sup_{0\le t\le T} \|F_n(t)\|_{H^m}. \tag{691}$$

The constants $A_{n,m}$ are distinguished from the initial fields $a_n$ in (679). Using $2ab \le a^2 + b^2$ in (690) yields

$$\sup_{0\le t\le T} \|V_n(t)\|_{H^m} \le A_{n,m}. \tag{692}$$

The estimate includes the endpoint through the continuous trace. Its constant depends on the indicated finite rectangle.

## 23 Critical production and velocity bounds

The two heights $1/2$ and $3/2$ enter together because viscosity pairs the critical velocity energy with one additional spatial derivative. We now extract both bounds from one finite rectangle and substitute them into the nonlinear current. Select $R = R_{0,1}$, and write $D = \|u_0\|^2_{H^1}$. Integrating the trace identity from time zero gives

$$\sup_t \|u\|^2_{H^1} \le D + R, \qquad \|u\|^2_{L^2H^2} \le R.$$

Since positive homogeneous derivative norms are bounded by these inhomogeneous norms,

$$\sup_t \|u\|_{\dot{H}^{1/2}} \le \sqrt{D+R}, \qquad \|u\|^2_{L^2\dot{H}^{3/2}} \le R. \tag{693}$$

The sharp fractional Sobolev inequality gives

$$\|v\|_3 \le C_3 \left\|\Lambda^{1/2} v\right\|_2, \qquad C_3 = 2^{-1/6}\pi^{-1/3}$$

[11, 17]. In dimension three at order $1/2$, the squared form has coefficient $2^{1/3}\pi^{2/3}$ on $\|v\|_3^2$. The constant also applies to Euclidean vectors and Frobenius tensors: the positive scalar kernel of $\Lambda^{-1/2}$ gives $|\Lambda^{-1/2}F| \le \Lambda^{-1/2}|F|$, followed by the scalar inequality. This introduces no factor for the number of components. Hölder applied to $\mathcal{P}_{1/2} = -\langle \Lambda u, (u\cdot\nabla)u\rangle$ and this embedding on its three factors gives

$$|\mathcal{P}_{1/2}| \le C_3^3 \|u\|_{\dot{H}^{1/2}} \|u\|^2_{\dot{H}^{3/2}}.$$

Substitution of (693) yields

$$\int_0^T |\mathcal{P}_{1/2}|\,\mathrm{d}t \le \frac{\sqrt{D+R}\,R}{\sqrt{2}\pi} < \infty. \tag{694}$$

For rest data on $(0,1)$, this reads

$$\sup_t E_{1/2}(t) \le R/2, \qquad \int_0^1 E_{3/2}(t)\,\mathrm{d}t \le R/2, \qquad \int_0^1 |\mathcal{P}_{1/2}|\,\mathrm{d}t \le \frac{R^{3/2}}{\sqrt{2}\pi}.$$

The first factor in production is uniformly bounded, and its two higher spatial factors have a finite time integral. No comparison between the size of this bound and viscosity is needed.

The forcing contribution is $W_{1/2} = \langle \Lambda^{-1/2} f, \Lambda^{3/2} u\rangle$, so

$$\int_0^T |W_{1/2}|\,\mathrm{d}t \le \|f\|_{L^2\dot{H}^{-1/2}} \sqrt{R} < \infty. \tag{695}$$

For completeness, the source's negative homogeneous norm is finite at low frequencies as well as high frequencies. Splitting the Fourier integral at $|\xi| = 1$, using $\left\|\widehat{f}\right\|_\infty \le (2\pi)^{-3/2}\|f\|_1$, gives

$$\left\|\Lambda^{-1/2} f\right\|_2^2 \le \frac{1}{4\pi^2}\|f\|_1^2 + \|f\|_2^2.$$

Both terms are uniformly finite under the source assumptions. The dual pairing places the velocity at order $3/2$, the order generated by viscous dissipation from the critical energy at $1/2$. Thus forcing enters the critical pair used in Part III: source regularity controls one factor, and the rectangle controls the velocity factor. The higher differentiated contributions remain those in (682) and (683). Substituting the two integrable currents into the critical balance gives

$$\frac{1}{2}\left\|\Lambda^{1/2}u(t)\right\|_2^2 + \nu\int_0^t \left\|\Lambda^{3/2}u\right\|_2^2 \mathrm{d}s = \frac{1}{2}\left\|\Lambda^{1/2}u_0\right\|_2^2 + \int_0^t (\mathcal{P}_{1/2} + W_{1/2})\,\mathrm{d}s.$$

Write $F_- = \|f\|_{L^2\dot{H}^{-1/2}}$. Substituting (694) and (695) into this identity bounds the complete forced critical balance:

$$\begin{aligned} E_{1/2}(t) + 2\nu\int_0^t E_{3/2}(s)\,\mathrm{d}s &\le E_{1/2}(0) \\ &\quad + \frac{\sqrt{D+R}\,R}{\sqrt{2}\pi} + F_-\sqrt{R}. \end{aligned} \tag{696}$$

For rest data, $D = E_{1/2}(0) = 0$, so the right side is $R^{3/2}/(\sqrt{2}\pi) + F_-\sqrt{R}$. Here $R$ bounds the two critical norms independently of the production estimate. The calculation requires no absorption of nonlinear production or source work into viscous dissipation.

### 23.1 A uniform velocity bound

The intersection also contains $R_{0,2} < \infty$. This finite selection already controls the speed of the forced fluid, before any estimate of the individual momentum terms is needed. Put

$$a = \|u\|_{L^2H^3}\,, \qquad b = \|u_t\|_{L^2H^1}\,, \qquad R_{0,2} = a^2 + b^2.$$

Integrating the Hilbert-pair identity at $m = 2$ from the prescribed initial velocity gives

$$\begin{aligned} \|u(t)\|_{H^2}^2 &= \|u_0\|_{H^2}^2 \\ &\quad + 2\,\mathrm{Re}\int_0^t \left\langle (1-\Delta)^{3/2}u, (1-\Delta)^{1/2}u_t \right\rangle \mathrm{d}s, \\ \sup_{0\le t\le T}\|u(t)\|_{H^2}^2 &\le \|u_0\|_{H^2}^2 + 2ab \\ &\le \|u_0\|_{H^2}^2 + R_{0,2}. \end{aligned} \tag{697}$$

Use the unitary Fourier transform and $\|v\|_{H^s}^2 = \int \langle\xi\rangle^{2s}|\widehat{v}(\xi)|^2\,\mathrm{d}\xi$. Since

$$\int_{\mathbb{R}^3}(1+|\xi|^2)^{-2}\,\mathrm{d}\xi = \pi^2,$$

Fourier inversion and Cauchy–Schwarz give

$$\|\partial_x^\alpha v\|_\infty \le c_\infty \|v\|_{H^{|\alpha|+2}}\,, \qquad c_\infty = \frac{1}{\sqrt{8\pi}}. \tag{698}$$

This holds for vector fields with their Euclidean norm.

Applying (698) to the velocity yields

$$\sup_{0\le t\le T}\|u(t)\|_\infty^2 \le \frac{\|u_0\|_{H^2}^2 + R_{0,2}}{8\pi}. \tag{699}$$

In particular, for a fluid initially at rest,

$$\sup_{0\le t\le T}\|u(t)\|_\infty^2 \le \frac{R_{0,2}}{8\pi} < \infty.$$

The endpoint belongs to the continuous $H^2$ trace established in the intersection trace calculation. Thus the intersection supplies a uniform velocity bound through the entire finite interval. The additional estimates below resolve how acceleration, transport, pressure, and viscosity behave individually within that velocity history.

### 23.2 Higher spatial energies from the critical coordinate

The critical coordinate also controls the growth of every higher spatial energy. The homogeneous commutator inequality proved in Lemma 18.2 is

$$|\mathcal{P}_m| \le K_m \left\|\Lambda^{3/2}u\right\|_2 \left\|\Lambda^{m+1}u\right\|_2 \|\Lambda^m u\|_2\,, \qquad m \ge 1. \tag{700}$$

Here $m$ is an integer and $K_m$ is finite for each fixed order. The inequality applies to the complete spatially differentiated energy row, with its multinomial weights. It holds for a divergence-free field

independently of the force: the pressure gradient remains orthogonal to the energy row, and the convection expression is unchanged.

Write

$$X_m = \|\Lambda^m u\|_2^2, \qquad Y_m = \left\|\Lambda^{m+1} u\right\|_2^2, \qquad a = \left\|\Lambda^{3/2} u\right\|_2, \qquad b_m = \left\|\Lambda^{m-1}\mathbb{P} f\right\|_2.$$

The forced balance is

$$\frac{1}{2} X_m' + \nu Y_m = \mathcal{P}_m + W_m, \qquad W_m = \left\langle \Lambda^{m+1} u, \Lambda^{m-1}\mathbb{P} f \right\rangle.$$

Young's inequality allocates a quarter of the dissipation to each pairing:

$$|\mathcal{P}_m| \le \frac{\nu}{4} Y_m + \frac{K_m^2}{\nu} a^2 X_m, \tag{701}$$

$$|W_m| \le \frac{\nu}{4} Y_m + \frac{1}{\nu} b_m^2. \tag{702}$$

Their substitution gives

$$X_m' + \nu Y_m \le \frac{2K_m^2}{\nu} a^2 X_m + \frac{2}{\nu} b_m^2. \tag{703}$$

The current energy enters linearly. Its coefficient has a finite time integral, since (693) gives $\int_0^T a(t)^2 \,\mathrm{d}t \le R$.

Set

$$A_m(t) = \frac{2K_m^2}{\nu} \int_0^t a(s)^2 \,\mathrm{d}s, \qquad \Phi_m = \int_0^T b_m(t)^2 \,\mathrm{d}t.$$

Multiplying (703) by $e^{-A_m(t)}$ and integrating gives

$$e^{-A_m(t)} X_m(t) + \nu \int_0^t e^{-A_m(s)} Y_m(s) \,\mathrm{d}s \le X_m(0) + \frac{2}{\nu} \int_0^t e^{-A_m(s)} b_m(s)^2 \,\mathrm{d}s.$$

Since $A_m$ is nondecreasing and $A_m(T) \le 2K_m^2 R/\nu$,

$$\begin{aligned} X_m(t) + \nu \int_0^t Y_m(s) \,\mathrm{d}s &\le C_m^f(R), \\ C_m^f(R) &:= \exp\!\left(\frac{2K_m^2 R}{\nu}\right) \left[\|\Lambda^m u_0\|_2^2 + \frac{2}{\nu}\Phi_m\right]. \end{aligned} \tag{704}$$

The source assumptions give $\Phi_m < \infty$, with no restriction on its size. Taking the supremum of the energy and the terminal limit of the dissipation separately bounds each by $C_m^f(R)$. The ordinary energy bound supplies the low frequencies. Higher temporal rows remain derivatives of this velocity and are generated by (675) from these spatial bounds and the prescribed source jets.

The factor controlling growth is the time integral of $\left\|\Lambda^{3/2} u\right\|_2^2$. Once this critical quantity is finite, the source contributes the finite additive term $\Phi_m$, and Gronwall's inequality bounds every fixed higher spatial order.

## 24 Bounds on each momentum term

A bounded sum in the momentum equation does not bound its summands: large inertial, pressure, and viscous terms can cancel. The mixed Sobolev estimates give stronger information. By selecting finitely many additional derivatives of $u$, we can bound each term before forming the sum.

## 24.1 Spatial product bounds

For $s \geq 1$,

$$\langle \xi \rangle^s \leq 2^{s-1} \left( \langle \eta \rangle^s + \langle \xi - \eta \rangle^s \right).$$

Apply this to the convolution formula for $\widehat{a \otimes b}$, use Young's convolution inequality, and use $\|\widehat{a}\|_1 \leq \pi \|a\|_{H^2}$. The result is

$$\|a \otimes b\|_{H^s} \leq 2^{s-1} c_\infty \left( \|a\|_{H^s} \|b\|_{H^2} + \|a\|_{H^2} \|b\|_{H^s} \right). \tag{705}$$

For tensors use the Frobenius norm. Divergence is bounded from $H^{m+1}$ to $H^m$ with constant one. If $\nabla \cdot a = 0$, transport is the divergence of the appropriate tensor product. Thus

$$\|(a \cdot \nabla) b\|_{H^m} \leq C_m \|a\|_{H^{m+2}} \|b\|_{H^{m+2}}, \qquad C_m = 2^{m+1} c_\infty, \quad m \geq 0. \tag{706}$$

## 24.2 The complete momentum row

For each temporal row and Sobolev height, define

$$S_{n,m} := C_m \sum_{a=0}^{n} \binom{n}{a} A_{a,m+2} A_{n-a,m+2}. \tag{707}$$

The finite sum retains both transport factors in every binomial summand.

**Theorem 24.1** (Individual mixed-term bounds)**.** *Assume* (669)*, the source regularity above, and* (I)*. For every $n, m \geq 0$, each term in the differentiated momentum equation satisfies*

$$\sup_t \|V_{n+1}(t)\|_{H^m} \leq A_{n+1,m}, \tag{708}$$

$$\sup_t \|B_n(t)\|_{H^m} \leq S_{n,m}, \tag{709}$$

$$\sup_t \|\nabla Q_n(t)\|_{H^m} \leq L_{n,m} + S_{n,m}, \tag{710}$$

$$\sup_t \|\nu \Delta V_n(t)\|_{H^m} \leq \nu A_{n,m+2}. \tag{711}$$

*For $|\alpha| = k$, their mixed spatial derivatives satisfy*

$$\|\partial_x^\alpha V_{n+1}\|_{L^\infty_{t,x}} \leq c_\infty A_{n+1,k+2}, \tag{712}$$

$$\|\partial_x^\alpha B_n\|_{L^\infty_{t,x}} \leq c_\infty S_{n,k+2}, \tag{713}$$

$$\|\partial_x^\alpha \nabla Q_n\|_{L^\infty_{t,x}} \leq c_\infty (L_{n,k+2} + S_{n,k+2}), \tag{714}$$

$$\|\nu \partial_x^\alpha \Delta V_n\|_{L^\infty_{t,x}} \leq \nu c_\infty A_{n,k+4}. \tag{715}$$

*Every constant on the right is finite and comes from finitely many velocity intersection coordinates and prescribed force coordinates.*

*Proof.* The first bound is (692) for temporal row $n+1$. Apply (706) to each summand of (674) and then apply (692); this gives (709). The Fourier multiplier $I - \mathbb{P}$ has operator norm at most one in every $H^m$. Its application to (677) proves (710), including the forcing contribution. The multiplier bound $|\xi|^2 \leq \langle \xi \rangle^2$ proves (711). Applying (698) to these four fields gives (712)–(715). □

For the undifferentiated momentum equation, the calculation is particularly concrete. The four individual bounds are

$$\begin{aligned}
\|u_t\|_{L^\infty_{t,x}} &\le c_\infty A_{1,2},\\
\|(u\cdot\nabla)u\|_{L^\infty_{t,x}} &\le c_\infty C_2 A_{0,4}^2 = \frac{A_{0,4}^2}{\pi},\\
\|\nabla p\|_{L^\infty_{t,x}} &\le c_\infty L_{0,2} + \frac{A_{0,4}^2}{\pi},\\
\|\nu\Delta u\|_{L^\infty_{t,x}} &\le \nu c_\infty A_{0,4}.
\end{aligned}$$

Here $C_2 = 8c_\infty$. These estimates bound each summand before the summands are added. The signs in (669) play no role in their finiteness.

### 24.3 Pressure completion and material acceleration

Pressure reconstruction separates the material acceleration into orthogonal solenoidal and gradient components. Let $\mathbb{Q} = I - \mathbb{P}$, let $B_0 = (u\cdot\nabla)u$, and write $a = D_t u$. Pressure reconstruction in (671) gives

$$\begin{aligned}
\nabla p &= \mathbb{Q}(f - B_0),\\
a &= \nu\Delta u + \mathbb{P}f + \mathbb{Q}B_0,\\
\mathbb{P}a &= \nu\Delta u + \mathbb{P}f, \qquad \mathbb{Q}a = \mathbb{Q}B_0.
\end{aligned} \tag{716}$$

The pressure response removes the gradient component of the applied force and fixes the gradient component of material acceleration through the velocity. The viscous contribution is divergence free. In every Fourier Sobolev inner product, these two components are orthogonal:

$$\|a\|_{H^s}^2 = \|\nu\Delta u + \mathbb{P}f\|_{H^s}^2 + \|\mathbb{Q}B_0\|_{H^s}^2 . \tag{717}$$

This identity also holds for homogeneous norms wherever the displayed quantities are finite. For $f = 0$, it reads

$$\|D_t u\|_{\dot H^s}^2 = \nu^2 \|u\|_{\dot H^{s+2}}^2 + \|\nabla p\|_{\dot H^s}^2 .$$

Pressure cannot cancel the viscous contribution in this global norm. With forcing retained, cancellation inside the divergence-free component is governed by the complete sum $\nu\Delta u + \mathbb{P}f$.

The critical rectangle gives a quantitative bound for these components. Put $h = \|u\|_{\dot H^{1/2}}$, $d = \|u\|_{\dot H^{3/2}}$, and $c_* = (\sqrt{2}\pi)^{-1}$. The Sobolev inequality used in the critical production calculation and its dual give

$$\|B_0\|_{\dot H^{-1/2}} \le C_3 \|u\|_3 \|\nabla u\|_3 \le c_* h d.$$

The projector $\mathbb{Q}$ has norm one, so this also bounds $\mathbb{Q}B_0$. There is a stronger spatial estimate for that gradient component. Expanding its divergence and using $\nabla\cdot u = 0$ puts one derivative on each factor:

$$\operatorname{div} B_0 = \partial_i u_j \, \partial_j u_i.$$

The Fourier symbol of $\mathbb{Q}$ and the dual Sobolev inequality now imply

$$\begin{aligned}
\|\mathbb{Q}B_0\|_{\dot H^{1/2}} &= \|\operatorname{div} B_0\|_{\dot H^{-1/2}}\\
&\le C_3 \left\|\operatorname{tr}[(\nabla u)^2]\right\|_{3/2}\\
&\le C_3 \|\nabla u\|_3^2 \le c_* d^2.
\end{aligned} \tag{718}$$

This estimate controls the nonlinear pressure gradient $\nabla p - \mathbb{Q}f = -\mathbb{Q}B_0$. It retains the prescribed source part of pressure separately.

Use the already selected $R = R_{0,1}$, $D = \|u_0\|_{H^1}^2$, and set $F_-^\sigma = \|\mathbb{P}f\|_{L^2\dot{H}^{-1/2}} \le F_-$. Substituting (693) into (717) and (718) yields

$$\begin{aligned} \|D_t u\|_{L^2\dot{H}^{-1/2}}^2 &\le \left(\nu\sqrt{R} + F_-^\sigma\right)^2 + \frac{(D+R)R}{2\pi^2}, \\ \|\nabla p - \mathbb{Q}f\|_{L^1\dot{H}^{1/2}} &\le \frac{R}{\sqrt{2}\pi}. \end{aligned} \tag{719}$$

These are finite consequences of the two critical coordinates. The first bound uses orthogonality to add the squared component norms; the second uses incompressibility to control the pressure contribution at a higher spatial order. The pointwise acceleration bound below uses a stronger finite rectangle.

Ordinary temporal differentiation preserves the two projections:

$$\partial_t^n a = \nu\Delta V_n + \mathbb{P}F_n + \mathbb{Q}B_n.$$

The orthogonal identity follows at each Eulerian temporal row. Repeated material differentiation also differentiates the transport operator; the commutators retained in Appendix J give those higher parcel derivatives.

### 24.4 Equivalent critical continuation conditions

The projected acceleration and critical dissipation are linked by an exact identity. Combined with the spatial energy estimate, it characterizes continuation through $T$ in terms of several equivalent quantities. Define

$$\mathcal{B}_T = \int_0^T \|u\|_{\dot{H}^{3/2}}^2 \, \mathrm{d}t.$$

Projection of the material momentum law gives the exact identity

$$\mathcal{B}_T = \nu^{-2} \left\|\mathbb{P}(D_t u - f)\right\|_{L^2(0,T;\dot{H}^{-1/2})}^2. \tag{720}$$

The identity follows from $\mathbb{P}(D_t u - f) = \nu\Delta u$ and Plancherel. It holds on each strict interval and passes to the nonnegative terminal integrals, including the value $+\infty$.

For the stated smooth datum and source, the following assertions about a classical forced history on $[0,T)$ are equivalent:

$$\begin{aligned} &u \in \mathscr{I}_T; \qquad \mathcal{B}_T < \infty; \qquad \mathbb{P}D_t u \in L^2(0,T;\dot{H}^{-1/2}); \\ &\widehat{C}_1[u] \in L^1(0,T); \qquad \sup_{t<T} \int_0^t \mathcal{P}_{1/2}(s)\,\mathrm{d}s < \infty. \end{aligned} \tag{721}$$

Here $\widehat{C}_1$ is the original completion formula evaluated on this forced velocity, as in (685). The source has a finite $L^2\dot{H}^{-1/2}$ norm, so subtraction of $\mathbb{P}f$ preserves the projected-acceleration condition. Equation (686) identifies the completed-coordinate condition with $\mathcal{B}_T < \infty$.

For the remaining implications, first suppose $\mathcal{B}_T < \infty$. At spatial height one, differentiation and incompressibility give

$$|\mathcal{P}_1| \le \|\nabla u\|_3 \|\nabla u\|_2 \|\nabla u\|_6 \le K_1 \|u\|_{\dot{H}^{3/2}} \|\nabla u\|_2 \|\Delta u\|_2.$$

Young's inequality in the forced energy balance gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\|\nabla u\|_2^2+\nu\|\Delta u\|_2^2\le\frac{2K_1^2}{\nu}\|u\|_{\dot H^{3/2}}^2\|\nabla u\|_2^2+\frac{2}{\nu}\|\mathbb{P}f\|_2^2.$$

Thus Gronwall bounds the $H^1$ norm uniformly through $T$. Local classical existence in $H^1$, with the prescribed smooth source, has a lifespan bounded below by its $H^1$ bound and a source bound. Restarting sufficiently near $T$ extends the history past $T$. Uniqueness identifies that extension on its overlap. Higher spatial differentiation and the momentum recurrence then supply every fixed spatial and temporal Sobolev norm on the closed horizon. Equivalently, the explicit higher-energy estimates of Section 23.2 perform this reconstruction. This proves $u\in\mathscr{I}_T$. The reverse implication follows by selecting the finite spatial coordinate.

Finally, the work $W_{1/2}=\langle u,\Lambda\mathbb{P}f\rangle$ is uniformly bounded by kinetic energy and a prescribed source norm. The exact critical balance is

$$E_{1/2}(t)+\nu\int_0^t\|u\|_{\dot H^{3/2}}^2\,\mathrm{d}s=E_{1/2}(0)+\int_0^t\mathcal{P}_{1/2}\,\mathrm{d}s+\int_0^t W_{1/2}\,\mathrm{d}s.$$

A uniform upper bound for the signed production integral controls both nonnegative terms on the left, including $\mathcal{B}_T$. Conversely, membership in $\mathscr{I}_T$ gives the absolute production bound (694). This proves all the equivalences.

These equivalences characterize full-interval regularity of a given forced solution. For the global solution of Part III, Corollary 20.1 supplies all these conditions and every finite-rectangle bound.

### 24.5 From spatial-height integrals to the forced mixed intersection

Let $(u,p)$ be the classical forced history on every strict interval $[0,S]$, $S<T<\infty$, with fixed $\nu>0$, $u_0\in H^\infty_\sigma(\mathbb{R}^3)$, and the prescribed smooth force whose mixed derivatives decay rapidly in space uniformly through $[0,T]$. Assume explicitly that every zeroth-row completed height has its whole-interval integral:

$$\int_0^T C^f_{0,k}(t)\,\mathrm{d}t=\frac12\int_0^T\left\|\Lambda^{k+1/2}u(t)\right\|_2^2\,\mathrm{d}t<\infty,\qquad k=0,1,2,\dots.\tag{722}$$

The forced momentum equation carries this spatial information to the temporal rows. The first step is to recover the low frequencies omitted by the homogeneous heights. Put

$$K_T=\|u_0\|_2+\int_0^T\|f(t)\|_2\,\mathrm{d}t,\qquad \mathsf{S}_m=\|u\|_{L^2(0,T;H^m)}.$$

The kinetic bound gives $\sup_{t<T}\|u(t)\|_2\le K_T$. For each integer $m\ge0$, split the Fourier integral at $|\xi|=1$. Below that radius use the kinetic bound; above it use $(1+|\xi|^2)^m\le2^m|\xi|^{2m+1}$. This gives

$$\mathsf{S}_m^2\le2^mTK_T^2+2^{m+1}\int_0^T C^f_{0,m}(t)\,\mathrm{d}t<\infty.\tag{723}$$

These time integrals bound the total change in each spatial norm. Indeed, the projected equation is

$$u_t=\nu\Delta u-\mathbb{P}[(u\cdot\nabla)u]+\mathbb{P}f.$$

The transport estimate (706) places its nonlinear term in $L^1(0,T;H^m)$, with norm at most $C_m\mathsf{S}_{m+2}^2$. Its viscous term has $L^1H^m$ norm at most $\nu\sqrt{T}\,\mathsf{S}_{m+2}$. Integrating the equation from the initial datum now yields

$$\begin{aligned}\sup_{t<T}\|u(t)\|_{H^m} &\le \mathsf{U}_{0,m},\\ \mathsf{U}_{0,m} &:= \|u_0\|_{H^m}+\nu\sqrt{T}\,\mathsf{S}_{m+2}+C_m\mathsf{S}_{m+2}^2+\|\mathbb{P}f\|_{L^1(0,T;H^m)}<\infty.\end{aligned}\tag{724}$$

The integral of this $L^1H^m$ right side supplies the continuous representative, so the bound does not assume a terminal trace.

Every finite spatial order is now available in the base row. Differentiating the forced equation retains every transport product:

$$\begin{aligned}V_{n+1}&=\nu\Delta V_n-\mathbb{P}B_n+\mathbb{P}F_n,\\ B_n&=\sum_{a=0}^{n}\binom{n}{a}(V_a\cdot\nabla)V_{n-a},\qquad V_n=\partial_t^n u,\quad F_n=\partial_t^n f.\end{aligned}\tag{725}$$

Suppose that rows $0,\dots,n$ have uniform bounds at every finite spatial order. Applying (706) term by term in this finite sum gives the next constants:

$$\begin{aligned}\mathsf{U}_{n+1,m}&:=\nu\mathsf{U}_{n,m+2}\\ &\quad+C_m\sum_{a=0}^{n}\binom{n}{a}\mathsf{U}_{a,m+2}\mathsf{U}_{n-a,m+2}+\|\mathbb{P}F_n\|_{L^\infty(0,T;H^m)}.\end{aligned}\tag{726}$$

The prescribed force makes the last term finite. Induction on $n$ proves $\sup_{t<T}\|V_n(t)\|_{H^m}\le\mathsf{U}_{n,m}<\infty$ for every fixed pair $n,m$. At each step only finitely many higher spatial orders occur, and each of them was already available in the earlier rows. Integrating these uniform bounds gives

$$\sum_{n=0}^{r}\|V_n\|_{L^2(0,T;H^m)}^2\le T\sum_{n=0}^{r}\mathsf{U}_{n,m}^2<\infty,\qquad u\in\mathscr{I}_T=\bigcap_{r,m\ge0}H^r(0,T;H^m).\tag{727}$$

Conversely, choosing spatial order $k+1$ in this intersection gives (722). The full spatial-height domain and the mixed intersection are equivalent for this forced history.

Pressure completes the momentum rows without changing the prescribed force. Write $Q_n=\partial_t^n p$, and use the decaying pressure normalization

$$\begin{aligned}\phi_n&=-(-\Delta)^{-1}\nabla\cdot F_n,\\ Q_n&=R_iR_j\sum_{a=0}^{n}\binom{n}{a}V_{a,i}V_{n-a,j}+\phi_n,\\ \nabla Q_n&=(I-\mathbb{P})(F_n-B_n),\end{aligned}\tag{728}$$

where $R_i$ are the Riesz transforms and repeated spatial indices are summed. The force potential is finite in every $H^M$. At low frequencies, $\widehat{F}_n$ is uniformly bounded and its inverse-derivative multiplier is controlled by $\int_{|\xi|\le1}|\xi|^{-2}\,\mathrm{d}\xi=4\pi$. At high frequencies, the prescribed Sobolev norms suffice. The smooth time dependence of the source gives $\phi_n\in C([0,T];H^M)$ for every fixed $n,M$.

The Riesz transforms are bounded on $H^M$, and $H^{M+1}\cdot H^{M+1}\subset H^M$ in three dimensions. For every integer $M\ge 0$, these product bounds and Cauchy–Schwarz in time give

$$\begin{aligned}&\|Q_n\|_{L^1H^M}+\|\nabla Q_n\|_{L^1H^{M-1}}\\&\quad\le c_M\sum_{a=0}^{n}\binom{n}{a}\|V_a\|_{L^2H^{M+1}}\|V_{n-a}\|_{L^2H^{M+1}}\\&\qquad+\|\phi_n\|_{L^1H^M}+\|\nabla\phi_n\|_{L^1H^{M-1}}<\infty.\end{aligned}\tag{729}$$

Here all time norms are over $(0,T)$. In particular, $M=0$ uses $H^{-1}$ for the gradient norm. The velocity part of each finite rectangle satisfies

$$R_{N,M}\le T\sum_{n=0}^{N}\left(\mathsf{U}_{n,M+1}^2+\mathsf{U}_{n+1,\max\{M-1,0\}}^2\right)<\infty.\tag{730}$$

Each prescribed $F_n$ belongs to $L^2(0,T;H^{M-1})$. Together with $Q_0,\dots,Q_N$ in the spaces just proved and $F_0,\dots,F_N$, this retains the full forced rectangle $(V_0,\dots,V_{N+1};Q_0,\dots,Q_N;F_0,\dots,F_N)$. Every retained row obeys

$$V_{n+1}+B_n+\nabla Q_n-\nu\Delta V_n=F_n,\qquad \nabla\cdot V_n=0.$$

Restriction to a smaller rectangle preserves all derivatives, pressure terms, and force coordinates because these entries come from the given history.

The next temporal row also controls approach to the endpoint. For $s,t<T$, integration of $\partial_t V_n=V_{n+1}$ gives

$$\|V_n(t)-V_n(s)\|_{H^m}\le|t-s|\,\mathsf{U}_{n+1,m}.\tag{731}$$

Each row has an $H^m$ limit at $T$. If a trace is taken first in a higher Sobolev space, its restriction is this limit by uniqueness of the $H^m$ limit. The traces assemble into one $H^\infty$ jet, with $u_*=V_0(T)\in H^\infty_\sigma$. Product continuity and (728) give the pressure traces and preserve the complete terminal equation:

$$V_{n+1}(T)+B_n(T)+\nabla Q_n(T)-\nu\Delta V_n(T)=F_n(T).$$

Choosing $m>k+3/2$ in (731) gives uniform convergence of every spatial derivative of order at most $k$. The finite products in the pressure formula give the corresponding limits for pressure and its gradient.

If the force is prescribed smoothly beyond $T$, local forced existence restarts the equation from $u_*$. The outgoing initial jet is fixed by (725) and (728), so it agrees with the incoming jet. Local uniqueness joins these histories into a smooth solution through $T$.

### 24.6 Material derivatives of incompressibility

Material acceleration is generally not divergence free, even when the velocity is. Write $U_{ij}=\partial_j u_i$. Differentiating $\nabla\cdot u=0$ along a trajectory gives

$$0=D_t(\nabla\cdot u)=\nabla\cdot(D_tu)-\operatorname{tr}(U^2).$$

The pressure equation consequently agrees with

$$\nabla\cdot(D_tu)=\operatorname{tr}(U^2)=-\Delta p+\nabla\cdot f.$$

Every ordinary mixed derivative of $\nabla \cdot u$ vanishes. Every material derivative must retain the transport commutator. With $A_k = D_t^k u$ and $g_k = \nabla \cdot A_k$, this reads

$$g_0 = 0, \qquad g_{k+1} = D_t g_k + \partial_i u_j \, \partial_j (A_k)_i.$$

Thus the acceleration, jerk, and higher parcel responses belong to one constraint hierarchy. Appendix J derives its first three rows and the complete conversion to mixed Eulerian derivatives.

For the deformation gradient $F = \partial_\ell X$, the volume ratio $j = \det F$ obeys $j_t = 0$ and

$$j_{tt} = \big[\nabla \cdot (D_t u) - \operatorname{tr}(U^2)\big](X,t) j = 0.$$

The term $\operatorname{tr}(U^2)$ accounts for the changing deformation of the parcel. Volume preservation is consistent with a nonzero divergence of acceleration. Quantitative bounds on the acceleration require Sobolev control of the velocity, as in the following corollary.

**Corollary 24.2** (Momentum and acceleration bounds from $R_{1,2}$)**.** *For a locally smooth solution of* (669) *with the stated force, the single condition $R_{1,2} < \infty$ suffices for uniform bounds on all four undifferentiated momentum terms. Set*

$$R := R_{1,2}, \qquad A := \sqrt{(1 + T^{-1})R}, \qquad F := L_{0,2}, \qquad G := \sup_t \|\mathbb{P} f(t)\|_{H^2}.$$

*Then*

$$\begin{aligned} \|u_t\|_{L^\infty_{t,x}} &\le c_\infty A, \\ \|(u \cdot \nabla) u\|_{L^\infty_{t,x}} &\le A^2/(8\pi), \\ \|\nabla p\|_{L^\infty_{t,x}} &\le A^2/(2\pi) + c_\infty F, \\ \|\nu \Delta u\|_{L^\infty_{t,x}} &\le A^2/(2\pi) + c_\infty (A + G). \end{aligned} \tag{732}$$

*The total material acceleration obeys*

$$\|D_t u\|_{L^\infty_{t,x}} \le c_\infty A + \frac{A^2}{8\pi}. \tag{733}$$

*Along each material trajectory,*

$$|u(X(b,t),t)| \le |u_0(b)| + t \left( c_\infty A + \frac{A^2}{8\pi} \right).$$

*Also $R_{0,1} \le R$, so* (694) *bounds the integrated critical production by $\sqrt{D + R}\, R/(\sqrt{2}\pi)$, with $D = \|u_0\|_{H^1}^2$.*

*Proof.* The rectangle contains $u, u_t \in L^2(0,T;H^3)$. The time trace in $H^3$ gives

$$\sup_t \|u(t)\|_{H^3}^2 \le T^{-1} \|u\|_{L^2 H^3}^2 + 2 \|u\|_{L^2 H^3} \|u_t\|_{L^2 H^3} \le A^2.$$

The adjacent pair $u_t \in L^2 H^3$, $u_{tt} \in L^2 H^1$ gives $\sup_t \|u_t\|_{H^2} \le A$ by (690). For $B = (u \cdot \nabla) u$, pointwise embedding gives $\|B\|_\infty \le c_\infty^2 A^2$. The weighted Fourier convolution used in (705), applied directly to this contraction, also gives

$$\|B\|_{H^2} \le 4 c_\infty \|u\|_{H^2} \|\nabla u\|_{H^2} \le 4 c_\infty A^2.$$

Euclidean and Frobenius norms bound the contraction with constant one. Finally retain both projected identities,

$$\nabla p = (I - \mathbb{P})(f - B), \qquad \nu \Delta u = u_t + \mathbb{P}B - \mathbb{P}f.$$

Their $H^2$ bounds and (698) prove (732). Adding $u_t + B$ proves (733). The bounded $H^3$ velocity supplies bounded speed and spatial gradient, hence trajectories throughout the interval. Integration of their acceleration proves the last bound. The critical estimate follows by restriction to $R_{0,1}$. □

The finite condition $R_{1,2} < \infty$ controls both the Eulerian momentum terms and the acceleration along trajectories, without a smallness restriction. Higher material derivatives are polynomials in finitely many mixed Eulerian derivatives. Appendix J proves the two-way conversion between these derivative families, gives an explicit rectangle for each requested order, and derives their endpoint traces and convergence rates. General production, pressure, and acceleration estimates are developed in Appendix G.

## 25 Exact cancellation in the prescribed exterior

The proposed construction combines a concentrating core, an exact swirling exterior, and spatial cutoffs. We begin with the exterior, where all correction fields vanish and the velocity and pressure have explicit formulas [18, p. 116, Step 4; p. 138, Lemma A.6]. This region already gives positive contributions to the Sobolev norms used above. After calculating its momentum balance and integer-order growth, we obtain fractional lower bounds that remain valid for any interior completion agreeing with this exterior.

Work at viscosity one. Write $\tau = 1 - t$, $A = \frac{1}{2} + h$, $D = \frac{1}{2} - h$, with $0 < h < 1/100$, and $b = 2^A \kappa > 0$. The amplitude $\kappa$ is the constant denoted by $c_\infty$ in that construction, unrelated to our embedding constant. Define

$$H(Z) = \frac{1}{\Gamma(1+h)} \int_0^\infty e^{-v} v^h (1 + Zv)^{-h} \, \mathrm{d}v, \qquad K(r,t) = b r^{-2A} H(4\tau / r^2). \tag{734}$$

On its exterior the velocity and pressure are

$$u = K(r,t) e_\theta, \qquad p = -\int_r^\infty K(\rho, t)^2 \frac{\mathrm{d}\rho}{\rho}. \tag{735}$$

These fields are smooth for $r > 0, t < 1$, and the pressure integral converges since its integrand decays as $\rho^{-3-4h}$.

**Proposition 25.1** (Exact exterior cancellation and failure of a finite rectangle)**.** *The exterior* (735) *has zero momentum residual. On the shrinking annuli specified below, all four individual momentum terms grow without bound. Any whole-space velocity agreeing with this exterior on those annuli satisfies*

$$R_{0,1} = \int_0^1 \left( \|u\|_{H^2}^2 + \|u_t\|_2^2 \right) \mathrm{d}t = \infty.$$

*The localized field prescribed in [18] agrees with the exterior on these annuli. Its material acceleration also fails to belong to $L^2_{t,x}$, although the exterior trajectories preserve volume and satisfy every differentiated material incompressibility identity.*

*Proof.* Differentiation under the integral, justified by an integrable exponential majorant, gives

$$H'(Z) = -\frac{h}{\Gamma(1+h)} \int_0^\infty e^{-v} v^{h+1} (1+Zv)^{-h-1} \, \mathrm{d}v < 0.$$

Thus $H(0) = 1$, $H'(0) = -h(1+h)$, and $H$ is positive. Integration by parts in $v$, with vanishing boundary terms, gives

$$Z^2 H'' + [1 + 2(1+h)Z] H' + h(1+h) H = 0.$$

Substitution into the radial derivatives of $K$ proves $K_t = (\partial_{rr} + r^{-1}\partial_r - r^{-2})K$. The angular unit vector contributes the last term. Computing transport and the pressure derivative now gives the four signed momentum terms:

$$\begin{aligned} u_t &= K_t e_\theta, & (u \cdot \nabla) u &= -K^2 r^{-1} e_r, \\ \nabla p &= K^2 r^{-1} e_r, & -\Delta u &= -K_t e_\theta. \end{aligned} \tag{736}$$

Their sum is zero on this open region, as is every fixed mixed derivative of their sum.

We next locate an open region where the exact localized field has these formulas. Its coordinates satisfy $\tau = q(1-\eta^2)$, $z = q^D \eta$, and $X = r^2/(2q)$ [18, (3.2)]. Fix $0 < d < 1$ and choose $c > 0$ with $c^2(1-d^2)/2 > X_{\mathrm{ext}}$. Set

$$\mathcal{E}_\tau = \{ c\sqrt{\tau} < r < 2c\sqrt{\tau}, \quad |z| < d\tau^D \}. \tag{737}$$

Since $q \geq \tau$, throughout this region

$$|\eta| \leq d, \qquad \tau \leq q \leq \tau/(1-d^2), \qquad X \geq c^2(1-d^2)/2 > X_{\mathrm{ext}}.$$

All corrections and the vector potential vanish there [18, p. 116, Step 4]. For sufficiently small $\tau$, $q < q_*$, $2c\sqrt{\tau} < r_0/2$, $d\tau^D < z_0/2$, and $\tau < \tau_0/2$. Both cutoffs in [18, (10.3)–(10.5)] equal one. This proves agreement with (735) throughout $\mathcal{E}_\tau$.

Take the Eulerian derivatives at fixed spatial position before evaluating at $r = y\sqrt{\tau}$. The positive functions

$$k(y) = b y^{-2A} H(4/y^2), \qquad j(y) = -4b y^{-2A-2} H'(4/y^2)$$

satisfy

$$\begin{aligned} |u| &= k(y) \tau^{-1/2-h}, \\ |u_t| = |\Delta u| &= j(y) \tau^{-3/2-h}, \\ |(u \cdot \nabla) u| = |\nabla p| &= k(y)^2 y^{-1} \tau^{-3/2-2h}. \end{aligned} \tag{738}$$

Every coefficient is strictly positive. The individual terms diverge while cancelling exactly as in (736).

Finally integrate over the annulus, rather than infer a Sobolev norm from one point. Cylindrical integration and $r = y\sqrt{\tau}$ give

$$\begin{aligned} \int_{\mathcal{E}_\tau} |u_t|^2 \, \mathrm{d}x &= \int_{\mathcal{E}_\tau} |\Delta u|^2 \, \mathrm{d}x \\ &= C_{c,d} \tau^{-3/2-3h}, \qquad C_{c,d} = 4\pi d \int_c^{2c} j(y)^2 y \, \mathrm{d}y > 0. \end{aligned} \tag{739}$$

The coefficient is finite and positive by continuity and strict positivity on the fixed compact interval. Since $\|u\|_{H^2}^2 \geq \|\Delta u\|_2^2$, for any sufficiently small $\varepsilon > 0$ and $0 < a < \varepsilon$,

$$\int_{1-\varepsilon}^{1-a} (\|u\|_{H^2}^2 + \|u_t\|_2^2) \, \mathrm{d}t \geq \frac{2C_{c,d}}{1/2 + 3h} \left( a^{-1/2-3h} - \varepsilon^{-1/2-3h} \right). \tag{740}$$

Letting $a \downarrow 0$ proves the claimed rectangle divergence.

Material differentiation retains the rotation of the cylindrical basis. Writing $a_{\mathrm{mat}} = D_t u$, we obtain

$$\begin{aligned} a_{\mathrm{mat}} &= -K^2 r^{-1} e_r + K_t e_\theta, \\ D_t a_{\mathrm{mat}} &= -3K K_t r^{-1} e_r + (K_{tt} - K^3 r^{-2}) e_\theta. \end{aligned} \tag{741}$$

The radial and angular acceleration components are orthogonal. Hence

$$\begin{aligned} \int_{\mathcal{E}_\tau} |a_{\mathrm{mat}}|^2 \,\mathrm{d}x &= C_{c,d} \tau^{-3/2-3h} + B_{c,d} \tau^{-3/2-5h}, \\ B_{c,d} &= 4\pi d \int_c^{2c} \frac{k(y)^4}{y} \,\mathrm{d}y > 0. \end{aligned} \tag{742}$$

Both powers are nonintegrable at zero. This proves the failure of the material $L^2_{t,x}$ coordinate without identifying it with $u_t$.

To verify volume preservation, follow a trajectory segment that stays inside this open exterior. With initial coordinates $(r_*, \theta_*, z_*)$, its exact motion is

$$r(t) = r_*, \qquad z(t) = z_*, \qquad \theta(t) = \theta_* + \Phi(r_*, t), \qquad \Phi(r,t) = \int_{t_0}^t \frac{K(r,s)}{r} \,\mathrm{d}s.$$

Its cylindrical coordinate Jacobian is one; the unchanged radius makes its Cartesian volume Jacobian one as well. Every derivative of the volume of a transported parcel vanishes. In particular, with $U_{ij} = \partial_j u_i$, direct differentiation gives

$$\nabla \cdot a_{\mathrm{mat}} = -\frac{2KK_r}{r} = \mathrm{tr}(U^2).$$

This is precisely the acceleration constraint derived in Appendix J. Higher orders follow by differentiating the identically constant volume Jacobian, or by its material divergence recurrence. The growing acceleration norm violates none of these identities. □

The region's kinetic energy instead equals

$$\int_{\mathcal{E}_\tau} |u|^2 \,\mathrm{d}x = 4\pi d \left( \int_c^{2c} k(y)^2 y \,\mathrm{d}y \right) \tau^{1/2-3h} \longrightarrow 0.$$

The shrinking volume permits large speeds and derivatives without requiring large kinetic energy in that region.

The cancellation has a direct mechanical meaning: $D_t u = \Delta u - \nabla p$ in this exterior. The inward acceleration is supplied by pressure, and the change in swirl speed is supplied by viscous diffusion. A zero external force does not make either contribution zero. Moving them to one side of the momentum law produces the two cancelling pairs in (736).

The annulus $\mathcal{E}_\tau$ is not a transported parcel: its radii and height shrink while exterior parcels retain their own radii and heights. Its decreasing volume and growing norms sample different particle labels. Diffusion can increase velocity at a point by transporting momentum from surrounding fluid; its dissipation of total energy is an integral statement.

This exterior occupies a region with a moving inner boundary. It is not a global unforced history on $\mathbb{R}^3$, so applying the unforced theorem to this restriction would change that theorem's hypotheses. The whole candidate includes an interior and cutoff forcing. The exterior calculation depends only on the formulas in the displayed region; the interior residual estimate enters the source calculation below.

## 25.1 Kinetic energy, strain, and critical concentration

The exterior's acceleration cost can be compared with its strain and kinetic costs on the explicit annulus (737). For the azimuthal velocity $K(r,t)e_\theta$, the symmetric velocity gradient $S = (\nabla u + \nabla u^{\mathsf{T}})/2$ has squared norm

$$|S|^2 = \tfrac{1}{2}(K_r - K/r)^2.$$

Use $K = \tau^{-A}k(r/\sqrt{\tau})$ and cylindrical volume measure $r\,dr\,d\theta\,dz$. Direct substitution gives

$$\begin{aligned}
\int_{\mathcal{E}_\tau} |S|^2\,dx &= C_S\tau^{-1/2-3h},\\
C_S &= 2\pi d\int_c^{2c} (k'(y) - k(y)/y)^2 y\,dy > 0,\\
\int_{\mathcal{E}_\tau} |u|^2\,dx &= 4\pi d\int_c^{2c} |k(y)|^2 y\,dy\;\tau^{1/2-3h},\\
\int_{\mathcal{E}_\tau} |u|^3\,dx &= 4\pi d\int_c^{2c} |k(y)|^3 y\,dy\;\tau^{-4h}.
\end{aligned} \tag{743}$$

Here $C_S > 0$ because the analytic, decaying heat profile cannot obey $k' = k/y$ throughout an open annulus. The first time integral is finite since $h < 1/6$, and the kinetic contribution tends to zero. The $L^3$ contribution grows without bound. This explains how finite kinetic energy and finite viscous action coexist with critical concentration in the displayed region. Their bounds do not substitute for the critical coordinates controlled by the full intersection.

## 25.2 Positive lower bounds at fractional Sobolev orders

The exterior also determines lower bounds at the precise fractional heights used by the intersection argument. These bounds use pairs of points lying entirely in that region. No choice of the unknown interior can cancel their contribution.

**Theorem 25.2** (Critical mixed-row lower bounds)**.** *Suppose a smooth whole-space velocity agrees with* (735) *on the prescribed exterior for all sufficiently late times. For every fixed pair of integers $n, m \ge 0$, there are $c_{n,m} > 0$ and $\delta_{n,m} > 0$ such that*

$$\|\partial_t^n u(1-\tau)\|^2_{\dot H^{m+1/2}} \ge c_{n,m}\tau^{-2n-m-3h} \qquad (0 < \tau < \delta_{n,m}). \tag{744}$$

*The constants are allowed to depend on the two derivative orders. The bound also holds, with changed constants, after subtracting any history smooth through time one whose finite mixed Sobolev norms are bounded. In particular,*

$$\begin{aligned}
\|u(1-\tau)\|^2_{\dot H^{1/2}} &\ge c_0\tau^{-3h},\\
\|u(1-\tau)\|^2_{\dot H^{3/2}} &\ge c_1\tau^{-1-3h}.
\end{aligned} \tag{745}$$

*Thus the critical dissipation integral diverges at the proposed singular time.*

*Proof.* For a scalar function $F$ on $\mathbb{R}^3$, the positive fractional seminorm formula is

$$\|F\|^2_{\dot H^{1/2}} = \frac{1}{2\pi^2}\iint_{\mathbb{R}^3\times\mathbb{R}^3} \frac{|F(x)-F(y)|^2}{|x-y|^4}\,dx\,dy. \tag{746}$$

It follows by Plancherel and $\int_{\mathbb{R}^3}(1-\cos(\xi\cdot z))|z|^{-4}dz = \pi^2|\xi|$. In particular any selected set of pairs supplies a lower bound.

Differentiate the heat profile in (735). On the exterior,

$$\partial_t^n u = \tau^{-A-n} k_n(r/\sqrt{\tau}) e_\theta,$$

where

$$k_n(\rho) = \frac{b 4^n (h)_n}{\Gamma(1+h)} \rho^{-2A-2n} \times \int_0^\infty e^{-v} v^{h+n} (1 + 4v/\rho^2)^{-h-n} \, dv.$$

Here $(a)_n$ denotes the rising factorial, with $(a)_0 = 1$. For each fixed $m$,

$$k_n^{(m)}(\rho) = (-1)^m d_{n,m} \rho^{-2A-2n-m} \big(1 + O_{n,m}(\rho^{-2})\big),$$
$$d_{n,m} = b 4^n (h)_n (1+h)_n (2A+2n)_m > 0.$$

The expansion, including its derivatives, follows from the exponentially integrable profile formula. Choose a fixed sufficiently large $c$. At transverse points $(c, 0)$ and $(2c, 0)$, the second component of $\partial_1^m [k_n(r) e_\theta]$ equals $k_n^{(m)}(c)$ and $k_n^{(m)}(2c)$. Their difference is nonzero. Two small fixed transverse patches retain a strictly positive gap.

Scale both patches by $\sqrt{\tau}$. Choose the first axial coordinate in an interval of length proportional to $\tau^D$, and the second within a distance proportional to $\sqrt{\tau}$ of it. Since $D < 1/2$, both axial coordinates remain in the exterior slab for small $\tau$. The transverse patches can be placed in the exact exterior by taking $c$ large, and the physical localization cutoffs equal one there for sufficiently small $\tau$. The selected pair set has measure bounded below by a positive constant times $\tau^{5/2+D}$. Its distances are at most a fixed multiple of $\sqrt{\tau}$. For $F = \partial_1^m \partial_t^n u_2$, the squared difference on those pairs is at least $c' \tau^{-2A-2n-m}$. Formula (746) now gives

$$\|F\|_{\dot{H}^{1/2}}^2 \geq c'' \tau^{5/2+D-2-2A-2n-m} = c'' \tau^{-2n-m-3h}.$$

The Fourier multiplier inequality $|\xi_1|^{2m} |\xi| \leq |\xi|^{2m+1}$ proves (744). Finally, the triangle inequality shows that subtracting a history with bounded mixed Sobolev norms preserves each divergent lower bound for small enough $\tau$. □

The first pressure-complete nonlinear prefix must also grow. Define

$$P_{1/2}(t) = -\left\langle \Lambda^{1/2} u, \Lambda^{1/2} \mathbb{P}((u \cdot \nabla) u) \right\rangle.$$

For $t_0 < 1$ fixed sufficiently late, the exact critical balance gives

$$\int_{t_0}^t P_{1/2}(s) \, ds = E_{1/2}(t) - E_{1/2}(t_0) + \nu \int_{t_0}^t \|u\|_{\dot{H}^{3/2}}^2 \, ds - \int_{t_0}^t \langle u, \Lambda \mathbb{P} f \rangle \, ds.$$

The last integral is bounded by the kinetic budget and the smooth source. Inserting (745) yields constants $c_* > 0$ and $C_* < \infty$ such that

$$\int_{t_0}^{1-\tau} P_{1/2}(s) \, ds \geq c_* \tau^{-3h} - C_*. \tag{747}$$

Every completion satisfying the stated forced equation must obey this lower bound. Its signed critical production consequently has an unbounded time integral, contradicting the finite production bound for a solution in the full intersection. The exterior calculation establishes the lower bound independently of that membership premise.

## 25.3 Divergence of the positive time convolutions

The mixed-row lower bound also supplies the integrated divergence required in Section C.2. Write $H_r = E_{r,1/2} = \frac{1}{2}\|\partial_t^r u\|^2_{\dot{H}^{1/2}}$. For a constant $d_r > 0$, Theorem 25.2 gives

$$H_r(1-\tau) \geq d_r \tau^{-2r-3h}$$

when $\tau$ is sufficiently small. Let $I^r$ denote the $r$-fold integral from time zero, defined in (841). For $r \geq 1$, its convolution kernel is positive. Restrict its lag $\ell = (1-\tau) - a$ to $[\tau, 2\tau]$. On that interval, $1 - a = \tau + \ell \leq 3\tau$, the kernel is at least $\tau^{r-1}/(r-1)!$, and the interval has length $\tau$. Thus

$$I^r H_r(1-\tau) \geq \frac{d_r 3^{-2r-3h}}{(r-1)!}\tau^{-r-3h} \longrightarrow \infty. \tag{748}$$

The case $r = 0$ uses the original pointwise lower bound. The source primitive and initial polynomial in the odd-row identity stay bounded. Consequently every odd completed height requires unbounded upper nonlinear production for a whole-space solution with this exterior. The conclusion uses the positive convolution explicitly; it does not infer integrated divergence from pointwise growth alone.

## 25.4 A material trajectory in the vortex core

The exterior annuli sample different particles as they shrink. Inside the core, the stated formulas permit a stronger calculation: one particle approaches the origin while its velocity and every material derivative increase without bound. We can recover that trajectory directly from the velocity on the axis.

The leading construction fixes $U(0,\eta) = 4\eta + j_0$, with $0 < j_0 \leq 0.05$ [18, p. 37; (B.1), (B.12)]. Every positive-order axial profile has zero axis value [18, p. 47, Lemma 5.1]. The cutoffs depend on $q(z,t)$; their additional curl terms are radial and leave this axial value unchanged. The wave and mean corrections vanish near the axis at each $t < 1$ [18, (5.45); pp. 114–115]. Thus the candidate's exact axis trace, where the final cutoffs equal one, is

$$u(0,0,z,t) = q^{-A}(4\eta + j_0)e_z, \qquad z = q^D\eta, \quad 1 - t = q(1-\eta^2). \tag{749}$$

Define $H_c(\eta) = D\eta + (1-\eta^2)(4\eta + j_0)$. Its values at $-j_0/4$ and zero have opposite signs, and $H_c' > 0$ between them. Let $\eta_c$ be its unique zero in that interval, and put

$$d_c = 1 - \eta_c^2, \qquad U_c = 4\eta_c + j_0 = -\frac{D\eta_c}{d_c} > 0.$$

For sufficiently late $t_0 < 1$, the curve

$$x_c(t) = \big(0, 0, \eta_c[(1-t)/d_c]^D\big), \qquad t_0 \leq t < 1, \tag{750}$$

lies inside the cutoff plateaux and local domain. Differentiating its axial component gives

$$\dot{z}_c = -D\eta_c d_c^{-D}(1-t)^{-A} = d_c^A U_c (1-t)^{-A} = u_z(x_c(t), t).$$

The curve is a material trajectory, rather than a choice of spatial sampling points. Along it, with $C_c = d_c^A U_c > 0$,

$$(D_t^n u)(x_c(t), t) = C_c(A)_n (1-t)^{-A-n} e_z, \qquad n \geq 0, \tag{751}$$

where $(A)_0 = 1$ and $(A)_n = A(A+1)\cdots(A+n-1)$. Acceleration, jerk, and every fixed higher material derivative diverge along this particle. Yet its position has a finite limit, since $A < 1$.

This growth remains compatible with local volume preservation. Ordinary axial differentiation of (749) yields

$$\partial_z u_z(x_c(t),t) = \frac{\beta_c}{1-t}, \qquad \beta_c = \frac{d_c[-2A\eta_c U_c + 4d_c]}{1-2h\eta_c^2} > 0.$$

Axis symmetry and incompressibility give transverse contraction rates $-\beta_c/[2(1-t)]$, with an additional transverse rotation. The linearized flow from $t_0$ has axial stretch $[(1-t_0)/(1-t)]^{\beta_c}$ and transverse length factors $[(1-t)/(1-t_0)]^{\beta_c/2}$. Rotation changes neither factor's determinant. Their volume product is exactly one at every $t < 1$. Unbounded stretch and vanishing transverse widths coexist with constant material volume; differentiating that constant produces zero at every order.

For the source along this trajectory, assume the core residual estimate [18, (9.20)]: every finite Eulerian derivative of $f$ is $O(q^N)$ for every $N$. Along (750), each fixed derivative of $x_c$ has only a finite power singularity. The chain rule gives

$$\frac{\mathrm{d}^k}{\mathrm{d}t^k} f(x_c(t),t) = O((1-t)^N) \quad \text{for every fixed } k, N \geq 0.$$

Under this residual estimate, every fixed material derivative of the force approaches zero along the particle, while (751) diverges. The momentum law permits this distinction because pressure and viscous stresses also contribute to the acceleration.

# 26 Admission of the candidate force to the forced theorem

Part III applies after the momentum residual of the proposed field has been placed in its prescribed force class. The cutoff calculation below keeps the full localization error, proves uniform bounds for every fixed mixed derivative, and produces compatible compactly supported terminal jets.

## 26.1 The localization residual

Multiplication by a spatial cutoff changes the fluid's velocity gradients. The resulting force must include those changes, including the correction needed to preserve incompressibility. OpenAI's local representation is $v = \operatorname{curl} \mathcal{A} + Be_\theta$, with $B$ independent of the angle. Its localization is exactly

$$u = cv + w, \qquad w = \nabla c \times \mathcal{A}, \qquad p = c\pi, \qquad c = \chi_x \chi_t. \tag{752}$$

Here $\pi$ is the local pressure, and $c$ is axisymmetric. This is [18, (10.4)]: the curl acts on $c\mathcal{A}$, while the swirl is multiplied directly by $c$. In particular, $\nabla \cdot u = 0$. Introducing a vector potential for the exterior swirl and localizing that additional potential would change the field.

Write $\mathcal{R}_\nu(v,\pi) = v_t + (v\cdot\nabla)v - \nu\Delta v + \nabla\pi$. Expanding (752) gives the entire localization error:

$$\begin{aligned} f - c\mathcal{R}_\nu(v,\pi) = {} & (c_t - \nu\Delta c)v - 2\nu \sum_{i=1}^{3} (\partial_i c)\partial_i v + \pi\nabla c \\ & + (c^2 - c)(v\cdot\nabla)v + c(v\cdot\nabla c)v \\ & + w_t - \nu\Delta w + c(v\cdot\nabla)w \\ & + c(w\cdot\nabla)v + (w\cdot\nabla)w. \end{aligned} \tag{753}$$

The apparent additional product $(w\cdot\nabla c)v$ vanishes because $(\nabla c \times \mathcal{A})\cdot\nabla c = 0$. Every remaining term belongs to the prescribed force. Where both cutoffs are constant at one, $w = 0$ and the error

vanishes. Where they vanish on a neighborhood, the localized fields vanish too. The extra force is confined to the transitions between these regions.

In the heat exterior, near time one, $\mathcal{A} = 0$ and $v = Ke_\theta$. The formula becomes a concrete force field:

$$\begin{aligned} f = & \left[\chi_x(1-\chi_x)\frac{K^2}{r} + \pi\partial_r\chi_x\right] e_r + \pi\partial_z\chi_x e_z \\ & - \nu\left[2(\partial_r\chi_x)K_r + K\left(\partial_{rr}\chi_x + r^{-1}\partial_r\chi_x + \partial_{zz}\chi_x\right)\right] e_\theta. \end{aligned} \tag{754}$$

The radial force accounts for the changed centrifugal balance and the pressure cutoff. The axial term comes from the pressure cutoff alone. The angular force accounts for diffusion across the tapered swirl. All three vanish where $\chi_x$ is constant at one. The bounds below control every finite derivative on their transition support.

For a space–time region $\Sigma$, define the finite mixed-jet seminorm

$$\mathcal{J}_m(F;\Sigma) = \sum_{|\gamma|\le m} \frac{\|\partial^\gamma F\|_{L^\infty(\Sigma)}}{\gamma!}, \qquad \gamma \in \mathbb{N}_0^4.$$

The four entries of $\gamma$ count three spatial derivatives and one temporal derivative. The multinomial product rule gives $\mathcal{J}_m(FG) \le \mathcal{J}_m(F)\mathcal{J}_m(G)$, also for dot and cross products. Each coordinate derivative obeys $\mathcal{J}_m(\partial_i F) \le (m+1)\mathcal{J}_{m+1}(F)$.

On a transition region put

$$\begin{aligned} C_m &= \mathcal{J}_m(c), \quad V_m = \mathcal{J}_m(v), \\ A_m &= \mathcal{J}_m(\mathcal{A}), \quad P_m = \mathcal{J}_m(\pi). \end{aligned} \qquad U_m = C_m V_m + 3(m+1)C_{m+1}A_m.$$

Then $\mathcal{J}_m(u) \le U_m$ and $\mathcal{J}_m(p) \le C_m P_m$. Differentiating the full momentum residual gives the explicit force-jet bound

$$\begin{aligned} \mathcal{J}_m(f) \le & (m+1)U_{m+1} + 3(m+1)U_m U_{m+1} \\ & + 3\nu(m+1)(m+2)U_{m+2} + 3(m+1)C_{m+1}P_{m+1}. \end{aligned} \tag{755}$$

This bounds every fixed temporal and spatial source derivative once the local field jets on the transition have been bounded. It requires no constant shared by infinitely many orders.

### 26.2 Uniform derivative bounds on the cutoff transitions

The relevant bounds follow from the actual support geometry. The cutoff in [18, p. 118] equals one for $r \le r_0/2$, $|z| \le z_0/2$, and has compact support inside $r < r_0$, $|z| < z_0$. Every spatial cutoff error in (753), including $c^2 - c$ near time one, is supported where

$$r \ge r_0/2 \quad \text{or} \quad |z| \ge z_0/2.$$

The construction's coordinates satisfy

$$\tau = q(1-\eta^2), \quad z = q^D\eta, \quad X = \frac{r^2}{2q}, \quad |\eta| \le 1, \qquad D = \frac{1}{2} - h > 0.$$

On the axial transition, $q \ge |z|^{1/D}$ yields

$$q \ge q_z := (z_0/2)^{1/D} > 0.$$

This remains true as $r$ tends to zero. On the radial transition, any correction inside $X \le X_{\mathrm{ext}}$ satisfies

$$q = \frac{r^2}{2X} \ge q_r := \frac{r_0^2}{8X_{\mathrm{ext}}} > 0.$$

Thus every interior correction meeting a spatial transition has $q \ge \min(q_z, q_r) > 0$. The cutoff choices also ensure $q < q_*/2$, by [18, (10.3)]. This is precisely the closed scale range covered by Theorem 3.1(ii) of that paper.

The one-sided interior bound has a concrete finite-order meaning. On $q \ge \delta > 0$, only finitely many summands in [18, (9.21)] survive, since their cutoff scales tend to infinity. The finite representatives, fixed phase labels, and closed parameter bounds are those specified in Proposition 9.9, Step 3. Coordinate conversion has denominator $1 - 2h\eta^2 \ge 1 - 2h > 0$. At the axis, the potential is represented in Cartesian coordinates as $a(r^2, z, t)(-x_2, x_1, 0)$, as in [18, (10.1)]. These are the stated interior inputs to the finite numbers $V_m, A_m, P_m$ in (755).

If $q$ approaches zero on the remaining radial transition, the field is instead in $X \ge X_{\mathrm{ext}}$. There $\mathcal{A} = 0$, $v = Ke_\theta$, and $\pi = -\int_r^\infty K^2 \,\mathrm{d}\rho/\rho$. The heat-profile formula gives

$$|\partial_\tau^j K| \le C_j r^{-1-2h-2j}, \qquad |\partial_\tau^j \pi| \le \frac{C_j}{2 + 4h + 2j} r^{-2-4h-2j}.$$

Their further spatial derivatives are bounded at $r \ge r_0/2$. Temporal cutoff derivatives occur only where $\tau_0/2 \le \tau \le \tau_0$, which gives $q \ge \tau_0/2$. Every transition thus supplies finite jets for (755).

Near the proposed singular point both cutoffs equal one. The source is then the local residual itself. Theorem 3.1(iii) supplies its all-orders flatness on $X \le X_{\mathrm{ext}}$; beyond that region the residual is exactly zero. Combining this input with the transition bounds proves

$$\sup_{x \in \mathbb{R}^3,\, 0 \le t < 1} |\partial_x^\alpha \partial_t^j f(x,t)| < \infty \qquad \text{for every finite } \alpha, j. \tag{756}$$

One additional time derivative in this bound makes each preceding jet uniformly Cauchy as $t \uparrow 1$. Uniform spatial derivative limits give compatible smooth terminal fields with fixed compact support. Let $K \subset \mathbb{R}^3$ be a fixed compact set containing these supports. The source's finite Sobolev coordinates can now be bounded explicitly. Put

$$M_{N,M} = \max_{0 \le j \le N,\ |\alpha| \le M} \sup_{x,t<1} |\partial_x^\alpha \partial_t^j f(x,t)|.$$

Expanding the integer Fourier weight $(1 + |\xi|^2)^M$, whose nonnegative multinomial coefficients sum to $4^M$, gives

$$\sum_{j=0}^{N} \left\| \partial_t^j f \right\|_{L^2(0,1;H^M)}^2 \le (N+1) 4^M |K| M_{N,M}^2 < \infty. \tag{757}$$

Thus the actual localized source belongs to every finite mixed Sobolev coordinate under the stated local inputs. The next construction extends its terminal fields through time one.

Localization thus preserves source smoothness under the interior hypotheses of [18, Theorem 3.1]. The local residual estimate [18, (3.4)], based on the finite-stage estimate [18, (9.18)], is an input to this conclusion; the calculation above bounds the additional terms produced by the cutoffs.

### 26.3 Continuing the terminal force jets

Write $F_j \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, with $\operatorname{supp} F_j \subset K$, for the terminal field obtained from $\partial_t^j f(\cdot, t)$ as $t \uparrow 1$. The source vanishes near $t = 0$, and all its left terminal jets have support in one fixed compact

set $K$. These jets can be continued through time one without imposing a bound common to all derivative orders.

Choose $\chi_0 \in C_c^\infty(\mathbb{R})$, equal to one near zero and zero for arguments at least one. For $\sigma = t-1 \geq 0$, define

$$\widetilde{f}_{\mathrm{OA}}(x, 1+\sigma) = \sum_{j=0}^{\infty} \chi_0(b_j\sigma)\frac{\sigma^j}{j!}F_j(x). \tag{758}$$

For $0 \leq t < 1$, set $\widetilde{f}_{\mathrm{OA}}(x,t) = f(x,t)$, the full localized momentum residual. Formula (758) is the extension used in [18, (10.11)]. Its full product rule is

$$\partial_\sigma^m\left(\chi_0(b_j\sigma)\frac{\sigma^j}{j!}\right) = \sum_{\ell=\max(0,m-j)}^{m}\binom{m}{\ell}b_j^\ell\chi_0^{(\ell)}(b_j\sigma)\frac{\sigma^{j-m+\ell}}{(j-m+\ell)!}.$$

On a summand's support, $\sigma \leq b_j^{-1}$, so every product $b_j^\ell\sigma^{j-m+\ell}$ is at most $b_j^{m-j}$. Writing

$$C_{j,m} = \sum_{\ell=\max(0,m-j)}^{m}\binom{m}{\ell}\frac{\left\|\chi_0^{(\ell)}\right\|_\infty}{(j-m+\ell)!},$$

we obtain

$$\left\|\partial_x^\alpha\partial_\sigma^m\left[\chi_0(b_j\sigma)\frac{\sigma^j}{j!}F_j\right]\right\|_\infty \leq C_{j,m}b_j^{m-j}\left\|F_j\right\|_{C^{|\alpha|}}. \tag{759}$$

For each $j \geq 1$, choose $b_j > b_{j-1}$, with $b_0 = 1$, so that this bound is at most $2^{-j}$ for all $|\alpha|+m \leq \lfloor j/2 \rfloor$. There are finitely many constraints at that step, and $m-j<0$ in each one. Such a choice exists for every finite collection of norms of $F_j$.

For fixed $\alpha, m$, the differentiated tail beginning at $j \geq \max\{1, 2(|\alpha|+m)\}$ is uniformly summable. At $\sigma = 0$, the $m$-th temporal derivative of the series receives a contribution only from $j = m$, and that contribution is $F_m$. All left and right jets agree. Each term retains support in $K$ and vanishes for $\sigma \geq 1$. Uniform differentiated convergence gives smoothness at the joining time and at the outer support boundary. Hence

$$\widetilde{f}_{\mathrm{OA}} \in C_c^\infty(\mathbb{R}^3 \times (0,\infty); \mathbb{R}^3) \subset C^\infty([0,\infty); \mathcal{S}(\mathbb{R}^3; \mathbb{R}^3)), \tag{760}$$

and its restriction to $t < 1$ is exactly the localized momentum residual. For the quantitative integrals below, write

$$f_{\mathrm{OA}} := \widetilde{f}_{\mathrm{OA}}\big|_{[0,1)} = f. \tag{761}$$

Its support lies in $K \times [0,2]$, and it retains the prescribed initial interval of rest. A different smooth continuation beyond time one has an identical restriction on every $[0,S]$, $S<1$, so strict-subinterval uniqueness gives that comparison below.

## 27 Strict-subinterval uniqueness and refutation

A regular solution constructed from the prescribed data constrains every classical candidate for those data. This follows from uniqueness before any terminal bound on the candidate is assumed.

**Theorem 27.1** (Exclusion of the proposed finite-time blowup)**.** *No classical pair with the zero datum, positive viscosity, prescribed force $\widetilde{f}_{\mathrm{OA}}$, and exterior asserted in [18, Theorem 1.1] can solve that forced initial-value problem on $[0,1)$.*

*Proof.* Apply Corollary 20.1 to $(0, \nu, \widetilde{f}_{\mathrm{OA}})$, and let $U, P$ be its regular solution on $[0,1]$. Let $v, p$ be a proposed classical solution on $[0,1)$, with identical initial datum, prescribed force, viscosity, and spatial domain. For the compactly supported candidate in OpenAI's Theorem 1.1, every spatial Sobolev norm is finite on each strict interval $[0,S]$, $S<1$. Put $z = v - U$, $\pi = p - P$. Subtracting their momentum equations gives

$$\begin{aligned} z_t + (v\cdot\nabla)z + (z\cdot\nabla)U + \nabla\pi = \nu\Delta z, \\ \nabla\cdot z = 0, \qquad z(0) = 0. \end{aligned} \tag{762}$$

The prescribed force cancels because both histories solve that one forced problem. Pairing with $z$ eliminates the pressure and the divergence-free transport term. With $L(t) = \|\nabla U(t)\|_\infty$,

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|z\|_2^2 + \nu\|\nabla z\|_2^2 = -\int z_i z_j \partial_j U_i \,\mathrm{d}x \le L(t)\,\|z\|_2^2 .$$

The intersection for $U$ supplies $L \in L^1(0,1)$. Gronwall's inequality on $[0,S]$ yields

$$\|z(t)\|_2^2 \le \|z(0)\|_2^2 \exp\left(2\int_0^t L(s)\,\mathrm{d}s\right) = 0.$$

Since $S<1$ was arbitrary, $v = U$ throughout $[0,1)$. Every estimate for $U$, including its mixed and material derivative estimates, consequently applies to $v$. No terminal regularity of $v$ was used to identify it with $U$.

For rest data, the finite rectangle $R_{0,2}[U]$ and (699) give

$$\begin{aligned} \sup_{0\le t<1}\|v(t)\|_\infty^2 &= \sup_{0\le t<1}\|U(t)\|_\infty^2 \\ &\le \frac{R_{0,2}[U]}{8\pi} < \infty. \end{aligned}$$

This bound excludes the velocity growth in [18, Theorem 1.1]. Section 24.3 also bounds the acceleration, transport term, pressure gradient, and viscous term individually. Each bound follows from finite mixed norms of $U$, so it applies irrespective of cancellation among the momentum terms.

The prescribed exterior already lies outside a lower-order rectangle. Equation (740) gives a positive contribution whose time integral is infinite, so $R_{0,1} = \infty$. Every velocity in $\mathscr{I}_1$ instead has $R_{0,1} < \infty$. No velocity containing that exterior belongs to the full-interval compatible intersection.

Material coordinates preserve the conclusion. Equation (742) shows a divergent material-acceleration integral, while the exact axis trajectory in Section 25.4 has unbounded velocity and every fixed higher material derivative. Incompressibility is retained through the volume Jacobian and its differentiated identities. The resulting norm blowup is consequently present in both the Eulerian and material descriptions of the proposed flow.

Equation (760) gives the compactly supported smooth continuation of the candidate's residual. The initial datum is zero and the viscosity is positive, so Corollary 20.1 gives the regular solution $U$ used above. In particular,

$$\sup_{t<1}\|v(t)\|_\infty^2 \le R_{0,2}[U]/(8\pi) < \infty$$

whereas the candidate has unbounded velocity. Its divergent critical coordinate in Theorem 25.2 gives a second contradiction. Both comparisons use uniqueness on strict subintervals; the terminal bounds follow from the global forced theorem proved in Part III. □

### 27.1 Quantitative source criteria and conservative forcing

The appendices provide two quantitative source criteria for continuation. They also reduce conservative forcing to the unforced problem proved in Part II. Each result provides a comparison solution under its own hypotheses. For the rest datum and unit terminal time, the two quantitative hypotheses are, respectively,

$$\begin{aligned} G_{\mathrm{OA}} &:= \int_0^1 \|\mathbb{P} f_{\mathrm{OA}}(t)\|_{H^3}\, \mathrm{d}t < \frac{\pi\sqrt{\nu}}{8}, \\ B_{\mathrm{OA}} &:= \int_0^1 \|\mathbb{P} f_{\mathrm{OA}}(t)\|_{\dot{H}^{-1/2}}^2\, \mathrm{d}t < \frac{4\pi^2\nu^3}{3}. \end{aligned} \tag{763}$$

Either inequality would independently give the displayed finite-horizon estimates. The explicit source fails both tests. Appendix M checks its actual residual directly and proves

$$\begin{aligned} G_{\mathrm{OA}} &\geq \frac{\sqrt{\pi\nu}}{2} > \frac{\pi\sqrt{\nu}}{8}, \\ B_{\mathrm{OA}} &> \frac{4\pi^2\nu^3}{3}. \end{aligned} \tag{764}$$

The first bound follows from the $H^3$ energy balance and the exterior's growing spatial norm. The second uses the critical energy balance up to its first threshold crossing, followed by the nonzero source in the fixed cutoff region. Neither calculation assumes terminal regularity of the velocity or global smooth continuation of the source. Thus the prescribed force lies outside both quantitative source classes. Equation (962) also excludes the conservative-force reduction. Failure of these sufficient conditions does not affect Corollary 20.1, which follows from the unrestricted forced theorem of Part III. The first-rectangle criterion in Section F also fails for this candidate: its conclusion would give the finite rectangle contradicted by (740). In each case the unbounded candidate norms prevent application of the corresponding sufficient condition.

Theorem 27.1 completes the formal refutation. The appendices that follow retain the two proof-lane coordinate catalogues and the auxiliary source, endpoint, material, and quantitative calculations used in Part IV.

## Appendix map

The first catalogue records the canonical coordinates of the unforced proof; the second records the distinct canonical coordinates of the forced proof. The remaining appendices belong to the refutation and supply its source-work, terminal-domain, parcel, and quantitative estimates. Their placement after the refutation does not move any appendix calculation upstream of the theorem that uses it.

## A Unforced canonical coordinate catalogue

The estimates in Section 7 are organized by their role in the argument. For reference, the following theorem collects their canonical-coordinate forms in one place.

**Theorem A.1** (Canonical finite-horizon coordinate estimates)**.** *Under the hypotheses and notation of Definition 7.2, let* $J_{n,\alpha} := \partial_x^\alpha V_n$ *and* $\Pi_{n,\alpha} := \partial_x^\alpha Q_n$*, as in Theorems 7.10 and 7.11. Let*

*$n, s, k, q \in \mathbb{N}_0$, $\alpha \in \mathbb{N}_0^3$, and set $r := s + |\alpha|$. Then*

$$\|J_{n,\alpha}\|_{L^2(I_T;H^{s+1})} \le \mathfrak{X}_{n,r+1}, \tag{765}$$

$$\|J_{n+1,\alpha}\|_{L^2(I_T;H^{s-1})} \le \mathfrak{X}_{n+1,r-1}, \tag{766}$$

$$\sup_{0\le t\le T} \|J_{n,\alpha}(t)\|_{H^s} \le \mathfrak{Z}_{n,r}, \tag{767}$$

$$\left\|J_{n,\alpha}^T\right\|_{H^s} \le \mathfrak{Z}_{n,r}, \tag{768}$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^{s-1}} \le \min\left\{|t-r_0|^{1/2}\mathfrak{X}_{n+1,r-1}, |t-r_0|\mathfrak{Z}_{n+1,r}\right\}, \tag{769}$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^{s}} \le \min\left\{|t-r_0|^{1/2}\mathfrak{X}_{n+1,r}, |t-r_0|\mathfrak{Z}_{n+1,r}\right\}, \tag{770}$$

*for $0 \le r_0 \le t \le T$.*
*For $0 \le \theta \le 1$,*

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^{s-\theta}} \le 2^{1-\theta}\mathfrak{Z}_{n,r}^{1-\theta}\mathfrak{X}_{n+1,r-1}^{\theta}|t-r_0|^{\theta/2}. \tag{771}$$

*If $1 \le a \le \infty$, $2 \le b \le \infty$, and $q > k + \frac{3}{2} - \frac{3}{b}$, then*

$$\left\|\nabla^k J_{n,\alpha}\right\|_{L^a(I_T;L^b)} \le S_{q,k,b}T^{1/a}\mathfrak{Z}_{n,q+|\alpha|}, \tag{772}$$

$$\left\|\nabla^k J_{n,\alpha}\right\|_{L^2(I_T;L^b)} \le S_{q,k,b}\mathfrak{X}_{n,q+|\alpha|}, \tag{773}$$

*where $T^{1/\infty} = 1$.*

*The pressure rows obey*

$$\|\Pi_{n,\alpha}\|_{L^1(I_T;H^s)} + \|\nabla\Pi_{n,\alpha}\|_{L^1(I_T;H^{s-1})} \le \mathfrak{Q}_{n,r}^{(1)}, \tag{774}$$

$$\|\Pi_{n,\alpha}\|_{L^2(I_T;H^s)} \le \mathfrak{Q}_{n,r}^{(2)}, \tag{775}$$

$$\sup_{0\le t\le T} \|\Pi_{n,\alpha}(t)\|_{H^s} \le \mathfrak{Q}_{n,r}^{(\infty)}, \tag{776}$$

$$\left\|\Pi_{n,\alpha}^T\right\|_{H^s} \le \mathfrak{Q}_{n,r}^{(\infty)}, \tag{777}$$

$$\|\Pi_{n,\alpha}(t) - \Pi_{n,\alpha}(r_0)\|_{H^s} \le \min\left\{|t-r_0|^{1/2}\mathfrak{Q}_{n,r}^{(\mathrm{mod})}, |t-r_0|\mathfrak{Q}_{n+1,r}^{(\infty)}\right\}. \tag{778}$$

*Under the preceding restrictions on $a, b, k, q$,*

$$\left\|\nabla^k \Pi_{n,\alpha}\right\|_{L^a(I_T;L^b)} \le S_{q,k,b}T^{1/a}\mathfrak{Q}_{n,q+|\alpha|}^{(\infty)}. \tag{779}$$

*Let $r_1 \in \mathbb{N}$ and $K \ge 1$. Then*

$$\|\mathsf{P}_{\ge K}J_{n,\alpha}\|_{L^2(I_T;H^s)} \le K^{-r_1}\mathfrak{X}_{n,s+r_1+|\alpha|}, \tag{780}$$

$$\sup_{0\le t\le T} \|\mathsf{P}_{\ge K}J_{n,\alpha}(t)\|_{H^s} \le K^{-r_1}\mathfrak{Z}_{n,s+r_1+|\alpha|}, \tag{781}$$

$$\|\mathsf{P}_{\ge K}\Pi_{n,\alpha}\|_{L^1(I_T;H^s)} \le K^{-r_1}\mathfrak{Q}_{n,s+r_1+|\alpha|}^{(1)}, \tag{782}$$

$$\sup_{0\le t\le T} \|\mathsf{P}_{\ge K}\Pi_{n,\alpha}(t)\|_{H^s} \le K^{-r_1}\mathfrak{Q}_{n,s+r_1+|\alpha|}^{(\infty)}. \tag{783}$$

*For $j \in \mathbb{N}_0$,*

$$\sup_{0\le t\le T} E_{j+1/2}(t) \le \frac{1}{2}\mathfrak{Z}_{0,j+1}^2, \tag{784}$$

$$\int_0^T E_{j+1/2}(t)\,\mathrm{d}t \le \frac{1}{2}\mathfrak{X}_{0,j+1}^2. \tag{785}$$

*If $m > j + \frac{1}{2}$ is an integer, then*

$$\sup_{0\le t\le T} E_{j+1/2,\ge K}(t) \le \frac{1}{2}K^{2j+1-2m}\mathfrak{Z}_{0,m}^2, \tag{786}$$

$$\int_0^T E_{j+1/2,\ge K}(t)\,\mathrm{d}t \le \frac{1}{2}K^{2j+1-2m}\mathfrak{X}_{0,m}^2. \tag{787}$$

*For $k \in \mathbb{N}$,*

$$\left\|H^{(k)}\right\|_{L^1(0,T)} \le \frac{1}{2}\sum_{a=0}^{k}\binom{k}{a}\mathfrak{X}_{a,1}\mathfrak{X}_{k-a,1}. \tag{788}$$

*For $j, \ell \in \mathbb{N}_0$, define*

$$\begin{aligned}
\mathfrak{E}_{j,\ell}^{[1]} &:= \frac{1}{2}\sum_{a=0}^{\ell}\binom{\ell}{a}\mathfrak{X}_{a,j+1}\mathfrak{X}_{\ell-a,j+1},\\
\mathfrak{E}_{j,\ell}^{[\infty]} &:= \frac{1}{2}\sum_{a=0}^{\ell}\binom{\ell}{a}\mathfrak{Z}_{a,j+1}\mathfrak{Z}_{\ell-a,j+1},
\end{aligned} \tag{789}$$

$$\begin{aligned}
\mathfrak{C}_{j,\ell}^{[1]} &:= \mathfrak{E}_{j,\ell+1}^{[1]} + 2\nu\mathfrak{E}_{j+1,\ell}^{[1]},\\
\mathfrak{C}_{j,\ell}^{[\infty]} &:= \mathfrak{E}_{j,\ell+1}^{[\infty]} + 2\nu\mathfrak{E}_{j+1,\ell}^{[\infty]}.
\end{aligned} \tag{790}$$

*Then the complete differentiated energy and production rows obey*

$$\left\|\partial_t^\ell E_{j+1/2}\right\|_{L^1(0,T)} \le \mathfrak{E}_{j,\ell}^{[1]}, \qquad \sup_{0\le t\le T}\left|\partial_t^\ell E_{j+1/2}(t)\right| \le \mathfrak{E}_{j,\ell}^{[\infty]}, \tag{791}$$

$$\left\|\partial_t^\ell \mathcal{P}_{j+1/2}\right\|_{L^1(0,T)} \le \mathfrak{C}_{j,\ell}^{[1]}, \qquad \sup_{0\le t\le T}\left|\partial_t^\ell \mathcal{P}_{j+1/2}(t)\right| \le \mathfrak{C}_{j,\ell}^{[\infty]}. \tag{792}$$

*Every current in* (75) *now has a finite-order majorant retaining temporal depth, spatial height, viscosity, and horizon. In particular,*

$$\left\|H'\right\|_{L^1(0,T)} \le \mathfrak{X}_{1,1}\mathfrak{X}_{0,1}, \tag{793}$$

$$\int_0^T \left\|\Lambda^{3/2}u(t)\right\|_2^2\,\mathrm{d}t \le \mathfrak{X}_{0,2}^2, \tag{794}$$

$$\left\|\mathcal{P}_{1/2}\right\|_{L^1(0,T)} \le \mathfrak{X}_{1,1}\mathfrak{X}_{0,1} + \nu\mathfrak{X}_{0,2}^2, \tag{795}$$

$$|H(T) - H(t)| \le \min\left\{\mathfrak{Z}_{0,1}\mathfrak{X}_{1,1}(T-t)^{1/2}, \mathfrak{Z}_{0,1}\mathfrak{Z}_{1,1}(T-t)\right\}, \qquad 0 \le t \le T. \tag{796}$$

*Finally,*

$$\|u\|_{L^2(I_T;L^\infty)} \le S_{2,0,\infty}\mathfrak{X}_{0,2}, \tag{797}$$

$$\int_0^T \|\nabla u(t)\|_\infty \,\mathrm{d}t \le S_{3,1,\infty}T^{1/2}\mathfrak{X}_{0,3}, \tag{798}$$

$$\int_0^T \|\nabla \times u(t)\|_\infty \,\mathrm{d}t \le 2S_{3,1,\infty}T^{1/2}\mathfrak{X}_{0,3}, \tag{799}$$

$$\sup_{0\le t\le T}\|u(t)\|_{H^m}^2 \le \|u_0\|_{H^m}^2 \exp\Big(2K_m S_{3,1,\infty}T^{1/2}\mathfrak{X}_{0,3}\Big), \qquad m\ge 3. \tag{800}$$

*Every right-hand side is an explicit finite expression in the displayed analytic constants, the initial-jet recursion* (188)–(189)*, and the finite collection of canonical rectangle-norm majorants* $\mathfrak{B}_{a,b}[u_0,\nu;T]$ *selected by the stated indices. Equations* (187) *and* (190)–(227) *specify every intermediate operation. Equation* (216) *bounds each such rectangle norm through the finite height recursion* (206)–(214)*. Together with the reverse bound in* (208)*, this identifies the mixed-row and height descriptions of the original history at every finite order. This is an exact coordinate-level catalogue for general data; each finite entry is obtained from the compatible completed-height and mixed-row coordinates selected by its displayed indices.*

*Proposition 7.6 displays the exact powers of* $\nu$ *and the rescaled horizon* $\nu T$*. For every fixed choice of the displayed indices, the definitions select only finitely many rectangle-norm majorants. Under the explicit critical smallness condition* (278)*, Proposition 8.4 eliminates those majorants and gives finite formulas solely in the displayed datum norms,* $\nu$*,* $T$*,* $N$*, and* $M$*.*

*Proof.* The Fourier multiplier inequality

$$\|\partial_x^\alpha f\|_{H^\ell} \le \|f\|_{H^{\ell+|\alpha|}},$$

followed by Lemma 7.8, proves (765)–(768). The Banach identity

$$J_{n,\alpha}(t) - J_{n,\alpha}(r_0) = \int_{r_0}^t J_{n+1,\alpha}(\tau)\,\mathrm{d}\tau$$

in the two indicated Sobolev spaces and Cauchy–Schwarz prove the square-root branches of (769) and (770). This identity, followed by the pointwise bound (221) for $V_{n+1}$, proves their linear branches. Interpolation between the low modulus and

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^s} \le 2\mathfrak{Z}_{n,r}$$

proves (771). The fixed Sobolev embedding constant in (180) proves (772) and (773).

Apply the recurrence

$$Q_n = R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i} V_{n-a,j}$$

term by term at order $r$, using (178). Cauchy–Schwarz in time gives (774); placing either factor in $L_t^\infty H_x^{r+1}$ gives (775); placing both factors there gives (776) and (777). After subtracting the recurrence at times $t$ and $r_0$, split each product difference into two terms and apply (770); this gives the square-root branch of (778). Since $\partial_t \Pi_{n,\alpha} = \Pi_{n+1,\alpha}$, the Banach identity and (776) at row $n+1$ give its linear branch. Sobolev embedding proves (779).

For every $f \in H^{\ell+r_1}$,

$$\|\mathsf{P}_{\ge K} f\|_{H^\ell} \le K^{-r_1}\,\|f\|_{H^{\ell+r_1}}.$$

Apply this inequality at spatial order $s + r_1$ to (220), (221), (774), and (776). The four resulting estimates are (780)–(783).

The inequalities

$$\|f\|_{\dot H^{j+1/2}} \le \|f\|_{H^{j+1}}\,, \qquad |\xi|^{2j+1} \le K^{2j+1-2m}|\xi|^{2m} \quad (|\xi| \ge K),$$

give (784)–(787). The exact identity

$$H^{(k)}(t) = \frac{1}{2}\sum_{a=0}^{k}\binom{k}{a}\operatorname{Re}\left\langle \Lambda^{1/2}V_a(t), \Lambda^{1/2}V_{k-a}(t)\right\rangle_{L^2}$$

and Cauchy–Schwarz in time give (788) and (793). More generally,

$$\partial_t^\ell E_{j+1/2} = \frac{1}{2}\sum_{a=0}^{\ell}\binom{\ell}{a}\operatorname{Re}\left\langle \Lambda^{j+1/2}V_a, \Lambda^{j+1/2}V_{\ell-a}\right\rangle_{L^2}.$$

Cauchy–Schwarz in time and the pointwise estimates (220)–(221) prove (791). Differentiating $\mathcal{P}_{j+1/2} = \partial_t E_{j+1/2} + 2\nu E_{j+3/2}$ then proves (792). The next two estimates follow from

$$\mathcal{P}_{1/2} = H' + \nu\left\|\Lambda^{3/2}u\right\|_2^2.$$

For the terminal critical-height modulus,

$$|H(T) - H(t)| \le \frac{1}{2}\big(\|u(T)\|_{\dot H^{1/2}} + \|u(t)\|_{\dot H^{1/2}}\big)\,\|u(T) - u(t)\|_{\dot H^{1/2}}\,,$$

and the Banach identity in $H^1$ prove the square-root branch of (796).
The identity

$$H' = \operatorname{Re}\left\langle \Lambda^{1/2}V_1, \Lambda^{1/2}V_0\right\rangle_{L^2},$$

together with (221), proves its linear branch.

Sobolev embedding and (220) prove (797)–(799). Lemma 2.8 gives

$$\frac{d}{dt}\|u\|_{H^m}^2 \le 2K_m\,\|\nabla u\|_\infty\,\|u\|_{H^m}^2\,.$$

Gronwall's inequality and (798) prove (800). □

# B Forced canonical coordinate catalogue

The estimates in Section 17 are organized by their role in the argument. For reference, the following theorem collects their canonical-coordinate forms in one place.

**Theorem B.1** (Canonical finite-horizon coordinate estimates)**.** *Under the hypotheses and notation of Definition 17.2, let $J_{n,\alpha} := \partial_x^\alpha V_n$ and $\Pi_{n,\alpha} := \partial_x^\alpha Q_n$, as in Theorems 17.10 and 17.11. Let $n, s, k, q \in \mathbb{N}_0$, $\alpha \in \mathbb{N}_0^3$, and set $r := s + |\alpha|$. Then*

$$\|J_{n,\alpha}\|_{L^2(I_T;H^{s+1})} \le \mathfrak{X}_{n,r+1}, \tag{801}$$

$$\|J_{n+1,\alpha}\|_{L^2(I_T;H^{s-1})} \le \mathfrak{X}_{n+1,r-1}, \tag{802}$$

$$\sup_{0\le t\le T}\|J_{n,\alpha}(t)\|_{H^s} \le \mathfrak{Z}_{n,r}, \tag{803}$$

$$\left\|J_{n,\alpha}^T\right\|_{H^s} \le \mathfrak{Z}_{n,r}, \tag{804}$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^{s-1}} \le \min\left\{|t-r_0|^{1/2}\mathfrak{X}_{n+1,r-1}, |t-r_0|\mathfrak{Z}_{n+1,r}\right\}, \tag{805}$$

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^{s}} \le \min\left\{|t-r_0|^{1/2}\mathfrak{X}_{n+1,r}, |t-r_0|\mathfrak{Z}_{n+1,r}\right\}, \tag{806}$$

*for* $0 \le r_0 \le t \le T$.
*For* $0 \le \theta \le 1$,

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^{s-\theta}} \le 2^{1-\theta}\mathfrak{Z}_{n,r}^{1-\theta}\mathfrak{X}_{n+1,r-1}^{\theta}|t-r_0|^{\theta/2}. \tag{807}$$

*If* $1 \le a \le \infty$, $2 \le b \le \infty$, *and* $q > k + \frac{3}{2} - \frac{3}{b}$, *then*

$$\left\|\nabla^k J_{n,\alpha}\right\|_{L^a(I_T;L^b)} \le S_{q,k,b}T^{1/a}\mathfrak{Z}_{n,q+|\alpha|}, \tag{808}$$

$$\left\|\nabla^k J_{n,\alpha}\right\|_{L^2(I_T;L^b)} \le S_{q,k,b}\mathfrak{X}_{n,q+|\alpha|}, \tag{809}$$

*where* $T^{1/\infty} = 1$.

*The pressure rows obey*

$$\|\Pi_{n,\alpha}\|_{L^1(I_T;H^s)} + \|\nabla\Pi_{n,\alpha}\|_{L^1(I_T;H^{s-1})} \le \mathfrak{Q}_{n,r}^{(1)}, \tag{810}$$

$$\|\Pi_{n,\alpha}\|_{L^2(I_T;H^s)} \le \mathfrak{Q}_{n,r}^{(2)}, \tag{811}$$

$$\sup_{0\le t\le T}\|\Pi_{n,\alpha}(t)\|_{H^s} \le \mathfrak{Q}_{n,r}^{(\infty)}, \tag{812}$$

$$\left\|\Pi_{n,\alpha}^T\right\|_{H^s} \le \mathfrak{Q}_{n,r}^{(\infty)}, \tag{813}$$

$$\|\Pi_{n,\alpha}(t) - \Pi_{n,\alpha}(r_0)\|_{H^s} \le \min\left\{|t-r_0|^{1/2}\mathfrak{Q}_{n,r}^{(\mathrm{mod})}, |t-r_0|\mathfrak{Q}_{n+1,r}^{(\infty)}\right\}. \tag{814}$$

*Under the preceding restrictions on* $a, b, k, q$,

$$\left\|\nabla^k \Pi_{n,\alpha}\right\|_{L^a(I_T;L^b)} \le S_{q,k,b}T^{1/a}\mathfrak{Q}_{n,q+|\alpha|}^{(\infty)}. \tag{815}$$

*Let* $r_1 \in \mathbb{N}$ *and* $K \ge 1$. *Then*

$$\|\mathsf{P}_{\ge K}J_{n,\alpha}\|_{L^2(I_T;H^s)} \le K^{-r_1}\mathfrak{X}_{n,s+r_1+|\alpha|}, \tag{816}$$

$$\sup_{0\le t\le T}\|\mathsf{P}_{\ge K}J_{n,\alpha}(t)\|_{H^s} \le K^{-r_1}\mathfrak{Z}_{n,s+r_1+|\alpha|}, \tag{817}$$

$$\|\mathsf{P}_{\ge K}\Pi_{n,\alpha}\|_{L^1(I_T;H^s)} \le K^{-r_1}\mathfrak{Q}_{n,s+r_1+|\alpha|}^{(1)}, \tag{818}$$

$$\sup_{0\le t\le T}\|\mathsf{P}_{\ge K}\Pi_{n,\alpha}(t)\|_{H^s} \le K^{-r_1}\mathfrak{Q}_{n,s+r_1+|\alpha|}^{(\infty)}. \tag{819}$$

*For* $j \in \mathbb{N}_0$,

$$\sup_{0\le t\le T}E_{j+1/2}(t) \le \frac{1}{2}\mathfrak{Z}_{0,j+1}^2, \tag{820}$$

$$\int_0^T E_{j+1/2}(t)\,\mathrm{d}t \le \frac{1}{2}\mathfrak{X}_{0,j+1}^2. \tag{821}$$

*If* $m > j + \frac{1}{2}$ *is an integer, then*

$$\sup_{0\le t\le T}E_{j+1/2,\ge K}(t) \le \frac{1}{2}K^{2j+1-2m}\mathfrak{Z}_{0,m}^2, \tag{822}$$

$$\int_0^T E_{j+1/2,\ge K}(t)\,\mathrm{d}t \le \frac{1}{2}K^{2j+1-2m}\mathfrak{X}_{0,m}^2. \tag{823}$$

*For $k \in \mathbb{N}$,*

$$\left\|H^{(k)}\right\|_{L^1(0,T)} \le \frac{1}{2}\sum_{a=0}^{k}\binom{k}{a}\mathfrak{X}_{a,1}\mathfrak{X}_{k-a,1}. \tag{824}$$

*For $j,\ell \in \mathbb{N}_0$, define*

$$\begin{aligned}
\mathfrak{E}^{[1]}_{j,\ell} &:= \frac{1}{2}\sum_{a=0}^{\ell}\binom{\ell}{a}\mathfrak{X}_{a,j+1}\mathfrak{X}_{\ell-a,j+1},\\
\mathfrak{E}^{[\infty]}_{j,\ell} &:= \frac{1}{2}\sum_{a=0}^{\ell}\binom{\ell}{a}\mathfrak{Z}_{a,j+1}\mathfrak{Z}_{\ell-a,j+1},
\end{aligned} \tag{825}$$

$$\begin{aligned}
\mathfrak{W}^{[1]}_{j,\ell} &:= \sum_{a=0}^{\ell}\binom{\ell}{a}\mathfrak{X}_{a,j+1}\mathfrak{G}^{(2)}_{\ell-a,j+1},\\
\mathfrak{W}^{[\infty]}_{j,\ell} &:= \sum_{a=0}^{\ell}\binom{\ell}{a}\mathfrak{Z}_{a,j+1}\mathfrak{G}^{(\infty)}_{\ell-a,j+1},
\end{aligned} \tag{826}$$

$$\begin{aligned}
\mathfrak{C}^{[1]}_{j,\ell} &:= \mathfrak{E}^{[1]}_{j,\ell+1} + 2\nu\mathfrak{E}^{[1]}_{j+1,\ell} + \mathfrak{W}^{[1]}_{j,\ell},\\
\mathfrak{C}^{[\infty]}_{j,\ell} &:= \mathfrak{E}^{[\infty]}_{j,\ell+1} + 2\nu\mathfrak{E}^{[\infty]}_{j+1,\ell} + \mathfrak{W}^{[\infty]}_{j,\ell}.
\end{aligned} \tag{827}$$

*Then the complete differentiated energy and production rows obey*

$$\left\|\partial_t^{\ell}E_{j+1/2}\right\|_{L^1(0,T)} \le \mathfrak{E}^{[1]}_{j,\ell}, \qquad \sup_{0\le t\le T}\left|\partial_t^{\ell}E_{j+1/2}(t)\right| \le \mathfrak{E}^{[\infty]}_{j,\ell}, \tag{828}$$

$$\left\|\partial_t^{\ell}\mathcal{W}_{j+1/2}\right\|_{L^1(0,T)} \le \mathfrak{W}^{[1]}_{j,\ell}, \qquad \sup_{0\le t\le T}\left|\partial_t^{\ell}\mathcal{W}_{j+1/2}(t)\right| \le \mathfrak{W}^{[\infty]}_{j,\ell}, \tag{829}$$

$$\left\|\partial_t^{\ell}\mathcal{P}_{j+1/2}\right\|_{L^1(0,T)} \le \mathfrak{C}^{[1]}_{j,\ell}, \qquad \sup_{0\le t\le T}\left|\partial_t^{\ell}\mathcal{P}_{j+1/2}(t)\right| \le \mathfrak{C}^{[\infty]}_{j,\ell}. \tag{830}$$

*Every current in* (406) *now has a finite-order majorant retaining temporal depth, spatial height, viscosity, and horizon. In particular,*

$$\left\|H'\right\|_{L^1(0,T)} \le \mathfrak{X}_{1,1}\mathfrak{X}_{0,1}, \tag{831}$$

$$\int_0^T \left\|\Lambda^{3/2}u(t)\right\|_2^2 \,\mathrm{d}t \le \mathfrak{X}_{0,2}^2, \tag{832}$$

$$\left\|\mathcal{P}_{1/2}\right\|_{L^1(0,T)} \le \mathfrak{X}_{1,1}\mathfrak{X}_{0,1} + \nu\mathfrak{X}_{0,2}^2 + \mathfrak{X}_{0,1}\mathfrak{G}^{(2)}_{0,1}, \tag{833}$$

$$|H(T)-H(t)| \le \min\left\{\mathfrak{Z}_{0,1}\mathfrak{X}_{1,1}(T-t)^{1/2}, \mathfrak{Z}_{0,1}\mathfrak{Z}_{1,1}(T-t)\right\}, \qquad 0\le t\le T. \tag{834}$$

*Finally,*

$$\|u\|_{L^2(I_T;L^\infty)} \le S_{2,0,\infty}\mathfrak{X}_{0,2}, \tag{835}$$

$$\int_0^T \|\nabla u(t)\|_\infty \,\mathrm{d}t \le S_{3,1,\infty}T^{1/2}\mathfrak{X}_{0,3}, \tag{836}$$

$$\int_0^T \|\nabla\times u(t)\|_\infty \,\mathrm{d}t \le 2S_{3,1,\infty}T^{1/2}\mathfrak{X}_{0,3}, \tag{837}$$

$$\sup_{0\le t\le T}\|u(t)\|_{H^m}^2 \le \left(\|u_0\|_{H^m} + \mathfrak{G}^{(1)}_{0,m}\right)^2 \exp\left(2K_mS_{3,1,\infty}T^{1/2}\mathfrak{X}_{0,3}\right), \qquad m\ge 3. \tag{838}$$

*Every right-hand side is an explicit finite expression in the displayed analytic constants, the prescribed-force coordinates, the initial-jet recursion* (531)–(532)*, and the finite collection of canonical rectangle-norm majorants* $\mathfrak{B}_{a,b}[u_0,\nu,f;T]$ *selected by the stated indices. Equations* (528) *and* (533)–(570) *specify every intermediate operation. Equation* (559) *bounds each such rectangle norm through the finite height recursion* (549)–(557)*. Together with the reverse bound in* (551)*, this identifies the mixed-row and height descriptions of the original forced history at every finite order. This is the exact coordinate-level catalogue for general data: every finite quantity is expressed through the compatible completed-height and mixed-row coordinates selected by its displayed indices.*

*Proposition 17.6 displays the exact powers of* $\nu$ *and the rescaled horizon* $\nu T$*. For every fixed choice of the displayed indices, the definitions select only finitely many rectangle-norm majorants. Under the explicit critical smallness condition* (617)*, Proposition 18.4 eliminates those majorants and gives finite formulas solely in the displayed velocity-datum and prescribed-force norms,* $\nu$*,* $T$*,* $N$*, and* $M$*.*

*Proof.* The Fourier multiplier inequality

$$\|\partial_x^\alpha f\|_{H^\ell} \le \|f\|_{H^{\ell+|\alpha|}} ,$$

followed by Lemma 17.8, proves (801)–(804). The Banach identity

$$J_{n,\alpha}(t) - J_{n,\alpha}(r_0) = \int_{r_0}^{t} J_{n+1,\alpha}(\tau)\,\mathrm{d}\tau$$

in the two indicated Sobolev spaces and Cauchy–Schwarz prove the square-root branches of (805) and (806). This identity, followed by the pointwise bound (564) for $V_{n+1}$, proves their linear branches. Interpolation between the low modulus and

$$\|J_{n,\alpha}(t) - J_{n,\alpha}(r_0)\|_{H^s} \le 2\mathfrak{Z}_{n,r}$$

proves (807). The fixed Sobolev embedding constant in (521) proves (808) and (809).

Apply the recurrence

$$Q_n = R_i R_j \sum_{a=0}^{n} \binom{n}{a} V_{a,i} V_{n-a,j} - \Phi_n$$

term by term at order $r$, using (519). Cauchy–Schwarz in time gives (810); placing either factor in $L_t^\infty H_x^{r+1}$ gives (811); placing both factors there gives (812) and (813), with the corresponding $\mathfrak{L}$ coordinate retaining $-\Phi_n$ in each case. After subtracting the recurrence at times $t$ and $r_0$, split each product difference into two terms and apply (806); this gives the square-root branch of (814). Since $\partial_t \Pi_{n,\alpha} = \Pi_{n+1,\alpha}$, the Banach identity and (812) at row $n+1$ give its linear branch. Sobolev embedding proves (815).

For every $f \in H^{\ell+r_1}$,

$$\|\mathsf{P}_{\ge K} f\|_{H^\ell} \le K^{-r_1} \|f\|_{H^{\ell+r_1}} .$$

Apply this inequality at spatial order $s+r_1$ to (563), (564), (810), and (812). The four resulting estimates are (816)–(819).

The inequalities

$$\|f\|_{\dot H^{j+1/2}} \le \|f\|_{H^{j+1}} , \qquad |\xi|^{2j+1} \le K^{2j+1-2m}|\xi|^{2m} \quad (|\xi| \ge K),$$

give (820)–(823). The exact identity

$$H^{(k)}(t) = \frac{1}{2}\sum_{a=0}^{k} \binom{k}{a} \operatorname{Re}\left\langle \Lambda^{1/2} V_a(t), \Lambda^{1/2} V_{k-a}(t)\right\rangle_{L^2}$$

and Cauchy–Schwarz in time give (824) and (831). More generally,

$$\partial_t^{\ell} E_{j+1/2} = \frac{1}{2} \sum_{a=0}^{\ell} \binom{\ell}{a} \operatorname{Re} \left\langle \Lambda^{j+1/2} V_a, \Lambda^{j+1/2} V_{\ell-a} \right\rangle_{L^2} .$$

Cauchy–Schwarz in time and the pointwise estimates (563)–(564) prove (828). Leibniz' rule applied to $\mathcal{W}_{j+1/2}$ gives the majorants (826). Differentiating the forced identity

$$\mathcal{P}_{j+1/2} = \partial_t E_{j+1/2} + 2\nu E_{j+3/2} - \mathcal{W}_{j+1/2}$$

then proves (830). The next two estimates follow from

$$\mathcal{P}_{1/2} = H' + \nu \left\| \Lambda^{3/2} u \right\|_2^2 - \mathcal{W}_{1/2}.$$

For the terminal critical-height modulus,

$$|H(T) - H(t)| \le \frac{1}{2} \big( \|u(T)\|_{\dot{H}^{1/2}} + \|u(t)\|_{\dot{H}^{1/2}} \big) \, \|u(T) - u(t)\|_{\dot{H}^{1/2}} ,$$

and the Banach identity in $H^1$ prove the square-root branch of (834).
The identity

$$H' = \operatorname{Re} \left\langle \Lambda^{1/2} V_1, \Lambda^{1/2} V_0 \right\rangle_{L^2} ,$$

together with (564), proves its linear branch.

Sobolev embedding and (563) prove (835)–(837). Lemma 12.8, with the prescribed force retained, gives after regularization of the norm

$$\frac{d}{dt} \|u\|_{H^m} \le K_m \|\nabla u\|_{\infty} \|u\|_{H^m} + \|\mathbb{P} f\|_{H^m} .$$

The inhomogeneous Gronwall inequality and (836) prove (838). □

# C Source work and its iterated time integrals

The source work can be controlled further before invoking any higher velocity rectangle. For the base temporal row, write $W_s = W_{0,s}$ and put $h_s = \Lambda^{2s} \mathbb{P} f$. Moving the spatial derivatives onto the prescribed field gives $W_s = \langle u, h_s \rangle$. This pairing measures how the fluid responds to a smooth spatial weight. Kinetic energy alone bounds this pairing; no higher spatial norm of the velocity enters the estimate.

The energy balance implies

$$\|u(t)\|_2 \le K_T, \qquad K_T = \|u_0\|_2 + \int_0^T \|f(t)\|_2 \, \mathrm{d}t.$$

Let $S_s = \operatorname{Sym} \nabla h_s$. Substitution of the momentum equation into the derivative of $\langle u, h_s \rangle$ gives

$$W_s' = \langle u, (\partial_t + \nu \Delta) h_s \rangle + \int_{\mathbb{R}^3} u_i u_j (S_s)_{ij} \, \mathrm{d}x + \langle f, h_s \rangle . \tag{839}$$

Incompressibility removes the pressure pairing; integration by parts moves the transport and viscous derivatives onto $h_s$. Using the matrix operator norm for $S_s$, define

$$\begin{aligned}\mathsf{B}_s &:= K_T \|h_s\|_{L_t^\infty L_x^2}\,,\\ \mathsf{L}_s &:= K_T \|(\partial_t+\nu\Delta)h_s\|_{L_t^\infty L_x^2} + K_T^2 \|S_s\|_{L_{t,x}^\infty}\\ &\qquad + \|f\|_{L_t^\infty L_x^2}\|h_s\|_{L_t^\infty L_x^2}\,.\end{aligned} \tag{840}$$

Every fixed constant is finite under the source assumptions, and $\|W_s\|_\infty \le \mathsf{B}_s$, $\|W_s'\|_\infty \le \mathsf{L}_s$. The projection need not preserve rapid spatial decrease; its boundedness on Sobolev spaces supplies the norms used here.

These two bounds control an integrated form of the complete source correction at every spatial height. Define

$$\begin{aligned}\mathcal{S}_{s,k} &:= \sum_{j=0}^{k-1}(-2\nu)^j \partial_t^{k-1-j} W_{s+j}, \qquad k\ge 2,\\ I^0 &:= \mathrm{Id}, \qquad (I^a g)(t) := \frac{1}{(a-1)!}\int_0^t (t-v)^{a-1} g(v)\,\mathrm{d}v, \quad a\ge 1.\end{aligned} \tag{841}$$

**Proposition C.1** (Integrated source correction)**.** *For each fixed $s\ge 0$ and integer $k\ge 2$, $I^{k-2}\mathcal{S}_{s,k}$ is bounded on $(0,T)$ by the datum and source. If $u_0=0$ and the force vanishes near $t=0$, then*

$$\begin{aligned} I^{k-2}\mathcal{S}_{s,k} &= W_s' - 2\nu W_{s+1} + \sum_{j=2}^{k-1}(-2\nu)^j I^{j-1} W_{s+j},\\ \left\| I^{k-2}\mathcal{S}_{s,k}\right\|_\infty &\le \mathsf{L}_s + \sum_{j=1}^{k-1}\frac{(2\nu)^j T^{j-1}}{(j-1)!}\mathsf{B}_{s+j}.\end{aligned} \tag{842}$$

*No source amplitude restriction or higher velocity estimate is used.*

*Proof.* Write $D=\partial_t$ and $m=k-2$. Repeated integration gives

$$I^m D^d W = \begin{cases} D^{d-m}W - \displaystyle\sum_{\ell=0}^{m-1} W^{(d-m+\ell)}(0)\frac{t^\ell}{\ell!}, & d\ge m,\\[2mm] I^{m-d}W - \displaystyle\sum_{\ell=0}^{d-1} W^{(\ell)}(0)\frac{t^{m-d+\ell}}{(m-d+\ell)!}, & d<m.\end{cases} \tag{843}$$

Empty sums vanish. In summand $j$ of $\mathcal{S}_{s,k}$, set $d=k-1-j=m+1-j$. The cases $j=0,1$ leave $W_s'$ and $-2\nu W_{s+1}$; the other cases leave the integrals in (842). Each initial correction is a polynomial whose coefficients are determined by

$$W_s^{(\ell)}(0) = \sum_{b=0}^{\ell}\binom{\ell}{b}\left\langle a_b, \partial_t^{\ell-b} h_s(0)\right\rangle.$$

They are finite by (679). Equation (843) specifies every correction for general datum. The inequality $\|I^a W\|_\infty \le T^a \|W\|_\infty / a!$ proves the bound. For rest datum and forcing that vanishes initially, local uniqueness gives a zero initial velocity history. All initial corrections vanish, yielding (842). □

The first instance bounds $W_s' - 2\nu W_{s+1}$ directly. The next bounds the time integral of $W_s'' - 2\nu W_{s+1}' + 4\nu^2 W_{s+2}$. At higher orders the bound uses exactly $k-2$ integrations. It controls the signed source contribution in that coordinate, rather than the absolute integral of each differentiated term. This specifies which part of the completed height calculation is bounded before using the velocity intersection.

## C.1 Source absorption in every temporal row

The kinetic estimate above bounds the pairing with $u$. At a higher temporal row, source work pairs the prescribed $F_r$ with the response $V_r$. Viscous dissipation controls one additional spatial derivative of $V_r$. Pairing that derivative with one lower source derivative allows part of the dissipation to absorb the source work. Since $V_r$ is divergence free,

$$\begin{aligned} W_{r,s} &= \operatorname{Re} \langle \Lambda^s V_r, \Lambda^s \mathbb{P} F_r \rangle \\ &= \operatorname{Re} \left\langle \Lambda^{s+1} V_r, \Lambda^{s-1} \mathbb{P} F_r \right\rangle . \end{aligned} \tag{844}$$

For the prescribed rapidly decreasing force, the second factor is finite at every $s \geq 0$. If $s < 1$, its Fourier weight has a singularity at zero, but $\widehat{F}_r$ is bounded there and the three-dimensional radial integral is $\int_0^1 \rho^{2s} \, \mathrm{d}\rho < \infty$. The smoothness of the force controls the high frequencies. All these bounds are uniform on the prescribed time interval.

Young's inequality now allocates one quarter of the viscous dissipation to source work:

$$\begin{aligned} |W_{r,s}| &\leq \frac{\nu}{4} \left\| \Lambda^{s+1} V_r \right\|_2^2 + \frac{1}{\nu} \left\| \Lambda^{s-1} \mathbb{P} F_r \right\|_2^2 \\ &= \frac{\nu}{2} E_{r,s+1} + \frac{1}{\nu} \left\| \Lambda^{s-1} \mathbb{P} F_r \right\|_2^2 . \end{aligned} \tag{845}$$

Substitution into the complete height identity gives

$$\partial_t E_{r,s} + \frac{3\nu}{2} E_{r,s+1} \leq \mathcal{P}_{r,s} + \frac{1}{\nu} \left\| \Lambda^{s-1} \mathbb{P} F_r \right\|_2^2 . \tag{846}$$

The nonlinear production retains its sign and its full binomial sum. The displayed source norm has a finite time integral determined by the prescribed force. An absolute integral of $W_{r,s}$ also requires control of the dissipation appearing in (845).

The argument can be written for the broader source class $\bigcap_{r,m} H_t^r H_x^m$, without a spatial-decay assumption. Use $J = (1-\Delta)^{1/2}$ and set

$$A_{r,s} = \tfrac{1}{2} \left\| J^s V_r \right\|_2^2 , \qquad D_{r,s} = \left\| \nabla J^s V_r \right\|_2^2 , \qquad \mathcal{P}_{r,s}^J = - \operatorname{Re} \langle J^s V_r, J^s B_r \rangle .$$

Pressure again pairs to zero. The identity $\left\| J^{s+1} V_r \right\|_2^2 = D_{r,s} + 2A_{r,s}$ and Young's inequality yield

$$A_{r,s}' + \frac{\nu}{2} D_{r,s} \leq \mathcal{P}_{r,s}^J + \nu A_{r,s} + \frac{1}{2\nu} \left\| J^{s-1} \mathbb{P} F_r \right\|_2^2 . \tag{847}$$

For $s < 1$, the multiplier defining $J^{s-1}$ is bounded at zero. Multiplying by $e^{-\nu t}$ and integrating controls the source contribution on every strict window with constants independent of the time cutoff. The resulting inequality still contains the signed integral of $\mathcal{P}_{r,s}^J$.

## C.2 The source primitive at every temporal row

For $h_s = \Lambda^{2s}\mathbb{P}f$, the kinetic bound also controls the full direct source current at every finite temporal row. Set

$$M_{s,k}(t) = \mathrm{Re}\langle u(t), \partial_t^k h_s(t)\rangle.$$

The exact binomial transfer is

$$W_{r,s} = \sum_{b=0}^{r}(-1)^b\binom{r}{b}\partial_t^{r-b}M_{s,r+b}.$$

To verify all indices and signs, expand the derivative on the right. The coefficient of $\langle V_a, \partial_t^{2r-a}h_s\rangle$ is

$$\sum_{b=0}^{r-a}(-1)^b\binom{r}{b}\binom{r-b}{a} = \binom{r}{a}\sum_{b=0}^{r-a}(-1)^b\binom{r-a}{b}.$$

It is zero for $a < r$ and one for $a = r$. The surviving term is $\langle V_r, \partial_t^r h_s\rangle = W_{r,s}$.

The momentum calculation in (839) also applies to every prescribed test field $g_{s,k} = \partial_t^k h_s$. It gives

$$\begin{aligned} M'_{s,k} = {} & \mathrm{Re}\,\langle u, (\partial_t + \nu\Delta)g_{s,k}\rangle \\ & + \int_{\mathbb{R}^3} u_i u_j (\mathrm{Sym}\,\nabla g_{s,k})_{ij}\,\mathrm{d}x + \mathrm{Re}\,\langle f, g_{s,k}\rangle\,. \end{aligned} \tag{848}$$

Thus both the test pairing and its first time derivative are bounded by kinetic energy and prescribed source norms:

$$\begin{aligned} \|M_{s,k}\|_\infty &\le \mathsf{B}_{s,k} := K_T\,\|g_{s,k}\|_{L_t^\infty L_x^2}\,, \\ \|M'_{s,k}\|_\infty &\le \mathsf{L}_{s,k} := K_T\,\|(\partial_t + \nu\Delta)g_{s,k}\|_{L_t^\infty L_x^2} \\ &\quad + K_T^2\,\|\mathrm{Sym}\,\nabla g_{s,k}\|_{L^\infty_{t,x},\mathrm{op}} + \|f\|_{L_t^\infty L_x^2}\,\|g_{s,k}\|_{L_t^\infty L_x^2}\,. \end{aligned} \tag{849}$$

The additional derivative bound removes two integrations from the source primitive at every temporal row. Write the complete source correction as

$$\mathcal{F}_{r,s,n} := \sum_{j=0}^{n-1}(-2\nu)^j\partial_t^{n-1-j}W_{r,s+j}, \qquad n \ge 1.$$

**Proposition C.2** (Source correction with two fewer integrations)**.** *For every fixed $r \ge 0$, $n \ge 1$, and $s \ge 0$, the complete source correction satisfies*

$$\left\| I^{\max(n+r-2,0)}\mathcal{F}_{r,s,n}\right\|_{L^\infty(0,T)} < \infty. \tag{850}$$

*The bound uses $\nu, T$, the kinetic bound, prescribed force norms, and finitely many initial jets. No restriction on the force amplitude or higher velocity norm is required.*

*Proof.* Substitution of the binomial transfer into the complete source correction gives

$$\mathcal{F}_{r,s,n} = \sum_{j=0}^{n-1}\sum_{b=0}^{r}(-2\nu)^j(-1)^b\binom{r}{b}\partial_t^{n+r-1-j-b}M_{s+j,r+b}. \tag{851}$$

When $(r,n)=(0,1)$, this is $M_{s,0}$, already bounded by (849). Otherwise put $q=n+r-2$. For $0\le v\le q+1$, repeated integration gives

$$I^q\partial_t^{q+1-v}M=\mathcal{J}_{q,v}[M],\qquad \mathcal{J}_{q,v}[M]=\begin{cases} M'-\displaystyle\sum_{\ell=0}^{q-1}M^{(\ell+1)}(0)\frac{t^\ell}{\ell!}, & v=0,\\[2mm] M-\displaystyle\sum_{\ell=0}^{q-1}M^{(\ell)}(0)\frac{t^\ell}{\ell!}, & v=1,\\[2mm] I^{v-1}M-\displaystyle\sum_{\ell=0}^{q-v}M^{(\ell)}(0)\frac{t^{v-1+\ell}}{(v-1+\ell)!}, & v\ge 2. \end{cases}\tag{852}$$

Empty sums vanish. In summand $(j,b)$ of (851), take $v=j+b$. The first two branches are bounded by $\mathsf{L}_{s+j,r+b}$ and $\mathsf{B}_{s+j,r+b}$, together with their initial polynomials. For the third branch,

$$\left\|I^{v-1}M_{s,k}\right\|_\infty\le\frac{T^{v-1}}{(v-1)!}\mathsf{B}_{s,k}.$$

Every initial coefficient is finite: the product rule writes it as

$$M_{s,k}^{(\ell)}(0)=\sum_{c=0}^{\ell}\binom{\ell}{c}\operatorname{Re}\left\langle V_c(0),g_{s,k+\ell-c}(0)\right\rangle,$$

and (679) determines each $V_c(0)$ from the smooth datum and prescribed force jet. There are finitely many summands for each fixed $(r,s,n)$, proving (850). □

For the nonnegative integrated height identity below, it remains useful to write the source correction after $n+r$ integrations explicitly. Apply $I^{n+r}$ to $\mathcal{F}_{r,s,n}$. For $0\le j<n$ and $0\le b\le r$, put $q=j+b+1$. The integration formula (843), with total integration order $n+r$, gives

$$\begin{aligned} &I^{n+r}\partial_t^{n-1-j+r-b}M_{s+j,r+b}(t)\\ &\quad=I^qM_{s+j,r+b}(t)-\sum_{\ell=0}^{n+r-2-j-b}M_{s+j,r+b}^{(\ell)}(0)\frac{t^{q+\ell}}{(q+\ell)!}. \end{aligned}$$

An upper index of $-1$ means an empty sum. Multiplying the preceding identity by $(-2\nu)^j(-1)^b\binom{r}{b}$ and summing gives the entire source current, with no terminal derivative.

For an explicit bound, set

$$S_{s,k}=\sup_{0\le t\le T}\|\partial_t^kh_s(t)\|_2.$$

Then $|M_{s,k}|\le K_TS_{s,k}$ and

$$\sup_{t<T}|I^{n+r}\mathcal{F}_{r,s,n}(t)|\le B_{r,s,n},$$

where

$$\begin{aligned} B_{r,s,n}:=\sum_{j=0}^{n-1}\sum_{b=0}^{r}(2\nu)^j\binom{r}{b}\Bigg[&K_TS_{s+j,r+b}\frac{T^{j+b+1}}{(j+b+1)!}\\ &+\sum_{\ell=0}^{n+r-2-j-b}|M_{s+j,r+b}^{(\ell)}(0)|\frac{T^{j+b+1+\ell}}{(j+b+1+\ell)!}\Bigg]<\infty. \end{aligned}$$

Every initial jet is finite by the forced recurrence and smooth initial data. This establishes finite integrated direct forcing in every temporal row and at every spatial height, separately for each finite $r, s, n$.

Write $\mathcal{N}_{r,s,n} = \sum_{j=0}^{n-1} (-2\nu)^j \partial_t^{n-1-j} \mathcal{P}_{r,s+j}$. For odd $n$, integrating (681) $n + r$ times gives

$$I^r E_{r,s} + (2\nu)^n I^{n+r} E_{r,s+n} = I^r T_{r,s,n} + I^{n+r} \mathcal{N}_{r,s,n} + I^{n+r} \mathcal{F}_{r,s,n},$$

where $T_{r,s,n}(t) = \sum_{\ell=0}^{n-1} E_{r,s}^{(\ell)}(0) t^\ell / \ell!$. Both left-hand terms are nonnegative. Thus divergence of $I^r E_{r,s}$ requires an unbounded upper value of $I^{n+r} \mathcal{N}_{r,s,n}$ at every odd $n$. For $r > 0$, this conclusion requires divergence of the indicated time integral; pointwise divergence of $E_{r,s}$ alone is insufficient.

### C.3 Source work in the first two temporal rows

The source work has a useful bound before any higher velocity rectangle is read off. This includes mixed spatial derivatives and the first temporal row. Keep the actual forced velocity, set $V_r = \partial_t^r u$ and $F_r = \partial_t^r f$, and write

$$U_* := \|u_0\|_2 + \int_0^T \|f(t)\|_2 \, \mathrm{d}t.$$

The kinetic identity gives $\sup_{t<T} \|u(t)\|_2 \le U_*$. For a spatial multi-index $\alpha$ and a real height $s \ge 0$, introduce the nonnegative self-adjoint multiplier

$$M_{\alpha,s} = (-1)^{|\alpha|} \partial_x^{2\alpha} \Lambda^{2s}, \qquad \widehat{M_{\alpha,s} v}(\xi) = \xi^{2\alpha} |\xi|^{2s} \widehat{v}(\xi).$$

Spatial self-adjointness gives the exact source pairing

$$W_{r,\alpha,s} := \langle \Lambda^s \partial^\alpha V_r, \Lambda^s \partial^\alpha F_r \rangle = \langle V_r, M_{\alpha,s} F_r \rangle . \tag{853}$$

All pairings in this section mean their real parts. At $r = 0$, every spatial derivative can thus be carried by the prescribed source:

$$\int_0^T |W_{0,\alpha,s}| \, \mathrm{d}t \le U_* \int_0^T \|M_{\alpha,s} f\|_2 \, \mathrm{d}t < \infty. \tag{854}$$

The constant is finite at each chosen order. This estimate consumes no viscous dissipation and requires no higher velocity norm.

For $r = 1$, let $h = M_{\alpha,s} \mathbb{P} F_1$. This is a prescribed smooth solenoidal field. Pairing the original momentum equation with $h$, moving spatial derivatives to $h$, and using incompressibility gives

$$W_{1,\alpha,s} = \nu \langle u, \Delta h \rangle + \int_{\mathbb{R}^3} u_i u_j (\operatorname{Sym} \nabla h)_{ij} \, \mathrm{d}x + \langle f, h \rangle . \tag{855}$$

Pressure contributes zero because $\nabla \cdot h = 0$. The transport term has the displayed positive sign after integration by parts; contraction with $u \otimes u$ removes the antisymmetric part of $\nabla h$. Consequently,

$$\int_0^T |W_{1,\alpha,s}| \, \mathrm{d}t \le \nu U_* \int_0^T \|\Delta h\|_2 \, \mathrm{d}t + U_*^2 \int_0^T \|\operatorname{Sym} \nabla h\|_{L_x^\infty, \mathrm{op}} \, \mathrm{d}t + \int_0^T \|f\|_2 \|h\|_2 \, \mathrm{d}t < \infty. \tag{856}$$

Every quantity on the right is fixed by the datum, viscosity, and source. Sobolev boundedness of $\mathbb{P}$, followed by an adequately high spatial embedding, controls these norms. No spatial support or rapid-decay property of $\mathbb{P}f$ is required. The estimate imposes no restriction on force amplitude.

The higher temporal rows have an exact finite reduction. For $r \geq 1$, put

$$B_{r,\alpha,s} = \sum_{j=0}^{r-1} (-1)^j \langle V_{r-1-j}, M_{\alpha,s} F_{r+j} \rangle .$$

Differentiating this sum cancels adjacent terms and leaves

$$B'_{r,\alpha,s} = W_{r,\alpha,s} + (-1)^{r-1} \langle u, M_{\alpha,s} F_{2r} \rangle .$$

Hence

$$\int_0^t W_{r,\alpha,s} \, \mathrm{d}v = B_{r,\alpha,s}(t) - B_{r,\alpha,s}(0) + (-1)^r \int_0^t \langle u, M_{\alpha,s} F_{2r} \rangle \, \mathrm{d}v. \tag{857}$$

The terminal pairings involve temporal velocity rows strictly below $r$. If $K_j = \sup_{t<T} \|V_j(t)\|_2$ for $j < r$, the explicit bound is

$$\begin{aligned} \sup_{t<T} \left| \int_0^t W_{r,\alpha,s} \, \mathrm{d}v \right| \leq 2 \sum_{j=0}^{r-1} K_{r-1-j} \sup_{t \leq T} \|M_{\alpha,s} F_{r+j}(t)\|_2 \\ + U_* \int_0^T \|M_{\alpha,s} F_{2r}\|_2 \, \mathrm{d}t. \end{aligned} \tag{858}$$

At $r = 1$, only the kinetic bound is needed. At larger $r$, the stated lower temporal norms enter explicitly; the datum-generated intersection supplies them in the principal argument. Formula (858) controls a signed integral, while (856) controls its absolute integral at the first temporal row.

For completeness, with $E_{r,\alpha,s} = \|\Lambda^s \partial^\alpha V_r\|_2^2/2$, the corresponding energy identity can be written

$$\partial_t (E_{r,\alpha,s} - B_{r,\alpha,s}) = -\nu \|\Lambda^{s+1} \partial^\alpha V_r\|_2^2 + \mathcal{P}_{r,\alpha,s} + (-1)^r \langle u, M_{\alpha,s} F_{2r} \rangle .$$

Here $\mathcal{P}_{r,\alpha,s}$ retains the complete differentiated nonlinear row and its pressure completion. This identity changes the recorded scalar by an explicitly bounded pairing when the lower rows are bounded. It leaves the underlying velocity and its equation intact.

## C.4 Source insertions in the velocity rows

Write

$$g = \mathbb{P}f, \qquad g_j = \partial_t^j g, \qquad L = \nu\Delta, \qquad B(v,w) = \mathbb{P}((v \cdot \nabla)w),$$
$$Q(v,w) = B(v,w) + B(w,v), \qquad \mathcal{K}(u) = Lu - B(u,u), \qquad \mathcal{A}_u h = Lh - Q(u,h).$$

Here $\mathcal{K}$ is the spatial differential expression in the projected momentum equation, and $\mathcal{A}_u$ is its derivative with respect to the velocity argument. The Leray projection is bounded in every Sobolev norm used below.

The momentum equation is $V_1 = \mathcal{K}(u) + g$. To differentiate it correctly, use

$$\partial_t \mathcal{K}(u) = \mathcal{A}_u V_1, \qquad \partial_t (\mathcal{A}_u h) = \mathcal{A}_u \partial_t h - Q(V_1, h).$$

In particular, the coefficients of $\mathcal{A}_u$ also change with time. Carrying this term gives

$$V_2 = \mathcal{A}_u \mathcal{K}(u) + S_2,$$
$$S_2 = \mathcal{A}_u g + g_1 = Lg - Q(u,g) + g_1,$$

and then

$$V_3 = \mathcal{A}_u^2 \mathcal{K}(u) - Q(\mathcal{K}(u), \mathcal{K}(u)) + S_3,$$
$$S_3 = \mathcal{A}_u^2 g + \mathcal{A}_u g_1 - 2Q(\mathcal{K}(u), g) - Q(g, g) + g_2.$$

These are exact expansions of the full binomial rows. The coefficient two in the formula for $S_3$ comes from differentiating both the velocity-dependent operator and the velocity argument. The quantities $S_2, S_3$ collect source insertions into the velocity jet. They contain the prescribed source together with the velocity on which it acts.

The kinetic-energy inequality already bounds the first insertion through the terminal time. Put

$$U = \|u_0\|_2 + \|f\|_{L_t^1 L_x^2}, \qquad D^2 = \nu^{-1} \left( \tfrac{1}{2} \|u_0\|_2^2 + U \|f\|_{L_t^1 L_x^2} \right).$$

The actual forced history satisfies $\sup_{t<T} \|u(t)\|_2 \le U$ and $\|\nabla u\|_{L_t^2 L_x^2} \le D$. Since $B$ is followed by an $L^2$-contractive projection, the formula for $S_2$ gives

$$\begin{aligned} \|S_2\|_{L_t^2 L_x^2} &\le \nu \|\Delta g\|_{L_t^2 L_x^2} + \|g_1\|_{L_t^2 L_x^2} \\ &\quad + \|g\|_{L_{t,x}^\infty} D + \|\nabla g\|_{L_{t,x}^\infty} \sqrt{T}\, U < \infty. \end{aligned}$$

Thus this first force–velocity insertion has a terminal bound independent of the critical velocity rectangle.

The next insertion admits an $L_t^2 H_x^{-3}$ estimate from the kinetic bounds. The weaker spatial norm accommodates the additional derivatives in this row. Define

$$Z = \|u\|_{L_t^2 H_x^1} \le (TU^2 + D^2)^{1/2}, \qquad G = \max_{0 \le j \le 2} \sup_{t \le T} \|g_j(t)\|_{H^6}.$$

The divergence form and Sobolev duality give

$$\begin{aligned} \|Q(u, g)\|_{L^2 L^2} &\le CGZ, \\ \|B(u, u)\|_{L^2 H^{-3/2}} &\le CUZ, \\ \|Q(u, Q(u, g))\|_{L^2 H^{-3}} &\le CGUZ, \\ \|Q(Lu, g)\|_{L^2 H^{-2}} &\le C\nu GZ, \\ \|Q(B(u, u), g)\|_{L^2 H^{-5/2}} &\le CGUZ. \end{aligned}$$

For the second line use $u \otimes u \in L_t^2 L_x^{3/2}$, obtained from $u \in L_t^\infty L_x^2 \cap L_t^2 L_x^6$, and the dual of $H^{1/2} \hookrightarrow L^3$. For the third line, $u \otimes Q(u, g)$ belongs to $L_t^2 L_x^1$; pairing its divergence against $H^3$ is bounded because $H^2 \hookrightarrow L^\infty$. Smooth multiplication by $g$ and its derivatives gives the final two lines. Expanding $\mathcal{A}_u^2 g$ in the formula for $S_3$ now gives

$$\|S_3\|_{L_t^2 H_x^{-3}} \le C \left[ \sqrt{T} \big( (1 + \nu + \nu^2) G + G^2 \big) + G(1 + \nu + U) Z \right] < \infty.$$

The constant $C$ depends on fixed Sobolev product and multiplier constants. All velocity dependence is contained in the kinetic quantities $U$ and $Z$. The two estimates control $S_2$ and $S_3$ at the stated spatial heights.

The all-order version is also explicit. Define spatial jet polynomials by

$$\mathcal{J}_0(v) = v, \qquad \mathcal{J}_{n+1}(v) = D\mathcal{J}_n(v)[\mathcal{K}(v)].$$

Here $D\mathcal{J}_n(v)[h]$ means the derivative of the spatial polynomial with respect to $v$ in direction $h$. Along the actual forced history,

$$V_n = \mathcal{J}_n(u) + S_n, \qquad S_0 = 0,$$

$$S_{n+1} = D\mathcal{J}_n(u)[g] + \partial_t S_n.$$

This follows by the chain rule and $u_t = \mathcal{K}(u) + g$. Every $S_n$ is a finite differential expression with at least one source factor. The recurrence retains all velocity factors as well. The bounds above apply at the specified spatial heights.

### C.5 Time-integrated completion and terminal integrability

To apply Proposition C.1 to the completed-height identity (681), take $r = 0$, $k \geq 2$, $m = k - 2$, and $H = E_{0,1/2}$. Define

$$\mathcal{N}_{s,k} := \sum_{j=0}^{k-1} (-2\nu)^j \partial_t^{k-1-j} \mathcal{P}_{0,s+j}.$$

Applying $I^m$ to the full completion gives

$$\begin{aligned}(-2\nu)^k I^m C^f_{0,k} = H'' - \sum_{\ell=0}^{m-1} H^{(\ell+2)}(0)\frac{t^\ell}{\ell!} \\ - I^m \mathcal{N}_{1/2,k} - I^m \mathcal{S}_{1/2,k}.\end{aligned} \tag{859}$$

Proposition C.1 bounds the explicit source term on the last line. For the forced history in Corollary 20.1, restriction of its mixed jet to $(0, T)$ gives the unweighted terminal integrals at every finite height. For finite $r, k$, this jet gives $V_r \in L^2(0, T; \dot{H}^{k+1/2})$, so

$$\int_0^T C^f_{r,k}(t)\,\mathrm{d}t = \frac{1}{2}\int_0^T \|V_r(t)\|^2_{\dot{H}^{k+1/2}}\,\mathrm{d}t < \infty.$$

In particular, the full mixed product rule for $W_{r,s}$ contains both source and response factors, even when every source coordinate is finite. For $k \geq 4$, boundedness of $I^{k-2}C^f_{0,k}$ would itself give only a weighted terminal integral,

$$\int_0^T (T-t)^{k-3} C^f_{0,k}(t)\,\mathrm{d}t < \infty.$$

The positive Volterra kernel gives this statement by monotone convergence; its vanishing terminal weight does not give the unweighted integral above. At $k = 2, 3$, a bound on the full right side of (859) yields the unweighted integral directly. For larger $k$, that bound controls the weighted integral, while the constructed mixed jet supplies the unweighted one.

## D Stokes lifting and differentiated source interactions

The prescribed force determines a linear Stokes response with finite norms at every fixed order. Separating this response from the velocity makes the additional nonlinear interactions explicit. Let $z$ solve

$$z_t - \nu\Delta z = \mathbb{P}f, \qquad z(0) = 0, \qquad z(t) = \int_0^t e^{\nu(t-s)\Delta}\mathbb{P}f(s)\,ds. \tag{860}$$

The heat semigroup is a contraction in every $H^q$. For $Z_n = \partial_t^n z$,

$$\begin{aligned} Z_{n+1} &= \nu\Delta Z_n + \mathbb{P}F_n, \\ Z_n(0) &= \sum_{j=0}^{n-1} (\nu\Delta)^{n-1-j}\mathbb{P}F_j(0) \quad (n \geq 1), \\ \sup_{t\leq T} \|Z_n(t)\|_{H^q} &\leq \|Z_n(0)\|_{H^q} + \int_0^T \|F_n(s)\|_{H^q}\, ds. \end{aligned} \tag{861}$$

Every norm in this estimate depends on finitely many derivatives of the prescribed force. Set $v = u - z$. This decomposition separates the known Stokes response from the remainder of the forced solution.

With $B(a, b) = \mathbb{P}[(a \cdot \nabla)b]$, the equation for $v$ is

$$v_t - \nu\Delta v + B(v, v) + B(z, v) + B(v, z) + B(z, z) = 0. \tag{862}$$

Put $W_n = \partial_t^n v$, so $V_n = W_n + Z_n$. Every temporal row becomes

$$\begin{aligned} W_{n+1} - \nu\Delta W_n + \sum_{a=0}^{n} \binom{n}{a} \big\{ & B(W_a, W_{n-a}) + B(Z_a, W_{n-a}) \\ & + B(W_a, Z_{n-a}) + B(Z_a, Z_{n-a}) \big\} = 0. \end{aligned} \tag{863}$$

The binomial self-interaction is retained in full. The other three terms contain a prescribed lift factor. If $\nabla\phi = \mathbb{Q}f$ and $\pi = p - \phi$, pressure is retained by

$$-\Delta\partial_t^n \pi = \sum_{a=0}^{n} \binom{n}{a} \partial_i (W_a + Z_a)_j \, \partial_j (W_{n-a} + Z_{n-a})_i. \tag{864}$$

**Theorem D.1** (Finite-order estimate for the Stokes decomposition)**.** *For a fixed finite $N$ and real $s \geq 0$, define*

$$\begin{aligned} \mathscr{E} &= \tfrac{1}{2} \sum_{n=0}^{N} \|W_n\|_{H^s}^2 , & \mathscr{D} &= \sum_{n=0}^{N} \|\nabla W_n\|_{H^s}^2 , \\ \mathscr{Z} &= \sum_{a=0}^{N} \binom{N}{a} \|Z_a\|_{H^{s+2}} , & G_n &= \sum_{a=0}^{n} \binom{n}{a} B(Z_a, Z_{n-a}), \\ \mathscr{G}^2 &= \sum_{n=0}^{N} \|G_n\|_{H^{s-1}}^2 . \end{aligned} \tag{865}$$

*Let the signed self-production be*

$$\mathscr{P}_{N,s}(W) = -\sum_{n=0}^{N} \left\langle W_n, \sum_{a=0}^{n} \binom{n}{a} B(W_a, W_{n-a}) \right\rangle_{H^s}. \tag{866}$$

*On every strict classical interval,*

$$\mathscr{E}' + \frac{\nu}{2}\mathscr{D} \leq \mathscr{P}_{N,s}(W) + \left( \nu + \frac{8C_s^2}{\nu}\mathscr{Z}^2 \right) \mathscr{E} + \frac{\mathscr{G}^2}{\nu}, \tag{867}$$

*where $C_s = 2^{s/2}\pi(2\pi)^{-3/2}$. The two added coefficients have finite time integrals determined entirely by the prescribed force. No smallness condition is used.*

*Proof.* The Fourier weight inequality $\langle\xi\rangle^s \le 2^{s/2}\langle\eta\rangle^s\langle\xi-\eta\rangle^s$ and Cauchy–Schwarz give

$$\|ab\|_{H^s} \le C_s \|a\|_{H^{s+2}} \|b\|_{H^s},$$

using $\int\langle\eta\rangle^{-4}d\eta = \pi^2$. Since both factors are divergence-free, the divergence form of $B$ gives $\|B(a,b)\|_{H^{s-1}} \le \|a\otimes b\|_{H^s}$. In either cross term, the factor placed at height $s+2$ is $Z$.

Let $L_n$ be the sum of the two cross interactions in (863). Discrete convolution and $\binom{n}{a} \le \binom{N}{a}$ give

$$\left(\sum_{n=0}^{N} \|L_n\|_{H^{s-1}}^2\right)^{1/2} \le 2C_s\mathscr{Z}\sqrt{2\mathscr{E}}.$$

Pairing through heights $s-1$ and $s+1$, all added terms are bounded by

$$\sqrt{\mathscr{D}+2\mathscr{E}}\left(2C_s\mathscr{Z}\sqrt{2\mathscr{E}}+\mathscr{G}\right).$$

Young's inequality bounds this by $\frac{\nu}{2}\mathscr{D} + \nu\mathscr{E} + 8C_s^2\mathscr{Z}^2\mathscr{E}/\nu + \mathscr{G}^2/\nu$. The remaining pairing is exactly (866), proving (867). Finally, $\mathscr{G} \le C_s\mathscr{Z}(\sum_{n\le N}\|Z_n\|_{H^s}^2)^{1/2}$, so (861) bounds every added coefficient. □

The added interactions have coefficients determined by the source, and their contribution is at most linear in $\mathscr{E}$. At $N = s = 0$, incompressibility also cancels the self-production. At higher orders, the signed nonlinear term in (866) remains. The estimate controls the source-dependent perturbation of the energy equation without giving an independent bound for that nonlinear term.

The decomposition must also retain the original initial derivatives. For $a_n = V_n(0)$, the recurrence (679) reads

$$a_{n+1} = \nu\Delta a_n - \mathbb{P}\sum_{b=0}^{n}\binom{n}{b}(a_b\cdot\nabla)a_{n-b} + \mathbb{P}F_n(0). \tag{868}$$

Thus the initial derivatives remain fixed by $u_0$ and the source derivatives at zero. Full-interval regularity for arbitrary forcing is supplied by Corollary 20.1, as a direct consequence of the global forced theorem proved in Part III.

## D.1 The force inside differentiated nonlinear production

The explicit work term does not contain every occurrence of the source after time differentiation. The nonlinear production is evaluated on a forced velocity, so differentiating it inserts the force into each velocity slot. At the second critical-height row this entire insertion can also be controlled from the kinetic budget.

Write $E_s(u) = \frac{1}{2}\|u\|_{\dot H^s}^2$, $g = \mathbb{P}f$, $\mathcal{A}(u) = \nu\Delta u - B(u,u)$ and $\mathcal{P}_s(u) = -\langle\Lambda^s u, \Lambda^s B(u,u)\rangle$. The vector field $\mathcal{A}$ is the unforced right side evaluated at the current velocity; the actual trajectory obeys $u_t = \mathcal{A}(u) + g$. Let $\mathcal{J}_s = D\mathcal{P}_s(u)[g]$. Differentiating its three velocity slots gives

$$\begin{aligned}\mathcal{J}_s = &-\langle\Lambda^s g, \Lambda^s((u\cdot\nabla)u)\rangle\\ &-\langle\Lambda^s u, \Lambda^s((g\cdot\nabla)u)\rangle\\ &-\langle\Lambda^s u, \Lambda^s((u\cdot\nabla)g)\rangle.\end{aligned} \tag{869}$$

The projection has been removed only inside global pairings with solenoidal fields.

Define $U = \sup_{t<T}\|u(t)\|_2$, $D = \|\nabla u\|_{L^2(0,T;L^2)}$, and

$$a_\rho(t) = (2\pi)^{-3/2}\int_{\mathbb{R}^3}|\xi|^\rho|\widehat{g}(t,\xi)|\,d\xi, \qquad h(t) = \|u(t)\|_{\dot H^{1/2}}.$$

Kinetic energy bounds $U$ and $D$ from the prescribed data. Every $a_\rho$ used below is bounded by a sufficiently high source Sobolev norm.

**Proposition D.2** (Complete critical source insertion)**.** *Without a source smallness condition,*

$$|\mathcal{J}_{1/2}| \le 2a_1 h^2 + a_{3/2} U h + a_2 U^2. \tag{870}$$

*In particular, with* $a_\rho^* = \sup_{t\le T} a_\rho(t)$*,*

$$\left\|\mathcal{J}_{1/2}\right\|_{L^1(0,T)} \le 2a_1^* U\sqrt{T}\, D + a_{3/2}^* U^{3/2} T^{3/4} D^{1/2} + a_2^* T U^2. \tag{871}$$

*Proof.* In the first pairing of (869), move all spatial derivatives onto $g$; its absolute value is bounded by $a_2 U^2$. In the second pairing the undifferentiated transport cancels by $\operatorname{div} g = 0$. Its commutator obeys

$$\left\|[\Lambda^{1/2}, g\cdot\nabla]u\right\|_2 \le a_1 h,$$

because $\left||\xi|^{1/2} - |\eta|^{1/2}\right| |\eta| \le |\xi-\eta|\,|\eta|^{1/2}$ in its Fourier kernel. Young's convolution inequality proves the bound. For the final pairing, splitting the half-derivative Fourier weight gives

$$\left\|\Lambda^{1/2}((u\cdot\nabla)g)\right\|_2 \le a_1 h + a_{3/2} U.$$

Summing gives (870). Finally, Fourier Cauchy–Schwarz gives $h^2 \le U\,\|\nabla u\|_2$. Integrating this inequality and its square root proves (871). □

The direct work terms can also be estimated using only kinetic energy and source norms. Set $W_s = \langle u, \Lambda^{2s} g\rangle$ and $k_s = \Lambda^{2s} g$. Moving derivatives onto the prescribed test field yields

$$\begin{aligned} W_s' &= \langle u, (\partial_t + \nu\Delta) k_s\rangle \\ &\quad + \int_{\mathbb{R}^3} u_i u_j\, \partial_j (k_s)_i \, dx + \langle g, k_s\rangle. \end{aligned} \tag{872}$$

Each term is uniformly bounded from $U$ and smooth source norms. The pressure pairing vanishes because $k_s$ is divergence-free.

Let $H(u) = \frac{1}{2}\|u\|_{\dot H^{1/2}}^2$ and write $L_{\mathcal{A}}\Phi = D\Phi[\mathcal{A}]$ for differentiation of a scalar functional along the unforced vector field. Along the forced history the exact second-height equation is

$$\begin{aligned} &H'' = L_{\mathcal{A}}^2 H(u) + R_f, \\ &R_f = \mathcal{J}_{1/2} + W_{1/2}' - 2\nu W_{3/2}, \qquad R_f \in L^1(0,T). \end{aligned} \tag{873}$$

Thus the source correction hidden inside nonlinear differentiation is controlled along with the explicit work terms. The remaining functional is

$$L_{\mathcal{A}}^2 H(u) = 4\nu^2 E_{5/2}(u) + D\mathcal{P}_{1/2}(u)[\mathcal{A}(u)] - 2\nu\mathcal{P}_{3/2}(u). \tag{874}$$

The leading viscous coefficient remains positive. The remainder depends on the nonlinear evolution of $u$. Subtracting the double time integral of $R_f$ corrects the scalar energy identity, but the velocity still satisfies $u_t = \mathcal{A}(u) + g$. Thus neither this scalar correction nor the Stokes decomposition identifies the forced solution with an unforced one. The estimates establish bounded source contributions at the stated orders; full-interval control of the nonlinear remainder is a separate assertion.

## D.2 Graph domains of the Stokes linearization

The linearization around the smooth Stokes response has the parabolic graph domain of the heat operator. Write $w = z$ for the prescribed Stokes response in (860), and define

$$K_w v = \mathbb{P}[(w \cdot \nabla)v + (v \cdot \nabla)w], \qquad L_w = \partial_t - \nu\Delta + K_w.$$

For an integer $m \geq 0$, let

$$\mathcal{E}_m(0,S) = C([0,S]; H^m_\sigma) \cap L^2(0,S; H^{m+1}_\sigma), \qquad v_t \in L^2(0,S; H^{m-1}_\sigma).$$

The derivative condition is included in the definition of $\mathcal{E}_m$.

**Proposition D.3** (Linear graph domains)**.** *For $0 < S \leq T$, the initial-trace graph completions of $L_w$ and $L_0 = \partial_t - \nu\Delta$ both equal $\mathcal{E}_m(0,S)$. Their norms are equivalent with constants bounded independently of $S$:*

$$\begin{aligned} &\|v\|_{CH^m} + \|v\|_{L^2H^{m+1}} + \|v_t\|_{L^2H^{m-1}} \\ &\qquad \leq C_{m,\nu,T,w} \left( \|v(0)\|_{H^m} + \|L_w v\|_{L^2H^{m-1}} \right). \end{aligned} \tag{875}$$

*Every coefficient in the constant is determined by finitely many prescribed source norms, with no restriction on their size.*

*Proof.* Product differentiation and the divergence form at $m = 0$ give

$$\|K_w v\|_{H^{m-1}} \leq C_m \|w\|_{W^{m+1,\infty}} \|v\|_{H^m}.$$

Put $r = L_w v$, and pair the equation at height $m$, distributing the duality between $H^{m-1}$ and $H^{m+1}$. Since $\|\nabla v\|^2_{H^m} = \|v\|^2_{H^{m+1}} - \|v\|^2_{H^m}$, Young's inequality yields

$$\frac{\mathrm{d}}{\mathrm{d}t} \|v\|^2_{H^m} + \nu \|v\|^2_{H^{m+1}} \leq C_{m,\nu,w} \|v\|^2_{H^m} + C_\nu \|r\|^2_{H^{m-1}}.$$

Gronwall and integration give the first two norms in (875); the equation gives the third. The exponential factor is bounded by its value at $T$. Conversely, each member of $\mathcal{E}_m$ has $L_w v \in L^2H^{m-1}$ by the product bound. Approximation of its initial trace and residual, followed by the difference estimate, gives the graph completion. Existence for the approximating linear problems follows from the heat integral equation on a finite partition, on which its smooth bounded coefficients give contraction. The energy estimate joins those intervals. □

Temporal differentiation retains the earlier lift rows. If $W_n = \partial_t^n v$, $Z_a = \partial_t^a w$, and $h_w = -\mathbb{P}[(w \cdot \nabla)w]$, the exact equation is

$$L_w W_n + \sum_{a=0}^{n} \binom{n}{a} \mathbb{P}[(W_a \cdot \nabla)W_{n-a}] = \partial_t^n h_w - \sum_{a=1}^{n} \binom{n}{a} K_{Z_a} W_{n-a}.$$

For $c_{a,m} = C_m \sup_{t \leq T} \|Z_a\|_{W^{m+1,\infty}}$,

$$\|\partial_t^n (K_w v)\|_{L^2H^{m-1}} \leq \sum_{a=0}^{n} \binom{n}{a} c_{a,m} \|W_{n-a}\|_{L^2H^m}.$$

The adjacent pair has parabolic degree $2n + m + 1$; each velocity factor on this right side has degree at most $2n + m$. At the base corner, kinetic energy already supplies the lower row. These source contributions preserve the terminal linear domain. The nonlinear self-interaction remains in the displayed equation, so the domain estimate does not identify $v$ with an unforced solution.

# E Critical growth, source reserve, and chord density

The source work at critical order is $\langle u, \Lambda g\rangle$, where $g = \mathbb{P}f$. Its spatial derivatives can all be placed on $g$, so the kinetic bound controls how much this term can still add before a prescribed finite time. Keeping that remaining contribution with the critical height gives a positive quantity whose growth is governed by the nonlinear production of the velocity.

Let $u$ be a classical history on $[0,T)$, with the force smooth through $T$ as in Section 20. All calculations first take place on strict classical intervals. Write

$$\begin{aligned} \mathsf{K}_T &= \|u_0\|_2 + \int_0^T \|g(t)\|_2 \,\mathrm{d}t, \\ H(t) &= \tfrac{1}{2}\left\|\Lambda^{1/2}u(t)\right\|_2^2, \qquad\qquad D(t) = \left\|\Lambda^{3/2}u(t)\right\|_2^2. \end{aligned} \tag{876}$$

The kinetic work satisfies

$$\int_0^t \|g(s)\|_2 \,\|u(s)\|_2 \,\mathrm{d}s \le \int_0^t \mathsf{K}_s\mathsf{K}_s' \,\mathrm{d}s = \tfrac{1}{2}\big(\mathsf{K}_t^2 - \|u_0\|_2^2\big).$$

The norm inequality and the energy identity give

$$\|u(t)\|_2 \le \mathsf{K}_t, \qquad \|u(t)\|_2^2 + 2\nu\int_0^t \|\nabla u(s)\|_2^2 \,\mathrm{d}s \le \mathsf{K}_t^2 \le \mathsf{K}_T^2. \tag{877}$$

At critical order, put $P_H = \mathcal{P}_{0,1/2}$ and $W = \operatorname{Re}\langle u, \Lambda g\rangle$. The exact balance is

$$H' + \nu D = P_H + W, \qquad |W(t)| \le b_f(t) := \mathsf{K}_T\,\|\Lambda g(t)\|_2 .$$

The source assumptions give $b_f \in L^1(0,T)$. Its remaining integral decreases at the rate $b_f$, so define

$$\mathscr{R}_f(t) = \int_t^T b_f(s)\,\mathrm{d}s, \qquad \Phi = H + \mathscr{R}_f, \qquad Z_f = b_f - W. \tag{878}$$

Then

$$\Phi' + \nu D + Z_f = P_H, \qquad Z_f \ge 0, \qquad \Phi \ge H \ge 0. \tag{879}$$

The known function $\mathscr{R}_f$ is absolutely continuous and tends to zero at $T$. Neither statement requires a terminal value of $H$. The production in (879) is still evaluated on the forced velocity. The force and its pressure contribution remain in the momentum equation.

## E.1 Where critical growth is distributed

The critical norm measures velocity differences over every separation. For a center $z$, a chord $y$, and $0 \le \theta \le 1$, put

$$x_+ = z + (1-\theta)y, \qquad x_- = z - \theta y, \qquad \delta u_\theta = u(x_+) - u(x_-), \qquad \widehat{y} = y/|y|.$$

The positive tensor

$$\mathsf{T}_u(z) = \frac{1}{\pi^2}\int_0^1 \int_{\mathbb{R}^3} \frac{|\delta u_\theta(z,y)|^2}{|y|^4}\widehat{y}\otimes\widehat{y}\,\mathrm{d}y\,\mathrm{d}\theta, \qquad \tau = \operatorname{tr}\mathsf{T}_u \tag{880}$$

weights each chord orientation by the squared magnitude of its velocity difference. The fractional Fourier identity and the complete transport pairing give

$$\int_{\mathbb{R}^3} \tau \,\mathrm{d}z = 2\left\|\Lambda^{1/2}u\right\|_2^2 = 4H, \qquad P_H = -\int_{\mathbb{R}^3} S : \mathsf{T}_u \,\mathrm{d}z, \quad S = \tfrac{1}{2}(\nabla u + \nabla u^T). \tag{881}$$

The transport identity follows by integration in the chord variable and averaging the strain along that chord; the local derivation below retains the resulting redistribution flux.

Where $\tau > 0$, define $r = -S : \mathsf{T}_u/\tau$, and set $r = 0$ where $\tau = 0$. Positivity of $\mathsf{T}_u$ gives

$$|r| \le \|S\|_{\mathrm{op}} \le |S|_F, \qquad P_H = \int \tau r \,\mathrm{d}z.$$

Thus the production samples strain with the nonlocal density $\tau$. For $H > 0$, use the probability measure and its positive sampled rate

$$\mathrm{d}\mu_t = \frac{\tau(z,t)\,\mathrm{d}z}{4H(t)}, \qquad \overline{r}_+(t) = \int r_+ \,\mathrm{d}\mu_t.$$

Set $\overline{r}_+ = 0$ when $H = 0$. Finite energy then forces $u = 0$, so $\tau = P_H = D = W = 0$; no probability measure is assigned to zero mass. Equation (879) gives

$$\Phi' \le P_H \le \int \tau r_+ \,\mathrm{d}z = 4H\overline{r}_+ \le 4\Phi\overline{r}_+. \tag{882}$$

On an interval $[a, b]$ where $\Phi > 0$, a doubling requires

$$4\int_a^b \overline{r}_+(t)\,\mathrm{d}t \ge \log\frac{\Phi(b)}{\Phi(a)} \ge \log 2.$$

The rate itself has an ordinary-volume square bound. For each eigenvalue $\lambda_i$ of the trace-free strain $S$, the other two sum to $-\lambda_i$, so

$$|S|_F^2 \ge \lambda_i^2 + \tfrac{1}{2}\lambda_i^2 \quad \text{for every } i, \qquad |r|^2 \le \|S\|_{\mathrm{op}}^2 \le \tfrac{2}{3}|S|_F^2.$$

Incompressibility gives $\|S\|_2^2 = \|\nabla u\|_2^2/2$, hence

$$\int_0^T \int_{\mathbb{R}^3} |r|^2 \,\mathrm{d}z\,\mathrm{d}t \le \frac{\mathsf{K}_T^2}{6\nu}. \tag{883}$$

The integral through $T$ is the monotone limit of strict-subinterval integrals.

To compare this volume bound with the density that samples it, put

$$\rho_f = \frac{\tau}{4\Phi}, \qquad V_f^{-1}(t) = \int_{\{r>0\}} \rho_f(z,t)^2 \,\mathrm{d}z, \qquad G_f = \{t : \Phi'(t) > 0\}. \tag{884}$$

These definitions apply where $\Phi > 0$; set $\rho_f = 0$ when $\Phi = 0$. Its mass is $H/\Phi \le 1$. If $V_{\mathrm{work}}^{-1} = \int_{\{r>0\}} (\tau/(4H))^2 \,\mathrm{d}z$ for $H > 0$, then

$$V_f^{-1} = (H/\Phi)^2 V_{\mathrm{work}}^{-1}.$$

At $H = 0 < \Phi$, equation (879) gives $\Phi' = -b_f \le 0$, so such times add no positive growth.

On a positive interval, discard the negative part of the logarithmic growth and apply Cauchy–Schwarz in space and then in time:

$$\log \frac{\Phi(b)}{\Phi(a)} \le 4 \left( \int_{[a,b]\cap G_f} \int r_+^2 \, \mathrm{d}z \, \mathrm{d}t \right)^{1/2} \left( \int_{[a,b]\cap G_f} V_f^{-1} \, \mathrm{d}t \right)^{1/2}. \tag{885}$$

A finite inverse-volume integral on the growth set thus bounds $\Phi$ from any positive initial value, and bounds $H$ with it. For $N$ disjoint positive doubling intervals $I_j$, write

$$A_j = \int_{I_j\cap G_f} \int r_+^2 \, \mathrm{d}z \, \mathrm{d}t, \qquad C_j = \int_{I_j\cap G_f} V_f^{-1} \, \mathrm{d}t.$$

Then $A_j C_j \ge (\log 2)^2/16$. If $\mathsf{K}_T > 0$, a second Cauchy–Schwarz inequality and (883) give

$$\sum_{j=1}^{N} C_j \ge \frac{(\log 2)^2 N^2}{16 \sum_{j=1}^{N} A_j} \ge \frac{3\nu (\log 2)^2}{8\mathsf{K}_T^2} N^2. \tag{886}$$

Infinite actions are read in the extended sense. If $\mathsf{K}_T = 0$, the velocity is zero and no positive doubling occurs. Repeated growth requires increasing concentration of the density that samples strain. The evolution of that density must retain the endpoints of each chord.

## E.2 The local density equation and its source

Write $p = p_u + p_f$, with

$$-\Delta p_u = \partial_i u_j \, \partial_j u_i, \qquad \nabla p_f = (I - \mathbb{P}) f.$$

Combining $p_f$ with the prescribed force gives $D_t u = \nu \Delta u - \nabla p_u + g$. Define

$$a_\pm = u(x_\pm) - u(z), \qquad D_z = \partial_t + u(z) \cdot \nabla_z,$$

and use the chord integral

$$\mathcal{K}[F](z) = \frac{1}{\pi^2} \int_0^1 \int_{\mathbb{R}^3} \frac{F(z,\theta,y)}{|y|^4} \, \mathrm{d}y \, \mathrm{d}\theta.$$

A fixed chord translated by the center velocity does not move with either endpoint. Substituting the momentum equation at both endpoints gives

$$\begin{aligned} D_z \delta u_\theta &= \nu \Delta_z \delta u_\theta + \mathcal{R} + \delta g_\theta, \\ \mathcal{R} &= -\delta \nabla p_u - (a_+ \cdot \nabla) u(x_+) + (a_- \cdot \nabla) u(x_-). \end{aligned} \tag{887}$$

Since $\tau = \mathcal{K}[|\delta u_\theta|^2]$, the product rule yields

$$\begin{aligned} (D_z - \nu \Delta_z)\tau &= \mathcal{I}_\tau - 2\nu Q_\tau + \mathcal{F}_\tau, \\ \mathcal{I}_\tau &= 2\mathcal{K}[\delta u_\theta \cdot \mathcal{R}], \\ Q_\tau &= \mathcal{K}[|\delta \nabla u_\theta|^2], \qquad \mathcal{F}_\tau = 2\mathcal{K}[\delta u_\theta \cdot \delta g_\theta]. \end{aligned} \tag{888}$$

Pressure and endpoint transport remain together in $\mathcal{I}_\tau$. The direct force term has an absolute spatial bound. Cauchy–Schwarz over all centers, chords, and center parameters gives

$$\|\mathcal{F}_\tau(t)\|_1 \le 4 \left\| \Lambda^{1/2} u(t) \right\|_2 \left\| \Lambda^{1/2} g(t) \right\|_2 = 4\sqrt{2H(t)} \, \left\| \Lambda^{1/2} g(t) \right\|_2. \tag{889}$$

The time integral can be bounded before a supremum of $H$ is known. For $0 < s \le 1$, Fourier interpolation gives

$$E_s^{1/s} \le 2^{-1/s} \|u\|_2^{2(1-s)/s} \|\nabla u\|_2^2, \qquad E_s = \tfrac{1}{2} \|\Lambda^s u\|_2^2.$$

Using (877),

$$\int_0^T E_s^{1/s} \,\mathrm{d}t \le 2^{-1/s-1} \nu^{-1} \mathsf{K}_T^{2/s}, \qquad \int_0^T H^2 \,\mathrm{d}t \le \frac{\mathsf{K}_T^4}{8\nu}. \tag{890}$$

When $\mathsf{K}_T = 0$, both sides vanish. Applying Hölder to (889) now gives

$$\int_0^T \|\mathcal{F}_\tau(t)\|_1 \,\mathrm{d}t \le 2^{7/4} \mathsf{K}_T \nu^{-1/4} \left\| \Lambda^{1/2} g \right\|_{L_t^{4/3} L_x^2}. \tag{891}$$

The manuscript's smooth source class supplies this norm as well as the $L_t^1 H_x^1$ norm used in (878). The direct chord forcing has finite absolute mass through $T$, without a terminal critical-height bound.

The integrated density equation has the normalization

$$\int Q_\tau \,\mathrm{d}z = 2D, \qquad \int \mathcal{F}_\tau \,\mathrm{d}z = 4W, \qquad \int \mathcal{I}_\tau \,\mathrm{d}z = 4P_H. \tag{892}$$

For each fixed chord, $\delta u_\theta$ is divergence free in $z$, so its global pairing with $\delta \nabla p_u$ vanishes. Integrating (888) gives $4H' = 4P_H - 4\nu D + 4W$. Pressure can still redistribute density locally. To expose that redistribution, its flux must be retained before spatial integration.

### E.3 Intrinsic production and redistribution

Endpoint advection acts on the chord coordinates through

$$c_\theta \cdot \nabla_z + \delta u_\theta \cdot \nabla_y, \qquad c_\theta = \theta u(x_+) + (1-\theta) u(x_-).$$

Both coordinate divergences vanish by incompressibility. The intrinsic integrand in (888) can be written

$$\nabla_z \cdot \big[ (u(z) - c_\theta) |\delta u_\theta|^2 - 2\delta p_\theta \, \delta u_\theta \big] - \nabla_y \cdot \big( \delta u_\theta |\delta u_\theta|^2 \big),$$

where $\delta p_\theta = p_u(x_+) - p_u(x_-)$. The first term already has a flux in the center coordinate:

$$J_0 = \mathcal{K} \big[ (u(z) - c_\theta) |\delta u_\theta|^2 - 2\delta p_\theta \, \delta u_\theta \big]. \tag{893}$$

Integrating the second term against the radial kernel produces $-4|\delta u_\theta|^2 (\delta u_\theta \cdot \widehat{y})/|y|$. The velocity difference is the integral of its gradient along the chord. With $s(z,y) = \widehat{y} \cdot S(z) \widehat{y}$,

$$\frac{\delta u_\theta \cdot \widehat{y}}{|y|} = \int_0^1 s(z + (\alpha - \theta) y, y) \,\mathrm{d}\alpha.$$

Changing the center parameter preserves the endpoints:

$$\delta u_\alpha(z + (\alpha - \theta) y, y) = \delta u_\theta(z, y).$$

Put $F_\alpha(z,y) = |\delta u_\alpha(z,y)|^2 s(z,y)$. Its translation away from $z$ has the exact divergence form

$$F_\alpha(z+h, y) - F_\alpha(z, y) = \nabla_z \cdot \left[ h \int_0^1 F_\alpha(z + \lambda h, y) \,\mathrm{d}\lambda \right].$$

With $h_{\alpha\theta} = (\alpha - \theta)y$, define

$$L(z) = \frac{1}{\pi^2} \int_{\mathbb{R}^3} \frac{\mathrm{d}y}{|y|^4} \int_0^1 \int_0^1 h_{\alpha\theta} \left[ \int_0^1 F_\alpha(z + \lambda h_{\alpha\theta}, y)\, \mathrm{d}\lambda \right] \mathrm{d}\alpha\, \mathrm{d}\theta, \qquad J_{\mathrm{int}} = J_0 - 4L. \tag{894}$$

The untranslated part is $S : \mathsf{T}_u$, giving

$$\mathcal{I}_\tau = -4S : \mathsf{T}_u + \nabla_z \cdot J_{\mathrm{int}} = 4\tau r + \nabla_z \cdot J_{\mathrm{int}}. \tag{895}$$

The complete local equation is

$$\partial_t \tau + \nabla_z \cdot (u\tau - J_{\mathrm{int}}) - \nu \Delta_z \tau = 4\tau r - 2\nu Q_\tau + \mathcal{F}_\tau. \tag{896}$$

The strain at the center controls the displayed production, while $J_{\mathrm{int}}$ retains pressure and the change between chord centers.

These operations can first be performed with $\varepsilon < |y| < R$. The boundary remainder in the chord integration obeys

$$\|B_{\varepsilon,R}\|_1 \le C\big(\varepsilon \|\nabla u\|_3^3 + R^{-2} \|u\|_3^3\big) \longrightarrow 0.$$

No endpoint in $\theta$ or $\alpha$ is differentiated. On strict smooth time windows the paired differences also give two powers of $|y|$ near zero, and the radial kernel is integrable at infinity. These bounds justify the local identities in spatial distributions.

The flux itself is absolutely integrable in space on such a window. Write $U = \|u\|_2$, $A = \|\nabla u\|_2$, and $B = \|u\|_3$. Fractional Sobolev embedding and the Fourier kernel identity yield

$$\int_{\mathbb{R}^3} \frac{\|\delta_y u\|_3^2}{|y|^4}\, \mathrm{d}y \le C \int_{\mathbb{R}^3} \frac{\big\|\Lambda^{1/2} \delta_y u\big\|_2^2}{|y|^4}\, \mathrm{d}y \le CA^2.$$

Since $\|u - c_\theta\|_3 \le 2B$, the transport part of $J_0$ has $L^1$ norm at most $CBA^2$.

For $L$, split the chords at length $\ell > 0$. The estimate $\|\delta_y u\|_4^2 \le C|y|^{1/2} A^2$ gives $C\ell^{1/2} \|S\|_2 A^2$ for short chords. For long chords, average the squared endpoint velocities in the center parameter. For an endpoint fraction $0 < a < 1$, changing variables to $h = ay$ gives the kernel

$$k_\ell(h) = |h|^{-3} \min(1, |h|/\ell), \qquad \|k_\ell\|_{6/5}^{6/5} = \frac{40\pi}{3} \ell^{-3/5}.$$

Hölder and Young convolution inequalities bound this part by $C\ell^{-1/2} \|S\|_2 B^2$. Choosing $\ell = (B/A)^2$ for nonzero $u$ gives $\|L\|_1 \le C \|S\|_2 AB \le CBA^2$. For the zero field there is no division.

The pressure part of $J_0$ satisfies

$$\|J_{0,\mathrm{pressure}}\|_1 \le 4 \left\|\Lambda^{1/2} p_u\right\|_2 \left\|\Lambda^{1/2} u\right\|_2.$$

The pressure multiplier is bounded, and Fourier Cauchy–Schwarz for each product $u_i u_j$ uses

$$\int_{\mathbb{R}^3} \frac{\mathrm{d}\eta}{|\eta|^2 |\xi - \eta|^2} = C|\xi|^{-1}.$$

Multiplying by $|\xi|$, integrating in $\xi$, and applying Tonelli gives $\left\|\Lambda^{1/2} p_u\right\|_2 \le CA^2$. Finally $B \le CU^{1/2} A^{1/2}$ and $\left\|\Lambda^{1/2} u\right\|_2 \le U^{1/2} A^{1/2}$, so

$$\|J_{\mathrm{int}}\|_1 \le CU^{1/2} A^{5/2}, \qquad \int_0^T \|J_{\mathrm{int}}\|_1^{4/5}\, \mathrm{d}t \le C\nu^{-1} \mathsf{K}_T^{12/5}. \tag{897}$$

The time exponent is $4/5$. A terminal $L_t^1 L_z^1$ flux estimate does not follow from this bound.

## E.4 Diffusion of the normalized density

Diffusion gives a local constraint on how sharply the density varies. Cauchy–Schwarz inside the chord integral gives

$$|\nabla\tau|^2 \le 4\tau Q_\tau, \qquad -2\nu Q_\tau \le -\frac{\nu}{2}\frac{|\nabla\tau|^2}{\tau} \quad \text{where } \tau > 0.$$

The intrinsic production in (888) remains alongside this favorable term. Its time integral cannot be discarded when estimating the diffusion of $\rho_f$.

On an interval with $\Phi > 0$, put

$$\psi = \sqrt{\rho_f}, \qquad m = \int \rho_f \,\mathrm{d}z = \frac{H}{\Phi} \le 1, \qquad \mathcal{J} = \int |\nabla\psi|^2 \,\mathrm{d}z.$$

Differentiating the normalization gives

$$(D_z - \nu\Delta_z)\rho_f = \frac{\mathcal{I}_\tau + \mathcal{F}_\tau}{4\Phi} - \frac{\nu Q_\tau}{2\Phi} - \frac{\Phi'}{\Phi}\rho_f. \tag{898}$$

Where $\tau > 0$, its square root satisfies

$$(D_z - \nu\Delta_z)\psi = \frac{\mathcal{I}_\tau + \mathcal{F}_\tau}{8\Phi\psi} - \frac{\nu}{\psi}\left(\frac{Q_\tau}{4\Phi} - |\nabla\psi|^2\right) - \frac{\Phi'}{2\Phi}\psi. \tag{899}$$

The bracket is nonnegative. The derivative of $\Phi$ retains its full sign, and pressure and endpoint transport remain inside $\mathcal{I}_\tau$.

The norm derivative inequality extends across zeros by Sobolev approximation, and (892) gives $\mathcal{J} \le D/(2\Phi)$. Let $S_6$ be a constant in $\|h\|_6 \le S_6 \|\nabla h\|_2$. Interpolation gives

$$V_f^{-1} \le \|\psi\|_4^4 \le \min\left\{ S_6^3 m^{1/2} \mathcal{J}^{3/2}, \frac{S_6^2 \|\tau\|_{3/2}\, \mathcal{J}}{4\Phi} \right\}. \tag{900}$$

The first estimate uses the $L^2$ and $L^6$ norms of $\psi$; the second uses its $L^3$ and $L^6$ norms. For the chord density, $\|\tau\|_{3/2} \le C_\tau \|\nabla u\|_2^2$: Minkowski bounds this norm by $C\int |y|^{-4} \|\delta_y u\|_3^2 \,\mathrm{d}y$, which is controlled by the fractional estimate used for $J_0$. Substituting these bounds gives

$$V_f^{-1} \le \min\left\{ \frac{S_6^3 H^{1/2} D^{3/2}}{2^{3/2}\Phi^2}, \frac{C_\tau S_6^2 \|\nabla u\|_2^2\, D}{8\Phi^2} \right\}. \tag{901}$$

Neither time integral on the right is bounded by (877) and (891). Even a finite integral of $\mathcal{J}$ would give only

$$\int (V_f^{-1})^{2/3} \,\mathrm{d}t \le S_6^2 \int \mathcal{J} \,\mathrm{d}t,$$

which has a smaller time exponent than the action in (885).

An alternative sufficient criterion for this density calculation is

$$\int_{G_f} V_f^{-1}(t) \,\mathrm{d}t < \infty$$

with a bound derived from $u_0, \nu, f$. The source reserve and the absolute chord-source estimate control the direct external contribution. This criterion measures the accumulated sampling of the density by strain under the complete production and redistribution in (896). The rate bound (883), the flux bound (897), and the instantaneous diffusion estimates above do not supply that time-integrated control.

# F A sufficient condition from the squared momentum equation

The temporal and viscous terms can be separated directly in an integrated identity. Write

$$\mathcal{G} = \mathbb{P}f - \mathbb{P}[(u\cdot\nabla)u], \qquad u_t - \nu\Delta u = \mathcal{G}.$$

For a strict classical horizon $S < T$, put

$$X(S) = \int_0^S (\|u_t\|_2^2 + \nu^2 \|\Delta u\|_2^2)\,\mathrm{d}t, \qquad G_0 = \|\nabla u_0\|_2^2 .$$

The cross term in the squared equation is an endpoint energy:

$$X(S) + \nu\,\|\nabla u(S)\|_2^2 = \nu G_0 + \int_0^S \|\mathcal{G}\|_2^2 \,\mathrm{d}t. \tag{902}$$

Indeed, $-2\langle u_t, \Delta u\rangle = \partial_t \|\nabla u\|_2^2$. This identity retains the full projected nonlinear term.

Set

$$U = \|u_0\|_2 + \int_0^T \|f\|_2 \,\mathrm{d}t, \qquad L = TU^2 + U^2/\nu, \qquad \Phi = \int_0^T \|\mathbb{P}f\|_2^2 \,\mathrm{d}t.$$

The kinetic balance gives $\sup_t \|u\|_2 \le U$ and $\nu\int_0^S \|\nabla u\|_2^2 \,\mathrm{d}t \le U^2/2$. Using the Bessel Sobolev norm,

$$\|u\|_{H^2}^2 = \|u\|_2^2 + 2\,\|\nabla u\|_2^2 + \|\Delta u\|_2^2 .$$

Consequently the first rectangle $R_{0,1}(S) = \int_0^S (\|u\|_{H^2}^2 + \|u_t\|_2^2)\,\mathrm{d}t$ satisfies

$$\begin{aligned} R_{0,1}(S) &\le L + \max(1,\nu^{-2})X(S), \\ X(S) &\le \max(1,\nu^2)R_{0,1}(S). \end{aligned} \tag{903}$$

A bound for the full $\mathcal{G}$ integral thus supplies both velocity coordinates without first assuming that rectangle.

The gradient trace identity and Young's inequality also give

$$\sup_{t\le S}\|\nabla u(t)\|_2^2 \le G_0 + X(S)/\nu, \qquad \int_0^S \|u\|_{H^2}^2 \,\mathrm{d}t \le L + X(S)/\nu^2.$$

Fourier inversion gives $\|u\|_\infty^2 \le \|u\|_{H^2}^2/(8\pi)$, proved explicitly in (698). Since $\|(u\cdot\nabla)u\|_2 \le \|u\|_\infty \|\nabla u\|_2$, substitution into (902) yields

$$X(S) \le \nu G_0 + 2\Phi + \frac{(G_0 + X(S)/\nu)(L + X(S)/\nu^2)}{4\pi}. \tag{904}$$

This proves a quantified criterion directly from the data. Define

$$a = \frac{1}{4\pi\nu^3}, \qquad b = \frac{G_0/\nu^2 + L/\nu}{4\pi}, \qquad c = \nu G_0 + 2\Phi + \frac{G_0 L}{4\pi}.$$

If $b < 1$ and $(1-b)^2 > 4ac$, then

$$\sup_{S<T} X(S) \le \frac{1 - b - \sqrt{(1-b)^2 - 4ac}}{2a}. \tag{905}$$

To see this, rewrite (904) as $aX^2 + (b-1)X + c \geq 0$. Its two roots enclose a forbidden interval. The continuous function $X$, starting from zero, cannot cross that interval. For rest datum the criterion becomes

$$\frac{U^2(T+\nu^{-1})}{4\pi\nu} + \sqrt{\frac{2\Phi}{\pi\nu^3}} < 1.$$

The bound is uniform through $T$. Without the stated root separation, (904) does not supply an upper bound.

# G General bounds for production, pressure, and acceleration

The selected rectangles control more than the critical energy and the pointwise momentum terms. Each temporal row has its own nonlinear production and source work. Retaining the full binomial row lets us bound these quantities together, and then use the two pressure projections to resolve how transport contributes to material acceleration.

## G.1 The production and work of all temporal rows

Fix nonnegative integers $N, m$. The pressure work in row $r$ vanishes after pairing with the solenoidal field $V_r$. The product estimate (706) gives

$$|\mathcal{P}_{r,m}| \leq C_m \, \|V_r\|_{H^m} \sum_{a=0}^{r} \binom{r}{a} \|V_a\|_{H^{m+2}} \, \|V_{r-a}\|_{H^{m+2}} \,, \qquad C_m = 2^{m+1} c_\infty. \tag{906}$$

The two factors in every summand are retained: temporal differentiation does not preserve every cancellation of the zeroth row.

Write

$$D_{N,m} := \sum_{r=0}^{N} \|V_r(0)\|_{H^m}^2 \,.$$

Integrating the adjacent trace identity from time zero and using $2ab \leq a^2 + b^2$ gives

$$\max_{r \leq N} \|V_r\|_{L^\infty H^m} \leq \sqrt{D_{N,m} + R_{N,m}}. \tag{907}$$

For $x_a = \|V_a\|_{L^2 H^{m+2}}$, time Cauchy–Schwarz and the binomial symmetry give

$$\sum_{a=0}^{r} \binom{r}{a} x_a x_{r-a} \leq 2^r \sum_{a=0}^{N} x_a^2 \leq 2^r R_{N,m+1}.$$

We can now integrate (906) and sum the full nonlinear contribution through row $N$:

$$\sum_{r=0}^{N} \|\mathcal{P}_{r,m}\|_{L^1} \leq C_m (2^{N+1} - 1) \sqrt{D_{N,m} + R_{N,m}} \, R_{N,m+1}. \tag{908}$$

This bounds the absolute production in every selected row, including each signed prefix integral.

The source work admits a bound with one adjacent spatial derivative placed on each side of the pairing. Since

$$W_{r,m} = \operatorname{Re} \left\langle \Lambda^{m+1} V_r, \Lambda^{m-1} F_r \right\rangle ,$$

define

$$\mathcal{F}^{\mathrm{work}}_{N,m-1} := \left( \sum_{r=0}^{N} \left\| \Lambda^{m-1} F_r \right\|^2_{L^2L^2} \right)^{1/2}.$$

Cauchy–Schwarz first in time and then over the finite rows yields

$$\sum_{r=0}^{N} \|W_{r,m}\|_{L^1} \le \mathcal{F}^{\mathrm{work}}_{N,m-1} \sqrt{R_{N,m}}. \tag{909}$$

At $m = 0$, the source coordinate is $\dot{H}^{-1}$. The assumed spatial decay supplies its low frequencies. For each fixed temporal order $\ell$, differentiation of the pairing gives the further bound

$$\left\| \partial_t^{\ell} W_{r,m} \right\|_{L^1} \le \sum_{a=0}^{\ell} \binom{\ell}{a} \left\| \Lambda^{m+1} V_{r+a} \right\|_{L^2L^2} \left\| \Lambda^{m-1} F_{r+\ell-a} \right\|_{L^2L^2}. \tag{910}$$

Every factor is a finite coordinate of the velocity or the prescribed source. The constants may depend on $N, m, \ell$; the calculation requires no bound uniform over all derivative orders.

## G.2 Projected transport from products and adjacent rows

The pressure projection also gives direct integral estimates for the nonlinear field, before it is paired with velocity. Put

$$X_{a,j} := \|V_a\|_{L^2H^j}.$$

For every nonnegative integer $M$, choose the finite product constant $C^{\mathrm{alg}}_M$ in

$$\|v \otimes w\|_{H^M} \le C^{\mathrm{alg}}_M \|v\|_{H^{M+1}} \|w\|_{H^{M+1}}.$$

For $M = 0$, this follows from $H^1 \subset L^4$ and Hölder; at higher integer orders it follows by the finite Leibniz expansion and the Sobolev product bounds. Tensor divergence maps $H^M$ to $H^{M-1}$ with norm at most one, and both $\mathbb{P}$ and $\mathbb{Q} = I - \mathbb{P}$ are contractions in these Fourier Sobolev norms. Applying them to the full transport row gives

$$\begin{aligned} \|\mathbb{P}B_n\|_{L^1H^{M-1}}, \quad \|\mathbb{Q}B_n\|_{L^1H^{M-1}} &\le C^{\mathrm{alg}}_M \sum_{a=0}^{n} \binom{n}{a} X_{a,M+1} X_{n-a,M+1} \\ &\le C^{\mathrm{alg}}_M 2^n R_{N,M}, \qquad n \le N. \end{aligned} \tag{911}$$

Here and below the two quantities separated by a comma each obey the displayed bound.

Let $Q^u_n$ denote the normalized nonlinear pressure in row $n$, so that $\nabla Q^u_n = -\mathbb{Q}B_n$. Its scalar reconstruction is a double Riesz transform of the binomial tensor sum. The multiplier has norm at most one from the Frobenius tensor norm to the scalar norm. The product estimate consequently gives

$$\begin{aligned} \|Q^u_n\|_{L^1H^M} + \|\nabla Q^u_n\|_{L^1H^{M-1}} &\le 2C^{\mathrm{alg}}_M \sum_{a=0}^{n} \binom{n}{a} X_{a,M+1} X_{n-a,M+1} \\ &\le 2C^{\mathrm{alg}}_M 2^n R_{N,M}. \end{aligned} \tag{912}$$

The total pressure gradient includes $\mathbb{Q}F_n$ as well.

Using the next temporal coordinate gives a stronger time norm for the projected transport. The momentum row itself reads

$$\mathbb{P}B_n - \mathbb{P}F_n = \nu \Delta V_n - V_{n+1}.$$

Its two terms on the right belong to the adjacent rectangle, so

$$\|\mathbb{P}B_n - \mathbb{P}F_n\|_{L^2H^m} \le \nu X_{n,m+2} + X_{n+1,m}.$$

Scalar Cauchy–Schwarz with coefficients $(\nu, 1)$, followed by summation over $n$, gives the explicit complete-row bound

$$\left(\sum_{n=0}^{N} \|\mathbb{P}B_n - \mathbb{P}F_n\|_{L^2H^m}^2\right)^{1/2} \le \sqrt{\nu^2+1}\, R_{N,m+1}^{1/2}. \tag{913}$$

The right side retains the next temporal row. This is the direct bound of the nonlinear field participating in the equation along the selected history.

The spatial recovery in that equation is also explicit. The Fourier inequality $(1+|\xi|^2)^2 \le 2(1+|\xi|^4)$ implies

$$\|V_n\|_{H^{m+2}} \le \sqrt{2}\left[\|V_n\|_{H^m} + \nu^{-1}\|V_{n+1} + \mathbb{P}B_n - \mathbb{P}F_n\|_{H^m}\right]. \tag{914}$$

Both the temporal derivative and the complete projected nonlinear response remain in this recovery of two spatial derivatives.

## G.3 Higher pressure rows and their time norms

The pressure gain in (718) extends to every temporal row. For two solenoidal fields, expanding the divergence gives

$$\operatorname{div}((v\cdot\nabla)w) = \partial_i v_j\, \partial_j w_i.$$

This expression is symmetric under exchanging $v, w$. The gradient projection has the exact norm relation

$$\|\mathbb{Q}((v\cdot\nabla)w)\|_{\dot H^{1/2}} = \|\partial_i v_j\, \partial_j w_i\|_{\dot H^{-1/2}}.$$

With $c_* = (\sqrt{2}\pi)^{-1}$, the Sobolev inequality and its dual bound the right side by $c_*\|v\|_{\dot H^{3/2}}\|w\|_{\dot H^{3/2}}$. Apply this to every binomial summand:

$$\|\nabla Q_n^u\|_{\dot H^{1/2}} \le c_* \sum_{a=0}^{n} \binom{n}{a} \|V_a\|_{\dot H^{3/2}} \|V_{n-a}\|_{\dot H^{3/2}}. \tag{915}$$

The symmetry also gives

$$\|\mathbb{Q}((v\cdot\nabla)w)\|_{\dot H^{-1/2}} \le c_* \min\big\{ \|v\|_{\dot H^{1/2}}\|w\|_{\dot H^{3/2}},\ \|w\|_{\dot H^{1/2}}\|v\|_{\dot H^{3/2}}\big\}. \tag{916}$$

Either assignment of the two heights is available for the pressure component because its divergence retains a derivative on both factors.

For the zeroth row, put

$$h(t) = \|u(t)\|_{\dot H^{1/2}}, \qquad d(t) = \|u(t)\|_{\dot H^{3/2}}, \qquad H_* := \|h\|_{L^\infty}, \quad D_* := \|d\|_{L^2}.$$

The bounds $\|B_0\|_{\dot H^{-1/2}} \le c_* hd$ and $\|\mathbb{Q}B_0\|_{\dot H^{1/2}} \le c_* d^2$ give

$$\|\mathbb{P}B_0\|_{L^2\dot H^{-1/2}}, \quad \|\mathbb{Q}B_0\|_{L^2\dot H^{-1/2}} \le c_* H_* D_*, \quad \|\mathbb{Q}B_0\|_{L^1\dot H^{1/2}} \le c_* D_*^2. \tag{917}$$

Interpolating the two spatial norms at each time gives

$$\|\mathbb{Q}B_0(t)\|_2 \le c_* h(t)^{1/2} d(t)^{3/2}.$$

Raising this to the power $4/3$ and integrating yields

$$\|\mathbb{Q}B_0\|_{L^{4/3}L^2} \le c_* H_*^{1/2} D_*^{3/2}. \tag{918}$$

The critical rectangle supplies $H_* \le \sqrt{D+R}$ and $D_* \le \sqrt{R}$, so the last right side is $c_*(D+R)^{1/4}R^{3/4}$. A Sobolev and Hölder estimate also gives the $L^{4/3}L^2$ conclusion for full transport. The additional pressure regularity is its $L^1\dot{H}^{1/2}$ endpoint; total forced pressure still includes the prescribed gradient source.

## G.4 Material acceleration from the first rectangle

The orthogonal acceleration identity can be evaluated in $L^2_x$ using $R_{0,1}$. Set

$$\mathsf{A} := \|u\|_{L^2H^2}, \qquad \mathsf{B} := \|u_t\|_{L^2L^2}, \qquad \mathsf{Z} := \Big(\|u_0\|_{H^1}^2 + 2\mathsf{AB}\Big)^{1/2}.$$

The adjacent trace gives $\|u\|_{L^\infty H^1} \le \mathsf{Z}$. Use the homogeneous Sobolev constants

$$S_6 = \frac{2^{2/3}}{\sqrt{3}\pi^{2/3}}, \qquad S_3 = 2^{-1/6}\pi^{-1/3}.$$

Hölder in space and then in time gives

$$\begin{aligned} \|B_0\|_{L^2L^2} &\le \|u\|_{L^\infty L^6}\,\|\nabla u\|_{L^2L^3} \\ &\le S_6 S_3 \mathsf{ZA} = \frac{1}{\pi}\sqrt{\frac{2}{3}}\,\mathsf{ZA}. \end{aligned} \tag{919}$$

In unforced flow, viscosity supplies the solenoidal part of acceleration and the pressure response supplies its gradient part. Their squared norms add by (717). We obtain

$$\|D_t u\|_{L^2L^2} \le \mathsf{A}\sqrt{\nu^2 + \frac{2\mathsf{Z}^2}{3\pi^2}}, \qquad f = 0. \tag{920}$$

For the forced equation, retain the prescribed solenoidal source in (717). The resulting bound at this first rectangle is

$$\|D_t u\|_{L^2L^2}^2 \le (\nu\mathsf{A} + \|\mathbb{P}f\|_{L^2L^2})^2 + \frac{2}{3\pi^2}\mathsf{Z}^2\mathsf{A}^2. \tag{921}$$

The triangle inequality bounds the solenoidal component, while the preceding transport estimate bounds the orthogonal gradient component. The adjacent row also gives $\|\mathbb{P}B_0\|_{L^2L^2} \le \nu\mathsf{A} + \mathsf{B}$ when $f = 0$. These bounds use the first rectangle, while the earlier pointwise acceleration bound selects the stronger $R_{1,2}$.

At critical negative-half order, the orthogonal identity gives a particularly precise unforced coefficient:

$$\nu d(t) \le \|D_t u(t)\|_{\dot{H}^{-1/2}} \le \sqrt{\nu^2 + \frac{h(t)^2}{2\pi^2}}\, d(t), \qquad f = 0. \tag{922}$$

The lower bound retains the solenoidal viscous component, and the upper bound uses $\|\mathbb{Q}B_0\|_{\dot{H}^{-1/2}} \le c_* h d$. Integration gives

$$\|D_t u\|_{L^2\dot{H}^{-1/2}} \le \sqrt{\nu^2 + \frac{H_*^2}{2\pi^2}}\, D_*, \qquad f = 0.$$

For general forcing, the solenoidal component is $\nu\Delta u + \mathbb{P}f$; retaining it gives

$$\|D_t u\|^2_{L^2\dot{H}^{-1/2}} \le \left(\nu D_* + \|\mathbb{P}f\|_{L^2\dot{H}^{-1/2}}\right)^2 + c_*^2 H_*^2 D_*^2. \tag{923}$$

Its substitution from $R_{0,1}$ is (719).

## G.5 Acceleration at a higher adjacent height

To place the full acceleration in $L^2H^{1/2}$, select the stronger adjacent coordinates

$$u \in L^2H^{5/2}, \qquad u_t \in L^2H^{1/2}. \tag{924}$$

Higher integer rectangles supply both by Sobolev inclusion. The inhomogeneous pairing $2\left\langle (1-\Delta)^{5/4}u, (1-\Delta)^{1/4}u_t\right\rangle$ gives the continuous $H^{3/2}$ trace and

$$\sup_{0\le t\le T} \|u(t)\|^2_{H^{3/2}} \le \|u_0\|^2_{H^{3/2}} + 2\,\|u\|_{L^2H^{5/2}}\,\|u_t\|_{L^2H^{1/2}}\,. \tag{925}$$

For the pressure component, the negative-half estimate controls frequencies below one and the positive-half estimate controls those above one. Together they give a continuous quadratic map from $H^{3/2}$ into $H^{1/2}$. Thus $\mathbb{Q}B_0 \in C([0,T];H^{1/2})$, with

$$\|\mathbb{Q}B_0\|_{L^2H^{1/2}} \le C\,\|u\|_{L^\infty H^{3/2}}\,\|u\|_{L^2H^{3/2}}\,. \tag{926}$$

If $\mathbb{P}f \in L^2H^{1/2}$, all components of $a = \nu\Delta u + \mathbb{P}f + \mathbb{Q}B_0$ now belong to $L^2H^{1/2}$. The projected momentum row separately yields

$$\|\mathbb{P}B_0\|_{L^2H^{1/2}} \le \nu\,\|u\|_{L^2H^{5/2}} + \|u_t\|_{L^2H^{1/2}} + \|\mathbb{P}f\|_{L^2H^{1/2}}\,. \tag{927}$$

The $L^2H^{5/2}$ bound controls the viscous term in $L^2H^{1/2}$, while the adjacent temporal bound yields the continuous $H^{3/2}$ trace.

## G.6 Unforced action and spatial recovery

For $f = 0$, squaring the projected equation in $\dot{H}^{-1/2}$ retains an exact relation between the temporal and viscous actions. Since

$$\mathbb{P}B_0 = \nu\Delta u - u_t, \qquad \langle \Delta u, u_t\rangle_{\dot{H}^{-1/2}} = -\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}h^2,$$

we have

$$\nu\frac{\mathrm{d}}{\mathrm{d}t}h^2 + \nu^2 d^2 + \|u_t\|^2_{\dot{H}^{-1/2}} = \|\mathbb{P}B_0\|^2_{\dot{H}^{-1/2}}\,, \qquad f = 0. \tag{928}$$

Add $\|\mathbb{Q}B_0\|^2_{\dot{H}^{-1/2}}$ to both sides. Orthogonality of the two transport projections and the acceleration identity give

$$\nu\frac{\mathrm{d}}{\mathrm{d}t}h^2 + \|u_t\|^2_{\dot{H}^{-1/2}} + \|D_t u\|^2_{\dot{H}^{-1/2}} = \|B_0\|^2_{\dot{H}^{-1/2}}\,, \qquad f = 0. \tag{929}$$

These identities retain the complete nonlinear action on the right. Adding the pressure square also adds its contribution there; the operation supplies no additional sign for critical production.

The solenoidal acceleration determines the viscous spatial derivative when $\nu > 0$. In unforced flow, $\mathbb{P}a = \nu\Delta u$, so the Fourier estimate used in (914) gives

$$\|u\|_{H^{m+2}} \le \sqrt{2}\left(\|u\|_{H^m} + \nu^{-1}\,\|a\|_{H^m}\right), \qquad f = 0. \tag{930}$$

This last relation recovers the spatial coordinate from the full acceleration at that height, with the lower velocity norm retained.

# H Viscous dissipation and material deformation

The deformation of a material parcel is controlled in an integral sense by viscous dissipation. Let $X(a,t)$ follow the parcel initially labeled $a$, let $F = \nabla_a X$, and let $C = F^T F$. The positive matrix $C$ records the lengths and angles of infinitesimal material vectors. Incompressibility preserves volume, while the symmetric velocity gradient $S = (\nabla u + \nabla u^T)/2$ changes this metric:

$$\dot{F} = (\nabla u)(X,t)F, \qquad \det F = 1, \qquad \dot{C} = 2F^T S(X,t)F. \tag{931}$$

The full metric speed has the exact squared norm

$$\operatorname{tr}(C^{-1}\dot{C}\,C^{-1}\dot{C}) = 4|S(X,t)|^2. \tag{932}$$

Indeed, $C^{-1}\dot{C} = 2F^{-1}SF$, and cyclicity of the trace removes the two deformation matrices. After changing variables by the incompressible flow and integrating by parts in space,

$$\frac{\nu}{2}\int_0^t \int_{\mathbb{R}^3} \operatorname{tr}(C^{-1}\dot{C}\,C^{-1}\dot{C})\,da\,ds = \nu \int_0^t \|\nabla u(s)\|_2^2\,ds. \tag{933}$$

Here $\|S\|_2^2 = \frac{1}{2}\|\nabla u\|_2^2$. The dissipation integral consequently controls the rate of change of the material metric, including shape changes that preserve volume.

The energy budget makes this quantitative through the full time interval. Define

$$F_T = \int_0^T \|f\|_2\,dt, \quad M_T = \|u_0\|_2 + F_T, \quad K_T = \|u_0\|_2^2 + 2M_T F_T.$$

The energy identity gives $\sup_t \|u\|_2 \le M_T$ and $2\nu\int_0^T \|\nabla u\|_2^2 \le K_T$. If $\sigma_i$ are the singular values of $F$, then the derivatives of $\log\sigma_i$ are diagonal entries of $S$ in a suitable orthonormal frame. Their squared sum is bounded by $|S|^2$. Integrating from $F(a,0) = I$ gives

$$\|\log C(a,t)\|_F \le 2\int_0^t |S(X(a,s),s)|\,ds.$$

Repeated singular values are handled by diagonalizing the symmetric variation within each eigenspace. Cauchy–Schwarz and volume preservation now yield the maximal estimate

$$\int_{\mathbb{R}^3} \sup_{t<T} \|\log C(a,t)\|_F^2\,da \le \frac{TK_T}{\nu}. \tag{934}$$

This is a bound over original material labels with the time supremum inside the integral. It rules out infinite distortion on a set of labels of positive measure. It permits concentration onto label sets whose measure tends to zero. A uniform bound over all labels would require further control beyond (934).

# I Angular momentum and the role of symmetry

The distinction between the leading vortex and the completed velocity matters to the comparison. The background in OpenAI's construction is axisymmetric. Its wave potentials contain factors $e^{ikm\Phi}$, with $\Phi = p\theta + \varphi$ and nonzero integer angular frequency $kmp$ [18, (6.27), (7.3), (7.38)]. Taking their curls retains these harmonics. They enter the realized and localized velocity through [18,

(9.21), (10.4)]; the final spatial cutoff does not average them over the angle. The complete field is nonaxisymmetric.

One can see exactly why that distinction affects a force bound. Define angular momentum per unit mass about the axis by $\Gamma = ru_\theta$. The angular component of (669), including the derivatives of the cylindrical basis vectors, gives for $r > 0$

$$\begin{aligned}\left(\partial_t + u_r\partial_r + \frac{u_\theta}{r}\partial_\theta + u_z\partial_z\right)\Gamma = \nu\left(\partial_{rr} - \frac{1}{r}\partial_r + \frac{1}{r^2}\partial_{\theta\theta} + \partial_{zz}\right)\Gamma \\ - \partial_\theta p + \frac{2\nu}{r}\partial_\theta u_r + r f_\theta.\end{aligned} \tag{935}$$

Indeed, multiplying the angular momentum equation for $u_\theta$ by $r$ combines its geometric term $u_r u_\theta$ with the material derivative of $r$. The vector Laplacian contributes $2\nu r^{-1}\partial_\theta u_r$. Pressure supplies the torque $-\partial_\theta p$. These terms describe internal momentum transfer; they are present even when the prescribed force is small.

**Proposition I.1** (A force bound for axisymmetric angular momentum)**.** *Suppose $u, p, f$ are axisymmetric, solve* (669) *from rest, and $u$ is bounded on every $\mathbb{R}^3 \times [0, S]$, $S < T$. If $G(t) := \int_0^t \|r f_\theta(s)\|_\infty \, \mathrm{d}s$ is finite through $T$, then*

$$\|ru_\theta(t)\|_\infty \le G(t), \qquad 0 \le t < T. \tag{936}$$

*There is no source-amplitude restriction in this bound.*

*Proof.* Under these symmetry assumptions the angular derivatives in (935) vanish. The resulting scalar operator is

$$L = \partial_t + u_r\partial_r + u_z\partial_z - \nu(\partial_{rr} - r^{-1}\partial_r + \partial_{zz}).$$

Fix $S < T$ and let $M_S = \sup_{[0,S]\times\mathbb{R}^3} |u|$. For $\Psi = 1 + r^2 + z^2$ and $c = M_S + 2\nu + 1$, direct differentiation gives

$$L(e^{ct}\Psi) = e^{ct}\big(c\Psi + 2ru_r + 2zu_z - 2\nu\big) > 0.$$

Here $2M_S\sqrt{r^2+z^2} \le M_S\Psi$. Compare $\pm\Gamma$ with $G(t) + \epsilon e^{ct}\Psi$. The comparison functions dominate initially. At the axis, $\Gamma = 0$ by Cartesian smoothness; at spatial infinity their quadratic growth dominates $|\Gamma| \le M_S r$. Apply the scalar maximum principle on bounded cylinders away from the axis, using the integral of the maximum of $|rf_\theta|$ on each cylinder as the temporal barrier. That integral is at most $G$. For sufficiently large outer radius the quadratic barrier dominates the outer boundary values; at a sufficiently small excluded axial radius, $|\Gamma| \le M_S r$ is dominated by the positive barrier. Exhausting the domain gives $\pm\Gamma \le G + \epsilon e^{ct}\Psi$ on $[0, S]$. Let $\epsilon \downarrow 0$ and then use the arbitrariness of $S < T$. □

Smooth rapidly decreasing forcing satisfies the hypothesis on $G$. This estimate would exclude an axisymmetric realization of the exterior growth calculated in Section 25, since there $ru_\theta$ grows like $(1-t)^{-h}$. It does not assert global regularity for every axisymmetric velocity: a bound on $ru_\theta$ is not a uniform bound on $u_\theta$ near the axis.

Nor does angular averaging turn the completed nonaxisymmetric field into a solution of the axisymmetric equation. Write a bar for angular average and a prime for the fluctuation about that average. Averaging the conservative form of (935) gives exactly

$$\begin{aligned}\partial_t\bar\Gamma + \bar u_r\partial_r\bar\Gamma + \bar u_z\partial_z\bar\Gamma = \nu(\partial_{rr} - r^{-1}\partial_r + \partial_{zz})\bar\Gamma + r\bar f_\theta \\ - \frac{1}{r}\partial_r\left(r\overline{u_r'\Gamma'}\right) - \partial_z\overline{u_z'\Gamma'}.\end{aligned} \tag{937}$$

The pressure torque and angular viscous derivative average to zero, but the products in the transport flux do not. Two zero-mean harmonics can have a nonzero averaged product. OpenAI's covariance construction uses precisely these products to supply momentum flux [18, (7.23), (9.11)]. A force-only estimate for $\bar{\Gamma}$ would have to control the two fluctuation fluxes in (937). Dropping them would change the equation being compared.

# J Endpoint jets and material derivatives

The intersection also gives $V_n \in H^1(0,T;H^m)$ at every fixed $n,m$. Its continuous representative has a limit at $T$. Product continuity in the Sobolev spaces used above gives limits for $B_n$; (677) gives limits for $\nabla Q_n$. Viscosity and spatial differentiation are continuous between the stated spaces. Each momentum term thus extends to the endpoint in every finite spatial norm, and (698) gives uniform convergence of each fixed spatial derivative.

These limits have quantitative rates. Since $\partial_t B_n = B_{n+1}$, integrating the next row yields, for $|\alpha| = k$,

$$\|\partial_x^\alpha [V_{n+1}(t) - V_{n+1}(T)]\|_\infty \le c_\infty (T-t) A_{n+2,k+2}, \tag{938}$$

$$\|\partial_x^\alpha [B_n(t) - B_n(T)]\|_\infty \le c_\infty (T-t) S_{n+1,k+2}, \tag{939}$$

$$\|\partial_x^\alpha [\nabla Q_n(t) - \nabla Q_n(T)]\|_\infty \le c_\infty (T-t)(L_{n+1,k+2} + S_{n+1,k+2}), \tag{940}$$

$$\|\nu \partial_x^\alpha \Delta [V_n(t) - V_n(T)]\|_\infty \le \nu c_\infty (T-t) A_{n+1,k+4}. \tag{941}$$

To justify the integration, first apply the fundamental theorem on $[t,S]$, then let $S \uparrow T$ in the continuous Sobolev representatives. The pressure rate follows by applying $I - \mathbb{P}$ to $F_{n+1} - B_{n+1}$. The rectangle $R_{n+2,k+4}$, together with the displayed source norm, contains all coordinates used by these four rates.

The terminal fields also obey the differentiated equation:

$$V_{n+1}(T) + B_n(T) + \nabla Q_n(T) - \nu \Delta V_n(T) = F_n(T).$$

All its spatial derivatives agree with the limits of the corresponding spatially differentiated rows. The endpoint is a compatible velocity and pressure-gradient jet, rather than a collection of unrelated terminal fields.

## J.1 Material acceleration and its higher derivatives

The fluid-particle acceleration uses both the first temporal row and transport:

$$D_t u = V_1 + B_0 = -\nabla p + \nu \Delta u + f. \tag{942}$$

Its uniform bound follows from (733). The next parcel derivative differentiates both contributions:

$$\begin{aligned} D_t^2 u = {} & V_2 + 2(V_0 \cdot \nabla) V_1 + (V_1 \cdot \nabla) V_0 \\ & + (V_0 \cdot \nabla)\big[(V_0 \cdot \nabla) V_0\big]. \end{aligned} \tag{943}$$

The factor two collects the temporal derivative of transport and the transport of $V_1$. Jerk is already a composite of the mixed jet.

Write $D = D_t$, $a = Du$, and $U_{ij} = \partial_j u_i$. Ordinary derivatives commute, so $\nabla \cdot V_n = \partial_t^n (\nabla \cdot u) = 0$. Moving with the fluid introduces the velocity coefficient:

$$[D, \partial_i] w = -(\partial_i u_j) \partial_j w.$$

For a smooth scalar or vector field $w$, expanding the commutators gives

$$\begin{aligned}[D,\nabla]p &= -U^{\mathsf{T}}\nabla p,\\ [D,\Delta]w &= -(\Delta u)\cdot\nabla w - 2\sum_i (\partial_i u)\cdot\nabla\partial_i w.\end{aligned} \tag{944}$$

The jerk of the parcel is consequently

$$\begin{aligned}Da = \nu\Delta a - \nabla(Dp) + U^{\mathsf{T}}\nabla p + Df\\ -\nu\left[(\Delta u)\cdot\nabla u + 2\sum_i (\partial_i u)\cdot\nabla\partial_i u\right].\end{aligned} \tag{945}$$

Every commutator coefficient is a spatial derivative of this velocity. The pressure and viscous terms remain attached to the mixed jet. In particular,

$$\begin{aligned}D(\nabla\cdot u) = \nabla\cdot a - \operatorname{tr}(U^2) = 0,\\ \nabla\cdot a = \operatorname{tr}(U^2) = -\Delta p + \nabla\cdot f.\end{aligned} \tag{946}$$

Acceleration's divergence is determined by the velocity gradient and the pressure equation. Its sign need not be positive: $\operatorname{tr}(U^2)$ differs from $|U|_{\mathrm{F}}^2$. The pressure pairing in a material row cannot be deleted using the divergence-free property of $V_1$.

Here spatial divergence and unbounded magnitude are distinct questions: $\nabla\cdot a \neq 0$ does not assert that $|a|$ is infinite. Clay's incompressibility condition remains $\nabla\cdot u = 0$. Equation (946) follows by differentiating that condition and is already encoded in its pressure Poisson equation. Retaining this identity introduces no additional hypothesis on a smooth solution of the original system.

A material parcel still preserves its volume. Let $X(\ell,t)$ be the flow from the particle label $\ell$, and let $j(\ell,t) = \det\partial_\ell X(\ell,t)$, with $j(\ell,0) = 1$. Differentiating the deformation matrix gives $\partial_t(\partial_\ell X) = U(X,t)\partial_\ell X$, so

$$\partial_t j = (\nabla\cdot u)(X,t)j = 0, \qquad j(\ell,t) = 1.$$

For a bounded initial parcel $E$, its volume is $|X(E,t)| = \int_E j(\ell,t)\,\mathrm{d}\ell = |E|$. Every time derivative of that volume vanishes. At second order the Jacobian formula displays the cancellation explicitly:

$$\partial_t^2 j = \left[\nabla\cdot a - \operatorname{tr}(U^2) + (\nabla\cdot u)^2\right](X,t)j = 0. \tag{947}$$

The nonzero divergence of acceleration is precisely accompanied by the changing parcel geometry. It does not create an additional degree of freedom or violate incompressibility. With the rectangle and constant $A$ used in (733),

$$\|\nabla\cdot a\|_{L^\infty_{t,x}} \le \|\nabla u\|^2_{L^\infty_{t,x}} \le \frac{A^2}{8\pi}.$$

The constraint continues through every material order. Put $w_q = D^q u$ and $\delta_q = \nabla\cdot w_q$. Expanding the divergence of $Dw_q$ gives the exact recurrence

$$\delta_0 = 0, \qquad \delta_{q+1} = D\delta_q + \operatorname{tr}(U\nabla w_q). \tag{948}$$

For example, $DU = \nabla a - U^2$, so the jerk satisfies

$$\nabla\cdot(Da) = 3\operatorname{tr}(U\nabla a) - 2\operatorname{tr}(U^3).$$

One further differentiation gives, with $G = U$, $B_1 = \nabla a$, and $B_2 = \nabla(Da)$,

$$\nabla \cdot (D^3 u) = 4\,\mathrm{tr}(GB_2) + 3\,\mathrm{tr}(B_1^2) - 12\,\mathrm{tr}(G^2 B_1) + 6\,\mathrm{tr}(G^4).$$

This follows from $DB_k = B_{k+1} - B_k G$ and cyclicity of the trace.

The higher spatial divergences are fixed by the lower material and spatial derivatives. Along the flow map, these relations express the compatibility of acceleration, jerk, and every subsequent derivative with $j \equiv 1$.

For the full recurrence, put $\mathcal{L}_0 = \Delta$, $\mathcal{G}_0 = \nabla$, and define $\mathcal{L}_{q+1} = [D, \mathcal{L}_q]$, $\mathcal{G}_{q+1} = [D, \mathcal{G}_q]$. Then

$$\begin{aligned} D^{n+1}u = \nu \sum_{q=0}^{n} \binom{n}{q} \mathcal{L}_q D^{n-q} u \\ - \sum_{q=0}^{n} \binom{n}{q} \mathcal{G}_q D^{n-q} p + D^n f. \end{aligned} \tag{949}$$

Induction uses $D(Lw) = L(Dw) + [D, L]w$ and Pascal's identity. Every operator coefficient is generated by the velocity in (669). The first two material source derivatives are

$$\begin{aligned} Df &= f_t + u \cdot \nabla f, \\ D^2 f &= f_{tt} + 2u \cdot \nabla f_t + a \cdot \nabla f + \sum_{i,j} u_i u_j \partial_{ij} f. \end{aligned}$$

These derivatives measure the force encountered by the moving parcel. Their velocity factors are part of the coupled response, while the Eulerian derivatives of the prescribed field remain source data. Further material derivatives differentiate the velocity coefficients inside $D_t$. Put $J_{r,\alpha} = \partial_t^r \partial_x^\alpha u$. The exact reconstruction rule is

$$D_t J_{r,\alpha} = J_{r+1,\alpha} + \sum_{\ell=1}^{3} u_\ell J_{r,\alpha+e_\ell}. \tag{950}$$

Start with $D_t^0 u = J_{0,0}$, apply this rule to each factor, and use Leibniz's rule for every product. After $q$ steps, every component of $D_t^q u$ is a polynomial in mixed jet coordinates. Each monomial contains at most $q+1$ velocity factors, and the sum of their derivative orders $r + |\alpha|$ is exactly $q$. This proves the reconstruction at every finite order without introducing independent material rows.

The reverse construction is equally explicit. Define the material–spatial coordinates $M_{q,\alpha} = \partial_x^\alpha D_t^q u$. Ordinary spatial differentiation increments $\alpha$. Expanding $\partial_x^\alpha (D_t w)$ for $w = D_t^q u$ gives

$$\partial_t M_{q,\alpha} = M_{q+1,\alpha} - \sum_{\beta \le \alpha} \binom{\alpha}{\beta} (M_{0,\beta} \cdot \nabla) M_{q,\alpha-\beta}. \tag{951}$$

Leibniz's rule now generates every ordinary mixed derivative from the material–spatial coordinates. These conversion rules relate the Eulerian and material derivative algebras. Their different operator orders are accounted for by the displayed commutator terms.

**Proposition J.1** (Equality of the complete derivative intersections)**.** *Let $0 < T < \infty$, and let $u$ be a smooth vector field on $[0,T) \times \mathbb{R}^3$. Define $D_t = \partial_t + u \cdot \nabla$ from this field. Then*

$$\begin{aligned} &\partial_t^r u \in L^2(0,T;H^m) \quad \text{for every } r, m \ge 0 \\ &\qquad\Longleftrightarrow \\ &D_t^q u \in L^2(0,T;H^m) \quad \text{for every } q, m \ge 0. \end{aligned} \tag{952}$$

*Here $r, q, m$ are integers. No commutation of $D_t$ with spatial derivatives is assumed.*

*Proof.* The Eulerian intersection gives $u \in H^1(0,T;H^m)$ at every spatial height, and gives this property for each temporal derivative as well. Hilbert-valued time embedding supplies their uniform $H^m$ bounds. The polynomial reconstruction (950), spatial Sobolev multiplication, and $T < \infty$ give every material coordinate in $L^2_t H^m_x$. For the reverse implication put $w_q = D_t^q u$ and $b_{q,m} = \|w_q\|_{L^2_t H^m_x}$. The exact identity

$$\partial_t w_q = w_{q+1} - (u \cdot \nabla) w_q$$

first gives a time-integrable derivative. For integer $m \geq 0$, Leibniz's rule and $H^2 \hookrightarrow L^\infty$ give a finite constant $C_m$ such that

$$\|v \cdot \nabla w\|_{H^m} \leq C_m \|v\|_{H^{m+2}} \|w\|_{H^{m+2}}.$$

Indeed, in each derivative of the product, the differentiated $v$ factor is bounded in $L^\infty$ using two additional derivatives, and the differentiated $\nabla w$ factor is bounded in $L^2$. Cauchy–Schwarz in time now yields

$$\|\partial_t w_q\|_{L^1_t H^m_x} \leq \sqrt{T}\, b_{q+1,m} + C_m b_{0,m+2} b_{q,m+2}.$$

Thus $w_q \in W^{1,1}(0,T;H^m)$. Select a time where its $H^m$ norm is at most $T^{-1/2} b_{q,m}$, or take a limiting sequence of such times, and integrate its derivative. This gives the finite-coordinate bound

$$\begin{aligned} &\sup_{t<T} \|w_q(t)\|_{H^m} \leq L_{q,m}, \\ &\qquad L_{q,m} := T^{-1/2} b_{q,m} + \sqrt{T}\, b_{q+1,m} \\ &\qquad\qquad + C_m b_{0,m+2} b_{q,m+2}. \end{aligned} \tag{953}$$

Every material row now has uniform spatial bounds at every finite height. The product reconstruction (951) expresses each Eulerian mixed derivative using finitely many of those rows and spatial derivatives. Sobolev multiplication bounds each product in $L^\infty_t H^m_x$; integration on the finite interval gives its $L^2_t H^m_x$ norm. This proves the reverse implication. □

The proposition identifies the complete intersection independently of construction order. Its reverse trace estimate also shows how a finite collection of material coordinates supplies an Eulerian bound. For an explicit bound in the other direction, set

$$K_q = \max_{r+|\alpha| \leq q} \|\partial_t^r \partial_x^\alpha u\|_{L^\infty_{t,x}} < \infty.$$

At the next material differentiation, a product of at most $q+1$ factors produces at most $4(q+1)$ terms: one time derivative and three convective derivatives per factor. Starting with one term at $q = 0$ gives the sufficient componentwise coefficient bound $4^q q!$. Hence

$$\|D_t^q u\|_{L^\infty_{t,x}} \leq \sqrt{3}\, 4^q q! \max(1, K_q)^{q+1} < \infty. \tag{954}$$

The bound is finite for every fixed order $q$. More generally, applying $D_t^q$ to a vector field $w$ produces products with one mixed derivative of $w$ and at most $q$ velocity factors. The identical counting argument bounds it by $\sqrt{3}\, 4^q q! \max(1, K_q)^q K_q^w$, where $K_q^w$ is the maximum of its mixed derivative norms through order $q$. For the prescribed source, set $K_q^f = \max_{r+|\alpha| \leq q} \|\partial_t^r \partial_x^\alpha f\|_{L^\infty_{t,x}}$. This gives

$$\|D_t^q f\|_{L^\infty_{t,x}} \leq \sqrt{3}\, 4^q q! \max(1, K_q)^q K_q^f.$$

These formulas preserve the dependence of a material force derivative on the fluid moving through its prescribed spatial field.

The bounded velocity and its bounded spatial gradient give a unique fluid trajectory through each initial point on the closed interval. Along that trajectory, ordinary time derivatives of $u(X(t),t)$ are the material derivatives just bounded. This connects the Eulerian rows to acceleration and its higher material derivatives without identifying $\partial_t^n u$ with $D_t^n u$.

### J.2 Explicit rectangles for finite material orders

The finite footprint of the material jet gives an explicit answer when the desired derivative orders are specified. Fix integers $N, K, m \ge 0$ and put

$$J_* = N + K, \qquad q = \max(m, 2), \qquad Q = q + K, \qquad R = R_{J_*,Q}.$$

Define

$$d_0 = \max_{0\le j\le J_*} \|V_j(0)\|_{H^Q}, \qquad X = \sqrt{R}, \qquad Z = \sqrt{d_0^2 + R}.$$

The initial recurrence (679) includes the initial source derivatives and makes $d_0$ finite. Each temporal row in this selection has a continuous $H^Q$ representative, with

$$\begin{aligned} \sup_{0\le t\le T} \|V_j(t)\|_{H^Q}^2 &\le \|V_j(0)\|_{H^Q}^2 + 2\|V_j\|_{L^2H^{Q+1}} \|V_{j+1}\|_{L^2H^{Q-1}} \\ &\le d_0^2 + R = Z^2. \end{aligned} \tag{955}$$

This follows by pairing the two adjacent Sobolev coordinates when differentiating the squared $H^Q$ norm. The Hilbert triple $H^{Q+1} \subset H^Q \subset H^{Q-1}$ also supplies continuity.

Every scalar component of $D_t^k u$ is a sum of monomials

$$c \prod_{\ell=1}^{r} \partial_x^{\beta_\ell} V_{j_\ell}, \quad 1 \le r \le k+1, \quad \sum_\ell j_\ell = k + 1 - r, \quad \sum_\ell |\beta_\ell| = r - 1.$$

A temporal derivative increments one temporal index. Transport adds one velocity factor and one spatial derivative. This proves the displayed footprint by induction. After $n$ ordinary temporal derivatives, its temporal total is $n + k + 1 - r$. The unique linear term is $V_{n+k}$; every nonlinear term has strictly lower temporal indices.

For integer $q \ge 2$, a scalar Sobolev algebra constant is

$$K_q = 2^q (2\pi)^{-3/2} \left( \pi^{3/2} \frac{\Gamma(q - 3/2)}{\Gamma(q)} \right)^{1/2}.$$

Indeed, the weight inequality $\langle \xi \rangle^q \le 2^{q-1}(\langle \eta \rangle^q + \langle \xi - \eta \rangle^q)$, Fourier convolution, and Cauchy–Schwarz with $\int \langle \xi \rangle^{-2q} d\xi = \pi^{3/2}\Gamma(q - 3/2)/\Gamma(q)$ give $\|vw\|_{H^q} \le K_q \|v\|_{H^q} \|w\|_{H^q}$.

For each component, the total absolute monomial coefficient in $D_t^k u$ is at most $4^k k!$. A monomial with $r$ factors produces at most $r$ temporal terms and $3r$ transport terms at the next material derivative. Applying $n$ further temporal derivatives costs at most $(k+1)^n$. Set

$$L_{n,k,m} = \sqrt{3}\, 4^k k! (k+1)^n \max(1, K_q)^k.$$

The factor $\sqrt{3}$ combines the three scalar components. Since each spatial factor has order at most $K$, its $H^q$ norm is bounded by its parent's $H^Q$ norm. Place one factor in $L_t^2 H_x^q$ and all others in $L_t^\infty H_x^q$. For every $n \le N$, $k \le K$, this proves

$$\begin{aligned} \|\partial_t^n D_t^k u\|_{L^2(0,T;H^m)} &\le L_{n,k,m} X \max(1, Z)^k, \\ \sup_{0\le t\le T} \|\partial_t^n D_t^k u(t)\|_{H^m} &\le L_{n,k,m} Z \max(1, Z)^k. \end{aligned} \tag{956}$$

Continuity of the factors and of multiplication in $H^q$ gives a continuous $H^m$ terminal trace. Replacing $m$ by $m+b$ also controls spatial derivatives through order $b$. Every desired finite material order thus has a specified finite rectangle within the full intersection. The constants may depend on those requested orders.

# K Conservative forcing and pressure adjustment

A conservative force can be absorbed entirely into pressure. Part II supplies the global unforced history, so this reduction gives forced continuation at arbitrary amplitude for conservative sources. Part III treats arbitrary prescribed smooth forcing directly on its own forced history.

**Proposition K.1** (Conservative forcing)**.** *Suppose $\mathbb{P}f = 0$. Any smooth unforced history with datum $u_0$ gives a forced history with identical velocity and full mixed pressure-gradient regularity on $[0,T]$.*

*Proof.* Define the potential by

$$\widehat{\phi}(\xi,t) = -\frac{i\xi\cdot\widehat{f}(\xi,t)}{|\xi|^2}.$$

Its gradient is $f$. The inverse derivative at zero frequency is legitimate in three dimensions for the stated source class. In fact, for every integer $m \geq 0$,

$$\|\phi\|_{H^m}^2 \leq \frac{2^{m-1}}{\pi^2}\,\|f\|_1^2 + \|f\|_{H^m}^2 .$$

Split the Fourier integral at radius one, use $|\widehat{f}| \leq (2\pi)^{-3/2}\|f\|_1$, and evaluate $\int_{|\xi|\leq 1}|\xi|^{-2}\,\mathrm{d}\xi = 4\pi$. The high-frequency part is bounded by the displayed $H^m$ norm. Every temporal derivative satisfies this estimate as well.

If $q$ is the unforced pressure, set $p = q + \phi$. The forced momentum equation follows directly, and all velocity coordinates are those of the unforced history. The source potential supplies the additional pressure jet. □

The restriction on the force is also necessary if the velocity is to be retained. To see the precise obstruction, fix a smooth time and allow the pressure to vary. For any pressure $q$ with an admissible Sobolev gradient, its unforced momentum residual is

$$\mathcal{R}_q := u_t + (u\cdot\nabla)u - \nu\Delta u + \nabla q = f + \nabla(q-p).$$

The solenoidal part is independent of that choice:

$$\mathbb{P}\mathcal{R}_q = \mathbb{P}f.$$

Orthogonality of gradient and solenoidal fields in $H^m$ gives

$$\begin{aligned} \|\mathcal{R}_q\|_{H^m}^2 &= \|\mathbb{P}f\|_{H^m}^2 + \|(I-\mathbb{P})f + \nabla(q-p)\|_{H^m}^2 , \\ \inf_q \|\mathcal{R}_q\|_{H^m} &= \|\mathbb{P}f\|_{H^m} . \end{aligned} \tag{957}$$

The infimum is attained at $q = p - \phi$, where $\nabla\phi = (I-\mathbb{P})f$. The Fourier formula above, applied to the original $f$, produces this potential. Thus pressure can convert the forced velocity into an

unforced solution exactly when $\mathbb{P}f = 0$. Subtracting a source coordinate from a completed row does not change this residual of the velocity itself.

Pressure normalization also matters in comparisons with periodic forcing results. Tao's Proposition 1.7 equates unforced and forced periodic regularity when nonperiodic pressure is permitted [15, pp. 28–29]. Its accelerated Galilean transformation does not preserve finite energy or $H^1(\mathbb{R}^3)$, as stated on page 43. That transformation is not the route used here: Part III proves the compact smoothly forced problem directly without importing an unforced velocity.

Proposition K.1 also applies on a final interval if $\mathbb{P}f$ vanishes there: use the smooth velocity at its initial time as the datum for the unforced continuation, and join by local uniqueness. Vanishing of force derivatives at a single spatial point does not meet this hypothesis. The next calculation distinguishes those two properties in the actual localized source.

# L The terminal rotational force in OpenAI's construction

The localized velocity and pressure in [18, Section 10] define the candidate source through the momentum equation. On the regular exterior and cutoffs, these formulas determine its terminal rotational component directly.

First set viscosity to one. In the paper's heat exterior, write $\tau = 1-t$, $A = 1/2+h$, $0 < h < 1/100$. Where the compactly supported vector-potential correction is absent, its localized field is

$$u = \chi(r,z)K(r,t)e_\theta, \qquad p = \chi(r,z)p_{\mathrm{ext}}(r,t), \qquad K = 2^A\kappa r^{-2A}H(4\tau/r^2), \tag{958}$$

with $\kappa > 0$. This amplitude is unrelated to our Sobolev constant $c_\infty$. The time cutoff equals one near $t = 1$. The heat profile obeys

$$\partial_t K = (\partial_{rr} + r^{-1}\partial_r - r^{-2})K.$$

The radial heat expansion in [18, Appendix A, (A.32)–(A.35)] gives, with $(b)_j$ denoting the rising factorial,

$$K_j(r) := \partial_t^j K(r,1) = 2^A\kappa\, 4^j (h)_j (1+h)_j\, r^{-1-2h-2j} > 0. \tag{959}$$

The spatial cutoff satisfies $0 \le \chi \le 1$, equals one for $r \le r_0/2$, $|z| \le z_0/2$, and has support compactly contained in $r < r_0$, $|z| < z_0$ [18, Section 10, (10.3)–(10.5)]. There is a connected rectangle

$$\Omega = (r_0/4, 2r_0) \times (-\varepsilon, \varepsilon)$$

lying in the heat exterior for all $t$ sufficiently close to one. To verify this region, use the source bound $q \le C_0(\tau + |z|^{1/D})$, $D = 1/2 - h > 0$, and the exterior condition $X = r^2/(2q) \ge X_{\mathrm{ext}}$. Here $q$ is the positive scale function used by that construction; $C_0, q_*, X_{\mathrm{ext}}$ are its fixed comparison and exterior constants. Choose

$$q_b = \min\{q_*/2, r_0^2/(64X_{\mathrm{ext}})\}, \qquad C_0(\eta + \varepsilon^{1/D}) < q_b, \quad \varepsilon < z_0/2.$$

Then $X > 2X_{\mathrm{ext}}$ on $\Omega$ for $1-\eta < t < 1$, as required by the exact exterior construction [18, Section 10, Step 4, p. 116].

The uncut swirl satisfies its azimuthal heat equation. Multiplication by $\chi$ creates spatial gradients at the edge of the cutoff. Pressure has no azimuthal component, and axisymmetric pure swirl has radial transport acceleration. The entire azimuthal residual is thus

$$f_\theta = -K(\chi_{rr} + r^{-1}\chi_r + \chi_{zz}) - 2K_r\chi_r. \tag{960}$$

This is a direct momentum calculation. It identifies the force needed to maintain the cutoff of a diffusing swirl.

**Proposition L.1** (Nonzero projected terminal jets)**.** *For the localized source defined above, each finite terminal temporal jet $F_j = \partial_t^j f(\cdot, 1)$ has*

$$\mathbb{P}F_j \not\equiv 0, \qquad j = 0, 1, 2, \dots.$$

*This persists under the viscosity rescaling in the source paper.*

*Proof.* Differentiate (960) and use $K_j'/K_j = -(1 + 2h + 2j)/r$. On $\Omega$,

$$(F_j)_\theta = -K_j \left( \chi_{rr} + \chi_{zz} - \frac{1 + 4h + 4j}{r} \chi_r \right). \tag{961}$$

If this component vanished throughout $\Omega$, the cutoff would solve the uniformly elliptic equation

$$\left( \partial_{rr} + \partial_{zz} - \frac{1 + 4h + 4j}{r} \partial_r \right) \chi = 0$$

on that connected rectangle. Its coefficients are bounded there, since $r \geq r_0/4$. The cutoff attains its maximum one at the interior point $(3r_0/8, 0)$, and vanishes at the interior point $(3r_0/2, 0)$. The strong maximum principle forbids both properties. Consequently $(F_j)_\theta$ is nonzero somewhere in $\Omega$.

Choose a smooth compactly supported $\psi(r, z)$ in a region where this component has strict sign. The three-dimensional test field $w = \psi(r, z) e_\theta$ is smooth, supported away from the axis, and divergence free. With its sign chosen appropriately,

$$\langle \mathbb{P}F_j, w \rangle = \langle F_j, w \rangle = 2\pi \int (F_j)_\theta \psi(r, z)\, r \, \mathrm{d}r \, \mathrm{d}z > 0.$$

This proves the projected nonvanishing. The rescaling $f_\nu(x, t) = \sqrt{\nu}\, f(x/\sqrt{\nu}, t)$ [18, Section 10, (10.22)] commutes with the Leray projection and preserves that conclusion. □

The source paper makes its residual flat at the spatial origin at the terminal time. Proposition L.1 locates nonzero terminal force elsewhere. Smooth flatness at the proposed singular point is compatible with a nonzero field in the surrounding cutoff region. The exterior test also gives a quantitative obstruction to that pressure conversion, without requiring convergence of the force on the whole space. Use the field $w$ from the proof at $j = 0$, and set

$$d := 2\pi \int (F_0)_\theta \psi(r, z)\, r \, \mathrm{d}r \, \mathrm{d}z > 0.$$

On the fixed support of $w$, the heat formula gives $\langle f(t), w \rangle \to d$. Choose $\eta_1 > 0$ so that this pairing is at least $d/2$ for $1 - \eta_1 < t < 1$. Since every gradient pairs to zero with $w$, Sobolev duality and (957) give, for each integer $m \geq 0$,

$$\inf_q \|\mathcal{R}_q(t)\|_{H^m} = \|\mathbb{P}f(t)\|_{H^m} \geq \frac{d}{2\,\|w\|_{H^{-m}}} > 0, \qquad 1 - \eta_1 < t < 1. \tag{962}$$

The positive lower bound prevents any pressure adjustment from turning this velocity into an unforced solution on a final interval. It is an obstruction to the conservative-force reduction, not a sign condition on the total work of the source.

# M A direct bound from the critical source norm

The source can also be tested before velocity-intersection membership is known. Begin with a classical forced history on its interval of existence and retain the nonlinear current in the critical energy balance. Put

$$y = \|u\|_{\dot H^{1/2}}, \qquad z = \|u\|_{\dot H^{3/2}}, \qquad \beta = \|\mathbb{P} f\|_{\dot H^{-1/2}}, \qquad C_* = \frac{1}{\sqrt{2}\pi}.$$

The product estimate used in (694), applied before substituting any rectangle bound, gives

$$y y' + (\nu - C_* y) z^2 \le \beta z. \tag{963}$$

This calculation is on a strict classical interval. It assumes no bound on a velocity coordinate at the terminal time.

As long as $y < \nu / C_*$, completing the square in $z$ yields

$$y y' \le \frac{\beta^2}{4(\nu - C_* y)}.$$

Multiplication by the positive denominator makes the left side an exact derivative. Define

$$\Psi(s) = 2\nu s^2 - \frac{4}{3} C_* s^3, \qquad B(t) = \int_0^t \beta(s)^2 \, \mathrm{d}s.$$

Then

$$\Psi(y(t)) \le \Psi(y(0)) + B(t), \qquad \Psi'(s) = 4(\nu - C_* s) s. \tag{964}$$

The function $\Psi$ is strictly increasing between zero and $\nu / C_*$. At the upper endpoint it takes the value

$$\Psi(\nu / C_*) = \frac{2\nu^3}{3 C_*^2} = \frac{4\pi^2}{3} \nu^3.$$

**Theorem M.1** (Critical source criterion)**.** *Under the datum and source assumptions of this paper, suppose $y_0 = \|u_0\|_{\dot H^{1/2}} < \nu / C_*$ and*

$$B(T) < \frac{4\pi^2}{3} \nu^3 - \Psi(y_0). \tag{965}$$

*The forced history extends smoothly through $T$ and belongs to $\mathscr{I}_T$. In particular, for rest data the sufficient condition is $B(T) < 4\pi^2 \nu^3 / 3$.*

*Proof.* Let $r < \nu / C_*$ be the unique nonnegative number satisfying $\Psi(r) = \Psi(y_0) + B(T)$. If $y$ first exceeded $r$ before reaching $\nu / C_*$, (964) would fail. Continuity thus gives $y \le r$ throughout classical existence up to $T$. At a zero of $y$, the energy identity for $y^2$ and absolute continuity justify the integrated inequality without dividing by $y$.

Write $a = \nu - C_* r > 0$. Young's inequality in (963) now gives

$$y y' + \frac{a}{2} z^2 \le \frac{\beta^2}{2a}, \qquad \int_0^S z^2 \, \mathrm{d}t \le \frac{y_0^2}{a} + \frac{B(T)}{a^2} \quad (S < T).$$

These are bounds obtained from the source and initial datum. To see directly how they prevent an endpoint loss, put $Y = \|\nabla u\|_2$ and $D = \|\Delta u\|_2$. Differentiation of the momentum equation, integration by parts, and incompressibility give

$$\frac{1}{2}(Y^2)' + \nu D^2 \le \|\nabla u\|_3 \, \|\nabla u\|_2 \, \|\nabla u\|_6 + \|\mathbb{P} f\|_2 \, D.$$

Let $C_S$ be a finite constant in $\|\nabla u\|_6 \le C_S \, \|\Delta u\|_2$, and let $C_3 = 2^{-1/6}\pi^{-1/3}$ as above. Since $\|\nabla u\|_3 \le C_3 z$, two applications of Young's inequality give

$$(Y^2)' + \nu D^2 \le \frac{2(C_S C_3)^2}{\nu} z^2 Y^2 + \frac{2}{\nu} \, \|\mathbb{P} f\|_2^2 \, .$$

Gronwall's inequality bounds $Y$ uniformly. The ordinary energy identity also gives $\|u(t)\|_2 \le \|u_0\|_2 + \int_0^t \|\mathbb{P} f(s)\|_2 \, \mathrm{d}s$. Thus the full $H^1$ norm remains bounded.

For completeness, this bound supplies a uniform local continuation time. The estimate $\|(v \cdot \nabla) w\|_{3/2} \le C \, \|v\|_{H^1} \, \|w\|_{H^1}$, the boundedness of $\mathbb{P}$ on $L^{3/2}$, and the heat estimates from $L^{3/2}$ to $H^1$ bound the bilinear mild map on a time interval of length $h$ by

$$C_\nu (h^{1/4} + h^{3/4}) \, \|v\|_{C_t H^1} \, \|w\|_{C_t H^1} \, .$$

The two powers integrate the heat factors $t^{-3/4}$ and $t^{-1/4}$. A contraction ball of fixed radius now has a fixed positive existence time, using also the bounded $H^1$ norm of the source. Restarting near a proposed finite endpoint continues the history beyond it. Heat smoothing at positive elapsed time and the smooth source preserve every finite spatial order; differentiation of the equation then preserves the full temporal jet. Together with the smooth initial datum and source, this proves $\mathscr{I}_T$ membership and the endpoint conclusions. □

The threshold is unchanged by the Navier–Stokes rescaling $u_\lambda(x,t) = \lambda u(\lambda x, \lambda^2 t)$, $f_\lambda(x,t) = \lambda^3 f(\lambda x, \lambda^2 t)$. The squared $\dot{H}^{-1/2}$ source norm gains a factor $\lambda^2$, which its time integral cancels.

## M.1 A necessary $H^3$ source budget

For a candidate that starts from rest, write

$$Y(t) = \|u(t)\|_{H^3} \, , \qquad D(t) = \|\nabla u(t)\|_{H^3} \, , \qquad g(t) = \|\mathbb{P} f(t)\|_{H^3} \, , \qquad G_T = \int_0^T g(t) \, \mathrm{d}t.$$

On each strict classical interval, the momentum equation and (705) give

$$YY' + \nu D^2 \le 8 c_\infty Y^2 D + gY, \qquad c_\infty = (8\pi)^{-1/2}.$$

Completing the entire viscous square yields

$$8 c_\infty Y^2 D - \nu D^2 \le \frac{16 c_\infty^2}{\nu} Y^4 = \frac{2}{\pi \nu} Y^4.$$

Thus $Y' \le 2Y^3/(\pi\nu) + g$ wherever $Y > 0$. At zeros, use $Y_\varepsilon = (Y^2 + \varepsilon^2)^{1/2}$; its derivative obeys $Y_\varepsilon' \le 2Y_\varepsilon^3/(\pi\nu) + g$.

If $G_T = \infty$, the finite source condition already fails. Otherwise put

$$Z_\varepsilon(t) = Y_\varepsilon(t) + \int_t^T g(s) \, \mathrm{d}s.$$

Since $Y_\varepsilon \le Z_\varepsilon$,

$$Z_\varepsilon' \le \frac{2}{\pi\nu} Z_\varepsilon^3, \qquad Z_\varepsilon(0) = G_T + \varepsilon.$$

Integrating this scalar comparison and taking $\varepsilon \downarrow 0$ gives

$$Y(t) \le \frac{G_T}{\sqrt{1 - 4G_T^2 t/(\pi\nu)}} \quad \text{if} \quad G_T < \sqrt{\frac{\pi\nu}{4T}}. \tag{966}$$

This bound uses the finite source budget and strict-interval equations; it assumes neither a terminal velocity norm nor smooth continuation of force through $T$.

The calculated exterior has an unbounded $L^3$ norm as $t \uparrow 1$. Since $H^3$ controls $\dot{H}^{1/2}$, which embeds into $L^3$, its $H^3$ norm is unbounded as well. Equation (966) thus proves the following lower bound for its actual residual:

$$G_{\mathrm{OA}} \ge \frac{\sqrt{\pi\nu}}{2}. \tag{967}$$

## M.2 Strict failure of the critical source condition

For the prescribed exterior, cylindrical integration on $\mathcal{E}_\tau$ gives an explicit critical lower bound:

$$\int_{\mathcal{E}_\tau} |u|^3 \,\mathrm{d}x = C_L \tau^{-4h}, \qquad C_L = 4\pi d \int_c^{2c} k(y)^3 y \,\mathrm{d}y > 0.$$

The Sobolev inequality implies

$$\|u\|_{\dot{H}^{1/2}} \ge C_3^{-1} C_L^{1/3} \tau^{-4h/3} \longrightarrow \infty.$$

The viscosity rescaling preserves this divergence. Continuity gives a first time $S < 1$ at which $y(S) = \nu/C_*$, with $y(t) < \nu/C_*$ for $0 \le t < S$. Integrate (964) before $S$, then pass to $S$ by continuity. The rest datum gives

$$B(S) \ge \Psi(\nu/C_*) = \frac{4\pi^2\nu^3}{3}. \tag{968}$$

The bound follows from the explicit exterior and momentum equation on strict subintervals of $[0, 1)$.

The residual continues to contribute a positive source norm after this crossing. Use the divergence-free test field $w$ from the fixed exterior calculation leading to (962), and write $\gamma_w := \lim_{t\uparrow 1} \langle f(t), w\rangle > 0$. For some $\eta_1 > 0$, local convergence on its support gives $\langle f(t), w\rangle \ge \gamma_w/2$ for $1 - \eta_1 < t < 1$, independently of terminal convergence elsewhere. Homogeneous Sobolev duality gives

$$\frac{\gamma_w}{2} \le \langle \mathbb{P} f(t), w\rangle \le \|\mathbb{P} f(t)\|_{\dot{H}^{-1/2}} \|w\|_{\dot{H}^{1/2}}.$$

The test norm is finite and positive because $w$ is smooth, compactly supported, and nonzero. Let $a = \max\{S, 1 - \eta_1\} < 1$. Integrating over $(a, 1)$, which is disjoint from $(0, S)$, yields

$$\begin{aligned} B(1) &\ge \frac{4\pi^2\nu^3}{3} + \frac{(1-a)\gamma_w^2}{4\|w\|_{\dot{H}^{1/2}}^2} \\ &> \frac{4\pi^2\nu^3}{3}. \end{aligned} \tag{969}$$

For general viscosity, use the rescaled exterior and test field; the source budget and positive correction both scale by $\nu^3$. The calculation also permits $B(1) = \infty$. Thus no global terminal source regularity is needed to establish failure of this sufficient criterion. The source may be smooth and have finite norms, but these particular quantitative estimates do not cover its magnitude. This failure does not restrict the global theorem of Part III.

## Final conclusion

The datum-generated compatible intersection proves global smoothness and uniqueness first for the unforced equations and then, on a separate history carrying its complete source jet, for the forced equations. The localized residual of the field proposed in OpenAI's *Finite Time Blowup for Navier–Stokes* belongs to the forced theorem's source class. The theorem produces a global smooth solution for those exact prescribed data, while strict-subinterval uniqueness identifies any classical candidate with that solution on $[0, 1)$. Its finite velocity and mixed-rectangle bounds are incompatible with the proposed field's unbounded velocity and with the divergent positive exterior contribution to $R_{0,1}$ calculated above. The proposed field does not solve the stated forced initial-value problem.

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